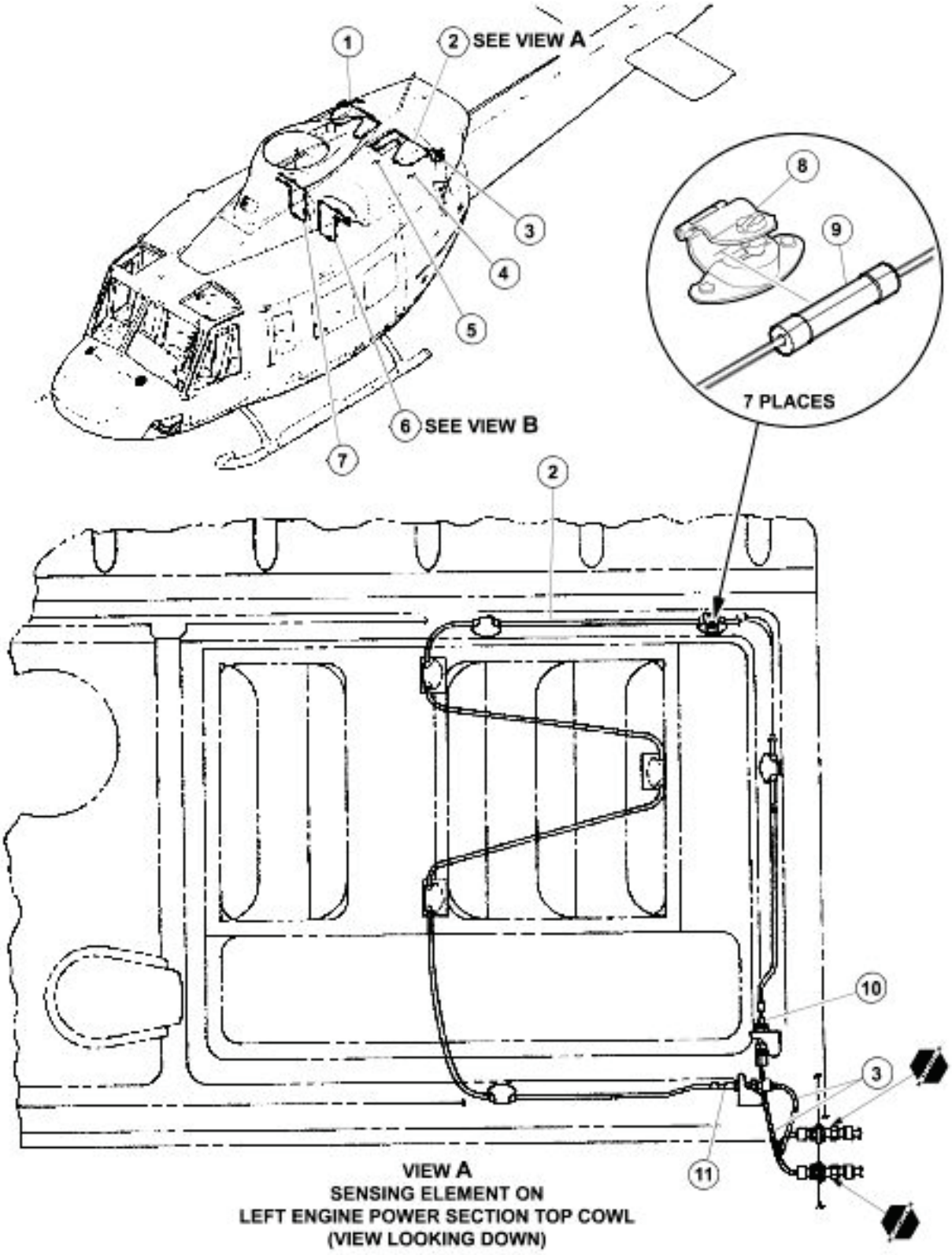
Engine Firewall Fire Detection Sensing Elements - Removal

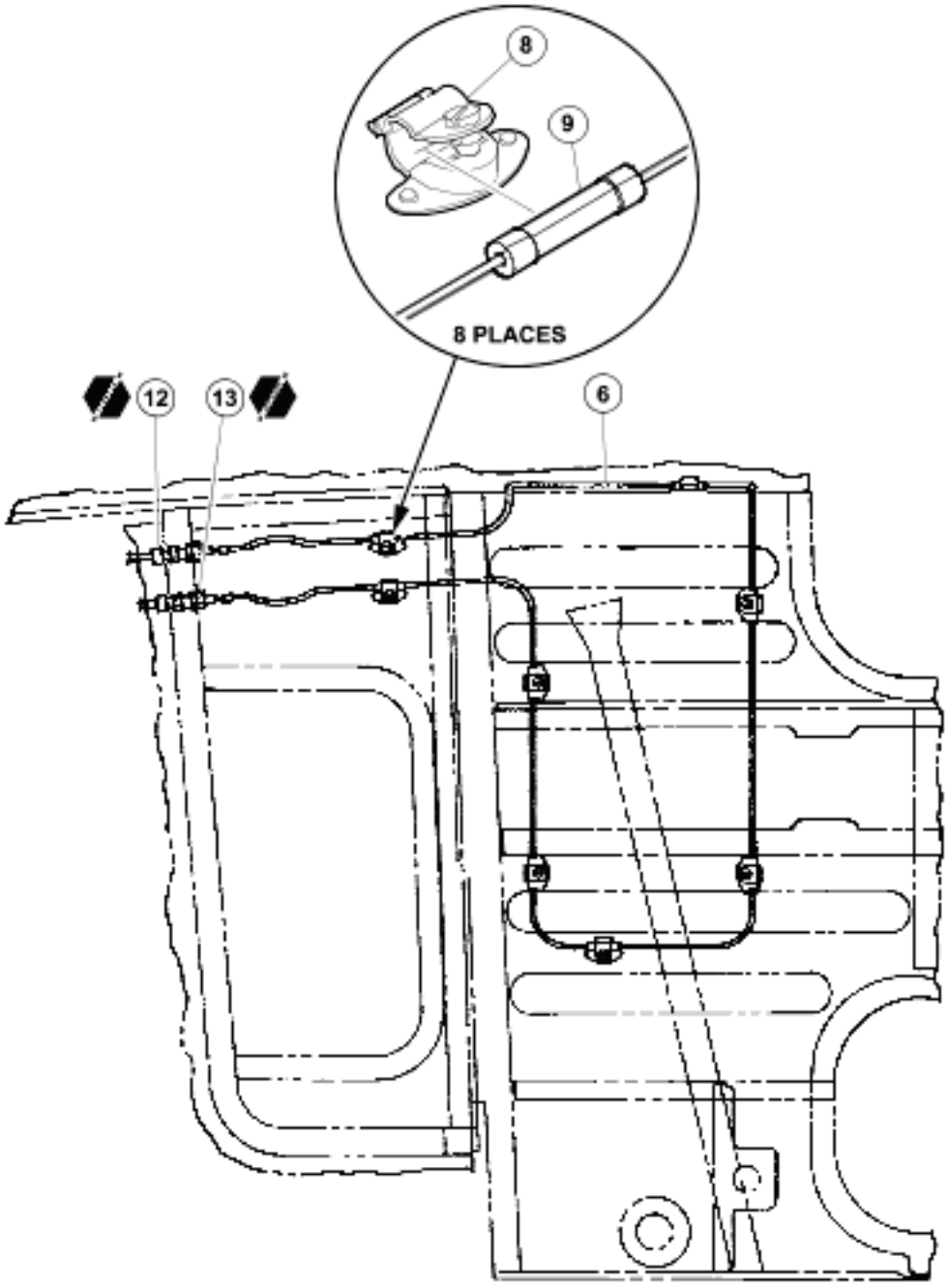
Procedure

1. Verify all electrical power is removed from the helicopter.
2. Disconnect the electrical connectors (12 and 13, Figure 1).
3. Untighten the studs of the clamps (8).
4. Spread the clamps (8) as little as possible and remove the sensing element (6).



ICN-412-A-261000-B-97499-00001-A-001-01

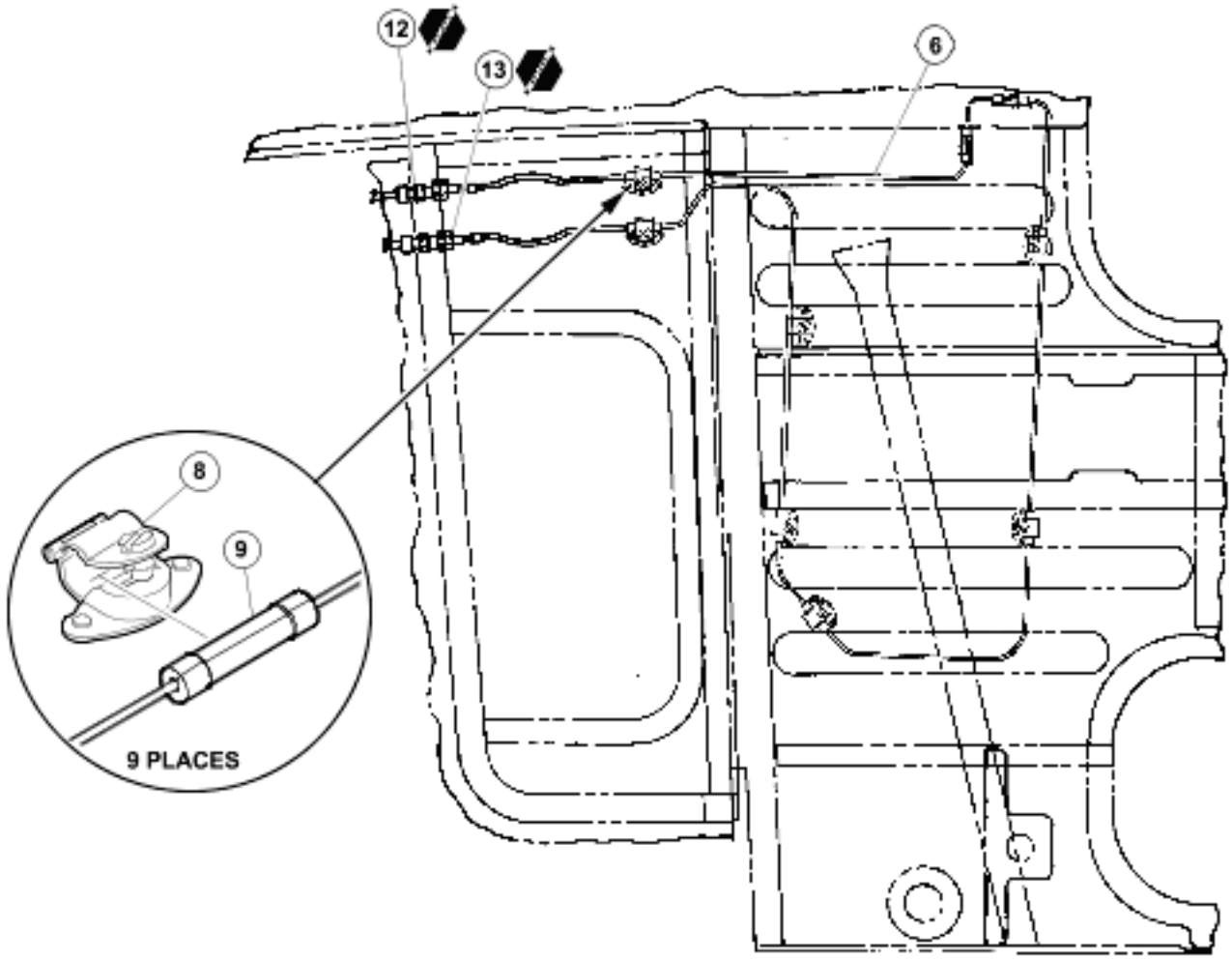
Figure 1. Engine Compartment Fire Detection Sensing Element (Sheet 1 of 3)



VIEW B
SENSING ELEMENT ON
LEFT ENGINE POWER SECTION FIREWALL
(VIEW LOOKING FORWARD)
(S/N 33001 THROUGH 36693)

ICN-412-A-261000-B-97499-00002-A-001-01

Figure 1. Engine Compartment Fire Detection Sensing Element (Sheet 2 of 3)



VIEW B
SENSING ELEMENT ON LEFT ENGINE
POWER SECTION FIREWALL
(S/N 36694 AND SUBSEQUENT)

1. Sensing element (right engine power section top cowl)
2. Sensing element (left engine power section top cowl)
3. Flex cable
4. Upper engine cowling
5. Engine cowl top panel
6. Sensing element (left power section firewall)
7. Sensing element (right power section firewall)
8. Clamp
9. Grommet
10. Electrical connector (8Z7P1)
11. Electrical connector (8Z7P2)
12. Electrical connector (8Z9P1)
13. Electrical connector (8Z9P2)

LOCKWIRE (C-405)

ICN-412-A-261000-B-97499-00003-A-001-01

Figure 1. Engine Compartment Fire Detection Sensing Element (Sheet 3 of 3)

Engine Firewall Fire Detection Sensing Elements - Inspection and Repair

Procedure

1. Refer to 412-A-96-00-00-00A-009A-A / 01870 for the inspection of the sensing elements. Replace the elements and/or retaining hardware if unserviceable.

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Engine Firewall Fire Detection Sensing Elements - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

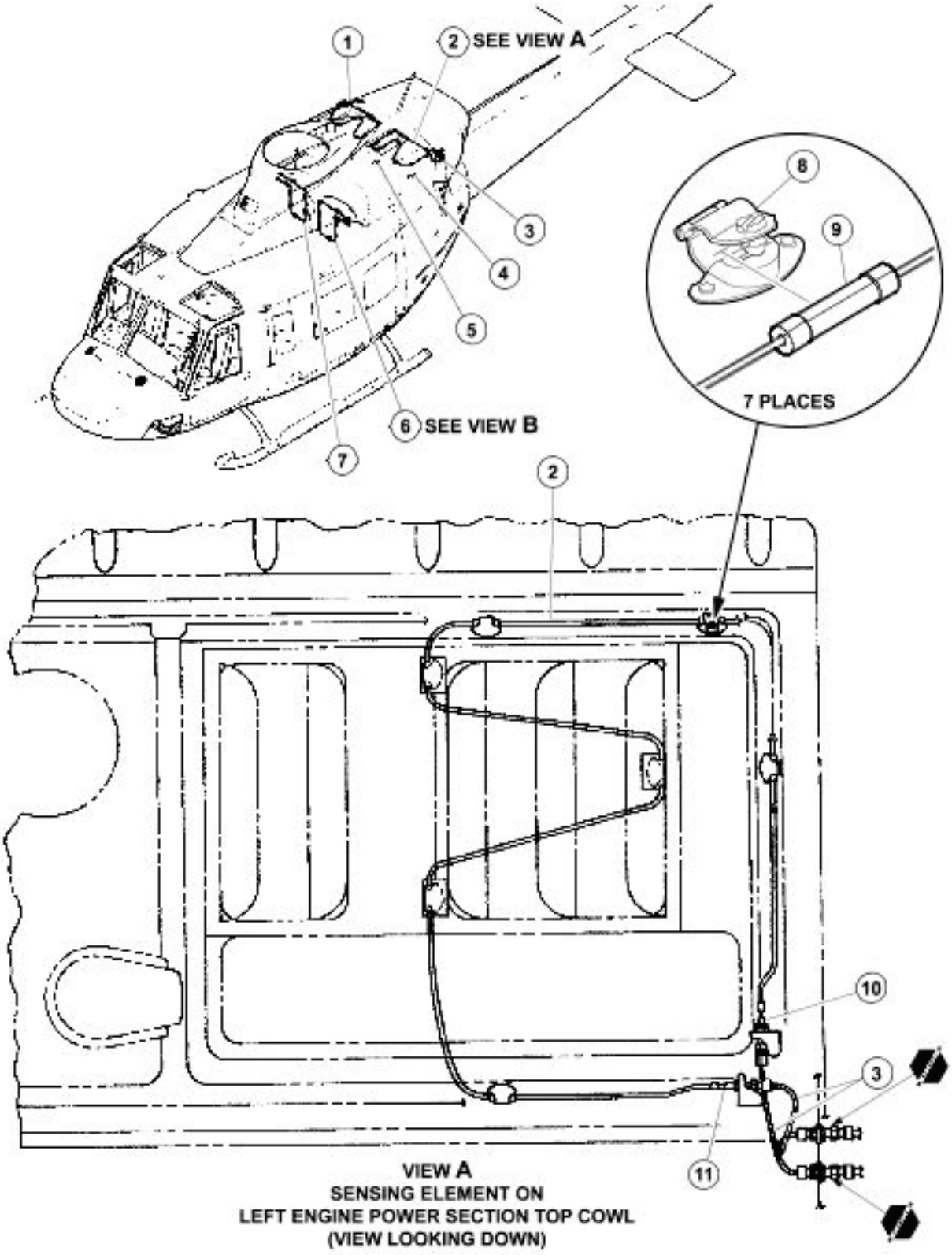
1. Verify all electrical power is removed from the helicopter.

2. Position the sensing element (6, Figure 11) in the clamps (8).

3. Tighten the studs in the clamps (8).

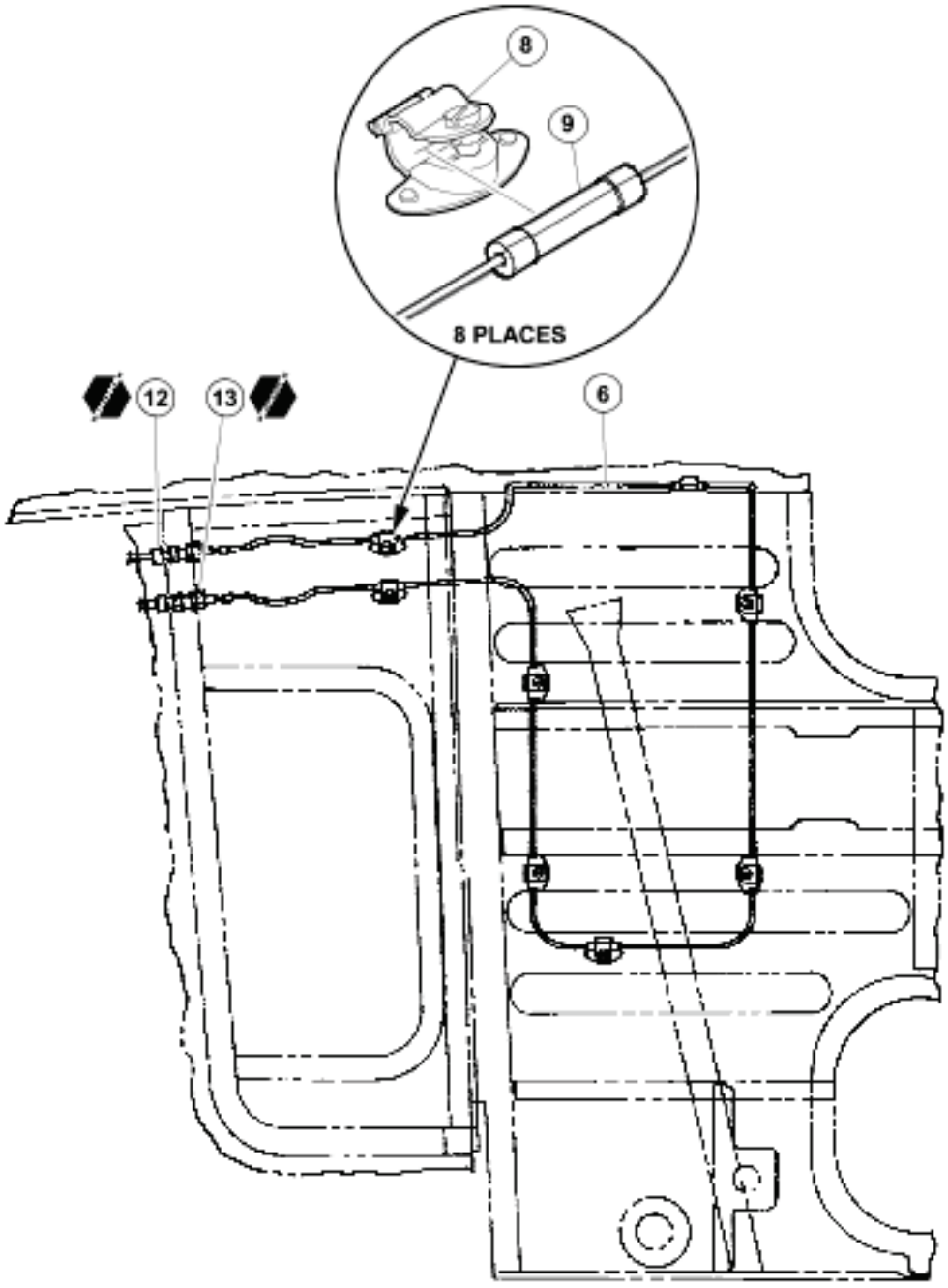
4. Connect the electrical connectors (12 and 13). Secure with lockwire (C-405).

5. Perform an operational check (412-A-26-10-00-02A-320A-A / 00302).



ICN-412-A-261000-B-97499-00001-A-001-01

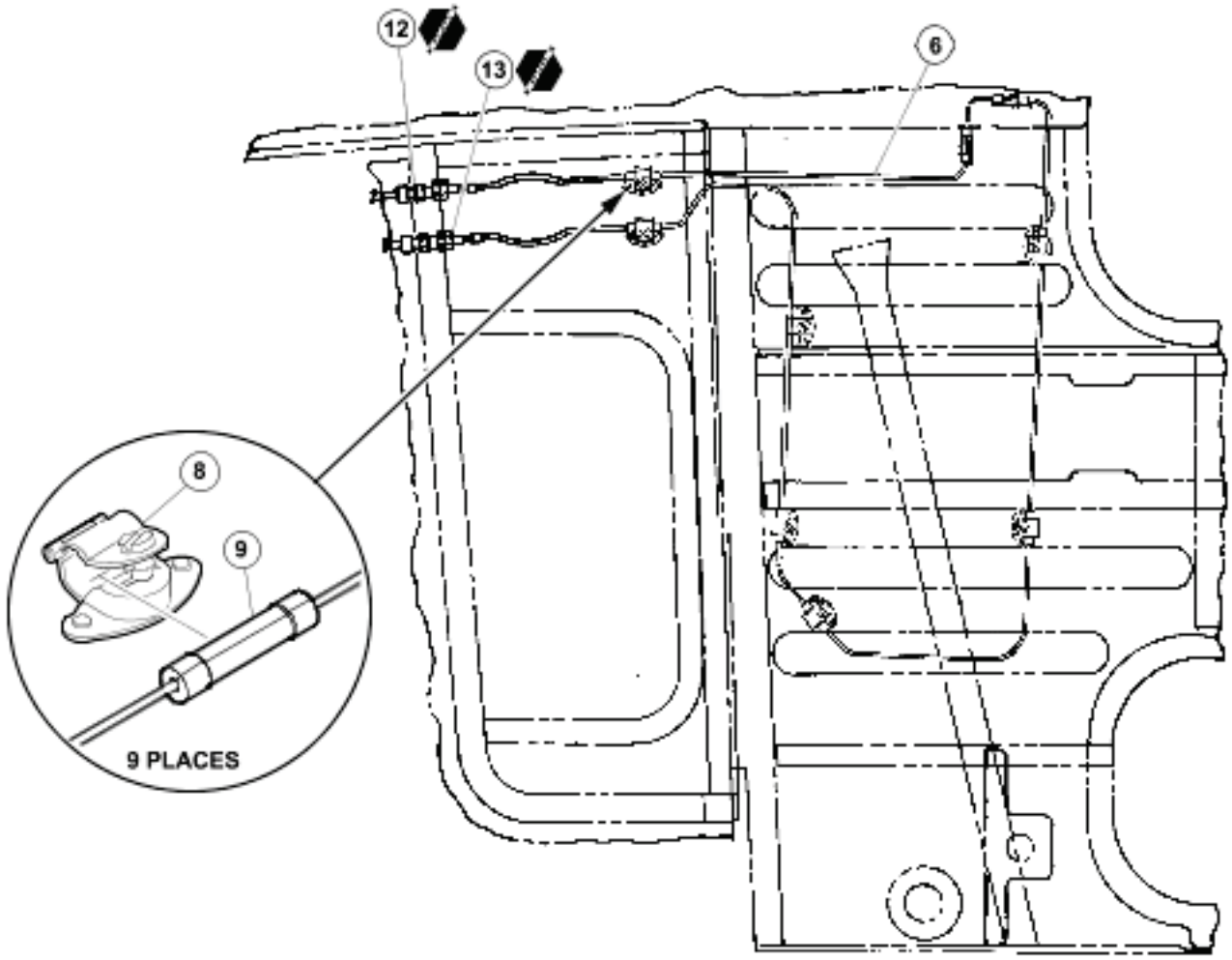
Figure 1. Engine Compartment Fire Detection Sensing Element (Sheet 1 of 3)



VIEW B
SENSING ELEMENT ON
LEFT ENGINE POWER SECTION FIREWALL
(VIEW LOOKING FORWARD)
(S/N 33001 THROUGH 36693)

ICN-412-A-261000-B-97499-00002-A-001-01

Figure 1. Engine Compartment Fire Detection Sensing Element (Sheet 2 of 3)



VIEW B
SENSING ELEMENT ON LEFT ENGINE
POWER SECTION FIREWALL
(S/N 36694 AND SUBSEQUENT)

1. Sensing element (right engine power section top cowl)
2. Sensing element (left engine power section top cowl)
3. Flex cable
4. Upper engine cowling
5. Engine cowl top panel
6. Sensing element (left power section firewall)
7. Sensing element (right power section firewall)
8. Clamp
9. Grommet
10. Electrical connector (8Z7P1)
11. Electrical connector (8Z7P2)
12. Electrical connector (8Z9P1)
13. Electrical connector (8Z9P2)

LOCKWIRE (C-405)

ICN-412-A-261000-B-97499-00003-A-001-01

Figure 1. Engine Compartment Fire Detection Sensing Element (Sheet 3 of 3)

Engine Firewall Fire Detection Sensing Elements - Operational Check

Procedure

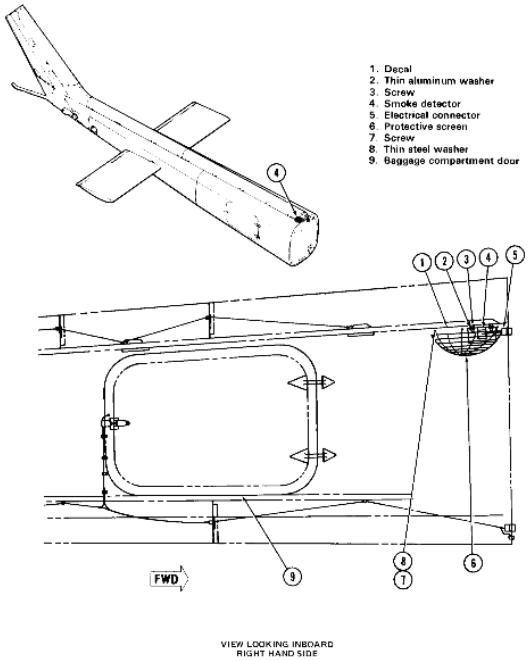
1. Apply 28 VDC to the helicopter electrical system.
2. Close the ENG 1 FIRE DET circuit breaker.
3. Press and hold the FIRE EXT PRESS TO TEST switch.
4. Verify the FIRE 1 PULL handle illuminates.
5. Release the PRESS TO TEST switch.
6. Verify the FIRE 1 PULL handle extinguishes.
7. Open the ENG 1 FIRE DET circuit breaker.
8. Close the ENG 2 FIRE DET circuit breaker.
9. Press and hold the PRESS TO TEST switch.
10. Verify the FIRE 2 PULL handle illuminates.
11. Release the PRESS TO TEST switch.
12. Verify the ENG 2 PULL handle extinguishes.
13. Remove the electrical power from helicopter.

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Baggage Compartment Smoke Detection System - Description

The baggage compartment smoke detector is a closed assembly housing solid-state electronic components and a light sensitive detector. The smoke detector (4, Figure 1) is located in the forward end of the baggage compartment roof. A protective screen is installed over the detector to provide protection from items stowed in the compartment. The instant smoke reduces light transmission in the baggage compartment 30 to 35% below that of clear air, the detector causes the BAGGAGE FIRE warning light on the instrument panel to flash intermittently. For a more detailed description and wiring diagrams, refer to 412-A-96-00-00-00A-009A-A / 01870 and 412-A-98-00-00-18B-051A-A / 02578, or 412-A-98-00-00-19B-051A-A / 02579.

The BAGGAGE FIRE TEST switch provides a means to test the baggage smoke detection system.



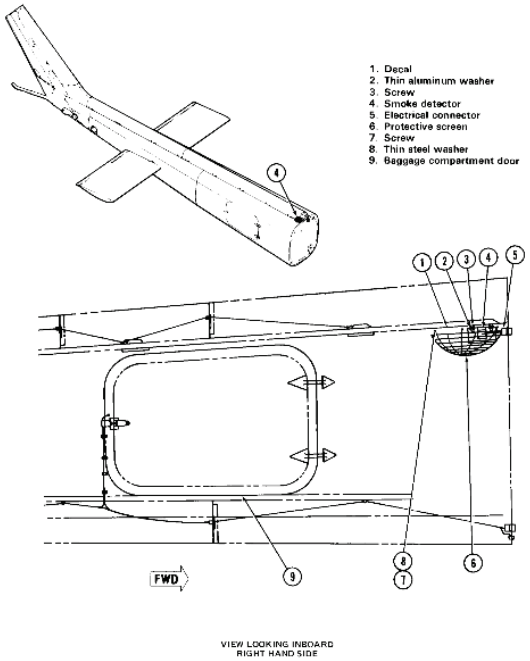
ICN-412-A-261000-B-97499-00004-A-001-01

Figure 1. Baggage Compartment Smoke Detector (Sheet 1 of 1)

Baggage Compartment Smoke Detector - Removal

Procedure

1. Remove all electrical power from helicopter.
2. Open baggage compartment door (9, Figure 1).
3. Remove screws (7) and protective screen (6).
4. Disconnect electrical connector (5).
5. Remove screws (3) and thin aluminum washers (2). Remove smoke detector (4).



ICN-412-A-261000-B-97499-00004-A-001-01

Figure 1. Baggage Compartment Smoke Detector (Sheet 1 of 1)

Baggage Compartment Smoke Detector - Inspection

Procedure

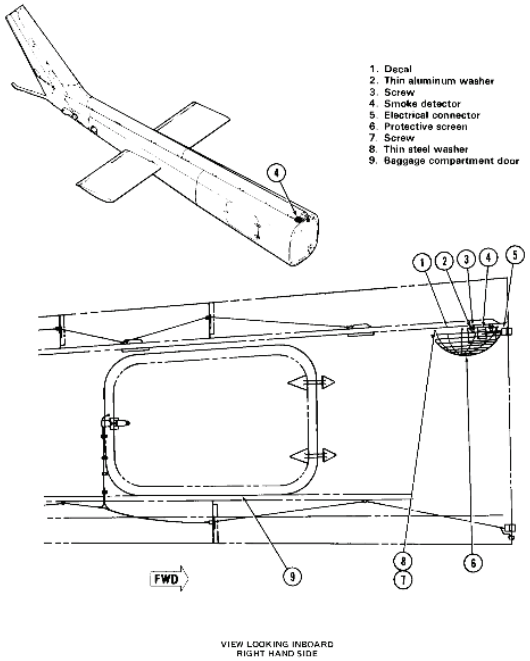
1. Inspect smoke detector for physical damage.
2. Inspect electrical connector for pin alignment and contact retention.

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Baggage Compartment Smoke Detector - Installation

Procedure

1. Align mounting holes in smoke detector (4, Figure 1) with airframe mounting holes.
2. Install screws (3) and thin aluminum washers (2).
3. Connect electrical connector (5).
4. Install protective screen (6) with screws (7) and thin steel washers (8).
5. Close baggage compartment door (9).
6. Perform operational check (412-A-26-10-00-03A-320A-A / 00307).



ICN-412-A-261000-B-97499-00004-A-001-01

Figure 1. Baggage Compartment Smoke Detector (Sheet 1 of 1)

Baggage Compartment Smoke Detector - Operational Check

Procedure

1. Apply 28 VDC power to the helicopter.
2. Close BAG COMPT FIRE circuit breaker.
3. Press and hold BAGGAGE FIRE TEST switch.
4. Verify BAGGAGE FIRE warning light flashes.
5. Release BAGGAGE FIRE TEST switch.
6. Verify BAGGAGE FIRE warning light extinguished.
7. Introduce smoke into baggage compartment.
8. Verify BAGGAGE FIRE warning light illuminates.
9. Remove smoke from baggage compartment.
10. Verify warning light extinguishes.

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Fire Extinguishers - Description

The helicopter is equipped with two engine compartment fire extinguishers and two portable, cabin mounted fire extinguishers.

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Engine Compartment Fire Extinguishers - Description

WARNING

THE FIRE EXTINGUISHER CONTAINERS ARE PRESSURIZED AND EQUIPPED WITH PYROTECHNIC CARTRIDGES. ANY TIME CONNECTING WIRES ARE DISCONNECTED, JUMPER WIRES (SHORTING DEVICES) (49 AND 50, Figure 2) SHALL BE INSTALLED TO PREVENT INADVERTENT DETONATION.

DO NOT USE A VOLTMETER, FLASHLIGHT BATTERY, CONTINUITY LIGHT, OR SIMILAR DEVICE TO TEST CARTRIDGES.

DO NOT LOOSEN FITTINGS AT DISCHARGE OUTLETS (6 OR 30) , PRESSURE INDICATOR GAUGE (26), OR FILL AND THERMAL RELIEF FITTING (15) WHEN CONTAINER IS PRESSURIZED.

MAKE SURE THAT NO POWER IS APPLIED TO HELICOPTER ELECTRICAL SYSTEMS AND HELICOPTER IS PROPERLY GROUNDED WHILE PERFORMING MAINTENANCE ON FIRE EXTINGUISHING SYSTEM.

NOTE

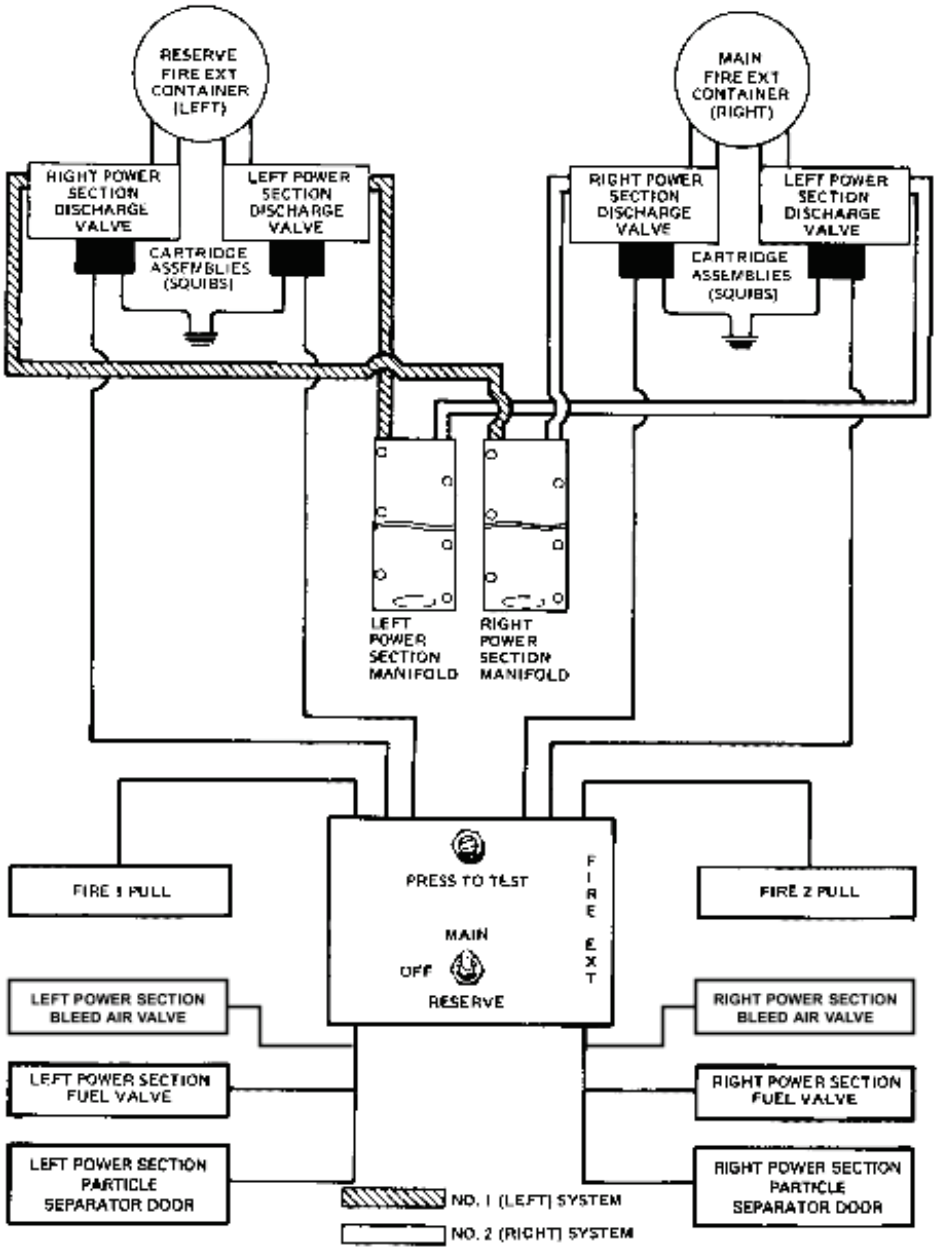
Maintenance procedures for both fire extinguisher containers are the same.

The two engine compartment fire extinguishing bottles are interconnected by tubing to allow either container to be discharged into either power section compartment (Figure 1). Therefore, in the event that discharging the first (main) bottle fails to extinguish a fire, the second (reserve) bottle may be discharged into the same compartment.

The containers (Figure 2) are spherical steel assemblies charged with freon and nitrogen to approximately 600 PSI. If the container is discharged, no special cleaning is required because of the non-corrosive properties of the extinguishing agent. Each container has two outlets each equipped with electrically actuated pyrotechnic cartridges, a guarded pressure gauge, and a fill and thermal relief fitting connected to a discharge indicator disc. A pressure/temperature correction decal is attached to each cylinder to aid during inspection and servicing.

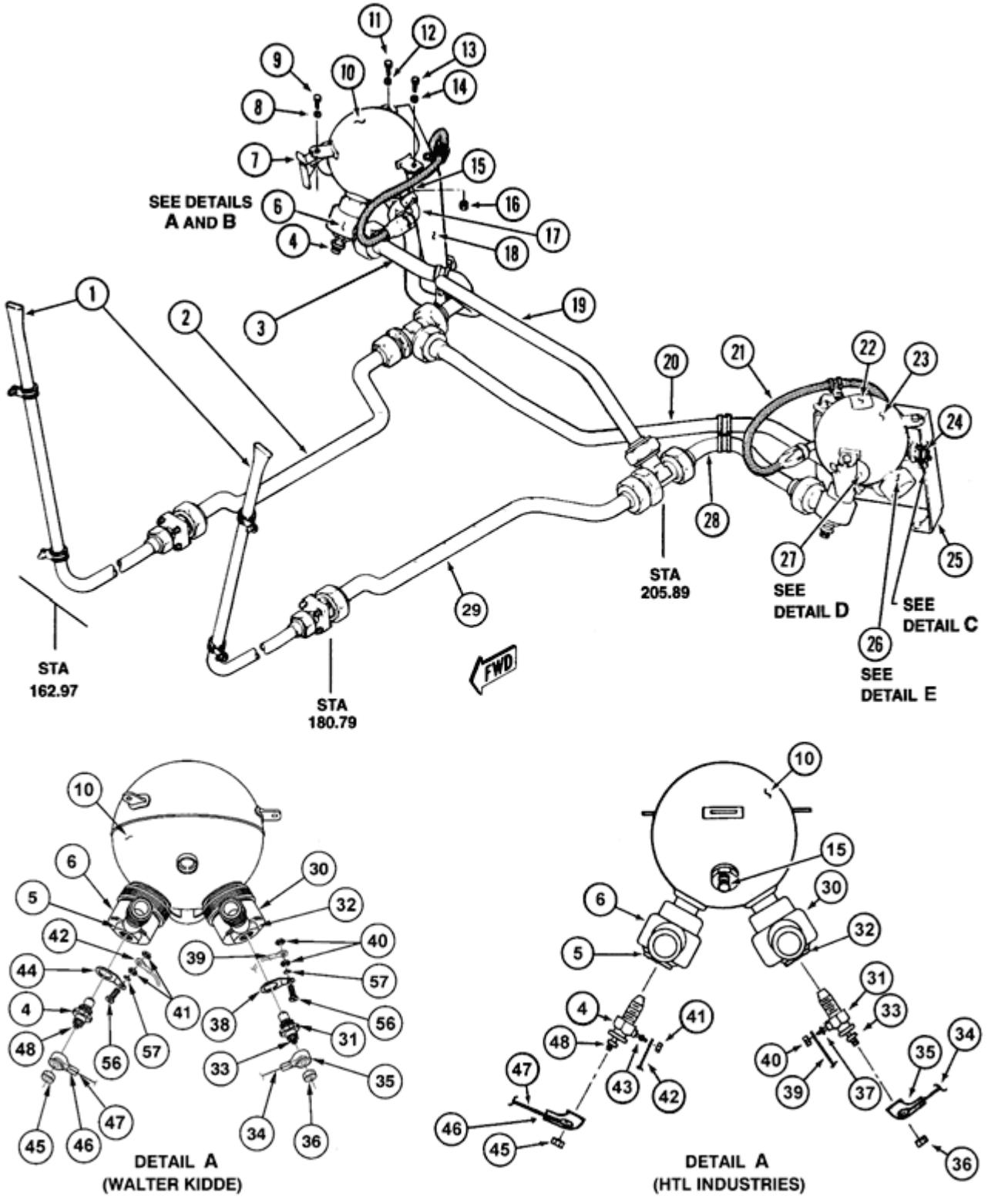
In the event a container is subjected to excessive heat, the fill and thermal relief fitting will open and allow the extinguishing agent to escape. During this process, the discharge indicator disc will be blown out to provide a visual indication of the event.

For further information, refer to 412-A-96-00-00-00A-009A-A / 01870 and 412-A-98-00-00-33A-051A-A / 02493, or 412-A-98-00-00-34A-051A-A / 02494.



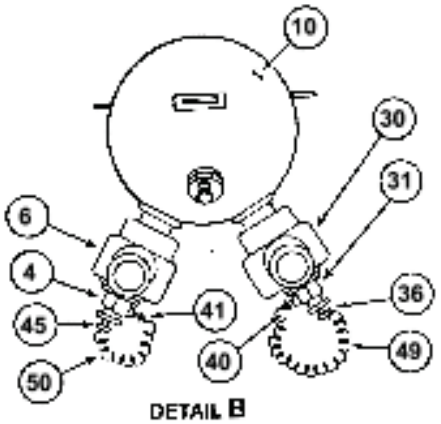
ICN-412-A-262000-B-97499-00001-A-001-01

Figure 1. Engine Compartment Fire Extinguisher Schematic (Sheet 1 of 1)

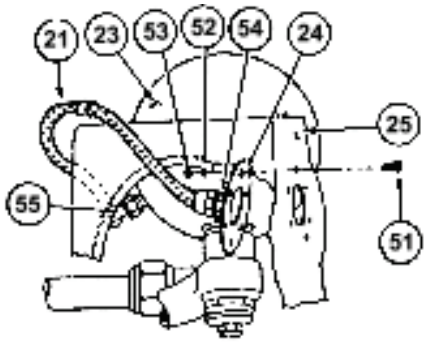


ICN-412-A-262000-B-97499-00002-A-001-01

Figure 2. Engine Compartment Fire Extinguisher (Sheet 1 of 2)



DETAIL B



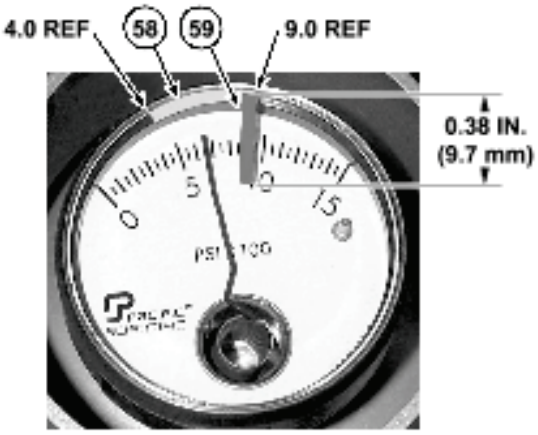
Left discharge indicator disc (24) installation shown.
Right discharge indicator disc installation is similar.

DETAIL C

TEMPERATURE/PRESSURE
CORRECTION TABLE

Temperature Degrees F	(C)	Pressure Minimum	Maximum
-65	(-54)	271	344
-60	(-51)	275	350
-40	(-40)	292	370
-20	(-29)	320	400
0	(-18)	355	437
+20	(-7)	396	485
+40	(4)	449	540
+60	(16)	518	618
+80	(27)	593	702
+100	(38)	691	784
+125	(52)	785	902

DETAIL D



DETAIL E

1. Forward tube
2. Manifold
3. Tube
4. Cartridge
5. Housing assembly
6. Discharge outlet
7. Support
8. Thin aluminum washer
9. Bolt
10. Fire extinguisher container (right)
11. Bolt
12. Thin aluminum washer
13. Bolt
14. Thin aluminum washer
15. Fill and thermal relief fitting
16. Nut
17. Hose assembly
18. Support
19. Tube
20. Tube
21. Hose
22. Name plate
23. Fire extinguisher container (left)
24. Discharge indicator disc
25. Support
26. Pressure indicator
27. Temperature/pressure correction table
28. Tube
29. Manifold
30. Discharge outlet
31. Cartridge
32. Housing assembly
33. Terminal (hot wire)
34. Electrical wire
35. Electrical terminal nipple
36. Nut
37. Terminal (ground wire)
38. Ground washer
39. Electrical wire
40. Nut
41. Nut
42. Electrical wire
43. Terminal (ground wire)
44. Ground washer
45. Nut
46. Electrical terminal nipple
47. Electrical wire
48. Terminal (hot wire)
49. Jumper wire (shorting device)
50. Jumper wire (shorting device)
51. Screw
52. Washer (AN960C6L)
53. Nut
54. Nipple
55. Fill and thermal relief fitting
56. Screw
57. Washer
58. Green range mark (31-113-1D)
59. Red range mark (31-113-6A)

ICN-412-A-262000-B-97499-00003-A-001-01

Figure 2. Engine Compartment Fire Extinguisher (Sheet 2 of 2)

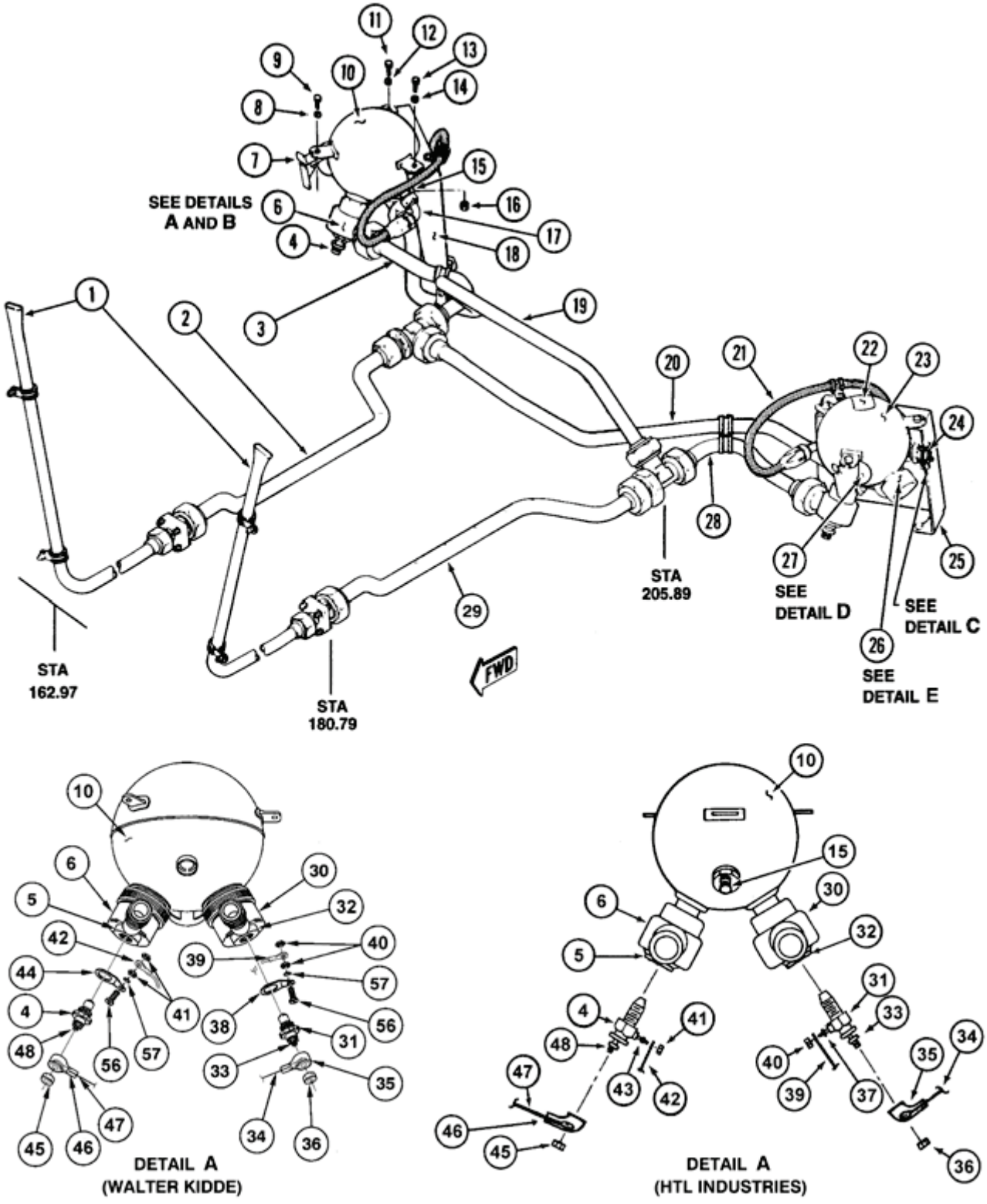
Engine Compartment Fire Extinguishers - Removal

Procedure

WARNING

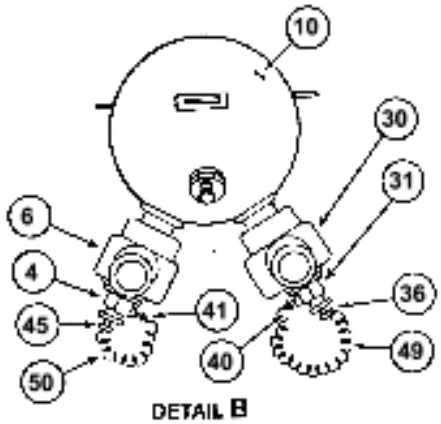
REMOVE HELICOPTER WIRING CONNECTORS ONE AT A TIME AND INSTALL JUMPER WIRE (SHORTING DEVICE) AT VACANT TERMINAL PRIOR TO REMOVING FURTHER WIRING. TAG DISCONNECTED WIRES FOR REINSTALLATION IN PROPER LOCATION.

1. Verify all electrical power is removed from helicopter and helicopter is properly grounded.
2. Open engine cowling to gain access to extinguisher bottles.
3. Remove nut (40, Figure 1). Tag and remove electrical wire (39).
4. Install jumper wire (shorting device) (49) on terminal (ground wire) (37) and install nut (40) (HTL Industries container), or on ground washer (38) and install nut (40) (Walter Kidde container).
5. Remove nut (36) and electrical wire (34) with electrical terminal nipple (35). Tag electrical wire.
6. Install loose end of jumper wire (shorting device) (49) on terminal (hot wire) (33) with nut (36).
7. Remove nut (45) and electrical wire (47) with electrical terminal nipple (46). Tag electrical wire.
8. Install jumper wire (shorting device) (50) on terminal (hot wire) (48) and install nut (45).
9. Remove nut (41) and electrical wire (42). Tag electrical wire.
10. Install loose end of jumper wire (shorting device) (50) on terminal (ground wire) (43) with nut (41) (HTL Industries), or on ground washer (44) and install nut (41) (Walter Kidde).
11. Disconnect tube (3 or 20) from discharge outlet (6).
12. Disconnect tube (19 or 28) from discharge outlet (30).
13. Remove hose assembly (17 or 21) from fill and thermal relief fitting (15 or 55).
14. Remove bolts (9 and 11) and thin aluminum washers (8 and 12).
15. Remove bolt (13), thin aluminum washer (14), and nut (16). Remove container from supports (7 and 18).

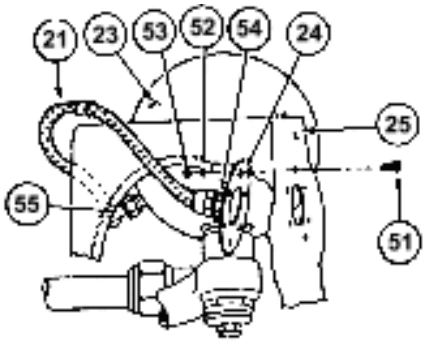


ICN-412-A-262000-B-97499-00002-A-001-01

Figure 1. Engine Compartment Fire Extinguisher (Sheet 1 of 2)



DETAIL B



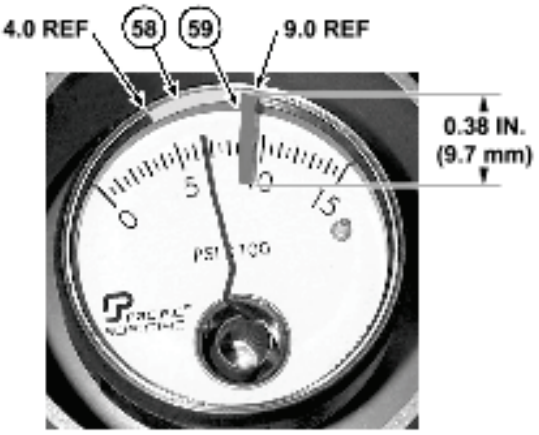
Left discharge indicator disc (24) installation shown.
Right discharge indicator disc installation is similar.

DETAIL C

TEMPERATURE/PRESSURE
CORRECTION TABLE

Temperature Degrees F	(C)	Pressure Minimum	Maximum
-65	(-54)	271	344
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+20	(-7)	396	485
+40	(4)	449	540
+60	(16)	518	618
+80	(27)	593	702
+100	(38)	691	784
+125	(52)	785	902

DETAIL D



DETAIL E

1. Forward tube
2. Manifold
3. Tube
4. Cartridge
5. Housing assembly
6. Discharge outlet
7. Support
8. Thin aluminum washer
9. Bolt
10. Fire extinguisher container (right)
11. Bolt
12. Thin aluminum washer
13. Bolt
14. Thin aluminum washer
15. Fill and thermal relief fitting
16. Nut
17. Hose assembly
18. Support
19. Tube
20. Tube
21. Hose
22. Name plate
23. Fire extinguisher container (left)
24. Discharge indicator disc
25. Support
26. Pressure indicator
27. Temperature/pressure correction table
28. Tube
29. Manifold
30. Discharge outlet
31. Cartridge
32. Housing assembly
33. Terminal (hot wire)
34. Electrical wire
35. Electrical terminal nipple
36. Nut
37. Terminal (ground wire)
38. Ground washer
39. Electrical wire
40. Nut
41. Nut
42. Electrical wire
43. Terminal (ground wire)
44. Ground washer
45. Nut
46. Electrical terminal nipple
47. Electrical wire
48. Terminal (hot wire)
49. Jumper wire (shorting device)
50. Jumper wire (shorting device)
51. Screw
52. Washer (AN960C6L)
53. Nut
54. Nipple
55. Fill and thermal relief fitting
56. Screw
57. Washer
58. Green range mark (31-113-1D)
59. Red range mark (31-113-6A)

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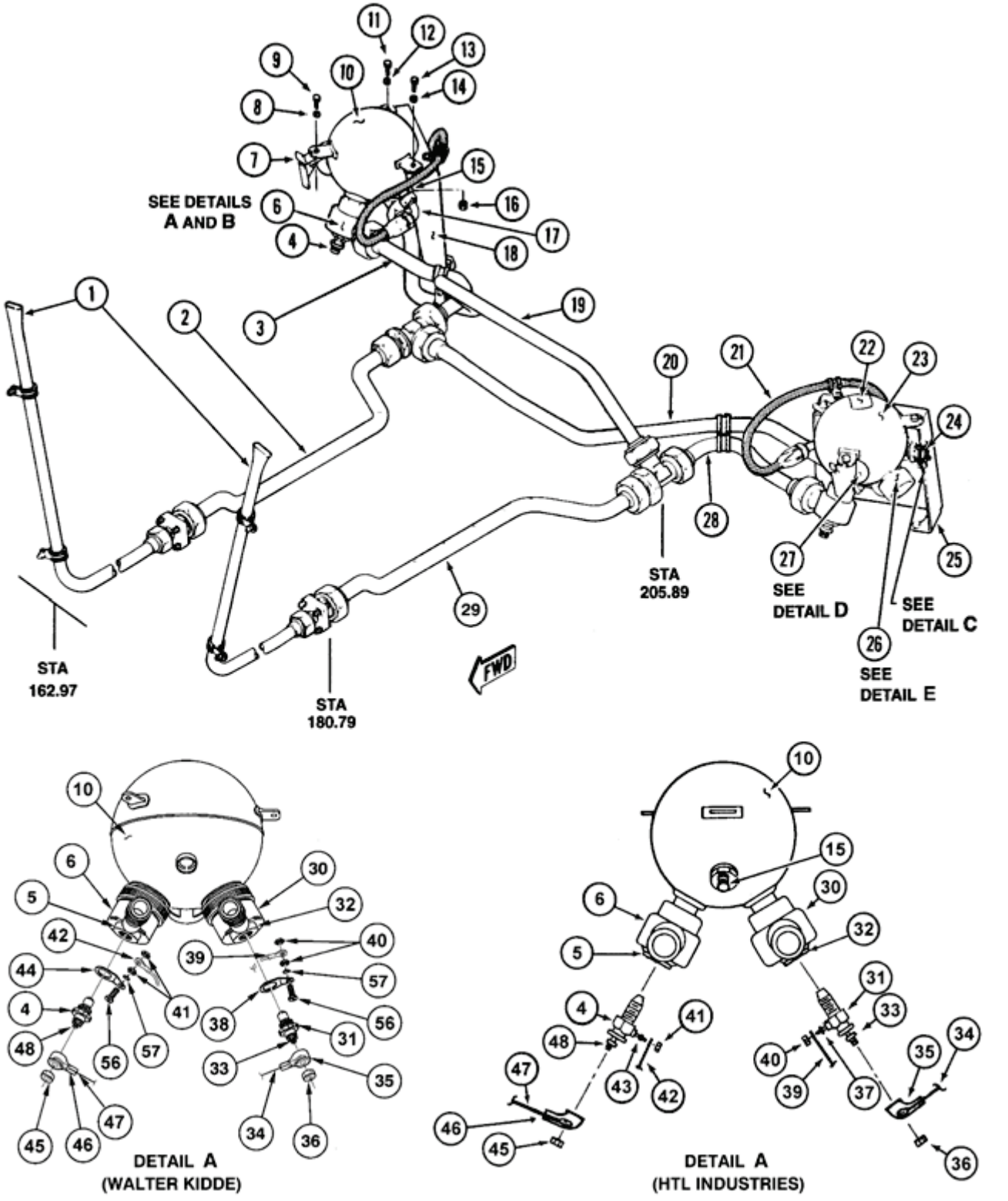
Figure 1. Engine Compartment Fire Extinguisher (Sheet 2 of 2)

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Engine Compartment Fire Extinguishers - Inspection

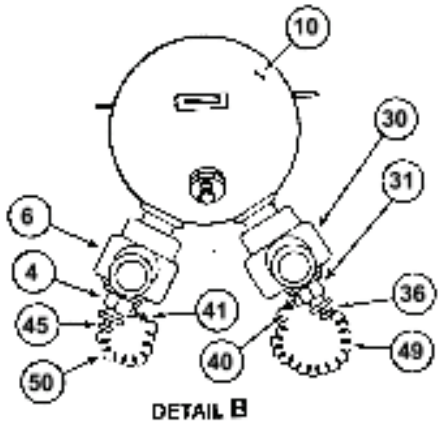
Procedure

1. Inspect container for obvious damage. Superficial damage that does not affect container structural integrity is acceptable without repair. If damage is such that loss of structural integrity is suspected, container shall be replaced.
2. Inspect temperature/pressure correction decal for security and legibility.
3. Inspect pressure indicator gauge (26, Figure 1) for acceptable range as shown on temperature/pressure correction table decal.
4. Inspect decals located on both pressure indicator gauges (26) for condition and security (412-A-11-00-00-00A-067A-A / 00124).
5. Replace or recharge container, if empty.
6. Inspect manifolds (2 and 29) and forward tubes (1) for obstructions or restrictions in holes provided for discharge.
7. Visually inspect the cartridge (4 or 31) for distortion of threads, cracks (non-permitted), security, mechanical damage, dents or corrosion.

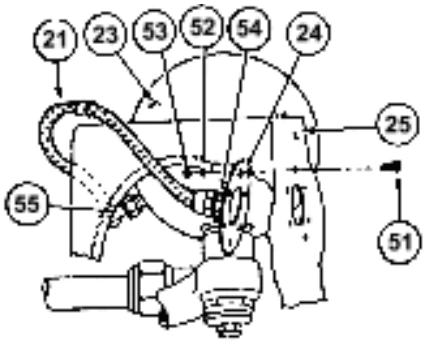


ICN-412-A-262000-B-97499-00002-A-001-01

Figure 1. Engine Compartment Fire Extinguisher (Sheet 1 of 2)



DETAIL B



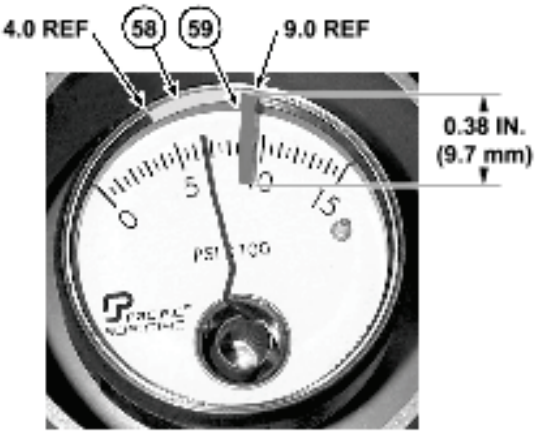
Left discharge indicator disc (24) installation shown.
Right discharge indicator disc installation is similar.

DETAIL C

TEMPERATURE/PRESSURE
CORRECTION TABLE

Temperature Degrees F	(C)	Pressure Minimum	Maximum
-65	(-54)	271	344
-60	(-51)	275	350
-40	(-40)	292	370
20	(-29)	320	400
0	(-18)	355	437
+20	(-7)	396	485
+40	(4)	449	540
+60	(16)	518	618
+80	(27)	593	702
+100	(38)	691	784
+125	(52)	785	902

DETAIL D



DETAIL E

1. Forward tube
2. Manifold
3. Tube
4. Cartridge
5. Housing assembly
6. Discharge outlet
7. Support
8. Thin aluminum washer
9. Bolt
10. Fire extinguisher container (right)
11. Bolt
12. Thin aluminum washer
13. Bolt
14. Thin aluminum washer
15. Fill and thermal relief fitting
16. Nut
17. Hose assembly
18. Support
19. Tube
20. Tube
21. Hose
22. Name plate
23. Fire extinguisher container (left)
24. Discharge indicator disc
25. Support
26. Pressure indicator
27. Temperature/pressure correction table
28. Tube
29. Manifold
30. Discharge outlet
31. Cartridge
32. Housing assembly
33. Terminal (hot wire)
34. Electrical wire
35. Electrical terminal nipple
36. Nut
37. Terminal (ground wire)
38. Ground washer
39. Electrical wire
40. Nut
41. Nut
42. Electrical wire
43. Terminal (ground wire)
44. Ground washer
45. Nut
46. Electrical terminal nipple
47. Electrical wire
48. Terminal (hot wire)
49. Jumper wire (shorting device)
50. Jumper wire (shorting device)
51. Screw
52. Washer (AN960C6L)
53. Nut
54. Nipple
55. Fill and thermal relief fitting
56. Screw
57. Washer
58. Green range mark (31-113-1D)
59. Red range mark (31-113-6A)

ICN-412-A-262000-B-97499-00003-A-001-01

Figure 1. Engine Compartment Fire Extinguisher (Sheet 2 of 2)

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Engine Compartment Fire Extinguishers - Repair/Replacement

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure



THE FIRE EXTINGUISHER CONTAINERS ARE PRESSURIZED AND EQUIPPED WITH PYROTECHNIC CARTRIDGES. ANY TIME CONNECTING WIRES ARE DISCONNECTED, JUMPER WIRES (SHORTING DEVICES) (49 AND 50, Figure 1) SHALL BE INSTALLED TO PREVENT INADVERTENT DETONATION.

DO NOT USE A VOLTMETER, FLASHLIGHT BATTERY, CONTINUITY LIGHT, OR SIMILAR DEVICE TO TEST CARTRIDGES.

DO NOT LOOSEN FITTINGS AT DISCHARGE OUTLETS (6 OR 30) , PRESSURE INDICATOR GAUGE (26), OR FILL AND THERMAL RELIEF FITTING (15) WHEN CONTAINER IS PRESSURIZED.



MAKE SURE THAT NO POWER IS APPLIED TO HELICOPTER ELECTRICAL SYSTEMS AND HELICOPTER IS PROPERLY GROUNDED WHILE PERFORMING MAINTENANCE ON FIRE EXTINGUISHING SYSTEM.

1. Repair of the fire extinguisher system components is limited to replacement of unserviceable parts. For recharging or leak testing of fire extinguisher containers, refer to appropriate vendors manuals. Recharge fire extinguisher containers as required at a facility equipped and authorized to service extinguisher containers.

NOTE

No repairs to cartridges are permitted.

2. Refer to 412-A-04-00-00-00A-288A-A / 00013 for cartridge retirement and replacement schedule. For cartridge testing, refer to appropriate vendors manuals. Replace cartridges as required; see Step 4..

3. Replace container temperature/pressure table decal if illegible or otherwise unserviceable.

4. For the replacement of cartridges, perform the following:



REMOVE HELICOPTER WIRING CONNECTORS ONE AT A TIME AND INSTALL JUMPER WIRE (SHORTING DEVICE) AT VACANT TERMINAL PRIOR TO REMOVING FURTHER WIRING. TAG DISCONNECTED WIRES FOR REINSTALLATION IN PROPER LOCATION.

THE CARTRIDGE IS A PYROTECHNIC DEVICE. INADVERTENT DETONATION OF THE CARTRIDGE CAN CAUSE PERSONAL INJURY. FOR SAFE HANDLING, THE ELECTRICAL CONNECTION PINS MUST BE SHORTED TOGETHER. THE SHORTING DEVICE MUST BE REMOVED BEFORE TESTING OR CONNECTING THE CARTRIDGE TO THE HELICOPTER WIRING.

NOTE

Replacement procedures for fire extinguisher cartridges are typical. The cartridges can be removed or installed with the container pressurized.

NOTE

For HTL Industries cartridges replacement, refer to Step 4.1.. For Walter Kidde cartridges replacement, refer to Step 4.2..

HTL Industries and Walter Kidde recharge kit and cartridges are not interchangeable.

4.1. Replace (HTL Industries) cartridges (4 and 31, Figure 1) in fire extinguisher containers (10 and 23) as follows:

4.1.1. Disconnect battery and verify all electrical power is removed from helicopter and that helicopter is properly grounded.

4.1.2. Open engine cowling to gain access to fire extinguisher containers.

4.1.3. Remove nut (40). Tag and remove electrical wire (39) from terminal (ground wire) (37).

4.1.4. Install jumper wire (shorting device) (49) on terminal (ground wire) (37) and install nut (40).

4.1.5. Install loose end of jumper wire (shorting device) (49, Detail B) on terminal (hot wire) (33) with nut (36). The jumper wire (shorting device) must remain connected at all times when the cartridge (31) is disconnected from the helicopter wiring.

4.1.6. Remove nut (45) and electrical wire (47) with electrical terminal nipple (46). Tag electrical wire.

4.1.7. Install jumper wire (shorting device) (50) on terminal (hot wire) (48) and install nut (45).

4.1.8. Remove nut (41) and electrical wire (39). Tag electrical wire.

4.1.9. Install loose end of jumper wire (shorting device) (50) on terminal (43) with nut (41). The jumper wire (shorting device) must remain connected at all times when the cartridge (4) is disconnected from the helicopter wiring.



TO PREVENT ACCIDENTAL DISCHARGE OF CONTAINER DURING CARTRIDGE (4 AND 31) REPLACEMENT, THE LARGE (1 1/4 INCHES) HEX HOUSING ASSEMBLIES (5 AND 32) MUST BE HELD TO PREVENT TURNING.

4.1.10. Remove lockwire from housing assemblies (5 and 32) and cartridges (4 and 31).



DO NOT LOOSEN THE HOUSING ASSEMBLIES (5 AND 32). THIS MAY CAUSE THE EXTINGUISHER TO DISCHARGE AND CAUSE SERIOUS INJURY.

4.1.11. Remove cartridges (4 and 31) from housing assemblies (5 and 32) by unscrewing the cartridge while holding the hexagonal head of the housing assemblies, to prevent accidental discharge. Discard seal.

4.1.12. Install new seal on cartridges (4 and 31). Install cartridges in housing assemblies (5 and 32). Hold hexagonal head of housing assemblies with a wrench while tightening cartridges to prevent excessive torque on housing assemblies. Torque cartridges to 70 to 80 inch-pounds (7.91 to 9.04 Nm).

4.1.13. Install lockwire (C-405) on housing assemblies (5 and 32) and cartridges (4 and 31).

NOTE

Refer to wiring diagrams (Figure 2) prior to connecting wires to cartridges (4 and 31, Figure 1). Make sure that wires are installed on correct terminals.

Remove ends of jumper wires one at a time and install helicopter wiring to each terminal as jumper wire is removed.

4.1.14. Remove one end of jumper wire (shorting device) (50). Install correct helicopter wire using attaching nut. Remove remaining end of jumper wire and install correct helicopter wire. Verify attaching hardware is installed and secure.

4.1.15. Repeat Step 4.1.14. for jumper wire (shorting device) (49).

4.1.16. Connect battery.

4.2. Replace Walter Kidde Co. Inc. cartridges (4 and 31, Figure 1) in fire extinguisher containers (10 and 23) as follows:

4.2.1. Disconnect battery and verify all electrical power is removed from helicopter and that helicopter is properly grounded.

4.2.2. Open engine cowling to gain access to fire extinguisher containers.

4.2.3. Remove nut (40). Tag and remove electrical wire (39) from ground washer (38).

4.2.4. Install jumper wire (shorting device) (49, Detail B) on ground washer (38) and install nut (40).

4.2.5. Install loose end of jumper wire (shorting device) (49) on terminal (hot wire) (33) with nut (36). The jumper wire (shorting device) must remain connected at all times when the cartridge (31) is disconnected from the helicopter wiring.

4.2.6. Remove nut (45) and electrical wire (47) with electrical terminal nipple (46). Tag electrical wire.

4.2.7. Install jumper wire (shorting device) (50) on terminal (hot wire) (48) and install nut (45).

4.2.8. Remove nut (41) and electrical wire (42). Tag electrical wire.

4.2.9. Install loose end of jumper wire (shorting device) (50) on ground wire (44) with nut (41). The jumper wire (shorting device) must remain connected at all times when the cartridge (4) is disconnected from the helicopter wiring.



TO PREVENT ACCIDENTAL DISCHARGE OF CONTAINER DURING CARTRIDGE (4 AND 31) REPLACEMENT, THE LARGE (1 1/4 INCHES) HEX HOUSING ASSEMBLIES (5 AND 32) MUST BE HELD TO PREVENT TURNING.

4.2.10. Remove lockwire from housing assemblies (5 and 32) and cartridges (4 and 31).



DO NOT LOOSEN THE HOUSING ASSEMBLIES (5 AND 32). THIS MAY CAUSE THE EXTINGUISHER TO DISCHARGE AND CAUSE SERIOUS INJURY.

4.2.11. Remove cartridges (4 and 31) from housing assemblies (5 and 32) by unscrewing the cartridge while holding the hexagonal head of the housing assemblies to prevent accidental discharge. Discard seal.

4.2.12. Install new seal on cartridges (4 and 31). Install cartridges in housing assemblies (5 and 32). Hold hexagonal head of housing assemblies with a wrench while tightening cartridges to prevent excessive torque on housing assemblies. Torque cartridges to 150 to 200 inch-pounds (16.95 to 22.60 Nm).

4.2.13. Install lockwire (C-405) on housing assemblies (5 and 32) and cartridges (4 and 31).

NOTE

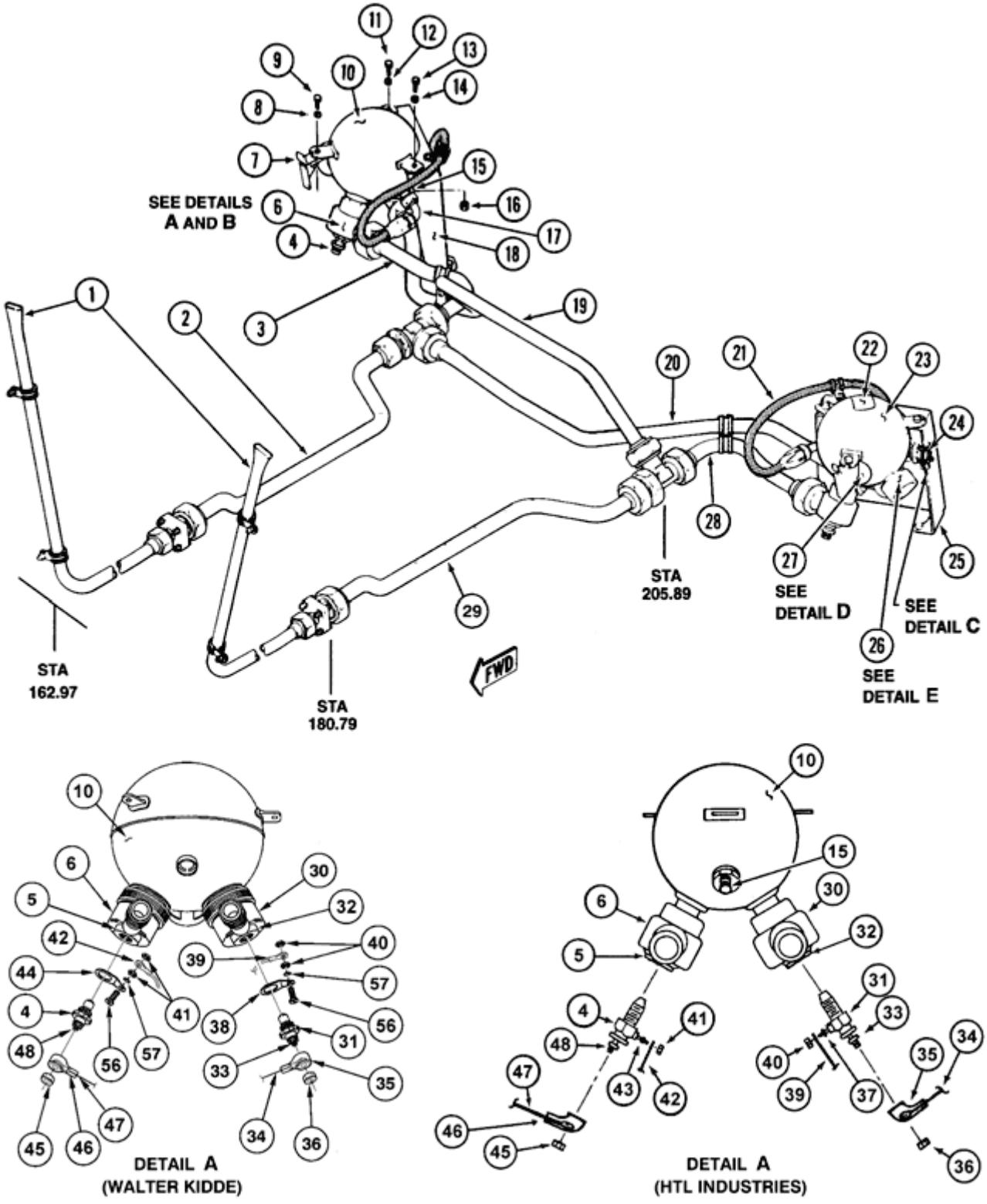
Refer to wiring diagrams (Figure 2) prior to connecting wires to cartridges (4 and 31, Figure 1). Make sure that wires are installed on correct terminals.

Remove ends of jumper wires one at a time and install helicopter wiring to each terminal as jumper wire is removed.

4.2.14. Remove one end of jumper wire (shorting device) (50). Install correct helicopter wire using attaching nut. Remove remaining end of jumper wire and install correct helicopter wire. Verify attaching hardware is installed and secure.

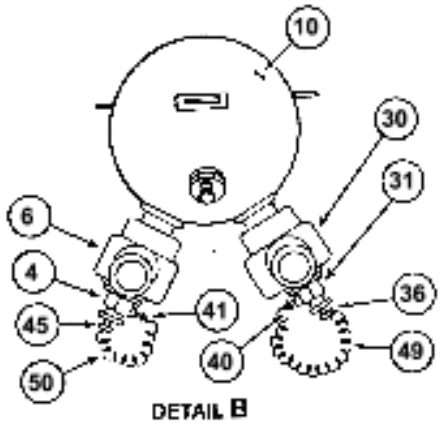
4.2.15. Repeat Step 4.2.14. for jumper wire (shorting device) (49).

4.2.16. Connect battery.

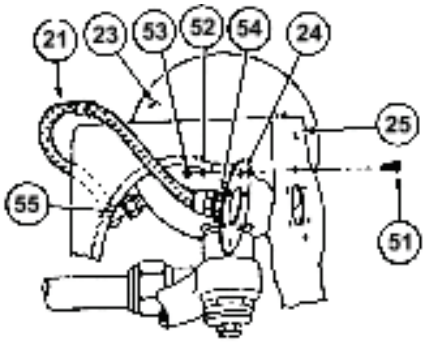


ICN-412-A-262000-B-97499-00002-A-001-01

Figure 1. Engine Compartment Fire Extinguisher (Sheet 1 of 2)



DETAIL B



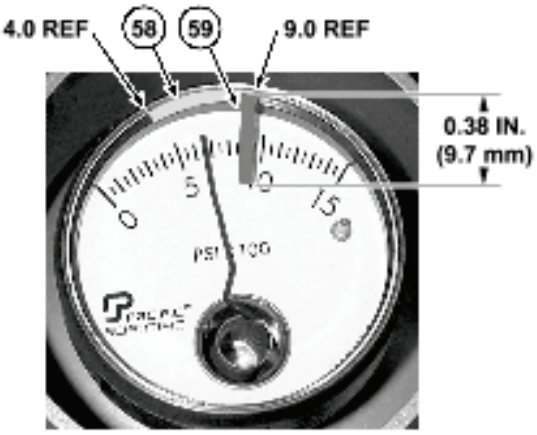
Left discharge indicator disc (24) installation shown.
Right discharge indicator disc installation is similar.

DETAIL C

TEMPERATURE/PRESSURE
CORRECTION TABLE

Temperature Degrees F	(C)	Pressure Minimum	Maximum
-65	(-54)	271	344
-60	(-51)	275	350
-40	(-40)	292	370
20	(-29)	320	400
0	(-18)	355	437
+20	(-7)	396	485
+40	(4)	449	540
+60	(16)	518	618
+80	(27)	593	702
+100	(38)	691	784
+125	(52)	785	902

DETAIL D

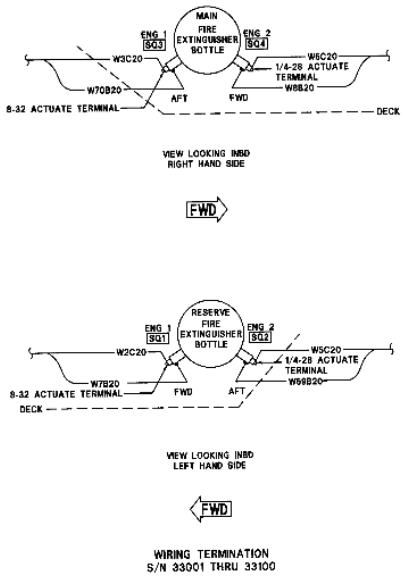


DETAIL E

1. Forward tube
2. Manifold
3. Tube
4. Cartridge
5. Housing assembly
6. Discharge outlet
7. Support
8. Thin aluminum washer
9. Bolt
10. Fire extinguisher container (right)
11. Bolt
12. Thin aluminum washer
13. Bolt
14. Thin aluminum washer
15. Fill and thermal relief fitting
16. Nut
17. Hose assembly
18. Support
19. Tube
20. Tube
21. Hose
22. Name plate
23. Fire extinguisher container (left)
24. Discharge indicator disc
25. Support
26. Pressure indicator
27. Temperature/pressure correction table
28. Tube
29. Manifold
30. Discharge outlet
31. Cartridge
32. Housing assembly
33. Terminal (hot wire)
34. Electrical wire
35. Electrical terminal nipple
36. Nut
37. Terminal (ground wire)
38. Ground washer
39. Electrical wire
40. Nut
41. Nut
42. Electrical wire
43. Terminal (ground wire)
44. Ground washer
45. Nut
46. Electrical terminal nipple
47. Electrical wire
48. Terminal (hot wire)
49. Jumper wire (shorting device)
50. Jumper wire (shorting device)
51. Screw
52. Washer (AN960C6L)
53. Nut
54. Nipple
55. Fill and thermal relief fitting
56. Screw
57. Washer
58. Green range mark (31-113-1D)
59. Red range mark (31-113-6A)

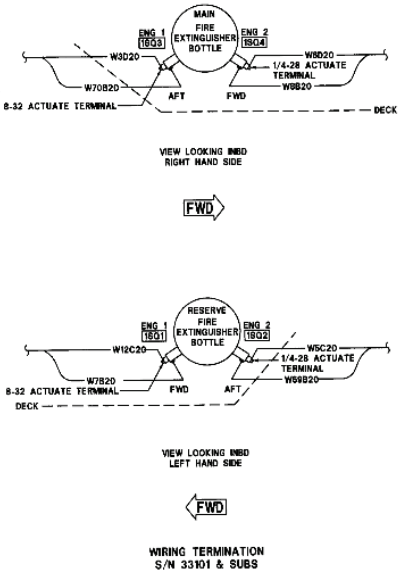
ICN-412-A-262000-B-97499-00003-A-001-01

Figure 1. Engine Compartment Fire Extinguisher (Sheet 2 of 2)



ICN-412-A-262000-B-97499-00004-A-001-01

Figure 2. Fire Extinguisher Cartridge Wiring Diagrams (Sheet 1 of 2)



ICN-412-A-262000-B-97499-00005-A-001-01

Figure 2. Fire Extinguisher Cartridge Wiring Diagrams (Sheet 2 of 2)

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Engine Compartment Fire Extinguishers - Installation

Procedure

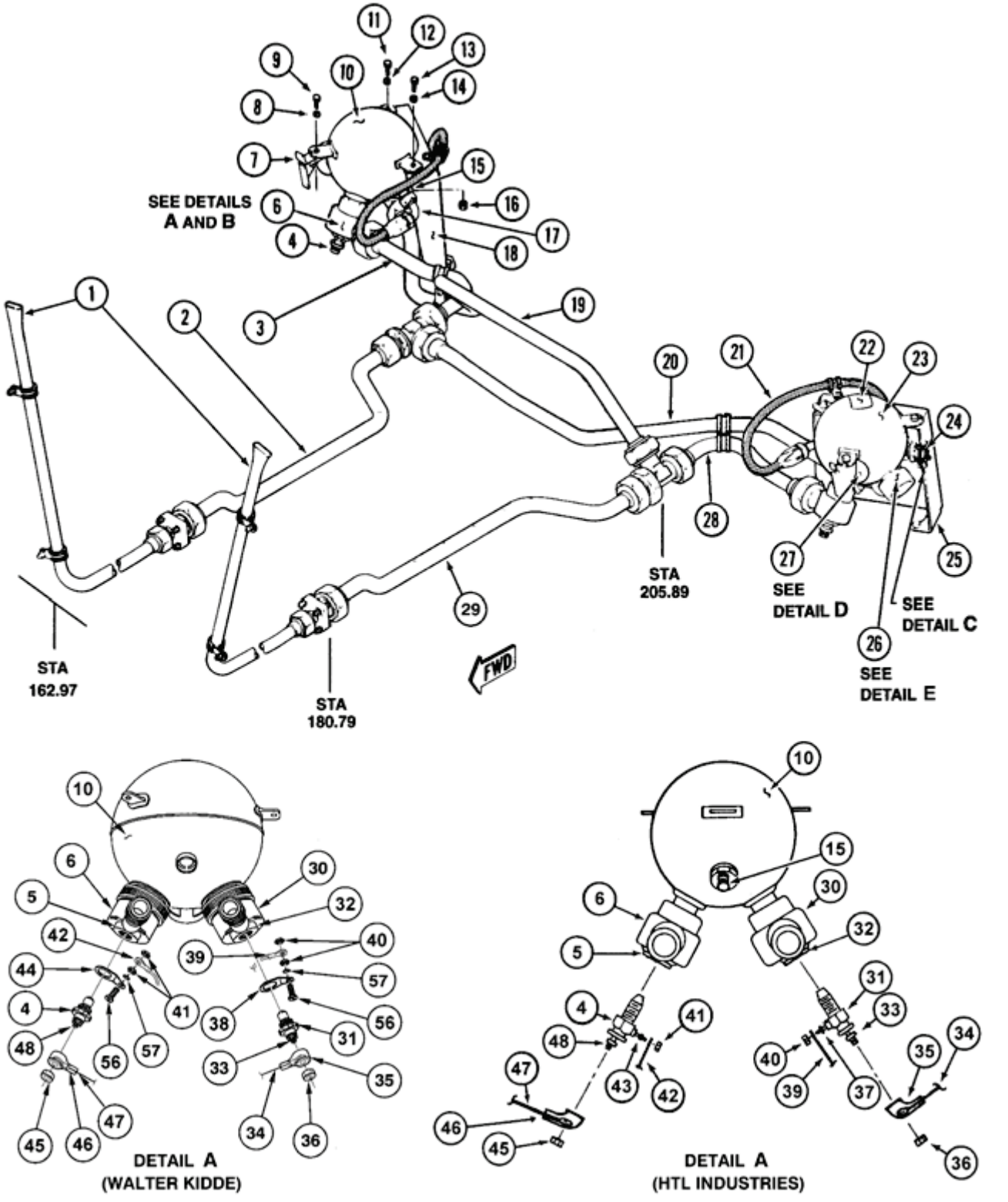
1. If not previously accomplished, install green range marking (58) and red range marking (59) on face of pressure indicator (26).
2. Position fire extinguisher container (10, Figure 1) in supports (7 and 18).
3. Install bolts (9 and 11) with thin aluminum washers (8 and 12).
4. Install bolt (13), thin aluminum washer (14), and nut (16).
5. Install hose assembly (17 or 21) on fill and thermal relief fitting (15).
6. Install tube (3 or 20) on discharge outlet (6).
7. Install tube (19 or 28) on discharge outlet (30).

NOTE

Refer to wiring diagrams (Figure 2) prior to connecting wires to cartridges (4 and 31, Figure 1). Make sure that wires are installed on correct terminals.

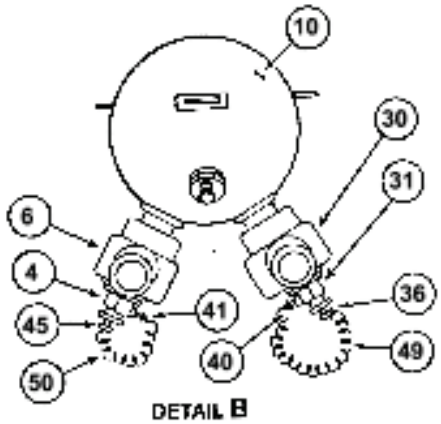
Remove ends of jumper wires one at a time and install helicopter wiring to each terminal as jumper wire is removed.

8. Remove one end of jumper wire (shorting device) (50, Figure 1). Install correct helicopter wire using attaching nut. Remove remaining end of jumper wire and install correct helicopter wire. Verify attaching hardware is secure.
9. Repeat Step 8. for jumper wire (shorting device) (49).

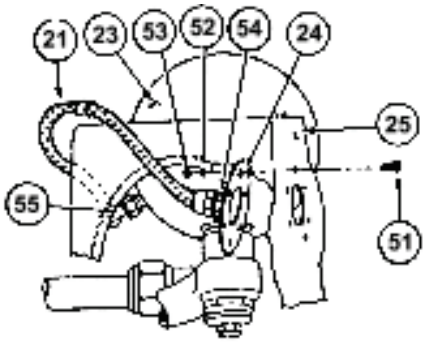


ICN-412-A-262000-B-97499-00002-A-001-01

Figure 1. Engine Compartment Fire Extinguisher (Sheet 1 of 2)



DETAIL B



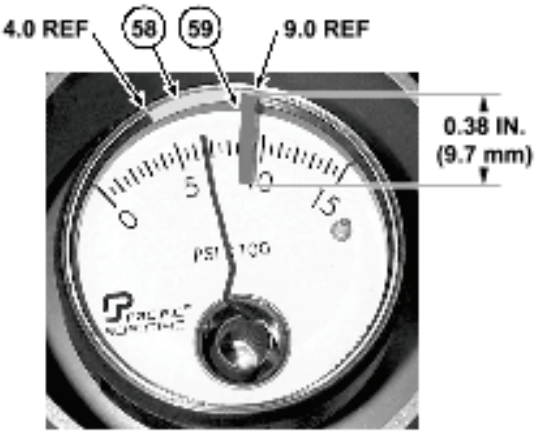
Left discharge indicator disc (24) installation shown.
Right discharge indicator disc installation is similar.

DETAIL C

TEMPERATURE/PRESSURE
CORRECTION TABLE

Temperature Degrees F	(C)	Pressure Minimum	Maximum
-65	(-54)	271	344
-60	(-51)	275	350
-40	(-40)	292	370
20	(-29)	320	400
0	(-18)	355	437
+20	(-7)	396	485
+40	(4)	449	540
+60	(16)	518	618
+80	(27)	593	702
+100	(38)	691	784
+125	(52)	785	902

DETAIL D

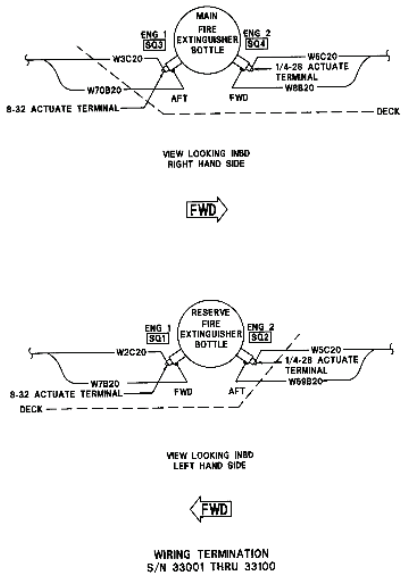


DETAIL E

1. Forward tube
2. Manifold
3. Tube
4. Cartridge
5. Housing assembly
6. Discharge outlet
7. Support
8. Thin aluminum washer
9. Bolt
10. Fire extinguisher container (right)
11. Bolt
12. Thin aluminum washer
13. Bolt
14. Thin aluminum washer
15. Fill and thermal relief fitting
16. Nut
17. Hose assembly
18. Support
19. Tube
20. Tube
21. Hose
22. Name plate
23. Fire extinguisher container (left)
24. Discharge indicator disc
25. Support
26. Pressure indicator
27. Temperature/pressure correction table
28. Tube
29. Manifold
30. Discharge outlet
31. Cartridge
32. Housing assembly
33. Terminal (hot wire)
34. Electrical wire
35. Electrical terminal nipple
36. Nut
37. Terminal (ground wire)
38. Ground washer
39. Electrical wire
40. Nut
41. Nut
42. Electrical wire
43. Terminal (ground wire)
44. Ground washer
45. Nut
46. Electrical terminal nipple
47. Electrical wire
48. Terminal (hot wire)
49. Jumper wire (shorting device)
50. Jumper wire (shorting device)
51. Screw
52. Washer (AN960C6L)
53. Nut
54. Nipple
55. Fill and thermal relief fitting
56. Screw
57. Washer
58. Green range mark (31-113-1D)
59. Red range mark (31-113-6A)

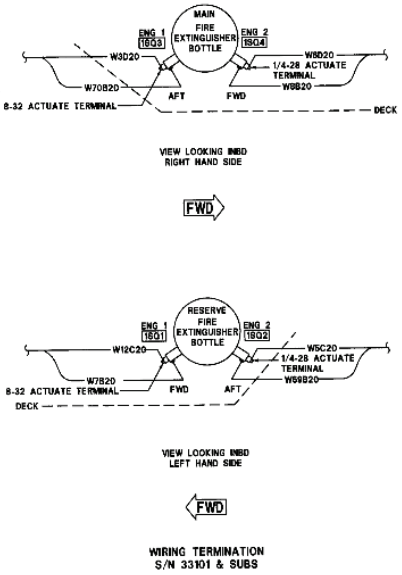
ICN-412-A-262000-B-97499-00003-A-001-01

Figure 1. Engine Compartment Fire Extinguisher (Sheet 2 of 2)



ICN-412-A-262000-B-97499-00004-A-001-01

Figure 2. Fire Extinguisher Cartridge Wiring Diagrams (Sheet 1 of 2)



ICN-412-A-262000-B-97499-00005-A-001-01

Figure 2. Fire Extinguisher Cartridge Wiring Diagrams (Sheet 2 of 2)

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Discharge Indicator Disc - Description



MAKE SURE THAT NO POWER IS APPLIED TO HELICOPTER ELECTRICAL SYSTEMS AND HELICOPTER IS PROPERLY GROUNDED WHILE PERFORMING MAINTENANCE ON FIRE EXTINGUISHING SYSTEM.

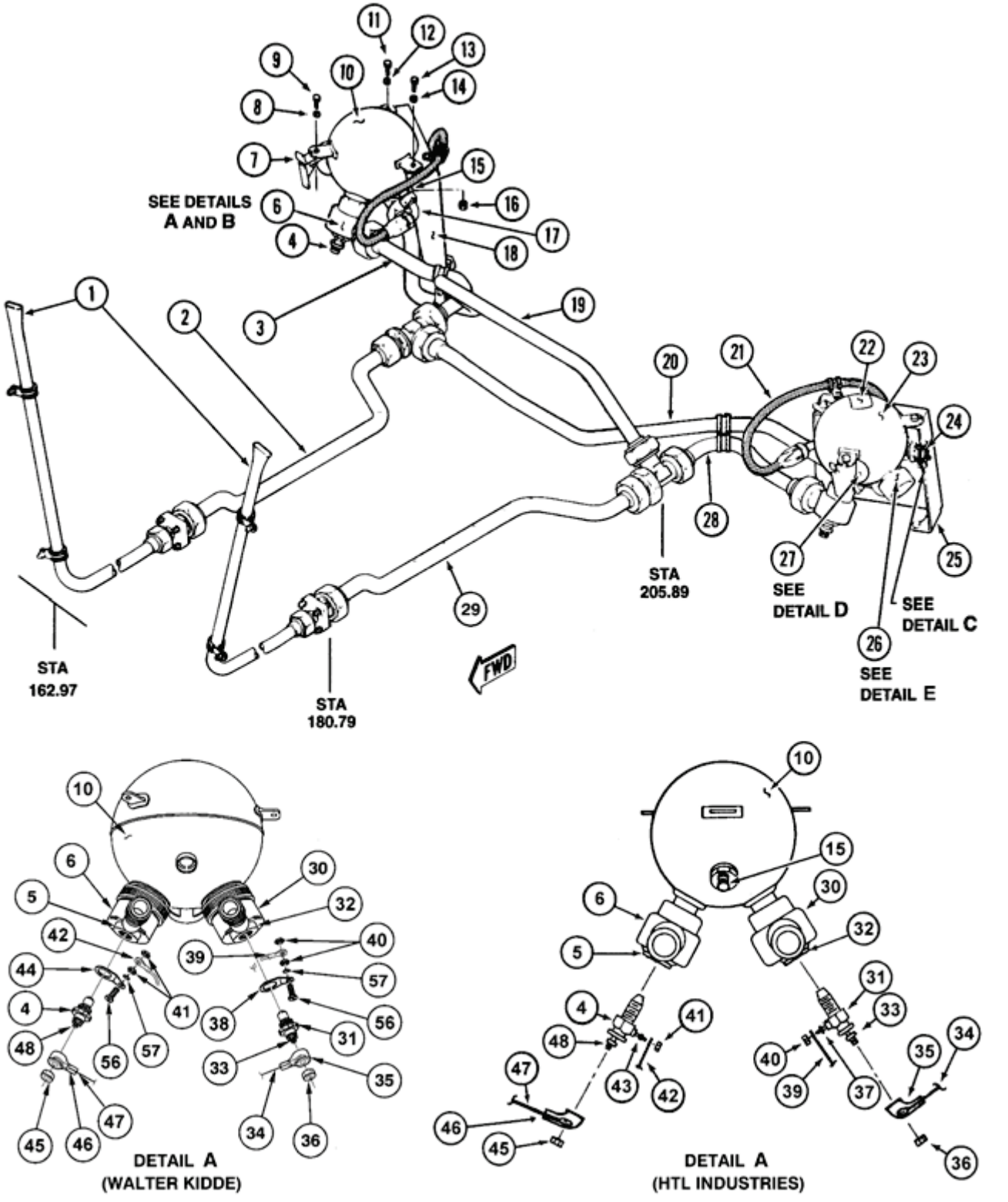
Maintenance procedures for both discharge indicator discs are typical.

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Discharge Indicator Disc - Removal

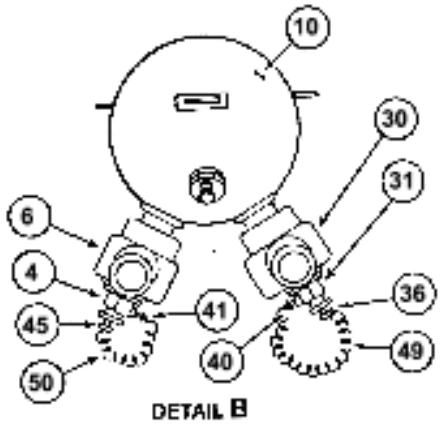
Procedure

1. Remove all electrical power from helicopter.
2. Disconnect hose (21, Figure 1) from nipple (54).
3. Remove screws (51), washers (52), and nuts (53).
4. Remove discharge indicator disc (24).

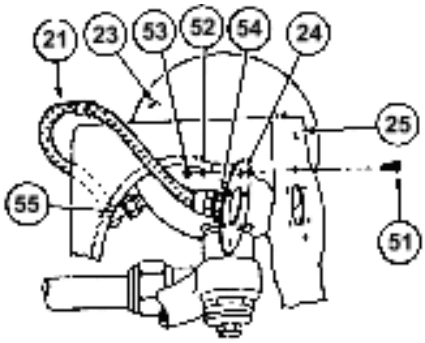


ICN-412-A-262000-B-97499-00002-A-001-01

Figure 1. Engine Compartment Fire Extinguisher (Sheet 1 of 2)



DETAIL B



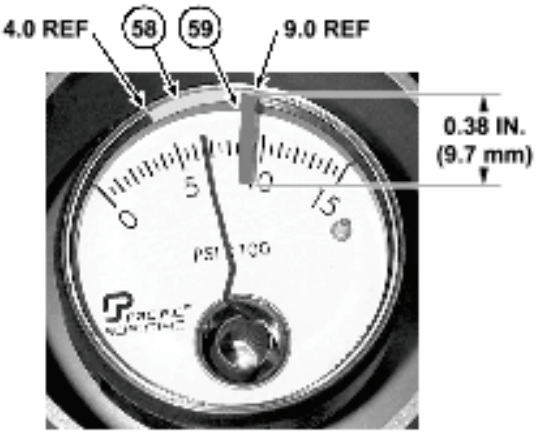
Left discharge indicator disc (24) installation shown.
Right discharge indicator disc installation is similar.

DETAIL C

TEMPERATURE/PRESSURE
CORRECTION TABLE

Temperature Degrees F	(C)	Pressure Minimum	Maximum
-65	(-54)	271	344
-60	(-51)	275	350
-40	(-40)	292	370
20	(-29)	320	400
0	(-18)	355	437
+20	(-7)	396	485
+40	(4)	449	540
+60	(16)	518	618
+80	(27)	593	702
+100	(38)	691	784
+125	(52)	785	902

DETAIL D



DETAIL E

1. Forward tube
2. Manifold
3. Tube
4. Cartridge
5. Housing assembly
6. Discharge outlet
7. Support
8. Thin aluminum washer
9. Bolt
10. Fire extinguisher container (right)
11. Bolt
12. Thin aluminum washer
13. Bolt
14. Thin aluminum washer
15. Fill and thermal relief fitting
16. Nut
17. Hose assembly
18. Support
19. Tube
20. Tube
21. Hose
22. Name plate
23. Fire extinguisher container (left)
24. Discharge indicator disc
25. Support
26. Pressure indicator
27. Temperature/pressure correction table
28. Tube
29. Manifold
30. Discharge outlet
31. Cartridge
32. Housing assembly
33. Terminal (hot wire)
34. Electrical wire
35. Electrical terminal nipple
36. Nut
37. Terminal (ground wire)
38. Ground washer
39. Electrical wire
40. Nut
41. Nut
42. Electrical wire
43. Terminal (ground wire)
44. Ground washer
45. Nut
46. Electrical terminal nipple
47. Electrical wire
48. Terminal (hot wire)
49. Jumper wire (shorting device)
50. Jumper wire (shorting device)
51. Screw
52. Washer (AN960C6L)
53. Nut
54. Nipple
55. Fill and thermal relief fitting
56. Screw
57. Washer
58. Green range mark (31-113-1D)
59. Red range mark (31-113-6A)

ICN-412-A-262000-B-97499-00003-A-001-01

Figure 1. Engine Compartment Fire Extinguisher (Sheet 2 of 2)

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Discharge Indicator Disc - Inspection

Procedure

1. Inspect discharge indicator discs on both containers for missing red indicator, secure installation, and obvious damage. A missing red disc indicates container may have discharged through fill and thermal relief fitting.

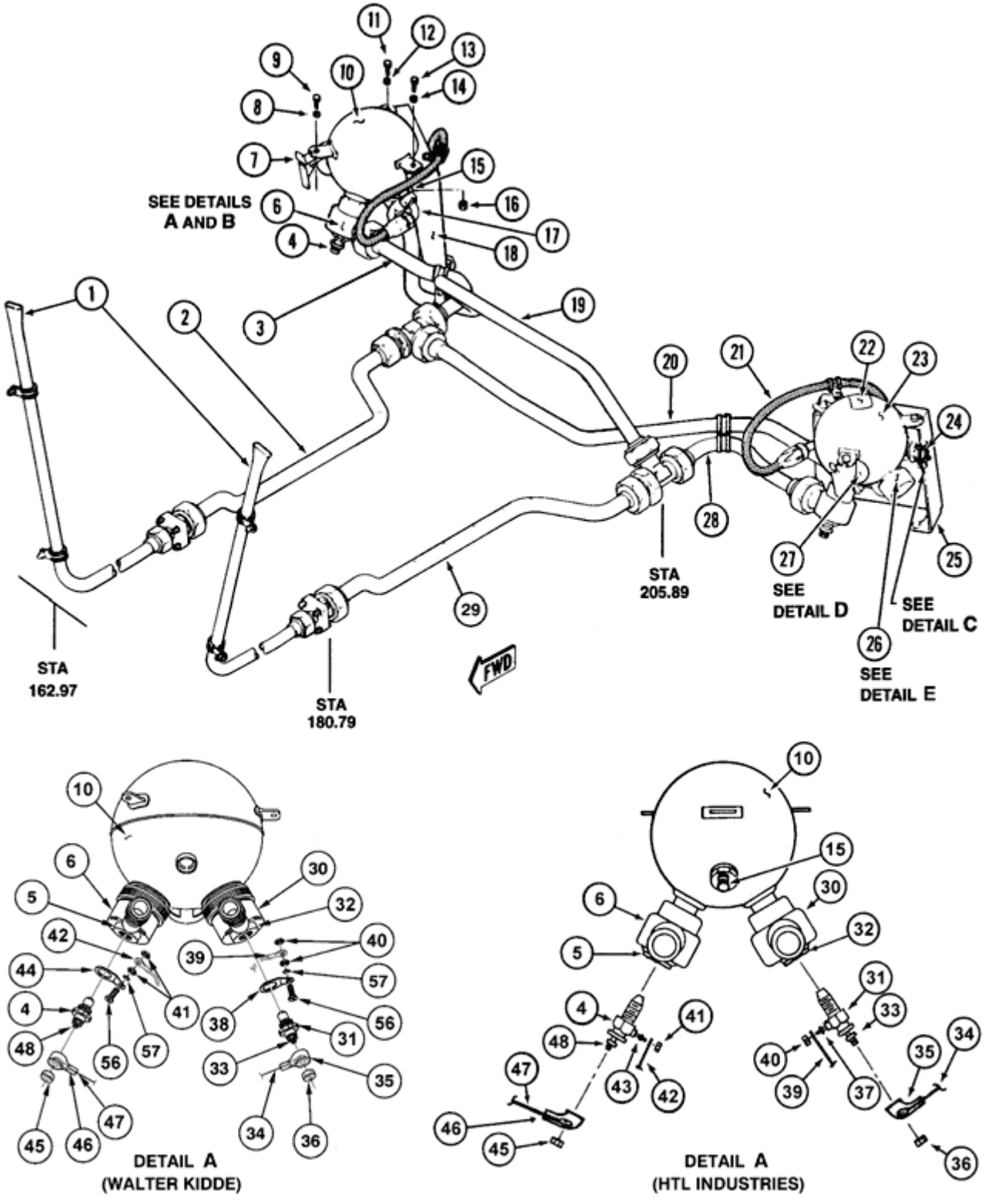
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Applicable to: ALL
ECCN EAR99

Discharge Indicator Disc - Installation

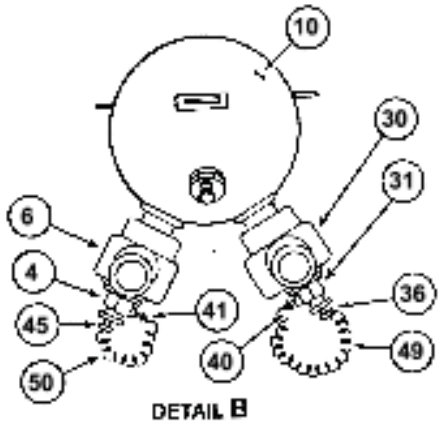
Procedure

1. Position discharge indicator disc (24, Figure 1) in support (25).
2. Install screws (51), washers (52), and nuts (53).
3. Install hose (21) on nipple (54).
4. Install hose (21) on fill and thermal relief fitting (55).

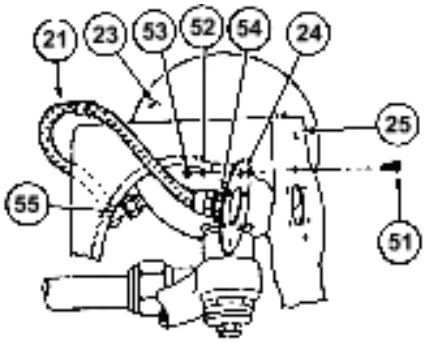


ICN-412-A-262000-B-97499-00002-A-001-01

Figure 1. Engine Compartment Fire Extinguisher (Sheet 1 of 2)



DETAIL B



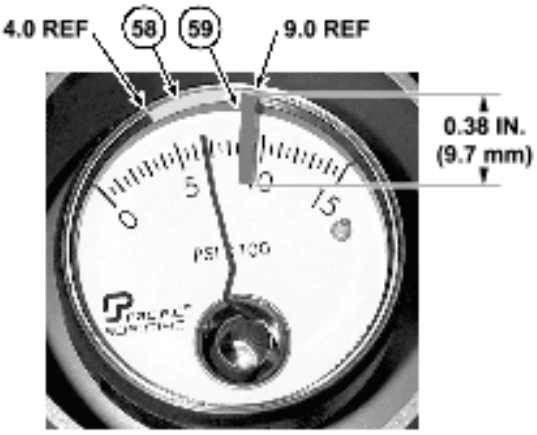
Left discharge indicator disc (24) installation shown.
Right discharge indicator disc installation is similar.

DETAIL C

TEMPERATURE/PRESSURE
CORRECTION TABLE

Temperature Degrees F	(C)	Pressure Minimum	Maximum
-65	(-54)	271	344
-60	(-51)	275	350
-40	(-40)	292	370
20	(-29)	320	400
0	(-18)	355	437
+20	(-7)	396	485
+40	(4)	449	540
+60	(16)	518	618
+80	(27)	593	702
+100	(38)	691	784
+125	(52)	785	902

DETAIL D



DETAIL E

1. Forward tube
2. Manifold
3. Tube
4. Cartridge
5. Housing assembly
6. Discharge outlet
7. Support
8. Thin aluminum washer
9. Bolt
10. Fire extinguisher container (right)
11. Bolt
12. Thin aluminum washer
13. Bolt
14. Thin aluminum washer
15. Fill and thermal relief fitting
16. Nut
17. Hose assembly
18. Support
19. Tube
20. Tube
21. Hose
22. Name plate
23. Fire extinguisher container (left)
24. Discharge indicator disc
25. Support
26. Pressure indicator
27. Temperature/pressure correction table
28. Tube
29. Manifold
30. Discharge outlet
31. Cartridge
32. Housing assembly
33. Terminal (hot wire)
34. Electrical wire
35. Electrical terminal nipple
36. Nut
37. Terminal (ground wire)
38. Ground washer
39. Electrical wire
40. Nut
41. Nut
42. Electrical wire
43. Terminal (ground wire)
44. Ground washer
45. Nut
46. Electrical terminal nipple
47. Electrical wire
48. Terminal (hot wire)
49. Jumper wire (shorting device)
50. Jumper wire (shorting device)
51. Screw
52. Washer (AN960C6L)
53. Nut
54. Nipple
55. Fill and thermal relief fitting
56. Screw
57. Washer
58. Green range mark (31-113-1D)
59. Red range mark (31-113-6A)

ICN-412-A-262000-B-97499-00003-A-001-01

Figure 1. Engine Compartment Fire Extinguisher (Sheet 2 of 2)

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Portable Fire Extinguishers and Brackets - Description

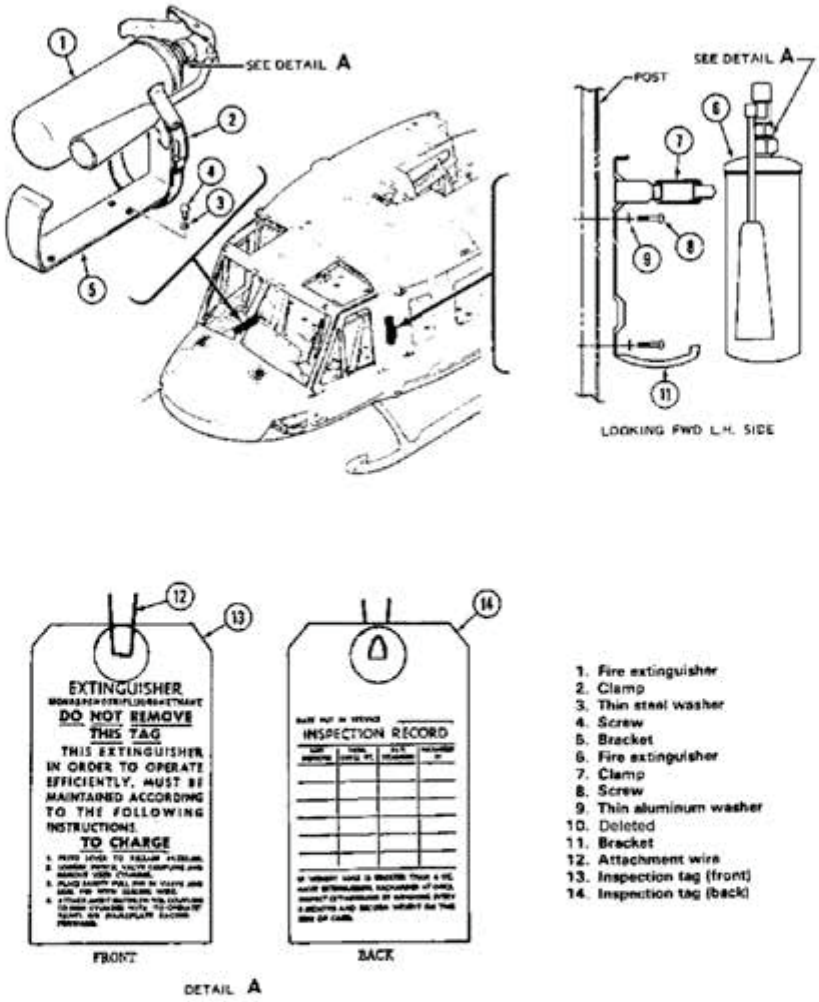
NOTE

Maintenance procedures for both fire extinguishers and brackets are identical.

NOTE

The relocation of hand-held fire extinguisher from the post installation to the under seat location is contained within modification 412-570-003-101. The modification is part of airline seat kit assembly P/N 412-706-028-101/-109. The relocation information is contained in BHT-412-SI-43, Airline Seat Kit.

Two hand-held, portable fire extinguishers are installed in quick-release brackets at the locations illustrated in Figure 1. These extinguishers are filled with a monobromotrifluoromethane extinguishing agent.



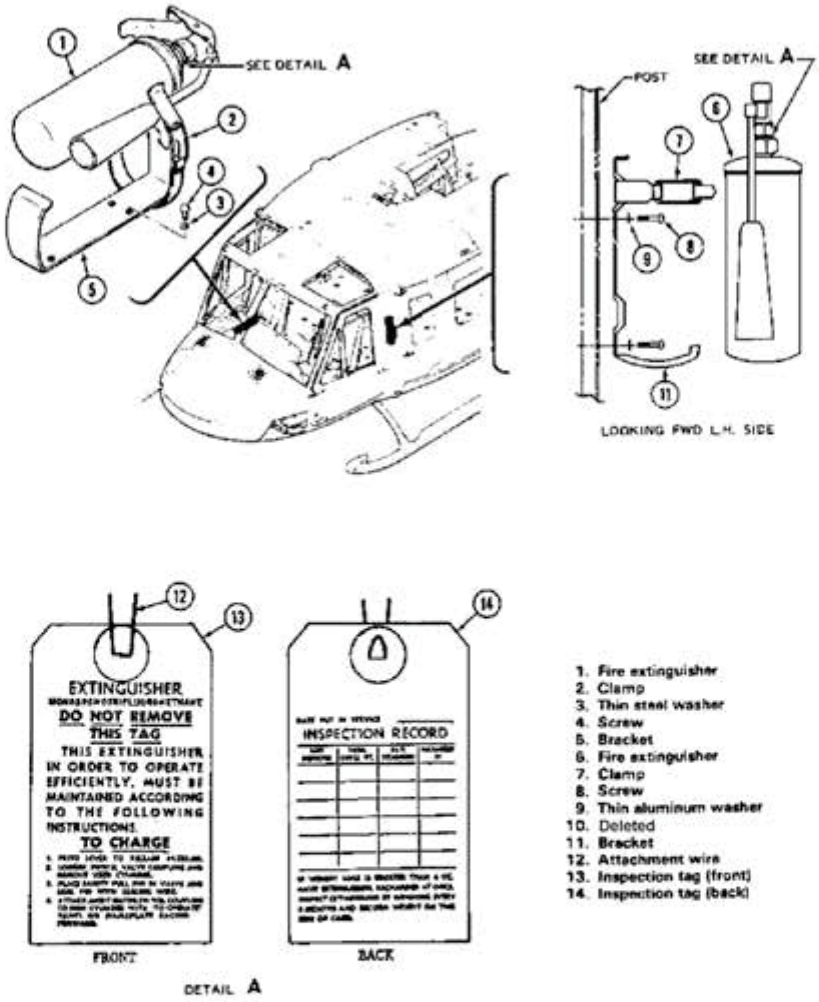
ICN-412-A-262000-B-97499-00006-A-002-01

Figure 1. Portable Fire Extinguishers and Brackets (Sheet 1 of 1)

Portable Fire Extinguishers and Brackets - Removal

Procedure

1. *Release the clamps (7, Figure 1) and remove fire extinguisher (6).*
2. *Remove screws (8) and thin aluminum washers (9). Remove the bracket (11).*



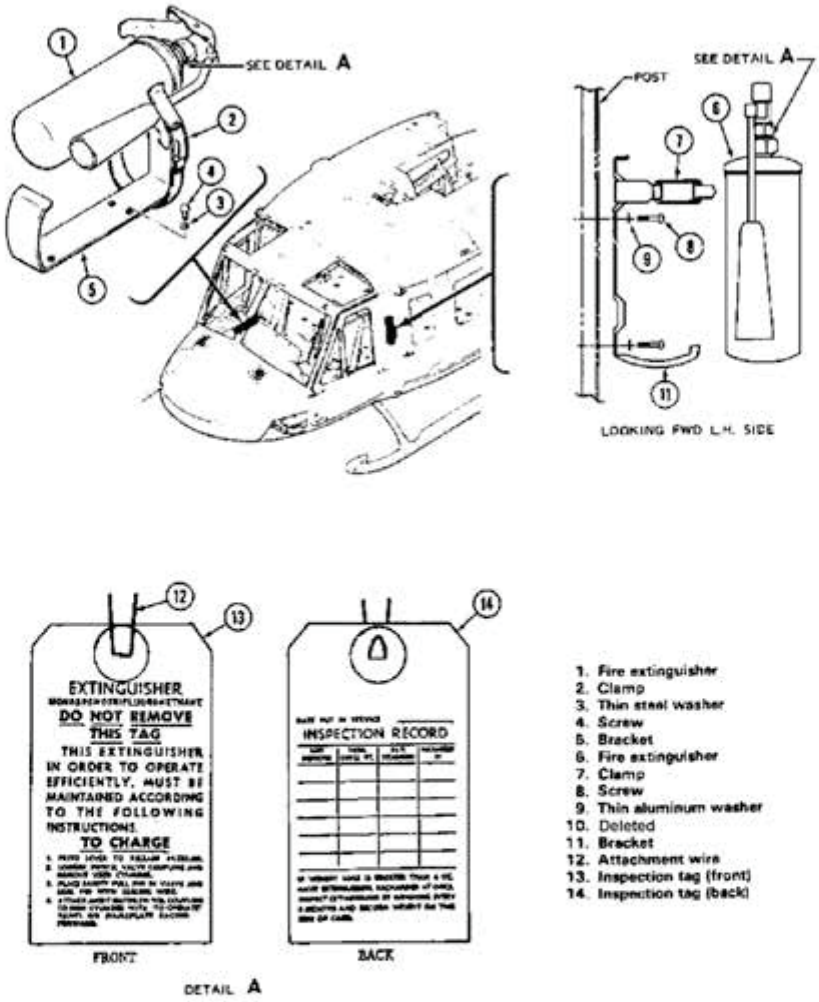
ICN-412-A-262000-B-97499-00006-A-002-01

Figure 1. Portable Fire Extinguishers and Brackets (Sheet 1 of 1)

Portable Fire Extinguishers and Brackets - Inspection

Procedure

1. Inspect fire extinguisher (6, Figure 1) for state of charge by weight (or by visual indicator, if installed).
2. Record inspection on tag attached to fire extinguisher (6, Detail A).
3. Inspect bracket (5) and attaching hardware for obvious damage, security of installation, and serviceability.



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Figure 1. Portable Fire Extinguishers and Brackets (Sheet 1 of 1)

Portable Fire Extinguishers and Brackets - Repair

Procedure

1. Recharge fire extinguisher as required at a facility equipped and authorized to service monobromotrifluoromethane type extinguishers.

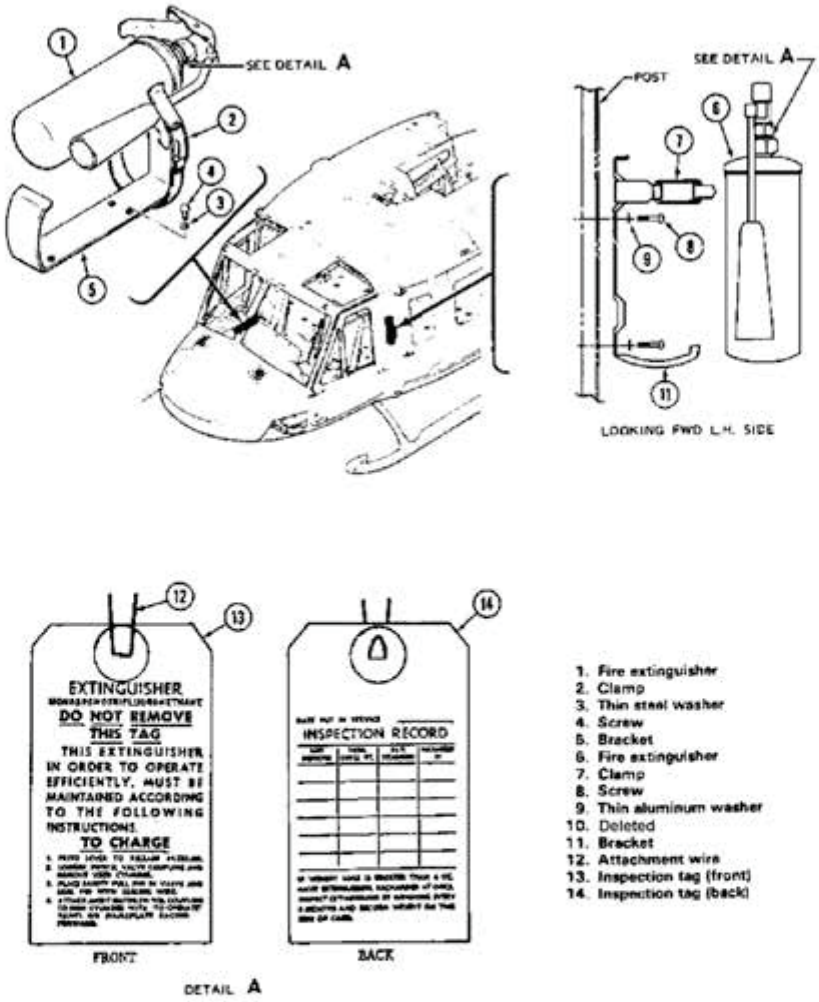
2. Replace unserviceable bracket parts and retaining hardware.

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Portable Fire Extinguishers and Brackets - Installation

Procedure

1. *Put the bracket (11, Figure 1) in its position on the mounting surface.*
2. *Install screws (8) and thin aluminum washers (9). Tighten screws (8).*
3. *Open retaining clamps (7) and position fire extinguisher (6) in bracket (11). Close retaining clamp (7).*
4. *Make sure that the bracket (11) is securely installed in helicopter and fire extinguisher (6) is securely installed in bracket.*



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Figure 1. Portable Fire Extinguishers and Brackets (Sheet 1 of 1)

CHAPTER 28
FUEL SYSTEM

MAINTENANCE MANUAL

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List of Effective Data Modules

The listed documents are included in issue 007, dated 2025-11-18, of this publication.

C = Changed data module, N = New data module

Document title	Data module code / Seq number	Issue date	Applicable to
Fuel System - General Description	DMC-412-A-28-00-00-00A-028A-A / 00324	2023-05-31	ALL
Fuel System - Safety Precautions	DMC-412-A-28-00-00-00A-012A-A / 00325	2023-05-31	ALL
Fuel Storage and Distribution System - Description and Operation	DMC-412-A-28-10-00-00A-042A-A / 00326	2023-05-31	SN: 33001-33107
Fuel Storage and Distribution System - Pressure Testing	DMC-412-A-28-10-00-00A-362A-A / 00327	2023-05-31	SN: 33001-33107
Forward Interconnect Valve - Description	DMC-412-A-28-10-00-01A-042A-A / 00328	2023-05-31	SN: 33001-33107
Forward Interconnect Valve - Removal	DMC-412-A-28-10-00-01A-520A-A / 00329	2023-05-31	SN: 33001-33107
Forward Interconnect Valve - Installation	DMC-412-A-28-10-00-01A-720A-A / 00330	2023-05-31	SN: 33001-33107
Aft Interconnect Valve - Description	DMC-412-A-28-10-00-02A-042A-A / 00331	2023-05-31	SN: 33001-33107
Aft Interconnect Valve - Removal	DMC-412-A-28-10-00-02A-520A-A / 00332	2023-05-31	SN: 33001-33107
Aft Interconnect Valve - Installation	DMC-412-A-28-10-00-02A-720A-A / 00333	2023-05-31	SN: 33001-33107
Sump Assemblies - Description	DMC-412-A-28-10-00-03A-042A-A / 00334	2023-05-31	SN: 33001-33107
Sump Assemblies - Removal	DMC-412-A-28-10-00-03A-520A-A / 00335	2023-05-31	SN: 33001-33107
Sump Assemblies - Inspection	DMC-412-A-28-10-00-03A-280A-A / 00336	2023-05-31	SN: 33001-33107
Sump Assemblies - Installation	DMC-412-A-28-10-00-03A-720A-A / 00337	2023-05-31	SN: 33001-33107
Low Fuel Level Warning Switches - Description	DMC-412-A-28-10-00-04A-042A-A / 00338	2023-05-31	SN: 33001-33107
Low Fuel Level Warning Switches - Removal	DMC-412-A-28-10-00-04A-520A-A / 00339	2023-05-31	SN: 33001-33107
Low Fuel Level Warning Switches - Installation	DMC-412-A-28-10-00-04A-720A-A / 00340	2023-05-31	SN: 33001-33107
Defueling Valves - Description	DMC-412-A-28-10-00-05A-042A-A / 00341	2023-05-31	SN: 33001-33107
Defueling Valves - Removal	DMC-412-A-28-10-00-05A-520A-A / 00342	2023-05-31	SN: 33001-33107
Defueling Valves - Installation	DMC-412-A-28-10-00-05A-720A-A / 00343	2023-05-31	SN: 33001-33107
Boost Pumps - Description	DMC-412-A-28-10-00-06A-042A-A / 00344	2023-05-31	SN: 33001-33107
Boost Pumps - Removal	DMC-412-A-28-10-00-06A-520A-A / 00345	2023-05-31	SN: 33001-33107
Boost Pumps - Installation	DMC-412-A-28-10-00-06A-720A-A / 00346	2023-05-31	SN: 33001-33107

Document title	Data module code / Seq number	Issue date	Applicable to
Canister Type Boost Pumps - Description	DMC-412-A-28-10-00-07A-042A-A / 00347	2023-05-31	SN: 33001-33107
Canister Type Boost Pumps - Removal	DMC-412-A-28-10-00-07A-520A-A / 00348	2023-05-31	SN: 33001-33107
Canister Type Boost Pumps - Installation	DMC-412-A-28-10-00-07A-720A-A / 00349	2023-05-31	SN: 33001-33107
Canister Type Boost Pump Cartridge - Description	DMC-412-A-28-10-00-08A-042A-A / 00350	2023-05-31	SN: 33001-33107
Sump Interconnect Valves - Description	DMC-412-A-28-10-00-09A-042A-A / 00351	2023-05-31	SN: 33001-33107
Sump Interconnect Valves - Removal	DMC-412-A-28-10-00-09A-520A-A / 00352	2023-05-31	SN: 33001-33107
Sump Interconnect Valves - Installation	DMC-412-A-28-10-00-09A-720A-A / 00353	2023-05-31	SN: 33001-33107
Sump Drain Valves - Description	DMC-412-A-28-10-00-10A-042A-A / 00354	2023-05-31	SN: 33001-33107
Sump Drain Valves - Removal	DMC-412-A-28-10-00-10A-520A-A / 00355	2023-05-31	SN: 33001-33107
Sump Drain Valves - Installation	DMC-412-A-28-10-00-10A-720A-A / 00356	2023-05-31	SN: 33001-33107
Flow Switches - Description	DMC-412-A-28-10-00-11A-042A-A / 00357	2023-05-31	SN: 33001-33107
Flow Switches - Removal	DMC-412-A-28-10-00-11A-520A-A / 00358	2023-05-31	SN: 33001-33107
Flow Switches - Installation	DMC-412-A-28-10-00-11A-720A-A / 00359	2023-05-31	SN: 33001-33107
Forward Cell Inlet/Outlet Fittings - Description	DMC-412-A-28-10-00-12A-042A-A / 00360	2023-05-31	SN: 33001-33107
Forward Cell Inlet/Outlet Fittings - Removal	DMC-412-A-28-10-00-12A-520A-A / 00361	2023-05-31	SN: 33001-33107
Forward Cell Inlet/Outlet Fittings - Installation	DMC-412-A-28-10-00-12A-720A-A / 00362	2023-05-31	SN: 33001-33107
Auxiliary Fuel Capped Connector - Description	DMC-412-A-28-10-00-13A-042A-A / 00363	2023-05-31	SN: 33001-33107
Auxiliary Fuel Capped Connector - Removal	DMC-412-A-28-10-00-13A-520A-A / 00364	2023-05-31	SN: 33001-33107
Auxiliary Fuel Capped Connector - Installation	DMC-412-A-28-10-00-13A-720A-A / 00365	2023-05-31	SN: 33001-33107
Flapper Valves - Description	DMC-412-A-28-10-00-14A-042A-A / 00366	2023-05-31	SN: 33001-33107
Flapper Valves - Removal	DMC-412-A-28-10-00-14A-520A-A / 00367	2023-05-31	SN: 33001-33107
Flapper Valves - Inspection	DMC-412-A-28-10-00-14A-280A-A / 00368	2023-05-31	SN: 33001-33107
Flapper Valves - Installation	DMC-412-A-28-10-00-14A-720A-A / 00369	2023-05-31	SN: 33001-33107
Ejector Pumps - Description	DMC-412-A-28-10-00-15A-042A-A / 00370	2023-05-31	SN: 33001-33107
Ejector Pumps - Removal	DMC-412-A-28-10-00-15A-520A-A / 00371	2023-05-31	SN: 33001-33107

Document title	Data module code / Seq number	Issue date	Applicable to
Ejector Pumps - Cleaning	DMC-412-A-28-10-00-15A-251A-A / 00372	2023-05-31	SN: 33001-33107
Ejector Pumps - Installation	DMC-412-A-28-10-00-15A-720A-A / 00373	2023-05-31	SN: 33001-33107
Ejector Pump Supports - Description	DMC-412-A-28-10-00-16A-042A-A / 00374	2023-05-31	SN: 33001-33107
Ejector Pump Supports - Removal	DMC-412-A-28-10-00-16A-520A-A / 00375	2023-05-31	SN: 33001-33107
Ejector Pump Supports - Installation	DMC-412-A-28-10-00-16A-720A-A / 00376	2023-05-31	SN: 33001-33107
Forward Cell Crash Resistant Vent Valves - Description	DMC-412-A-28-10-00-17A-042A-A / 00377	2023-05-31	SN: 33001-33107
Forward Cell Crash Resistant Vent Valves - Removal	DMC-412-A-28-10-00-17A-520A-A / 00378	2023-05-31	SN: 33001-33107
Forward Cell Crash Resistant Vent Valves - Installation	DMC-412-A-28-10-00-17A-720A-A / 00379	2023-05-31	SN: 33001-33107
Forward Cells - Description	DMC-412-A-28-10-00-18A-042A-A / 00380	2023-05-31	SN: 33001-33107
Forward Cells - Removal	DMC-412-A-28-10-00-19A-520A-A / 00381	2023-05-31	SN: 33001-33107
Forward Cells - Cleaning	DMC-412-A-28-10-00-19A-251A-A / 00382	2023-05-31	SN: 33001-33107
Forward Cells - Inspection	DMC-412-A-28-10-00-19A-280A-A / 00383	2023-05-31	SN: 33001-33107
Forward Cells - Installation	DMC-412-A-28-10-00-19A-720A-A / 00384	2023-05-31	SN: 33001-33107
Siphon Breaker Valves - Description	DMC-412-A-28-10-00-20A-042A-A / 00385	2023-05-31	SN: 33001-33107
Siphon Breaker Valves - Removal	DMC-412-A-28-10-00-20A-520A-A / 00386	2023-05-31	SN: 33001-33107
Siphon Breaker Valves - Installation	DMC-412-A-28-10-00-20A-720A-A / 00387	2023-05-31	SN: 33001-33107
Shutoff Valves - Description	DMC-412-A-28-10-00-21A-042A-A / 00388	2023-05-31	SN: 33001-33107
Shutoff Valves - Removal	DMC-412-A-28-10-00-21A-520A-A / 00389	2023-05-31	SN: 33001-33107
Shutoff Valves - Installation	DMC-412-A-28-10-00-21A-720A-A / 00390	2023-05-31	SN: 33001-33107
Crossfeed Valve - Description	DMC-412-A-28-10-00-22A-042A-A / 00391	2023-05-31	SN: 33001-33107
Crossfeed Valve - Removal	DMC-412-A-28-10-00-22A-520A-A / 00392	2023-05-31	SN: 33001-33107
Crossfeed Valve - Installation	DMC-412-A-28-10-00-22A-720A-A / 00393	2023-05-31	SN: 33001-33107
Fuel Pressure Transmitters - Description	DMC-412-A-28-10-00-23A-042A-A / 00394	2023-05-31	SN: 33001-33107
Fuel Pressure Transmitters - Removal	DMC-412-A-28-10-00-23A-520A-A / 00395	2023-05-31	SN: 33001-33107
Fuel Pressure Transmitters - Installation	DMC-412-A-28-10-00-23A-720A-A / 00396	2023-05-31	SN: 33001-33107

Document title	Data module code / Seq number	Issue date	Applicable to
Center Aft Fuel Cell Outlet Fittings - Description	DMC-412-A-28-10-00-24A-042A-A / 00397	2023-05-31	SN: 33001-33107
Center Aft Fuel Cell Outlet Fittings - Removal	DMC-412-A-28-10-00-24A-520A-A / 00398	2023-05-31	SN: 33001-33107
Center Aft Fuel Cell Outlet Fittings - Installation	DMC-412-A-28-10-00-24A-720A-A / 00399	2023-05-31	SN: 33001-33107
Center Aft Fuel Cell Vent Cover - Description	DMC-412-A-28-10-00-25A-042A-A / 00400	2023-05-31	SN: 33001-33107
Center Aft Fuel Cell Vent Cover - Removal	DMC-412-A-28-10-00-25A-520A-A / 00401	2023-05-31	SN: 33001-33107
Center Aft Fuel Cell Vent Cover - Installation	DMC-412-A-28-10-00-25A-720A-A / 00402	2023-05-31	SN: 33001-33107
Aft Fuel Cell Vent Fittings - Description	DMC-412-A-28-10-00-26A-042A-A / 00403	2023-05-31	SN: 33001-33107
Aft Fuel Cell Vent Fittings - Removal	DMC-412-A-28-10-00-26A-520A-A / 00404	2023-05-31	SN: 33001-33107
Aft Fuel Cell Vent Fittings - Installation	DMC-412-A-28-10-00-26A-720A-A / 00405	2023-05-31	SN: 33001-33107
Center Aft Fuel Cell Rear Cover - Removal	DMC-412-A-28-10-00-27A-520A-A / 00406	2023-05-31	SN: 33001-33107
Center Aft Fuel Cell Rear Cover - Installation	DMC-412-A-28-10-00-27A-720A-A / 00407	2023-05-31	SN: 33001-33107
Governor Bleed Line Check Valves - Description	DMC-412-A-28-10-00-28A-042A-A / 00408	2023-05-31	SN: 33001-33107
Governor Bleed Line Check Valves - Removal	DMC-412-A-28-10-00-28A-520A-A / 00409	2023-05-31	SN: 33001-33107
Governor Bleed Line Check Valves - Installation	DMC-412-A-28-10-00-28A-720A-A / 00410	2023-05-31	SN: 33001-33107
Aft Cell Outlet Fittings - Description	DMC-412-A-28-10-00-29A-042A-A / 00411	2023-05-31	SN: 33001-33107
Aft Cell Outlet Fittings - Removal	DMC-412-A-28-10-00-29A-520A-A / 00412	2023-05-31	SN: 33001-33107
Aft Cell Outlet Fittings - Installation	DMC-412-A-28-10-00-29A-720A-A / 00413	2023-05-31	SN: 33001-33107
Filler Cap and Adapter Assembly - Description	DMC-412-A-28-10-00-30A-042A-A / 00414	2023-05-31	TB 412-89-71: Pre
Filler Cap and Adapter Assembly - Removal	DMC-412-A-28-10-00-30A-520A-A / 00415	2023-05-31	TB 412-89-71: Pre
Filler Cap and Adapter Assembly - Installation	DMC-412-A-28-10-00-30A-720A-A / 00416	2023-05-31	TB 412-89-71: Pre
Improved Filler Cap and Adapter Assembly - Description	DMC-412-A-28-10-00-31A-042A-A / 00417	2023-05-31	TB 412-89-71: Post
Improved Filler Cap and Adapter Assembly - Removal	DMC-412-A-28-10-00-31A-520A-A / 00418	2023-05-31	TB 412-89-71: Post
Improved Filler Cap and Adapter Assembly - Installation	DMC-412-A-28-10-00-31A-720A-A / 00419	2023-05-31	TB 412-89-71: Post
Aft Cells - Description	DMC-412-A-28-10-00-32A-042A-A / 00420	2023-05-31	SN: 33001-33107
Aft Cells - Removal	DMC-412-A-28-10-00-32A-520A-A / 00421	2023-05-31	SN: 33001-33107

Document title	Data module code / Seq number	Issue date	Applicable to
Aft Cells - Cleaning	DMC-412-A-28-10-00-32A-251A-A / 00422	2023-05-31	SN: 33001-33107
Aft Cells - Inspection	DMC-412-A-28-10-00-32A-280A-A / 00423	2023-05-31	SN: 33001-33107
Aft Cells - Installation	DMC-412-A-28-10-00-32A-720A-A / 00424	2023-05-31	SN: 33001-33107
Center Aft Cell - Description	DMC-412-A-28-10-00-33A-042A-A / 00425	2023-05-31	SN: 33001-33107
Center Aft Cell - Removal	DMC-412-A-28-10-00-33A-520A-A / 00426	2023-05-31	SN: 33001-33107
Center Aft Cell - Cleaning	DMC-412-A-28-10-00-33A-251A-A / 00427	2023-05-31	SN: 33001-33107
Center Aft Cell - Inspection	DMC-412-A-28-10-00-33A-280A-A / 00428	2023-05-31	SN: 33001-33107
Center Aft Cell - Installation	DMC-412-A-28-10-00-33A-720A-A / 00429	2023-05-31	SN: 33001-33107
Fuel Quantity Probes - Description	DMC-412-A-28-10-00-34A-042A-A / 00430	2023-05-31	SN: 33001-33107
Forward Cell Fuel Quantity Probes - Description	DMC-412-A-28-10-00-35A-042A-A / 00431	2023-05-31	SN: 33001-33107
Forward Cell Fuel Quantity Probes - Removal	DMC-412-A-28-10-00-35A-520A-A / 00432	2023-05-31	SN: 33001-33107
Forward Cell Fuel Quantity Probes - Installation	DMC-412-A-28-10-00-35A-720A-A / 00433	2023-05-31	SN: 33001-33107
Center Aft Cell Fuel Quantity Probe - Description	DMC-412-A-28-10-00-36A-042A-A / 00434	2023-05-31	SN: 33001-33107
Center Aft Cell Fuel Quantity Probe - Removal	DMC-412-A-28-10-00-36A-520A-A / 00435	2023-05-31	SN: 33001-33107
Center Aft Cell Fuel Quantity Probe - Installation	DMC-412-A-28-10-00-36A-720A-A / 00436	2023-05-31	SN: 33001-33107
Fuel Storage and Distribution System - Description and Operation	DMC-412-A-28-10-00-00A-042B-A / 00437	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Fuel Storage and Distribution System - Pressure Testing	DMC-412-A-28-10-00-00A-362B-A / 00438	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent
Lower Forward Cells - Description	DMC-412-A-28-10-00-37A-042A-A / 00439	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Lower Forward Cells - Removal	DMC-412-A-28-10-00-37A-520A-A / 00440	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Lower Forward Cells - Cleaning	DMC-412-A-28-10-00-37A-251A-A / 00441	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent

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Lower Forward Cells - Inspection	DMC-412-A-28-10-00-37A-280A-A / 00442	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent
Lower Forward Cells - Installation	DMC-412-A-28-10-00-37A-720A-A / 00443	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Fuel Vent Valves - Description	DMC-412-A-28-10-00-38A-042A-A / 00444	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent
Fuel Vent Manifolds - Removal	DMC-412-A-28-10-00-38A-520A-A / 00445	2023-05-31	SN: 33076 or 33108-33167
Fuel Vent Manifolds - Installation	DMC-412-A-28-10-00-38A-720A-A / 00446	2023-05-31	SN: 33076 or 33108-33167 TB 412-90-88: Pre
Fuel Vent Manifolds - Removal	DMC-412-A-28-10-00-38A-520B-A / 00447	2023-05-31	SN: 33168-Subsequent or 36001-Subsequent or TB 412-90-88: Post
Fuel Vent Manifolds - Installation	DMC-412-A-28-10-00-38A-720B-A / 00448	2023-05-31	SN: 33168-Subsequent or 36001-Subsequent or TB 412-90-88: Post
Lower Forward Cell Vent Hoses - Description	DMC-412-A-28-10-00-39A-042A-A / 00449	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent
Lower Forward Cell Vent Hoses - Removal	DMC-412-A-28-10-00-39A-520A-A / 00450	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Lower Forward Cell Vent Hoses - Installation	DMC-412-A-28-10-00-39A-720A-A / 00451	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Lower Forward Cell Interconnect Hose and Fitting - Description	DMC-412-A-28-10-00-40A-042A-A / 00452	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent

Document title	Data module code / Seq number	Issue date	Applicable to
Lower Forward Cell Interconnect Hose and Fitting - Removal	DMC-412-A-28-10-00-40A-520A-A / 00453	2023-05-31	SN: 33008-33213 or 36001-36999
Lower Forward Cell Interconnect Hose and Fitting - Installation	DMC-412-A-28-10-00-40A-720A-A / 00454	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Lower Forward Cell Sump Assemblies - Description	DMC-412-A-28-10-00-41A-042A-A / 00455	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Lower Forward Cell Sump Assemblies - Removal	DMC-412-A-28-10-00-41A-520A-A / 00456	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Lower Forward Cell Sump Assemblies - Inspection	DMC-412-A-28-10-00-41A-280A-A / 00457	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Lower Forward Cell Sump Assemblies - Installation	DMC-412-A-28-10-00-41A-720A-A / 00458	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Transfer Pumps - Description	DMC-412-A-28-10-00-42A-042A-A / 00459	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Transfer Pumps - Removal	DMC-412-A-28-10-00-42A-520A-A / 00460	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Transfer Pumps - Installation	DMC-412-A-28-10-00-42A-720A-A / 00461	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Canister Type Transfer Pumps - Description	DMC-412-A-28-10-00-43A-042A-A / 00462	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Canister Type Transfer Pumps - Removal	DMC-412-A-28-10-00-43A-520A-A / 00463	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Canister Type Transfer Pumps - Installation	DMC-412-A-28-10-00-43A-720A-A / 00464	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Canister Type Transfer Pump Cartridge - Description	DMC-412-A-28-10-00-44A-042A-A / 00465	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent

Document title	Data module code / Seq number	Issue date	Applicable to
Sump Drain Valves - Description	DMC-412-A-28-10-00-45A-042A-A / 00466	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent
Sump Drain Valves - Removal	DMC-412-A-28-10-00-45A-520A-A / 00467	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Sump Drain Valves - Installation	DMC-412-A-28-10-00-45A-720A-A / 00468	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Lower Forward Cell Check Valves - Description	DMC-412-A-28-10-00-46A-042A-A / 00469	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent
Lower Forward Cell Check Valves - Removal	DMC-412-A-28-10-00-46A-520A-A / 00470	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Lower Forward Cell Check Valves - Installation	DMC-412-A-28-10-00-46A-720A-A / 00471	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Lower Forward Cell/Lower Middle Cell Interconnect Fitting - Description	DMC-412-A-28-10-00-47A-042A-A / 00472	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent
Lower Forward Cell/Lower Middle Cell Interconnect Fitting - Removal	DMC-412-A-28-10-00-47A-520A-A / 00473	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Lower Forward Cell/Lower Middle Cell Interconnect Fitting - Installation	DMC-412-A-28-10-00-47A-720A-A / 00474	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Lower Middle Cells and Lower Main Cells - Description	DMC-412-A-28-10-00-48A-042A-A / 00475	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-37999
Lower Middle Cells and Lower Main Cells - Removal	DMC-412-A-28-10-00-48A-520A-A / 00476	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Lower Middle Cells and Lower Main Cells - Cleaning	DMC-412-A-28-10-00-48A-251A-A / 00477	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent

Document title	Data module code / Seq number	Issue date	Applicable to
Lower Middle Cells and Lower Main Cells - Inspection	DMC-412-A-28-10-00-48A-280A-A / 00478	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-37999
Lower Middle Cells and Lower Main Cells - Installation	DMC-412-A-28-10-00-48A-720A-A / 00479	2024-05-09	SN: 33108-Subsequent or 36001-Subsequent
Forward to Middle Fuel Cell - Leak Check	DMC-412-A-28-10-00-00A-364A-A / 00480	2025-05-13	SN: 33108-33213 or 36001-Subsequent
Lower Middle Cell Vent Hoses - Description	DMC-412-A-28-10-00-49A-042A-A / 00481	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent
Lower Middle Cell Vent Hoses - Removal	DMC-412-A-28-10-00-49A-520A-A / 00482	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Lower Middle Cell Vent Hoses - Installation	DMC-412-A-28-10-00-49A-720A-A / 00483	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Lower Middle Cell Interconnect Hose and Fitting - Description	DMC-412-A-28-10-00-50A-042A-A / 00484	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Lower Middle Cell Interconnect Hose and Fitting - Removal	DMC-412-A-28-10-00-50A-520A-A / 00485	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Lower Middle Cell Interconnect Hose and Fitting - Installation	DMC-412-A-28-10-00-50A-720A-A / 00486	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Lower Middle Cell Sump Assemblies - Description	DMC-412-A-28-10-00-51A-042A-A / 00487	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent
Lower Middle Cell Sump Assemblies - Removal	DMC-412-A-28-10-00-51A-520A-A / 00488	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Lower Middle Cell Sump Assemblies - Inspection and Repair	DMC-412-A-28-10-00-51A-280A-A / 00489	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent

Document title	Data module code / Seq number	Issue date	Applicable to
Lower Middle Cell Sump Assemblies - Installation	DMC-412-A-28-10-00-51A-720A-A / 00490	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Ejector Pumps - Description	DMC-412-A-28-10-00-52A-042A-A / 00491	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Ejector Pumps - Removal	DMC-412-A-28-10-00-52A-520A-A / 00492	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Ejector Pumps - Cleaning	DMC-412-A-28-10-00-52A-251A-A / 00493	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent
Ejector Pumps - Inspection	DMC-412-A-28-10-00-52A-280A-A / 00494	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent
Ejector Pumps - Installation	DMC-412-A-28-10-00-52A-720A-A / 00495	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Sump Drain Valves - Description	DMC-412-A-28-10-00-53A-042A-A / 00496	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent
Sump Drain Valves - Removal	DMC-412-A-28-10-00-53A-520A-A / 00497	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Sump Drain Valves - Installation	DMC-412-A-28-10-00-53A-720A-A / 00498	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Lower Middle Cell/Lower Main Cell Interconnect Fittings - Description	DMC-412-A-28-10-00-54A-042A-A / 00499	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent
Lower Middle Cell/Lower Main Cell Interconnect Fittings - Removal	DMC-412-A-28-10-00-54A-520A-A / 00500	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Lower Middle Cell/Lower Main Cell Interconnect Fittings - Installation	DMC-412-A-28-10-00-54A-720A-A / 00501	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent

Document title	Data module code / Seq number	Issue date	Applicable to
Lower Main Cell Vent Valves and Hoses - Description	DMC-412-A-28-10-00-55A-042A-A / 00502	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent
Lower Main Cell Vent Valves and Hoses - Removal	DMC-412-A-28-10-00-55A-520A-A / 00503	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Lower Main Cell Vent Valves and Hoses - Installation	DMC-412-A-28-10-00-55A-720A-A / 00504	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Lower Main Cell Interconnect Hoses - Description	DMC-412-A-28-10-00-56A-042A-A / 00505	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent
Lower Main Cell Interconnect Hoses - Removal	DMC-412-A-28-10-00-56A-520A-A / 00506	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Lower Main Cell Interconnect Hoses - Installation	DMC-412-A-28-10-00-56A-720A-A / 00507	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Lower Main Cell Sump Assemblies - Description	DMC-412-A-28-10-00-57A-042A-A / 00508	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent
Lower Main Cell Sump Assemblies - Removal	DMC-412-A-28-10-00-57A-520A-A / 00509	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Lower Main Cell Sump Assemblies - Inspection	DMC-412-A-28-10-00-57A-280A-A / 00510	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent
Lower Main Cell Sump Assemblies - Installation	DMC-412-A-28-10-00-57A-720A-A / 00511	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Boost Pumps - Description	DMC-412-A-28-10-00-58A-042A-A / 00512	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Boost Pumps - Removal	DMC-412-A-28-10-00-58A-520A-A / 00513	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent

Document title	Data module code / Seq number	Issue date	Applicable to
Boost Pumps - Installation	DMC-412-A-28-10-00-58A-720A-A / 00514	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Canister Type Boost Pump - Description	DMC-412-A-28-10-00-59A-042A-A / 00515	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Canister Type Boost Pump - Removal	DMC-412-A-28-10-00-59A-520A-A / 00516	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Canister Type Boost Pump - Installation	DMC-412-A-28-10-00-59A-720A-A / 00517	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Canister Type Boost Pump Cartridge - Description	DMC-412-A-28-10-00-60A-042A-A / 00518	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent
Canister Type Boost Pump Cartridge - Removal	DMC-412-A-28-10-00-60A-520A-A / 00519	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Canister Type Boost Pump Cartridge - Cleaning	DMC-412-A-28-10-00-60A-251A-A / 00520	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent
Canister Type Boost Pump Cartridge - Installation	DMC-412-A-28-10-00-60A-720A-A / 00521	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Flow Switch Check Valves - Description	DMC-412-A-28-10-00-61A-042A-A / 00522	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Flow Switch Check Valves - Removal	DMC-412-A-28-10-00-61A-520A-A / 00523	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Flow Switch Check Valves - Installation	DMC-412-A-28-10-00-61A-720A-A / 00524	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Sump Drain Valves - Description	DMC-412-A-28-10-00-62A-042A-A / 00525	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent

Document title	Data module code / Seq number	Issue date	Applicable to
Sump Drain Valves - Removal	DMC-412-A-28-10-00-62A-520A-A / 00526	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Sump Drain Valves - Installation	DMC-412-A-28-10-00-62A-720A-A / 00527	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Lower Main Cell Inlet/Outlet Fittings - Description	DMC-412-A-28-10-00-63A-042A-A / 00528	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent
Lower Main Cell Inlet/Outlet Fittings - Removal	DMC-412-A-28-10-00-63A-520A-A / 00529	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Lower Main Cell Inlet/Outlet Fittings - Installation	DMC-412-A-28-10-00-63A-720A-A / 00530	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Interconnect Valve - Description	DMC-412-A-28-10-00-64A-042A-A / 00531	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent
Interconnect Valve - Removal	DMC-412-A-28-10-00-64A-520A-A / 00532	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Interconnect Valve - Installation	DMC-412-A-28-10-00-64A-720A-A / 00533	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Upper Forward Center Cell - Description	DMC-412-A-28-10-00-65A-042A-A / 00534	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent
Upper Forward Center Cell - Removal	DMC-412-A-28-10-00-65A-520A-A / 00535	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Upper Forward Center Cell - Cleaning	DMC-412-A-28-10-00-65A-251A-A / 00536	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent
Upper Forward Center Cell - Inspection	DMC-412-A-28-10-00-65A-280A-A / 00537	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent

Document title	Data module code / Seq number	Issue date	Applicable to
Upper Forward Center Cell - Installation	DMC-412-A-28-10-00-65A-720A-A / 00538	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Upper Forward Center Cell Bottom Cover (Short Standpipe) Assembly - Description	DMC-412-A-28-10-00-66A-042A-A / 00539	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Upper Forward Center Cell Bottom Cover (Short Standpipe) Assembly - Removal	DMC-412-A-28-10-00-66A-520A-A / 00540	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Upper Forward Center Cell Bottom Cover (Short Standpipe) Assembly - Installation	DMC-412-A-28-10-00-66A-720A-A / 00541	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Upper Forward Center Cell Bottom Cover (Tall Standpipe) Assembly - Description	DMC-412-A-28-10-00-67A-042A-A / 00542	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Upper Forward Center Cell Bottom Cover (Tall Standpipe) Assembly - Removal	DMC-412-A-28-10-00-67A-520A-A / 00543	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Upper Forward Center Cell Bottom Cover (Tall Standpipe) Assembly - Installation	DMC-412-A-28-10-00-67A-720A-A / 00544	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Upper Forward Center Cell Interconnect Fittings - Description	DMC-412-A-28-10-00-68A-042A-A / 00545	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent
Upper Forward Center Cell Interconnect Fittings - Removal	DMC-412-A-28-10-00-68A-520A-A / 00546	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Upper Forward Center Cell Interconnect Fittings - Installation	DMC-412-A-28-10-00-68A-720A-A / 00547	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Upper Forward Center Cell Top Cover Assembly - Description	DMC-412-A-28-10-00-69A-042A-A / 00548	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent
Upper Forward Center Cell Top Cover Assembly - Removal	DMC-412-A-28-10-00-69A-520A-A / 00549	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Upper Forward Center Cell Top Cover Assembly - Installation	DMC-412-A-28-10-00-69A-720A-A / 00550	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent

Document title	Data module code / Seq number	Issue date	Applicable to
Upper Aft Center Cell - Description	DMC-412-A-28-10-00-70A-042A-A / 00551	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Upper Aft Center Cell - Removal	DMC-412-A-28-10-00-70A-520A-A / 00552	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Upper Aft Center Cell - Cleaning	DMC-412-A-28-10-00-70A-251A-A / 00553	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent
Upper Aft Center Cell - Inspection	DMC-412-A-28-10-00-70A-280A-A / 00554	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent
Upper Aft Center Cell - Installation	DMC-412-A-28-10-00-70A-720A-A / 00555	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Upper Aft Center Cell Bottom Cover Assembly and Fitting - Description	DMC-412-A-28-10-00-71A-042A-A / 00556	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent
Upper Aft Center Cell Bottom Cover Assembly and Fitting - Removal	DMC-412-A-28-10-00-71A-520A-A / 00557	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Upper Aft Center Cell Bottom Cover Assembly and Fitting - Installation	DMC-412-A-28-10-00-71A-720A-A / 00558	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Upper Aft Center Cell Bottom Fitting - Description	DMC-412-A-28-10-00-72A-042A-A / 00559	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent
Upper Aft Center Cell Bottom Fitting - Removal	DMC-412-A-28-10-00-72A-520A-A / 00560	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Upper Aft Center Cell Bottom Fitting - Installation	DMC-412-A-28-10-00-72A-720A-A / 00561	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Upper Aft Center Cell Top Cover Assembly - Description	DMC-412-A-28-10-00-73A-042A-A / 00562	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent

Document title	Data module code / Seq number	Issue date	Applicable to
Upper Aft Center Cell Top Cover Assembly - Removal	DMC-412-A-28-10-00-73A-520A-A / 00563	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Upper Aft Center Cell Top Cover Assembly - Installation	DMC-412-A-28-10-00-73A-720A-A / 00564	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Upper Left Cell and Upper Right Cell - Description	DMC-412-A-28-10-00-74A-042A-A / 00565	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent
Upper Left Cell and Upper Right Cell - Removal	DMC-412-A-28-10-00-74A-520A-A / 00566	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Upper Left Cell and Upper Right Cell - Cleaning	DMC-412-A-28-10-00-74A-251A-A / 00567	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Upper Left Cell and Upper Right Cell - Inspection	DMC-412-A-28-10-00-74A-280A-A / 00568	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Upper Left Cell and Upper Right Cell - Installation	DMC-412-A-28-10-00-74A-720A-A / 00569	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Upper Left Cell and Upper Right Cell Outlet Fittings - Description	DMC-412-A-28-10-00-75A-042A-A / 00570	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Upper Left Cell and Upper Right Cell Outlet Fittings - Removal	DMC-412-A-28-10-00-75A-520A-A / 00571	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Upper Left Cell and Upper Right Cell Outlet Fittings - Installation	DMC-412-A-28-10-00-75A-720A-A / 00572	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Upper Left Cell and Upper Right Cell Top Cover Assemblies - Description	DMC-412-A-28-10-00-76A-042A-A / 00573	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent
Upper Left Cell and Upper Right Cell Top Cover Assemblies - Removal	DMC-412-A-28-10-00-76A-520A-A / 00574	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Upper Left Cell and Upper Right Cell Top Cover Assemblies - Installation	DMC-412-A-28-10-00-76A-720A-A / 00575	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent

Document title	Data module code / Seq number	Issue date	Applicable to
Filler Cap and Adapter Assembly - Description	DMC-412-A-28-10-00-77A-042A-A / 00576	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Filler Cap and Adapter Assembly - Removal	DMC-412-A-28-10-00-77A-520A-A / 00577	2023-05-31	ALL
Filler Cap and Adapter Assembly - Installation	DMC-412-A-28-10-00-77A-720A-A / 00578	2023-05-31	ALL
Improved Filler Cap and Adapter Assembly - Description	DMC-412-A-28-10-00-78A-042A-A / 00579	2023-05-31	SN: 33168-Subsequent or 36001-Subsequent or (SN: 33001-33167 TB 412-89-71: Post)
Improved Filler Cap and Adapter Assembly - Removal	DMC-412-A-28-10-00-78A-520A-A / 00580	2023-05-31	SN: 33168-Subsequent or 36001-Subsequent or (SN: 33001-33167 TB 412-89-71: Post)
Improved Filler Cap and Adapter Assembly - Installation	DMC-412-A-28-10-00-78A-720A-A / 00581	2023-05-31	SN: 33168-Subsequent or 36001-Subsequent or (SN: 33001-33167 TB 412-89-71: Post)
Engine Fuel Feed Line Pressure Switches - Description	DMC-412-A-28-10-00-79A-042A-A / 00582	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent
Engine Fuel Feed Line Pressure Switches - Removal	DMC-412-A-28-10-00-79A-520A-A / 00583	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Engine Fuel Feed Line Pressure Switches - Installation	DMC-412-A-28-10-00-79A-720A-A / 00584	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Fuel Feed Line Check Valves - Description	DMC-412-A-28-10-00-80A-042A-A / 00585	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Fuel Feed Line Check Valves - Removal	DMC-412-A-28-10-00-80A-520A-A / 00586	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent

Document title	Data module code / Seq number	Issue date	Applicable to
Fuel Feed Line Check Valves - Installation	DMC-412-A-28-10-00-80A-720A-A / 00587	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Shutoff Valves - Description	DMC-412-A-28-10-00-81A-042A-A / 00588	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Shutoff Valves - Removal	DMC-412-A-28-10-00-81A-520A-A / 00589	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Shutoff Valves - Installation	DMC-412-A-28-10-00-81A-720A-A / 00590	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Crossfeed Valve - Description	DMC-412-A-28-10-00-82A-042A-A / 00591	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Crossfeed Valve - Removal	DMC-412-A-28-10-00-82A-520A-A / 00592	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Crossfeed Valve - Installation	DMC-412-A-28-10-00-82A-720A-A / 00593	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Fuel Pressure Transmitters - Description	DMC-412-A-28-10-00-83A-042A-A / 00594	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Fuel Pressure Transmitters - Removal	DMC-412-A-28-10-00-83A-520A-A / 00595	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Fuel Pressure Transmitters - Installation	DMC-412-A-28-10-00-83A-720A-A / 00596	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Fuel Quantity Probes - Description	DMC-412-A-28-10-00-84A-042A-A / 00597	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Lower Forward Cell Fuel Quantity Probes - Description	DMC-412-A-28-10-00-85A-042A-A / 00598	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent
Lower Forward Cell Fuel Quantity Probes - Removal	DMC-412-A-28-10-00-85A-520A-A / 00599	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent

Document title	Data module code / Seq number	Issue date	Applicable to
Lower Forward Cell Fuel Quantity Probes - Installation	DMC-412-A-28-10-00-85A-720A-A / 00600	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Lower Middle Cell Fuel Quantity Probes - Description	DMC-412-A-28-10-00-86A-042A-A / 00601	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Lower Middle Cell Fuel Quantity Probes - Removal	DMC-412-A-28-10-00-86A-520A-A / 00602	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Lower Middle Cell Fuel Quantity Probes - Installation	DMC-412-A-28-10-00-86A-720A-A / 00603	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Lower Main Cell Fuel Quantity Probes - Description	DMC-412-A-28-10-00-87A-042A-A / 00604	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Lower Main Cell Fuel Quantity Probes - Removal	DMC-412-A-28-10-00-87A-520A-A / 00605	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Lower Main Cell Fuel Quantity Probes - Installation	DMC-412-A-28-10-00-87A-720A-A / 00606	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Upper Left and Upper Right Cell Fuel Quantity Probes - Description	DMC-412-A-28-10-00-88A-042A-A / 00607	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent or 37002-Subsequent
Upper Left and Upper Right Cell Fuel Quantity Probes - Removal	DMC-412-A-28-10-00-88A-520A-A / 00608	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent
Upper Left and Upper Right Cell Fuel Quantity Probes - Installation	DMC-412-A-28-10-00-88A-720A-A / 00609	2023-05-31	SN: 33108-Subsequent or 36001-Subsequent

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Fuel System - General Description

For ease of use, this chapter is divided into two parts. Before proceeding with any maintenance on the helicopter fuel system, carefully read the general safety precautions (412-A-28-00-00-00A-012A-A / 00325).

Applicable: SN: 33001-33107:

The first part provides complete maintenance instructions.

Applicable: SN: 33108-33213 or 36001-36999 or 37002-37999:

The second part provides the same depth of coverage.

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ONLY APPROVED, EXPLOSION PROOF TYPE LIGHTS ARE TO BE USED NEAR OPEN FUEL CELLS AND FUEL LINES.



HANDLE CELLS WITH EXTREME CARE DURING REMOVAL AND INSTALLATION TO PREVENT DAMAGE TO CELLS. DO NOT ATTEMPT TO REMOVE, FOLD, OR INSTALL A CELL WHEN CELL TEMPERATURE IS BELOW 65ºF (18ºC).



ALL FUELING AND DEFUELING OPERATIONS SHALL BE CONDUCTED IN AREAS WHERE FIRE HAZARDS ARE REDUCED TO A MINIMUM. FOR COMPLETE DEFUELING/REFUELING PROCEDURES, READ AND FOLLOW INSTRUCTIONS CONTAINED IN 412-A-12-10-01-00A-913A-A / 00129.



TO AVOID DAMAGE TO THE AUTOMATIC FUEL CONTROL UNIT (AFCU) DRIVE BEARING WHEN DRAINING THE FUEL FILTER AFTER FUEL FILTER INSPECTION OR REPLACEMENT, THE FUEL FILTER DRAIN SHALL BE OPEN FOR A MAXIMUM OF 30 SECONDS WHILE DRAINING ONE ENGINE AT A TIME.

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Fuel Storage and Distribution System - Description and Operation

Applicable: SN: 33001-33107:

The following paragraphs provide a functional description of the components which store and deliver fuel to the engine power sections.

The fuel system (Figure 1 and Figure 2) is comprised of fuel storage cells, transfer system, venting system, dual engine feed system, and electrical control and warning systems. The system is designed to provide fuel containment in the event of an accident or incident during which fuel system rupture might occur. The cells are of a crash-resistant, laminated fabric and rubber construction. The cells are suspended in the fuselage cavities by nylon cord laced between hangers attached to the cells and fuselage structure.

The fuel system has a maximum capacity of 214.2 U.S. gallons (211.4 U.S. gallons usable). Fuel is contained in five interconnected cells (1, 8, 13, 19, and 33, Figure 1). The two forward cells (19 and 33), located under the cabin floor, are supplied continuously by gravity feed from the three aft cells (1, 8, and 13). Under normal conditions, each forward cell supplies fuel to the power section located on the same side of the helicopter. However, if boost pump (16 or 28) in either cell ceases to function, crossfeed valve (4) opens automatically to permit flow from remaining pump to be directed to both power sections.

Forward cells (19 and 33) are divided into forward and aft compartments by baffles (17 and 34). The aft compartments contain sump assemblies (16 and 28) equipped with electrically operated boost pumps, crashworthy defueling valves, flow switches connected to the FUEL BOOST caution panel segments, sump drain valves, sump interconnect valves, and low fuel level warning switches connected to the FUEL LOW caution panel segments. The forward cells are connected by forward and aft interconnect lines which are normally closed by electrically controlled interconnect valves (24 and 31). These valves may be opened by the pilot to permit flow between cells. Fuel quantity probes (tank units) (25, 26, 29, and 35) are installed in each compartment. Flapper valves (18 and 30) in baffles (17 and 34) between compartments allow front to back flow. Ejector pumps (20 and 32) mounted on cell front walls continuously send fuel back over baffles (17 and 34) to aft compartments to ensure maximum usability of fuel at all helicopter attitudes.

The three aft cells (1, 8, and 13) are aligned laterally across the fuselage. The right aft cell (1) supplies the right forward cell (33), and the left aft cell (8) supplies the left forward cell (19). Interconnect fittings (11 and 22) connect the center aft cell (13) to right and left aft cells (1 and 8) to maintain the same fuel level in all three aft cells. The helicopter is fueled through a filler cap and adapter assembly (2) is installed on the right aft cell (1) for system servicing. A fuel quantity probe (23) is installed in center cell (13). Capped connections for use with auxiliary fuel tanks are provided on system vent lines and on fittings located aft of each forward cell (19 and 33).

The center aft cell (13) is vented through a line attached to vent cover (5). Right and left aft cells (1 and 8) are vented through fittings (7 and 41) attached to ports in top of cells. Crashworthy vent valves (27 and 36) are used to vent forward cells (19 and 33).

A cover (9) on rear of center aft cell (13) provides main point of access to cell interior. Vent cover (5) may also be removed for access when necessary. The rear cover contains two governor bleed line check valves (10) that enable excess fuel from each engine fuel control to return to center aft cell (13).

Fuel is drained from center aft cell (13) through outlet fittings (12 and 38) mounted on underside of cell. Similarly, fuel is drained from right and left aft cells (1 and 8) through outlet fittings (14 and 39) attached to underside of cells. Fuel enters and leaves forward cells (19 and 33) through inlet/outlet fittings (15 and 37) installed in rear walls of aft compartment of cells.

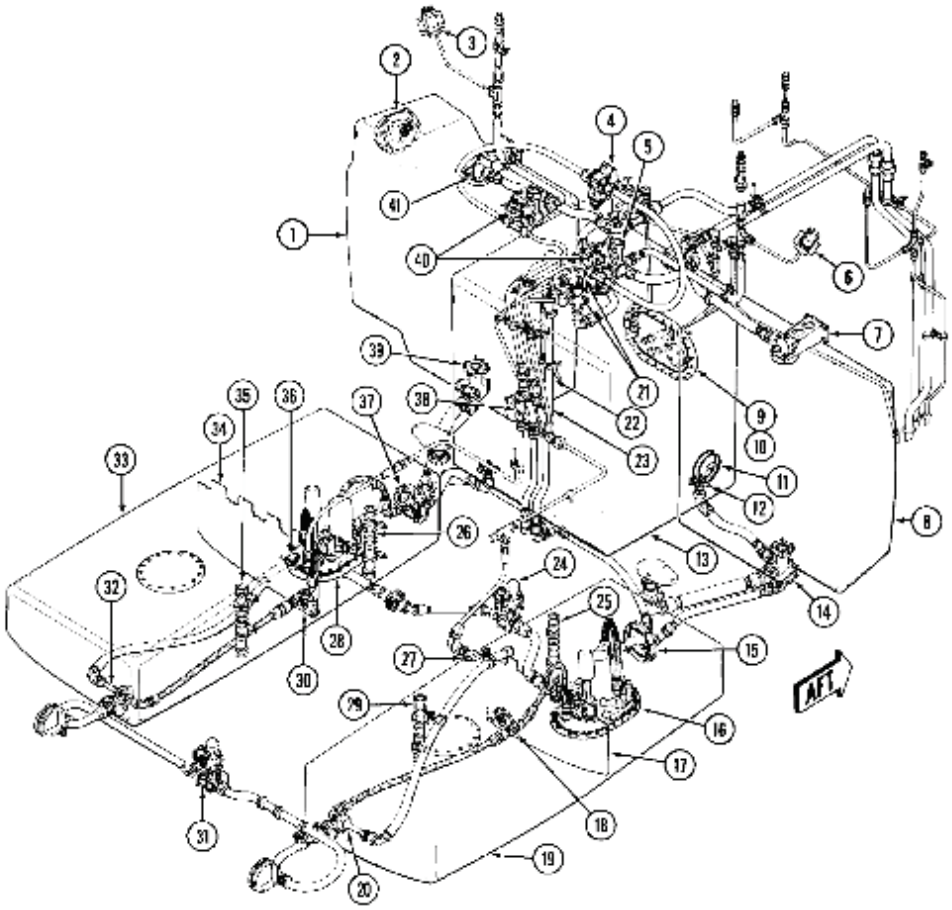
Two siphon breaker valves (21) are installed in vent lines near left shutoff valve (40) to prevent fuel loss due to siphoning through vent tubes.

Engine fuel feed line pressure (boost pump pressure) is sensed by fuel pressure transmitters (3 and 6), and corresponding electrical signals are transmitted to pressure (FUEL PRESS) indicators located in instrument panel.

Refer to 412-A-52-00-00-00A-028A-A / 00775 for locations of doors, panels, cowlings, and covers to be removed for access to fuel system components.

Table 1. Fuel System Troubleshooting (S/N 33001 through 33107)

INDICATION OF TROUBLE	PROBABLE CAUSE	CORRECTIVE ACTION
FUEL FILTER caution panel segment illuminated.	Filter element clogged.	Remove and inspect filter element. Clean or replace filter element. (Refer to engine maintenance manual.)
FUEL BOOST caution panel segment illuminated.	Obstructed pump line through flow switch and ejector pump.	Replace flow switch (412-A-28-10-00-11A-042A-A / 00357), ejector pump (412-A-28-10-00-15A-042A-A / 00370), or obstructed hoses.
	No electrical power to boost pump.	Repair wiring or replace faulty units in circuit.
	Defective boost pump.	Replace boost pump (412-A-28-10-00-06A-042A-A / 00344).
Shutoff valve inoperative.	Shutoff valve faulty or lacking electrical power.	Repair circuit.
		Replace shutoff valve (412-A-28-10-00-21A-042A-A / 00388).
Interconnect valve (forward, aft, or crossfeed) inoperative.	Interconnect valve faulty or lacking electrical power.	Repair circuit.
		Replace interconnect valve (412-A-28-10-00-01A-042A-A / 00328, 412-A-28-10-00-02A-042A-A / 00331, and 412-A-28-10-00-09A-042A-A / 00351).
FUEL LOW caution panel segment illuminated when fuel is not low or fails to illuminate when fuel is low.	Defective float switch or wiring.	Replace float switch. Repair wiring.



1. Right outboard aft fuel cell
2. Filler cap and adapter assembly
3. Right fuel pressure transmitter
4. System crossfeed valve
5. Center aft fuel cell vent cover
6. Left fuel pressure transmitter
7. Left outboard aft fuel cell vent fitting
8. Left outboard aft fuel cell
9. Center aft fuel cell rear cover
10. Governor bleed line check valve
11. Center aft fuel cell left interconnect fitting
12. Center aft fuel cell left outlet fitting
13. Center aft fuel cell
14. Left outboard aft fuel cell outlet fitting
15. Left forward fuel cell inlet/outlet fitting

(S/N 33001 33107)

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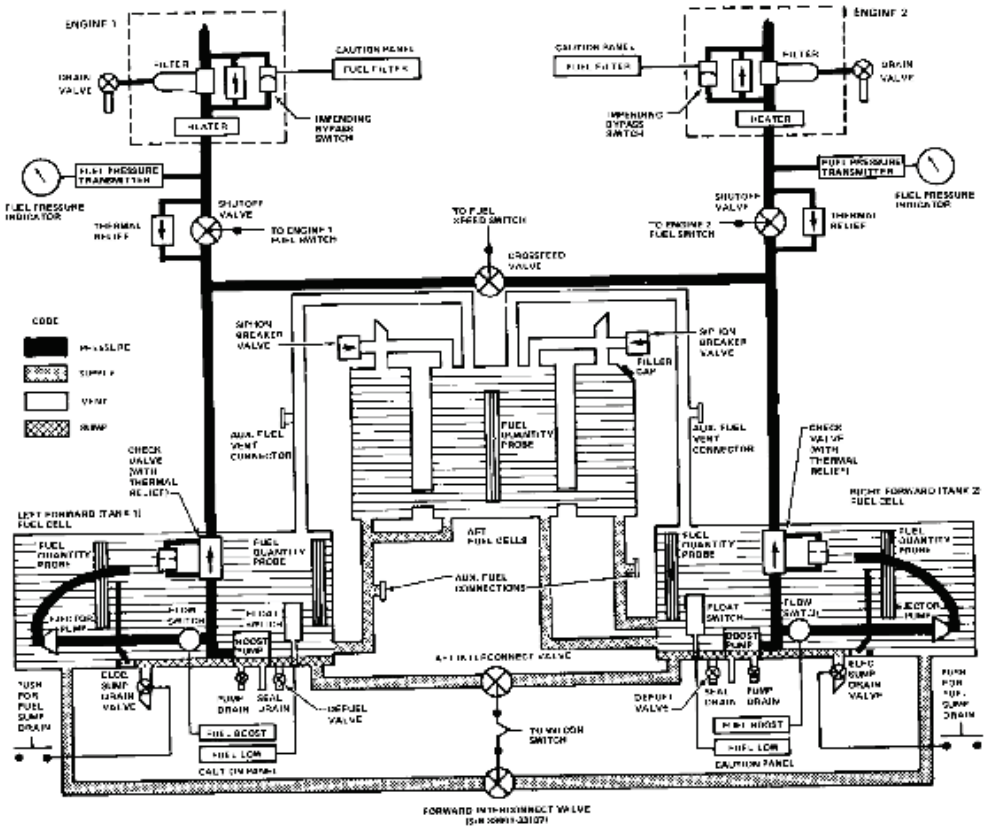
Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 1 of 2)

16. Left forward fuel cell fuel sump area consisting of:
    - Fuel sump assembly
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
17. Left forward fuel cell baffle
18. Left forward fuel cell flapper valve
19. Left forward fuel cell
20. Left forward fuel cell ejector pump and support
21. Siphon breaker valves
22. Center aft fuel cell right interconnect fitting
23. Aft fuel cell fuel quantity probe
24. Aft interconnect valve
25. Left forward fuel cell aft compartment fuel quantity probe
26. Right forward fuel cell aft compartment fuel quantity probe
27. Left forward fuel cell crashworthy vent valve
28. Right forward fuel cell fuel sump area consisting of:
    - Fuel sump
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
29. Left forward fuel cell forward compartment fuel quantity probe
30. Right forward fuel cell flapper valve
31. Forward interconnect valve
32. Right forward fuel cell ejector pump and support
33. Right forward fuel cell
34. Right forward fuel cell baffle
35. Right forward fuel cell forward compartment fuel quantity probe
36. Right forward fuel cell crashworthy vent valve
37. Right forward fuel cell inlet/outlet fitting
38. Center aft fuel cell right outlet fitting
39. Right outboard aft fuel cell outlet fitting
40. Shutoff valve
41. Right outboard aft fuel cell vent fitting

(S/N 33001 33107)

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Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 2 of 2)



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Figure 2. Fuel System Schematic (S/N 33001 through 33107) (Sheet 1 of 1)

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Fuel Storage and Distribution System - Pressure Testing

Procedure

1. Cap main fuel line and vent connections.



ENSURE FUEL CELL ACCESS PANELS ARE INSTALLED PRIOR TO APPLYING INFLATION PRESSURE.

NOTE

For field pressure test of the fuel system, fill the helicopter fuel cells with fuel. Inspect fuel system and the area affected by the work accomplished for leaks. Repeat the inspection procedures several times as the fuel cells are being filled.

2. Use regulated, low pressure, filtered, compressed nitrogen or compressed air source with a manometer or accurate pressure gauge and a shutoff valve.



DO NOT APPLY MORE THAN 1.0 PSIG (6.9 KPA) PRESSURE TO FUEL SYSTEM. SEVERE DAMAGE TO CELLS AND STRUCTURE MAY RESULT.

3. Apply 0.5 to 1.0 PSIG (3.5 to 6.9 kPa) pressure to all cells and tubing of the fuel system. Shut off pressure source. Verify pressure does not decrease after 15 minutes.

4. If leakage is detected, apply a soap and water solution to all repaired areas and to areas suspected of leaking. Make repairs and repeat pressure test until no leaks are indicated.

NOTE

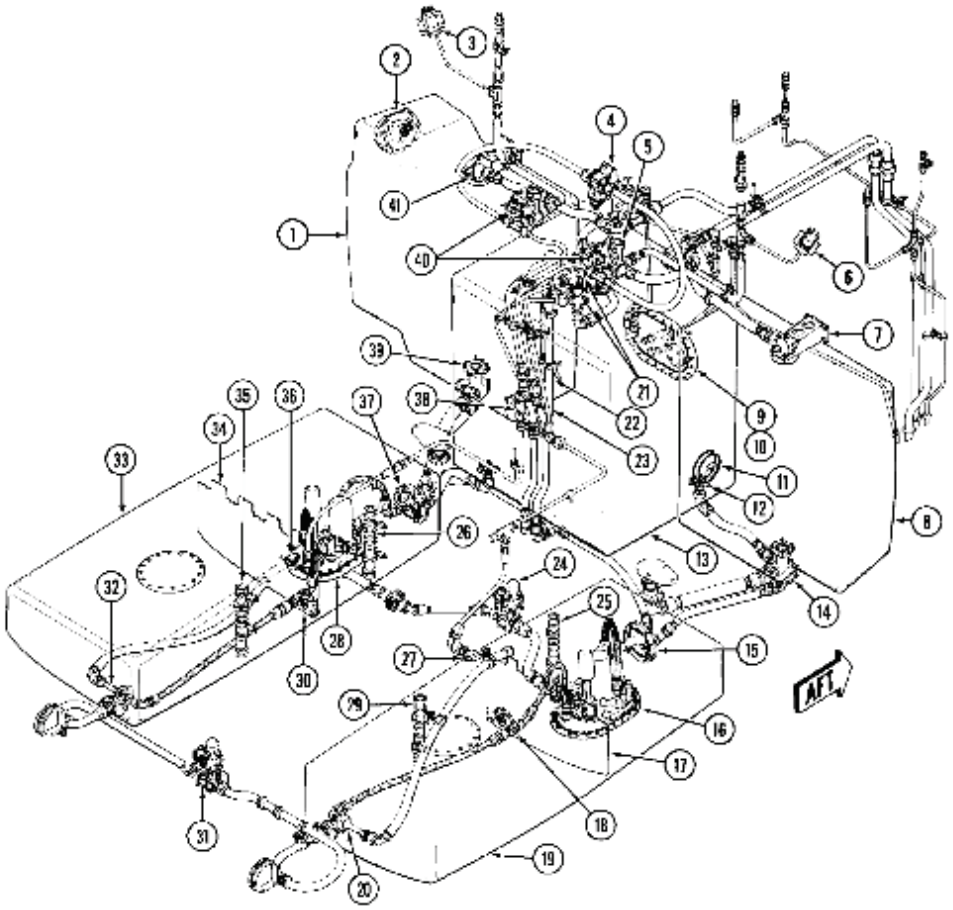
For field pressure test of the fuel system, fill the helicopter fuel cells with fuel. Inspect fuel system and the area affected by the work performed for leaks. Repeat the inspection procedures several times as the fuel cells are being filled.

5. Upon satisfactory conclusion of test, remove soap residue and remove all caps and plugs used to close off system.

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Forward Interconnect Valve - Description

The electrically controlled forward interconnect valve (31, Figure 1) is installed in an interconnect line between the forward compartments of the forward fuel cells. This valve is normally closed, but can be opened to permit crossflow between cells when fuel level becomes low.



1. Right outboard aft fuel cell
2. Filler cap and adapter assembly
3. Right fuel pressure transmitter
4. System crossfeed valve
5. Center aft fuel cell vent cover
6. Left fuel pressure transmitter
7. Left outboard aft fuel cell vent fitting
8. Left outboard aft fuel cell
9. Center aft fuel cell rear cover
10. Governor bleed line check valve
11. Center aft fuel cell left interconnect fitting
12. Center aft fuel cell left outlet fitting
13. Center aft fuel cell
14. Left outboard aft fuel cell outlet fitting
15. Left forward fuel cell inlet/outlet fitting

(S/N 33001 33107)

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Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 1 of 2)

16. Left forward fuel cell fuel sump area consisting of:
    - Fuel sump assembly
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
17. Left forward fuel cell baffle
18. Left forward fuel cell flapper valve
19. Left forward fuel cell
20. Left forward fuel cell ejector pump and support
21. Siphon breaker valves
22. Center aft fuel cell right interconnect fitting
23. Aft fuel cell fuel quantity probe
24. Aft interconnect valve
25. Left forward fuel cell aft compartment fuel quantity probe
26. Right forward fuel cell aft compartment fuel quantity probe
27. Left forward fuel cell crashworthy vent valve
28. Right forward fuel cell fuel sump area consisting of:
    - Fuel sump
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
29. Left forward fuel cell forward compartment fuel quantity probe
30. Right forward fuel cell flapper valve
31. Forward interconnect valve
32. Right forward fuel cell ejector pump and support
33. Right forward fuel cell
34. Right forward fuel cell baffle
35. Right forward fuel cell forward compartment fuel quantity probe
36. Right forward fuel cell crashworthy vent valve
37. Right forward fuel cell inlet/outlet fitting
38. Center aft fuel cell right outlet fitting
39. Right outboard aft fuel cell outlet fitting
40. Shutoff valve
41. Right outboard aft fuel cell vent fitting

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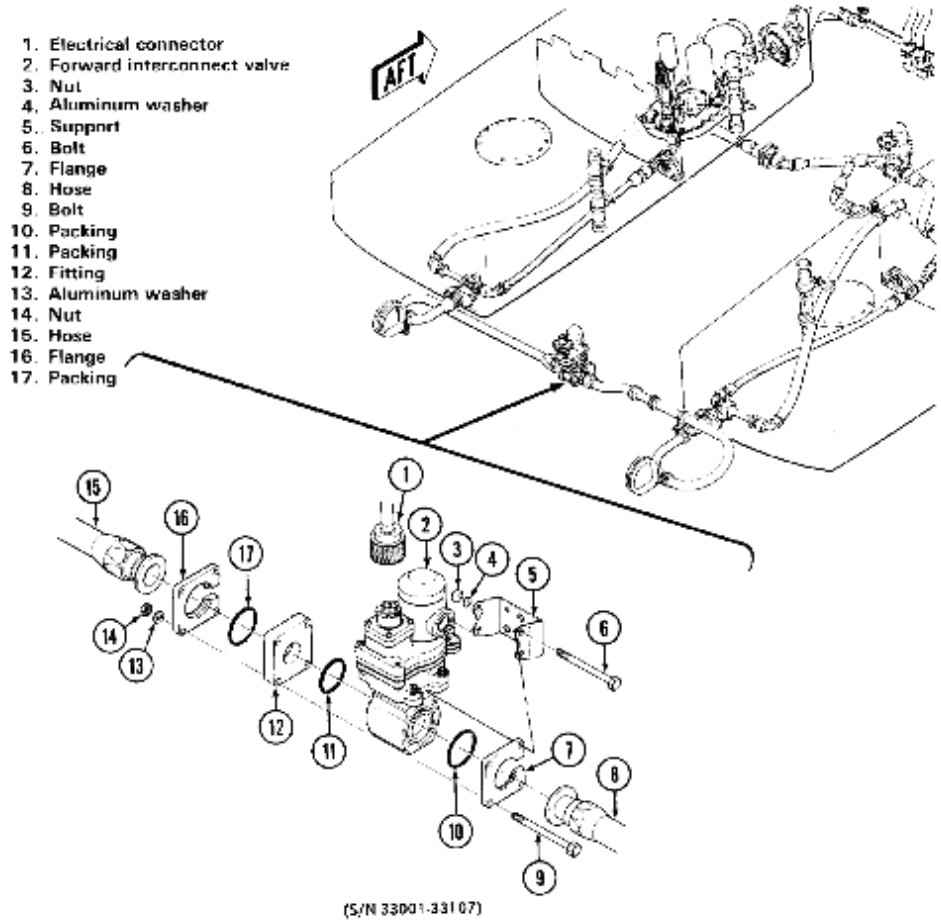
Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 2 of 2)

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Forward Interconnect Valve - Removal

Procedure

1. Disconnect electrical power.
2. Defuel system (412-A-12-00-00-00A-009A-A / 00125).
3. Remove cabin floor panel.
4. Disconnect electrical connector (1, Figure 1).
5. Remove bolts (6 and 9).
6. Separate forward interconnect valve (2) from support (5). Remove flanges (7 and 16), packings (10, 11, and 17), and fitting (12). Discard packings.
7. Cap or plug openings.



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Figure 1. Forward Interconnect Valve (Sheet 1 of 1)

Forward Interconnect Valve - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

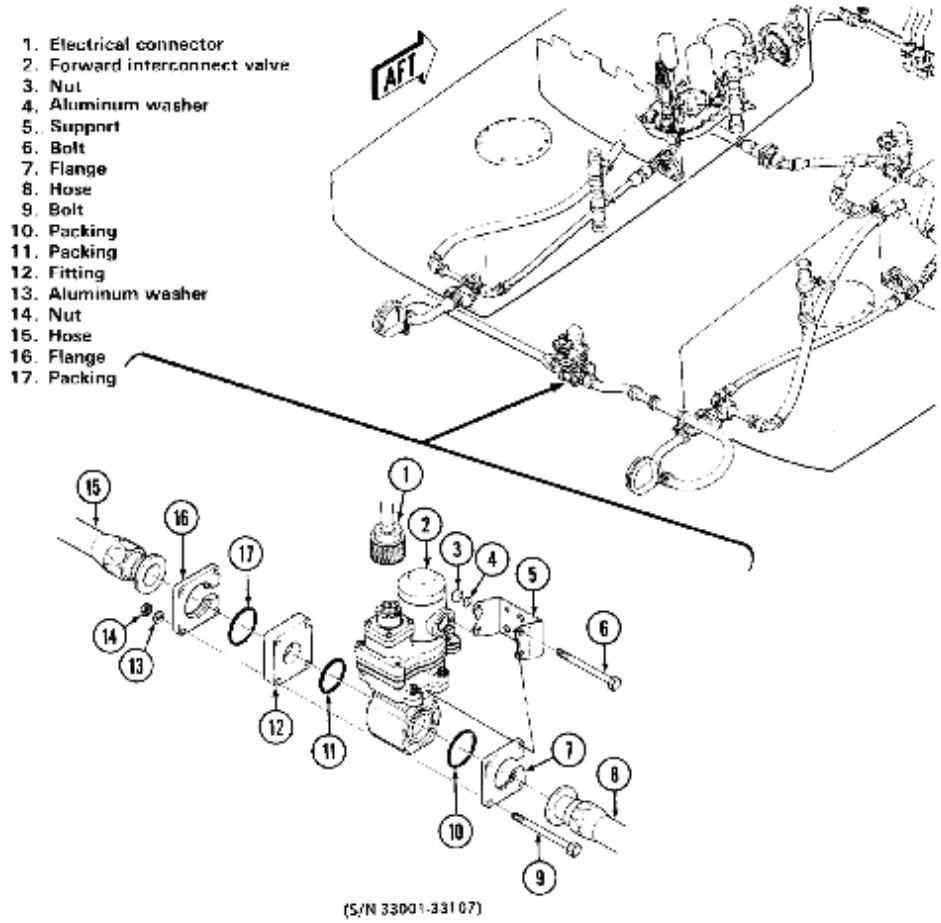
Procedure

1. Apply assembly fluid (C-024) to new packings (10, 11, and 17, Figure 1).
2. Position packings (10 and 11) in forward interconnect valve (2) and position packing (17) in groove of fitting (12).

NOTE

Fitting (12) has a plug installed in one side. Locate plug on bottom when installing fitting.

3. Place fitting (12) on forward interconnect valve (2) with packing (17) outboard.
4. Remove protective caps or plugs from hoses (8 and 15).
5. Place flanges (7 and 16) on hoses (8 and 15) with flange slots facing aft. Align mounting holes in parts and install two bolts (9), aluminum washers (13), and nuts (14). Do not tighten nuts at this time.
6. Position forward interconnect valve (2) in support (5). Install two bolts (6), aluminum washers (4), and nuts (3). Tighten nuts (3 and 14).
7. Connect electrical connector (1).
8. Install cabin floor panel.
9. Perform fuel system pressure test (412-A-28-10-00-00A-362A-A / 00327).
10. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).
11. Perform operational check of INTCON switch and forward and aft interconnect valves (412-A-96-03-01-20A-320A-A / 02073 and 412-A-96-04-00-06A-320A-A / 02117).
12. Check area for leaks.
13. Install sump access door.

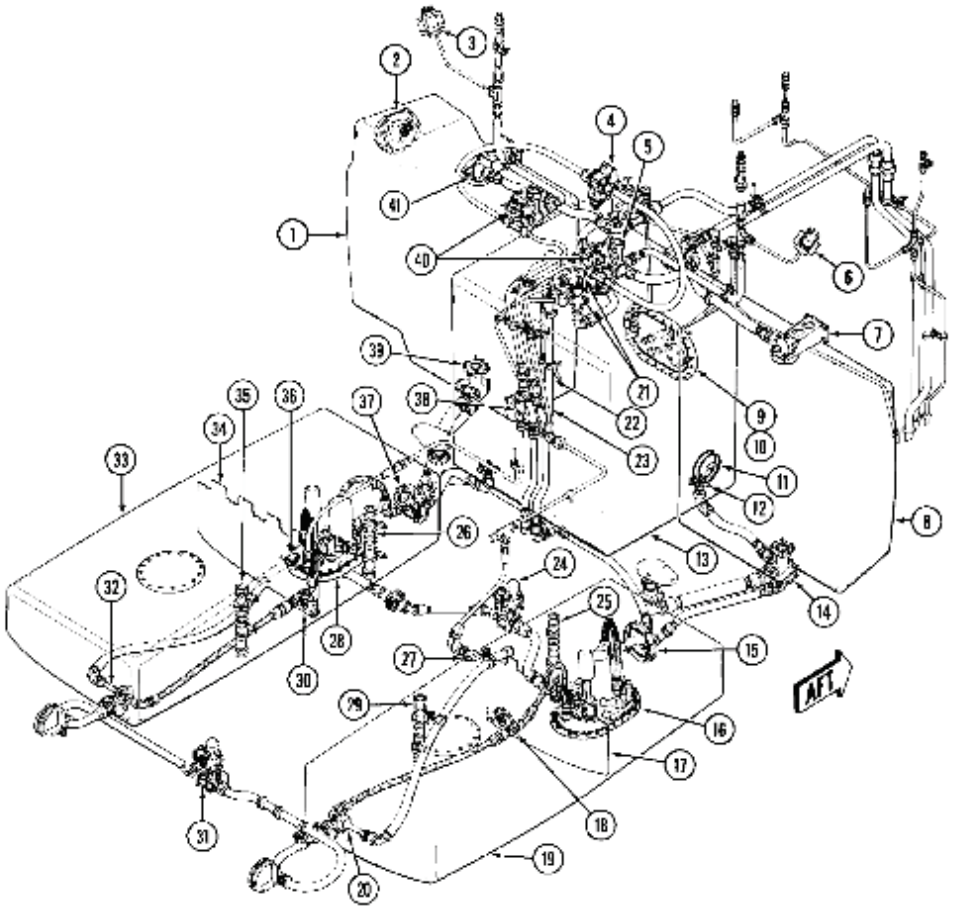


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Figure 1. Forward Interconnect Valve (Sheet 1 of 1)

Aft Interconnect Valve - Description

The electrically controlled aft interconnect valve (24, Figure 1) is installed in an interconnect line between aft compartments of aft fuel cells. This valve is normally closed, but may be opened to permit crossflow between cells.



1. Right outboard aft fuel cell
2. Filler cap and adapter assembly
3. Right fuel pressure transmitter
4. System crossfeed valve
5. Center aft fuel cell vent cover
6. Left fuel pressure transmitter
7. Left outboard aft fuel cell vent fitting
8. Left outboard aft fuel cell
9. Center aft fuel cell rear cover
10. Governor bleed line check valve
11. Center aft fuel cell left interconnect fitting
12. Center aft fuel cell left outlet fitting
13. Center aft fuel cell
14. Left outboard aft fuel cell outlet fitting
15. Left forward fuel cell inlet/outlet fitting

(S/N 33001 33107)

ICN-412-A-281000-B-97499-00001-A-001-01

Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 1 of 2)

16. Left forward fuel cell fuel sump area consisting of:
    - Fuel sump assembly
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
17. Left forward fuel cell baffle
18. Left forward fuel cell flapper valve
19. Left forward fuel cell
20. Left forward fuel cell ejector pump and support
21. Siphon breaker valves
22. Center aft fuel cell right interconnect fitting
23. Aft fuel cell fuel quantity probe
24. Aft interconnect valve
25. Left forward fuel cell aft compartment fuel quantity probe
26. Right forward fuel cell aft compartment fuel quantity probe
27. Left forward fuel cell crashworthy vent valve
28. Right forward fuel cell fuel sump area consisting of:
    - Fuel sump
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
29. Left forward fuel cell forward compartment fuel quantity probe
30. Right forward fuel cell flapper valve
31. Forward interconnect valve
32. Right forward fuel cell ejector pump and support
33. Right forward fuel cell
34. Right forward fuel cell baffle
35. Right forward fuel cell forward compartment fuel quantity probe
36. Right forward fuel cell crashworthy vent valve
37. Right forward fuel cell inlet/outlet fitting
38. Center aft fuel cell right outlet fitting
39. Right outboard aft fuel cell outlet fitting
40. Shutoff valve
41. Right outboard aft fuel cell vent fitting

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ICN-412-A-281000-B-97499-00002-A-001-01

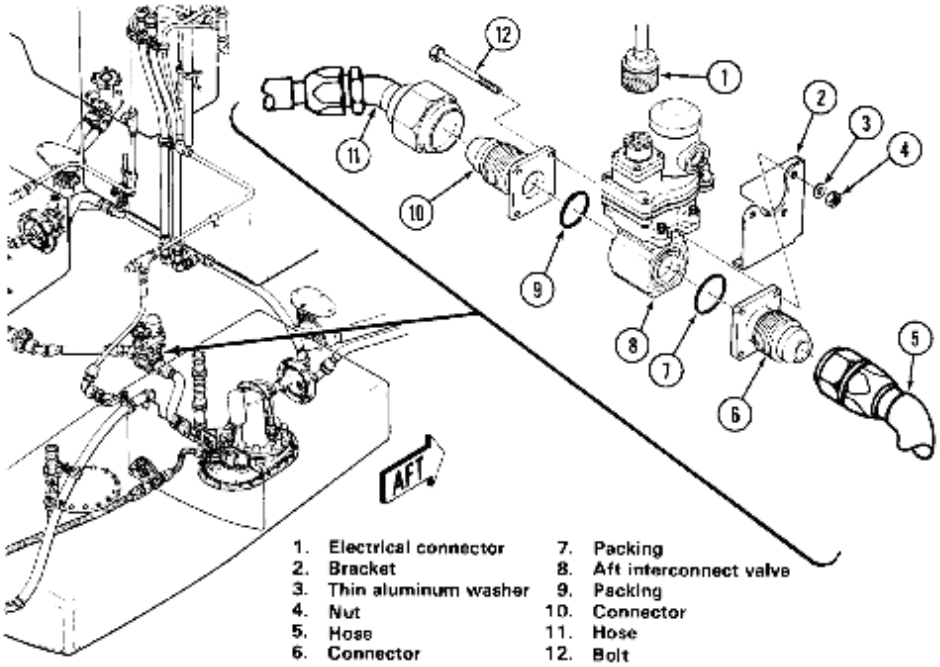
Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 2 of 2)

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Aft Interconnect Valve - Removal

Procedure

1. Disconnect electrical power.
2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).
3. Remove access door from underside of fuselage.
4. Disconnect electrical connector (1, Figure 1).
5. Disconnect hoses (5 and 11).
6. Remove bolts (12).
7. Remove aft interconnect valve (8), connectors (6 and 10), and packings (7 and 9). Discard packings.
8. Cap or plug openings.



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Figure 1. Aft Interconnect Valve (Sheet 1 of 1)

Aft Interconnect Valve - Installation

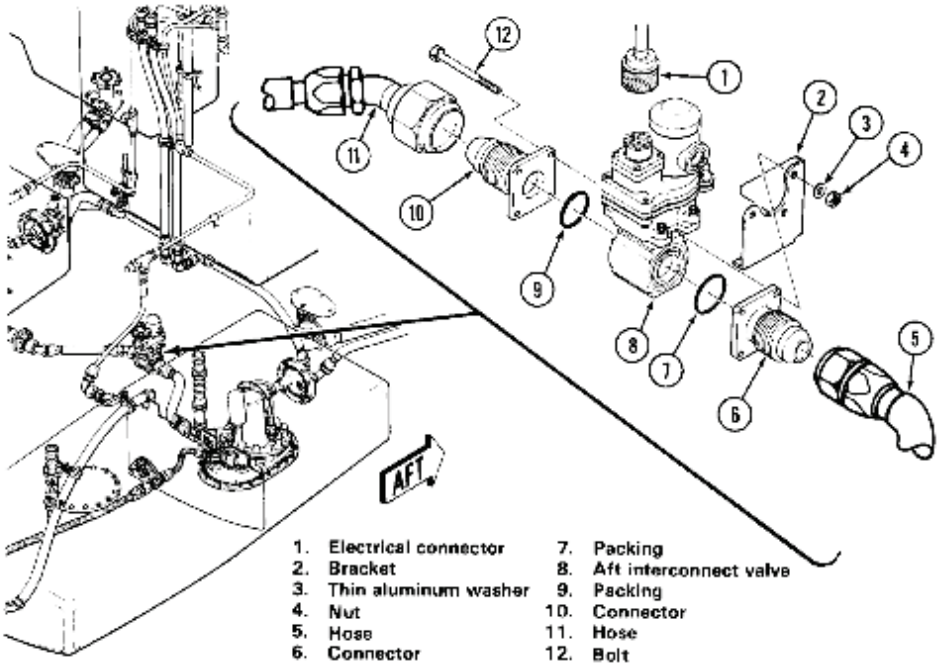
Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

1. Apply assembly fluid (C-024) to packings (7 and 9, Figure 1).
2. Place packings (7 and 9) and connectors (6 and 10) on aft interconnect valve (8).
3. Place aft interconnect valve (8) and connectors (6 and 10) on bracket (2).
4. Install bolts (12), thin aluminum washers (3), and nuts (4).
5. Connect hoses (5 and 11).
6. Connect electrical connector (1).
7. Perform fuel system pressure test (412-A-28-10-00-00A-362A-A / 00327).
8. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).
9. Perform operational check of aft interconnect valve (8) (412-A-96-04-00-06A-320A-A / 02117).
10. Check area for leaks.
11. Install access doors.



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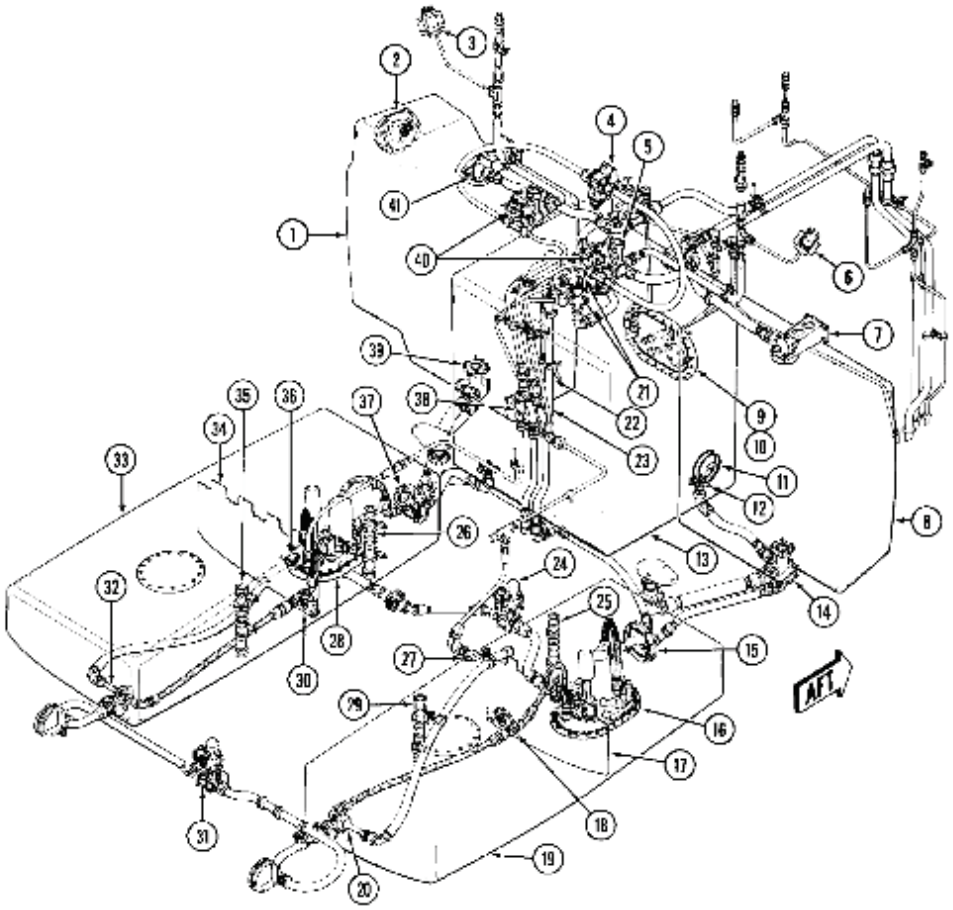
Figure 1. Aft Interconnect Valve (Sheet 1 of 1)

Sump Assemblies - Description

A sump assembly is mounted in the openings on the underside of the aft compartment of each forward cell (16 and 28, Figure 1). Removal of the sump assembly permits access for maintenance of boost pump, flow switch, defueling valve, sump drain valve, low fuel level warning switch, and sump interconnect valve. The boost pump and sump drain valve may be removed without removing the sump assembly.

NOTE

Maintenance procedures are the same for both sump assemblies.



1. Right outboard aft fuel cell
2. Filler cap and adapter assembly
3. Right fuel pressure transmitter
4. System crossfeed valve
5. Center aft fuel cell vent cover
6. Left fuel pressure transmitter
7. Left outboard aft fuel cell vent fitting
8. Left outboard aft fuel cell
9. Center aft fuel cell rear cover
10. Governor bleed line check valve
11. Center aft fuel cell left interconnect fitting
12. Center aft fuel cell left outlet fitting
13. Center aft fuel cell
14. Left outboard aft fuel cell outlet fitting
15. Left forward fuel cell inlet/outlet fitting

(S/N 33001 33107)

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Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 1 of 2)

16. Left forward fuel cell fuel sump area consisting of:
    - Fuel sump assembly
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
17. Left forward fuel cell baffle
18. Left forward fuel cell flapper valve
19. Left forward fuel cell
20. Left forward fuel cell ejector pump and support
21. Siphon breaker valves
22. Center aft fuel cell right interconnect fitting
23. Aft fuel cell fuel quantity probe
24. Aft interconnect valve
25. Left forward fuel cell aft compartment fuel quantity probe
26. Right forward fuel cell aft compartment fuel quantity probe
27. Left forward fuel cell crashworthy vent valve
28. Right forward fuel cell fuel sump area consisting of:
    - Fuel sump
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
29. Left forward fuel cell forward compartment fuel quantity probe
30. Right forward fuel cell flapper valve
31. Forward interconnect valve
32. Right forward fuel cell ejector pump and support
33. Right forward fuel cell
34. Right forward fuel cell baffle
35. Right forward fuel cell forward compartment fuel quantity probe
36. Right forward fuel cell crashworthy vent valve
37. Right forward fuel cell inlet/outlet fitting
38. Center aft fuel cell right outlet fitting
39. Right outboard aft fuel cell outlet fitting
40. Shutoff valve
41. Right outboard aft fuel cell vent fitting

(S/N 33001 33107)

ICN-412-A-281000-B-97499-00002-A-001-01

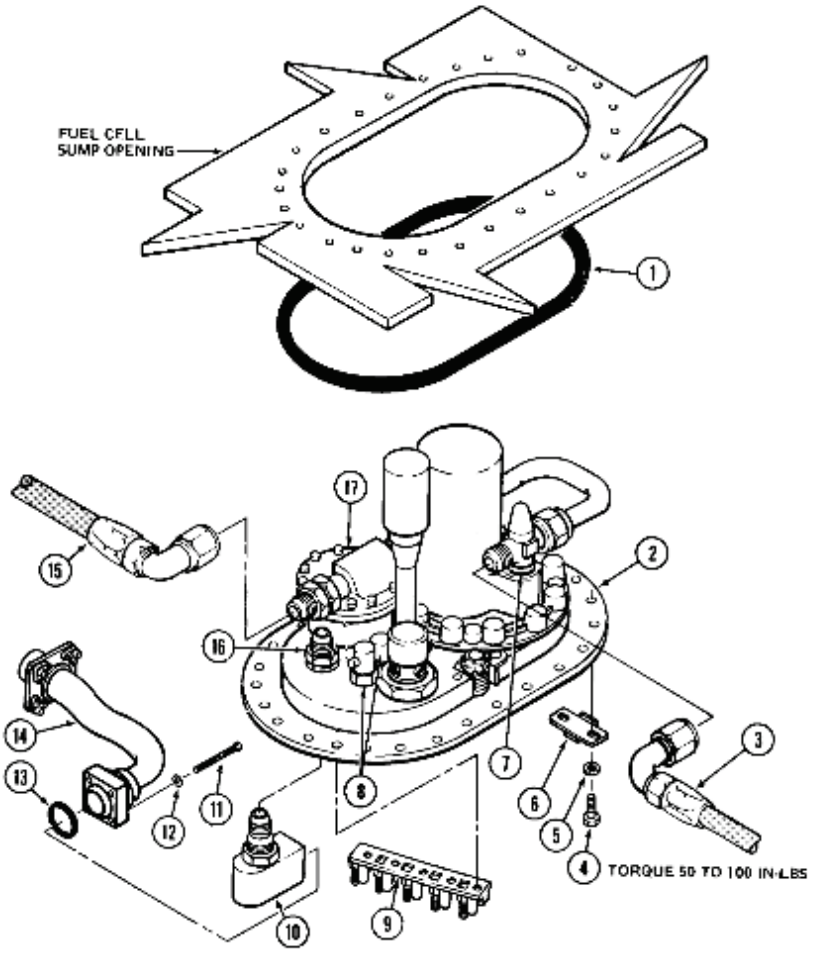
Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 2 of 2)

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Sump Assemblies - Removal

Procedure

1. Disconnect electrical power.
2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).
3. Disconnect electrical leads at terminal block (9, Figure 1) and fuel quantity probe connectors (8) on bottom of sump assembly (2).
4. Remove screws (11) and washers (12).
5. Separate hose (14) from fitting (10). Remove and discard packing (13). Remove bonding strap from sump assembly (2).
6. Remove bolts (4) and retainers (6).
7. Lower sump assembly (2) enough to reach through cell sump opening.
8. Disconnect hose (3).
9. Disconnect hose (15).
10. Disconnect wires from fuel quantity probe connectors (8).
11. Remove packing (1) from cell. Discard packing.
12. Cover fuel cell sump opening.



1. Packing
2. Sump assembly
3. Hose
4. Bolt
6. Thin aluminum washer
6. Retainer
7. Fuel outlet fitting
8. Fuel quantity probe connectors
9. Terminal block
10. Fitting
11. Screw
12. Washer
13. Packing
14. Hose
15. Hose
16. Valve (AE80854M)
17. Flow switch

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Figure 1. Sump Assembly (Sheet 1 of 1)

Sump Assemblies - Inspection

Procedure

1. Inspect area where sump assembly mates with cell for damage and cleanliness. Inspect packing groove.
2. Inspect components for security of mounting.
3. Inspect hoses, union, and fittings for contamination and thread damage. Contamination shall be removed. Thread damage that can be detected visually is not acceptable; part shall be replaced.
4. Inspect electrical leads and connectors for condition.
5. Replace unserviceable components.

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Sump Assemblies - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

1. Remove protective covering from fuel cell sump opening.

2. Coat packing (1, Figure 1) with assembly fluid (C-024) and place in groove around fuel cell sump opening.

3. Hold sump assembly (2) slightly below fuel cell sump opening.

4. Connect hose (15) to flow switch (17).

5. Connect hose (3) to fuel outlet fitting (7).

6. Connect wires to fuel quantity probe connectors (8).

NOTE

Retainers (6) are installed on bolts (4) at each side and each end of sump assembly (2).

7. Install bolts (4), washers (5), and retainers (6). Torque bolts 50 to 100 in.lbs. (5.65 to 11.30 N-m).

8. Connect hose (14) to fitting (10) with packing (13), screws (11), and washers (12).

9. Connect electrical leads at terminal block (9) and fuel quantity probe connectors (8) on bottom of sump assembly (2).

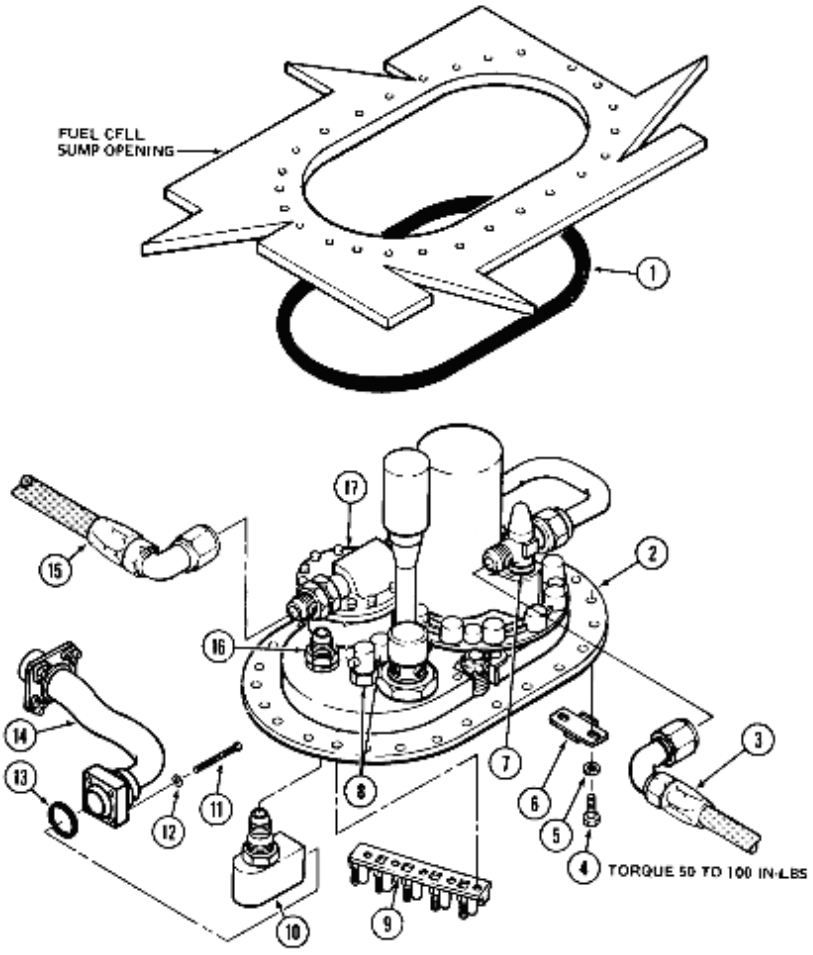
10. Perform fuel system pressure test (412-A-28-10-00-00A-362A-A / 00327).

11. Perform operational check of the fuel quantity indicating system before servicing the system (412-A-96-12-00-11A-273A-A / 02006).

12. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).

13. Check area for leaks.

14. Install sump access door.



1. Packing
2. Sump assembly
3. Hose
4. Bolt
6. Thin aluminum washer
6. Retainer
7. Fuel outlet fitting
8. Fuel quantity probe connectors
9. Terminal block
10. Fitting
11. Screw
12. Washer
13. Packing
14. Hose
15. Hose
16. Valve (AE80854M)
17. Flow switch

(S/N 33001-33107)

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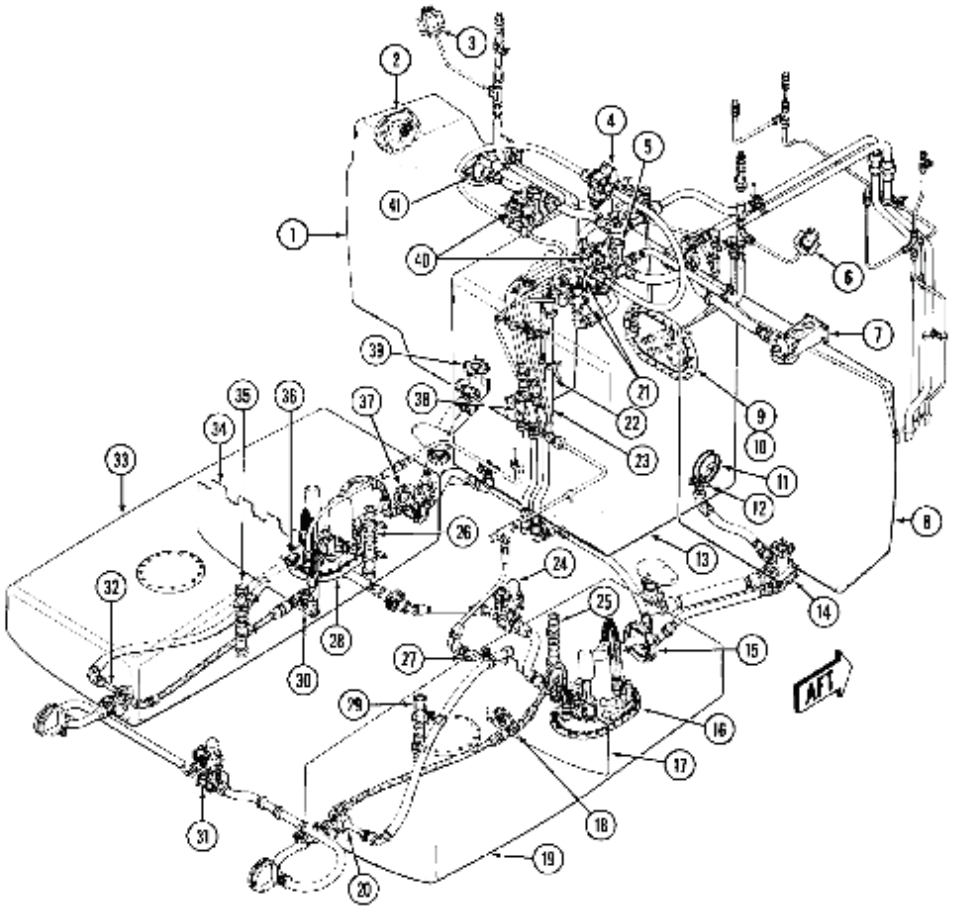
Figure 1. Sump Assembly (Sheet 1 of 1)

Low Fuel Level Warning Switches - Description

Float actuated low fuel level warning switches are connected to a FUEL LOW caution panel segment. The switches are installed on a support on each sump assembly (16 and 28, Figure 1).

NOTE

The maintenance procedures for left and right low fuel level warning switches are the same.



1. Right outboard aft fuel cell
2. Filler cap and adapter assembly
3. Right fuel pressure transmitter
4. System crossfeed valve
5. Center aft fuel cell vent cover
6. Left fuel pressure transmitter
7. Left outboard aft fuel cell vent fitting
8. Left outboard aft fuel cell
9. Center aft fuel cell rear cover
10. Governor bleed line check valve
11. Center aft fuel cell left interconnect fitting
12. Center aft fuel cell left outlet fitting
13. Center aft fuel cell
14. Left outboard aft fuel cell outlet fitting
15. Left forward fuel cell inlet/outlet fitting

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Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 1 of 2)

16. Left forward fuel cell fuel sump area consisting of:
    - Fuel sump assembly
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
17. Left forward fuel cell baffle
18. Left forward fuel cell flapper valve
19. Left forward fuel cell
20. Left forward fuel cell ejector pump and support
21. Siphon breaker valves
22. Center aft fuel cell right interconnect fitting
23. Aft fuel cell fuel quantity probe
24. Aft interconnect valve
25. Left forward fuel cell aft compartment fuel quantity probe
26. Right forward fuel cell aft compartment fuel quantity probe
27. Left forward fuel cell crashworthy vent valve
28. Right forward fuel cell fuel sump area consisting of:
    - Fuel sump
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
29. Left forward fuel cell forward compartment fuel quantity probe
30. Right forward fuel cell flapper valve
31. Forward interconnect valve
32. Right forward fuel cell ejector pump and support
33. Right forward fuel cell
34. Right forward fuel cell baffle
35. Right forward fuel cell forward compartment fuel quantity probe
36. Right forward fuel cell crashworthy vent valve
37. Right forward fuel cell inlet/outlet fitting
38. Center aft fuel cell right outlet fitting
39. Right outboard aft fuel cell outlet fitting
40. Shutoff valve
41. Right outboard aft fuel cell vent fitting

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Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 2 of 2)

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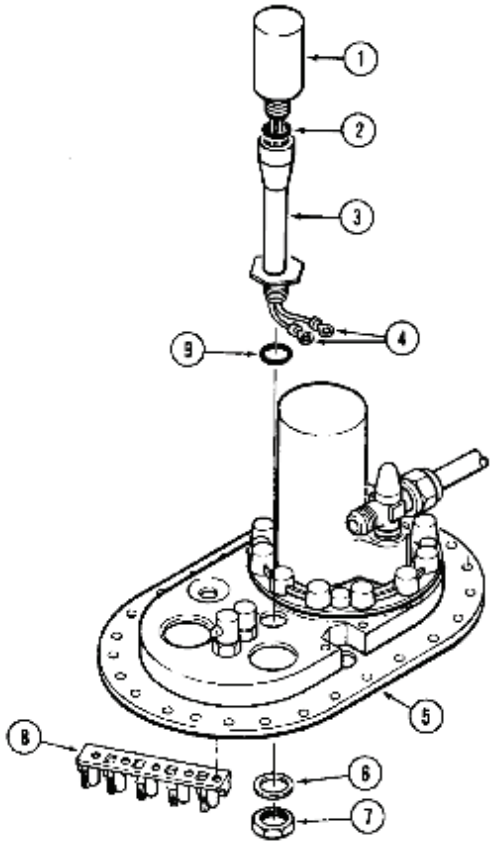
Low Fuel Level Warning Switches - Removal

Procedure

NOTE

The following removal procedures for left and right low fuel level warning switches are the same.

1. Remove sump assembly (412-A-28-10-00-03A-520A-A / 00335).
2. Disconnect electrical leads (4, Figure 1) from terminal block (8).
3. Hold support (3) and unscrew low fuel level warning switch (1).
4. Remove low fuel level warning switch (1) and attached electrical leads (4) from support (3). Remove and discard packing (2).
5. Remove nut (7). Lift support (3) from sump assembly (5).
6. Remove and discard packing (9).



1. Low fuel level warning switch
2. Packing
3. Support
4. Electrical leads
5. Sump assembly
6. Thin aluminum washer
7. Nut
8. Terminal block
9. Packing

(S/N 33001 — 33107)

ICN-412-A-281000-B-97499-00007-A-001-01

Figure 1. Low Fuel Level Warning Switch (Sheet 1 of 1)

Low Fuel Level Warning Switches - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

NOTE

The following installation procedures for left and right low fuel level warning switches are the same.

1. Place packing (2, Figure 1) on low fuel level warning switch (1).

2. Insert electrical leads (4) through support (3). Install and tighten low fuel level warning switch (1).

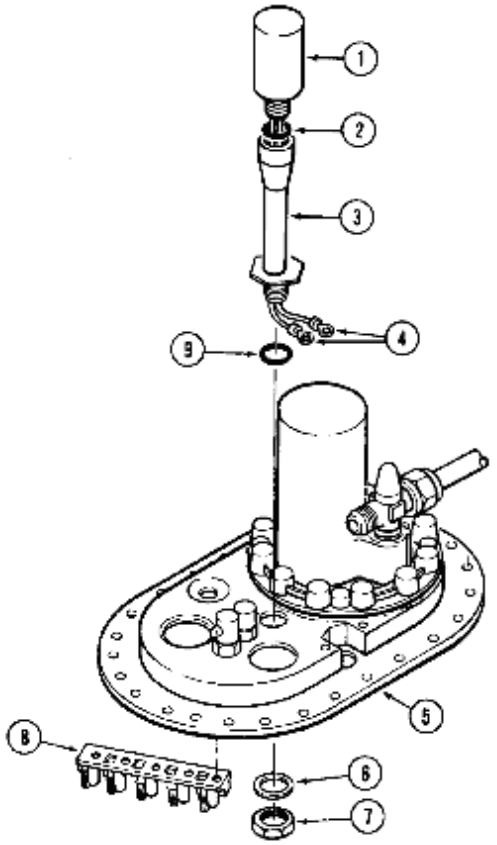
3. Place packing (9) on support (3).

4. Position support (3) on sump assembly (5).

5. Install washer (6) and nut (7). Torque nut (7) 15 to 25 in.lbs. (1.70 to 2.83 N-m) and secure with lockwire (C-405).

6. Connect electrical leads (4) to terminal block (8).

7. Install sump assembly (5) (412-A-28-10-00-03A-720A-A / 00337).



1. Low fuel level warning switch
2. Packing
3. Support
4. Electrical leads
5. Sump assembly
6. Thin aluminum washer
7. Nut
8. Terminal block
9. Packing

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Figure 1. Low Fuel Level Warning Switch (Sheet 1 of 1)

Defueling Valves - Description

Each sump assembly is provided with a crash-resistant defuel valve which is manually operated by insertion of an AN815-12 union to provide gravity defueling of cells.

NOTE

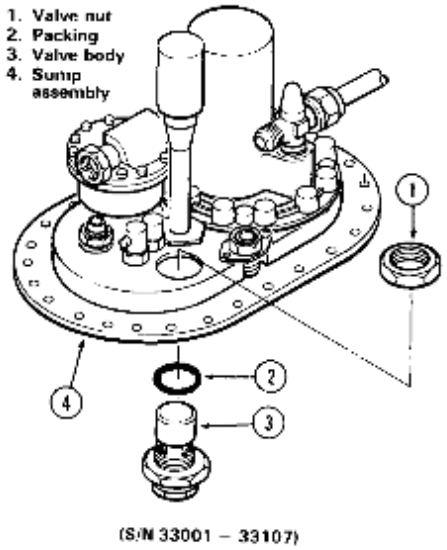
Maintenance procedures for defueling valve in each sump assembly are the same.

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Defueling Valves - Removal

Procedure

1. Remove sump assembly (412-A-28-10-00-03A-520A-A / 00335).
2. Remove nut (1, Figure 1), valve body (3), and packing (2). Discard packing.



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Figure 1. Defueling Valve (Sheet 1 of 1)

Defueling Valves - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

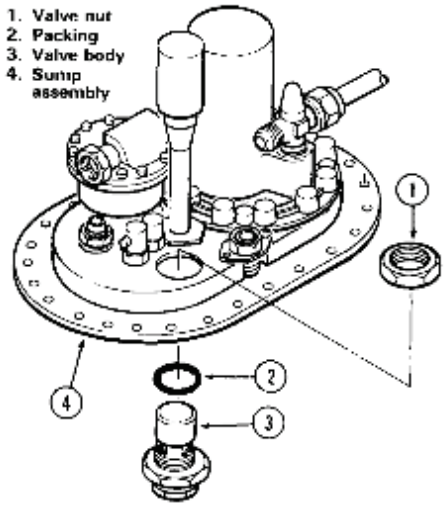
Procedure

1. Place packing (2, Figure 1) on valve body (3).

2. Insert valve body (3) through sump assembly (4). Install and tighten valve nut (1). Apply lockwire (C-405) to valve body.

3. Install sump assembly (4) (412-A-28-10-00-03A-720A-A / 00337).

4. Perform operational check on valve body (3) by verifying fuel gravity drains upon insertion of union and valve body does not leak following removal of union.



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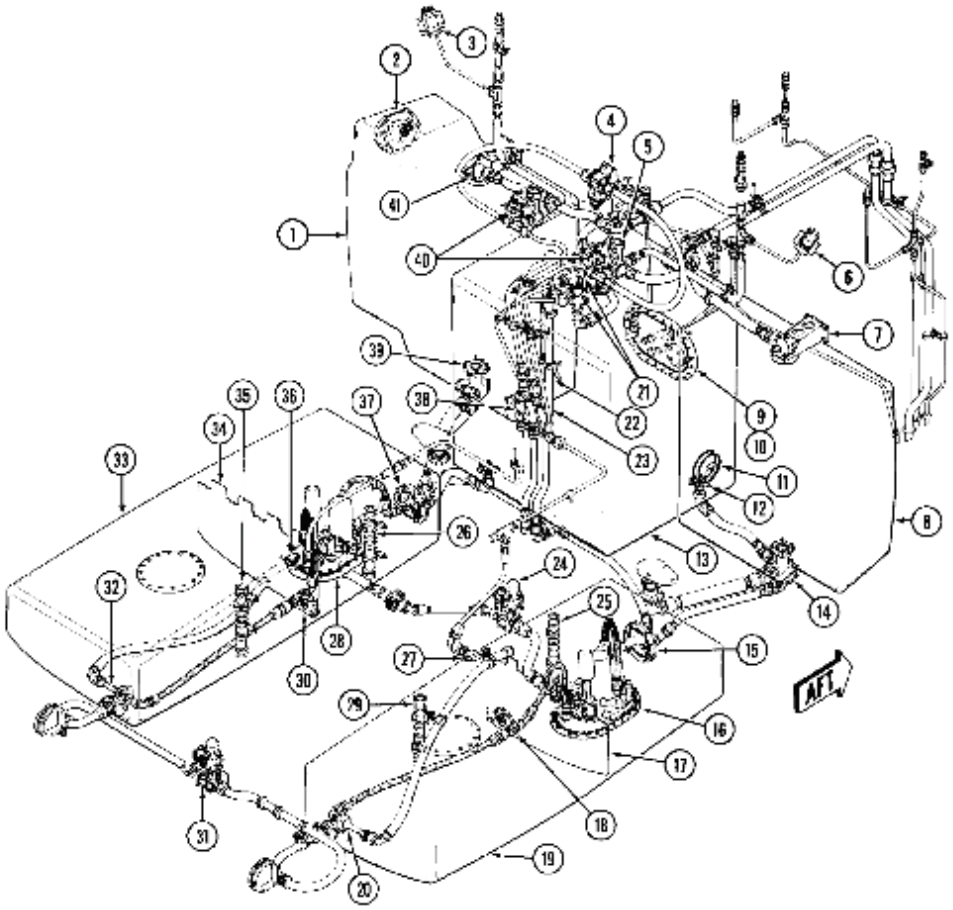
Figure 1. Defueling Valve (Sheet 1 of 1)

Boost Pumps - Description

The boost pumps (installed on sump assemblies) (16 and 28, Figure 1) are electrically operated by 28 Vdc circuits controlled by separate switches. Each boost pump is installed with a drain valve, a seal drain adapter, and a fuel outlet fitting.

NOTE

Maintenance procedures for left and right boost pumps are the same.



1. Right outboard aft fuel cell
2. Filler cap and adapter assembly
3. Right fuel pressure transmitter
4. System crossfeed valve
5. Center aft fuel cell vent cover
6. Left fuel pressure transmitter
7. Left outboard aft fuel cell vent fitting
8. Left outboard aft fuel cell
9. Center aft fuel cell rear cover
10. Governor bleed line check valve
11. Center aft fuel cell left interconnect fitting
12. Center aft fuel cell left outlet fitting
13. Center aft fuel cell
14. Left outboard aft fuel cell outlet fitting
15. Left forward fuel cell inlet/outlet fitting

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Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 1 of 2)

16. Left forward fuel cell fuel sump area consisting of:
    - Fuel sump assembly
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
17. Left forward fuel cell baffle
18. Left forward fuel cell flapper valve
19. Left forward fuel cell
20. Left forward fuel cell ejector pump and support
21. Siphon breaker valves
22. Center aft fuel cell right interconnect fitting
23. Aft fuel cell fuel quantity probe
24. Aft interconnect valve
25. Left forward fuel cell aft compartment fuel quantity probe
26. Right forward fuel cell aft compartment fuel quantity probe
27. Left forward fuel cell crashworthy vent valve
28. Right forward fuel cell fuel sump area consisting of:
    - Fuel sump
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
29. Left forward fuel cell forward compartment fuel quantity probe
30. Right forward fuel cell flapper valve
31. Forward interconnect valve
32. Right forward fuel cell ejector pump and support
33. Right forward fuel cell
34. Right forward fuel cell baffle
35. Right forward fuel cell forward compartment fuel quantity probe
36. Right forward fuel cell crashworthy vent valve
37. Right forward fuel cell inlet/outlet fitting
38. Center aft fuel cell right outlet fitting
39. Right outboard aft fuel cell outlet fitting
40. Shutoff valve
41. Right outboard aft fuel cell vent fitting

(S/N 33001 33107)

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Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 2 of 2)

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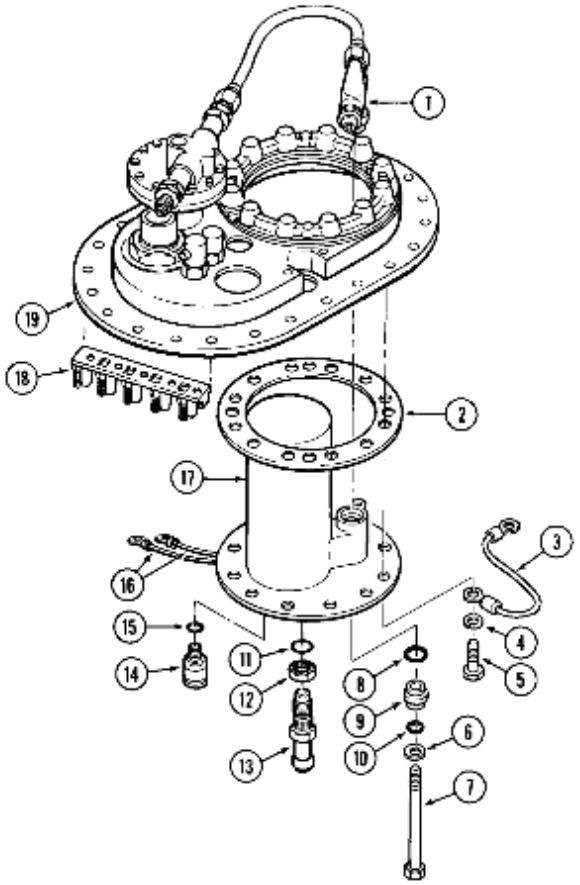
Boost Pumps - Removal

Procedure

NOTE

The following removal procedures for left and right boost pumps are the same.

1. Disconnect electrical power.
2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).
3. Drain trapped fuel through drain valve (14, Figure 1).
4. Disconnect electrical leads (16) from terminal block (18).
5. Remove bolt (7), washer (6), packings (8 and 10), and plug (9). Discard packings.
6. Remove bolts (5) and grounding jumper wire (3).
7. Remove boost pump (17) and gasket (2) from sump assembly (19). Discard gasket.
8. Loosen nut (12) and remove seal drain adapter (13) with packing (11). Discard packing.
9. Remove pump drain valve (14) and packing (15). Discard packing.
10. Cover opening in cell.



1. Fuel outlet fitting
2. Gasket
3. Grounding jumper wire
4. Washer
5. Bolt
6. Washer
7. Bolt
8. Packing
9. Plug
10. Packing
11. Packing
12. Nut
13. Seal drain adapter
14. Pump drain valve
15. Packing
16. Electrical leads
17. Boost pump
18. Terminal block
19. Sump assembly

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Figure 1. Boost Pump (Sheet 1 of 1)

Boost Pumps - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	
Lockwire	C-405	AR	

Procedure

NOTE

The following installation procedures for left and right boost pumps are the same.

1. Coat new packing (15, Figure 1) with assembly fluid (C-024) and install on pump drain valve (14). Install pump drain valve (14) in boost pump (17).

2. Apply assembly fluid (C-024) to packing (11). Install seal drain adapter (13) and nut (12) with packing (11).

3. Apply assembly fluid (C-024) to packings (8 and 10). Install washers (6) on bolt (7). Install packings (8 and 10) on plug (9) and then install plug (9) on bolt (7).

4. Install new gasket (2) on boost pump (17).

5. Remove protective cover from cell opening.

6. Position boost pump (17) on sump assembly (19). Align pump (17) with fuel outlet fitting (1). Install bolt (7) and lockwire (C-405).

NOTE

Install grounding jumper wire (3) on nearest bolt (5).

7. Install bolts (5).

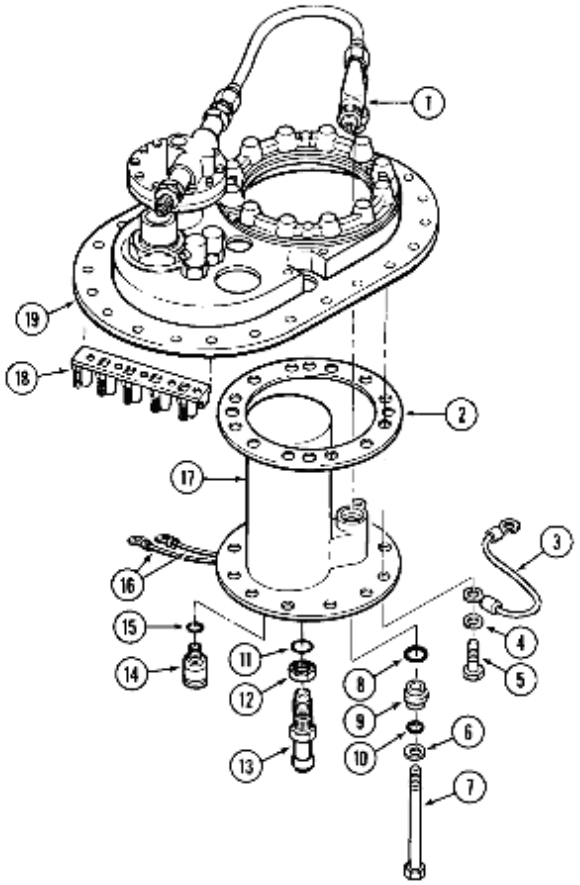
8. Perform system pressure test (412-A-28-10-00-00A-362A-A / 00327).

9. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).

10. Perform operational check on boost pump (412-A-96-04-00-02A-320B-A / 02105 and 412-A-96-04-00-02A-320C-A / 02106).

11. Check area for leaks.

12. Install sump access door.



1. Fuel outlet fitting
2. Gasket
3. Grounding jumper wire
4. Washer
5. Bolt
6. Washer
7. Bolt
8. Packing
9. Plug
10. Packing
11. Packing
12. Nut
13. Seal drain adapter
14. Pump drain valve
15. Packing
16. Electrical leads
17. Boost pump
18. Terminal block
19. Sump assembly

(S/N 33001 — 33107)

ICN-412-A-281000-B-97499-00009-A-001-01

Figure 1. Boost Pump (Sheet 1 of 1)

Canister Type Boost Pumps - Description

Applicable: SN: 33001-33075 or (SN: 33077-33107 TB 412-89-77: Post):

The canister type boost pumps are electrically operated by 28 Vdc circuits controlled by separate switches. Each boost pump incorporates an integral valve and filter assembly.

NOTE

Maintenance procedures for left and right canister type boost pumps are the same.

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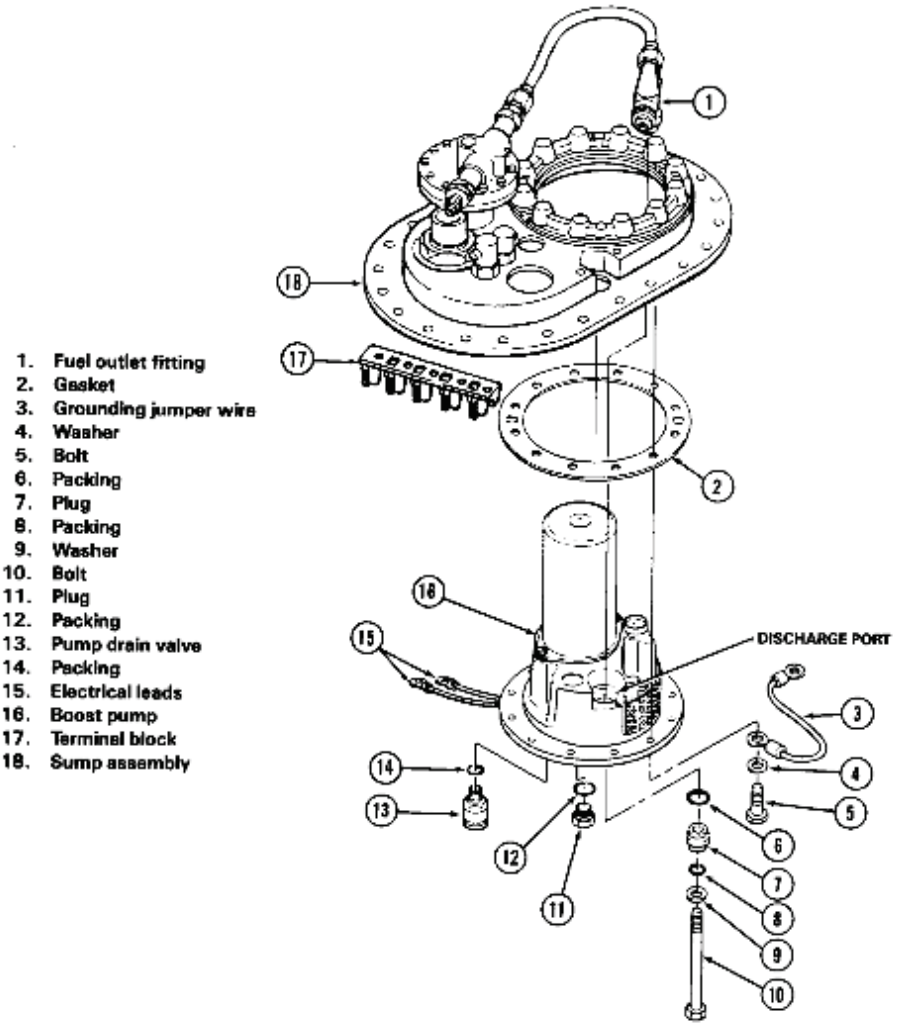
Canister Type Boost Pumps - Removal

Procedure

NOTE

The following removal procedures for left and right canister type boost pumps are the same.

1. Disconnect electrical power.
2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).
3. Drain trapped fuel through pump drain valve (13, Figure 1).
4. Disconnect electrical leads (15) from terminal block (17).
5. Remove bolt (10), washer (9), packings (6 and 8), and plug (7). Discard packings.
6. Remove bolts (5) and grounding jumper wire (3).
7. Remove boost pump (16) and gasket (2) from sump assembly (18). Discard gasket.
8. Remove pump drain valve (13) and packing (14). Discard packing.
9. Cover opening in cell.



ICN-412-A-281000-B-97499-00010-A-001-01

Figure 1. Canister Type Boost Pump (Sheet 1 of 1)

Canister Type Boost Pumps - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	
Lockwire	C-405	AR	

Procedure

NOTE

The following installation procedures for left and right canister type boost pumps are the same.

1. Lubricate new packing (14, Figure 1) with assembly fluid (C-024) and install on pump drain valve (13). Install pump drain valve in base of pump at cartridge drain port.

2. Lubricate packing (12) with assembly fluid (C-024) and install on plug (11). Install plug (11) in base of pump at seal drain port.

3. Coat gasket (2) with assembly fluid (C-024). Install gasket (2) on boost pump (16).

4. Lubricate packings (6 and 8) with assembly fluid (C-024) and install on plug (7).

5. Install washer (9) on bolt (10). Insert bolt (10) through plug (7) and into discharge port of pump.

6. Position boost pump (16) on sump assembly (18). Align boost pump with fuel outlet fitting (1). Tighten bolt (10) and ensure fitting seats in discharge port.

7. Install grounding jumper wire (3) on nearest bolt (5).

8. Install bolts (5) and washers (4).

9. Secure pump drain valve (13), plug (11), and bolt (10) with lockwire (C-405).

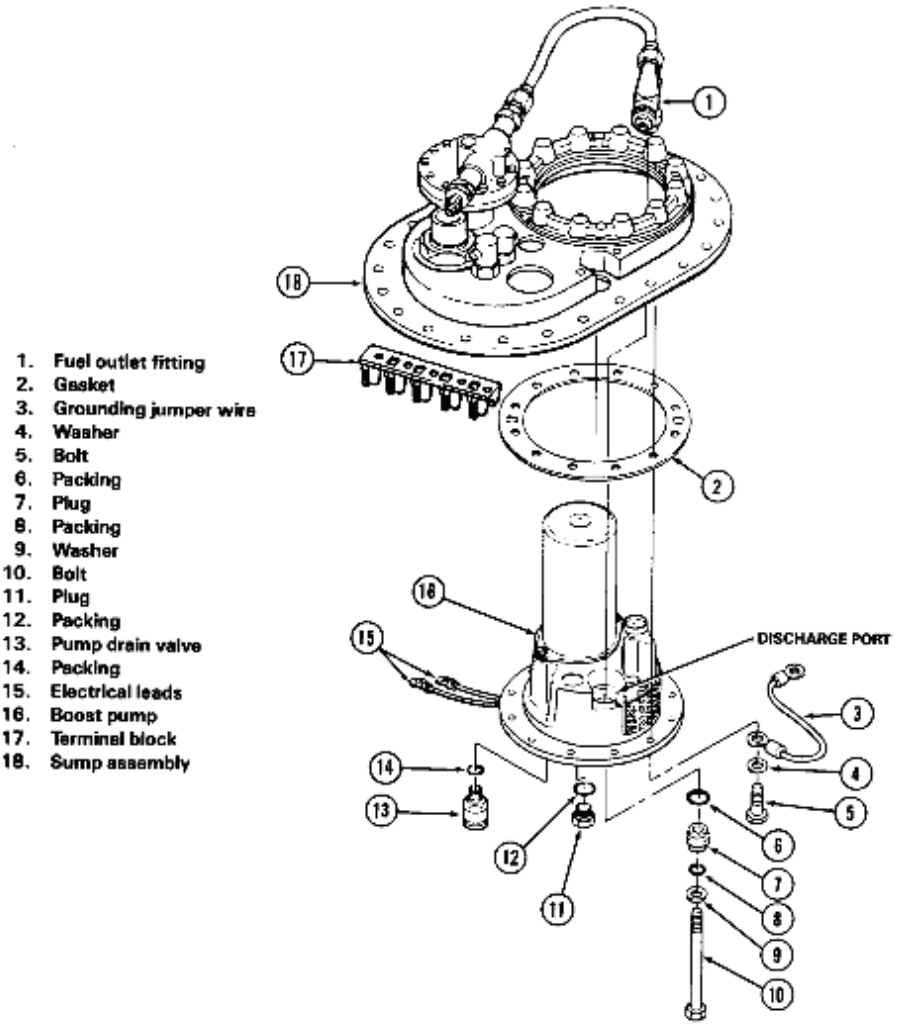
10. Perform system pressure test (412-A-28-10-00-00A-362A-A / 00327).

11. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).

12. Perform operational check on boost pump (412-A-96-04-00-02A-320B-A / 02105 and 412-A-96-04-00-02A-320C-A / 02106).

13. Check area for leaks.

14. Install sump access door.



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Figure 1. Canister Type Boost Pump (Sheet 1 of 1)

Canister Type Boost Pump Cartridge - Description

For removal, cleaning and installation procedures, refer to 412-A-28-10-00-60A-042A-A / 00518.

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Sump Interconnect Valves - Description

NOTE

Maintenance procedures for the left and right sump interconnect valves are the same.

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Sump Interconnect Valves - Removal

Procedure

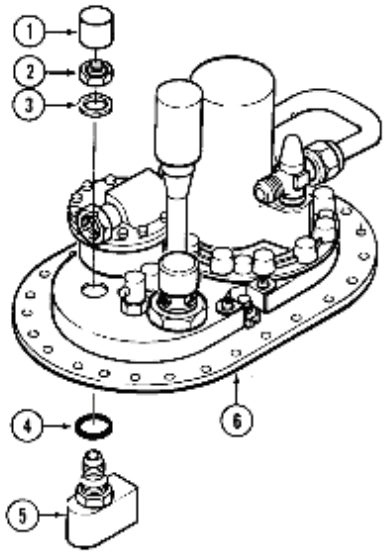
NOTE

The following removal procedure for the left and right sump interconnect valves are the same.

1. Remove sump assembly (412-A-28-10-00-03A-520A-A / 00335).

2. Remove valve (1, Figure 1).

3. Remove nut (2), washer (3), packing (4), and sump interconnect valve fitting (5). Discard packing.



1. Valve
2. Nut
3. Thin aluminum washer
4. Packing
5. Sump interconnect valve fitting
6. Sump assembly

(S/N 33001 – 33107)

ICN-412-A-281000-B-97499-00011-A-001-01

Figure 1. Sump Interconnect Valve (Sheet 1 of 1)

Sump Interconnect Valves - Installation

Procedure

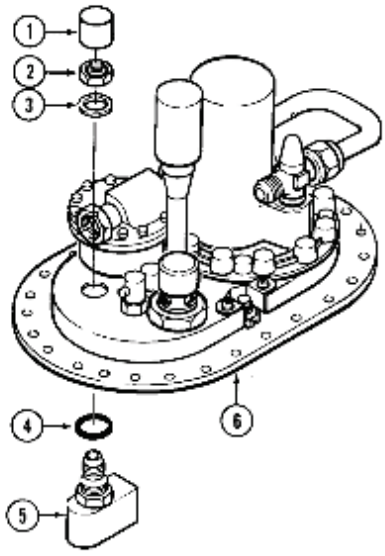
NOTE

The following installation procedure for the left and right sump interconnect valves are the same.

1. Install packing (4, Figure 1) on interconnect valve fitting (5).

2. Install interconnect valve fitting (5), washer (3), nut (2) and valve (1) in sump (6).

3. Install sump assembly (6) (412-A-28-10-00-03A-720A-A / 00337).



1. Valve
2. Nut
3. Thin aluminum washer
4. Packing
5. Sump interconnect valve fitting
6. Sump assembly

(S/N 33001 – 33107)

ICN-412-A-281000-B-97499-00011-A-001-01

Figure 1. Sump Interconnect Valve (Sheet 1 of 1)

Sump Drain Valves - Description

Each sump assembly is provided with a sump drain valve which can be operated manually or electrically to drain off accumulations of water or other contaminants.

NOTE

Maintenance procedures for left and right sump drain valves are the same.

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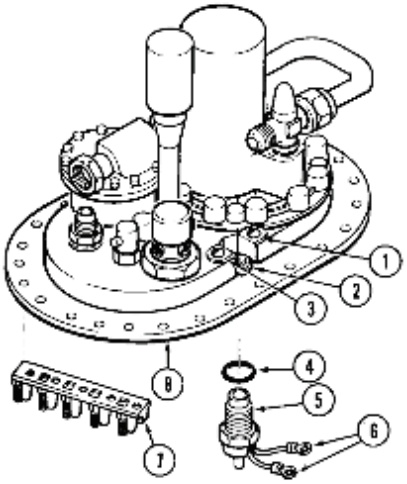
Sump Drain Valves - Removal

Procedure

NOTE

The following removal procedures for left and right sump drain valves are the same.

1. Disconnect electrical power.
2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).
3. Disconnect electrical leads (6, Figure 1) from terminal block (7).
4. Remove sump drain valve (5) and packing (4). Discard packing.
5. Cover or plug opening in sump assembly (8).



1. Retainer
2. Nut
3. Retainer
4. Packing
5. Sump drain valve
6. Electrical leads
7. Terminal block
8. Sump assembly

(S/N 33001 – 33107)

ICN-412-A-281000-B-97499-00012-A-001-01

Figure 1. Sump Drain Valve (Sheet 1 of 1)

Sump Drain Valves - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

NOTE

The following installation procedures for left and right sump drain valves are the same.

1. Install packing (4, Figure 1) on sump drain valve (5).

NOTE

Nut (2) is held in place by retainers (1 and 3). Carefully thread sump drain valve (5) into nut (2). Do not push against nut (2).

2. Remove protective cover or plug from sump assembly (8).

3. Install sump drain valve (5). Apply lockwire (C-405) to sump drain valve.

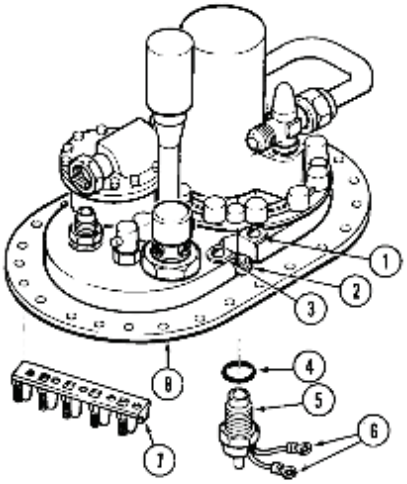
4. Connect electrical leads (6) to terminal block (7).

5. Perform fuel system pressure test (412-A-28-10-00-00A-362A-A / 00327).

6. Perform operational check on sump drain valve (412-A-96-04-00-09A-320A-A / 02125).

7. Check area for leaks.

8. Install sump access door.



1. Retainer
2. Nut
3. Retainer
4. Packing
5. Sump drain valve
6. Electrical leads
7. Terminal block
8. Sump assembly

(S/N 33001 – 33107)

ICN-412-A-281000-B-97499-00012-A-001-01

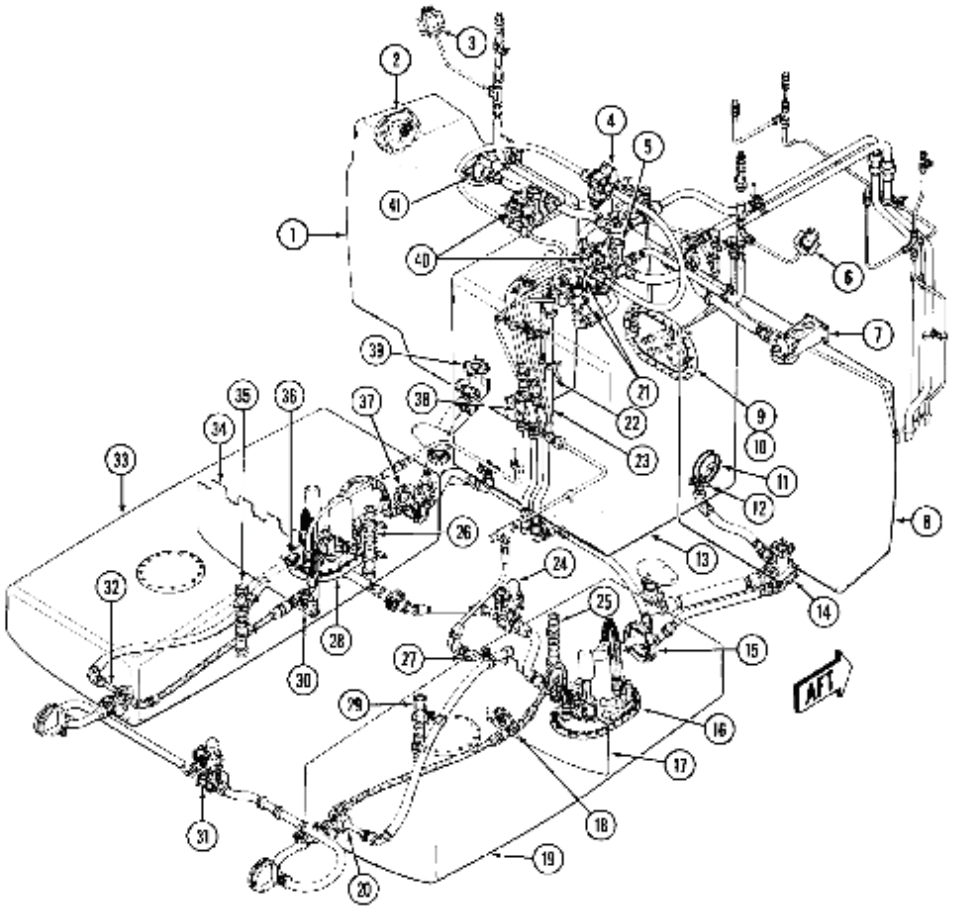
Figure 1. Sump Drain Valve (Sheet 1 of 1)

Flow Switches - Description

The flow switch is actuated by outlet flow from the boost pump (16 and 28, Figure 1). If fuel flow is interrupted, the flow switch will cause the respective caution panel FUEL BOOST segment to illuminate.

NOTE

Maintenance procedures for left and right flow switches are the same.



1. Right outboard aft fuel cell
2. Filler cap and adapter assembly
3. Right fuel pressure transmitter
4. System crossfeed valve
5. Center aft fuel cell vent cover
6. Left fuel pressure transmitter
7. Left outboard aft fuel cell vent fitting
8. Left outboard aft fuel cell
9. Center aft fuel cell rear cover
10. Governor bleed line check valve
11. Center aft fuel cell left interconnect fitting
12. Center aft fuel cell left outlet fitting
13. Center aft fuel cell
14. Left outboard aft fuel cell outlet fitting
15. Left forward fuel cell inlet/outlet fitting

(S/N 33001 33107)

ICN-412-A-281000-B-97499-00001-A-001-01

Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 1 of 2)

16. Left forward fuel cell fuel sump area consisting of:
    - Fuel sump assembly
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
17. Left forward fuel cell baffle
18. Left forward fuel cell flapper valve
19. Left forward fuel cell
20. Left forward fuel cell ejector pump and support
21. Siphon breaker valves
22. Center aft fuel cell right interconnect fitting
23. Aft fuel cell fuel quantity probe
24. Aft interconnect valve
25. Left forward fuel cell aft compartment fuel quantity probe
26. Right forward fuel cell aft compartment fuel quantity probe
27. Left forward fuel cell crashworthy vent valve
28. Right forward fuel cell fuel sump area consisting of:
    - Fuel sump
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
29. Left forward fuel cell forward compartment fuel quantity probe
30. Right forward fuel cell flapper valve
31. Forward interconnect valve
32. Right forward fuel cell ejector pump and support
33. Right forward fuel cell
34. Right forward fuel cell baffle
35. Right forward fuel cell forward compartment fuel quantity probe
36. Right forward fuel cell crashworthy vent valve
37. Right forward fuel cell inlet/outlet fitting
38. Center aft fuel cell right outlet fitting
39. Right outboard aft fuel cell outlet fitting
40. Shutoff valve
41. Right outboard aft fuel cell vent fitting

(S/N 33001 33107)

ICN-412-A-281000-B-97499-00002-A-001-01

Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 2 of 2)

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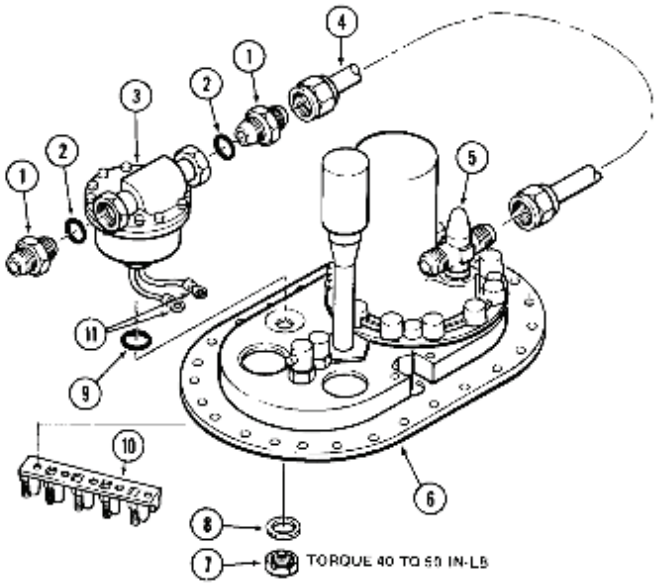
Flow Switches - Removal

Procedure

NOTE

The following removal procedures for left and right flow switches are the same.

1. Remove sump assembly (412-A-28-10-00-03A-520A-A / 00335).
2. Disconnect tube (4, Figure 1) from union (1) and fuel outlet fitting (5).
3. Disconnect electrical leads (11) from terminal block (10).
4. Remove nut (7), thin aluminum washer (8), packing (9), and flow switch (3). Discard packing.
5. Remove unions (1) and packings (2). Discard packings.
6. Install protective cover over sump opening.



1. Union
2. Packing
3. Flow switch
4. Tube
5. Fuel outlet fitting
6. Sump assembly
7. Nut
8. Thin aluminum washer
9. Packing
10. Terminal block
11. Electrical leads

(S/N 33001 – 33107)

ICN-412-A-281000-B-97499-00013-A-001-01

Figure 1. Flow Switch (Sheet 1 of 1)

Flow Switches - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	
Lockwire	C-405	AR	

Procedure

NOTE

The following installation procedures for left and right flow switches are the same.

1. Apply assembly fluid (C-024) to new packings (2, Figure 1). Install unions (1) and packings.

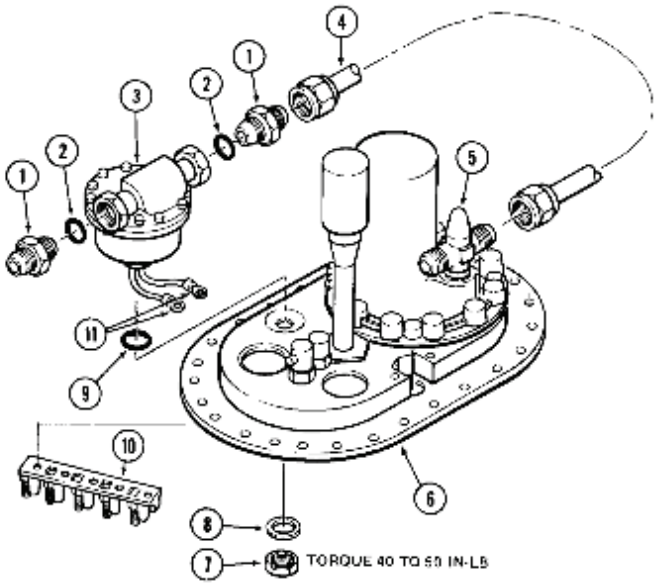
2. Install packing (9) on flow switch (3).

3. Position flow switch (3) on sump assembly (6). Install thin aluminum washer (8) and nut (7). Torque nut to 40 to 50 inch-pounds (4.52 to 5.65 Nm) and lockwire (C-405) nut (7).

4. Connect tube (4) to union (1) and fuel outlet fitting (5).

5. Connect electrical leads (11) to terminal block (10).

6. Install sump assembly (412-A-28-10-00-03A-720A-A / 00337).



1. Union
2. Packing
3. Flow switch
4. Tube
5. Fuel outlet fitting
6. Sump assembly
7. Nut
8. Thin aluminum washer
9. Packing
10. Terminal block
11. Electrical leads

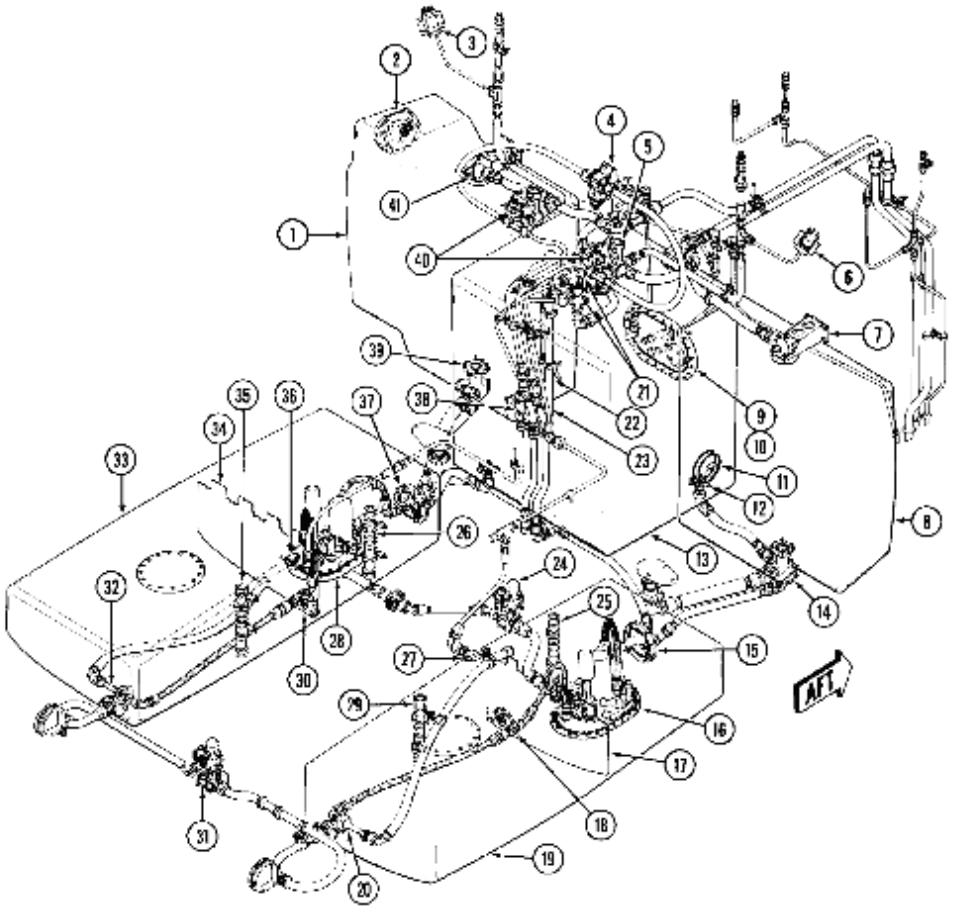
(S/N 33001 – 33107)

ICN-412-A-281000-B-97499-00013-A-001-01

Figure 1. Flow Switch (Sheet 1 of 1)

Forward Cell Inlet/Outlet Fittings - Description

An inlet/outlet fitting (15 and 37, Figure 1) is installed on the rear of each forward cell. The fitting provides an attachment point for hoses from aft cells and for a supply hose from boost pump to shutoff valve. A check valve installed on forward side of each fitting is connected to a hose from the boost pump.



1. Right outboard aft fuel cell
2. Filler cap and adapter assembly
3. Right fuel pressure transmitter
4. System crossfeed valve
5. Center aft fuel cell vent cover
6. Left fuel pressure transmitter
7. Left outboard aft fuel cell vent fitting
8. Left outboard aft fuel cell
9. Center aft fuel cell rear cover
10. Governor bleed line check valve
11. Center aft fuel cell left interconnect fitting
12. Center aft fuel cell left outlet fitting
13. Center aft fuel cell
14. Left outboard aft fuel cell outlet fitting
15. Left forward fuel cell inlet/outlet fitting

(S/N 33001 33107)

ICN-412-A-281000-B-97499-00001-A-001-01

Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 1 of 2)

16. Left forward fuel cell fuel sump area consisting of:
    - Fuel sump assembly
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
17. Left forward fuel cell baffle
18. Left forward fuel cell flapper valve
19. Left forward fuel cell
20. Left forward fuel cell ejector pump and support
21. Siphon breaker valves
22. Center aft fuel cell right interconnect fitting
23. Aft fuel cell fuel quantity probe
24. Aft interconnect valve
25. Left forward fuel cell aft compartment fuel quantity probe
26. Right forward fuel cell aft compartment fuel quantity probe
27. Left forward fuel cell crashworthy vent valve
28. Right forward fuel cell fuel sump area consisting of:
    - Fuel sump
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
29. Left forward fuel cell forward compartment fuel quantity probe
30. Right forward fuel cell flapper valve
31. Forward interconnect valve
32. Right forward fuel cell ejector pump and support
33. Right forward fuel cell
34. Right forward fuel cell baffle
35. Right forward fuel cell forward compartment fuel quantity probe
36. Right forward fuel cell crashworthy vent valve
37. Right forward fuel cell inlet/outlet fitting
38. Center aft fuel cell right outlet fitting
39. Right outboard aft fuel cell outlet fitting
40. Shutoff valve
41. Right outboard aft fuel cell vent fitting

(S/N 33001 33107)

ICN-412-A-281000-B-97499-00002-A-001-01

Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 2 of 2)

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Forward Cell Inlet/Outlet Fittings - Removal

Procedure

1. Disconnect electrical power.

2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).

3. Remove access door located on bottom skin aft of sump access door.

NOTE

Refer to Step 4.1. through Step 4.5. for removal of right forward cell inlet/outlet fitting and Step 5.1. through Step 5.5. for removal of left forward cell inlet/outlet fitting.

4. Remove inlet/outlet fitting from right forward cell:

4.1. Disconnect hoses (11, 14, and 17, Figure 1, Detail A).

4.2. Remove bolts (8), thin aluminum washers (7), and retainers (6).

4.3. Separate inlet/outlet fitting (5) from cell and disconnect hose (1). Remove and discard packing (4).

4.4. Remove check valve (2). Remove and discard packing (3).

4.5. Remove unions (10, 13, and 16) with packings (9, 12, and 15). Discard packings.

5. Remove inlet/outlet fitting from left forward cell:

5.1. Disconnect hoses (23 and 24, Detail B).

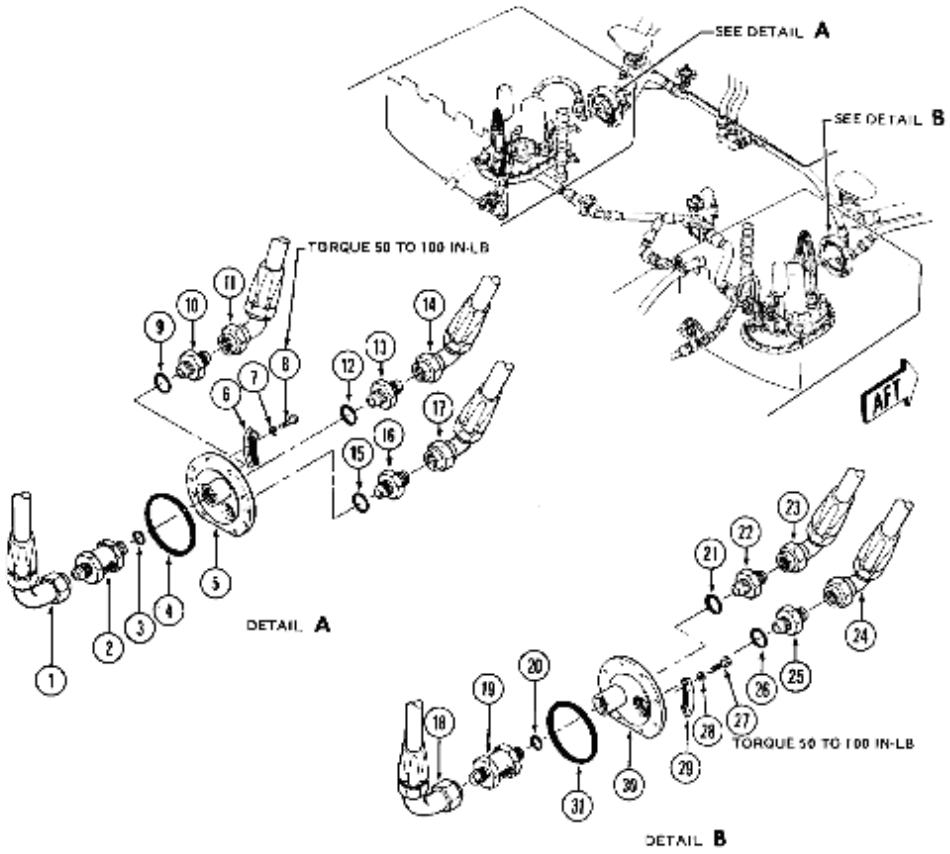
5.2. Remove bolts (27), thin aluminum washers (28), and retainers (29).

5.3. Separate inlet/outlet fitting (30) from cell and disconnect hose (18). Remove and discard packing (31).

5.4. Remove check valve (19) and packing (20). Discard packing.

5.5. Remove unions (22 and 25) with packings (21 and 26). Discard packings.

6. Cap or plug open lines and install protective cover over cell opening.



1. Hose
2. Check valve
3. Packing
4. Packing
5. Inlet/outlet fitting
6. Retainer
7. Thin aluminum washer
8. Bolt
9. Packing
10. Union
11. Hose
12. Packing
13. Union
14. Hose
15. Packing
16. Union
17. Hose
18. Hose
19. Check valve
20. Packing
21. Packing
22. Union
23. Hose
24. Hose
25. Union
26. Packing
27. Bolt
28. Thin aluminum washer
29. Retainer
30. Inlet/outlet fitting
31. Packing

S/N 33001 – 331071

ICN-412-A-281000-B-97499-00014-A-001-01

Figure 1. Forward Cell Inlet/Outlet Fitting (Sheet 1 of 1)

Forward Cell Inlet/Outlet Fittings - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

NOTE

Refer to Step 1.1. through Step 1.8. for installation of right forward cell inlet/outlet fitting and Step 2.1. through Step 2.9. for installation of left forward cell inlet/outlet fitting.

1. Remove protective caps and covers. Install inlet/outlet fitting on right forward cell as follows:

1.1. Apply assembly fluid (C-024) to packings (9, 12, and 15, Figure 1, Detail A). Install unions (10, 13, and 16) with packings.

1.2. Apply assembly fluid (C-024) to packing (3). Install check valve (2) and packing.

1.3. Place packing (4) in groove of cell. Packing may be retained in place with assembly fluid (C-024).

1.4. Hold inlet/outlet fitting (5) near cell port. Connect hose (1).

NOTE

The six bolts (8) used in top six holes of inlet/outlet fitting (5) with retainers (6) are longer than two bolts (8) used in bottom holes without retainers.

1.5. Position inlet/outlet fitting (5) over cell port.

1.6. Install longer bolts (8) with thin aluminum washers (7) and retainers (6) at the upper six holes in inlet/outlet fitting (5).

1.7. Install two shorter bolts (8) with thin aluminum washers (7) at two lower holes in inlet/outlet fitting (5). Torque bolts to 50 to 100 inch-pounds (5.65 to 11.30 Nm).

1.8. Connect hoses (11, 14, and 17).

2. Install inlet/outlet fitting (30) on left forward cell.

2.1. Apply assembly fluid (C-024) to packing (20, Detail B).

2.2. Install check valve (19) and packing (20).

2.3. Install unions (22 and 25) with packings (21 and 26).

2.4. Coat packing (31) with assembly fluid (C-024) and place packing (3) in groove of cell.

2.5. Hold inlet/outlet fitting (30) near cell port. Connect hose (18).

NOTE

Six bolts (27) used in top six holes of inlet/outlet fitting (30) with retainers (29) are longer than two bolts (27) used in bottom holes without retainers.

2.6. Position inlet/outlet fitting (30) over cell port.

2.7. Install longer bolts (27) with thin aluminum washers (28) and retainers (29) in upper six holes in inlet/outlet fitting (30).

2.8. Install shorter two bolts (27) with thin aluminum washers (28) in two lower holes in inlet/outlet fitting (30). Torque bolts to 50 to 100 inch-pounds (5.65 to 11.30 Nm).

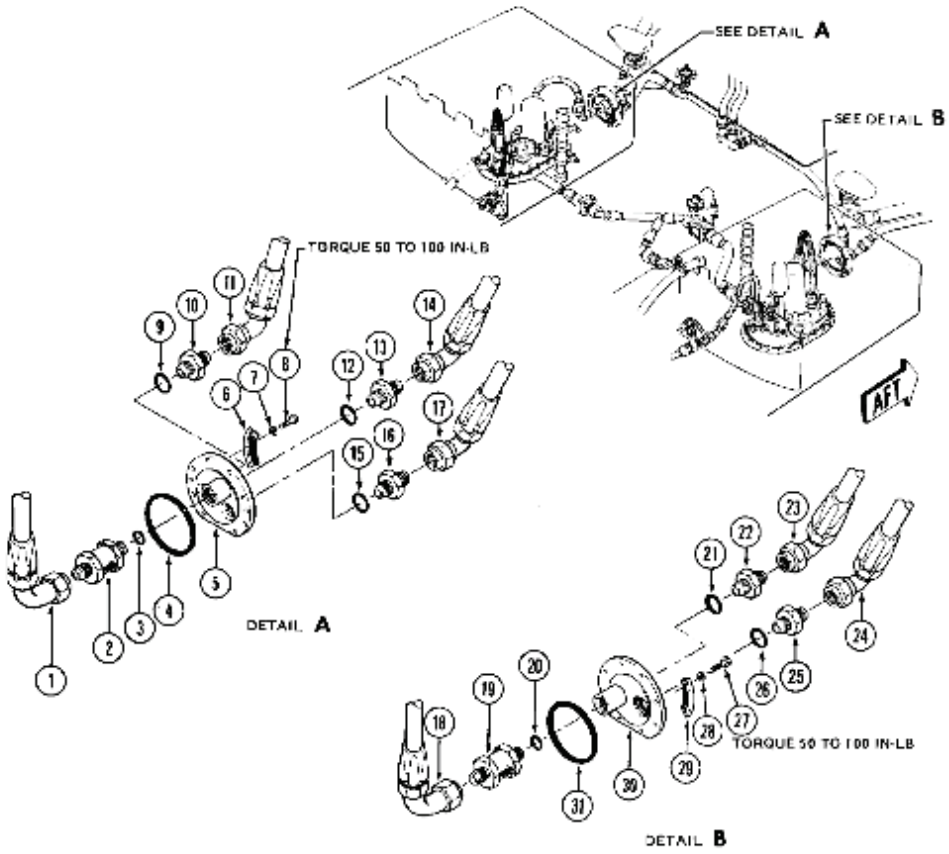
2.9. Connect hoses (23 and 24).

3. Perform fuel system pressure test (412-A-28-10-00-00A-362A-A / 00327).

4. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).

5. Check area for leaks.

6. Install access doors.



1. Hose
2. Check valve
3. Packing
4. Packing
5. Inlet/outlet fitting
6. Retainer
7. Thin aluminum washer
8. Bolt
9. Packing
10. Union
11. Hose
12. Packing
13. Union
14. Hose
15. Packing
16. Union
17. Hose
18. Hose
19. Check valve
20. Packing
21. Packing
22. Union
23. Hose
24. Hose
25. Union
26. Packing
27. Bolt
28. Thin aluminum washer
29. Retainer
30. Inlet/outlet fitting
31. Packing

(S/N 33001 – 33107)

ICN-412-A-281000-B-97499-00014-A-001-01

Figure 1. Forward Cell Inlet/Outlet Fitting (Sheet 1 of 1)

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Auxiliary Fuel Capped Connector - Description

NOTE

Maintenance procedures for left and right auxiliary fuel capped connectors are the same.

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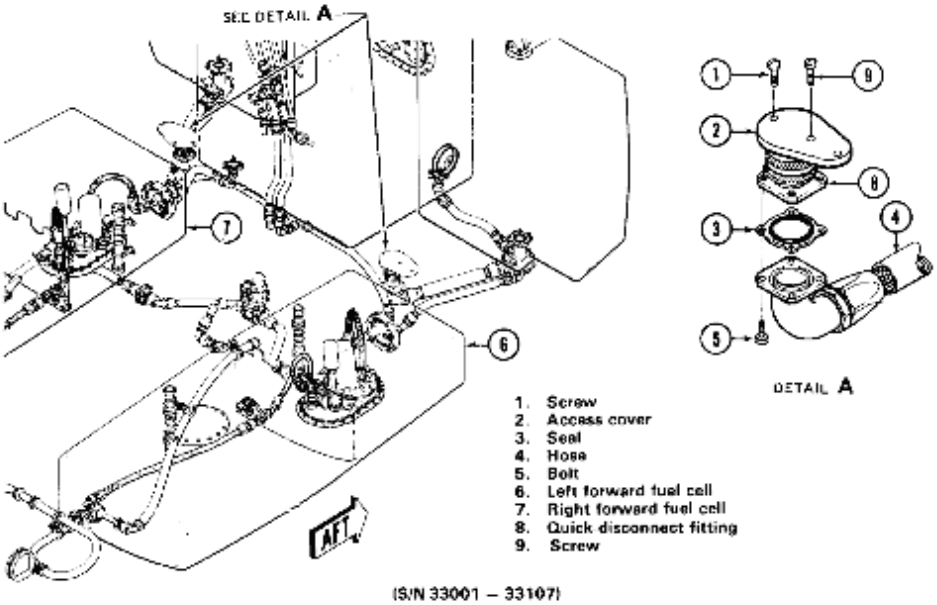
Auxiliary Fuel Capped Connector - Removal

Procedure

NOTE

The following removal procedure for left and right auxiliary fuel capped connectors are the same.

1. Disconnect electrical power.
2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).
3. Remove access door located on fuselage bottom aft of sump access door.
4. Remove two outer screws (1, Figure 1).
5. Pull access cover (2) outboard to release clip.
6. Lift access cover (2) and hold quick-disconnect fitting (8). Remove screw (9) to separate access cover from fitting.
7. Install protective cap or plug on open fuel line.
8. If required, remove bolts (5) to separate hose (4) from quick-disconnect fitting (8). Remove and discard seal (3).



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Figure 1. Auxiliary Fuel Capped Connector (Sheet 1 of 1)

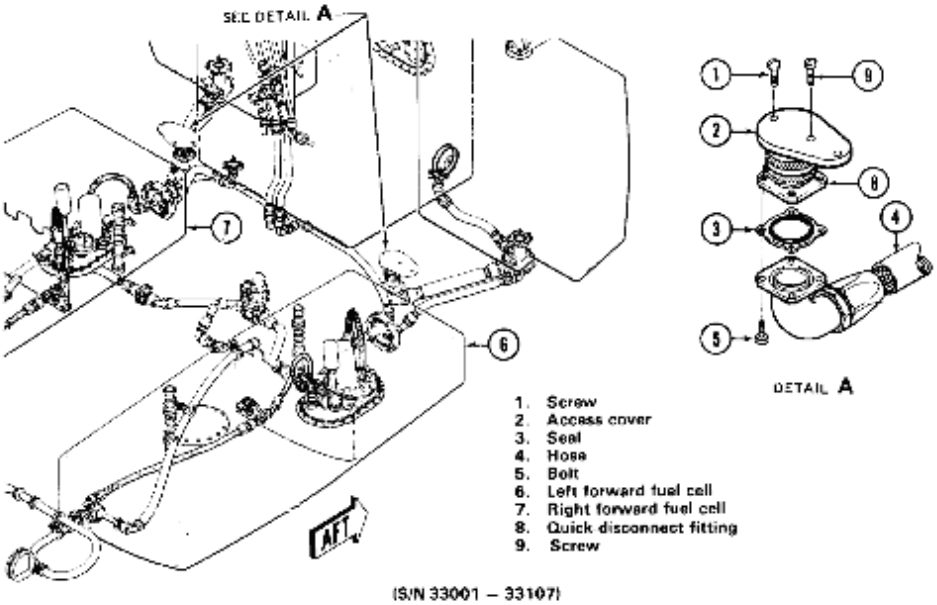
Auxiliary Fuel Capped Connector - Installation

Procedure

NOTE

The following installation procedure for left and right auxiliary fuel capped connectors are the same.

1. Remove protective cap or plug. Install seal (3, Figure 1) using bolts (5).
2. Ensure packing is installed inside fitting (8) and fitting (8) is tightened.
3. Secure access cover (2) to fitting (8) using screw (9).
4. Install access cover (2) by inserting clip on access over under floor panel structure.
5. Install screws (1).
6. Perform fuel system pressure test (412-A-28-10-00-00A-362A-A / 00327).
7. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).
8. Check area for leaks.
9. Install access doors.



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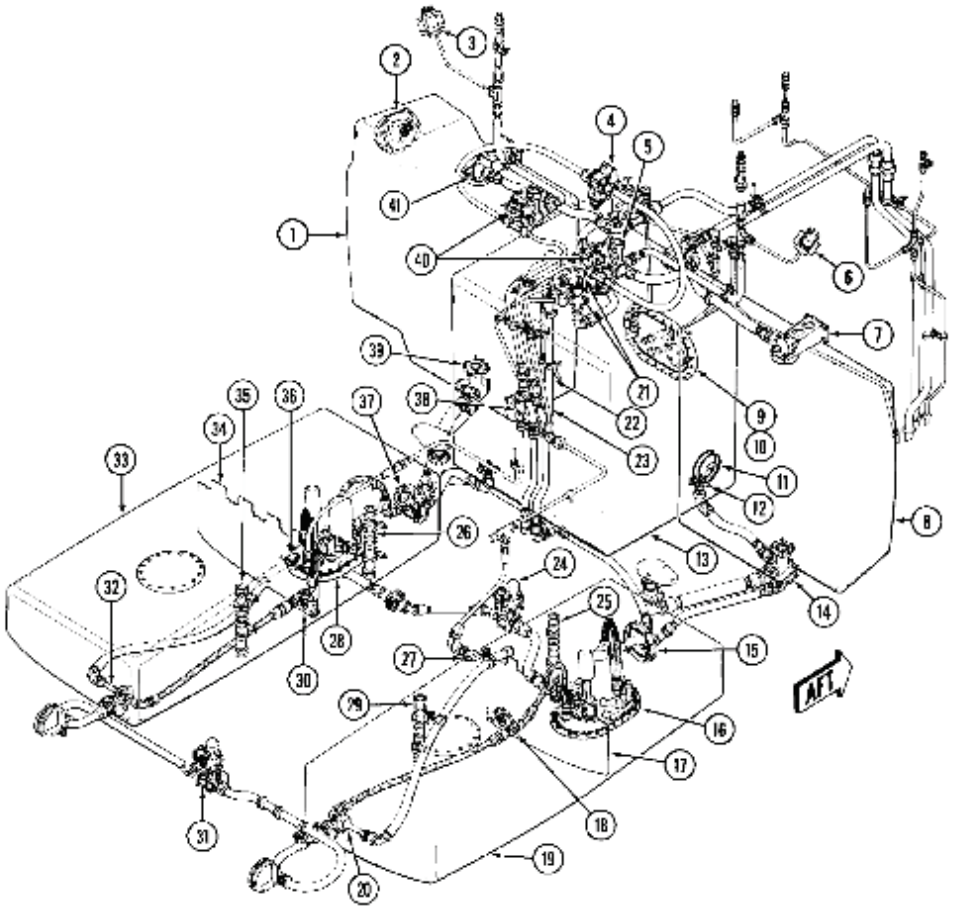
Figure 1. Auxiliary Fuel Capped Connector (Sheet 1 of 1)

Flapper Valves - Description

A flapper valve, consisting of two plates and a valve element, (18 and 30, Figure 1) is installed on baffle between front and aft compartments of forward cells. The flapper valve allows fuel to flow from front compartment to aft compartment but prevents flow in the opposite direction.

NOTE

Maintenance procedures for left and right flapper valves are the same.



1. Right outboard aft fuel cell
2. Filler cap and adapter assembly
3. Right fuel pressure transmitter
4. System crossfeed valve
5. Center aft fuel cell vent cover
6. Left fuel pressure transmitter
7. Left outboard aft fuel cell vent fitting
8. Left outboard aft fuel cell
9. Center aft fuel cell rear cover
10. Governor bleed line check valve
11. Center aft fuel cell left interconnect fitting
12. Center aft fuel cell left outlet fitting
13. Center aft fuel cell
14. Left outboard aft fuel cell outlet fitting
15. Left forward fuel cell inlet/outlet fitting

(S/N 33001 33107)

ICN-412-A-281000-B-97499-00001-A-001-01

Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 1 of 2)

16. Left forward fuel cell fuel sump area consisting of:
    - Fuel sump assembly
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
17. Left forward fuel cell baffle
18. Left forward fuel cell flapper valve
19. Left forward fuel cell
20. Left forward fuel cell ejector pump and support
21. Siphon breaker valves
22. Center aft fuel cell right interconnect fitting
23. Aft fuel cell fuel quantity probe
24. Aft interconnect valve
25. Left forward fuel cell aft compartment fuel quantity probe
26. Right forward fuel cell aft compartment fuel quantity probe
27. Left forward fuel cell crashworthy vent valve
28. Right forward fuel cell fuel sump area consisting of:
    - Fuel sump
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
29. Left forward fuel cell forward compartment fuel quantity probe
30. Right forward fuel cell flapper valve
31. Forward interconnect valve
32. Right forward fuel cell ejector pump and support
33. Right forward fuel cell
34. Right forward fuel cell baffle
35. Right forward fuel cell forward compartment fuel quantity probe
36. Right forward fuel cell crashworthy vent valve
37. Right forward fuel cell inlet/outlet fitting
38. Center aft fuel cell right outlet fitting
39. Right outboard aft fuel cell outlet fitting
40. Shutoff valve
41. Right outboard aft fuel cell vent fitting

(S/N 33001 33107)

ICN-412-A-281000-B-97499-00002-A-001-01

Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 2 of 2)

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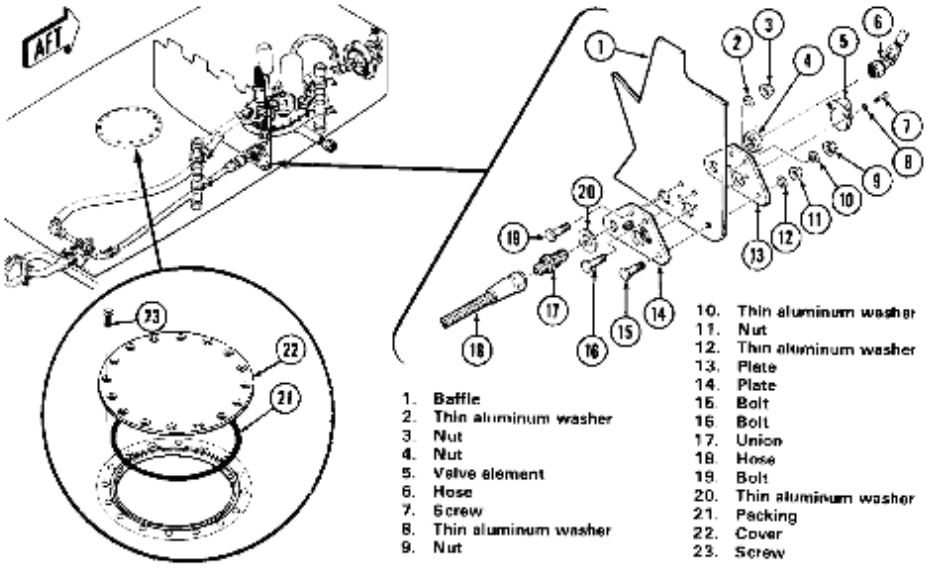
Flapper Valves - Removal

Procedure

NOTE

The following removal procedures for left and right flapper valves are the same.

1. Remove sump assembly (412-A-28-10-00-03A-520A-A / 00335).
2. Remove cabin floor access panel.
3. Remove screws (23, Figure 1), cover (22), and packing (21). Discard packing.
4. Remove screws (7), washers (8), and valve element (5).
5. Disconnect hoses (6 and 18). Remove nut (4), union (17), and washer (20).
6. Remove nuts (3, 9, and 11), washers (2, 10, and 12), and bolts (15, 16, and 19). Remove plates (13 and 14).
7. Install protective covers on cell openings.



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Figure 1. Flapper Valve (Sheet 1 of 1)

Flapper Valves - Inspection

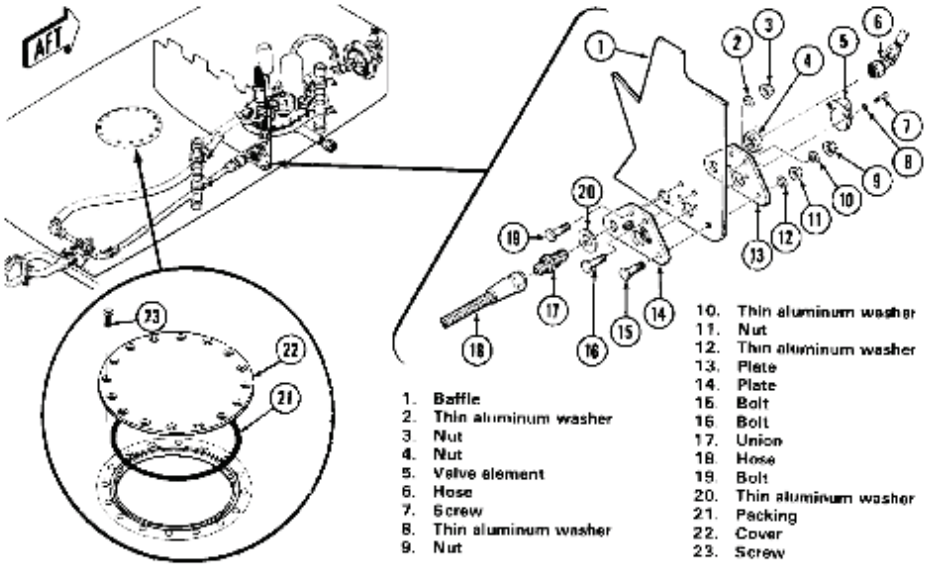
Procedure

NOTE

The following inspection procedures for left and right flapper valves are the same.

1. Inspect valve element (5, Figure 1) for cracks and nicks. Cracks and nicks which will affect valve function are not acceptable.

2. Inspect plates (13 and 14) for mechanical and corrosion damage. Mechanical damage in excess of a superficial type and/or any corrosion damage is not acceptable.



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Figure 1. Flapper Valve (Sheet 1 of 1)

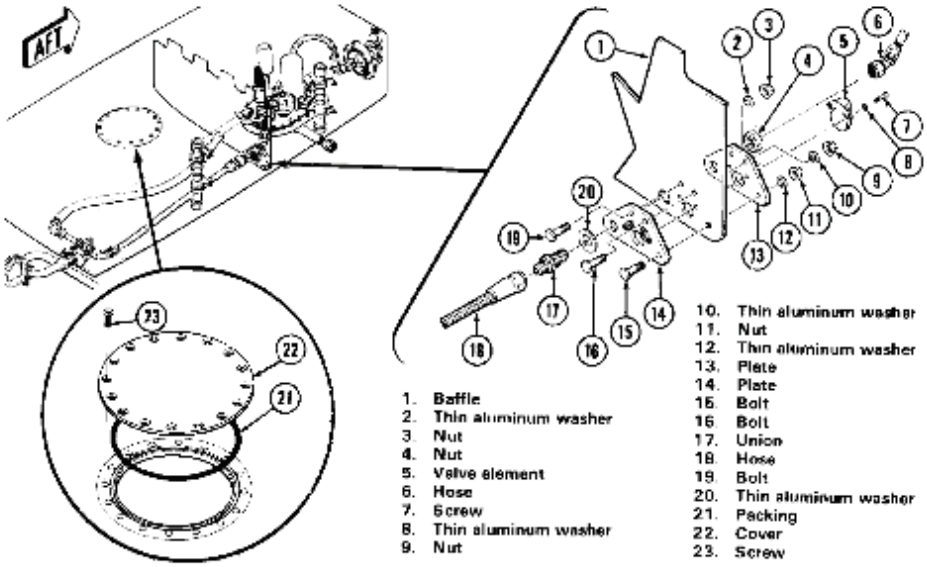
Flapper Valves - Installation

Procedure

NOTE

The following installation procedures for left and right flapper valves are the same.

1. Remove protective covers.
2. Position plates (13 and 14, Figure 1) on baffle (1).
3. Install bolts (15, 16, and 19) with washers (2, 10, and 12), and nuts (3, 9, and 11).
4. Install union (17), washer (20), and nut (4).
5. Install valve element (5) with screws (7) and washers (8).
6. Verify valve element (5) works freely and centers over opening in plate (13). If necessary, loosen screws (7) and reposition valve. Tighten screws.
7. Connect hoses (6 and 18).
8. Install packing (21), cover (22), and screws (23).
9. Install cabin floor access panel.
10. Install sump assembly (412-A-28-10-00-03A-720A-A / 00337).



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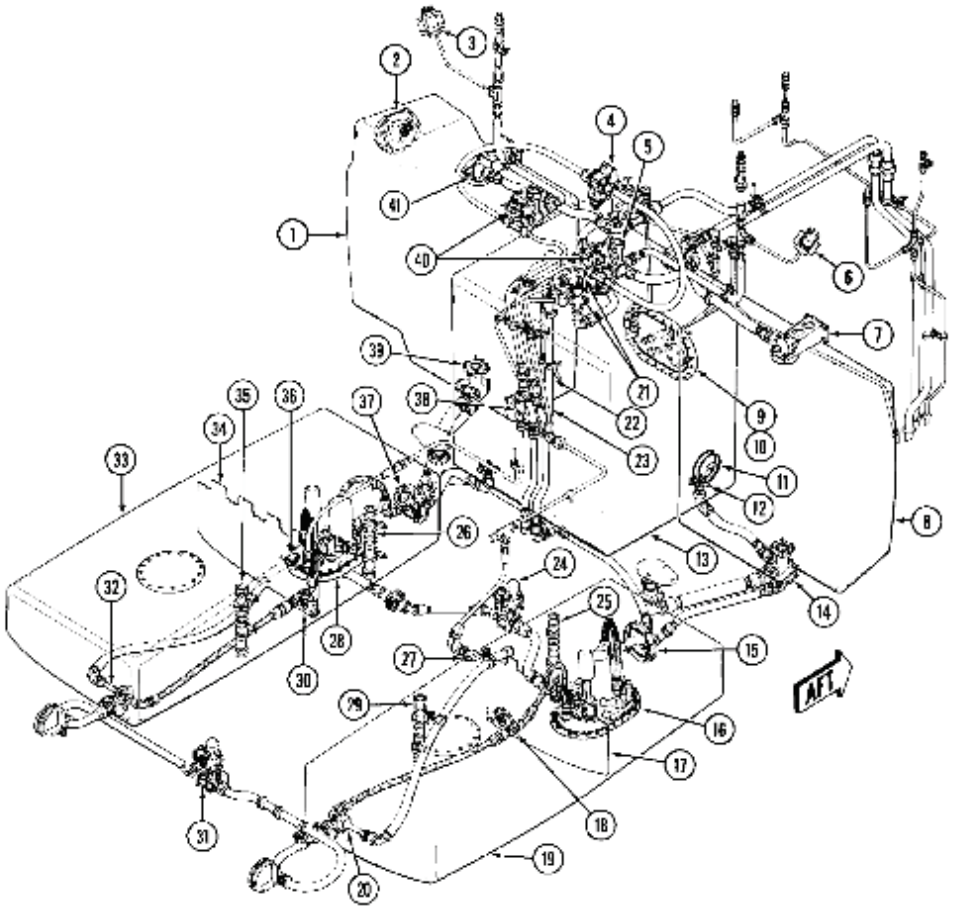
Figure 1. Flapper Valve (Sheet 1 of 1)

Ejector Pumps - Description

An ejector pump (20 and 32, Figure 1) is installed in the forward compartment of each forward cell. The pump is operated by fuel delivered through a hose from the boost pump. The ejector pump draws fuel from the forward compartment and returns fuel continuously to aft compartment of forward cell.

NOTE

Maintenance procedures for left and right ejector pumps are the same.



1. Right outboard aft fuel cell
2. Filler cap and adapter assembly
3. Right fuel pressure transmitter
4. System crossfeed valve
5. Center aft fuel cell vent cover
6. Left fuel pressure transmitter
7. Left outboard aft fuel cell vent fitting
8. Left outboard aft fuel cell
9. Center aft fuel cell rear cover
10. Governor bleed line check valve
11. Center aft fuel cell left interconnect fitting
12. Center aft fuel cell left outlet fitting
13. Center aft fuel cell
14. Left outboard aft fuel cell outlet fitting
15. Left forward fuel cell inlet/outlet fitting

(S/N 33001 33107)

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Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 1 of 2)

16. Left forward fuel cell fuel sump area consisting of:
    - Fuel sump assembly
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
17. Left forward fuel cell baffle
18. Left forward fuel cell flapper valve
19. Left forward fuel cell
20. Left forward fuel cell ejector pump and support
21. Siphon breaker valves
22. Center aft fuel cell right interconnect fitting
23. Aft fuel cell fuel quantity probe
24. Aft interconnect valve
25. Left forward fuel cell aft compartment fuel quantity probe
26. Right forward fuel cell aft compartment fuel quantity probe
27. Left forward fuel cell crashworthy vent valve
28. Right forward fuel cell fuel sump area consisting of:
    - Fuel sump
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
29. Left forward fuel cell forward compartment fuel quantity probe
30. Right forward fuel cell flapper valve
31. Forward interconnect valve
32. Right forward fuel cell ejector pump and support
33. Right forward fuel cell
34. Right forward fuel cell baffle
35. Right forward fuel cell forward compartment fuel quantity probe
36. Right forward fuel cell crashworthy vent valve
37. Right forward fuel cell inlet/outlet fitting
38. Center aft fuel cell right outlet fitting
39. Right outboard aft fuel cell outlet fitting
40. Shutoff valve
41. Right outboard aft fuel cell vent fitting

(S/N 33001 33107)

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Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 2 of 2)

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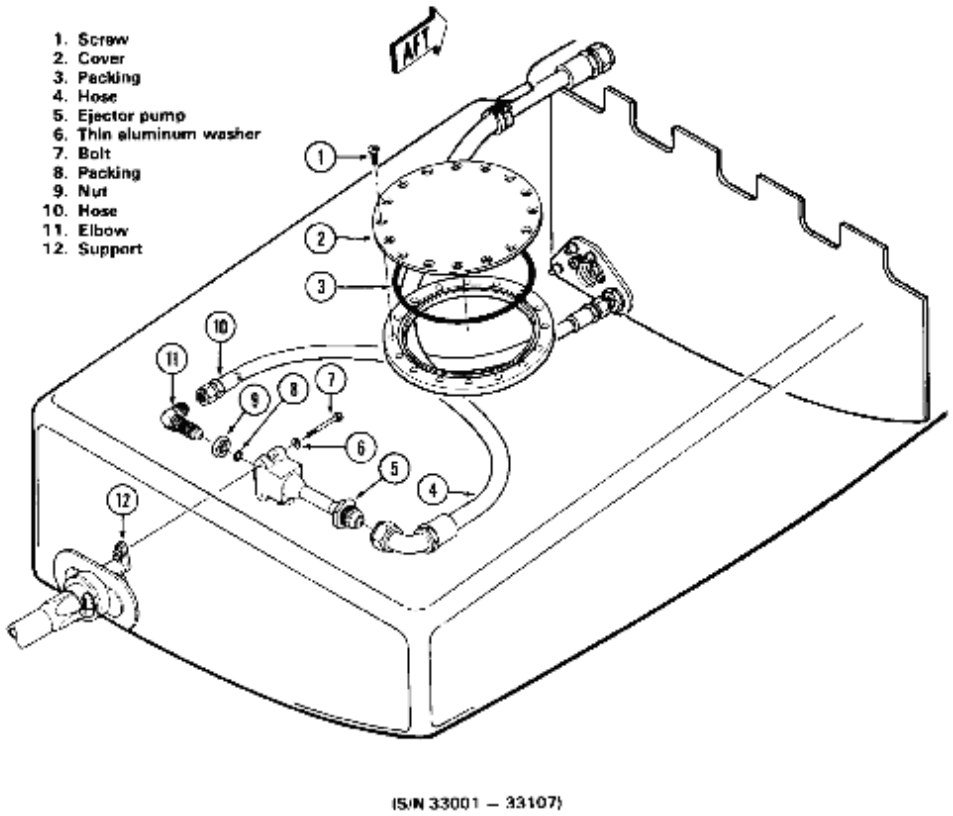
Ejector Pumps - Removal

Procedure

NOTE

The following removal procedures for left and right ejector pumps are the same.

1. Disconnect electrical power.
2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).
3. Remove cabin floor access panel above fuel cell.
4. Remove screws (1, Figure 1), cover (2), and packing (3). Discard packing.
5. Disconnect hose (4).
6. Disconnect hose (10).
7. Remove bolts (7) and washers (6). Remove ejector pump (5).
8. Remove elbow (11), nut (9), and packing (8). Discard packing.
9. Install protective cover over cell opening.



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Figure 1. Ejector Pump (Sheet 1 of 1)

Ejector Pumps - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

NOTE

The following cleaning procedures for left and right ejector pumps are the same.

1. Remove any obstructions from openings of ejector pump using compressed air.

2. If necessary, clean nozzle opening of ejector pump using lockwire (C-405). Do not enlarge nozzle in cleaning process.

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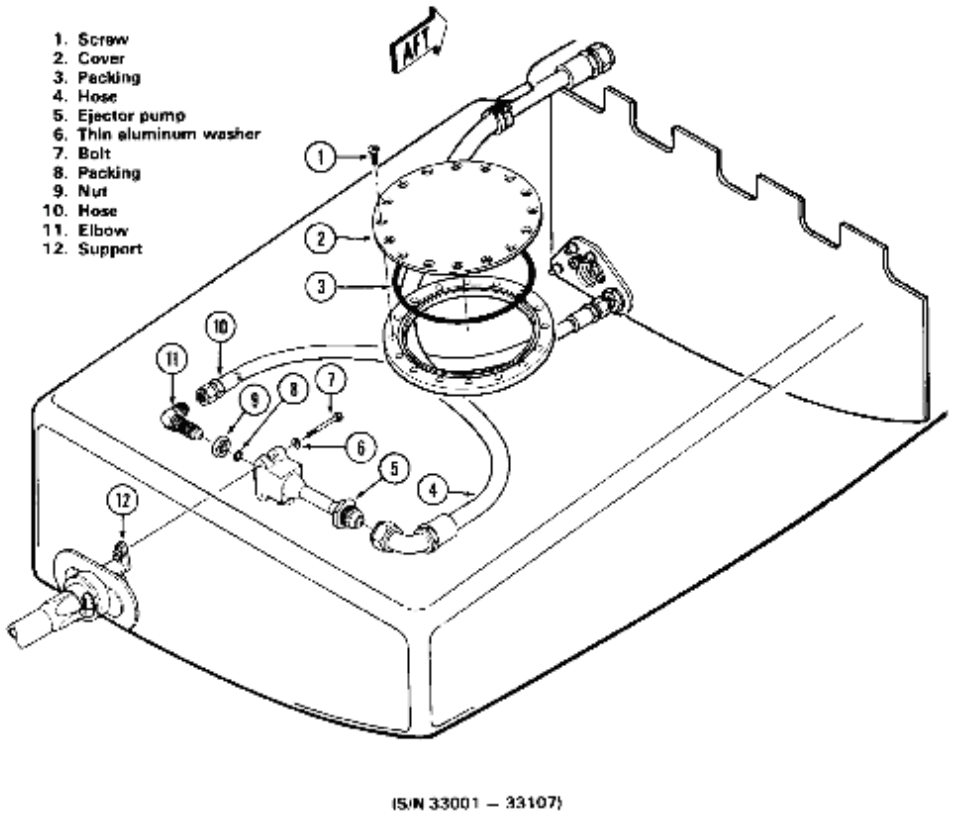
Ejector Pumps - Installation

Procedure

NOTE

The following installation procedures for left and right ejector pumps are the same.

1. Remove protective cover.
2. Install elbow (11, Figure 1) with packing (8) and nut (9).
3. Position ejector pump (5) on support (12). Install and tighten bolts (7) with washers (6).
4. Connect hose (10) to elbow (11).
5. Install new packing (3), cover (2), and screws (1).
6. Install cabin flow access panel.
7. Perform fuel system pressure test (412-A-28-10-00-00A-362A-A / 00327).
8. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).
9. Perform operational check of ejector pump (412-A-96-04-00-03A-042A-A / 02107).
10. Check area for leaks.



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Figure 1. Ejector Pump (Sheet 1 of 1)

Ejector Pump Supports - Description

NOTE

Maintenance procedures for left and right ejector pump supports are the same.

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Ejector Pump Supports - Removal

Procedure

NOTE

The following removal procedure for left and right ejector pump supports are the same.

1. Remove ejector pump (412-A-28-10-00-15A-520A-A / 00371).

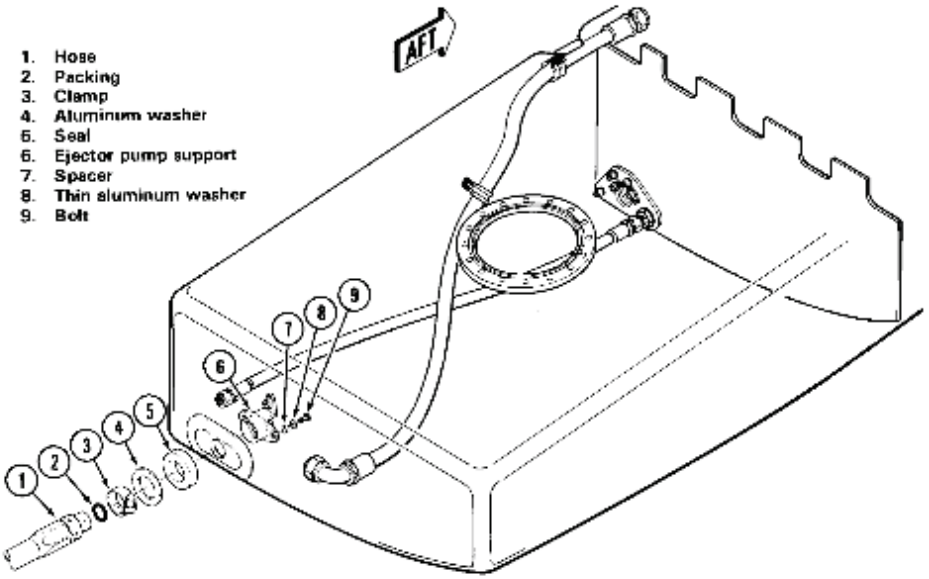
2. Disconnect hose (1, Figure 1) from forward interconnect valve (412-A-28-10-00-01A-042A-A / 00328).

3. Loosen clamp (3). Disconnect hose (1) from ejector pump support (6).

4. Remove washer (4), seal (5), and packing (2). Discard packing.

5. Remove bolts (9), washers (8), spacers (7), and ejector pump support (6).

6. Install protective covers over cell openings.



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Figure 1. Ejector Pump Support (Sheet 1 of 1)

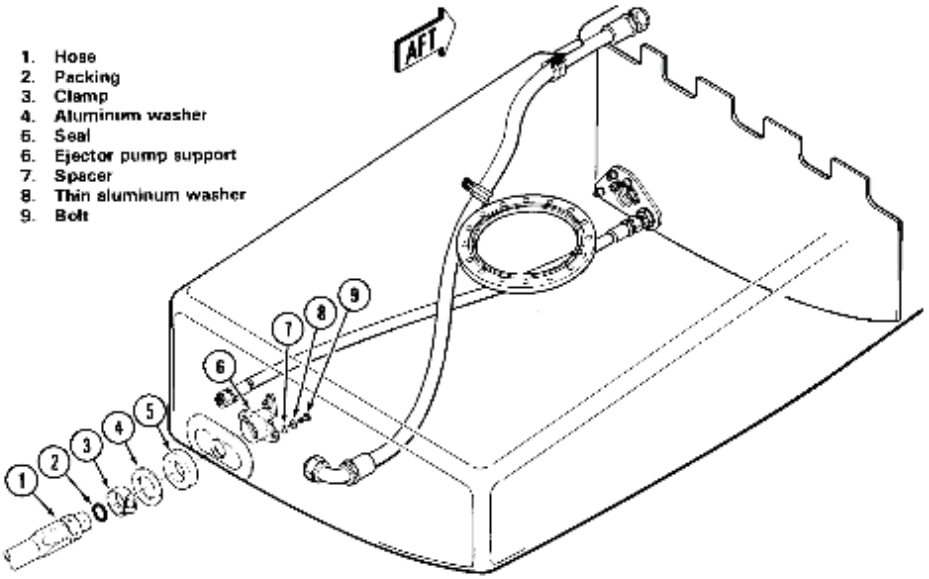
Ejector Pump Supports - Installation

Procedure

NOTE

The following installation procedure for left and right ejector pump supports are the same.

1. Remove protective covers.
2. Position ejector pump support (6, Figure 1) to port in cell. Install bolts (9) with washers (8) and spacers (7).
3. Place clamp (3), washers (4), seal (5) (if required), and packing (2) on hose (1). Connect hose (1) to ejector pump support (6).
4. Position clamp (3) over hex portion of fitting on hose (1) until seal (5) (if installed) is partially compressed. Tighten clamp (3).
5. Connect hose (1) to forward interconnect valve (412-A-28-10-00-01A-042A-A / 00328).
6. Install ejector pump (412-A-28-10-00-15A-720A-A / 00373).



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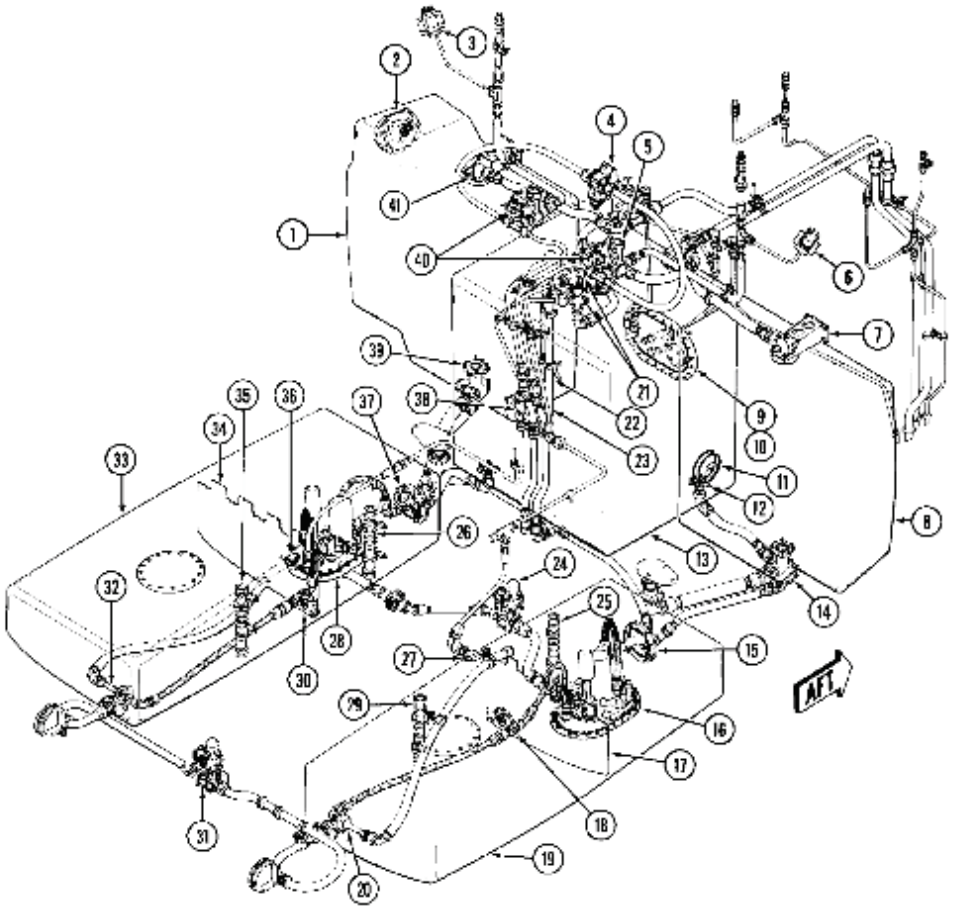
Figure 1. Ejector Pump Support (Sheet 1 of 1)

Forward Cell Crash Resistant Vent Valves - Description

A crash resistant vent valve (27 and 36, Figure 1) is installed on inboard side of each forward cell. These valves connect the forward cells into fuel vent system. In event of an impact severe enough to break cell loose from mounting points, the valve section will separate. The valve portion remaining attached to cell seals automatically and prevents fuel spillage.

NOTE

Maintenance procedures for left and right vent valves are the same.



1. Right outboard aft fuel cell
2. Filler cap and adapter assembly
3. Right fuel pressure transmitter
4. System crossfeed valve
5. Center aft fuel cell vent cover
6. Left fuel pressure transmitter
7. Left outboard aft fuel cell vent fitting
8. Left outboard aft fuel cell
9. Center aft fuel cell rear cover
10. Governor bleed line check valve
11. Center aft fuel cell left interconnect fitting
12. Center aft fuel cell left outlet fitting
13. Center aft fuel cell
14. Left outboard aft fuel cell outlet fitting
15. Left forward fuel cell inlet/outlet fitting

(S/N 33001 33107)

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Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 1 of 2)

16. Left forward fuel cell fuel sump area consisting of:
    - Fuel sump assembly
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
17. Left forward fuel cell baffle
18. Left forward fuel cell flapper valve
19. Left forward fuel cell
20. Left forward fuel cell ejector pump and support
21. Siphon breaker valves
22. Center aft fuel cell right interconnect fitting
23. Aft fuel cell fuel quantity probe
24. Aft interconnect valve
25. Left forward fuel cell aft compartment fuel quantity probe
26. Right forward fuel cell aft compartment fuel quantity probe
27. Left forward fuel cell crashworthy vent valve
28. Right forward fuel cell fuel sump area consisting of:
    - Fuel sump
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
29. Left forward fuel cell forward compartment fuel quantity probe
30. Right forward fuel cell flapper valve
31. Forward interconnect valve
32. Right forward fuel cell ejector pump and support
33. Right forward fuel cell
34. Right forward fuel cell baffle
35. Right forward fuel cell forward compartment fuel quantity probe
36. Right forward fuel cell crashworthy vent valve
37. Right forward fuel cell inlet/outlet fitting
38. Center aft fuel cell right outlet fitting
39. Right outboard aft fuel cell outlet fitting
40. Shutoff valve
41. Right outboard aft fuel cell vent fitting

(S/N 33001 33107)

ICN-412-A-281000-B-97499-00002-A-001-01

Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 2 of 2)

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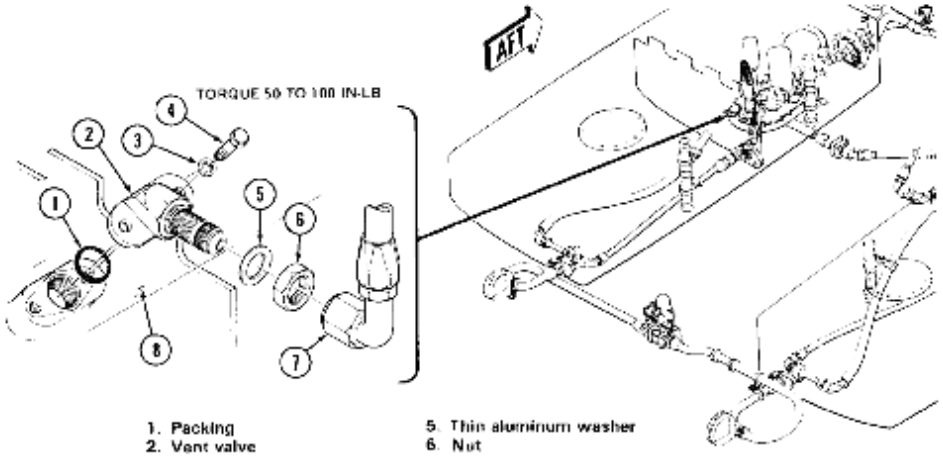
Forward Cell Crash Resistant Vent Valves - Removal

Procedure

NOTE

The following removal procedures for left and right vent valves are the same.

1. Disconnect electrical power.
2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).
3. Remove cabin floor panel above fuel cells.
4. Disconnect hose (7, Figure 1).
5. Remove nut (6) and washer (5).
6. Remove bolts (4), washers (3), valve (2), and packing (1). Discard packing.
7. Install protective cover over cell opening.



1. Packing
2. Vent valve
3. Thin aluminum washer
4. Bolt
5. Thin aluminum washer
6. Nut
7. Hose
8. Fuel cell

(S/N 33001 – 33107)

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Figure 1. Forward Cell Crash Resistant Vent Valve (Sheet 1 of 1)

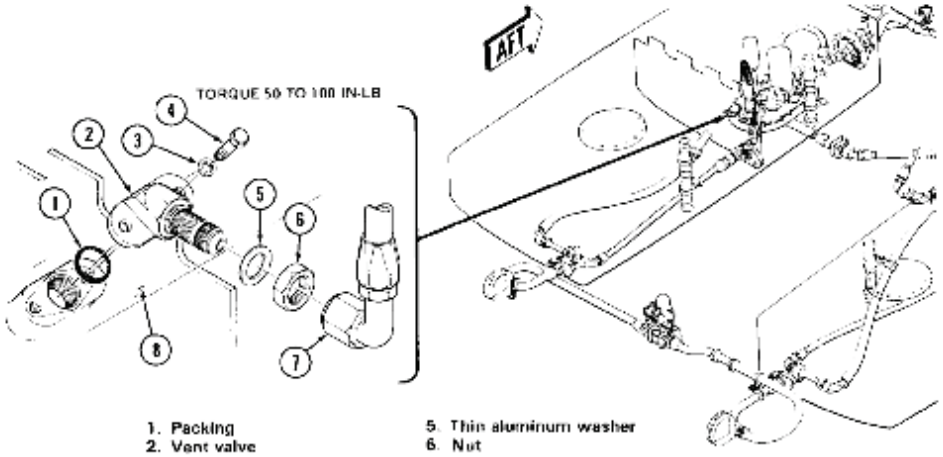
Forward Cell Crash Resistant Vent Valves - Installation

Procedure

NOTE

The following installation procedures for left and right vent valves are the same.

1. Remove protective cover.
2. Position packing (1, Figure 1) and vent valve (2) on cell (8).
3. Install bolts (4) and washers (3). Torque bolts (4) 50 to 100 in.lbs. (5.65 to 11.30 N-m).
4. Install washer (5) and nut (6).
5. Connect hose (7) to valve (2).
6. Install cabin floor panel.
7. Perform fuel system pressure test (412-A-28-10-00-00A-362A-A / 00327).
8. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).
9. Check area for leaks.
10. Install sump access door.



1. Packing
2. Vent valve
3. Thin aluminum washer
4. Bolt
5. Thin aluminum washer
6. Nut
7. Hose
8. Fuel cell

(S/N 33001 – 33107)

ICN-412-A-281000-B-97499-00019-A-001-01

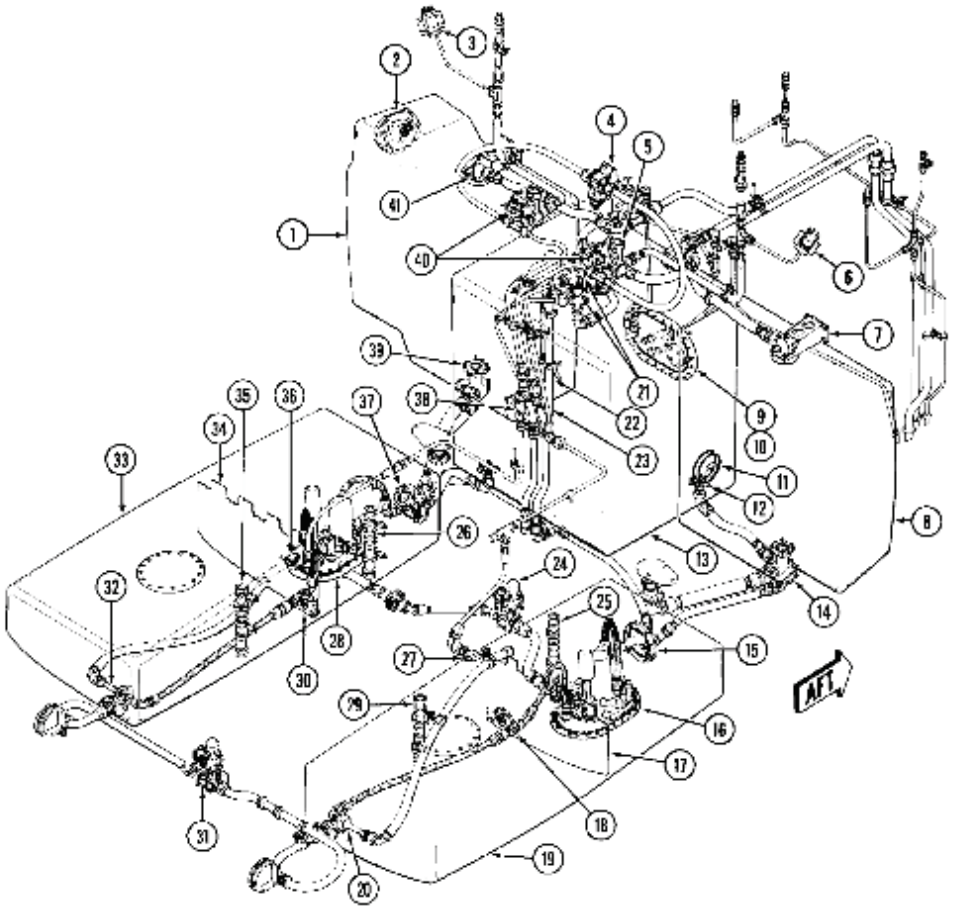
Figure 1. Forward Cell Crash Resistant Vent Valve (Sheet 1 of 1)

Forward Cells - Description

Two cells (19 and 33, Figure 1) are located under cabin floor outboard of B.L. 14 main longitudinal beams, between F.S.102 and 155. Each cell has a vent port on the upper inboard side, interconnection ports on forward and aft ends, and a large opening for sump assembly on lower side. A baffle divides each cell into forward and aft compartments and has a directional flow flapper valve. The sump opening provides access to the aft compartment. A covered round opening on the upper surface can be used as access to forward compartment. Cells are suspended in fuselage cavities by nylon cord laced between hangers on cells and on structure. Each cell has threaded inserts on inboard wall for attachment of two fuel quantity probes.

NOTE

Maintenance procedures for left and right forward cells are the same.



1. Right outboard aft fuel cell
2. Filler cap and adapter assembly
3. Right fuel pressure transmitter
4. System crossfeed valve
5. Center aft fuel cell vent cover
6. Left fuel pressure transmitter
7. Left outboard aft fuel cell vent fitting
8. Left outboard aft fuel cell
9. Center aft fuel cell rear cover
10. Governor bleed line check valve
11. Center aft fuel cell left interconnect fitting
12. Center aft fuel cell left outlet fitting
13. Center aft fuel cell
14. Left outboard aft fuel cell outlet fitting
15. Left forward fuel cell inlet/outlet fitting

(S/N 33001 33107)

ICN-412-A-281000-B-97499-00001-A-001-01

Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 1 of 2)

16. Left forward fuel cell fuel sump area consisting of:
    - Fuel sump assembly
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
17. Left forward fuel cell baffle
18. Left forward fuel cell flapper valve
19. Left forward fuel cell
20. Left forward fuel cell ejector pump and support
21. Siphon breaker valves
22. Center aft fuel cell right interconnect fitting
23. Aft fuel cell fuel quantity probe
24. Aft interconnect valve
25. Left forward fuel cell aft compartment fuel quantity probe
26. Right forward fuel cell aft compartment fuel quantity probe
27. Left forward fuel cell crashworthy vent valve
28. Right forward fuel cell fuel sump area consisting of:
    - Fuel sump
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
29. Left forward fuel cell forward compartment fuel quantity probe
30. Right forward fuel cell flapper valve
31. Forward interconnect valve
32. Right forward fuel cell ejector pump and support
33. Right forward fuel cell
34. Right forward fuel cell baffle
35. Right forward fuel cell forward compartment fuel quantity probe
36. Right forward fuel cell crashworthy vent valve
37. Right forward fuel cell inlet/outlet fitting
38. Center aft fuel cell right outlet fitting
39. Right outboard aft fuel cell outlet fitting
40. Shutoff valve
41. Right outboard aft fuel cell vent fitting

(S/N 33001 33107)

ICN-412-A-281000-B-97499-00002-A-001-01

Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 2 of 2)

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Forward Cells - Removal

Procedure

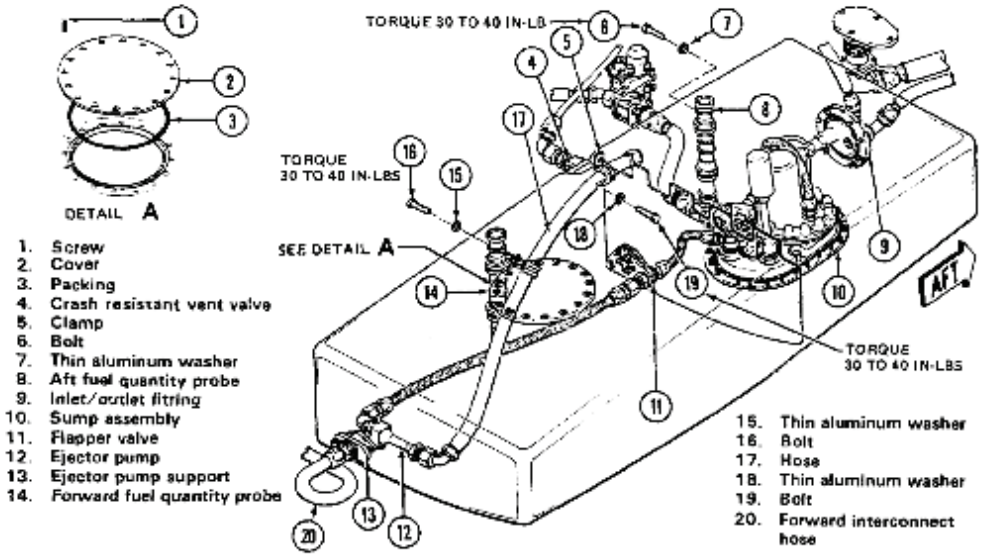
CAUTION

DO NOT HANDLE FUEL CELL WHEN CELL TEMPERATURE IS BELOW 65°F (18°C).

NOTE

The following removal procedures for left and right forward cells are the same.

1. Disconnect electrical power.
2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).
3. Remove cabin floor panels and access panels from fuselage underside for access to forward cell and connections.
4. Remove sump assembly (412-A-28-10-00-03A-520A-A / 00335).
5. Remove inlet/outlet fitting (9, Figure 1) (412-A-28-10-00-12A-520A-A / 00361), except hose inside cell may remain connected and be removed with fitting.
6. Remove screws (1), cover (2), and packing (3). Discard packing.
7. Remove aft and forward fuel quantity probes (8 and 14) (412-A-28-10-00-35A-520A-A / 00432 and 412-A-28-10-00-36A-520A-A / 00435).
8. Remove flapper valve (11) (412-A-28-10-00-14A-520A-A / 00367).
9. Remove bolt (19) and washer (18).
10. Remove ejector pump (12) (412-A-28-10-00-15A-520A-A / 00371), except hoses can remain connected and be removed with ejector pump.
11. Remove ejector pump support (13) (412-A-28-10-00-16A-520A-A / 00375).
12. Remove forward interconnect hose (20).
13. Remove bolts (6 and 16) with washers (7 and 15).
14. Remove vent valve (4) (412-A-28-10-00-17A-520A-A / 00378).
15. Remove nylon cord from hangers around top of cell and cavity.
16. Lift cell from cavity.
17. Install protective caps or plugs in open fuel lines and over cell openings.



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Figure 1. Forward Fuel Cell (Sheet 1 of 1)

Forward Cells - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Soap	100066	AR	Zodiac Aerospace (www.zodiacaerospace.com)
Cleaning Solution	601199	AR	Zodiac Aerospace www.zodiacaerospace.com
Clean cloth	C-516	AR	

Safety conditions



OBEY ALL THE SAFETY PRECAUTIONS WHEN YOU DO MAINTENANCE ON FUEL SYSTEM EQUIPMENT.



OBEY ALL THE FUEL CELL HANDLING PRECAUTIONS DURING CLEANING OR DAMAGE TO THE CELL MAY OCCUR.



DO NOT ALLOW SOLVENTS OR COMPOUNDS TO MAKE CONTACT WITH PARTS MADE OF ELASTOMERIC MATERIAL OR DAMAGE WILL OCCUR.

Procedure

NOTE

The following cleaning procedures for left and right forward cells are the same.

NOTE

Any evidence of fuel leakage must be identified before cleaning.

NOTE

Refer to the document "FUEL-F6101-SPM20-00" (Zodiac Aerospace), for additional information on the cleaning of the fuel cells.

1. Clean the internal surface of the fuel cell with a clean cloth (C-516) moistened with a cleaning solution (601199). Dry the liner with low-pressure filtered compressed air for 2 to 4 hours, at approximately 68°F (20°C).
2. Clean small areas on the external surface of the fuel cell with a clean cloth (C-516) moistened with a cleaning solution (601199). Dry with a clean cloth (C-516).
3. Clean an extensive area on the external surface of the fuel cell with a solution of soap (100066) and hot water. Rinse with water and dry with a clean cloth (C-516).
4. Dry clean the fittings of the fuel cell with filtered compressed air and a clean cloth (C-516) to remove any blockage or obstruction.
5. Clean fungus growth found on the fuel cell or on the fuel cell cavity per the instructions contained in the BHT-ALL-SPM, Chapter 5, Standard Practices Manual.

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Forward Cells - Inspection

Procedure

1. Inspect cell for cracks, scuffing, and general condition.
2. Inspect threaded inserts for serviceability. Inserts with thread damage which can be detected visually shall be replaced.
3. Inspect hoses for damage and deterioration. Hoses exhibiting damage or deterioration which can be detected visually shall be replaced.
4. Inspect removed components for damage.

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Forward Cells - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Talcum powder	C-408	AR	
Lacing cord	C-471	AR	

Safety conditions



DO NOT HANDLE FUEL CELL WHEN TEMPERATURE OF CELL IS BELOW 65°F (18°C).

Procedure

NOTE

The following installation procedures for left and right forward cells are the same.

1. Inspect cell cavity for clean, smooth condition of surfaces.

2. Apply talcum powder (C-408) to cavity surfaces.

3. Remove protective caps, plugs and covers. Place cell in cavity and align ports with fuselage openings.

4. Install vent valve (4, Figure 1) (412-A-28-10-00-17A-720A-A / 00379).

5. Install bolt (6) with washer (7) and bolt (16) with washer (15). Torque bolts (6 and 16) 30 to 40 in.lbs. (3.39 to 4.52 N-m).

6. Install ejector pump support (13) (412-A-28-10-00-16A-720A-A / 00376, steps 1. through 4.).

7. Install ejector pump (12) (412-A-28-10-00-15A-720A-A / 00373, steps 1. through 4.).

8. Install aft quantity probe (8) (412-A-28-10-00-36A-720A-A / 00436) and forward quantity probe (14) (412-A-28-10-00-35A-720A-A / 00433).

9. Install bolt (19) with washer (18) to secure clamp (5) on hose (17) to cell wall. Torque bolt (19) 30 to 40 in.lbs. (3.39 to 4.52 N-m).

10. Install flapper valve (11) (412-A-28-10-00-14A-720A-A / 00369, steps 1. through 6.).

11. Lace and tie a single length of lacing cord (C-471) between hangers around top of cell and cavity (Figure 2).

12. Install inlet/outlet fitting (9) (412-A-28-10-00-12A-720A-A / 00362, step 1. or step 2., as applicable).

13. Install sump assembly (10) (412-A-28-10-00-03A-720A-A / 00337).

14. Install packing (3), cover (2), and screws (1).

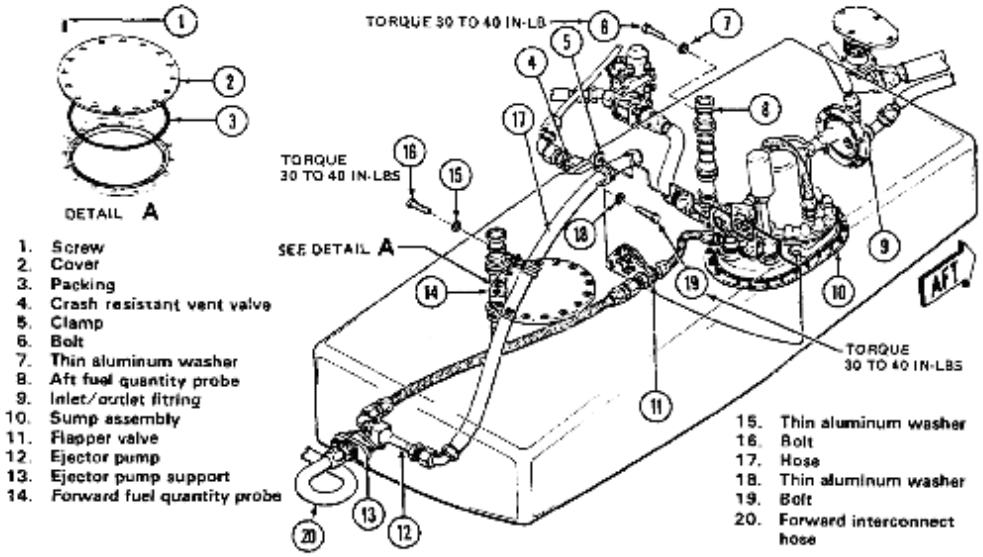
15. Install cabin floor panels and doors on underside of fuselage.

16. Perform fuel system pressure test (412-A-28-10-00-00A-362A-A / 00327).

17. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).

18. Perform fuel system operational check.

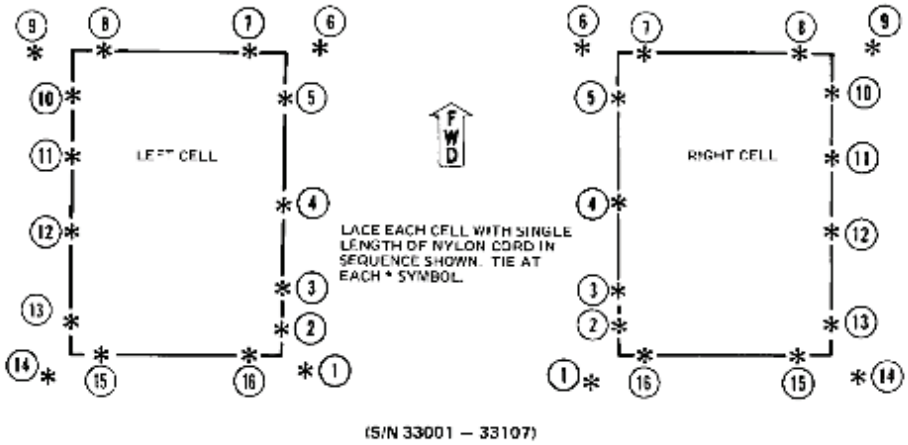
19. Check area for leaks.



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Figure 1. Forward Fuel Cell (Sheet 1 of 1)



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Figure 2. Forward Cell Lacing (Sheet 1 of 1)

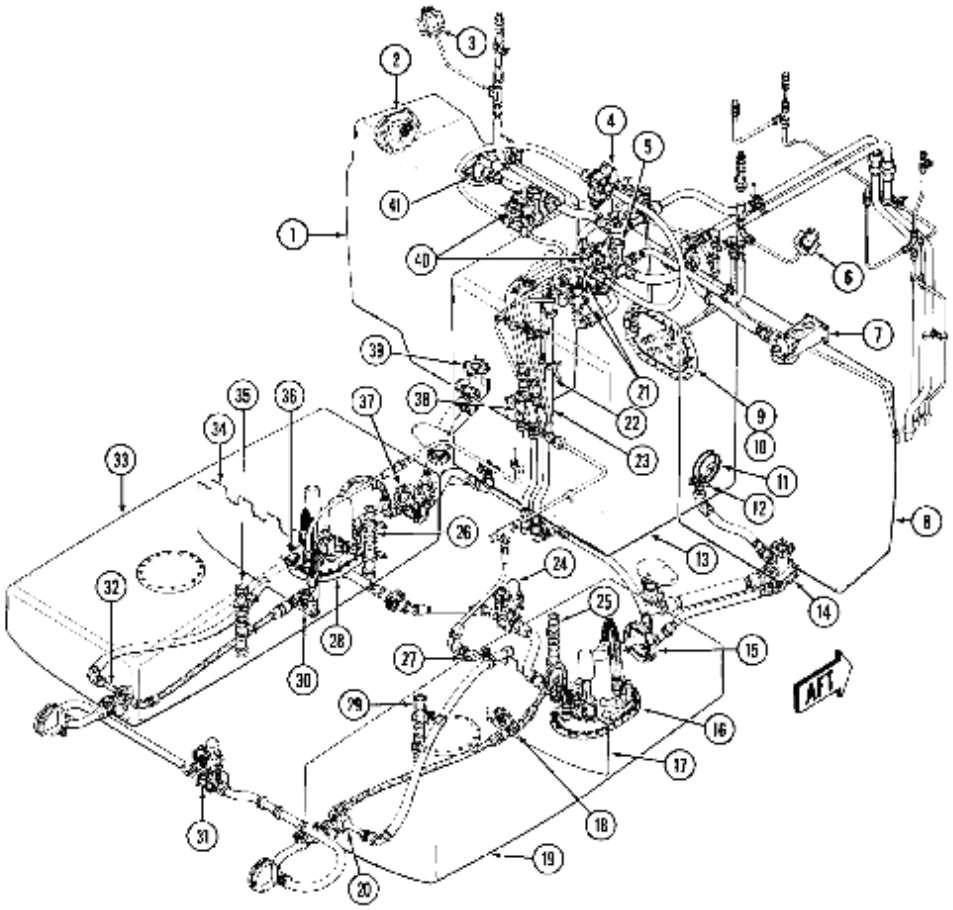
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Siphon Breaker Valves - Description

Two siphon breaker valves (check valves) (21, Figure 1) are installed in vent tube tees above the engine deck near left shutoff valve.

NOTE

Maintenance procedures for left and right siphon breaker valves are the same.



1. Right outboard aft fuel cell
2. Filler cap and adapter assembly
3. Right fuel pressure transmitter
4. System crossfeed valve
5. Center aft fuel cell vent cover
6. Left fuel pressure transmitter
7. Left outboard aft fuel cell vent fitting
8. Left outboard aft fuel cell
9. Center aft fuel cell rear cover
10. Governor bleed line check valve
11. Center aft fuel cell left interconnect fitting
12. Center aft fuel cell left outlet fitting
13. Center aft fuel cell
14. Left outboard aft fuel cell outlet fitting
15. Left forward fuel cell inlet/outlet fitting

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Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 1 of 2)

16. Left forward fuel cell fuel sump area consisting of:
    - Fuel sump assembly
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
17. Left forward fuel cell baffle
18. Left forward fuel cell flapper valve
19. Left forward fuel cell
20. Left forward fuel cell ejector pump and support
21. Siphon breaker valves
22. Center aft fuel cell right interconnect fitting
23. Aft fuel cell fuel quantity probe
24. Aft interconnect valve
25. Left forward fuel cell aft compartment fuel quantity probe
26. Right forward fuel cell aft compartment fuel quantity probe
27. Left forward fuel cell crashworthy vent valve
28. Right forward fuel cell fuel sump area consisting of:
    - Fuel sump
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
29. Left forward fuel cell forward compartment fuel quantity probe
30. Right forward fuel cell flapper valve
31. Forward interconnect valve
32. Right forward fuel cell ejector pump and support
33. Right forward fuel cell
34. Right forward fuel cell baffle
35. Right forward fuel cell forward compartment fuel quantity probe
36. Right forward fuel cell crashworthy vent valve
37. Right forward fuel cell inlet/outlet fitting
38. Center aft fuel cell right outlet fitting
39. Right outboard aft fuel cell outlet fitting
40. Shutoff valve
41. Right outboard aft fuel cell vent fitting

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Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 2 of 2)

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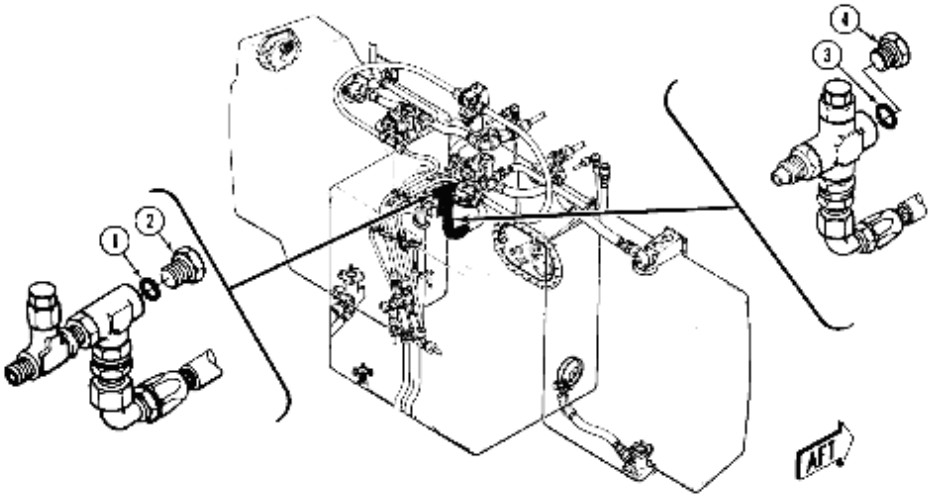
Siphon Breaker Valves - Removal

Procedure

NOTE

The following removal procedures for left and right siphon breaker valves are the same.

1. Disconnect electrical power.
2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).
3. Remove access door in left side of pylon support in cabin.
4. Remove siphon breaker valve (2 or 4, Figure 1) with packing (1 or 3). Discard packing.
5. Install protective cap or plug in open fuel line.



1. Packing
2. Siphon breaker valve
3. Packing
4. Siphon breaker valve

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Figure 1. Siphon Breaker Valve (Sheet 1 of 1)

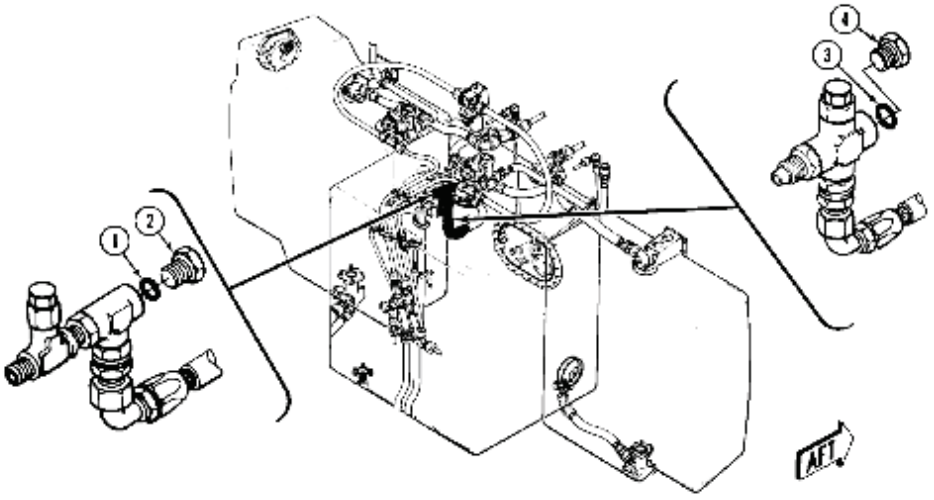
Siphon Breaker Valves - Installation

Procedure

NOTE

The following installation procedures for left and right siphon breaker valves are the same.

1. Remove protective cap or plug. Install siphon breaker valve (2 or 4, Figure 1) using packing (1 or 3).
2. Perform fuel system pressure test (412-A-28-10-00-00A-362A-A / 00327).
3. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).
4. Check fuel system for leaks.
5. Install access doors.



1. Packing
2. Siphon breaker valve
3. Packing
4. Siphon breaker valve

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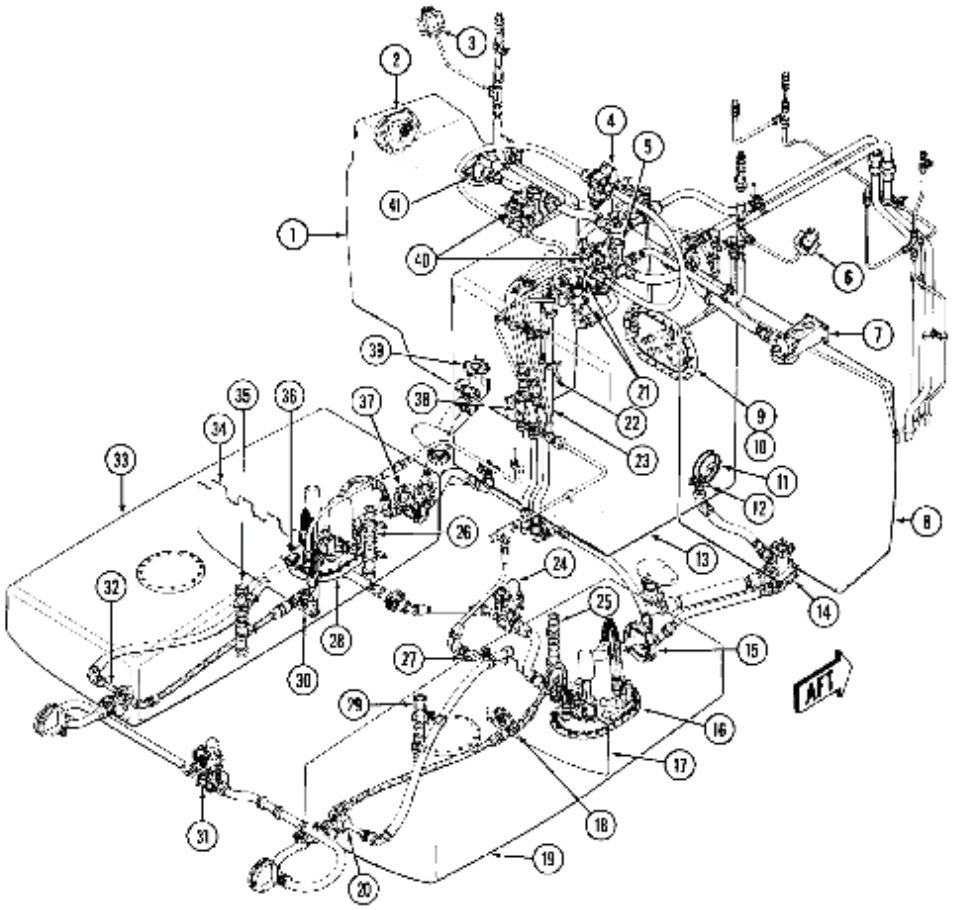
Figure 1. Siphon Breaker Valve (Sheet 1 of 1)

Shutoff Valves - Description

Two motor-operated shutoff valves (40, Figure 1), connected in main fuel pressure lines to engine power sections, are located on front of lower forward engine firewall. The valves are electrically controlled by ENGINE 1 FUEL and ENGINE 2 FUEL switches and have manual override handles which are also position indicators. Thermal relief valves are incorporated for internal bypass of trapped fuel when the system is inoperative.

NOTE

Maintenance procedures for left and right shutoff valves are the same.



1. Right outboard aft fuel cell
2. Filler cap and adapter assembly
3. Right fuel pressure transmitter
4. System crossfeed valve
5. Center aft fuel cell vent cover
6. Left fuel pressure transmitter
7. Left outboard aft fuel cell vent fitting
8. Left outboard aft fuel cell
9. Center aft fuel cell rear cover
10. Governor bleed line check valve
11. Center aft fuel cell left interconnect fitting
12. Center aft fuel cell left outlet fitting
13. Center aft fuel cell
14. Left outboard aft fuel cell outlet fitting
15. Left forward fuel cell inlet/outlet fitting

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Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 1 of 2)

16. Left forward fuel cell fuel sump area consisting of:
    - Fuel sump assembly
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
17. Left forward fuel cell baffle
18. Left forward fuel cell flapper valve
19. Left forward fuel cell
20. Left forward fuel cell ejector pump and support
21. Siphon breaker valves
22. Center aft fuel cell right interconnect fitting
23. Aft fuel cell fuel quantity probe
24. Aft interconnect valve
25. Left forward fuel cell aft compartment fuel quantity probe
26. Right forward fuel cell aft compartment fuel quantity probe
27. Left forward fuel cell crashworthy vent valve
28. Right forward fuel cell fuel sump area consisting of:
    - Fuel sump
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
29. Left forward fuel cell forward compartment fuel quantity probe
30. Right forward fuel cell flapper valve
31. Forward interconnect valve
32. Right forward fuel cell ejector pump and support
33. Right forward fuel cell
34. Right forward fuel cell baffle
35. Right forward fuel cell forward compartment fuel quantity probe
36. Right forward fuel cell crashworthy vent valve
37. Right forward fuel cell inlet/outlet fitting
38. Center aft fuel cell right outlet fitting
39. Right outboard aft fuel cell outlet fitting
40. Shutoff valve
41. Right outboard aft fuel cell vent fitting

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Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 2 of 2)

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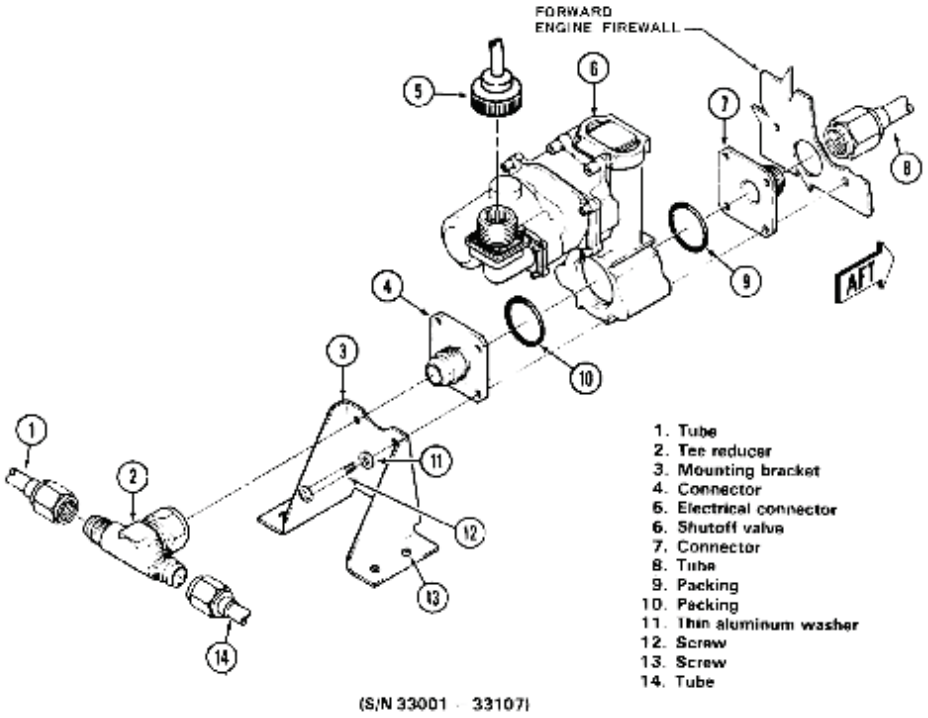
Shutoff Valves - Removal

Procedure

NOTE

The following removal procedures for left and right shutoff valves are the same.

1. Disconnect electrical power.
2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).
3. Open engine cowling.
4. Remove access door on side of pylon support in cabin.
5. Disconnect electrical connector (5, Figure 1).
6. Disconnect tubes (1 and 14).
7. Remove tee reducer (2) if connector (4) is to be replaced.
8. Disconnect tube (8).
9. Remove four screws (12) and washers (11).
10. Remove shutoff valve (6), connectors (4 and 7), and packings (9 and 10). Discard packings.
11. Cap or plug openings.



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Figure 1. Shutoff Valve (Sheet 1 of 1)

Shutoff Valves - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

NOTE

The following installation procedures for left and right shutoff valves are the same.

1. Remove protective caps or plugs.

2. Apply assembly fluid (C-024) to packings (9 and 10, Figure 1).

3. Position packings (9 and 10) and connectors (4 and 7) on body of shutoff valve (6). Insert two upper screws (12) with thin aluminum washers (11).

NOTE

Screws (13) securing mounting bracket (3) to deck may be loosened as necessary to provide clearance for installation of shutoff valve (6).

4. Position shutoff valve (6) between mounting bracket (3) and firewall.

5. Start upper screws (12) into firewall nutplates.

6. Install two lower screws (12) with thin aluminum washers (11). Tighten all four screws (12).

7. Tighten screws (13) if previously loosened.

8. If not previously accomplished, install tee reducer (2).

9. Connect tubes (1 and 14).

10. Connect tube (8).

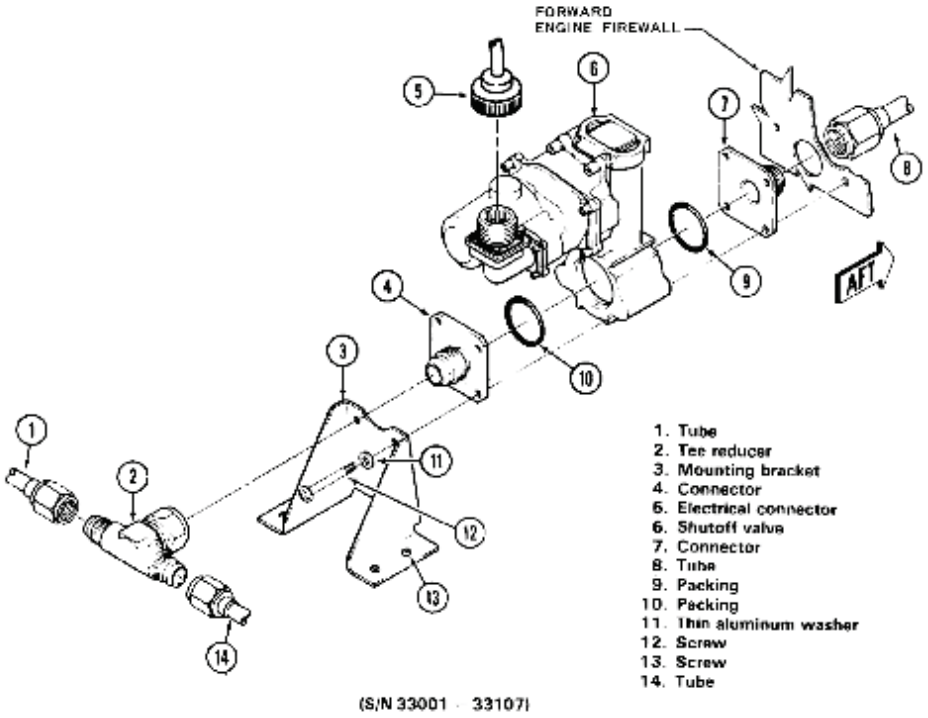
11. Connect electrical connector (5).

12. Perform fuel system pressure test (412-A-28-10-00-00A-362A-A / 00327).

13. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).

14. Check area for leaks.

15. Close engine cowling and install sump access door.

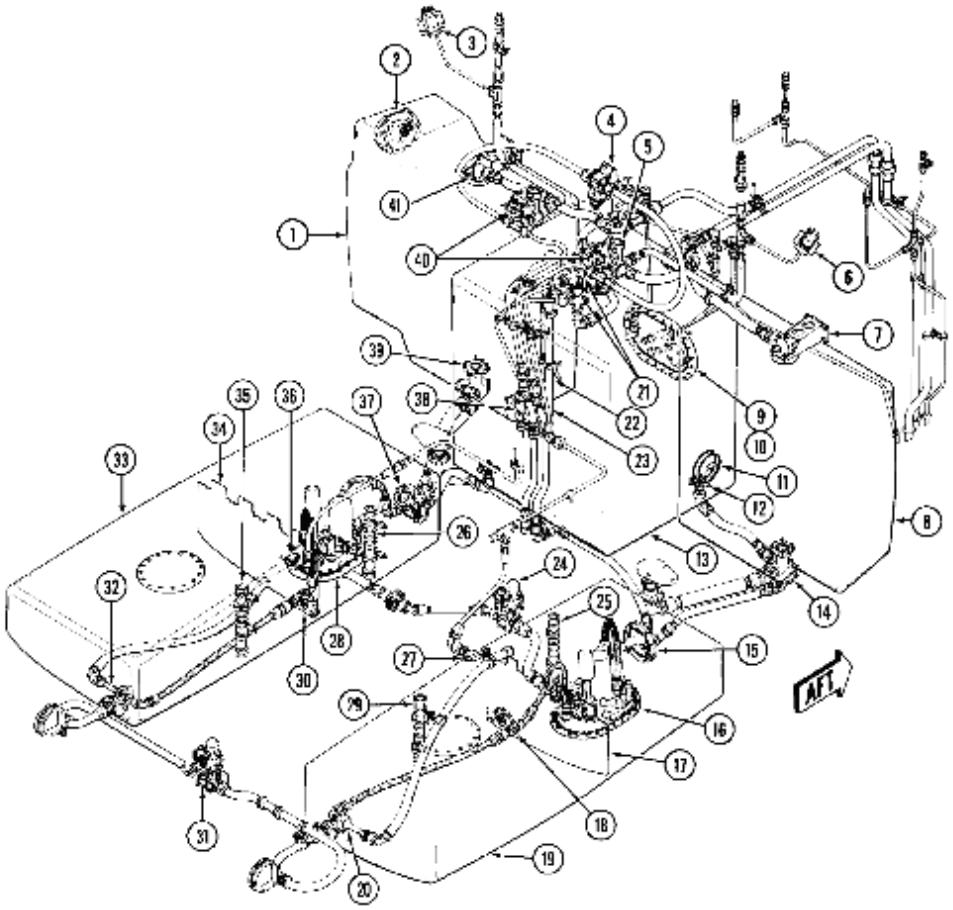


ICN-412-A-281000-B-97499-00023-A-001-01

Figure 1. Shutoff Valve (Sheet 1 of 1)

Crossfeed Valve - Description

The crossfeed valve (4, Figure 1) is located above and between the shutoff valves. Under normal conditions the crossfeed valve is closed and fuel is supplied to each power section from lower cell on same side of helicopter. However, if boost pump in either cell ceases to function, the crossfeed valve opens to permit flow from operating pump to both power sections.



1. Right outboard aft fuel cell
2. Filler cap and adapter assembly
3. Right fuel pressure transmitter
4. System crossfeed valve
5. Center aft fuel cell vent cover
6. Left fuel pressure transmitter
7. Left outboard aft fuel cell vent fitting
8. Left outboard aft fuel cell
9. Center aft fuel cell rear cover
10. Governor bleed line check valve
11. Center aft fuel cell left interconnect fitting
12. Center aft fuel cell left outlet fitting
13. Center aft fuel cell
14. Left outboard aft fuel cell outlet fitting
15. Left forward fuel cell inlet/outlet fitting

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Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 1 of 2)

16. Left forward fuel cell fuel sump area consisting of:
    - Fuel sump assembly
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
17. Left forward fuel cell baffle
18. Left forward fuel cell flapper valve
19. Left forward fuel cell
20. Left forward fuel cell ejector pump and support
21. Siphon breaker valves
22. Center aft fuel cell right interconnect fitting
23. Aft fuel cell fuel quantity probe
24. Aft interconnect valve
25. Left forward fuel cell aft compartment fuel quantity probe
26. Right forward fuel cell aft compartment fuel quantity probe
27. Left forward fuel cell crashworthy vent valve
28. Right forward fuel cell fuel sump area consisting of:
    - Fuel sump
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
29. Left forward fuel cell forward compartment fuel quantity probe
30. Right forward fuel cell flapper valve
31. Forward interconnect valve
32. Right forward fuel cell ejector pump and support
33. Right forward fuel cell
34. Right forward fuel cell baffle
35. Right forward fuel cell forward compartment fuel quantity probe
36. Right forward fuel cell crashworthy vent valve
37. Right forward fuel cell inlet/outlet fitting
38. Center aft fuel cell right outlet fitting
39. Right outboard aft fuel cell outlet fitting
40. Shutoff valve
41. Right outboard aft fuel cell vent fitting

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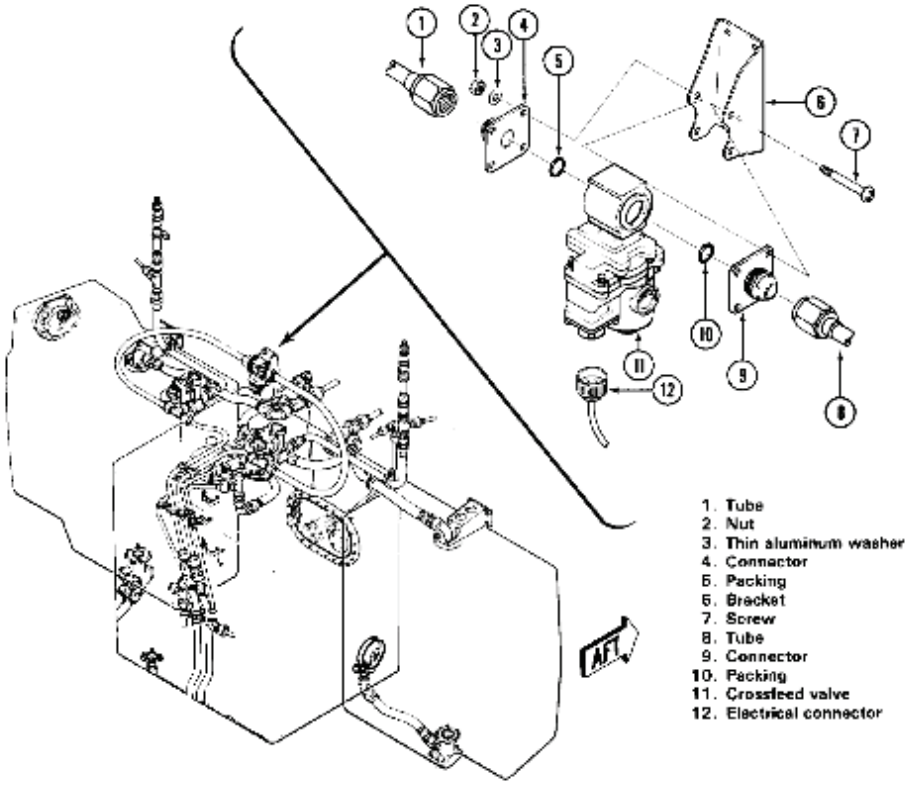
Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 2 of 2)

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Crossfeed Valve - Removal

Procedure

1. Disconnect electrical power.
2. Defuel system (412-A-12-00-00-00A-009A-A / 00125).
3. Remove transmission cowling.
4. Disconnect electrical connector (12, Figure 1).
5. Disconnect tubes (1 and 8).
6. Remove screws (7).
7. Remove crossfeed valve (11), packings (5 and 10), and connectors (4 and 9). Discard packings.
8. Cap or plug openings.



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Figure 1. Crossfeed Valve (Sheet 1 of 1)

Crossfeed Valve - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

1. Remove protective caps or plugs.

2. Apply assembly fluid (C-024) to packings (5 and 10, Figure 1).

3. Position packings (5 and 10) and connectors (4 and 9) on crossfeed valve (11). Install two forward screws (7), thin aluminum washers (3), and nuts (2). Do not tighten nuts at this time.

4. Position crossfeed valve (11) and connectors (4 and 9) in bracket (6). Install two aft screws (7), thin aluminum washers (3), and nuts (2). Tighten four nuts.

5. Connect tubes (1 and 8).

6. Connect electrical connector (12).

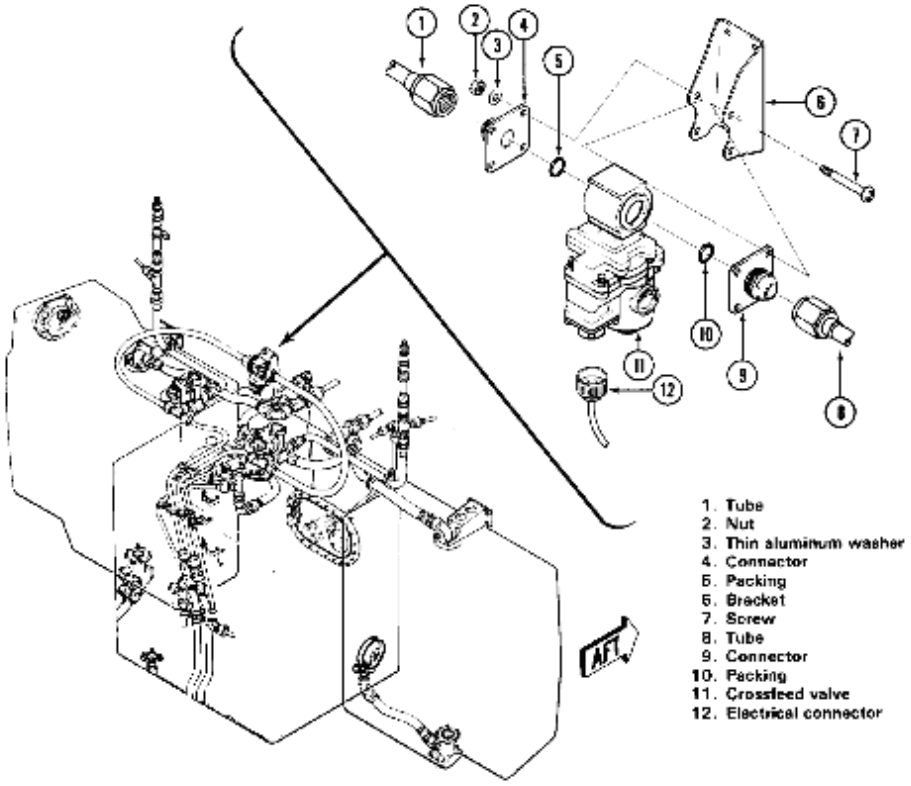
7. Perform fuel system pressure test (412-A-28-10-00-00A-362A-A / 00327).

8. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).

9. Perform operational check of crossfeed valve (412-A-96-04-00-07A-320A-A / 02119).

10. Check area for leaks.

11. Install transmission cowling and sump access door.



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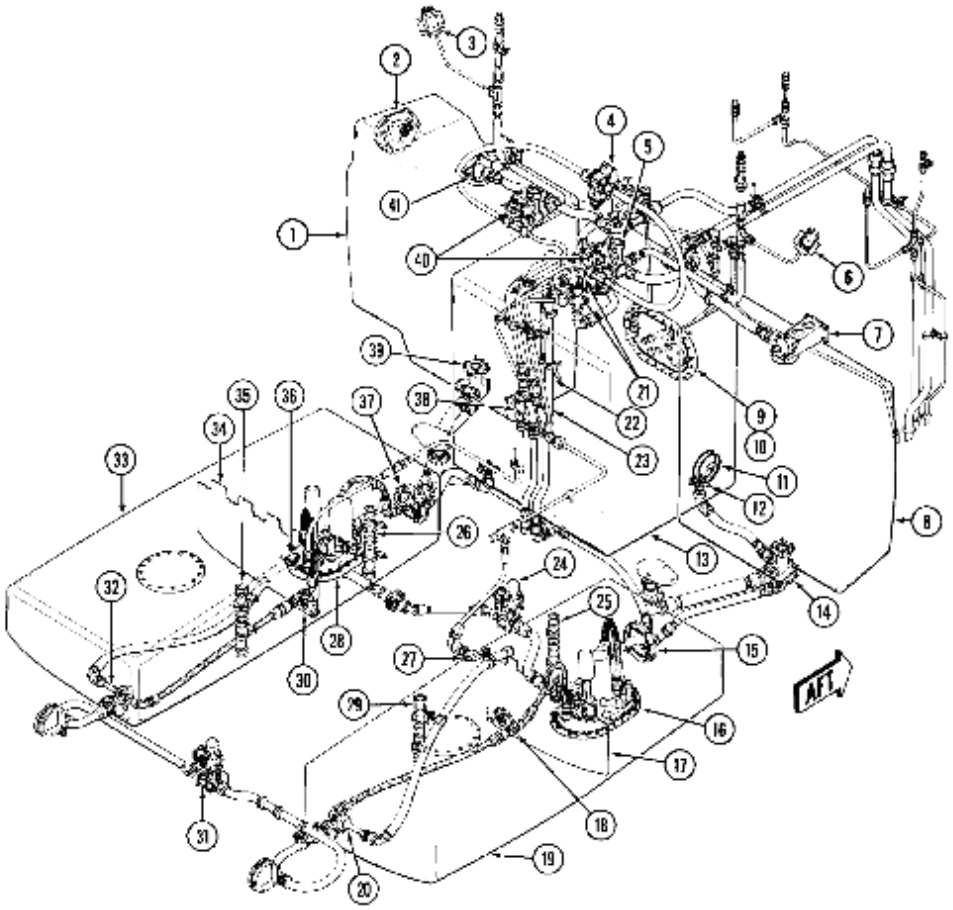
Figure 1. Crossfeed Valve (Sheet 1 of 1)

Fuel Pressure Transmitters - Description

Two electrical fuel pressure transmitters (3 and 6, Figure 1) for the fuel pressure indicating system are connected to main fuel lines to the engine power sections. The transmitters are mounted on left and right rear sides of forward engine firewall.

NOTE

Maintenance procedures for left and right fuel pressure transmitters are the same.



1. Right outboard aft fuel cell
2. Filler cap and adapter assembly
3. Right fuel pressure transmitter
4. System crossfeed valve
5. Center aft fuel cell vent cover
6. Left fuel pressure transmitter
7. Left outboard aft fuel cell vent fitting
8. Left outboard aft fuel cell
9. Center aft fuel cell rear cover
10. Governor bleed line check valve
11. Center aft fuel cell left interconnect fitting
12. Center aft fuel cell left outlet fitting
13. Center aft fuel cell
14. Left outboard aft fuel cell outlet fitting
15. Left forward fuel cell inlet/outlet fitting

(S/N 33001 33107)

ICN-412-A-281000-B-97499-00001-A-001-01

Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 1 of 2)

16. Left forward fuel cell fuel sump area consisting of:
    - Fuel sump assembly
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
17. Left forward fuel cell baffle
18. Left forward fuel cell flapper valve
19. Left forward fuel cell
20. Left forward fuel cell ejector pump and support
21. Siphon breaker valves
22. Center aft fuel cell right interconnect fitting
23. Aft fuel cell fuel quantity probe
24. Aft interconnect valve
25. Left forward fuel cell aft compartment fuel quantity probe
26. Right forward fuel cell aft compartment fuel quantity probe
27. Left forward fuel cell crashworthy vent valve
28. Right forward fuel cell fuel sump area consisting of:
    - Fuel sump
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
29. Left forward fuel cell forward compartment fuel quantity probe
30. Right forward fuel cell flapper valve
31. Forward interconnect valve
32. Right forward fuel cell ejector pump and support
33. Right forward fuel cell
34. Right forward fuel cell baffle
35. Right forward fuel cell forward compartment fuel quantity probe
36. Right forward fuel cell crashworthy vent valve
37. Right forward fuel cell inlet/outlet fitting
38. Center aft fuel cell right outlet fitting
39. Right outboard aft fuel cell outlet fitting
40. Shutoff valve
41. Right outboard aft fuel cell vent fitting

(S/N 33001 33107)

ICN-412-A-281000-B-97499-00002-A-001-01

Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 2 of 2)

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Fuel Pressure Transmitters - Removal

Procedure

NOTE

The following removal procedures for left and right fuel pressure transmitters are the same.

1. Disconnect electrical power.

2. Open engine cowling.

3. Disconnect electrical connector (1, Figure 1) from pressure transmitter (4) located on forward firewall.

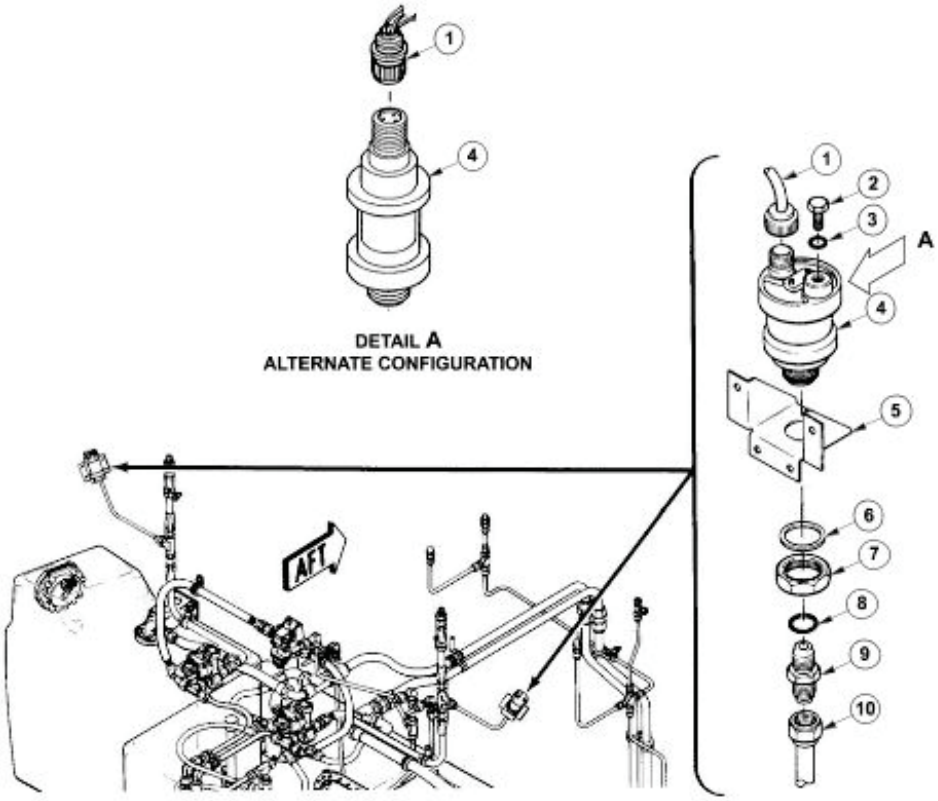
4. Disconnect tube (10).

5. Remove union (9) and packing (8). Discard packing.

6. Remove aluminum alloy nut (7) and corrosion resistant steel washer (6).

7. Remove pressure transmitter (4). If applicable, remove plug (2) and packing (3). Discard packing.

8. Cap or plug openings.



(S/N 33001 – 33107)

1. Electrical connector
2. Plug
3. Packing
4. Pressure transmitter
5. Bracket
6. Corrosion resistant steel washer
7. Aluminum alloy nut
8. Packing
9. Union
10. Tube

ICN-412-A-281000-B-97499-00025-A-001-01

Figure 1. Fuel Pressure Transmitter (Sheet 1 of 1)

Fuel Pressure Transmitters - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

NOTE

The following installation procedures for left and right fuel pressure transmitters are the same.

1. Apply assembly fluid (C-024) to packing (3, Figure 1).

2. Remove protective caps or plugs. If applicable, install plug (2) and packing (3).

3. Position pressure transmitter (4) on bracket (5). Install corrosion resistant steel washer (6) and aluminum alloy nut (7).

4. Install packing (8) and union (9).

5. Connect tube (10).

6. Connect electrical connector (1).

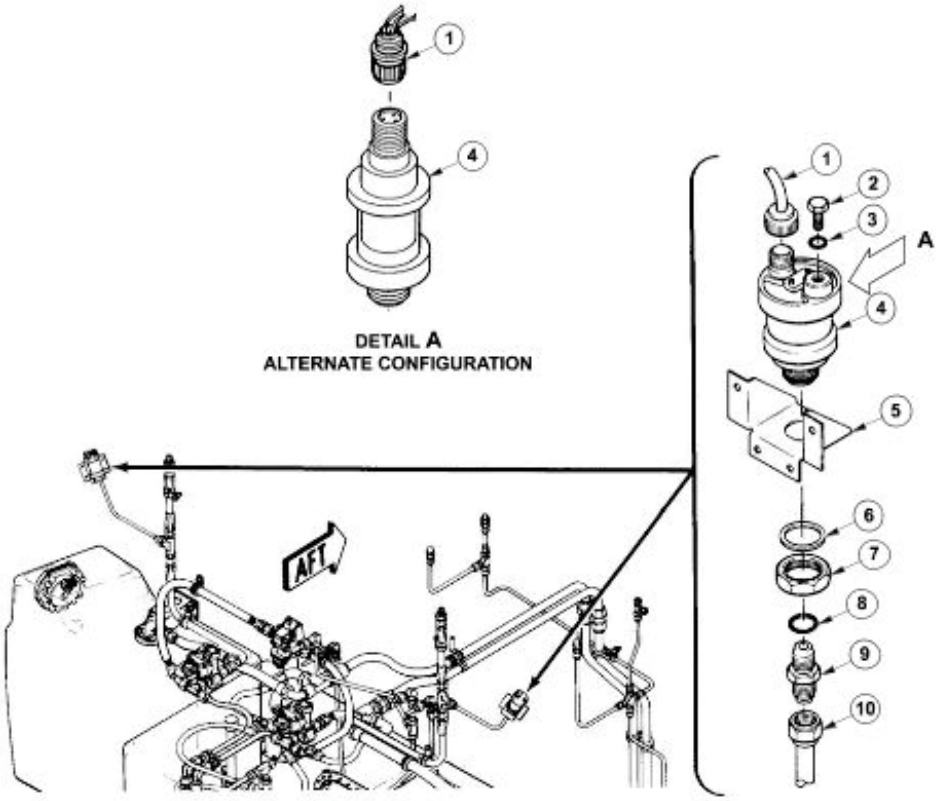
7. Perform fuel system pressure test (412-A-28-10-00-00A-362A-A / 00327).

8. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).

9. Perform operational check.

10. Check area for leaks.

11. Close engine cowling and install sump access door.



(S/N 33001 – 33107)

1. Electrical connector
2. Plug
3. Packing
4. Pressure transmitter
5. Bracket
6. Corrosion resistant steel washer
7. Aluminum alloy nut
8. Packing
9. Union
10. Tube

ICN-412-A-281000-B-97499-00025-A-001-01

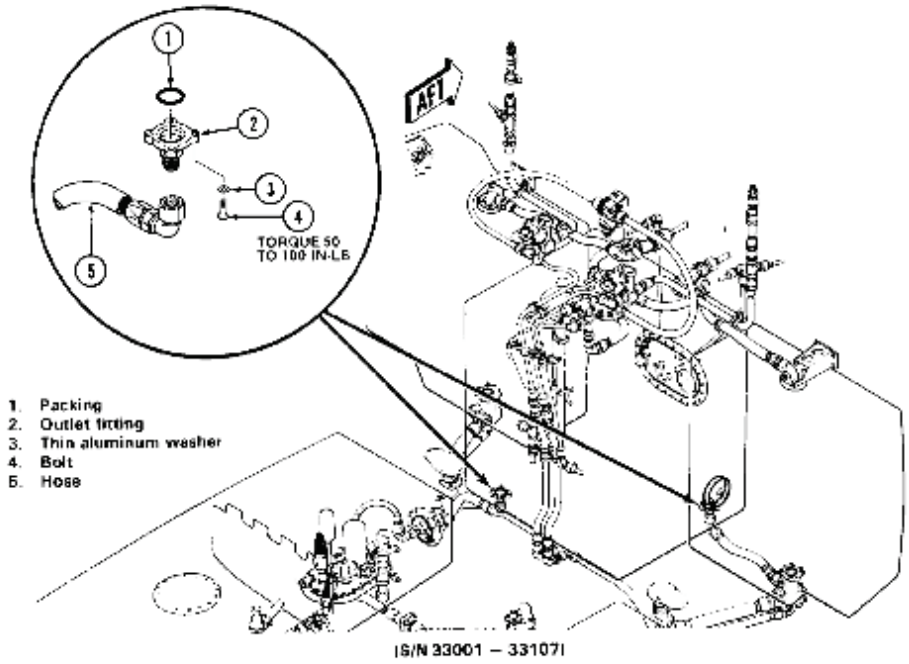
Figure 1. Fuel Pressure Transmitter (Sheet 1 of 1)

Center Aft Fuel Cell Outlet Fittings - Description

Two outlet fittings (Figure 1) are installed on bottom of center aft cell. The fittings provide attachment points for hoses which transfer fuel to forward cells.

NOTE

Maintenance procedures for left and right center aft cell outlet fittings are the same.



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Figure 1. Center Aft Cell Outlet Fitting (Sheet 1 of 1)

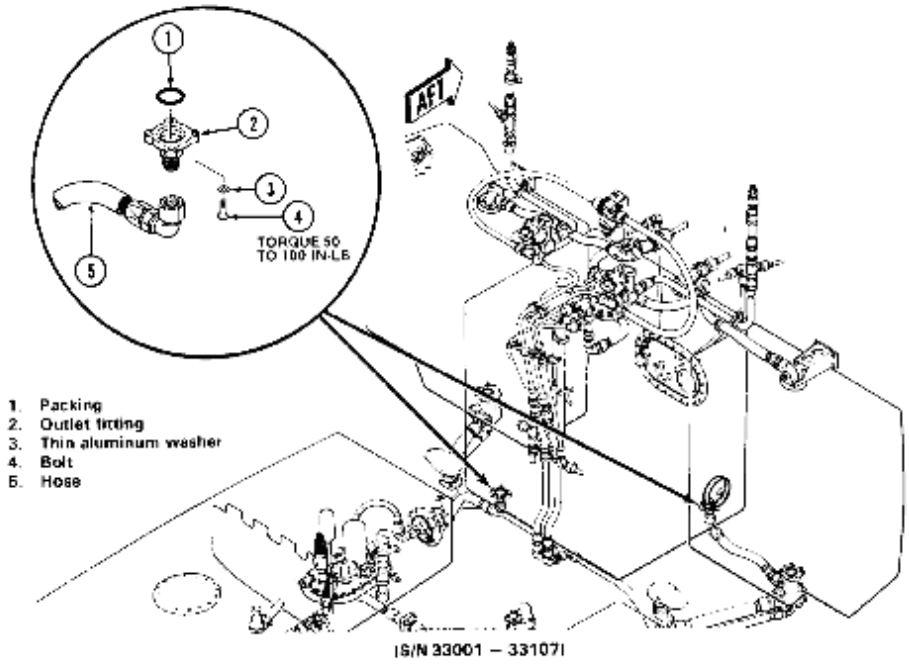
Center Aft Fuel Cell Outlet Fittings - Removal

Procedure

NOTE

The following removal procedures for left and right center aft cell outlet fittings are the same.

1. Disconnect electrical power.
2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).
3. Remove access door on underside of fuselage below outlet fitting (2, Figure 1).
4. Disconnect hose (5).
5. Remove bolts (4), outlet fitting (2), and packing (1). Discard packing.
6. Cap or plug open lines and cover cell opening.



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Figure 1. Center Aft Cell Outlet Fitting (Sheet 1 of 1)

Center Aft Fuel Cell Outlet Fittings - Installation

Preliminary requirements

Consumables, materials and expendables

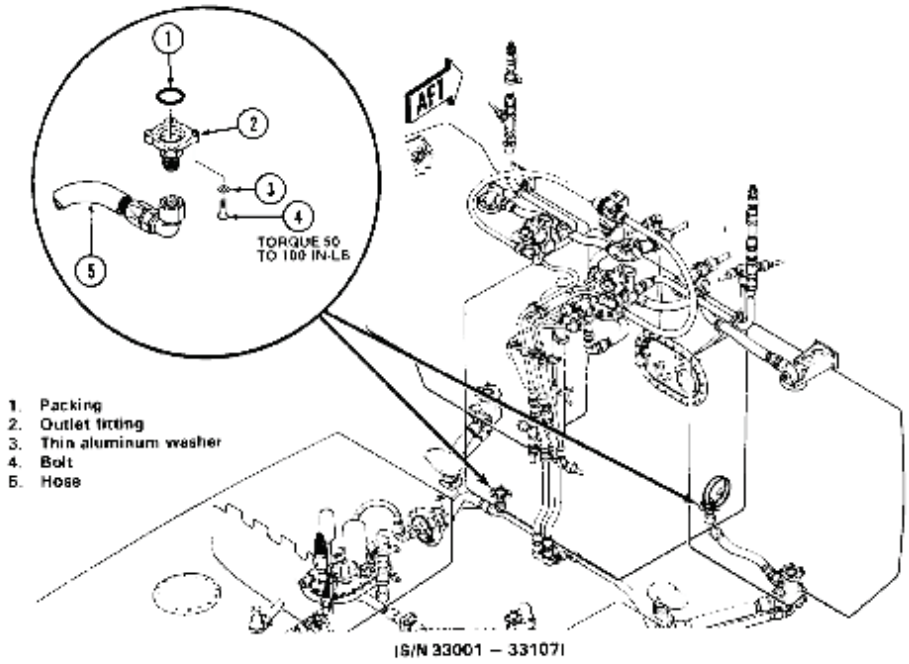
Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

NOTE

The following installation procedures for left and right center aft cell outlet fittings are the same.

1. Apply assembly fluid (C-024) to packing (1, Figure 1).
2. Remove protective caps, plugs, and cover.
3. Position packing (1) and outlet fitting (2) over cell port. Install bolts (4) with thin aluminum washers (3). Torque bolts to 50 to 100 inch-pounds (5.65 to 11.30 Nm).
4. Connect hose (5).
5. Perform fuel system pressure test (412-A-28-10-00-00A-362A-A / 00327).
6. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).
7. Check area for leaks.
8. Install access doors.

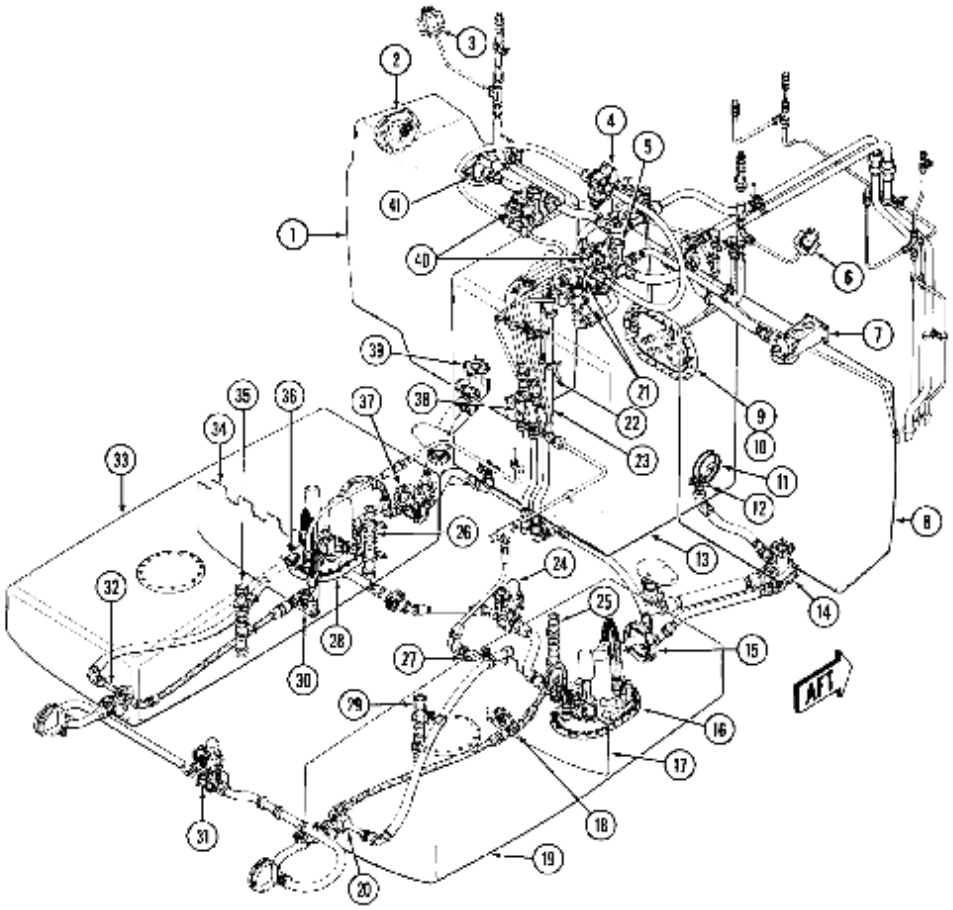


ICN-412-A-281000-B-97499-00026-A-001-01

Figure 1. Center Aft Cell Outlet Fitting (Sheet 1 of 1)

Center Aft Fuel Cell Vent Cover - Description

A vent cover located on top of center aft fuel cell (5, Figure 1) provides attachment points for four hoses in fuel vent system.



1. Right outboard aft fuel cell
2. Filler cap and adapter assembly
3. Right fuel pressure transmitter
4. System crossfeed valve
5. Center aft fuel cell vent cover
6. Left fuel pressure transmitter
7. Left outboard aft fuel cell vent fitting
8. Left outboard aft fuel cell
9. Center aft fuel cell rear cover
10. Governor bleed line check valve
11. Center aft fuel cell left interconnect fitting
12. Center aft fuel cell left outlet fitting
13. Center aft fuel cell
14. Left outboard aft fuel cell outlet fitting
15. Left forward fuel cell inlet/outlet fitting

(S/N 33001 33107)

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Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 1 of 2)

16. Left forward fuel cell fuel sump area consisting of:
    - Fuel sump assembly
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
17. Left forward fuel cell baffle
18. Left forward fuel cell flapper valve
19. Left forward fuel cell
20. Left forward fuel cell ejector pump and support
21. Siphon breaker valves
22. Center aft fuel cell right interconnect fitting
23. Aft fuel cell fuel quantity probe
24. Aft interconnect valve
25. Left forward fuel cell aft compartment fuel quantity probe
26. Right forward fuel cell aft compartment fuel quantity probe
27. Left forward fuel cell crashworthy vent valve
28. Right forward fuel cell fuel sump area consisting of:
    - Fuel sump
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
29. Left forward fuel cell forward compartment fuel quantity probe
30. Right forward fuel cell flapper valve
31. Forward interconnect valve
32. Right forward fuel cell ejector pump and support
33. Right forward fuel cell
34. Right forward fuel cell baffle
35. Right forward fuel cell forward compartment fuel quantity probe
36. Right forward fuel cell crashworthy vent valve
37. Right forward fuel cell inlet/outlet fitting
38. Center aft fuel cell right outlet fitting
39. Right outboard aft fuel cell outlet fitting
40. Shutoff valve
41. Right outboard aft fuel cell vent fitting

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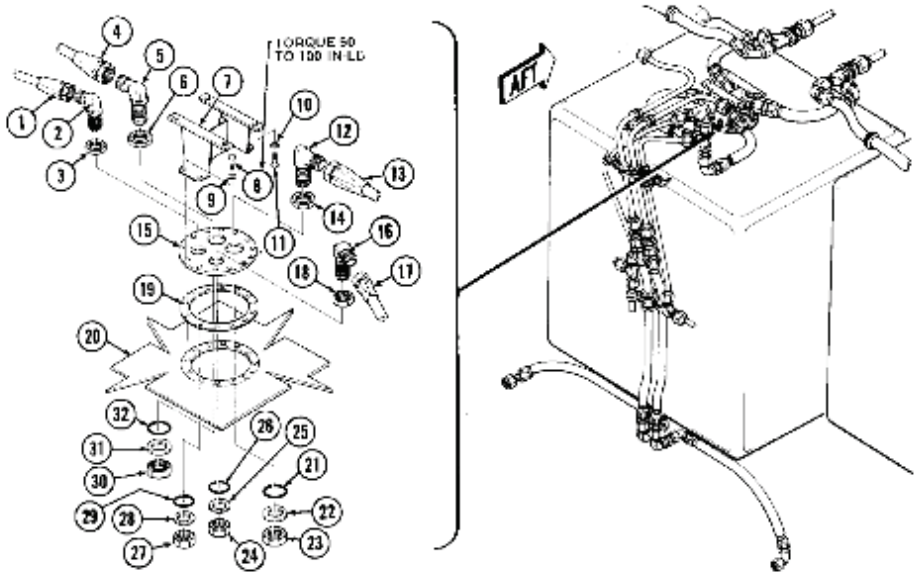
Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 2 of 2)

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Center Aft Fuel Cell Vent Cover - Removal

Procedure

1. Disconnect electrical power.
2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).
3. Remove large round access door from underside of fuselage aft of center aft fuel cell.
4. Remove center aft fuel cell rear cover (412-A-28-10-00-33A-520A-A / 00426).
5. Disconnect hoses (1, 4, 13, and 17, Figure 1).
6. Remove bolts (11) and thin aluminum washers (10).
7. Remove bolts (8) and thin aluminum washers (9).
8. Remove support (7), vent cover (15), and gasket (19). Discard gasket.
9. Remove aluminum elbow (2) by removing nut (27), thin aluminum washer (28), packing (29), and nut (3). Discard packing.
10. Remove aluminum elbow (5) by removing nut (30), thin aluminum washer (31), packing (32), and nut (6). Discard packing.
11. Remove aluminum elbow (12) by removing nut (23), thin aluminum washer (22), packing (21), and nut (14). Discard packing.
12. Remove aluminum elbow (16) by removing nut (24), thin aluminum washer (25), packing (26), and nut (18). Discard packing.
13. Cover openings with caps or plugs and covers.



1. Hose
2. Aluminum elbow
3. Nut
4. Hose
5. Aluminum elbow
6. Nut
7. Support
8. Bolt
9. Thin aluminum washer
10. Thin aluminum washer
11. Bolt
12. Aluminum elbow
13. Hose
14. Nut
15. Vent cover
16. Aluminum elbow
17. Hose
18. Nut
19. Gasket
20. Fuel cell
21. Packing
22. Thin aluminum washer
23. Nut
24. Nut
25. Thin aluminum washer
26. Packing
27. Nut
28. Thin aluminum washer
29. Packing
30. Nut
31. Thin aluminum washer
32. Packing

(S/N 33001 – 33107)

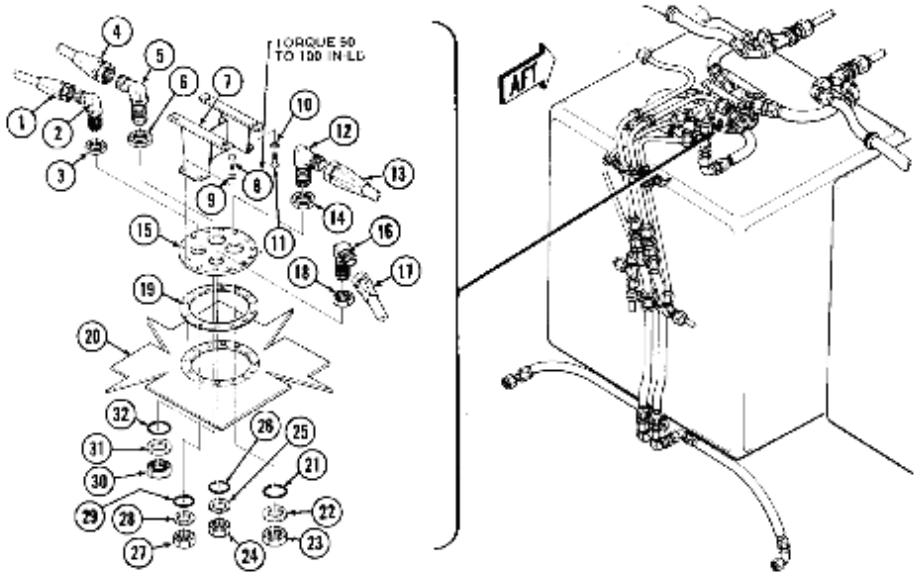
ICN-412-A-281000-B-97499-00027-A-001-01

Figure 1. Center Aft Cell Vent Cover (Sheet 1 of 1)

Center Aft Fuel Cell Vent Cover - Installation

Procedure

1. Remove protective caps, plugs, and covers.
2. Assemble aluminum elbow (2, Figure 1) to vent cover (15) with nut (3), packing (29), thin aluminum washer (28), and nut (27).
3. Assemble aluminum elbow (5) to vent cover (15) with nut (6), packing (32), thin aluminum washer (31), and nut (30).
4. Assemble aluminum elbow (12) to vent cover (15) with nut (14), packing (21), thin aluminum washer (22), and nut (23).
5. Assemble aluminum elbow (16) to vent cover (15) with nut (18), packing (26), thin aluminum washer (25), and nut (24).
6. Position gasket (19), vent cover (15), and support (7) on fuel cell (20). Install bolts (8) with thin aluminum washers (9). Torque bolts (8) to 50 to 100 inch-pounds (5.65 to 11.30 Nm).
7. Install bolts (11) with thin aluminum washers (10).
8. Connect hoses (1, 4, 13, and 17).
9. Install center aft fuel cell rear cover (412-A-28-10-00-27A-720A-A / 00407).



1. Hose
2. Aluminum elbow
3. Nut
4. Hose
5. Aluminum elbow
6. Nut
7. Support
8. Bolt
9. Thin aluminum washer
10. Thin aluminum washer
11. Bolt
12. Aluminum elbow
13. Hose
14. Nut
15. Vent cover
16. Aluminum elbow
17. Hose
18. Nut
19. Gasket
20. Fuel cell
21. Packing
22. Thin aluminum washer
23. Nut
24. Nut
25. Thin aluminum washer
26. Packing
27. Nut
28. Thin aluminum washer
29. Packing
30. Nut
31. Thin aluminum washer
32. Packing

(S/N 33001 – 33107)

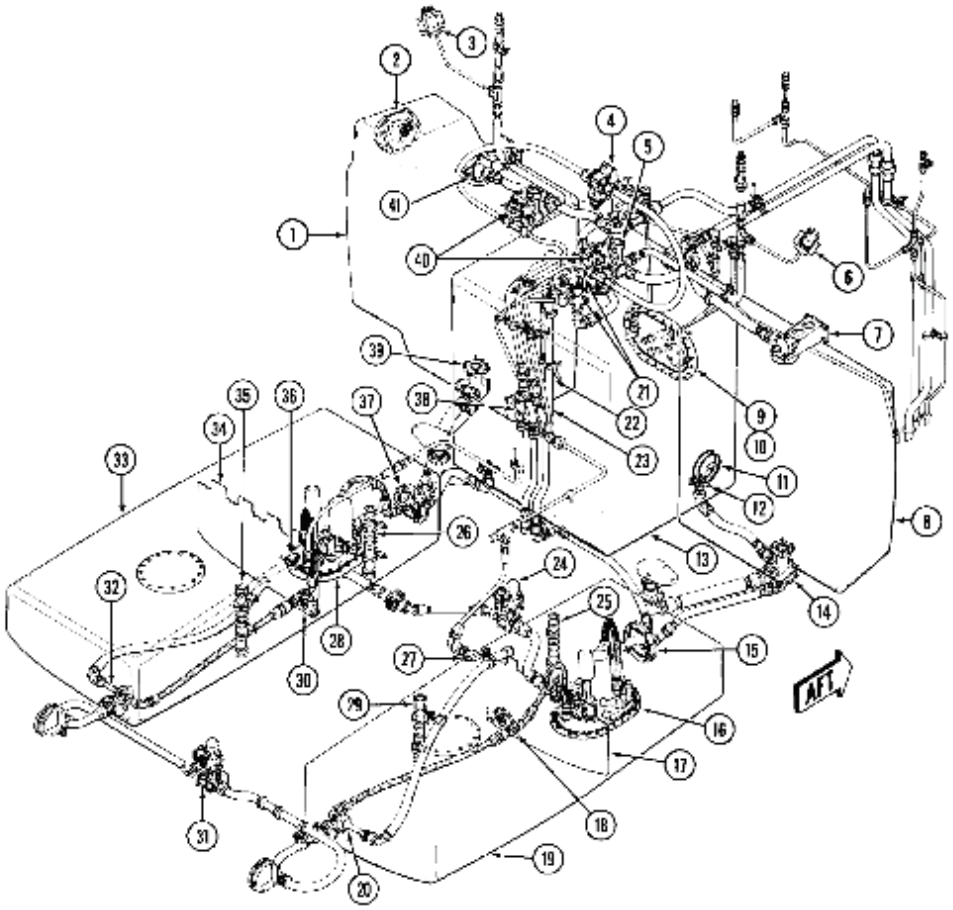
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Figure 1. Center Aft Cell Vent Cover (Sheet 1 of 1)

Aft Fuel Cell Vent Fittings - Description

NOTE

Maintenance procedures for left aft and right aft fuel cell vent fittings (7 and 41, Figure 1) are the same.



1. Right outboard aft fuel cell
2. Filler cap and adapter assembly
3. Right fuel pressure transmitter
4. System crossfeed valve
5. Center aft fuel cell vent cover
6. Left fuel pressure transmitter
7. Left outboard aft fuel cell vent fitting
8. Left outboard aft fuel cell
9. Center aft fuel cell rear cover
10. Governor bleed line check valve
11. Center aft fuel cell left interconnect fitting
12. Center aft fuel cell left outlet fitting
13. Center aft fuel cell
14. Left outboard aft fuel cell outlet fitting
15. Left forward fuel cell inlet/outlet fitting

(S/N 33001 33107)

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Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 1 of 2)

16. Left forward fuel cell fuel sump area consisting of:
    - Fuel sump assembly
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
17. Left forward fuel cell baffle
18. Left forward fuel cell flapper valve
19. Left forward fuel cell
20. Left forward fuel cell ejector pump and support
21. Siphon breaker valves
22. Center aft fuel cell right interconnect fitting
23. Aft fuel cell fuel quantity probe
24. Aft interconnect valve
25. Left forward fuel cell aft compartment fuel quantity probe
26. Right forward fuel cell aft compartment fuel quantity probe
27. Left forward fuel cell crashworthy vent valve
28. Right forward fuel cell fuel sump area consisting of:
    - Fuel sump
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
29. Left forward fuel cell forward compartment fuel quantity probe
30. Right forward fuel cell flapper valve
31. Forward interconnect valve
32. Right forward fuel cell ejector pump and support
33. Right forward fuel cell
34. Right forward fuel cell baffle
35. Right forward fuel cell forward compartment fuel quantity probe
36. Right forward fuel cell crashworthy vent valve
37. Right forward fuel cell inlet/outlet fitting
38. Center aft fuel cell right outlet fitting
39. Right outboard aft fuel cell outlet fitting
40. Shutoff valve
41. Right outboard aft fuel cell vent fitting

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Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 2 of 2)

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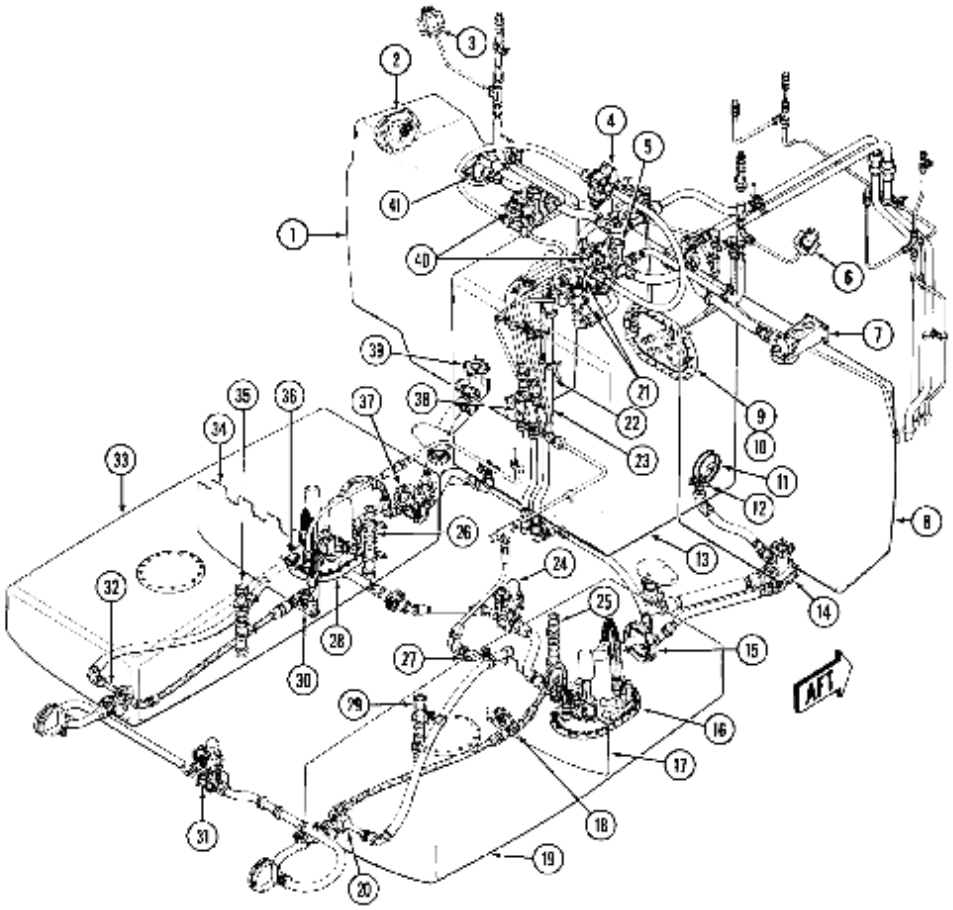
Aft Fuel Cell Vent Fittings - Removal

Procedure

NOTE

The following removal procedure for left aft and right aft fuel cell vent fittings (7 and 41, Figure 1) are the same.

1. Disconnect electrical power.
2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).
3. Remove deck panel above cell.
4. Disconnect hose (1, Figure 2).
5. Remove bolts (6) and thin aluminum washers (5).
6. Remove bolts (2) and thin aluminum washers (3). Remove bracket (4), plate (7), elbow (8), and packing (9). Discard packing.
7. Install protective cap or plug in vent line and cover cell opening.



1. Right outboard aft fuel cell
2. Filler cap and adapter assembly
3. Right fuel pressure transmitter
4. System crossfeed valve
5. Center aft fuel cell vent cover
6. Left fuel pressure transmitter
7. Left outboard aft fuel cell vent fitting
8. Left outboard aft fuel cell
9. Center aft fuel cell rear cover
10. Governor bleed line check valve
11. Center aft fuel cell left interconnect fitting
12. Center aft fuel cell left outlet fitting
13. Center aft fuel cell
14. Left outboard aft fuel cell outlet fitting
15. Left forward fuel cell inlet/outlet fitting

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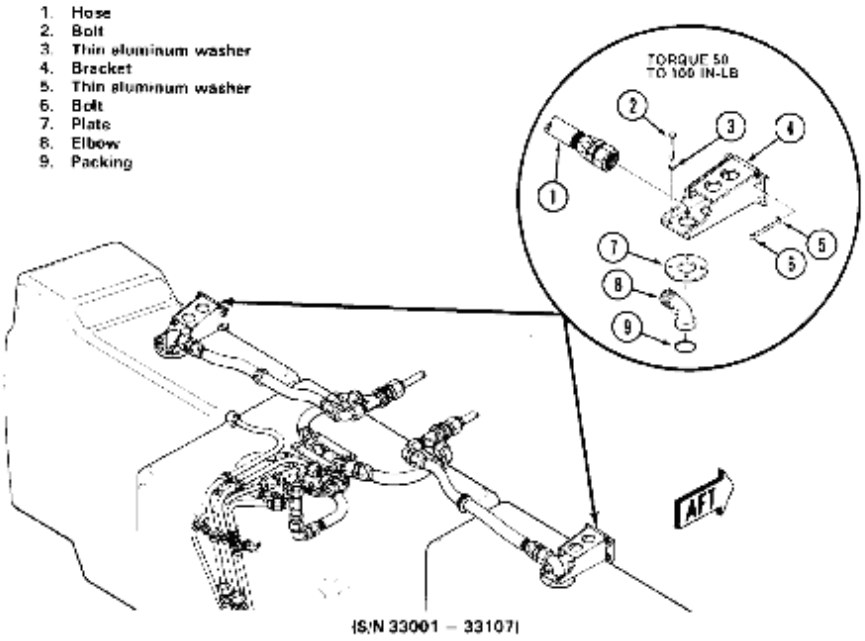
Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 1 of 2)

16. Left forward fuel cell fuel sump area consisting of:
    - Fuel sump assembly
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
17. Left forward fuel cell baffle
18. Left forward fuel cell flapper valve
19. Left forward fuel cell
20. Left forward fuel cell ejector pump and support
21. Siphon breaker valves
22. Center aft fuel cell right interconnect fitting
23. Aft fuel cell fuel quantity probe
24. Aft interconnect valve
25. Left forward fuel cell aft compartment fuel quantity probe
26. Right forward fuel cell aft compartment fuel quantity probe
27. Left forward fuel cell crashworthy vent valve
28. Right forward fuel cell fuel sump area consisting of:
    - Fuel sump
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
29. Left forward fuel cell forward compartment fuel quantity probe
30. Right forward fuel cell flapper valve
31. Forward interconnect valve
32. Right forward fuel cell ejector pump and support
33. Right forward fuel cell
34. Right forward fuel cell baffle
35. Right forward fuel cell forward compartment fuel quantity probe
36. Right forward fuel cell crashworthy vent valve
37. Right forward fuel cell inlet/outlet fitting
38. Center aft fuel cell right outlet fitting
39. Right outboard aft fuel cell outlet fitting
40. Shutoff valve
41. Right outboard aft fuel cell vent fitting

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Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 2 of 2)



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Figure 2. Left Aft and Right Aft Cell Vent Fitting (Sheet 1 of 1)

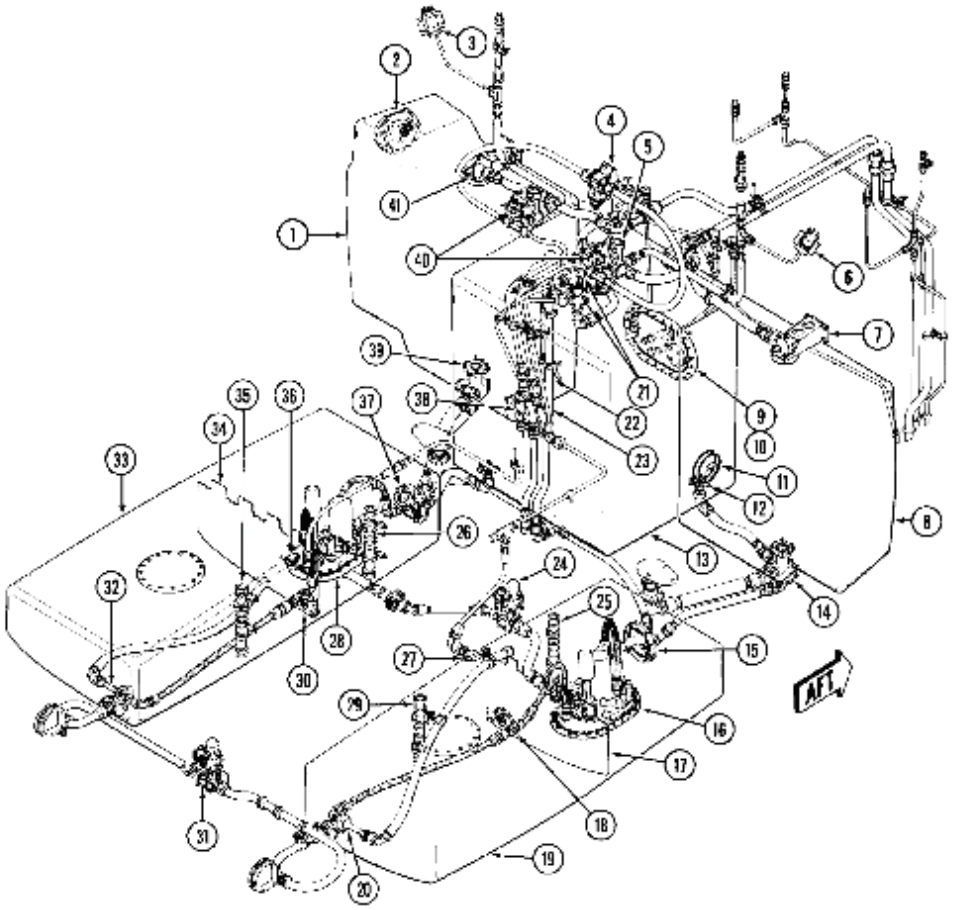
Aft Fuel Cell Vent Fittings - Installation

Procedure

NOTE

The following installation procedure for left aft and right aft fuel cell vent fittings (7 and 41, Figure 1) are the same.

1. Remove protective cap and cover.
2. Place packing (9, Figure 2) in groove around cell port.
3. Assemble elbow (8), plate (7), and bracket (4).
4. Position assembled parts on top of cell. Install bolts (2) with thin aluminum washers (3). Torque bolts to 50 to 100 inch-pounds (5.65 to 11.30 Nm).
5. Install bolts (6) with thin aluminum washers (5).
6. Connect hose (1) to elbow (8).
7. Perform fuel system pressure test (412-A-28-10-00-00A-362A-A / 00327).
8. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).
9. Check area for leaks.
10. Install sump access door and deck panel above cell.



1. Right outboard aft fuel cell
2. Filler cap and adapter assembly
3. Right fuel pressure transmitter
4. System crossfeed valve
5. Center aft fuel cell vent cover
6. Left fuel pressure transmitter
7. Left outboard aft fuel cell vent fitting
8. Left outboard aft fuel cell
9. Center aft fuel cell rear cover
10. Governor bleed line check valve
11. Center aft fuel cell left interconnect fitting
12. Center aft fuel cell left outlet fitting
13. Center aft fuel cell
14. Left outboard aft fuel cell outlet fitting
15. Left forward fuel cell inlet/outlet fitting

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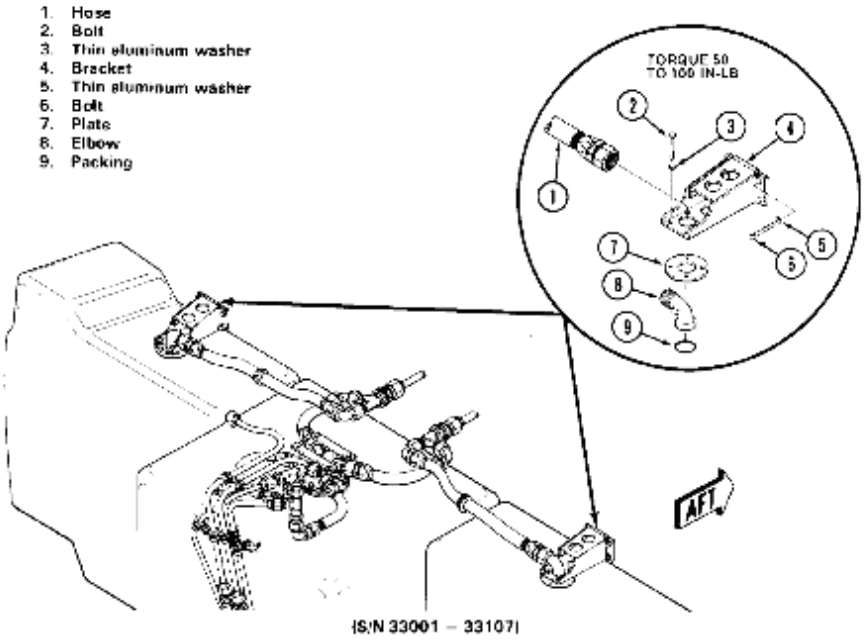
Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 1 of 2)

16. Left forward fuel cell fuel sump area consisting of:
    - Fuel sump assembly
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
17. Left forward fuel cell baffle
18. Left forward fuel cell flapper valve
19. Left forward fuel cell
20. Left forward fuel cell ejector pump and support
21. Siphon breaker valves
22. Center aft fuel cell right interconnect fitting
23. Aft fuel cell fuel quantity probe
24. Aft interconnect valve
25. Left forward fuel cell aft compartment fuel quantity probe
26. Right forward fuel cell aft compartment fuel quantity probe
27. Left forward fuel cell crashworthy vent valve
28. Right forward fuel cell fuel sump area consisting of:
    - Fuel sump
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
29. Left forward fuel cell forward compartment fuel quantity probe
30. Right forward fuel cell flapper valve
31. Forward interconnect valve
32. Right forward fuel cell ejector pump and support
33. Right forward fuel cell
34. Right forward fuel cell baffle
35. Right forward fuel cell forward compartment fuel quantity probe
36. Right forward fuel cell crashworthy vent valve
37. Right forward fuel cell inlet/outlet fitting
38. Center aft fuel cell right outlet fitting
39. Right outboard aft fuel cell outlet fitting
40. Shutoff valve
41. Right outboard aft fuel cell vent fitting

(S/N 33001 33107)

ICN-412-A-281000-B-97499-00002-A-001-01

Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 2 of 2)



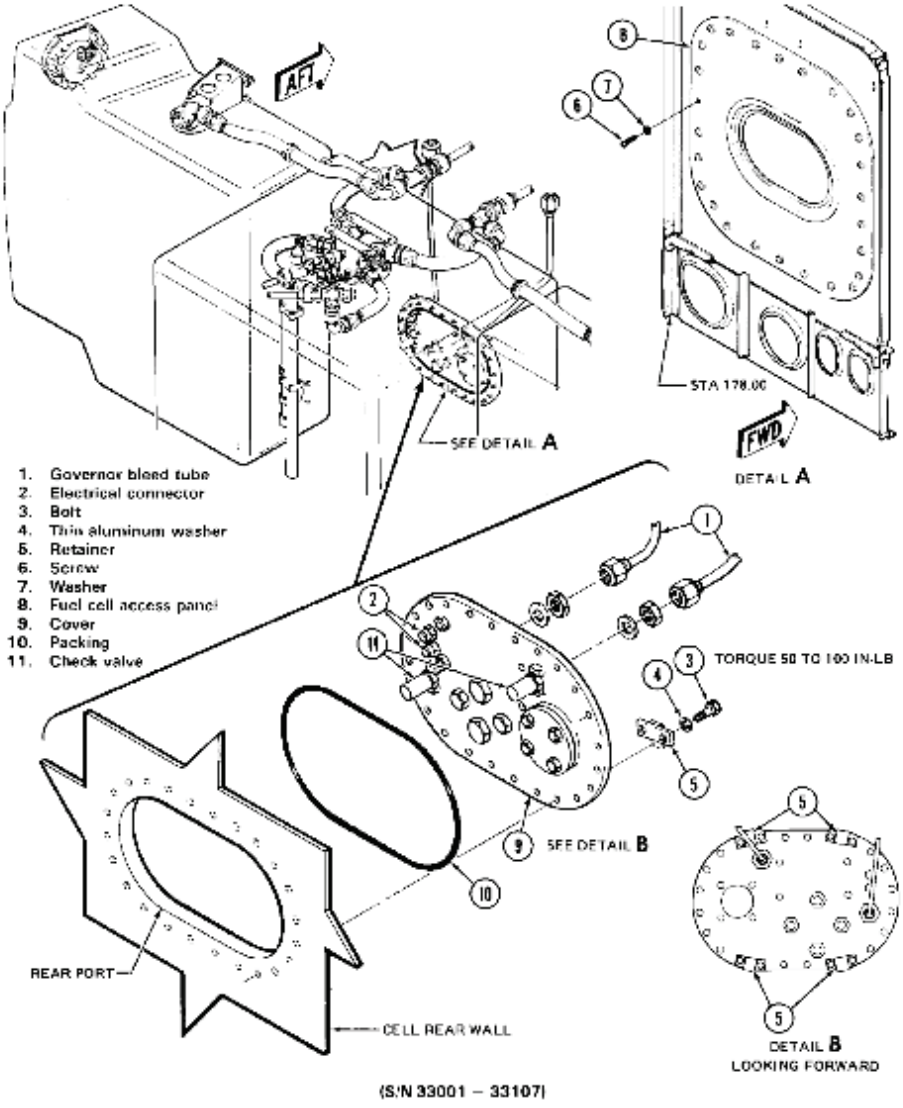
ICN-412-A-281000-B-97499-00028-A-001-01

Figure 2. Left Aft and Right Aft Cell Vent Fitting (Sheet 1 of 1)

Center Aft Fuel Cell Rear Cover - Removal

Procedure

1. Disconnect electrical power.
2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).
3. Remove large round access door from bottom of fuselage aft of center aft cell.
4. Remove governor bleed lines (tubes) (1, Figure 1).
5. Disconnect electrical wiring from electrical connectors (2).
6. Remove bolts (3) and washers (4) securing retainers (5).
7. Remove screws (6) and washers (7) and remove fuel cell access panel (8).
8. Remove remaining bolts (3) and washers (4). Separate cover (9) from cell.
9. Reach through cell rear port and disconnect fuel quantity probe wires from electrical connectors (2) on inside of cover (9).
10. Remove cover (9) and packing (10). Discard packing.
11. Cap governor bleed tube fittings on underside of engine deck. Cap removed tubing and install protective cover over cell opening.



ICN-412-A-281000-B-97499-00029-A-001-01

Figure 1. Center Aft Cell Rear Cover (Sheet 1 of 1)

Center Aft Fuel Cell Rear Cover - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

1. Remove protective cover from cell opening.

2. Coat packing (10, Figure 1) with assembly fluid (C-024) and place packing in groove around cell rear port opening.

3. Hold cover (9) close to cell rear port. Connect wires from fuel quantity probe to electrical connectors (2) on inside of cover.

4. Position cover (9) over cell rear port and install only those bolts (3) and washers (4) not used with the retainers (5). (Refer to Detail B for locations of hardware to be omitted at this time.)

5. Position fuel cell access panel (8) on bulkhead and install screws (6) and washers (7).

6. Position cover (9) against fuel cell access panel (8) and install retainers (5) with remaining bolts (3) and washers (4). Torque all bolts (3) to 50 to 100 inch-pounds (5.65 to 11.30 Nm).

7. Connect fuel quantity probe wires to electrical connectors (2).

8. Remove protective caps and plugs. Connect governor bleed tubes (1) between check valves (11) and fittings on underside of engine deck.

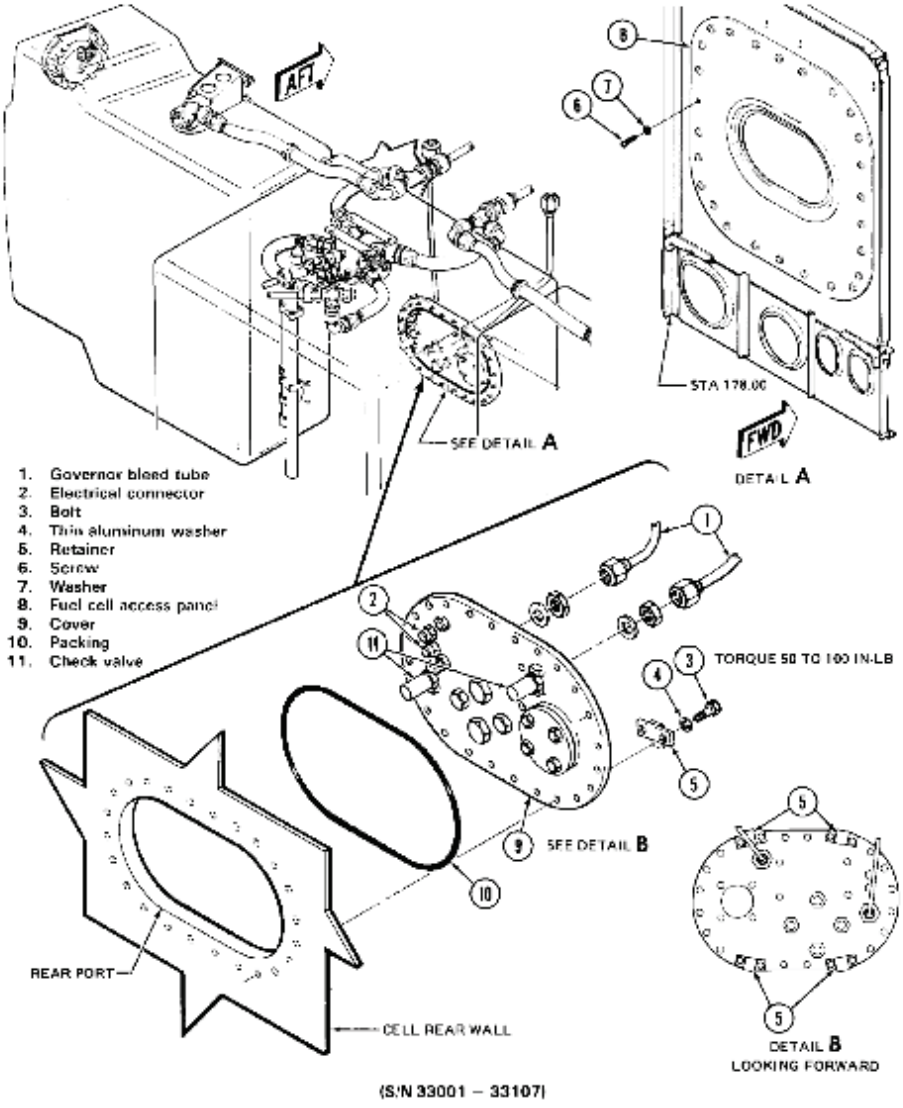
9. Perform fuel system pressure test (412-A-28-10-00-00A-362A-A / 00327).

10. Perform operational check of fuel quantity indicating system before servicing system (412-A-96-11-00-10A-273A-A / 01973).

11. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).

12. Check area for leaks.

13. Install access doors.



ICN-412-A-281000-B-97499-00029-A-001-01

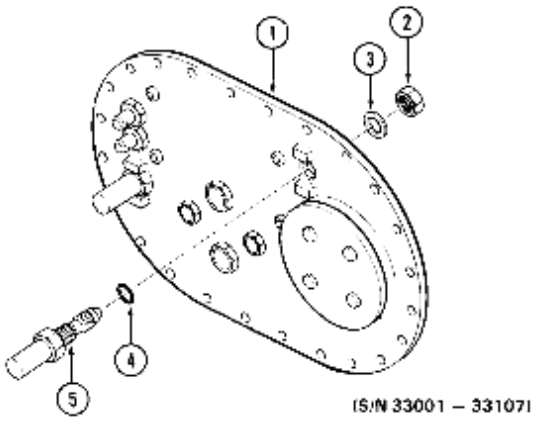
Figure 1. Center Aft Cell Rear Cover (Sheet 1 of 1)

Governor Bleed Line Check Valves - Description

Two directional flow check valves (Figure 1) are installed in the center aft cell rear cover. The check valves are connected to governor bleed lines which return excess fuel from each power section fuel control to the center aft cell.

NOTE

Maintenance procedures are the same for both governor bleed line check valves.



1. Cover
2. Nut
3. Thin aluminum washer
4. Packing
5. Check valve

ICN-412-A-281000-B-97499-00030-A-001-01

Figure 1. Governor Bleed Line Check Valve (Sheet 1 of 1)

Governor Bleed Line Check Valves - Removal

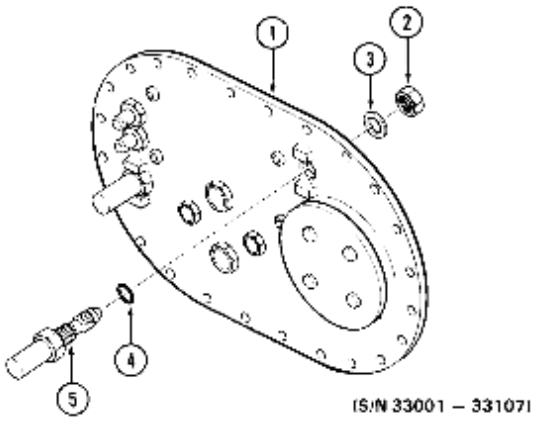
Procedure

NOTE

The following removal procedures are the same for both governor bleed line check valves.

1. Remove center aft cell rear cover (412-A-28-10-00-27A-520A-A / 00406).

2. Remove nut (2, Figure 1), thin aluminum washer (3), packing (4), and check valve (5). Discard packing.



1. Cover
2. Nut
3. Thin aluminum washer
4. Packing
5. Check valve

ICN-412-A-281000-B-97499-00030-A-001-01

Figure 1. Governor Bleed Line Check Valve (Sheet 1 of 1)

Governor Bleed Line Check Valves - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

NOTE

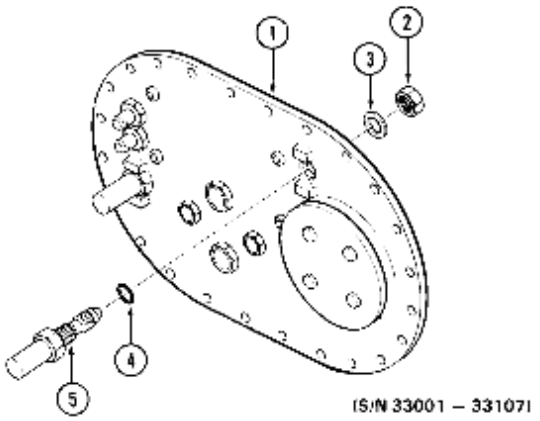
The following installation procedures are the same for both governor bleed line check valves.

1. Remove protective cover and cap from cell opening and tube.

2. Apply assembly fluid (C-024) to packing (4, Figure 1).

3. Install check valve (5), packing (4), thin aluminum washer (3), and nut (2).

4. Install center aft cell rear cover (412-A-28-10-00-27A-720A-A / 00407).



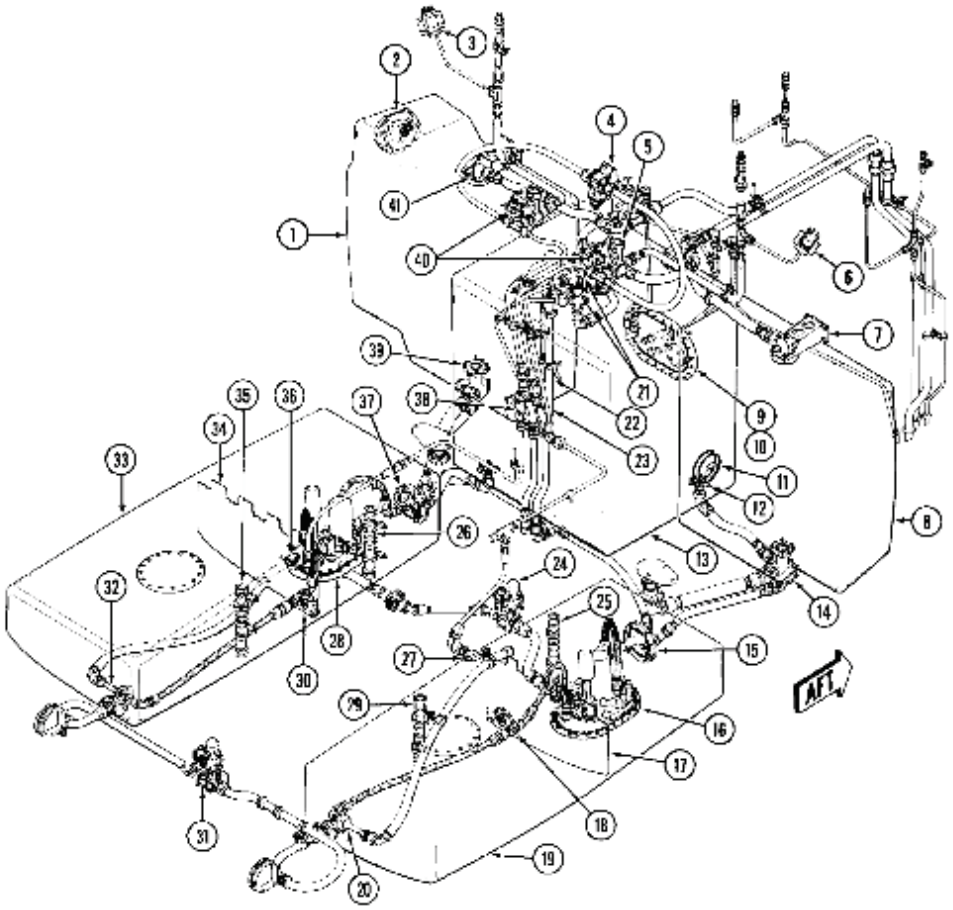
1. Cover
2. Nut
3. Thin aluminum washer
4. Packing
5. Check valve

ICN-412-A-281000-B-97499-00030-A-001-01

Figure 1. Governor Bleed Line Check Valve (Sheet 1 of 1)

Aft Cell Outlet Fittings - Description

An outlet fitting (14 and 39, Figure 1) is installed on the bottom of each outboard (left and right) aft cell. The fittings provide hose attachment points for the transfer of fuel from aft cells to forward cells.



1. Right outboard aft fuel cell
2. Filler cap and adapter assembly
3. Right fuel pressure transmitter
4. System crossfeed valve
5. Center aft fuel cell vent cover
6. Left fuel pressure transmitter
7. Left outboard aft fuel cell vent fitting
8. Left outboard aft fuel cell
9. Center aft fuel cell rear cover
10. Governor bleed line check valve
11. Center aft fuel cell left interconnect fitting
12. Center aft fuel cell left outlet fitting
13. Center aft fuel cell
14. Left outboard aft fuel cell outlet fitting
15. Left forward fuel cell inlet/outlet fitting

(S/N 33001 33107)

ICN-412-A-281000-B-97499-00001-A-001-01

Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 1 of 2)

16. Left forward fuel cell fuel sump area consisting of:
    - Fuel sump assembly
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
17. Left forward fuel cell baffle
18. Left forward fuel cell flapper valve
19. Left forward fuel cell
20. Left forward fuel cell ejector pump and support
21. Siphon breaker valves
22. Center aft fuel cell right interconnect fitting
23. Aft fuel cell fuel quantity probe
24. Aft interconnect valve
25. Left forward fuel cell aft compartment fuel quantity probe
26. Right forward fuel cell aft compartment fuel quantity probe
27. Left forward fuel cell crashworthy vent valve
28. Right forward fuel cell fuel sump area consisting of:
    - Fuel sump
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
29. Left forward fuel cell forward compartment fuel quantity probe
30. Right forward fuel cell flapper valve
31. Forward interconnect valve
32. Right forward fuel cell ejector pump and support
33. Right forward fuel cell
34. Right forward fuel cell baffle
35. Right forward fuel cell forward compartment fuel quantity probe
36. Right forward fuel cell crashworthy vent valve
37. Right forward fuel cell inlet/outlet fitting
38. Center aft fuel cell right outlet fitting
39. Right outboard aft fuel cell outlet fitting
40. Shutoff valve
41. Right outboard aft fuel cell vent fitting

(S/N 33001 33107)

ICN-412-A-281000-B-97499-00002-A-001-01

Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 2 of 2)

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Aft Cell Outlet Fittings - Removal

Procedure

1. Disconnect electrical power.

2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).

3. Remove lower access door.

4. Remove outlet fitting (4, Figure 1) from right aft cell as follows:

4.1. Remove bolts (13), thin aluminum washers (12), hose (14), and packing (11). Discard packing.

4.2. Remove bolts (15), thin aluminum washers (16), hose (17), and packing (18). Discard packing.

4.3. Remove bolts (10), thin aluminum washers (9), outlet fitting (4), and packing (3). Discard packing.

4.4. Remove plug (1) and packing (2). Discard packing.

4.5. Remove bolts (8), thin aluminum washer (7), cover (6), and packing (5). Discard packing.

5. Remove outlet fitting (23) from left aft cell as follows:

5.1. Remove bolts (31), thin aluminum washers (30), hose (32), and packing (33). Discard packing.

5.2. Disconnect hose (19).

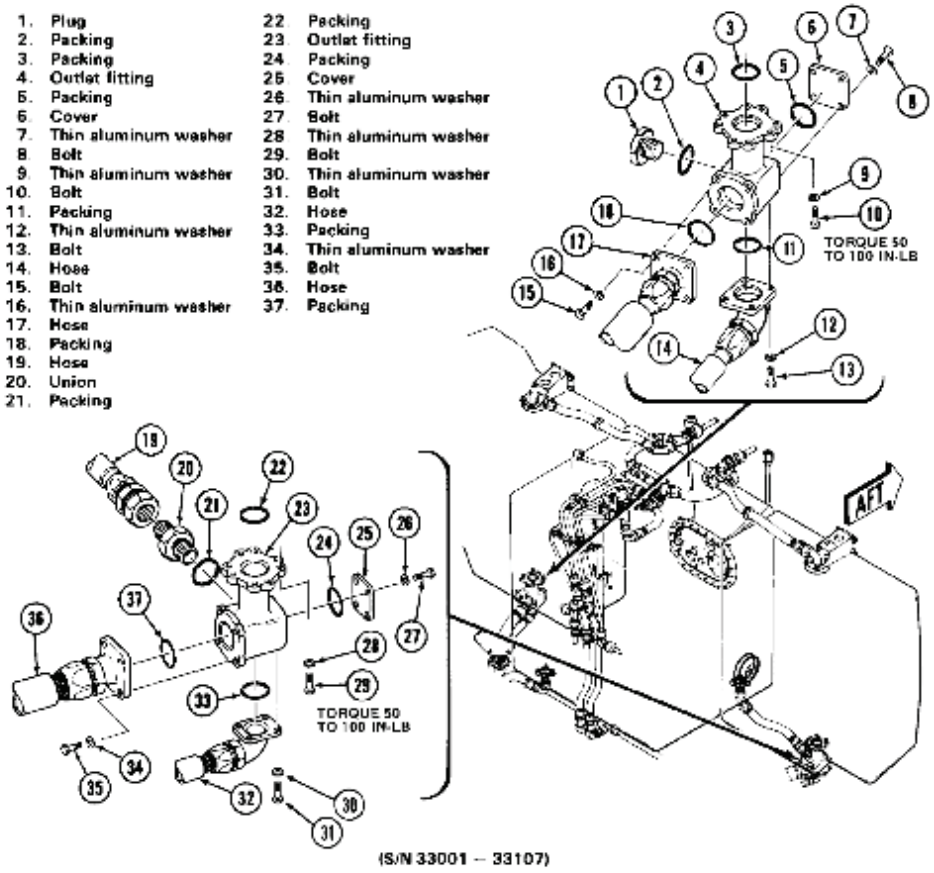
5.3. Remove bolts (35), thin aluminum washers (34), hose (36), and packing (37). Discard packing.

5.4. Remove bolts (29), thin aluminum washers (28), outlet fitting (23), and packing (22). Discard packing.

5.5. Remove union (20) and packing (21). Discard packing.

5.6. Remove bolts (27), thin aluminum washers (26), cover (25), and packing (24). Discard packing.

6. Cover cell opening and cap or plug open lines.



ICN-412-A-281000-B-97499-00031-A-001-01

Figure 1. Left Aft and Right Aft Cell Outlet Fitting (Sheet 1 of 1)

Aft Cell Outlet Fittings - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

1. Install outlet fitting (4, Figure 1) on right aft cell as follows:

1.1. Remove cover and protective caps.

1.2. Coat packing (5) with assembly fluid (C-024). Install cover (6), packing (5), bolts (8), and thin aluminum washers (7).

1.3. Apply assembly fluid (C-024) to packing (2).

1.4. Install plug (1) and packing (2).

1.5. Coat packing (3) with assembly fluid (C-024). Position packing (3) and outlet fitting (4) on bottom of fuel cell. Install bolts (10) and thin aluminum washers (9). Torque bolts (10) to 50 to 100 inch-pounds (5.65 to 11.30 Nm).

1.6. Coat packing (18) with assembly fluid (C-024). Install packing (11) and hose (14) with bolts (13) and thin aluminum washers (12).

1.7. Coat packing (11) with assembly fluid (C-024). Install packing (11) and hose (14) with bolts (13) and thin aluminum washers (12).

2. Install outlet fitting (23) on left aft cell as follows:

2.1. Remove cover and protective caps.

2.2. Apply assembly fluid (C-024) to packing (21).

2.3. Install union (20) with packing (21).

2.4. Coat packing (24) with assembly fluid (C-024).

2.5. Install cover (25), packing (24), bolts (27), and thin aluminum washers (26).

2.6. Coat packing (22) with assembly fluid (C-024).

2.7. Position packing (22) and outlet fitting (23) on bottom of fuel cell. Install bolts (29) and thin aluminum washers (28). Torque bolts (29) to 50 to 100 inch-pounds (5.65 to 11.30 Nm).

2.8. Coat packing (37) with assembly fluid (C-024).

2.9. Install packing (37) and hose (36) with bolts (35) and thin aluminum washers (34).

2.10. Connect hose (19).

2.11. Coat packing (33) with assembly fluid (C-024).

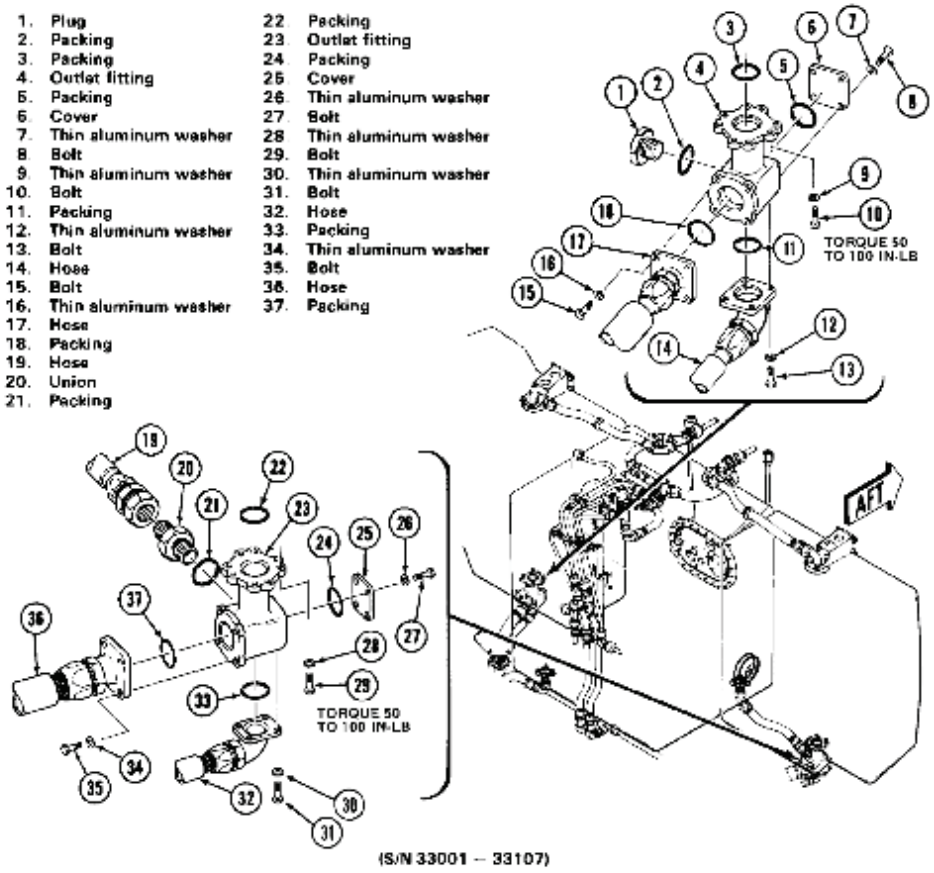
2.12. Install packing (33) and hose (32) with bolts (31) and thin aluminum washers (30).

3. Pressure test fuel system (412-A-28-10-00-00A-362A-A / 00327).

4. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).

5. Check area for leaks.

6. Install access doors.

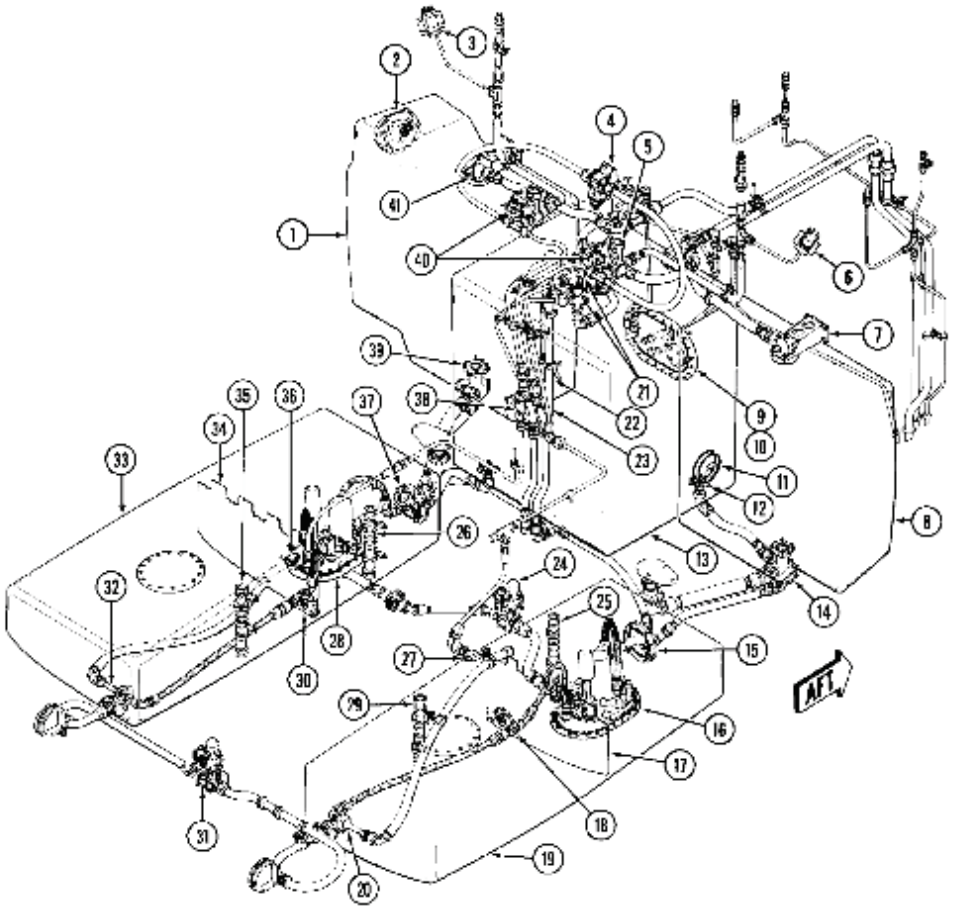


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Figure 1. Left Aft and Right Aft Cell Outlet Fitting (Sheet 1 of 1)

Filler Cap and Adapter Assembly - Description

The filler cap and adapter assembly (2, Figure 1) is located on the right aft cell. A spring-loaded plate inside of the assembly serves as a seal in the event that the cap opens accidentally.



1. Right outboard aft fuel cell
2. Filler cap and adapter assembly
3. Right fuel pressure transmitter
4. System crossfeed valve
5. Center aft fuel cell vent cover
6. Left fuel pressure transmitter
7. Left outboard aft fuel cell vent fitting
8. Left outboard aft fuel cell
9. Center aft fuel cell rear cover
10. Governor bleed line check valve
11. Center aft fuel cell left interconnect fitting
12. Center aft fuel cell left outlet fitting
13. Center aft fuel cell
14. Left outboard aft fuel cell outlet fitting
15. Left forward fuel cell inlet/outlet fitting

(S/N 33001 33107)

ICN-412-A-281000-B-97499-00001-A-001-01

Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 1 of 2)

16. Left forward fuel cell fuel sump area consisting of:
    - Fuel sump assembly
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
17. Left forward fuel cell baffle
18. Left forward fuel cell flapper valve
19. Left forward fuel cell
20. Left forward fuel cell ejector pump and support
21. Siphon breaker valves
22. Center aft fuel cell right interconnect fitting
23. Aft fuel cell fuel quantity probe
24. Aft interconnect valve
25. Left forward fuel cell aft compartment fuel quantity probe
26. Right forward fuel cell aft compartment fuel quantity probe
27. Left forward fuel cell crashworthy vent valve
28. Right forward fuel cell fuel sump area consisting of:
    - Fuel sump
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
29. Left forward fuel cell forward compartment fuel quantity probe
30. Right forward fuel cell flapper valve
31. Forward interconnect valve
32. Right forward fuel cell ejector pump and support
33. Right forward fuel cell
34. Right forward fuel cell baffle
35. Right forward fuel cell forward compartment fuel quantity probe
36. Right forward fuel cell crashworthy vent valve
37. Right forward fuel cell inlet/outlet fitting
38. Center aft fuel cell right outlet fitting
39. Right outboard aft fuel cell outlet fitting
40. Shutoff valve
41. Right outboard aft fuel cell vent fitting

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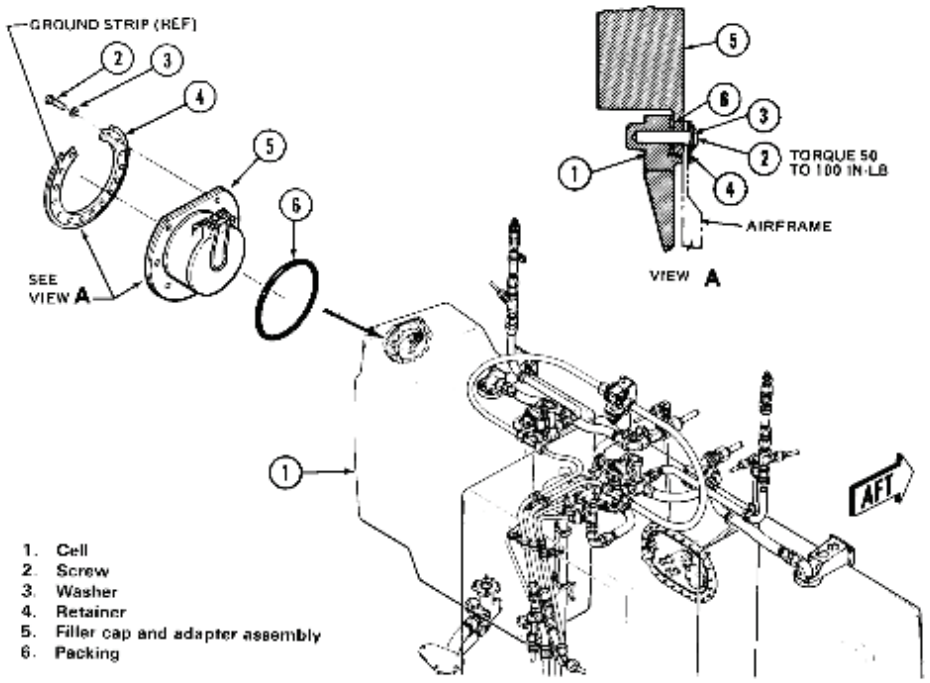
Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 2 of 2)

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Filler Cap and Adapter Assembly - Removal

Procedure

1. Disconnect electrical power.
2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).
3. Remove screws (2, Figure 1), washers (3), retainer (4), filler cap and adapter assembly (5), and packing (6). Discard packing.
4. Install protective cover over cell opening if not immediately replacing filler cap and adapter assembly (5).



(S/N 33001 – 33107)

ICN-412-A-281000-B-97499-00032-A-001-01

Figure 1. Filler Cap and Adapter Assembly (Helicopters not Modified by T.B. 412-89-71) (Sheet 1 of 1)

Filler Cap and Adapter Assembly - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

1. Remove protective cover.

2. Coat packing (6, Figure 1) with assembly fluid (C-024). Place packing in groove around cell opening.

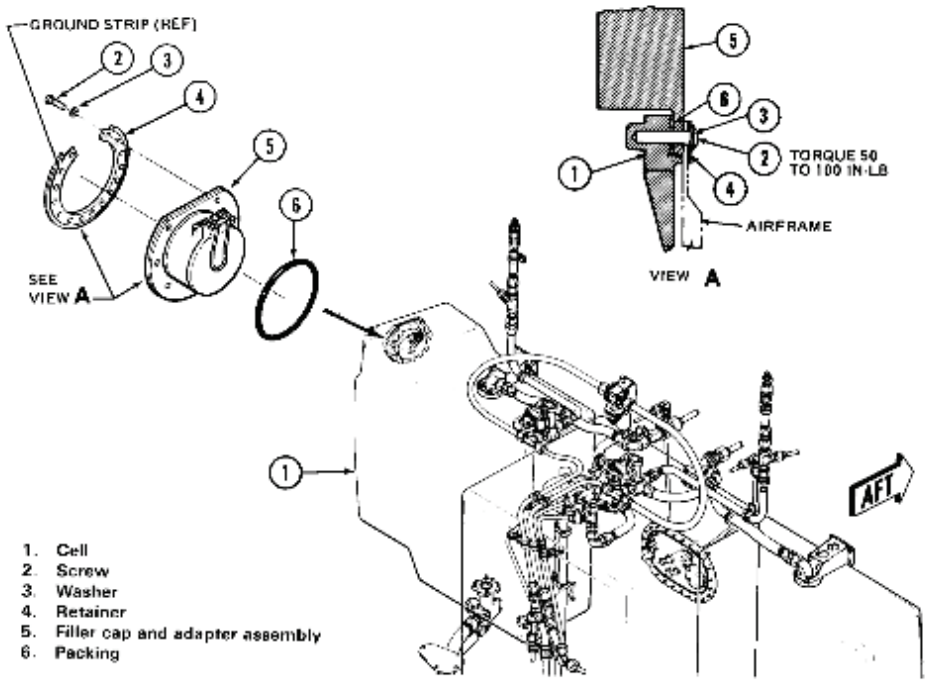
3. Taking care not to disturb packing (6), assemble filler cap and adapter assembly (5) and retainer (4) to cell (1), as shown in View A.

4. Install screws (2) with washers (3), installing ground strip at hole location indicated in Figure 1. Torque screws (2) 50 to 100 inch-pounds (5.65 to 11.30 Nm).

5. Pressure test fuel system (412-A-28-10-00-00A-362A-A / 00327).

6. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).

7. Check area for leaks.



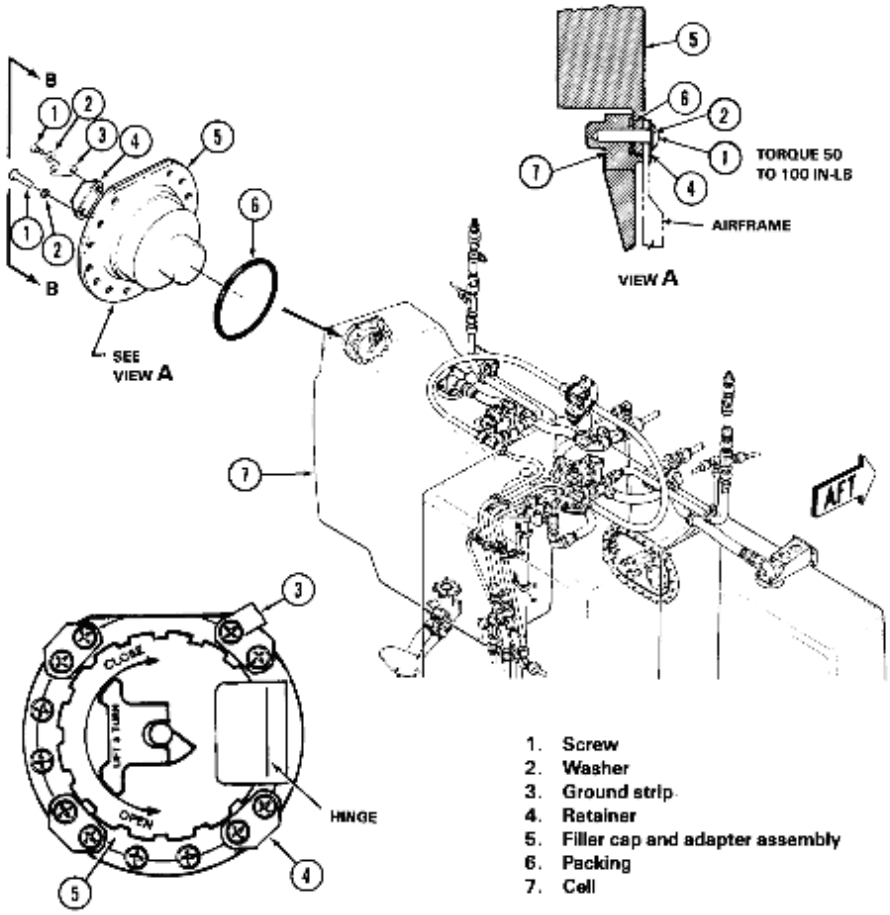
(S/N 33001 – 33107)

ICN-412-A-281000-B-97499-00032-A-001-01

Figure 1. Filler Cap and Adapter Assembly (Helicopters not Modified by T.B. 412-89-71) (Sheet 1 of 1)

Improved Filler Cap and Adapter Assembly - Description

The improved filler cap and adapter assembly (Figure 1) is located on the right aft cell. This cap incorporates a spring-loaded closure which retains the cap and minimizes fuel loss in the event that the cap is not secured.



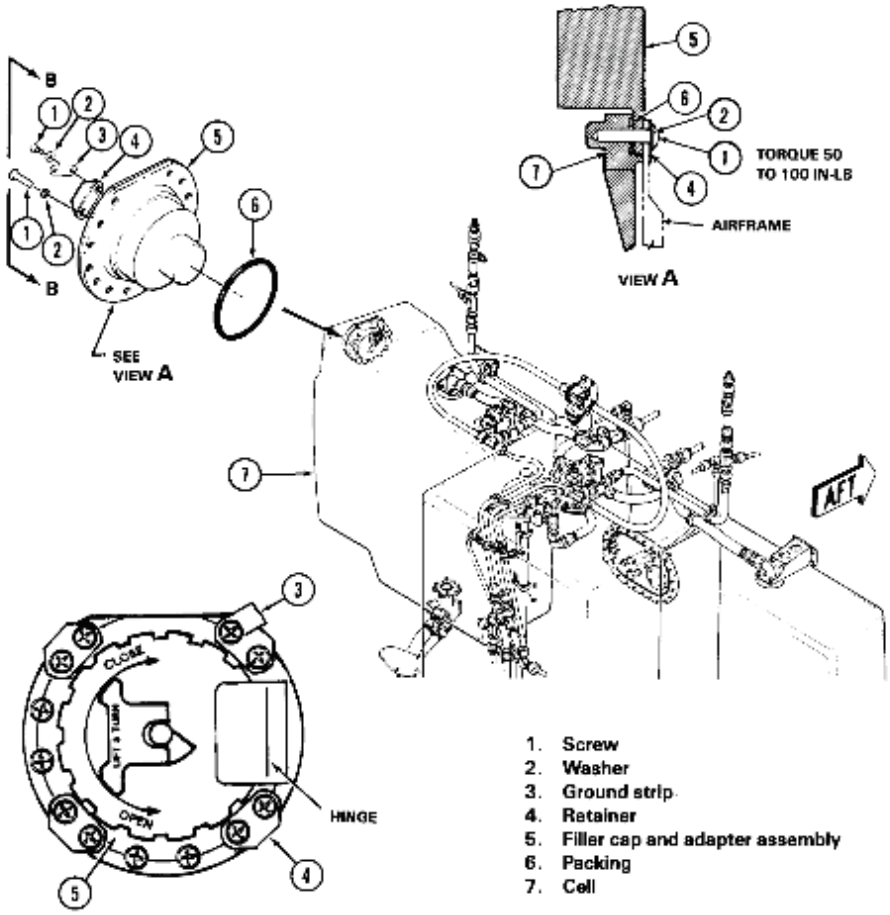
ICN-412-A-281000-B-97499-00033-A-001-01

Figure 1. Improved Filler Cap and Adapter Assembly (Sheet 1 of 1)

Improved Filler Cap and Adapter Assembly - Removal

Procedure

1. Disconnect electrical power.
2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).
3. Remove screws (1, Figure 1), washers (2), four retainers (4), filler cap and adapter assembly (5), and packing (6). Discard packing.
4. Install protective cover over cell opening if not immediately replacing filler cap and adapter assembly (5).



ICN-412-A-281000-B-97499-00033-A-001-01

Figure 1. Improved Filler Cap and Adapter Assembly (Sheet 1 of 1)

Improved Filler Cap and Adapter Assembly - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

1. Remove protective cover. Coat packing (6, Figure 1) with assembly fluid (C-024). Place packing in groove around cell opening.

2. Taking care not to disturb packing (6), install cap and adapter assembly (5) with retainers (4), washers (2), and screws (1). Locate retainers at the 2, 4, 8, and 10 o'clock position, as shown in Detail B.

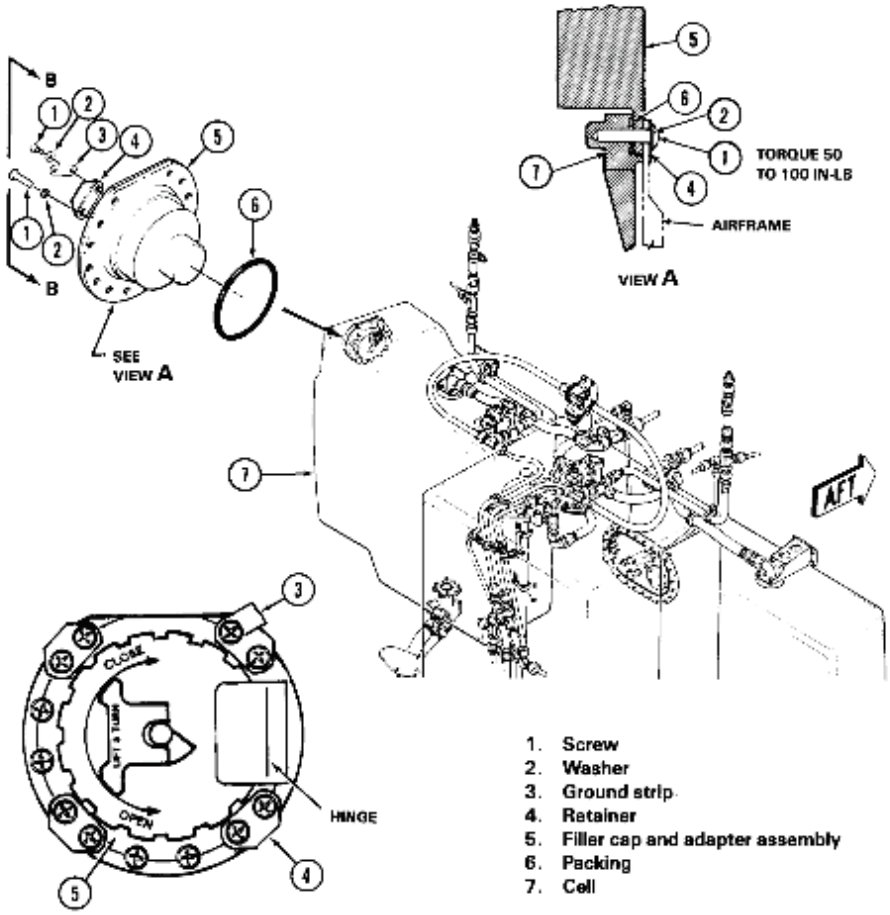
3. Connect ground strip (3) at the one o'clock position.

4. Torque all screws (1) to 50 to 100 inch-pounds (5.65 to 11.30 Nm).

5. Pressure test fuel system (412-A-28-10-00-00A-362A-A / 00327).

6. Service fuel system (412-A-12-00-00-00A-009A-A / 00125).

7. Check area for leaks.



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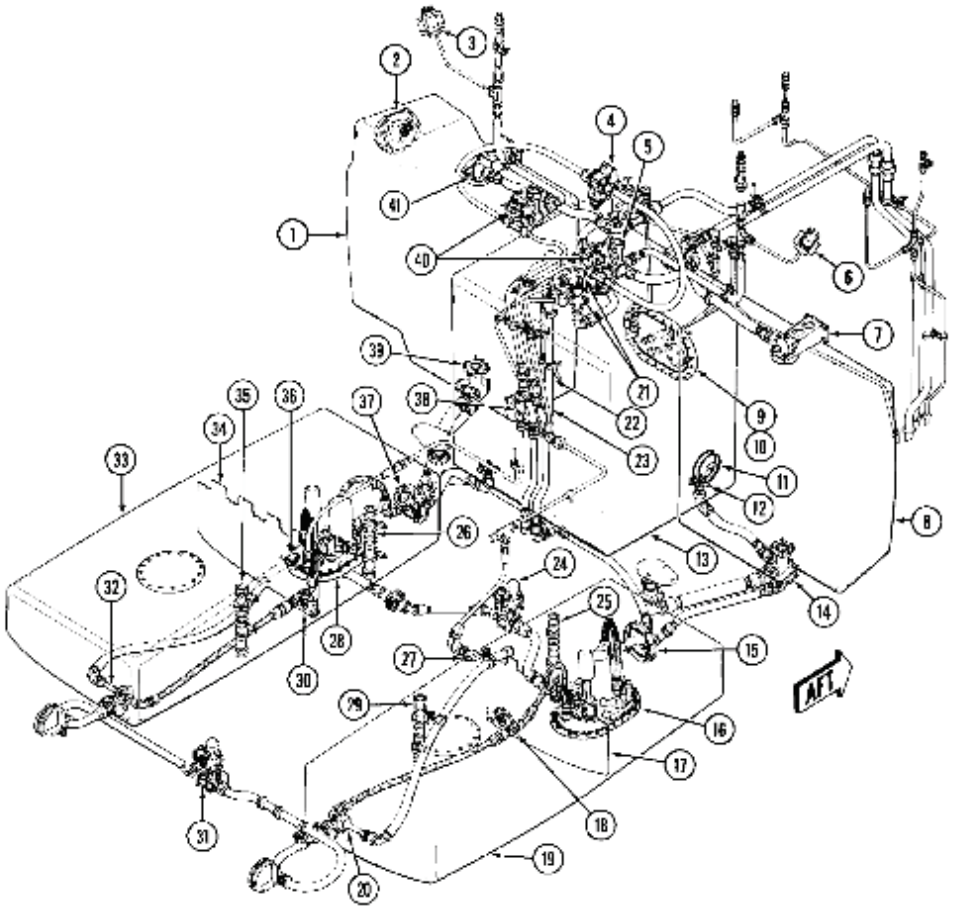
Figure 1. Improved Filler Cap and Adapter Assembly (Sheet 1 of 1)

Aft Cells - Description

NOTE

Maintenance procedures for left aft and right aft cells are the same.

The left aft and right aft cells (1 and 8, Figure 1) are outboard of main structural beams between FS 166 and 178. The cells have interconnection ports on the bottom and lower inboard sides and a vent port on top. The right aft cell has a port for the filler cap and adapter assembly.



1. Right outboard aft fuel cell
2. Filler cap and adapter assembly
3. Right fuel pressure transmitter
4. System crossfeed valve
5. Center aft fuel cell vent cover
6. Left fuel pressure transmitter
7. Left outboard aft fuel cell vent fitting
8. Left outboard aft fuel cell
9. Center aft fuel cell rear cover
10. Governor bleed line check valve
11. Center aft fuel cell left interconnect fitting
12. Center aft fuel cell left outlet fitting
13. Center aft fuel cell
14. Left outboard aft fuel cell outlet fitting
15. Left forward fuel cell inlet/outlet fitting

(S/N 33001 33107)

ICN-412-A-281000-B-97499-00001-A-001-01

Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 1 of 2)

16. Left forward fuel cell fuel sump area consisting of:
    - Fuel sump assembly
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
17. Left forward fuel cell baffle
18. Left forward fuel cell flapper valve
19. Left forward fuel cell
20. Left forward fuel cell ejector pump and support
21. Siphon breaker valves
22. Center aft fuel cell right interconnect fitting
23. Aft fuel cell fuel quantity probe
24. Aft interconnect valve
25. Left forward fuel cell aft compartment fuel quantity probe
26. Right forward fuel cell aft compartment fuel quantity probe
27. Left forward fuel cell crashworthy vent valve
28. Right forward fuel cell fuel sump area consisting of:
    - Fuel sump
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
29. Left forward fuel cell forward compartment fuel quantity probe
30. Right forward fuel cell flapper valve
31. Forward interconnect valve
32. Right forward fuel cell ejector pump and support
33. Right forward fuel cell
34. Right forward fuel cell baffle
35. Right forward fuel cell forward compartment fuel quantity probe
36. Right forward fuel cell crashworthy vent valve
37. Right forward fuel cell inlet/outlet fitting
38. Center aft fuel cell right outlet fitting
39. Right outboard aft fuel cell outlet fitting
40. Shutoff valve
41. Right outboard aft fuel cell vent fitting

(S/N 33001 33107)

ICN-412-A-281000-B-97499-00002-A-001-01

Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 2 of 2)

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Aft Cells - Removal

Procedure

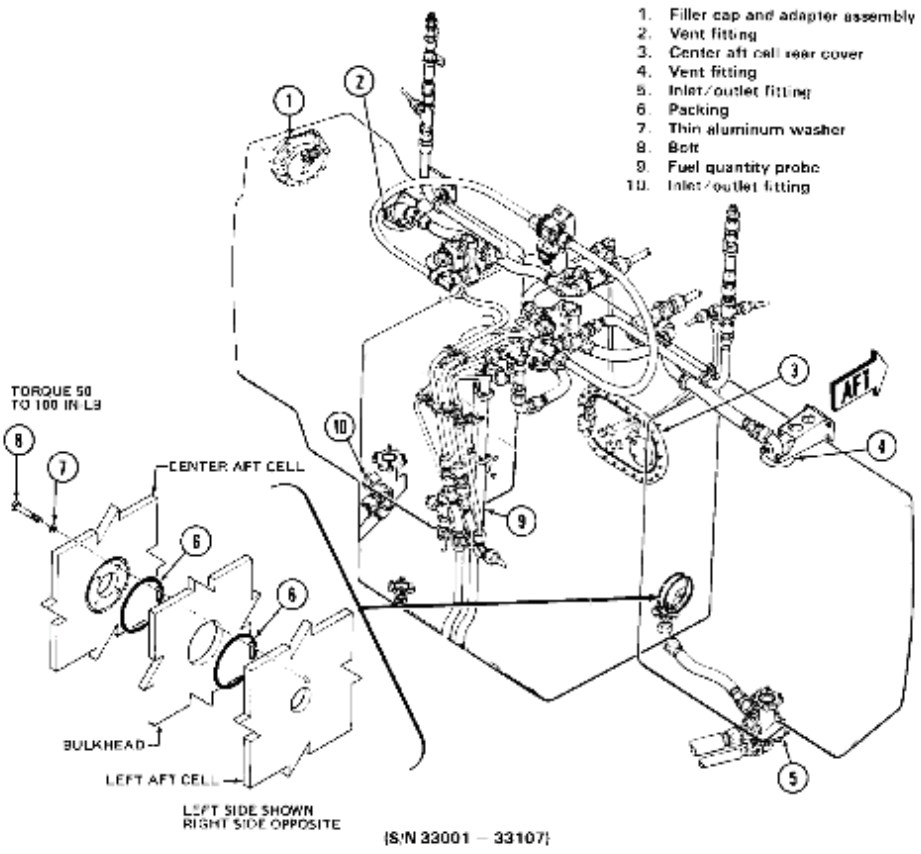


DO NOT HANDLE FUEL CELL WHEN AMBIENT TEMPERATURE IS BELOW 65°F (18°C).

NOTE

The following removal procedures for left aft and right aft cells are the same.

1. Disconnect electrical power.
2. Defuel system (412-A-12-00-00-00A-009A-A / 00125).
3. Remove deck panel above cell.
4. Remove vent fitting (2 or 4, Figure 1) (412-A-28-10-00-26A-520A-A / 00404).
5. Remove center aft cell rear cover (3) (412-A-28-10-00-27A-520A-A / 00406).
6. Working through center aft cell rear access port, remove bolts (8) and washers (7). If removing right aft cell, fuel quantity probe (9) is secured by these same bolts.
7. Remove inlet/outlet fitting (5 or 10) (412-A-28-10-00-12A-520A-A / 00361).
8. If removing right aft cell, remove filler cap and adapter assembly (1) (412-A-28-10-00-30A-520A-A / 00415 or 412-A-28-10-00-31A-520A-A / 00418).
9. Lift cell from cavity.
10. Cover cell openings and cap or plug open lines.



ICN-412-A-281000-B-97499-00034-A-001-01

Figure 1. Left Aft and Right Aft Cell (Sheet 1 of 1)

Aft Cells - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Soap	100066	AR	Zodiac Aerospace (www.zodiacaerospace.com)
Cleaning Solution	601199	AR	Zodiac Aerospace (www.zodiacaerospace.com)
Clean cloth	C-516	AR	

Procedure



OBEY ALL THE SAFETY PRECAUTIONS WHEN YOU DO MAINTENANCE ON FUEL SYSTEM EQUIPMENT.



OBEY ALL THE FUEL CELL HANDLING PRECAUTIONS DURING CLEANING OR DAMAGE TO THE CELL MAY OCCUR.



DO NOT ALLOW SOLVENTS OR COMPOUNDS TO MAKE CONTACT WITH PARTS MADE OF ELASTOMERIC MATERIAL OR DAMAGE WILL OCCUR.

NOTE

The following cleaning procedures for left aft and right aft cells are the same.

NOTE

Any evidence of fuel leakage must be identified before cleaning.

NOTE

Refer to the document "FUEL-F6101-SPM20-00" (Zodiac Aerospace), for additional information on the cleaning of the fuel cells.

1. Clean the internal surface of the fuel cell with a clean cloth (C-516) moistened with a cleaning solution (601199). Dry the liner with low-pressure filtered compressed air for 2 to 4 hours, at approximately 68°F (20°C).
2. Clean small areas on the external surface of the fuel cell with a clean cloth (C-516) moistened with a cleaning solution (601199). Dry with a clean cloth (C-516).
3. Clean an extensive area on the external surface of the fuel cell with a solution of soap (100066) and hot water. Rinse with water and dry with a clean cloth (C-516).
4. Dry clean the fittings of the fuel cell with filtered compressed air and a clean cloth (C-516) to remove any blockage or obstruction.
5. Clean fungus growth found on the fuel cell or on the fuel cell cavity per the instructions contained in the BHT-ALL-SPM, Chapter 5, Standard Practices Manual.

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Aft Cells - Inspection

Procedure



DO NOT HANDLE FUEL CELL WHEN AMBIENT TEMPERATURE IS BELOW 65°F (18°C).

NOTE

The following inspection procedures for left aft and right aft cells are the same.

1. Inspect cell for cracks and scuffing.

2. Inspect threaded insert for damage.

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Aft Cells - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	
Talcum powder	C-408	AR	

Procedure

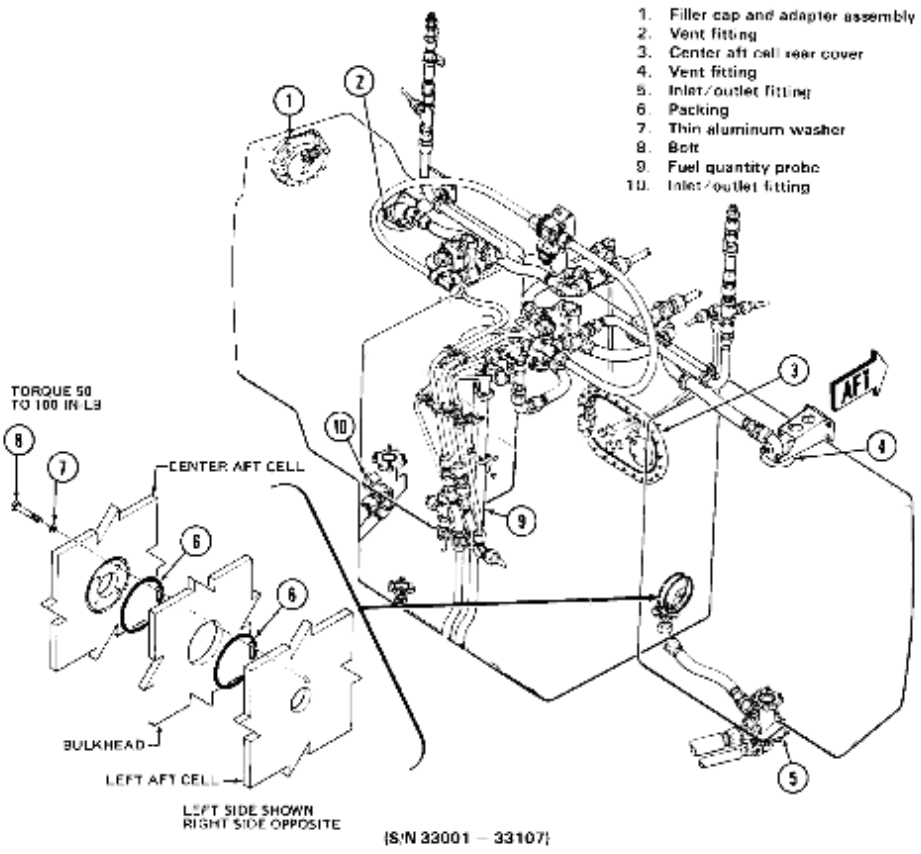


DO NOT HANDLE FUEL CELL WHEN AMBIENT TEMPERATURE IS BELOW 65°F (18°C).

NOTE

The following installation procedures for left aft and right aft cells are the same.

1. Inspect cell cavity for foreign objects and rough interior surfaces. Remove foreign objects and/or repair rough surfaces prior to installing cell.
2. Apply talcum powder (C-408) to interior surfaces of cavity.
3. Coat packings (6, Figure 1) with assembly fluid (C-024) and place packings in grooves of left aft or right aft and center aft cell.
4. Remove protective covers, caps, and plugs.
5. Insert cell into cavity. Using care to avoid damage to cell, align cell ports with fuselage openings.
6. If installing right aft cell, install filler cap and adapter assembly (1, Figure 1) (412-A-28-10-00-30A-720A-A / 00416 or 412-A-28-10-00-32A-042A-A / 00420).
7. Install inlet/outlet fitting (5 or 10) (412-A-28-10-00-24A-720A-A / 00399).
8. Working through access port at rear of center aft cell, ensure packings (6) are in place in grooves of center aft and left aft or right aft cell. If installing right aft cell, place fuel quantity probe (9) in position.
9. Install bolts (8) with washers (7). Torque bolts to 50 to 100 inch-pounds (5.65 to 11.30 Nm).
10. Install center aft cell rear cover (3) (412-A-28-10-00-27A-720A-A / 00407, step 1 through step 7 only).
11. Install vent fitting (2 or 4) (412-A-28-10-00-26A-720A-A / 00405, step 1 through step 6 only).
12. Perform operational check of fuel quantity indicating system before servicing system (412-A-96-11-00-10A-273A-A / 01973).
13. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).
14. Check area for leaks.
15. Install deck panel and access doors.

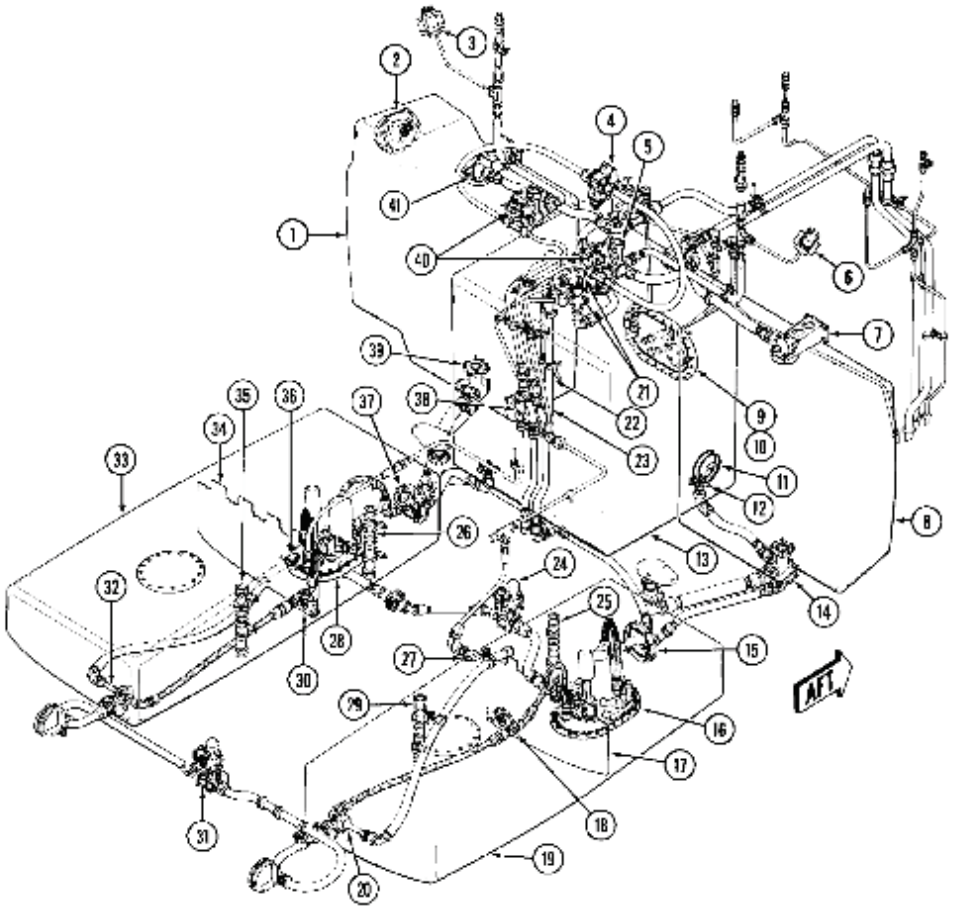


ICN-412-A-281000-B-97499-00034-A-001-01

Figure 1. Left Aft and Right Aft Cell (Sheet 1 of 1)

Center Aft Cell - Description

The center aft cell (13, Figure 1) is located between the main structure beams at BL 14, and between bulkheads at FS 155 and 178. The cell has interconnection ports on the bottom and on both sides. A vent fitting port is located on top, and a port for a large access cover is on the aft side.



1. Right outboard aft fuel cell
2. Filler cap and adapter assembly
3. Right fuel pressure transmitter
4. System crossfeed valve
5. Center aft fuel cell vent cover
6. Left fuel pressure transmitter
7. Left outboard aft fuel cell vent fitting
8. Left outboard aft fuel cell
9. Center aft fuel cell rear cover
10. Governor bleed line check valve
11. Center aft fuel cell left interconnect fitting
12. Center aft fuel cell left outlet fitting
13. Center aft fuel cell
14. Left outboard aft fuel cell outlet fitting
15. Left forward fuel cell inlet/outlet fitting

(S/N 33001 33107)

ICN-412-A-281000-B-97499-00001-A-001-01

Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 1 of 2)

16. Left forward fuel cell fuel sump area consisting of:
    - Fuel sump assembly
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
17. Left forward fuel cell baffle
18. Left forward fuel cell flapper valve
19. Left forward fuel cell
20. Left forward fuel cell ejector pump and support
21. Siphon breaker valves
22. Center aft fuel cell right interconnect fitting
23. Aft fuel cell fuel quantity probe
24. Aft interconnect valve
25. Left forward fuel cell aft compartment fuel quantity probe
26. Right forward fuel cell aft compartment fuel quantity probe
27. Left forward fuel cell crashworthy vent valve
28. Right forward fuel cell fuel sump area consisting of:
    - Fuel sump
    - Low fuel level warning switch
    - Crashworthy defuel valve
    - Fuel boost pump and nut flange
    - Sump interconnect valve
    - Sump drain valve
    - Flow switch
29. Left forward fuel cell forward compartment fuel quantity probe
30. Right forward fuel cell flapper valve
31. Forward interconnect valve
32. Right forward fuel cell ejector pump and support
33. Right forward fuel cell
34. Right forward fuel cell baffle
35. Right forward fuel cell forward compartment fuel quantity probe
36. Right forward fuel cell crashworthy vent valve
37. Right forward fuel cell inlet/outlet fitting
38. Center aft fuel cell right outlet fitting
39. Right outboard aft fuel cell outlet fitting
40. Shutoff valve
41. Right outboard aft fuel cell vent fitting

(S/N 33001 33107)

ICN-412-A-281000-B-97499-00002-A-001-01

Figure 1. Fuel Storage and Distribution System (S/N 33001 through 33107) (Sheet 2 of 2)

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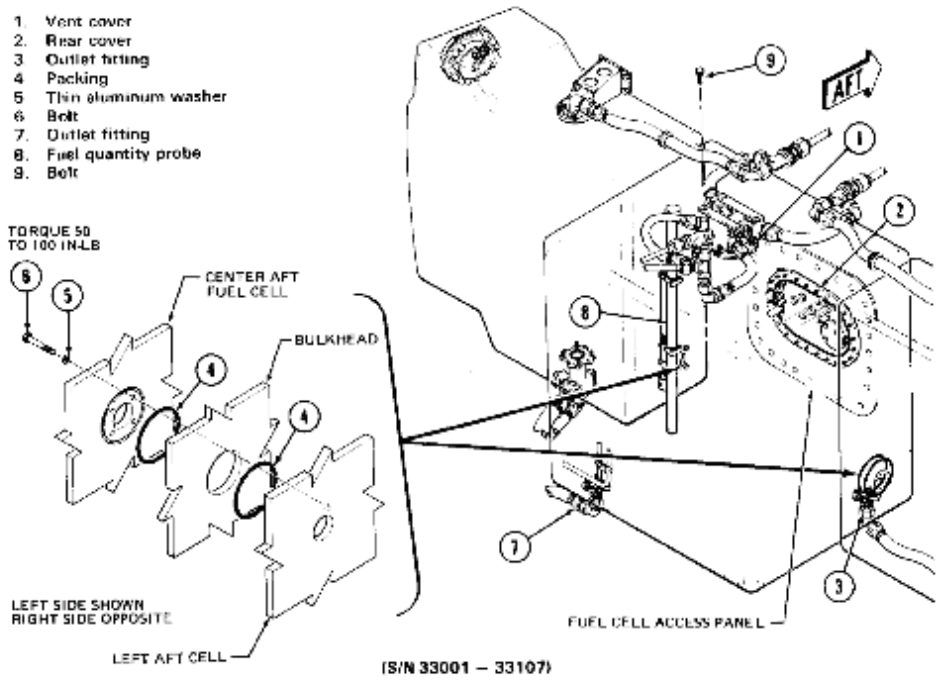
Center Aft Cell - Removal

Procedure



DO NOT HANDLE FUEL CELL WHEN AMBIENT TEMPERATURE IS BELOW 65ºF (18ºC).

1. Remove rear cover (2, Figure 1) (412-A-28-10-00-27A-520A-A / 00406).
2. Remove fuel quantity probe (8).
3. Working through access at rear of cell, remove bolts (6) and washers (5) from both sides of cell.
4. Remove outlet fittings (3 and 7) (412-A-28-10-00-29A-520A-A / 00412).
5. Collapse aft upper edge of cell downward and inward.
6. With cell held in this position, remove four bolts (9).
7. Disconnect four vent hoses from vent cover (1).
8. Position two webbing straps around cell. Run straps vertically around cell.
9. Tighten straps and remove cell from rear access opening.
10. Cover cell openings and restore to original shape. Store cell in heated area if ambient temperature is below 65ºF (18ºC).



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Figure 1. Center Aft Cell (Sheet 1 of 1)

Center Aft Cell - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Soap	100066	AR	Zodiac Aerospace (www.zodiacaerospace.com)
Cleaning Solution	601199	AR	Zodiac Aerospace (www.zodiacaerospace.com)
Clean cloth	C-516	AR	

Procedure



OBEY ALL THE SAFETY PRECAUTIONS WHEN YOU DO MAINTENANCE ON FUEL SYSTEM EQUIPMENT.



OBEY ALL THE FUEL CELL HANDLING PRECAUTIONS DURING CLEANING OR DAMAGE TO THE CELL MAY OCCUR.



DO NOT ALLOW SOLVENTS OR COMPOUNDS TO MAKE CONTACT WITH PARTS MADE OF ELASTOMERIC MATERIAL OR DAMAGE WILL OCCUR.

NOTE

Any evidence of fuel leakage must be identified before cleaning.

NOTE

Refer to the document "FUEL-F6101-SPM20-00" (Zodiac Aerospace), for additional information on the cleaning of the fuel cells.

1. Clean the internal surface of the fuel cell with a clean cloth (C-516) moistened with a cleaning solution (601199). Dry the liner with low-pressure filtered compressed air for 2 to 4 hours, at approximately 68°F (20°C).
2. Clean small areas on the external surface of the fuel cell with a clean cloth (C-516) moistened with a cleaning solution (601199). Dry with a clean cloth (C-516).
3. Clean an extensive area on the external surface of the fuel cell with a solution of soap (100066) and hot water. Rinse with water and dry with a clean cloth (C-516).
4. Dry clean the fittings of the fuel cell with filtered compressed air and a clean cloth (C-516) to remove any blockage or obstruction.
5. Clean fungus growth found on the fuel cell or on the fuel cell cavity per the instructions contained in the BHT-ALL-SPM, Chapter 5, Standard Practices Manual.

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Center Aft Cell - Inspection

Procedure

1. Inspect cell for cracks and scuffing. Cracks and severe scuffing are not acceptable.
2. Inspect threaded inserts for damaged threads. Damaged threads which can be detected visually are not acceptable.

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Center Aft Cell - Installation

Preliminary requirements

Consumables, materials and expendables

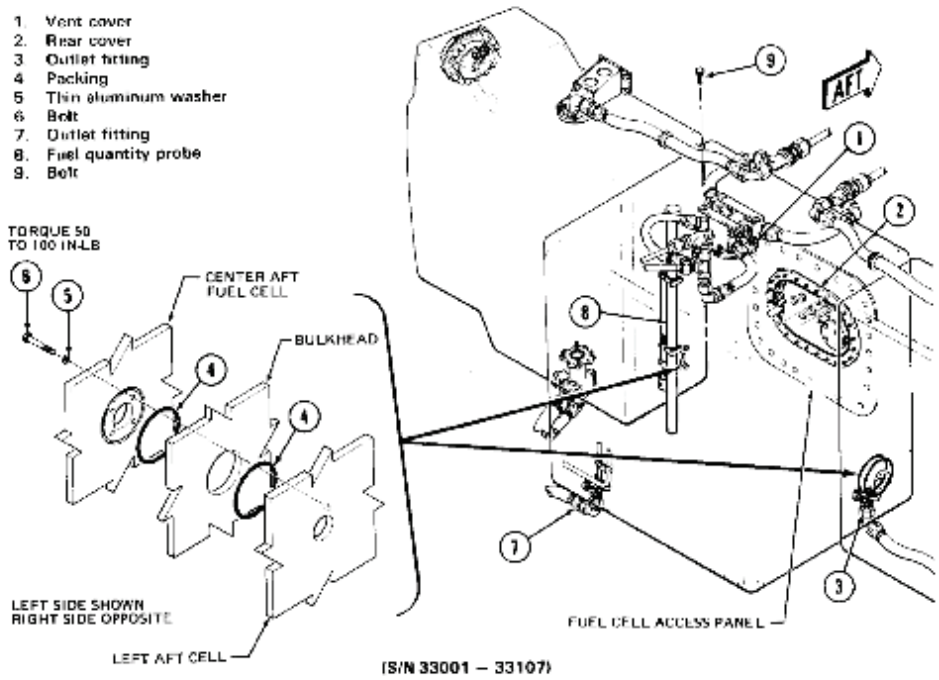
Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	
Talcum powder	C-408	AR	

Procedure



DO NOT HANDLE FUEL CELL WHEN AMBIENT TEMPERATURE IS BELOW 65°F (18°C).

1. Inspect cell cavity for cleanliness and smoothness of interior surfaces.
2. Apply talcum powder (C-408) to interior surfaces of cell cavity.
3. Collapse and fold cell. Secure with suitable webbing straps to a bundle which can pass through rear access opening of cavity.
4. Carefully insert cell into cavity. Remove straps and arrange in approximate position.
5. Install outlet fittings (3 and 7, Figure 1) (412-A-28-10-00-29A-720A-A / 00413).
6. Coat packings (4) with assembly fluid (C-024) and place packings in groove of two interconnection ports at sides of cell. Ensure packings are still in place on mating fitting of right aft and left aft cells.
7. Install bolts (6) with washers (5). On right side of cell, position mount of fuel quantity probe (8) under head of bolts. Torque bolts (6) 50 to 100 inch-pounds (5.65 to 11.30 Nm).
8. Install vent cover (1) (412-A-28-10-00-25A-720A-A / 00402).



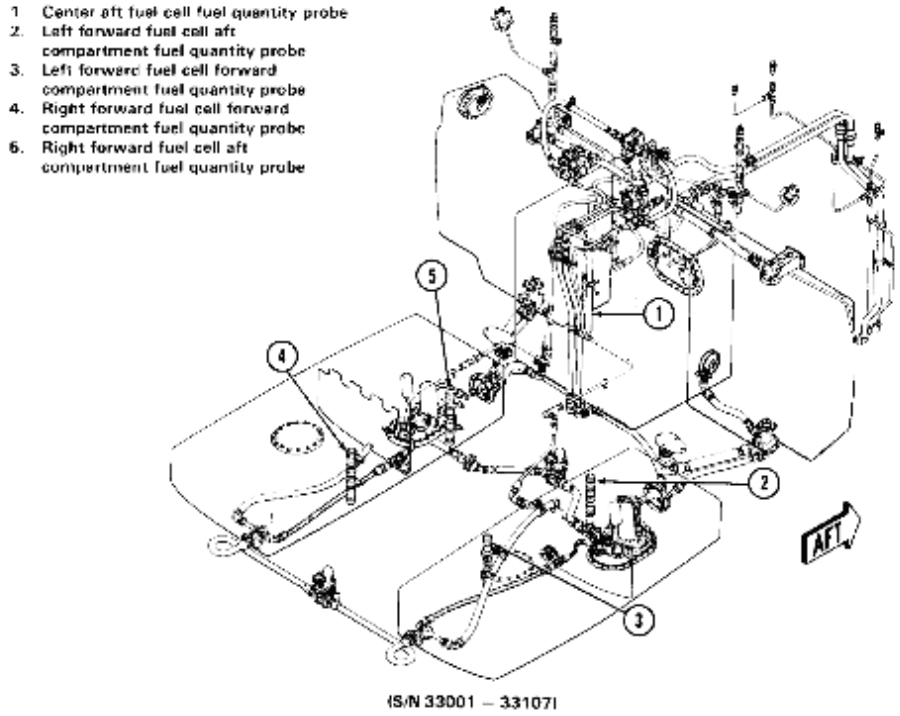
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Figure 1. Center Aft Cell (Sheet 1 of 1)

Fuel Quantity Probes - Description

Five cell mounted fuel quantity probes (tank units) (1 through 5, Figure 1) are used to measure quantity (weight) of fuel in the system. The probes are variable capacitors in which capacitance varies with changes in dielectric constant of fuel and air present between tubes (capacitor plates) of probes. The level and density of fuel in cells determine dielectric constant. Probe capacitance is transmitted to a self-balancing bridge network in the indicator.

Refer to 412-A-52-00-00-00A-028A-A / 00775 for location of the doors, panels, cowlings, and covers which must be removed to access the fuel system components.



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Figure 1. Fuel Quantity Probes (Sheet 1 of 1)

Forward Cell Fuel Quantity Probes - Description

NOTE

Maintenance procedures for forward left and right cell fuel quantity probes are the same.

Each compartment of the forward cells contains a fuel quantity probe.

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Forward Cell Fuel Quantity Probes - Removal

Procedure

NOTE

The following removal procedures for forward left and right cell fuel quantity probes are the same.

1. Remove sump assembly (412-A-28-10-00-03A-520A-A / 00335).

2. Remove forward fuel quantity probe (1, Figure 1), as follows:

2.1. Remove right or left floor panel, as applicable.

2.2. Remove screws (22) and cover (23).

2.3. Remove bolt (10), spacers (3 and 9), and washer (4).

2.4. Remove bolt (7) and washer (5).

2.5. Route electrical leads (2) over baffle (21) and remove forward fuel quantity probe (1).

2.6. Remove clamps (6) and (8).

3. Remove aft fuel quantity probe (12), as follows:

3.1. Remove bolt (20), spacer (14), and washer (15).

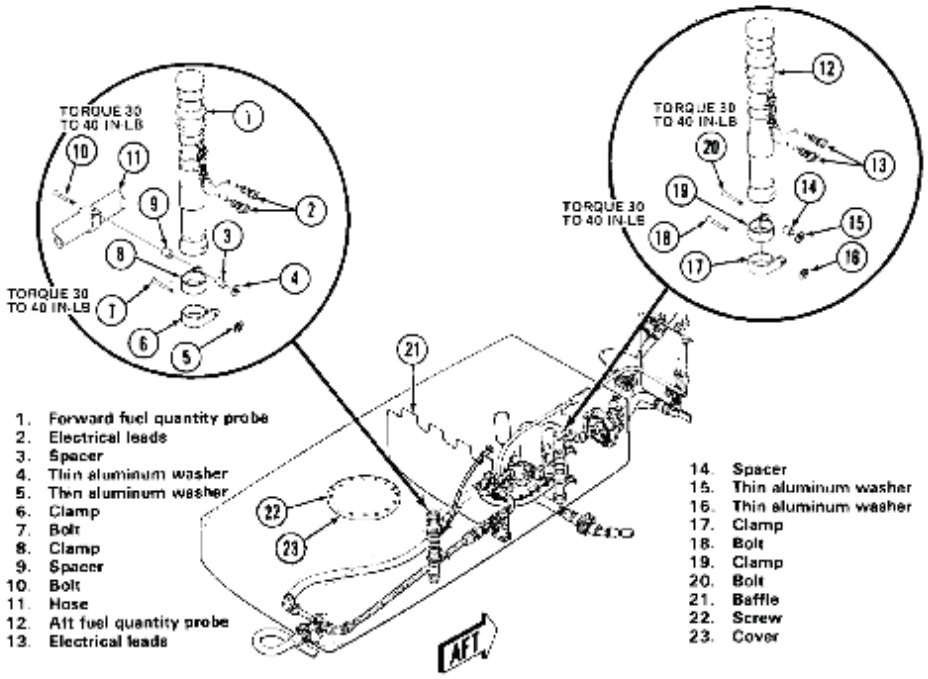
3.2. Remove bolt (18) and washer (16).

NOTE

Verify electrical leads (13) were disconnected when sump assembly was removed.

3.3. Remove aft fuel quantity probe (12) and clamps (17 and 19).

4. Install protective cover on cell sump opening.



(S/N 33001 — 33107)

ICN-412-A-281000-B-97499-00037-A-001-01

Figure 1. Forward Cell Fuel Quantity Probe (Sheet 1 of 1)

Forward Cell Fuel Quantity Probes - Installation

Procedure

NOTE

The following installation procedures for forward left and right cell fuel quantity probes are the same.

1. Install forward fuel quantity probe (1, Figure 1) as follows:

1.1. Remove protective cover from sump opening. Place clamps (6 and 8) on forward fuel quantity probe (1).

1.2. Align clamps (6 and 8) over attachment points in cell.

1.3. Place bolt (10) through clamp on hose (11). Use spacer (9), clamp (8), spacer (3), and washer (5). Install bolt (10) in cell. Torque bolt (10) 30 to 40 in.-lbs. (3.39 to 4.52 N-m).

1.4. Place bolt (7) through clamp (6) and washer (5). Install bolt (7) in cell. Torque bolt (7) 30 to 40 in.-lbs. (3.39 to 4.52 N-m).

1.5. Route electrical leads (2) over baffle (21).

1.6. Install cover (23) and screws (22).

1.7. Install floor panel.

2. Install aft fuel quantity probe (12) as follows:

2.1. Remove protective cover from sump opening. Place clamps (17 and 19) on aft fuel quantity probe (12). Align clamps (17 and 19) over attachment points in cell.

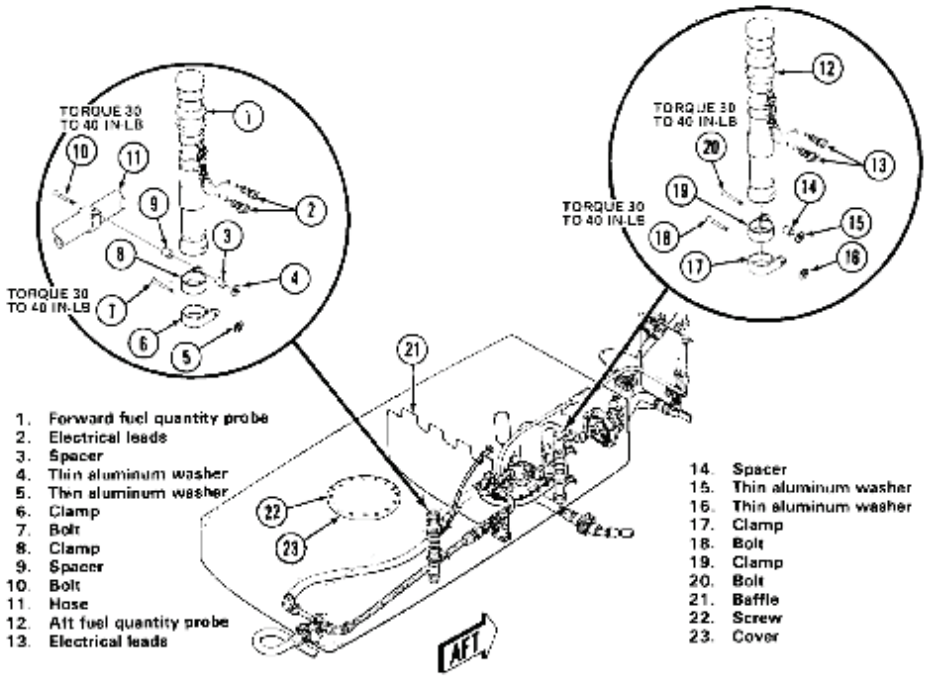
2.2. Place bolt (20) through clamp (19), spacer (14), and washer (15). Install bolt (20) in cell. Torque bolt (20) 30 to 40 in.-lbs. (3.39 to 4.52 N-m).

2.3. Place bolt (18) through clamp (17) and washer (16). Install bolt (18) in cell. Torque bolt (18) 30 to 40 in.-lbs. (3.39 to 4.52 N-m).

NOTE

Electrical leads (13) will be connected when sump is installed at Step 3.. below:

3. Install sump assembly (412-A-28-10-00-03A-720A-A / 00337).



(S/N 33001 — 33107)

ICN-412-A-281000-B-97499-00037-A-001-01

Figure 1. Forward Cell Fuel Quantity Probe (Sheet 1 of 1)

Center Aft Cell Fuel Quantity Probe - Description

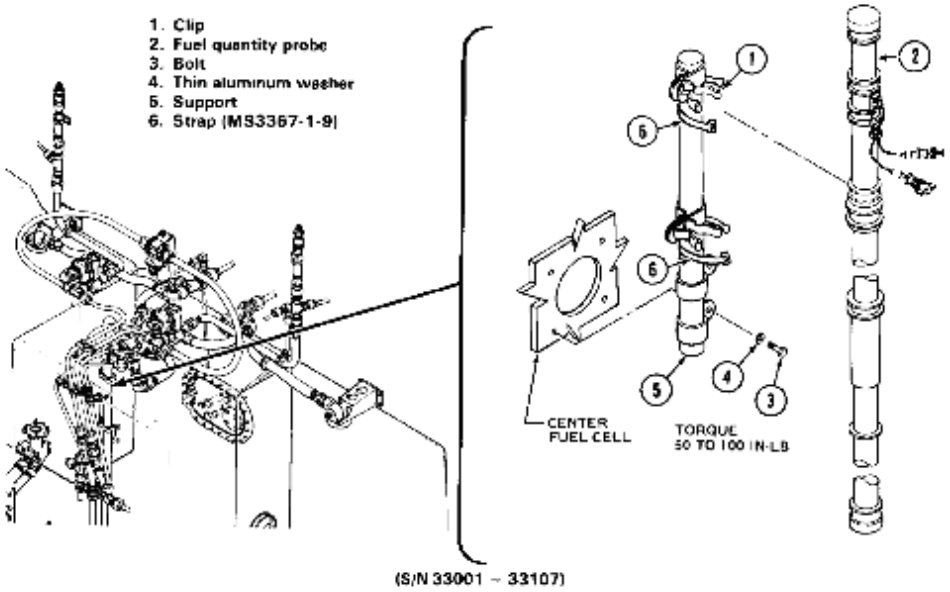
The center aft cell contains a single fuel quantity probe mounted on a support on right wall of cell.

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Center Aft Cell Fuel Quantity Probe - Removal

Procedure

1. Remove center aft cell rear cover (412-A-28-10-00-27A-520A-A / 00406). Do not install protective cover over cell opening at this time.
2. Remove straps (6, Figure 1). Remove fuel quantity probe (2).
3. Remove bolts (3), washers (4), and support (5).
4. Install protective cover over cell opening.



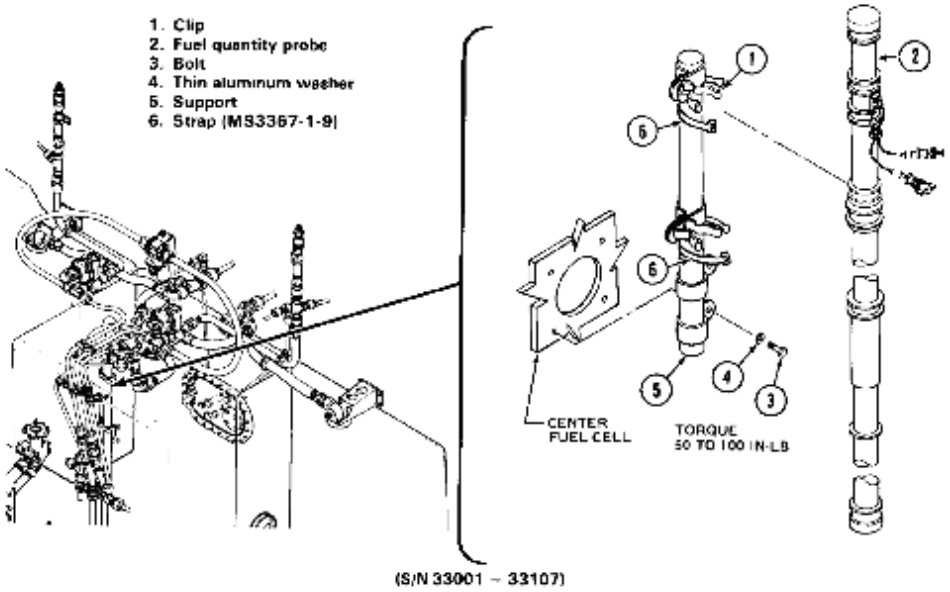
ICN-412-A-281000-B-97499-00038-A-001-01

Figure 1. Center Cell Fuel Quantity Probe (Sheet 1 of 1)

Center Aft Cell Fuel Quantity Probe - Installation

Procedure

1. Remove protective cover.
2. Position support (5, Figure 1) on wall of center aft fuel cell. Install volts (3) and washers (4). Torque bolts (3) 50 to 100 in.-lbs. (5.65 to 11.30 N-m).
3. Position fuel quantity probe (2) in clips (1). Install straps (6).
4. Connect electrical leads to quantity probe and install center aft cell rear cover (412-A-28-10-00-27A-720A-A / 00407).



ICN-412-A-281000-B-97499-00038-A-001-01

Figure 1. Center Cell Fuel Quantity Probe (Sheet 1 of 1)

Fuel Storage and Distribution System - Description and Operation

Applicable: SN: 33108-Subsequent or 36001-Subsequent:

The following paragraphs provide a functional description for the fuel system components for helicopters which store and deliver fuel to the engine power sections.

The fuel system (Figure 1 and Figure 2) is comprised of fuel storage cells, transfer system, venting system, dual engine feed system, and electrical control and warning systems. The system is designed to provide fuel containment in the event of an accident or incident where fuel system rupture might occur. The cells are of a lightweight, crash-resistant, laminated fabric and rubber construction. The cells are suspended in the fuselage cavities by nylon cord laced between hangers attached to the cells and the fuselage structure.

The ten interconnected cells (1, 5, 8, 9, 13, 16, 22, 26, 28, and 34, Figure 1) have a total capacity of 337.5 U.S. gallons (approximately 330.5 U.S. gallons usable). A quantity of fuel, 131 U.S. gallons (65.5 U.S. gallons per side), is reserved by use of internal barriers and an interconnect valve to provide an isolated fuel supply for each engine.

There are six cells below the cabin floor and four cells located below engine compartments aft of cabin and pylon. The helicopter is fueled through a filler cap and adapter assembly (2) in upper right cell (1). Fuel gravity feeds from upper cells into lower cells through interconnecting lines. Transfer pumps (located on left and right forward sump assemblies) are used to transfer fuel from lower forward cells (16 and 22) and lower middle cells (13 and 26) aft to lower main cells (9 and 28) and from there through feed lines to the two engine fuel inlets.

The lower forward cell sump areas (18 and 21) contain fuel sump assemblies equipped with transfer pumps and drain valves. The lower middle cell sump areas (12 and 25) contain sump assemblies equipped with drain valves. Ejector pumps (15 and 27) are installed near the lower middle cell sump assemblies. The lower main cell sump areas (10 and 29) contain sump assemblies equipped with defueling valves, drain valves, flow switch check valves, and boost pumps.

Sump drain valves in the lower main cells (9 and 28) have solenoids and may be opened by either electrical or manual actuation. Sump drain valves in other lower cells are manually operated, push-to-drain types. All six valves are spring-loaded, poppet type valves. Defueling valves located in the lower main cell sump assemblies require removal of a plug and insertion of a standard fitting to open the spring-loaded poppet.

A collective vent line on top of upper aft center cell connects the four upper cell vent spaces with two vertical lines (one behind upper right cell (1) and one behind upper aft cell (5)) which vent overboard beneath the fuselage. Each lower main cell vents through a line connecting into the upper forward center cell. A forward vent line on each side of the fuselage connects with the respective lower forward and lower middle cells. These lines rise through the doorposts to a waterline above the upper cells and then double back downward to vent overboard beneath the fuselage.

To enable fuel transfer, an interconnect system joins lower cells together and also joins lower cells with upper cells. Fuel drainage from upper cells to lower cells is controlled by standpipes that sequence depletion flow among cells to maintain desired center-of-gravity during in-flight consumption. Isolation barriers in the upper center cells and a normally closed interconnect valve in the interconnect line between lower main cells provide for isolation of individual power section fuel supplies.

Applicable: SN: 33168-Subsequent TB 412-92-106: Pre:

Flow of fuel from the lower forward cell transfer pumps provides motive force for lower middle cell ejector pumps to pump fuel from lower middle cells into lower main cells. Where fuel is pumped into aft center cell. Fuel level sensors, (thermistors) located in the base of lower forward cell fuel quantity probes (tank units), cause transfer pumps to shut off automatically when lower forward cells become empty. If a flow switch check valve senses fuel flow has stopped and fuel is still present in forward midtanks, the corresponding caution panel segment (NO 1 FUEL TRANS or NO 2 FUEL TRANS) is illuminated. However, if the interruption in flow is due to a cell (lower forward or lower middle) becoming empty, a thermistor in fuel quantity probe in the empty cell provides a ground to the appropriate signal conditioner to cancel segment illumination.

All fuel passes through lower main cells, from which the power sections are supplied. Check valves (17 and 23) in lower forward cells, flow switch check valves in lower main cells (9 and 28), and power section fuel feed line check valves prevent backflow of fuel during periods of shutdown or crossfeed operation. The power section fuel feed line check valves (33) and electrically operated shutoff valves (36) also provide thermal pressure relief during system shutdown. Shutoff valves isolate fuel from engine compartment during shutdown or in event of engine fire. Fuel pressure transmitters (3 and 6) sense fuel feed line pressure and transmit corresponding electrical signals to engine 1 and engine 2 fuel pressure (FUEL PRESS) indicators.

A crossfeed line and normally closed crossfeed valve (4) provide optional interconnection between left and right engine fuel feed lines. This crossfeed valve opens automatically in event failure of a boost pump is sensed by fuel pressure switch (32) in either feed line.

The capacitance type fuel quantity indicating system uses quantity probes, signal conditioners, and a dual pointer indicator with digital fuel totalizer. There are quantity probes in the upper left and upper right cells, and each lower cell. Fuel level signals from quantity probes are summed in signal conditioners and transmitted to the indicator. A switch near the indicator enables checking lower forward and lower middle cell fuel quantities separately to ensure proper balance is maintained during any shutdown period with partially empty tanks.

Boost pumps and transfer pumps are individually controlled by two-position BOOST PUMP switches located on engine and fuel control panel. On the same panel are four switches for valve control: ENGINE 1 FUEL and ENGINE 2 FUEL switches to control shutoff valves (36), FUEL XFEED switch to control crossfeed valve (4), and FUEL INTCON switch to control interconnect valve (31). All are two-position switches except FUEL INTCON switch, which is a three-position switch. The normal position of crossfeed valve and interconnect valve is closed; the interconnect valve also has an open position. The FUEL INTCON and FUEL XFEED switches have an override (OVRD CLOSE) position to enable valves (4 and 36) to be closed, if necessary, after being opened automatically.

Caution light segments in the caution panel annunciate conditions which may exist within the following functional areas in the fuel system: Fuel transfer (NO. 1 FUEL TRANS and NO. 2 FUEL TRANS), engine feed line fuel pressure (FUEL BOOST), fuel level (FUEL LOW), fuel filter (NO. 1 FUEL FILTER and NO. 2 FUEL FILTER), firewall shutoff (fuel) valves (NO 1. FUEL VALVE and NO. 2 FUEL VALVE), crossfeed valve (FUEL XFEED), and interconnect valve (FUEL INTCON).

Fuel burn sequence (Figure 3) is established by system design to maintain center-of-gravity within an acceptable range. The first burn uses fuel from all upper cells above tall standpipe in upper forward center cell. The second burn empties fuel from lower middle cells. The third burn uses fuel from all upper cells above short standpipe in upper forward center cell. The fourth burn empties fuel from lower forward cells. The fifth burn drops fuel level to the barriers in upper center cells that isolate engine fuel supplies, and depletes remainder of fuel in upper cells. The sixth burn uses fuel remaining in lower main cells.

A fuel level sensor (thermistor) in each lower forward cell quantity probe signals fuel depletion at the end of the fourth burn, thereby causing automatic transfer pump shutdown and cancellation of the respective FUEL TRANS caution light signal. A fuel level sensor (thermistor) in each upper outboard (left and right) cell fuel quantity probe signals depletion of fuel at end of fifth burn, causing the FUEL LOW caution light to illuminate.

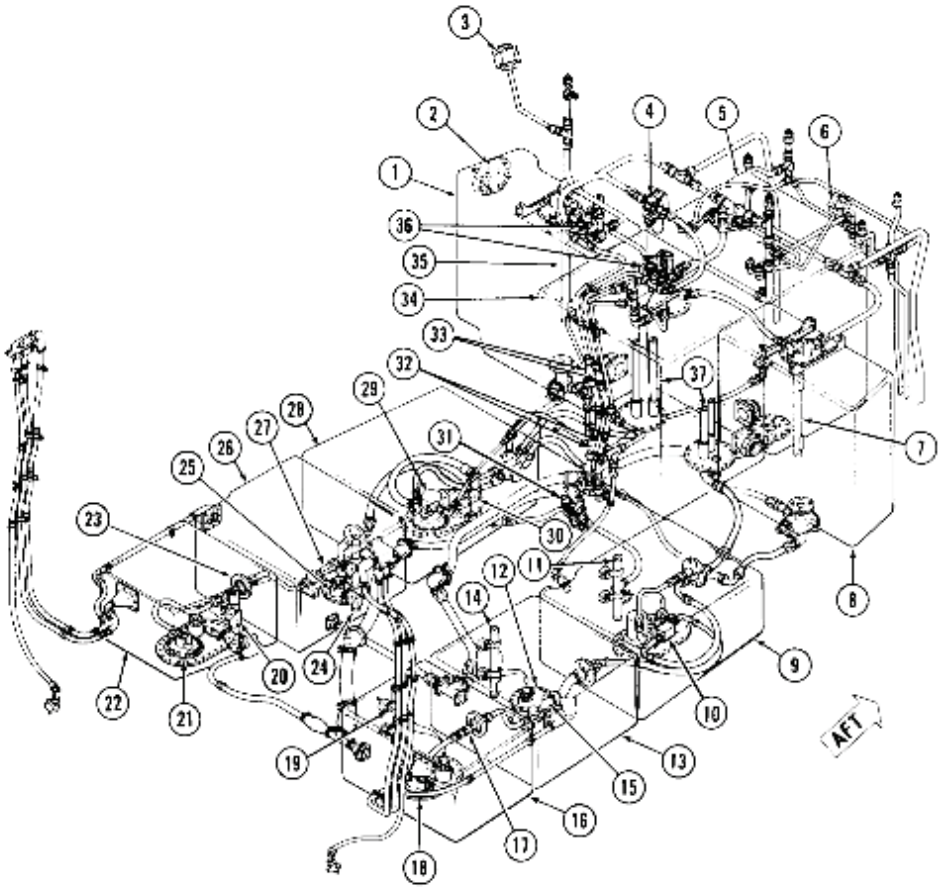
Refer to 412-A-96-00-01-00A-042A-A / 01872 for maintenance instructions for electrically operated components of the system and 412-A-98-00-00-00A-028A-A / 02455 for applicable wiring diagrams.

Table 1. Fuel System Troubleshooting (S/N 33108 and Sub.)

INDICATION OF TROUBLE	PROBABLE CAUSE	CORRECTIVE ACTION
FUEL FILTER caution segment illuminated.	Filter element clogged.	Remove and inspect filter element. Clean or replace filter element (refer to Engine Manual).
FUEL BOOST caution segment illuminated.	Defective pressure switch in pump outlet line or obstructed line.	Replace pressure switch, or obstructed hoses.
	No electrical power to pump.	Repair wiring or replace faulty units in circuit.
	Defective pump.	Replace pump.
Shutoff valve inoperative.	Valve faulty or lacking electrical power.	Repair circuit.
		Replace valve.
Interconnect valve inoperative.	Valve faulty or lacking electrical power.	Repair circuit.
		Replace valve.
Cross feed valve inoperative.	Valve faulty or lacking electrical power.	Repair circuit.
		Replace valve.
FUEL TRANS caution segment illuminated.	Defective pump.	Replace pump.
	No electrical power to pump.	Repair circuit.
	Defective thermistor or forward or mid tank quantity probe.	Replace forward or mid fuel quantity probe.

Table 1. Fuel System Troubleshooting (S/N 33108 and Sub.) (continued)

INDICATION OF TROUBLE	PROBABLE CAUSE	CORRECTIVE ACTION
	SN: 33108-33167 TB 412-92-106: Pre: Fuel in mid tank when fwd tank empty due to leaking check valve between aft and mid tank.	Replace check valve in aft fuel tank.
	Defective fuel quantity signal conditioner.	Replace signal conditioner.
FUEL LOW caution segment illuminated when fuel is not low or fails to illuminate when fuel is low.	Defective thermistor on upper quantity probe.	Replace probe.
	Defective signal conditioner.	Replace signal conditioner.
	Faulty wiring.	Repair wiring.



1. Upper right cell
2. Filler cap and adapter assembly
3. Right fuel pressure transmitter
4. Crossfeed valve
5. Upper aft center cell
6. Left fuel pressure transmitter
7. Upper left cell fuel quantity probe
8. Upper left cell
9. Lower left main cell
10. Lower left main cell sump area consisting of:
    - Fuel sump assembly
    - Defueling valve
    - Sump drain valve
    - Flow switch check valve
    - Boost pump
11. Lower left main cell fuel quantity probe

(S/N 33108 THRU 33213)
(S/N 36001 AND SUBSEQUENT)

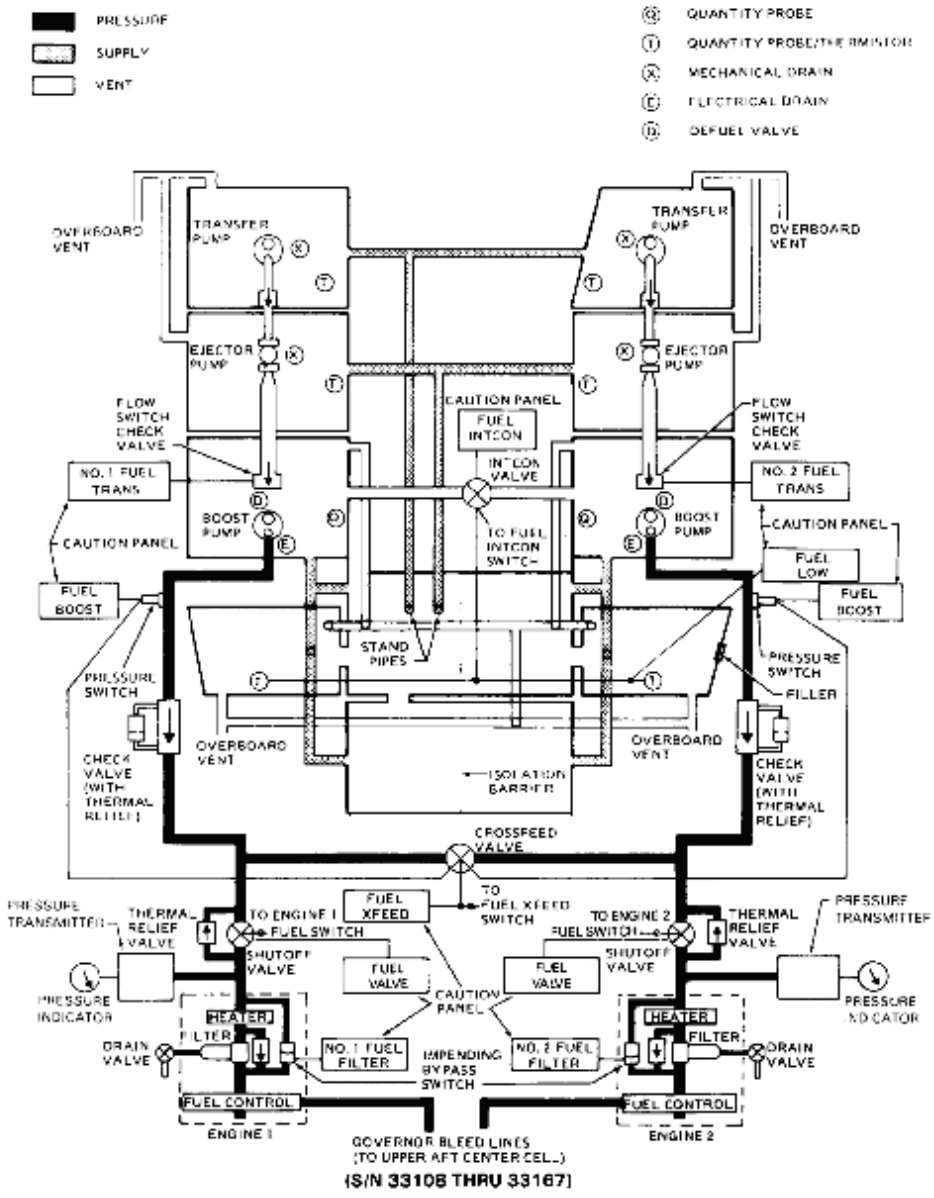
ICN-412-A-281000-B-97499-00039-A-001-01

Figure 1. Fuel Storage and Distribution System (S/N 33108 and Sub.) (Sheet 1 of 2)

12. Lower left middle cell sump area of:
    - Fuel sump assembly
    - Sump drain valve
13. Lower left middle cell
14. Lower left middle cell fuel quantity probe
15. Lower left middle cell ejector pump
16. Lower left forward cell
17. Lower left forward cell check valve
18. Lower left forward cell sump area consisting of:
    - Fuel sump assembly
    - Sump drain valve
    - Transfer pump
19. Lower left forward cell fuel quantity probe
20. Lower right forward cell fuel quantity probe
21. Lower right forward cell sump area consisting of:
    - Fuel sump assembly
    - Sump drain valve
    - Transfer pump
22. Lower right forward cell
23. Lower right forward cell check valve
24. Lower right middle cell fuel quantity probe
25. Lower right middle cell sump area consisting of:
    - Fuel sump assembly
    - Sump drain valve
26. Lower right middle cell
27. Lower right middle cell ejector pump
28. Lower right main cell
29. Lower right main cell sump area consisting of:
    - Fuel sump assembly
    - Defueling valve
    - Sump drain valve
    - Flow switch check valve
    - Boost pump
30. Lower right main cell fuel quantity probe
31. Interconnect valve
32. Engine fuel feed line fuel pressure switches
33. Engine fuel feed line check valves
34. Upper forward center cell
35. Upper right cell fuel quantity probe
36. Shutoff valves
37. Standpipes

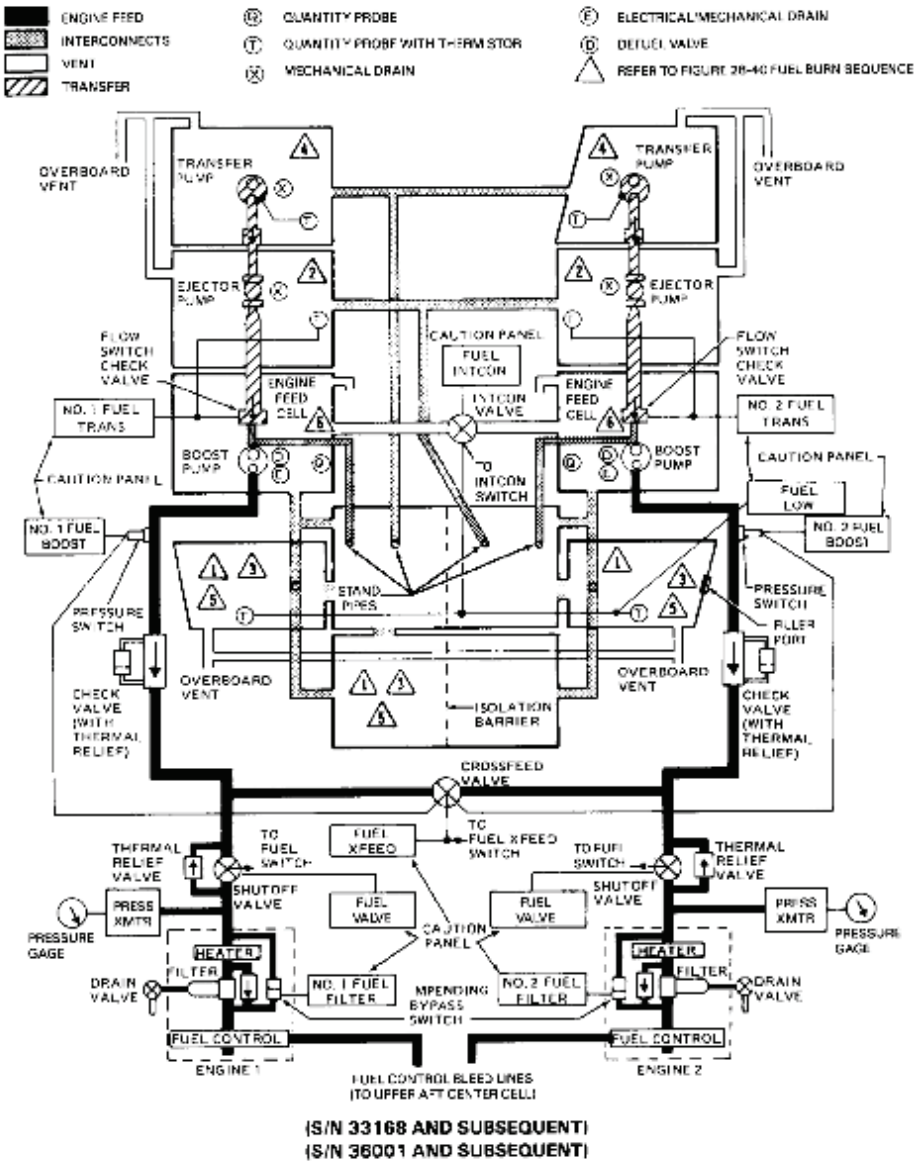
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Figure 1. Fuel Storage and Distribution System (S/N 33108 and Sub.) (Sheet 2 of 2)



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Figure 2. Fuel System Schematic (Sheet 1 of 2)



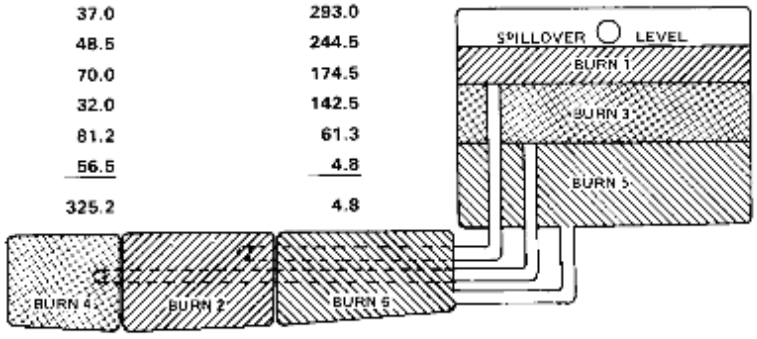
(S/N 33168 AND SUBSEQUENT)
(S/N 36001 AND SUBSEQUENT)

ICN-412-A-281000-B-97499-00042-A-001-01

Figure 2. Fuel System Schematic (Sheet 2 of 2)

CALCULATED FUEL QUANTITIES

BURN NO.	APPROXIMATE GALLONS CONSUMED	APPROXIMATE GALLONS REMAINING
0	0	330.0
1	37.0	293.0
2	48.5	244.5
3	70.0	174.5
4	32.0	142.5
5	81.2	61.3
6	56.5	4.8
TOTAL	325.2	4.8



(S/N 33108 THRU 33213)
(S/N 36001 AND SUBSEQUENT)

ICN-412-A-281000-B-97499-00043-A-001-01

Figure 3. Fuel Burn Sequence Diagram (Sheet 1 of 1)

Fuel Storage and Distribution System - Pressure Testing

Procedure

NOTE

For field pressure test of the fuel system, an alternate method is to fill the helicopter fuel cells with fuel. Inspect fuel system and the area affected by the work accomplished for leaks. Do the inspection procedure again to examine for leaks several times as the fuel cells are being filled.

1. Cap main fuel line and vent connections.



MAKE SURE FUEL CELL ACCESS PANELS ARE INSTALLED PRIOR TO APPLYING PRESSURE.

2. Use regulated, low pressure, filtered, compressed nitrogen or compressed air source with a manometer or accurate pressure gauge and a shutoff valve.



DO NOT APPLY MORE THAN 1.0 PSIG (6.9 KPA) PRESSURE TO CELLS. DAMAGE TO CELLS AND STRUCTURE MAY RESULT.

3. Apply 0.5 to 1.0 PSIG (3.5 to 6.9 kPa) pressure to all cells and tubing of the fuel system. Shut off pressure source. Make sure that pressure does not decrease during 15 minute time period.

4. If leakage is detected, apply a soap and water solution to all repaired areas and to areas suspected of leaking. Make repairs and do the pressure test again until no leaks are shown.

5. Upon satisfactory conclusion of test, remove soap residue and remove all caps and plugs used to close off system.

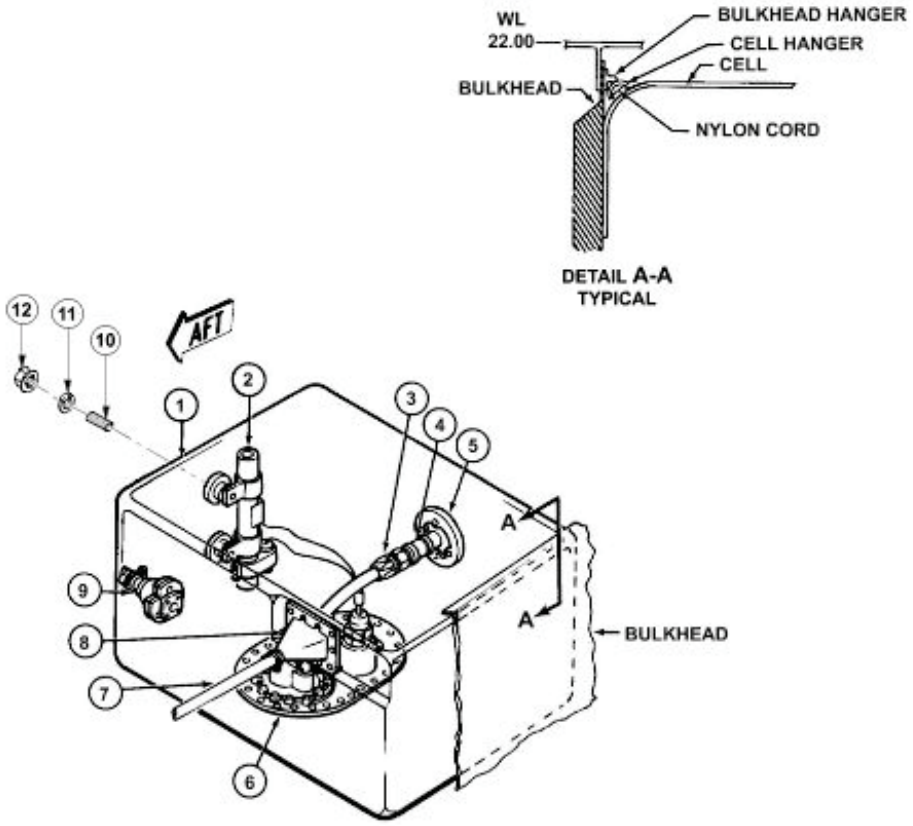
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Lower Forward Cells - Description

NOTE

Maintenance procedures for left and right forward cells are the same.

Two lower forward cells (1 and 28, Figure 1) are located under the cabin floor below WL 22, outboard of each main longitudinal beam (at left BL 14 and right BL 14), and between FS 84.5 and 102. Each lower forward cell has a vent port on the upper front side and interconnect ports on the inboard and aft sides. A sump opening on the bottom provides access for installation and maintenance of internally mounted components.



1. Lower forward cell
2. Fuel quantity probe
3. Hose
4. Check valve
5. Lower forward cell/lower middle cell interconnect fitting
6. Sump assembly
7. Vent hose
8. Retainer
9. Interconnect fitting
10. Setscrew
11. Washer
12. Nut

ICN-412-A-281000-B-97499-00044-A-001-01

Figure 1. Lower Forward Cell (Sheet 1 of 1)

Lower Forward Cells - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure



DO NOT HANDLE CELL WHEN AMBIENT TEMPERATURE IS BELOW 65°F (18°C).

NOTE

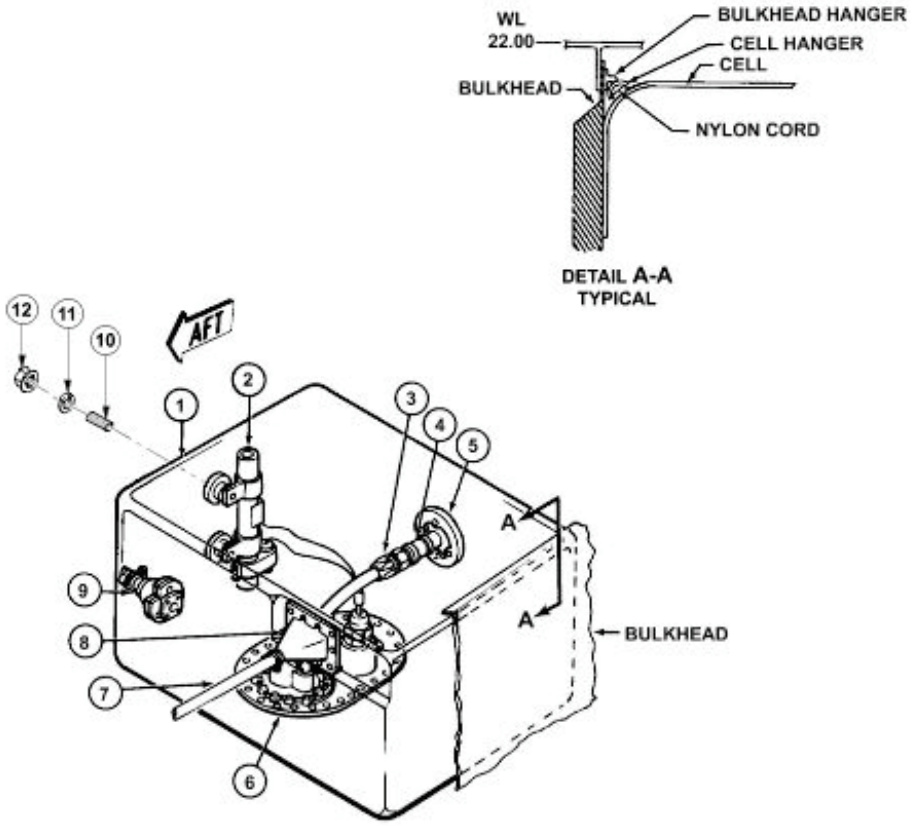
The following removal procedures for left and right forward cells are the same.

1. Disconnect electrical power.
2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).
3. Remove cabin floor panels above cell to be removed. Remove access doors on underside of helicopter directly below respective lower forward cell.
4. Remove sump assembly (6, Figure 1) (412-A-28-10-00-57A-520A-A / 00509).

NOTE

Check valve (4) and hose (3) can remain connected to interconnect fitting and be removed with fitting.

5. Remove interconnect fitting (5) (412-A-28-10-00-47A-520A-A / 00473).
6. Remove interconnect fitting (9) (412-A-28-10-00-40A-520A-A / 00453).
7. Remove vent hose (7) (412-A-28-10-00-39A-520A-A / 00450).
8. Remove nylon cord from hangers around top of cell and cell cavity.
9. While holding the setscrew (10) with an allen key, remove the nut (11) and washer (12).
10. With the allen key, remove the setscrew (10).
11. Lift cell from cavity.
12. Install caps or plugs (C-428) on all open fuel lines and over cell openings.



1. Lower forward cell
2. Fuel quantity probe
3. Hose
4. Check valve
5. Lower forward cell/lower middle cell interconnect fitting
6. Sump assembly
7. Vent hose
8. Retainer
9. Interconnect fitting
10. Setscrew
11. Washer
12. Nut

ICN-412-A-281000-B-97499-00044-A-001-01

Figure 1. Lower Forward Cell (Sheet 1 of 1)

Lower Forward Cells - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Soap	100066	AR	Zodiac Aerospace (www.zodiacaerospace.com)
Cleaning Solution	601199	AR	Zodiac Aerospace (www.zodiacaerospace.com)
Clean cloth	C-516	AR	

Safety conditions



OBEY ALL THE SAFETY PRECAUTIONS WHEN YOU DO MAINTENANCE ON FUEL SYSTEM EQUIPMENT.



OBEY ALL THE FUEL CELL HANDLING PRECAUTIONS DURING CLEANING OR DAMAGE TO THE CELL MAY OCCUR.



DO NOT ALLOW SOLVENTS OR COMPOUNDS TO MAKE CONTACT WITH PARTS MADE OF ELASTOMERIC MATERIAL OR DAMAGE WILL OCCUR.

Procedure

NOTE

The following cleaning procedures for left and right forward cells are the same.

NOTE

Any evidence of fuel leakage must be identified before cleaning.

NOTE

Refer to the document "FUEL-F6101-SPM20-00" (Zodiac Aerospace), for additional information on the cleaning of the fuel cells.

1. Clean the internal surface of the fuel cell with a clean cloth (C-516) moistened with a cleaning solution (601199). Dry the liner with low-pressure filtered compressed air for 2 to 4 hours, at approximately 68°F (20°C).
2. Clean small areas on the external surface of the fuel cell with a clean cloth (C-516) moistened with a cleaning solution (601199). Dry with a clean cloth (C-516).
3. Clean an extensive area on the external surface of the fuel cell with a solution of soap (100066) and hot water. Rinse with water and dry with a clean cloth (C-516).
4. Dry clean the fittings of the fuel cell with filtered compressed air and a clean cloth (C-516) to remove any blockage or obstruction.
5. Clean fungus growth found on the fuel cell or on the fuel cell cavity per the instructions contained in the BHT-ALL-SPM, Chapter 5, Standard Practices Manual.

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Lower Forward Cells - Inspection

Procedure

NOTE

The following inspection procedures for left and right forward cells are the same.

1. Inspect cell for cracks, scuffing, and general condition.

2. Inspect threaded inserts for serviceability. Any defect which can be detected visually is not acceptable.

3. Inspect hoses for damage and deterioration. Any hose which exhibits damage or deterioration shall be replaced.

4. Inspect removed components for damage.

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Lower Forward Cells - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Talcum powder	C-408	AR	
Lacing cord	C-471	AR	

Procedure

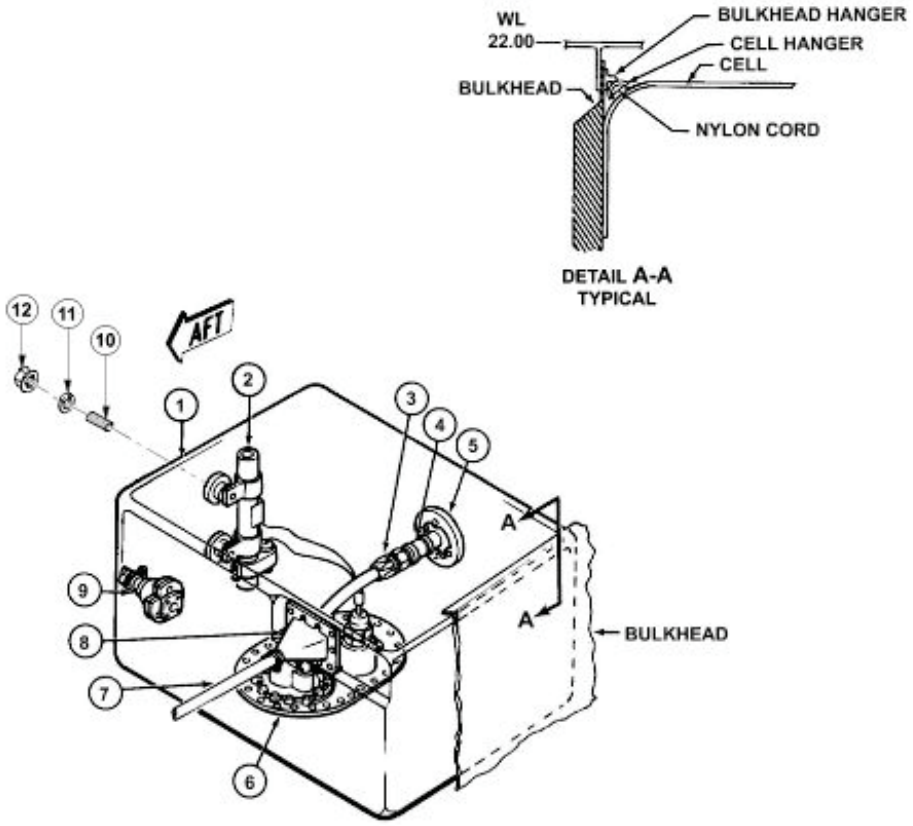


DO NOT HANDLE CELLS WHEN AMBIENT TEMPERATURE IS BELOW 65°F (18°C).

NOTE

The following installation procedures for left and right forward cells are the same.

1. Remove protective caps, plugs, and covers. Inspect cell cavity for clean, smooth condition of surfaces.
2. Apply talcum powder (C-408) to cavity surfaces.
3. Place cell in cavity and align interconnect port with port on lower middle cell and bulkhead opening.
4. With an allen key, install the setscrew (10, Figure 1) in the lower forward cell (1) through the bulkhead.
5. Make four turns of the setscrew (10) with the allen key.
6. While holding the setscrew (10) in place with the allen key, install the washer (11) and nut (12). Tighten the nut.
7. Install lower forward cell interconnect fitting (5) (412-A-28-10-00-47A-720A-A / 00474).
8. Install sump assembly (6) (412-A-28-10-00-41A-720A-A / 00458).
9. Lace and tie a single length of lacing cord (C-471) between bulkhead hangers around top of cell and top of cell cavity (Detail A-A).
10. Install interconnect fitting (9) (412-A-28-10-00-40A-720A-A / 00454).
11. Install vent hose (8) (412-A-28-10-00-39A-720A-A / 00451).
12. Install cabin floor panels. Install access doors on under side of helicopter.
13. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).
14. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).
15. Perform fuel system operational check.
16. Check area for leaks.



1. Lower forward cell
2. Fuel quantity probe
3. Hose
4. Check valve
5. Lower forward cell/lower middle cell interconnect fitting
6. Sump assembly
7. Vent hose
8. Retainer
9. Interconnect fitting
10. Setscrew
11. Washer
12. Nut

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Figure 1. Lower Forward Cell (Sheet 1 of 1)

Fuel Vent Valves - Description

NOTE

Maintenance procedures for the left and right fuel vent manifolds are the same, except where noted.

Fuel vent manifolds are located inside the top of the left and right doorposts behind the pilot and copilot seats. The fuel vent manifolds are attached to the lower forward cell, lower middle cell, and overboard vent hoses on each side of the helicopter.

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Fuel Vent Manifolds - Removal

Procedure

NOTE

The following removal procedures for the left and right fuel vent manifolds are the same, except where noted.

1. Disconnect electrical power.

2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).

NOTE

The upper inboard circular panel on the right side is attached with blind rivets.

3. Remove the upper inboard and upper aft doorpost circular access panels.

4. Remove straps (1, Figure 1) on boot (2).

5. Move boot (2) away from fuel vent manifold (8) to gain access to overboard vent hose (3).

6. Remove overboard vent hose (3).

7. Remove screws (14) and washers (13).

8. Move retainer (12) from boot (11).

9. Remove straps (15).

NOTE

There is faying surface sealant between the boot (11) and fuel vent manifold (8).

10. Move boot (11) away from fuel vent manifold (8) to gain access to fuel cell vent hoses (16 and 17).

11. Remove lower forward cell vent hose (17) from forward union (10) and identify for installation.

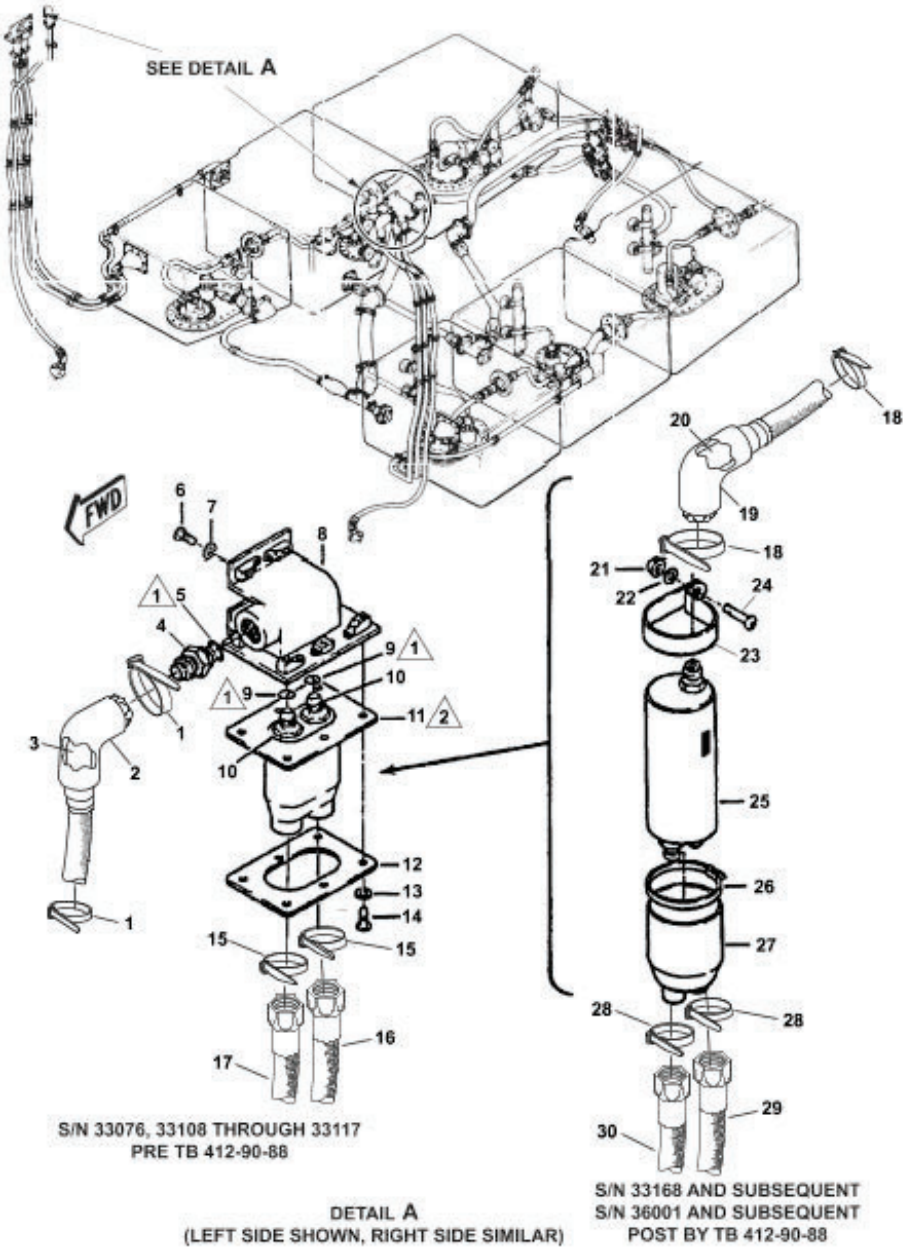
12. Remove lower middle cell vent hose (16) from aft union (10) and identify for installation.

13. Remove screws (6) and washers (7).

14. Remove fuel vent manifold (8).

15. Remove unions (4 and 10) and packings (5 and 9). Discard packings.

16. Install protective caps and covers over hose assemblies and openings.



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Figure 1. Fuel Vent Manifold (Sheet 1 of 2)

1. Strap
2. Boot
3. Overboard vent hose
4. Union
5. Packing
6. Screw
7. Washer
8. Fuel vent manifold
9. Packing
10. Union
11. Boot
12. Retainer
13. Washer
14. Screw
15. Strap
16. Lower middle cell vent hose
17. Lower forward cell vent hose
18. Strap
19. Boot
20. Overboard vent hose
21. Nut
22. Washer
23. Clamp
24. Screw
25. Fuel vent manifold
26. Strap
27. Boot
28. Strap
29. Lower middle cell vent hose
30. Lower forward cell vent hose

NOTES

1. Install packing with petrolatum (C-008).
2. Apply sealant (C-308) to boot faying surface before installing on manifold.

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Figure 1. Fuel Vent Manifold (Sheet 2 of 2)

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Fuel Vent Manifolds - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Petrolatum	C-008	AR	
Sealant	C-308	AR	

Procedure

NOTE

The following installation procedures for the left and right fuel vent manifolds are the same, except where noted.

1. Install packings (5 and 9, Figure 1) on unions (4 and 10) using petrolatum (C-008).

2. Install unions (4 and 10) in fuel vent manifold (8).

3. Remove protective caps and covers from hose assemblies and openings in doorpost.

4. Install fuel vent manifold (8) in doorpost using screws (6) and washers (7).

5. Ensure boot (11) and retainer (12) are located on lower forward cell vent hose (17) and lower middle cell vent hose (16).

6. Install lower forward cell vent hose (17) on forward union (10) of fuel vent manifold (8).

7. Install lower middle fuel cell vent hose (16) on aft union (10) of fuel vent manifold (8).

8. Apply sealant (C-308) to faying surface of boot (11).

9. Install retainer (12) and boot (11) on fuel vent manifold (8) with screws (14) and washers (13).

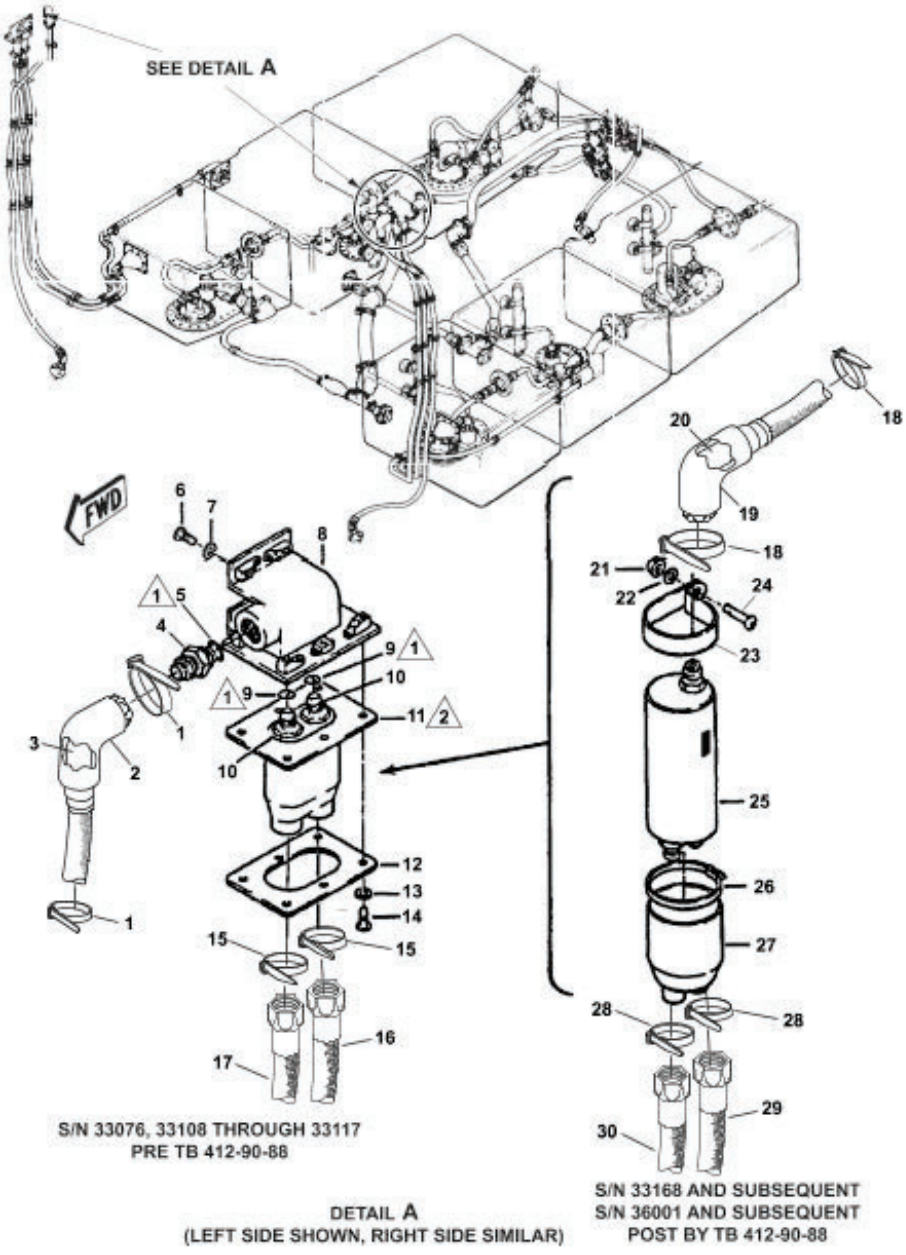
10. Install straps (15) on bottom of boot (11) around lower forward cell vent hose (17) and lower middle cell vent hose (16).

11. Ensure boot (2) is located on overboard vent hose (3).

12. Install overboard vent hose (3) on union (4) of fuel vent manifold (8).

13. Install boot (2) on fuel vent manifold (8) and overboard vent hose (3) with straps (1).

14. Install doorpost access panels.



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Figure 1. Fuel Vent Manifold (Sheet 1 of 2)

1. Strap
2. Boot
3. Overboard vent hose
4. Union
5. Packing
6. Screw
7. Washer
8. Fuel vent manifold
9. Packing
10. Union
11. Boot
12. Retainer
13. Washer
14. Screw
15. Strap
16. Lower middle cell vent hose
17. Lower forward cell vent hose
18. Strap
19. Boot
20. Overboard vent hose
21. Nut
22. Washer
23. Clamp
24. Screw
25. Fuel vent manifold
26. Strap
27. Boot
28. Strap
29. Lower middle cell vent hose
30. Lower forward cell vent hose

NOTES

1. Install packing with petrolatum (C-008).
2. Apply sealant (C-308) to boot faying surface before installing on manifold.

ICN-412-A-281000-B-97499-00046-A-001-01

Figure 1. Fuel Vent Manifold (Sheet 2 of 2)

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Fuel Vent Manifolds - Removal

Procedure

NOTE

The following removal procedures for the left and right fuel vent manifolds are the same, except where noted.

1. Disconnect electrical power.

2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).

NOTE

The upper inboard circular panel on the right side is attached with blind rivets.

3. Remove the upper inboard and upper aft doorpost circular access panels.

4. Remove straps (18, Figure 1) on boot (19).

5. Move boot (19) away from fuel vent manifold (25) to gain access to overboard vent hose (20).

6. Remove overboard vent hose (20).

7. Remove straps (26 and 28).

8. Move boot (27) away from fuel vent manifold (25) to gain access to fuel cell vent hoses (29 and 30).

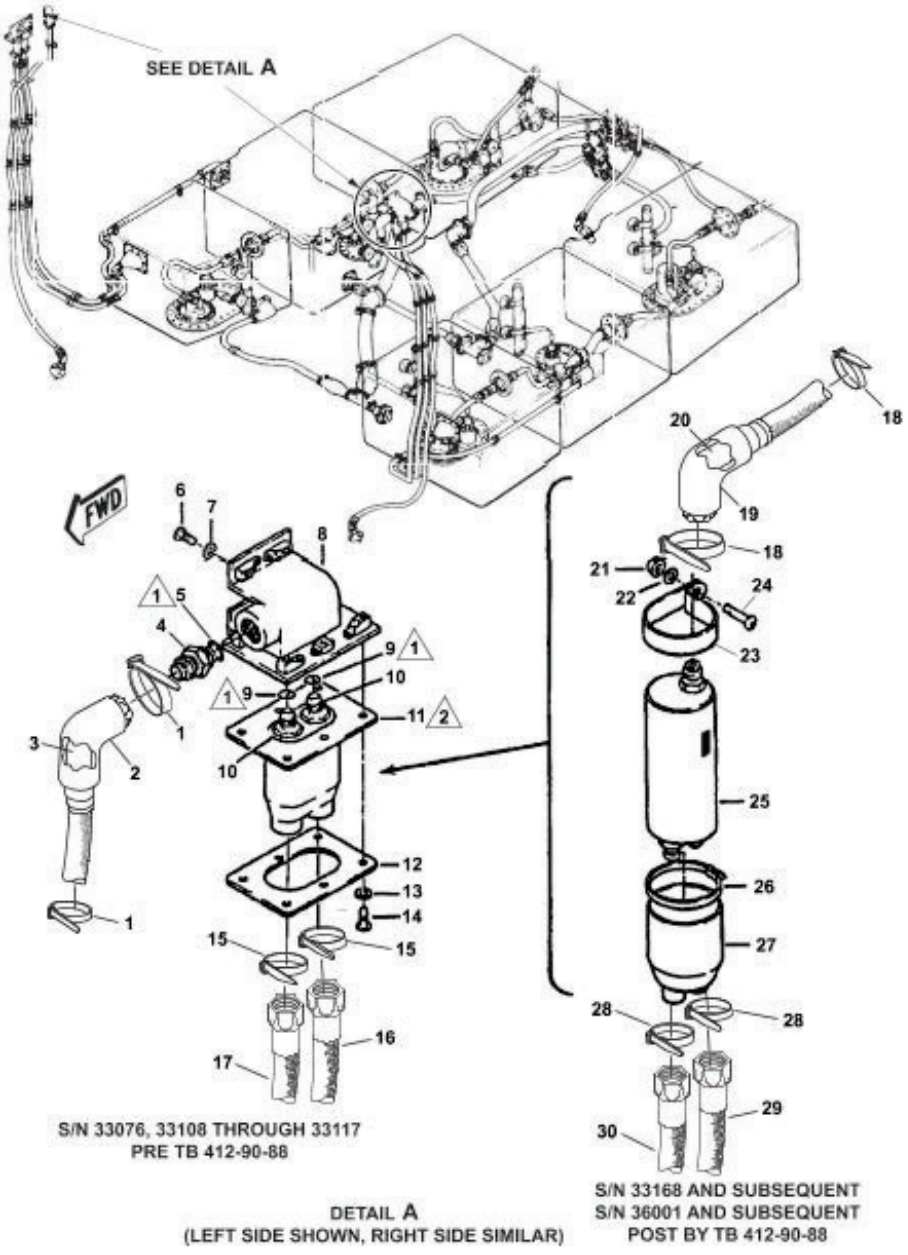
9. Remove lower forward cell vent hose (30) from forward union of fuel vent manifold (25) and identify for installation.

10. Remove lower middle cell vent hose (29) from aft union of fuel vent manifold (25) and identify for installation.

11. Remove nut (21), washer (22), screw (24), and clamp (23).

12. Remove fuel vent manifold (25).

13. Install protective caps and covers over hose assemblies and openings.



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Figure 1. Fuel Vent Manifold (Sheet 1 of 2)

1. Strap
2. Boot
3. Overboard vent hose
4. Union
5. Packing
6. Screw
7. Washer
8. Fuel vent manifold
9. Packing
10. Union
11. Boot
12. Retainer
13. Washer
14. Screw
15. Strap
16. Lower middle cell vent hose
17. Lower forward cell vent hose
18. Strap
19. Boot
20. Overboard vent hose
21. Nut
22. Washer
23. Clamp
24. Screw
25. Fuel vent manifold
26. Strap
27. Boot
28. Strap
29. Lower middle cell vent hose
30. Lower forward cell vent hose

NOTES

1. Install packing with petrolatum (C-008).
2. Apply sealant (C-308) to boot faying surface before installing on manifold.

ICN-412-A-281000-B-97499-00046-A-001-01

Figure 1. Fuel Vent Manifold (Sheet 2 of 2)

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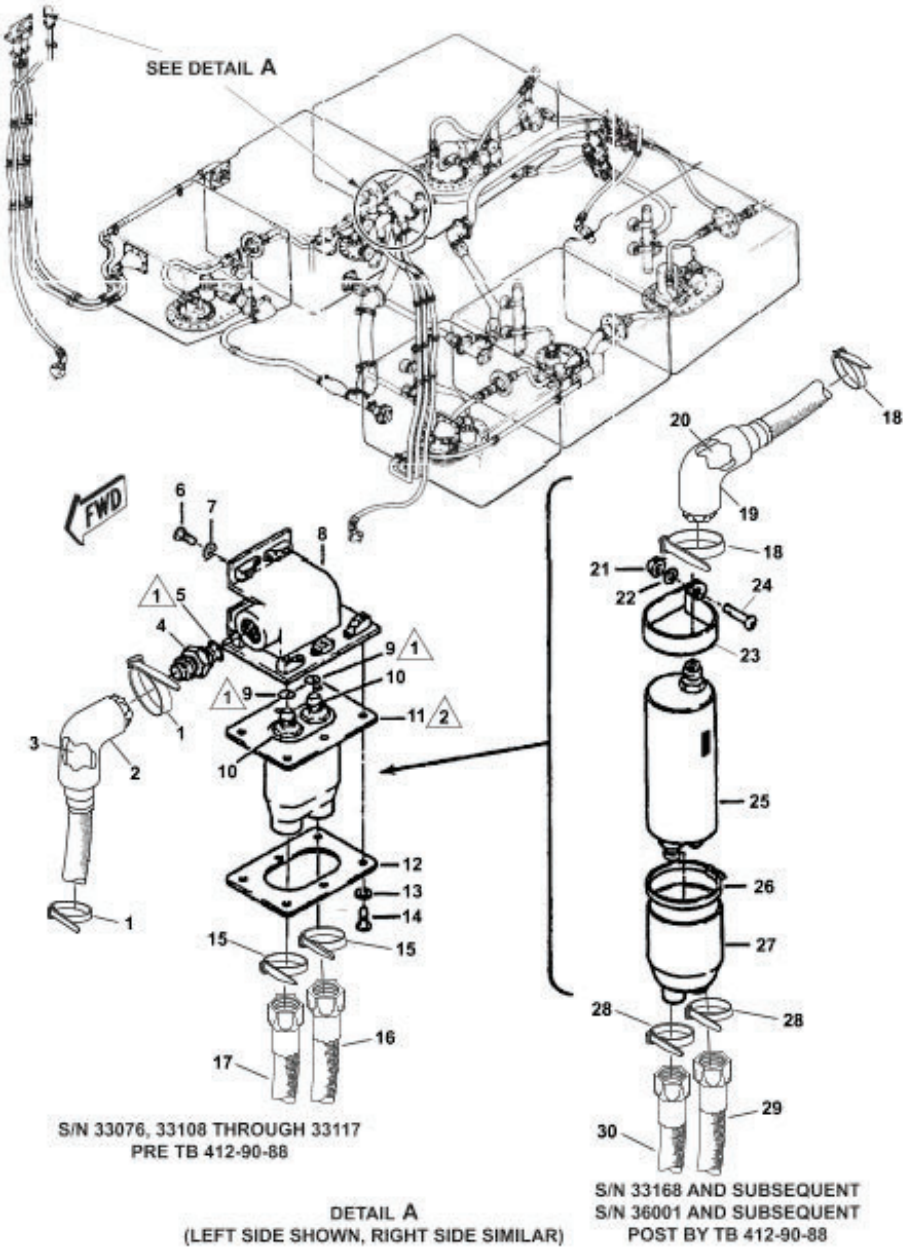
Fuel Vent Manifolds - Installation

Procedure

NOTE

The following installation procedures for the left and right fuel vent manifolds are the same, except where noted.

1. Remove protective caps and covers from hose assemblies and openings in doorpost.
2. Install fuel vent manifold (25, Figure 1) in doorpost using clamp (23), screw (24), nut (21), and washer (22).
3. Ensure unions on fuel vent manifold (25) align with vent hoses (20, 29, and 30).
4. Ensure boot (27) is located on lower forward cell vent hose (30) and lower middle cell vent hose (29).
5. Install lower forward cell vent hose (30) on forward union of fuel vent manifold (25).
6. Install lower middle cell vent hose (29) on aft union of fuel vent manifold (25).
7. Install boot (27) on fuel vent manifold (25) with strap (26).
8. Install straps (28) on bottom of boot (27) around lower forward cell vent hose (30) and lower middle cell vent hose (29).
9. Ensure boot (19) is located on overboard vent hose (20).
10. Install overboard vent hose (20) on upper union of fuel vent manifold (25).
11. Install boot (19) on fuel vent manifold (25) and overboard vent hose (20) with straps (18).
12. Install doorpost access panels.



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Figure 1. Fuel Vent Manifold (Sheet 1 of 2)

1. Strap
2. Boot
3. Overboard vent hose
4. Union
5. Packing
6. Screw
7. Washer
8. Fuel vent manifold
9. Packing
10. Union
11. Boot
12. Retainer
13. Washer
14. Screw
15. Strap
16. Lower middle cell vent hose
17. Lower forward cell vent hose
18. Strap
19. Boot
20. Overboard vent hose
21. Nut
22. Washer
23. Clamp
24. Screw
25. Fuel vent manifold
26. Strap
27. Boot
28. Strap
29. Lower middle cell vent hose
30. Lower forward cell vent hose

NOTES

1. Install packing with petrolatum (C-008).
2. Apply sealant (C-308) to boot faying surface before installing on manifold.

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Figure 1. Fuel Vent Manifold (Sheet 2 of 2)

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Lower Forward Cell Vent Hoses - Description

NOTE

Maintenance procedures for left and right lower forward cell vent hoses are the same.

Vent hoses are attached to vent ports on upper forward side of each lower forward cell. These vent hoses connect lower forward cells into forward venting system. Fume-tight boots are installed on hose connections to lower forward cell to prevent escape of fumes into surrounding areas.

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Lower Forward Cell Vent Hoses - Removal

Procedure

NOTE

The following removal procedures for left and right lower forward cell vent hoses are the same.

1. Disconnect electrical power.

2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).

3. Remove cabin floor panel.

4. Loosen and slip clamp (11, Figure 1) back on hose assembly (3).

5. Remove screws (1) and washers (2).

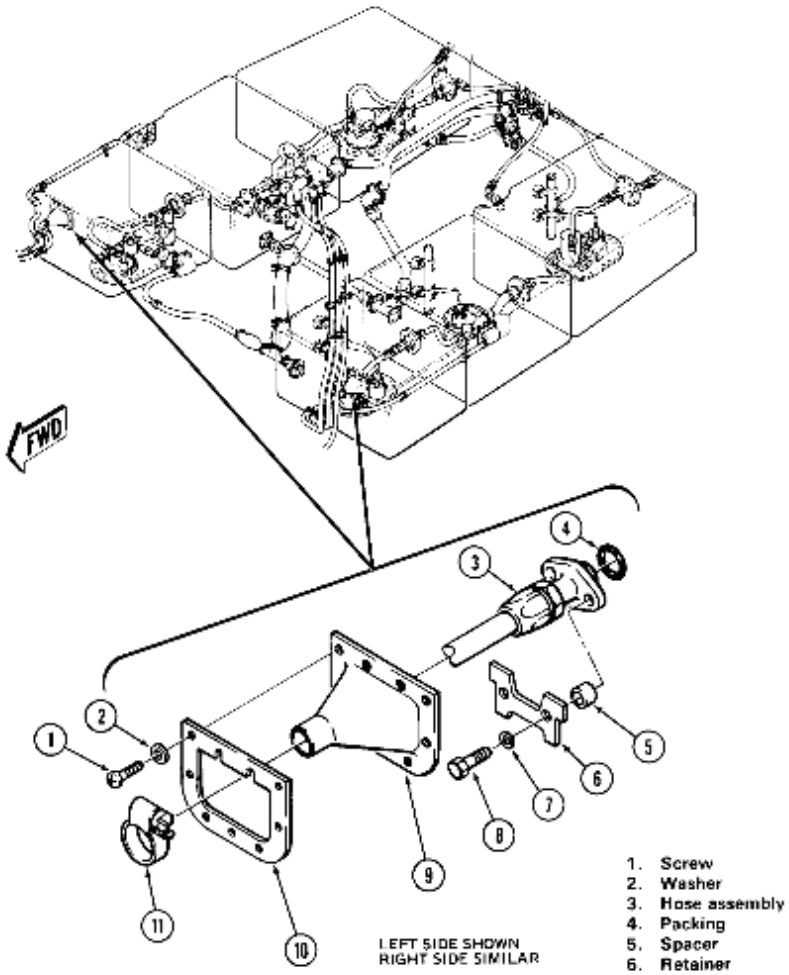
6. Loosen retainer (10) and slip back on hose assembly (3).

7. Loosen boot (9) and work back on hose assembly (3).

8. Remove bolts (8), washers (7), retainer (6), and spacers (5).

9. Remove hose assembly (3) and packing (4). Discard packing.

10. Install protective covers over cell opening and hose assembly (3).



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Figure 1. Lower Forward Cell Vent Hose (Sheet 1 of 1)

Lower Forward Cell Vent Hoses - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

NOTE

The following installation procedures for left and right lower forward cell vent hoses are the same.

1. Remove protective caps and covers.

2. Install packing (4, Figure 1) on end of hose assembly (3). If required, apply assembly fluid (C-024) to hold packing in place.

NOTE

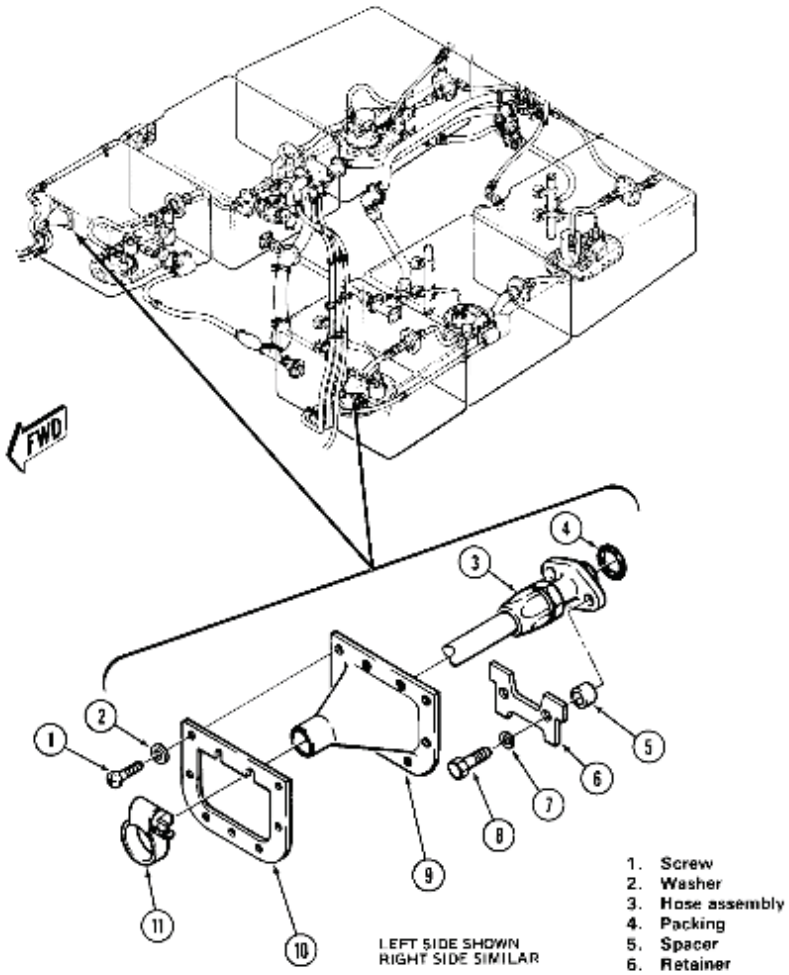
When installing new hose assembly (3), ensure boot (9), retainer (10), and clamp (11) are on hose assembly in proper order, and pushed back on hose assembly for access to end of hose.

3. Hold end of hose assembly (3) close to vent port of cell.

4. Install bolts (8), washers (7), retainer (6), spacers (5), and hose assembly (3) on face of cell vent port. Tighten bolts (8).

5. Slip boot (9) and retainer (10) in place over end of hose assembly (3). Install screws (1), washers (2), retainer (10), and boot (9) on vent port of cell.

6. Slip clamp (11) over back end of boot (9) and tighten around hose assembly (3).



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Figure 1. Lower Forward Cell Vent Hose (Sheet 1 of 1)

Lower Forward Cell Interconnect Hose and Fitting - Description

NOTE

Both the interconnect hose and fitting are included in removal and installation procedures. If both are not to be removed and/or installed, do only applicable steps.

A fitting is connected directly to the inboard interconnect port of the left lower forward cell. An interconnect hose is attached to one port of the fitting and to the inboard interconnect port of the right lower forward cell. An interconnect hose is attached to the remaining port of the fitting and to the upper forward center cell output port.

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Lower Forward Cell Interconnect Hose and Fitting - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure

1. Disconnect electrical power.

2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).

3. Remove cabin floor panels above left and right lower forward cells.

4. Remove strap (9, Figure 1) from boot (5).

5. Remove screws (8) and washers (7).

6. Loosen retainer (6) and slide back on hose assembly (2).

7. Loosen boot (5) and slide back on hose assembly (2).

8. Remove bolts (4) and washers (3). Remove the retainer (47) and spacer (48), if installed.

9. Remove hose assembly (2) from cell interconnect port. Remove and discard packing (1).

10. Remove straps (10 and 12).

11. Slip boot (11) over connection and back on hose assembly (2).

12. Remove bolts (13), washers (16), and nuts (17).

13. Remove hose assembly (2) and packing (14). Discard packing.

14. Remove straps (33 and 34).

15. Slip boot (35) over connection and back on hose assembly.

16. Remove bolts (18), washers (24), and nuts (25).

17. Remove hose assembly (26) and packing (23) from fitting (15). Discard packing.

18. Remove strap (36).

19. Remove the screws (37), washers (40), nuts (41), and doubler (38).

20. Slip the boot (39) forward on the hose assembly (26).

21. Remove the bolts (42), washers (43), and nuts (44).

22. Remove the hose assembly (26) and the packing (45) from the hose assembly (46). Discard the packing.

23. Remove strap (28).

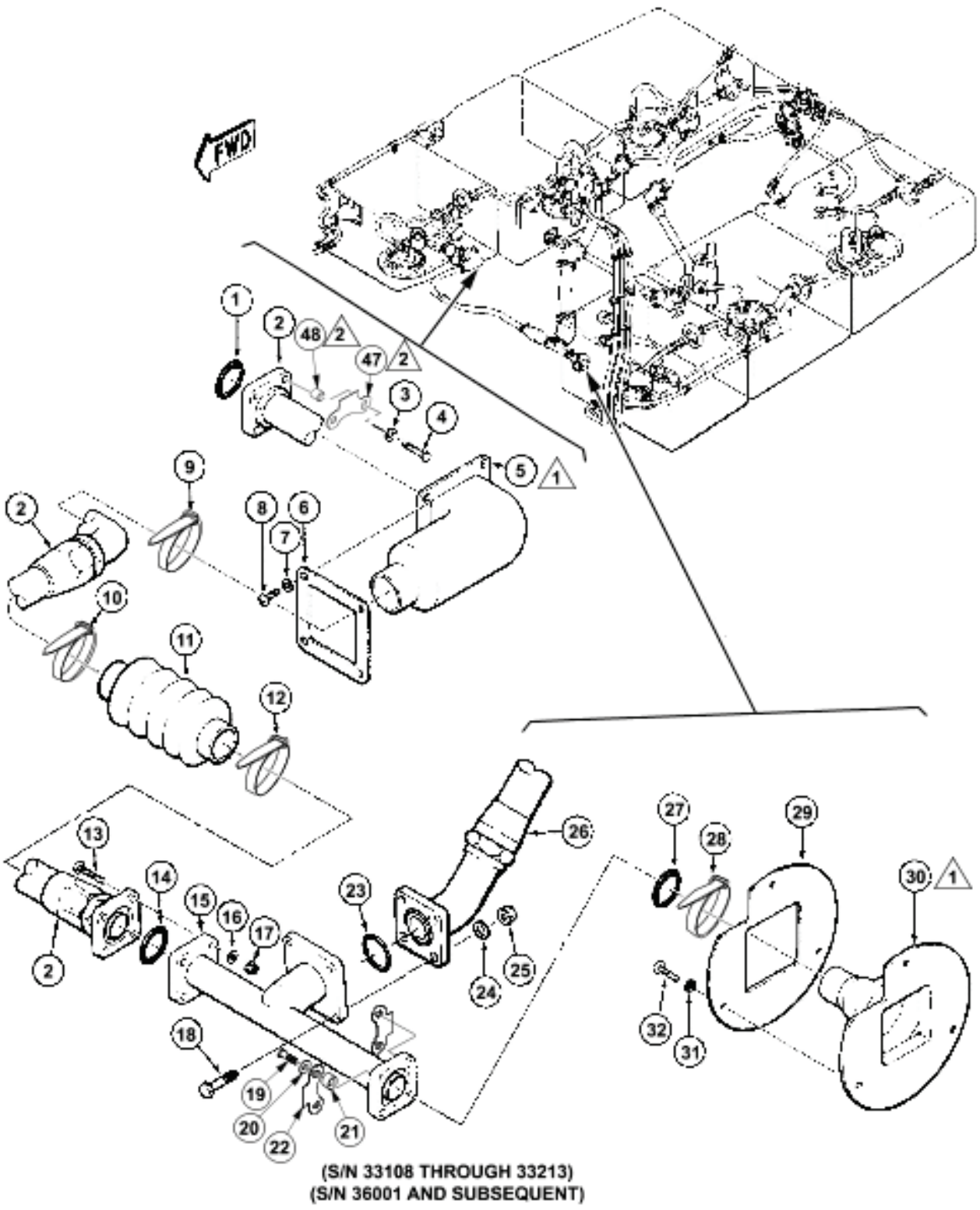
24. Remove screws (32) and washers (31).

25. Loosen retainer (29) and boot (30), and slide both back on fitting (15) for access to cell connection.

26. Remove bolts (19), washers (20), retainers (22), and spacers (21).

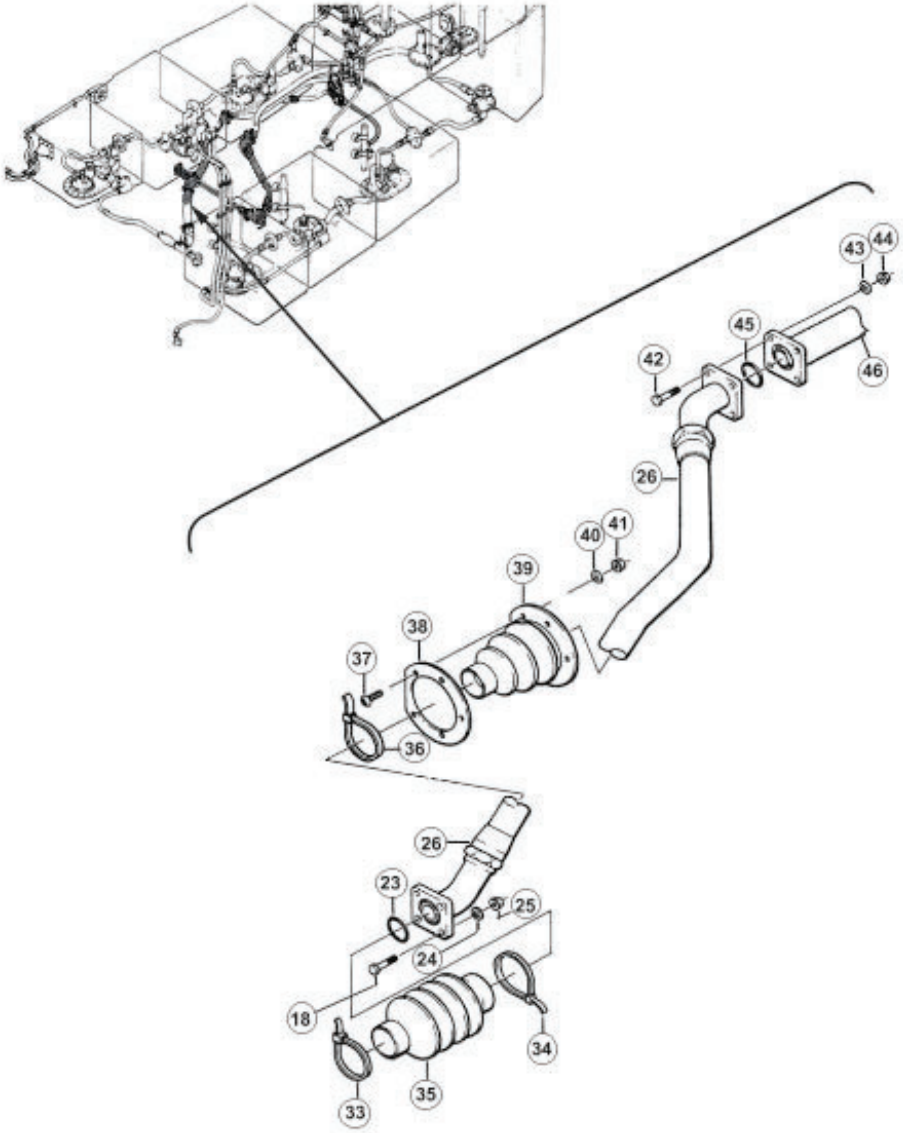
27. Remove fitting (15) and packing (27) from interconnect port of cell. Discard packing.

28. Install caps or plugs (C-428) over cell opening and open end of fitting (15).



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Figure 1. Lower Forward Cell Interconnect Hose and Fitting (Sheet 1 of 3)



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Figure 1. Lower Forward Cell Interconnect Hose and Fitting (Sheet 2 of 3)

1. Packing
2. Hose assembly
3. Washer
4. Bolt
5. Boot
6. Retainer
7. Washer
8. Screw
9. Strap
10. Strap
11. Boot
12. Strap
13. Bolt
14. Packing
15. Fitting
16. Washer
17. Nut
18. Bolt
19. Bolt
20. Washer
21. Spacer
22. Retainer
23. Packing
24. Washer
25. Nut
26. Hose assembly
27. Packing
28. Strap
29. Retainer
30. Boot
31. Washer
32. Screw
33. Strap
34. Strap
35. Boot
36. Strap
37. Screw
38. Doubler
39. Boot
40. Washer
41. Nut
42. Bolt
43. Washer
44. Nut
45. Packing
46. Hose assembly
47. Retainer
48. Spacer

NOTES

1. Seal between boot and structure with sealant (C-308).

2. Retainer (47) and spacer (48) is applicable if fuel cell P/N 412-361-706-101 is installed.

ICN-412-A-281000-B-97499-00050-A-001-01

Figure 1. Lower Forward Cell Interconnect Hose and Fitting (Sheet 3 of 3)

Lower Forward Cell Interconnect Hose and Fitting - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	
Sealant	C-308	AR	

Procedure

1. Apply assembly fluid (C-024) to packing (27, Figure 1).

2. Place packing (27) and fitting (15) on left forward cell interconnect port.

3. Place retainer (29) and boot (30) on fitting (15), and slide back on body of fitting.

4. Install bolts (19), washers (20), retainers (22), spacers (21), and fitting (15). Tighten bolts.

5. Apply sealant (C-308) to face of boot (30).

6. Slide boot (30) over end of fitting (15).

7. Install screws (32) and washers (31). Tighten screws.

8. Remove excess sealant (C-308) from around boot (30).

9. Install strap (28) on boot (30).

10. On the aft end of hose assembly (26), slide the doubler (38), and then the boot (39) on the hose assembly.

11. Put the hose assembly (26) in position in the helicopter.

12. Apply assembly fluid (C-024) to packing (45).

13. Place the packing (45) and flange of hose assembly (26) on the flange of hose assembly (46).

14. Install the bolts (42), washers (43), and nuts (44). Tighten the nuts.

15. Apply sealant (C-308) to face of boot (39).

16. Install boot (39) with doubler (38), screws (37), washers (40), and nuts (41). Tighten the nuts.

17. Remove the excess sealant from around the boot (39).

18. Install the strap (36) on the boot (39).

19. On the forward end of the hose assembly (26), slide the boot (35) on the hose assembly.

20. Apply assembly fluid (C-024) to packing (23).

21. Place packing (23) and flange of hose assembly (26) on flange of fitting (15).

22. Install bolts (18), washers (24), and nuts (25). Tighten bolts.

23. Slide the boot (35) forward over the hose assembly (26) to the fitting (15) connection.

24. Install straps (33 and 34) on boot (35).

25. Lubricate packing (14) with assembly fluid (C-024).

26. Ensure boot (11) is slid back on hose assembly (2).

27. Place packing (14) and flange of hose assembly (2) on flange of fitting (15).

28. Install bolts (13), washers (16), and nuts (17). Tighten bolts.

29. Slip end of boot (11) over connection and secure with strap (12).

30. Install strap (10) on other end of boot (11).

31. Ensure retainer (6), and boot (5) are over end of hose assembly (2) in proper order.

32. Apply assembly fluid (C-024) to packing (1).

33. Place packing (1) and flange of hose assembly (2) on surface of right lower forward cell interconnect port.

34. Install the spacer (48) and retainer (47), if applicable. Install the bolts (4) and washers (3). Tighten bolts (4).

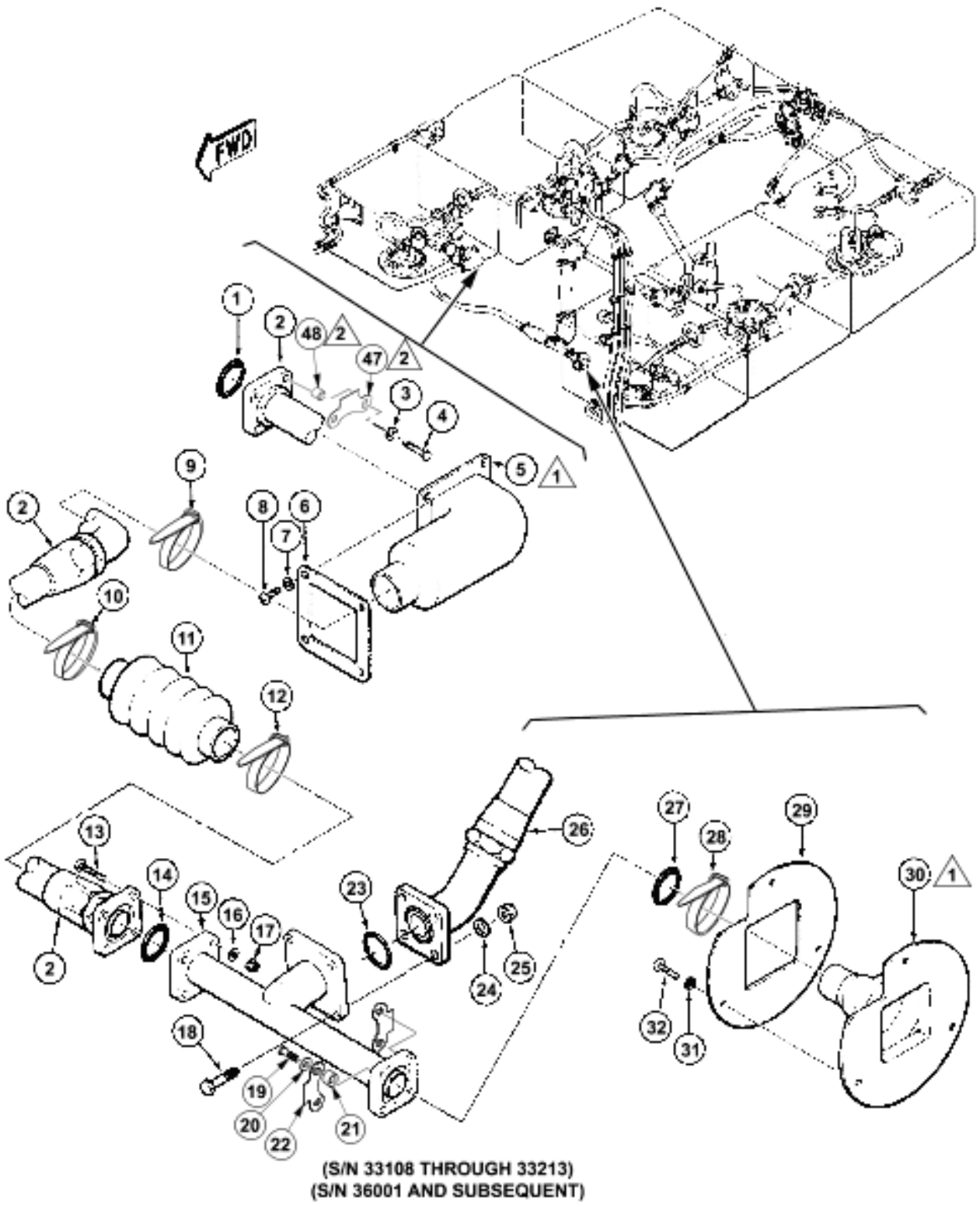
35. Apply sealant (C-308) to face of boot (5).

36. Slip boot (5) in place over connection. Place retainer (6) over flat mounting surface of boot (5).

37. Install screws (8) and washers (7). Tighten screws.

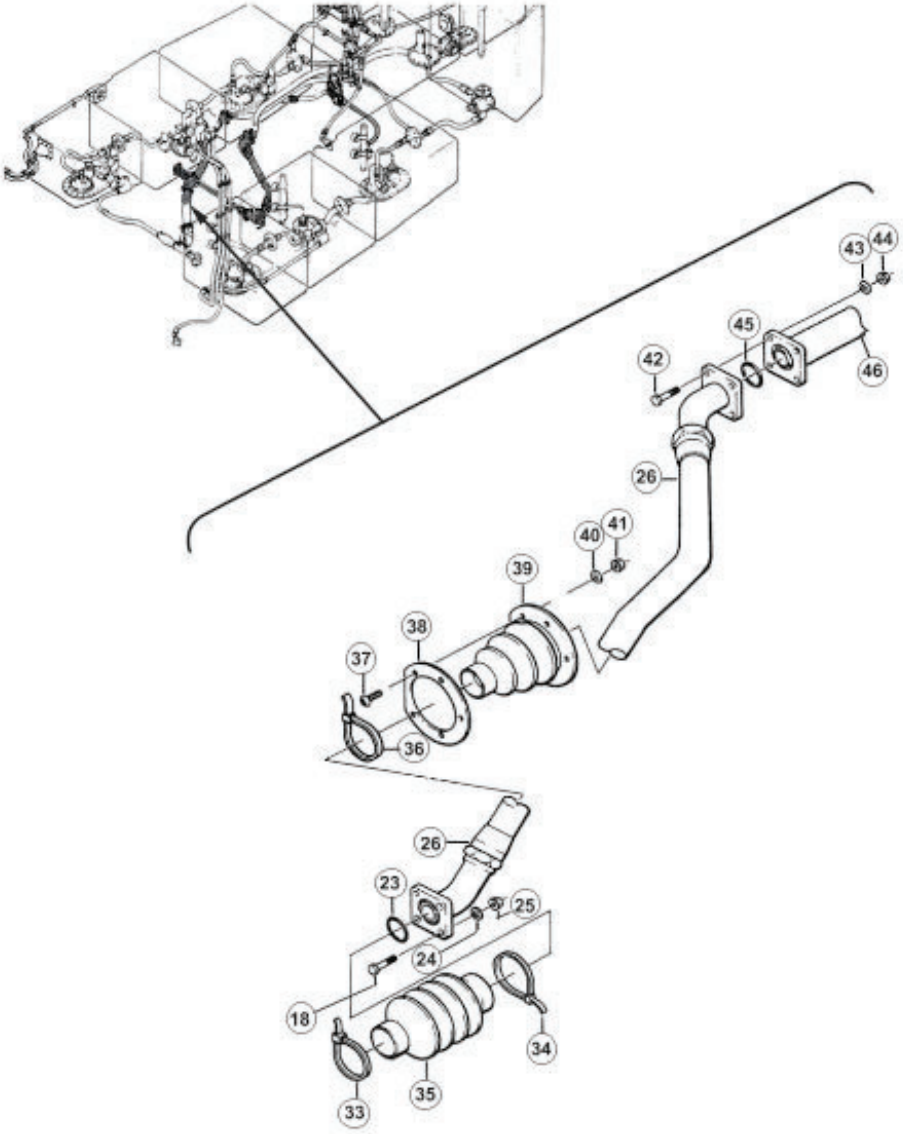
38. Remove excess sealant from around boot (5).

39. Install strap (9) on boot (5).



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Figure 1. Lower Forward Cell Interconnect Hose and Fitting (Sheet 1 of 3)



ICN-412-A-281000-B-97499-00049-A-001-01

Figure 1. Lower Forward Cell Interconnect Hose and Fitting (Sheet 2 of 3)

1. Packing
2. Hose assembly
3. Washer
4. Bolt
5. Boot
6. Retainer
7. Washer
8. Screw
9. Strap
10. Strap
11. Boot
12. Strap
13. Bolt
14. Packing
15. Fitting
16. Washer
17. Nut
18. Bolt
19. Bolt
20. Washer
21. Spacer
22. Retainer
23. Packing
24. Washer
25. Nut
26. Hose assembly
27. Packing
28. Strap
29. Retainer
30. Boot
31. Washer
32. Screw
33. Strap
34. Strap
35. Boot
36. Strap
37. Screw
38. Doubler
39. Boot
40. Washer
41. Nut
42. Bolt
43. Washer
44. Nut
45. Packing
46. Hose assembly
47. Retainer
48. Spacer

NOTES

1. Seal between boot and structure with sealant (C-308).

2. Retainer (47) and spacer (48) is applicable if fuel cell P/N 412-361-706-101 is installed.

ICN-412-A-281000-B-97499-00050-A-001-01

Figure 1. Lower Forward Cell Interconnect Hose and Fitting (Sheet 3 of 3)

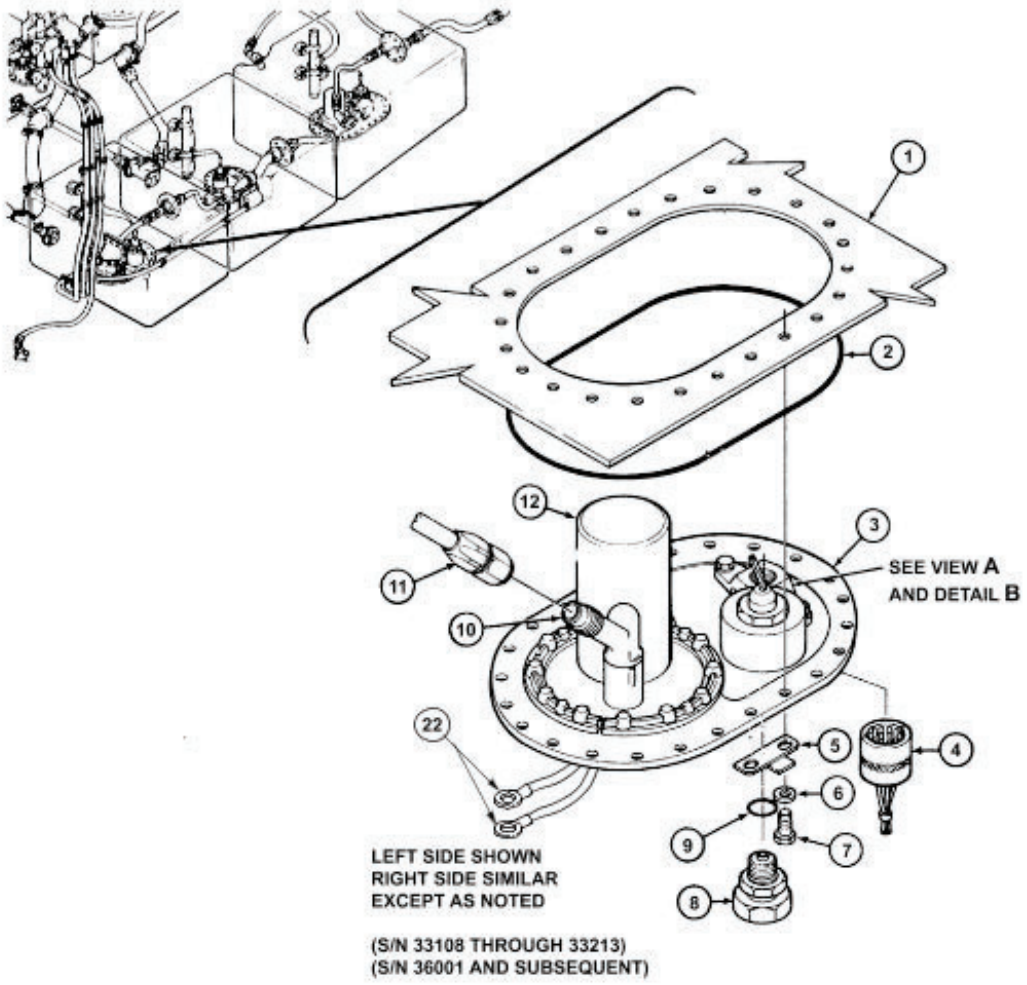
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Lower Forward Cell Sump Assemblies - Description

NOTE

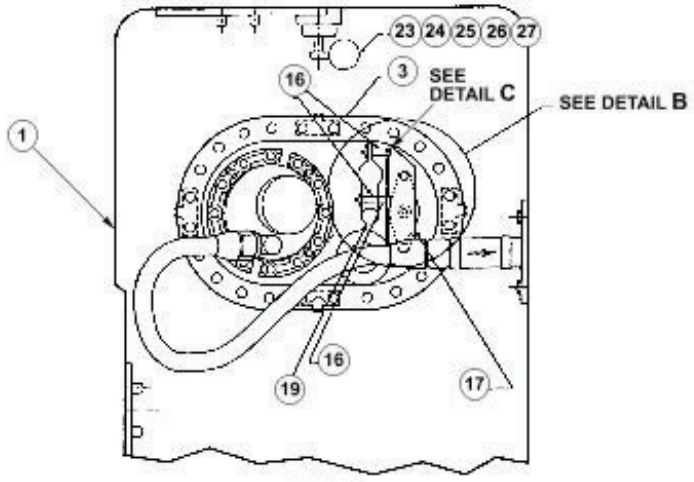
Maintenance procedures are the same for both lower forward cell sump assemblies except as noted in Figure 1.

Sump assemblies (2 and 29, Figure 2) are mounted over the opening on bottom of each lower forward cell. Removal of sump assembly provides access for maintenance of transfer pump, drain valve, and internally mounted components including check valve, fuel quantity probe, and fuel level sensor. The transfer pump and drain valve can both be removed without removing sump assembly.

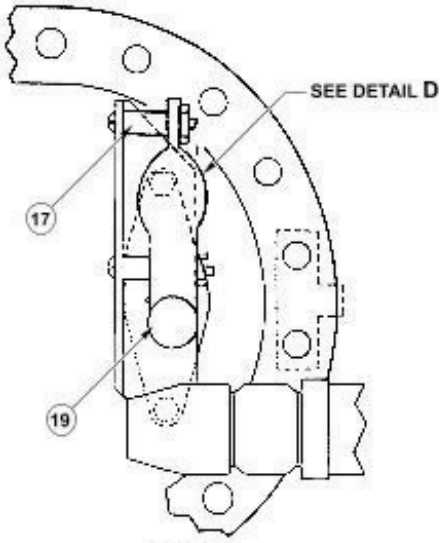


ICN-412-A-281000-B-97499-00051-A-001-01

Figure 1. Lower Forward Cell Sump Assembly (Sheet 1 of 3)



VIEW A
LOOKING DOWN (LHS SHOWN, RHS TYPICAL)
PRE TB 412-95-133



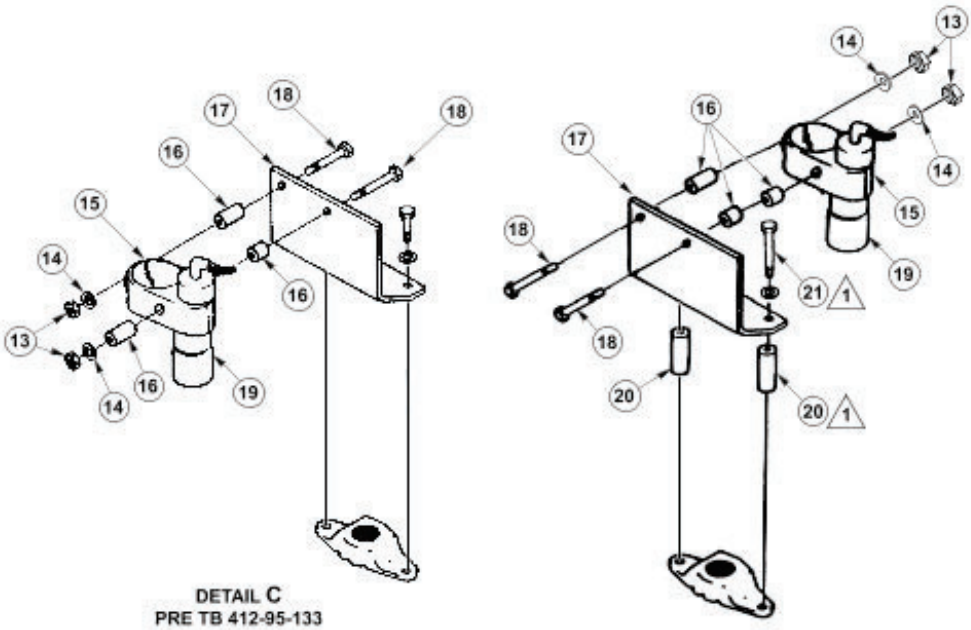
DETAIL B
S/N 36087, 36096 AND 36110 AND SUBSEQUENT
POST TB 412-95-133 (1)

NOTE

(1) On helicopters S/N 33108 through 33213, 36001 through 36086, 36088 through 36095, and 36097 through 36109.

ICN-412-A-281000-B-97499-00052-A-001-01

Figure 1. Lower Forward Cell Sump Assembly (Sheet 2 of 3)



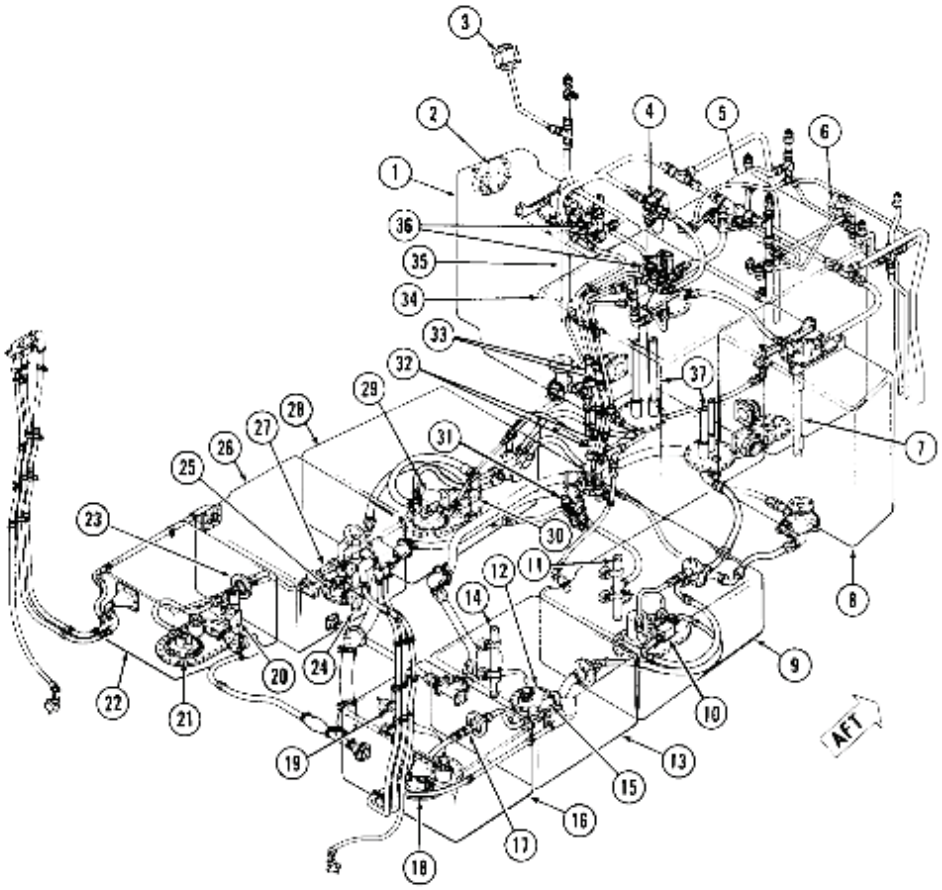
1. Lower forward cell
2. Packing
3. Sump assembly
4. Electrical connector
5. Retainer
6. Washer
7. Bolt
8. Drain valve
9. Packing
10. Fitting
11. Hose
12. Transfer pump
13. Nut
14. Washer
15. Clamp
16. Spacer
17. Bracket
18. Bolt
19. Fuel sensor (thermistor)
20. Spacer (NAS43DD3-96N)
21. Bolt (NAS6203.28)
22. Transfer pump electrical leads
23. Fuel level transmitter
24. Bolt
25. Washer
26. Spacer
27. Mounting bracket

NOTE

1 Right-hand installation only, Post TB 412-95-133.

ICN-412-A-281000-B-97499-00053-A-001-01

Figure 1. Lower Forward Cell Sump Assembly (Sheet 3 of 3)



1. Upper right cell
2. Filler cap and adapter assembly
3. Right fuel pressure transmitter
4. Crossfeed valve
5. Upper aft center cell
6. Left fuel pressure transmitter
7. Upper left cell fuel quantity probe
8. Upper left cell
9. Lower left main cell
10. Lower left main cell sump area consisting of:
    - Fuel sump assembly
    - Defueling valve
    - Sump drain valve
    - Flow switch check valve
    - Boost pump
11. Lower left main cell fuel quantity probe

(S/N 33108 THRU 33213)
(S/N 36001 AND SUBSEQUENT)

ICN-412-A-281000-B-97499-00039-A-001-01

Figure 2. Fuel Storage and Distribution System (S/N 33108 and Sub.) (Sheet 1 of 2)

12. Lower left middle cell sump area of:
    - Fuel sump assembly
    - Sump drain valve
13. Lower left middle cell
14. Lower left middle cell fuel quantity probe
15. Lower left middle cell ejector pump
16. Lower left forward cell
17. Lower left forward cell check valve
18. Lower left forward cell sump area consisting of:
    - Fuel sump assembly
    - Sump drain valve
    - Transfer pump
19. Lower left forward cell fuel quantity probe
20. Lower right forward cell fuel quantity probe
21. Lower right forward cell sump area consisting of:
    - Fuel sump assembly
    - Sump drain valve
    - Transfer pump
22. Lower right forward cell
23. Lower right forward cell check valve
24. Lower right middle cell fuel quantity probe
25. Lower right middle cell sump area consisting of:
    - Fuel sump assembly
    - Sump drain valve
26. Lower right middle cell
27. Lower right middle cell ejector pump
28. Lower right main cell
29. Lower right main cell sump area consisting of:
    - Fuel sump assembly
    - Defueling valve
    - Sump drain valve
    - Flow switch check valve
    - Boost pump
30. Lower right main cell fuel quantity probe
31. Interconnect valve
32. Engine fuel feed line fuel pressure switches
33. Engine fuel feed line check valves
34. Upper forward center cell
35. Upper right cell fuel quantity probe
36. Shutoff valves
37. Standpipes

ICN-412-A-281000-B-97499-00040-A-001-01

Figure 2. Fuel Storage and Distribution System (S/N 33108 and Sub.) (Sheet 2 of 2)

Lower Forward Cell Sump Assemblies - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure

NOTE

The following removal procedures are the same for both lower forward cell sump assemblies except as noted in Figure 1.

1. Disconnect electrical power.

2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).

3. Remove sump access door.

4. Disconnect electrical connector (4, Figure 1).

5. Disconnect transfer pump electrical leads (22).

6. Remove bolts (7), washers (6), and retainers (5).

7. Lower the lower forward cell sump assembly (3) for access.

8. Disconnect hose (11).

NOTE

TB 412-95-133: Pre or TB 412-95-133: Post:

Record position of fuel sensor thermistor in accordance with (Figure 1).

NOTE

Ensure hardware stack-up is retained.

9. Remove nuts (13), washers (14), spacers (16), and bolts (18).

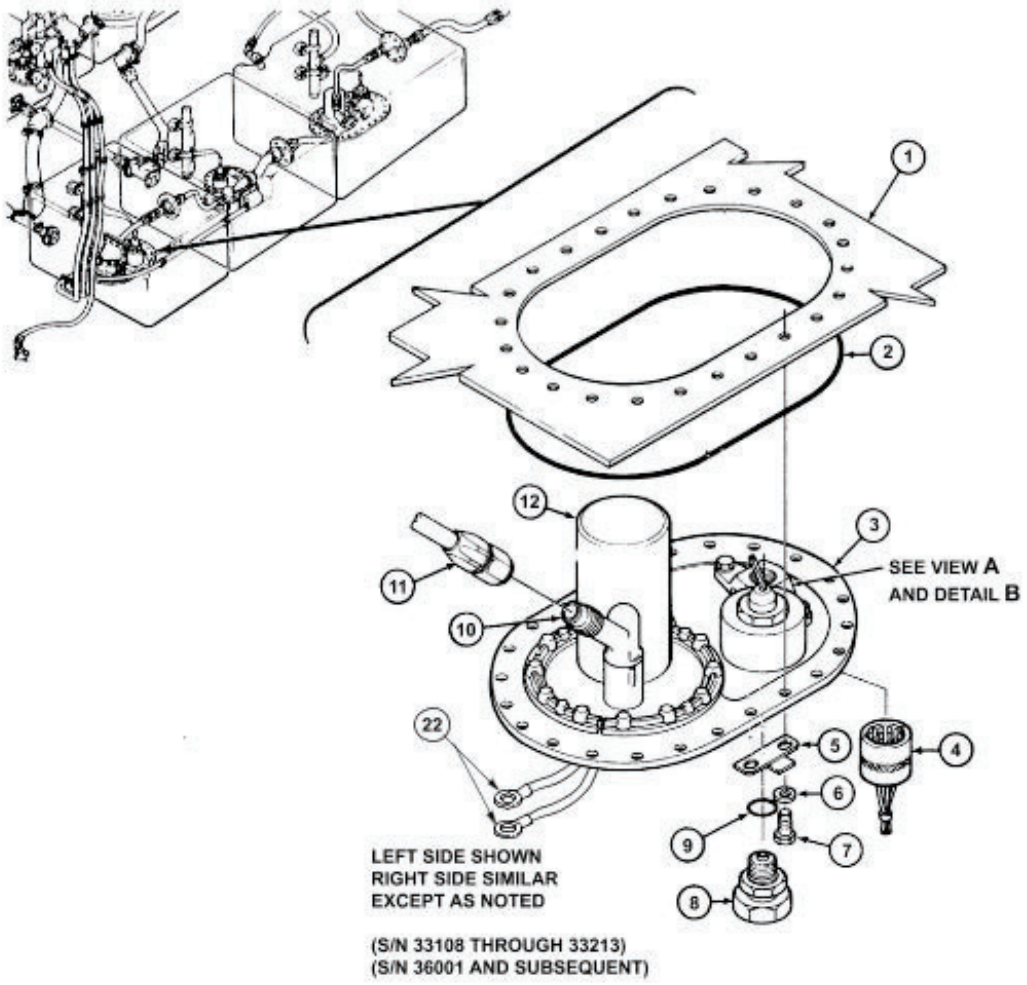
10. Remove fuel level sensor (thermistor) (19) from clamp (15) and lay fuel level sensor (thermistor) on the lower forward cell sump assembly (3).

11. Remove the bolts (24), washers (25), and spacers (26) from the mounting bracket (27).

12. Remove the fuel level transmitter (23) and lay on the lower forward fuel cell sump assembly (3).

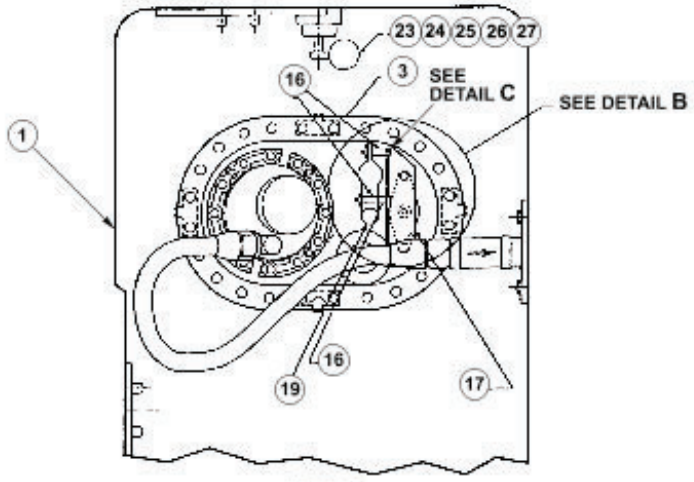
13. Remove the lower forward cell sump assembly (3), fuel level sensor (19), and fuel level transmitter (23).

14. Install caps or plugs (C-428) and covers over cell opening.

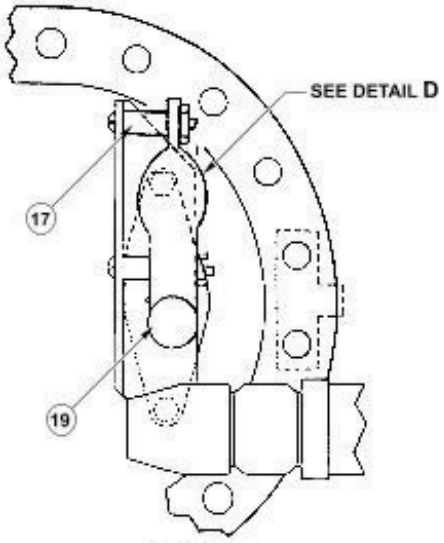


ICN-412-A-281000-B-97499-00051-A-001-01

Figure 1. Lower Forward Cell Sump Assembly (Sheet 1 of 3)



VIEW A
LOOKING DOWN (LHS SHOWN, RHS TYPICAL)
PRE TB 412-95-133



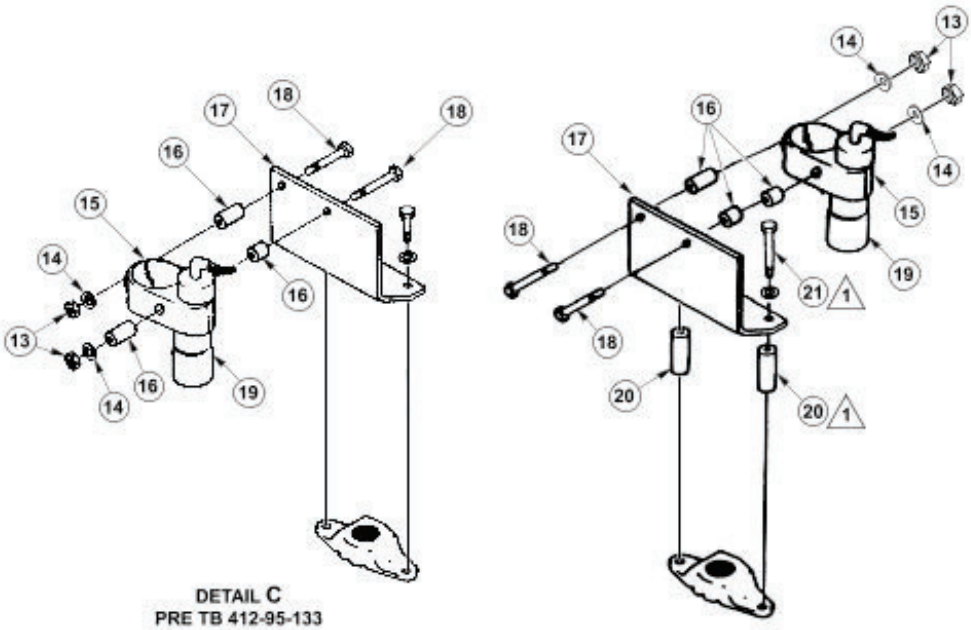
DETAIL B
S/N 36087, 36096 AND 36110 AND SUBSEQUENT
POST TB 412-95-133 (1)

NOTE

(1) On helicopters S/N 33108 through 33213, 36001 through 36086, 36088 through 36095, and 36097 through 36109.

ICN-412-A-281000-B-97499-00052-A-001-01

Figure 1. Lower Forward Cell Sump Assembly (Sheet 2 of 3)



1. Lower forward cell
2. Packing
3. Sump assembly
4. Electrical connector
5. Retainer
6. Washer
7. Bolt
8. Drain valve
9. Packing
10. Fitting
11. Hose
12. Transfer pump
13. Nut
14. Washer
15. Clamp
16. Spacer
17. Bracket
18. Bolt
19. Fuel sensor (thermistor)
20. Spacer (NAS43DD3-96N)
21. Bolt (NAS6203.28)
22. Transfer pump electrical leads
23. Fuel level transmitter
24. Bolt
25. Washer
26. Spacer
27. Mounting bracket

NOTE

1 Right-hand installation only, Post TB 412-95-133.

ICN-412-A-281000-B-97499-00053-A-001-01

Figure 1. Lower Forward Cell Sump Assembly (Sheet 3 of 3)

Lower Forward Cell Sump Assemblies - Inspection

Procedure

NOTE

The following inspection procedures are the same for both lower forward cell sump assemblies except as noted in Figure 1.

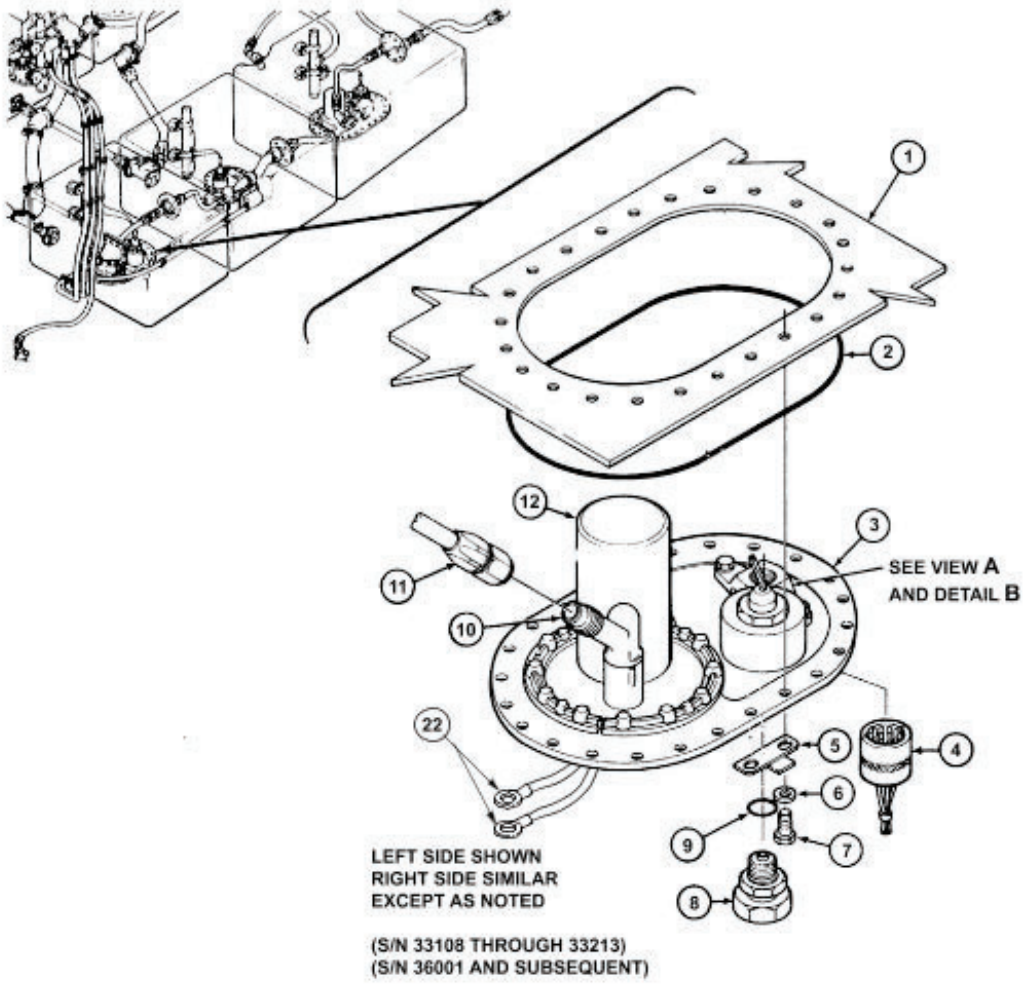
1. Inspect area where sump assembly (3, Figure 1) mates with cell for damage and cleanliness. Inspect packing groove.

2. Inspect components for security of mounting.

3. Inspect hoses and fittings for contamination. Contamination shall be removed. Damage which can be detected visually necessitates replacement.

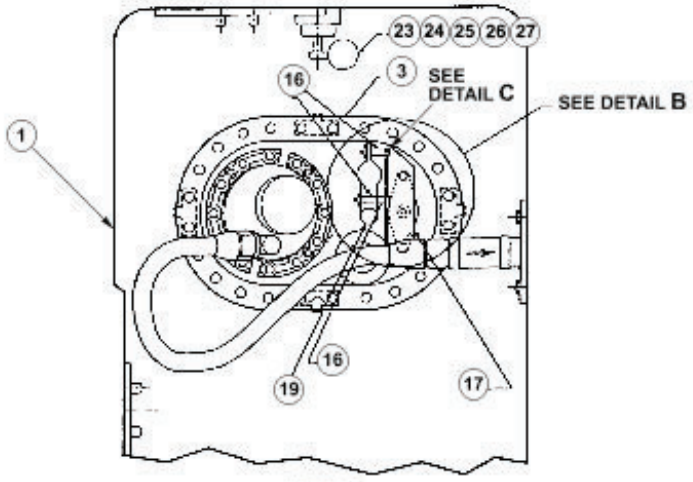
4. Inspect electrical wiring and connectors for condition.

5. Replace any components which fail to meet inspection requirements.

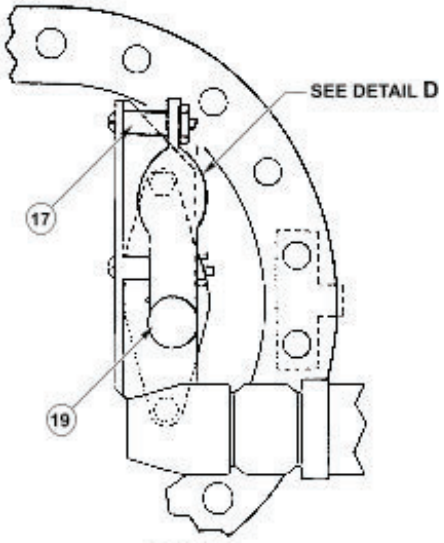


ICN-412-A-281000-B-97499-00051-A-001-01

Figure 1. Lower Forward Cell Sump Assembly (Sheet 1 of 3)



VIEW A
LOOKING DOWN (LHS SHOWN, RHS TYPICAL)
PRE TB 412-95-133



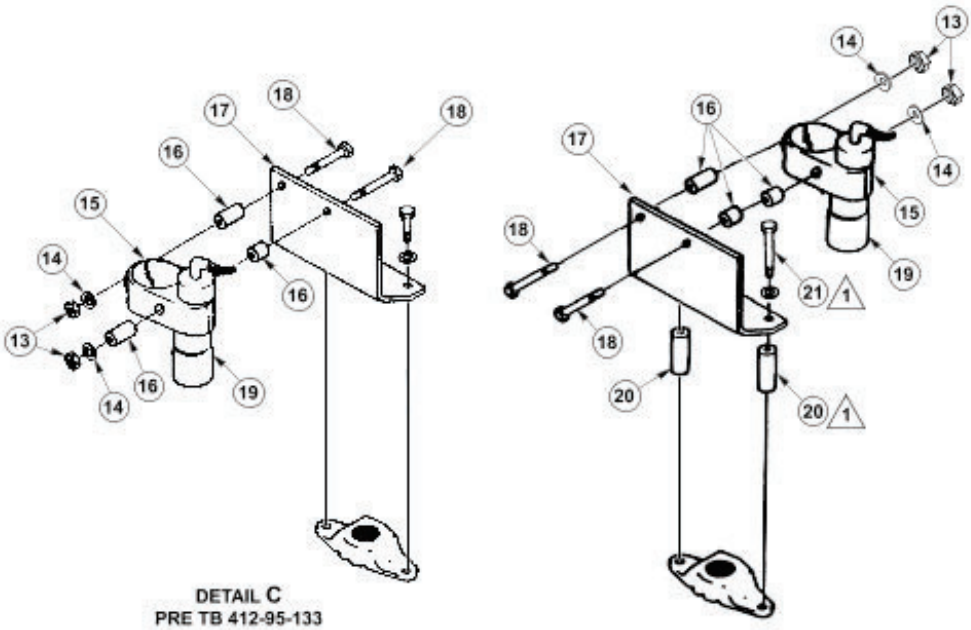
DETAIL B
S/N 36087, 36096 AND 36110 AND SUBSEQUENT
POST TB 412-95-133 (1)

NOTE

(1) On helicopters S/N 33108 through 33213, 36001 through 36086, 36088 through 36095, and 36097 through 36109.

ICN-412-A-281000-B-97499-00052-A-001-01

Figure 1. Lower Forward Cell Sump Assembly (Sheet 2 of 3)



1. Lower forward cell
2. Packing
3. Sump assembly
4. Electrical connector
5. Retainer
6. Washer
7. Bolt
8. Drain valve
9. Packing
10. Fitting
11. Hose
12. Transfer pump
13. Nut
14. Washer
15. Clamp
16. Spacer
17. Bracket
18. Bolt
19. Fuel sensor (thermistor)
20. Spacer (NAS43DD3-96N)
21. Bolt (NAS6203.28)
22. Transfer pump electrical leads
23. Fuel level transmitter
24. Bolt
25. Washer
26. Spacer
27. Mounting bracket

NOTE

1 Right-hand installation only, Post TB 412-95-133.

ICN-412-A-281000-B-97499-00053-A-001-01

Figure 1. Lower Forward Cell Sump Assembly (Sheet 3 of 3)

Lower Forward Cell Sump Assemblies - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

NOTE

The following installation procedures are the same for both lower forward cell sump assemblies except as noted in Figure 1.

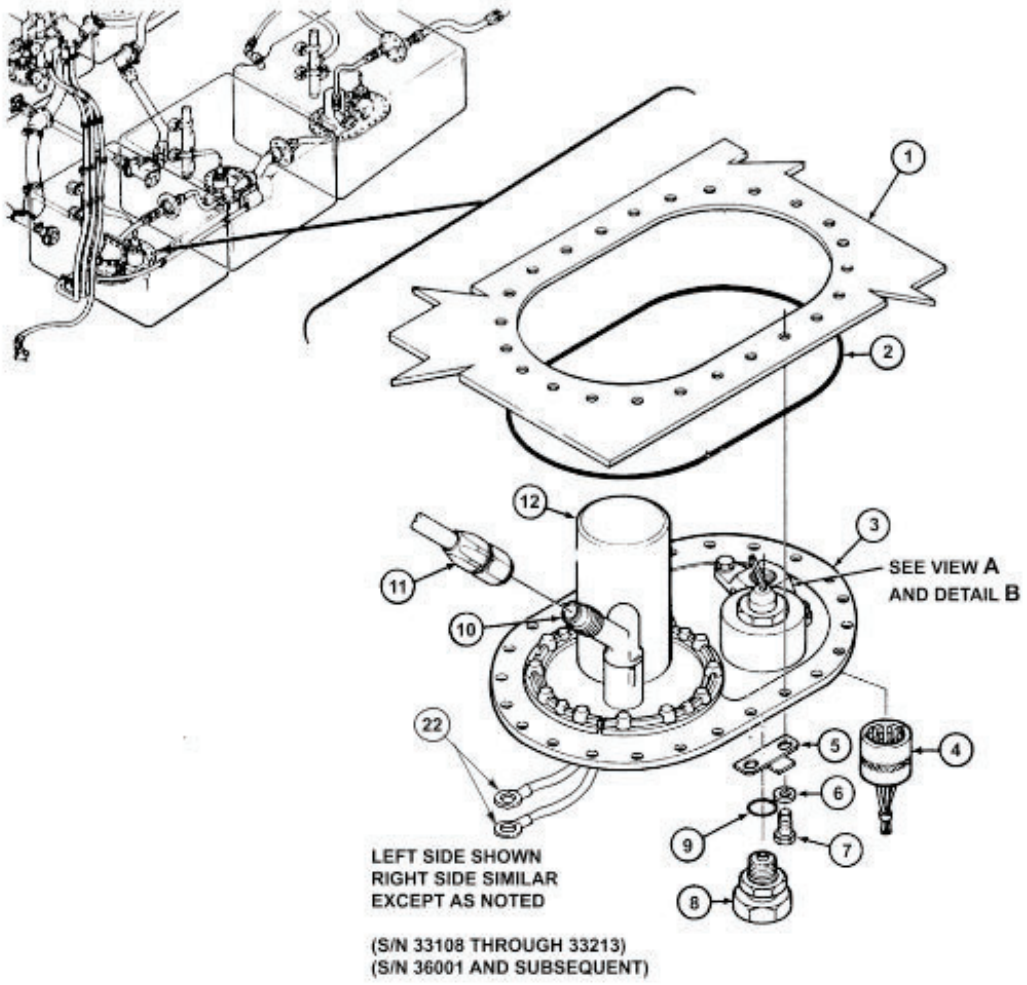
1. Remove protective cover from cell opening.
2. Coat packing (2, Figure 1) with assembly fluid (C-024). Place packing in groove around sump opening of lower forward cell (1).
3. Install fuel level sensor (thermistor) (19) in clamp (15).
4. Hold lower forward sump assembly (3) below sump opening of lower forward cell (1).

NOTE

TB 412-95-133: Pre or TB 412-95-133: Post:

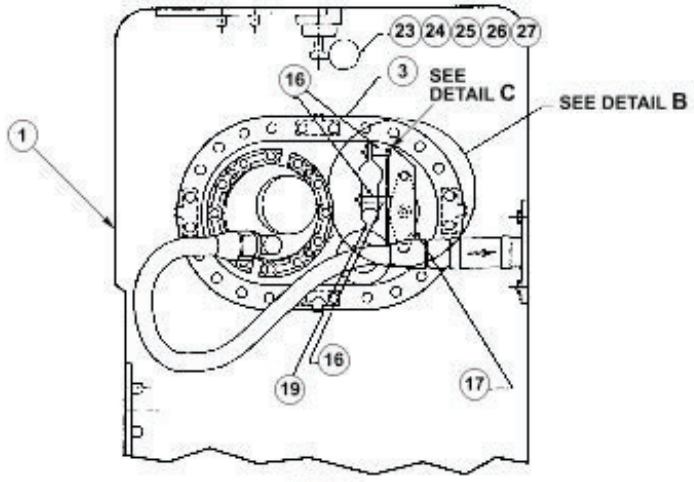
Ensure position of fuel level sensor (thermistor) (19) is in accordance with configuration (Figure 1, Detail C or D).

5. Install fuel level sensor (thermistor) (19) and clamp (15) in bracket (17) using nuts (13), washers (14), spacers (16), and bolts (18).
6. Install the bolts (24) through the fuel level transmitter mounting brackets (27).
7. Install the spacers (26) and the washers (25) on the bolts (24).
8. Install the fuel level transmitter (23) on the inboard wall of the lower forward fuel cell (1). Tighten the bolts (24).
9. Connect hose (11) to fitting (10).
10. Connect internal electrical connector.
11. Position lower forward cell sump assembly (3) in place over sump opening on bottom of lower forward cell (1).
12. Install mounting bolts (7), washers (6), and retainers (5).
13. Connect electrical connector (4) on bottom of lower forward cell sump assembly (3).
14. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).
15. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).
16. Perform fuel system operational check.
17. Check installation area for leaks.

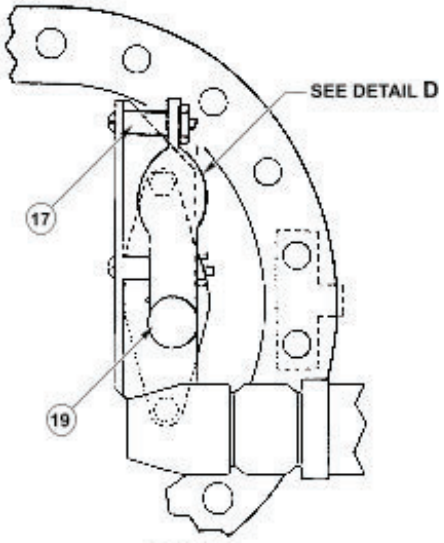


ICN-412-A-281000-B-97499-00051-A-001-01

Figure 1. Lower Forward Cell Sump Assembly (Sheet 1 of 3)



VIEW A
LOOKING DOWN (LHS SHOWN, RHS TYPICAL)
PRE TB 412-95-133



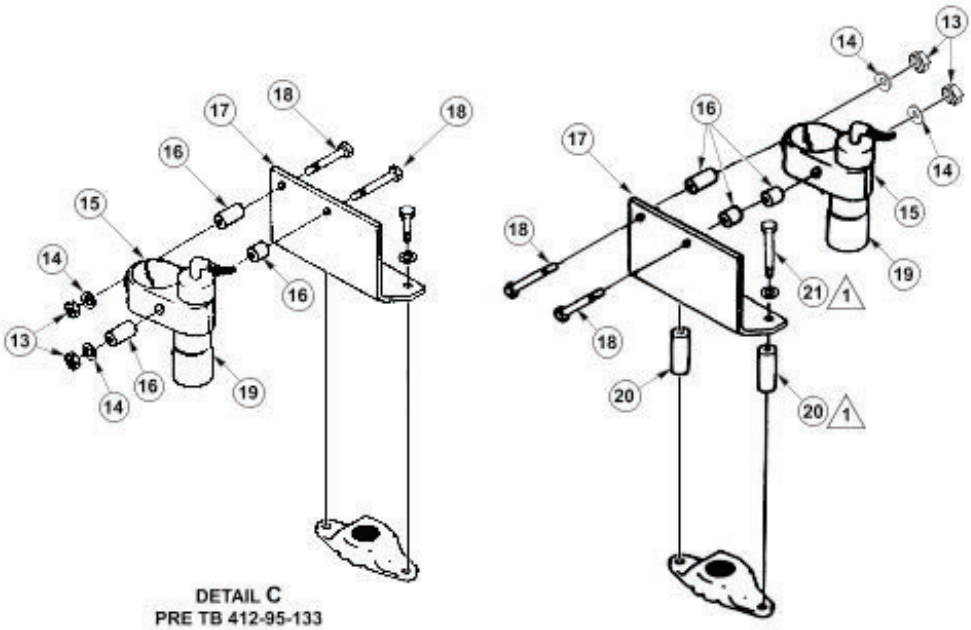
DETAIL B
S/N 36087, 36096 AND 36110 AND SUBSEQUENT
POST TB 412-95-133 (1)

NOTE

(1) On helicopters S/N 33108 through 33213, 36001 through 36086, 36088 through 36095, and 36097 through 36109.

ICN-412-A-281000-B-97499-00052-A-001-01

Figure 1. Lower Forward Cell Sump Assembly (Sheet 2 of 3)



1. Lower forward cell
2. Packing
3. Sump assembly
4. Electrical connector
5. Retainer
6. Washer
7. Bolt
8. Drain valve
9. Packing
10. Fitting
11. Hose
12. Transfer pump
13. Nut
14. Washer
15. Clamp
16. Spacer
17. Bracket
18. Bolt
19. Fuel sensor (thermistor)
20. Spacer (NAS43DD3-96N)
21. Bolt (NAS6203.28)
22. Transfer pump electrical leads
23. Fuel level transmitter
24. Bolt
25. Washer
26. Spacer
27. Mounting bracket

NOTE

1 Right-hand installation only, Post TB 412-95-133.

ICN-412-A-281000-B-97499-00053-A-001-01

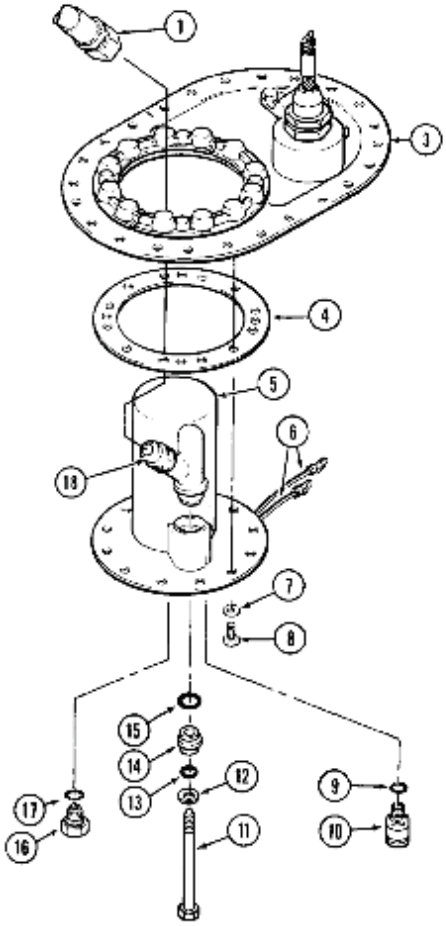
Figure 1. Lower Forward Cell Sump Assembly (Sheet 3 of 3)

Transfer Pumps - Description

NOTE

Maintenance procedures for the left and right transfer pumps are the same.

A transfer pump (Figure 1) is mounted on sump assembly of each lower forward cell. The transfer pumps provide fuel pressures required to transfer fuel from lower forward cells to lower main cells. The transfer pumps are electrically operated, and controlled by separate FUEL TRANS switches located on engine and fuel control panel and by fuel level sensors in lower forward cell fuel quantity probes.



(S/N 33108 THRU 33213)
(S/N 36001 AND SUBSEQUENT)

1. Hose
2. Deleted
3. Sump assembly
4. Gasket
5. Transfer pump
6. Electrical leads
7. Washer
8. Bolt
9. Packing
10. Pump drain valve
11. Bolt
12. Washer
13. Packing
14. Retainer
15. Packing
16. Plug
17. Packing
18. Fitting

ICN-412-A-281000-B-97499-00054-A-001-01

Figure 1. Transfer Pump (Sheet 1 of 1)

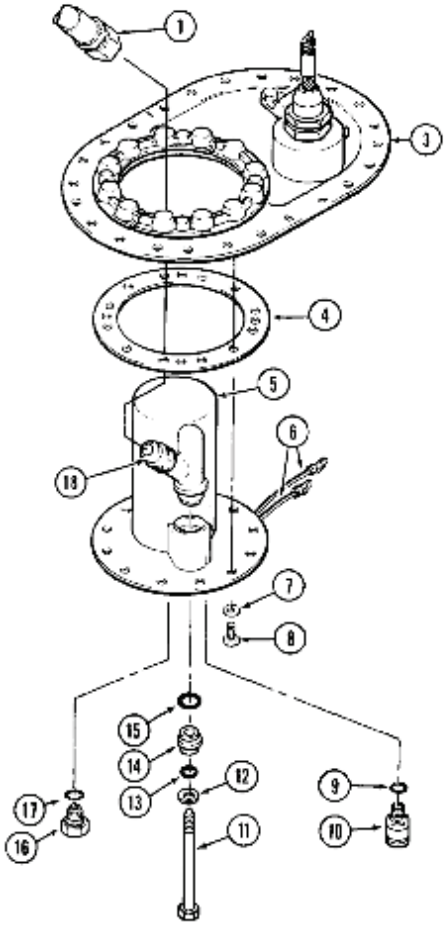
Transfer Pumps - Removal

Procedure

NOTE

The following removal procedures for the left and right transfer pumps are the same.

1. Disconnect electrical power.
2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).
3. Drain any trapped fuel through pump drain valve (10, Figure 1).
4. Disconnect transfer pump electrical leads (6).
5. Remove bolt (11), washer (12), packings (13 and 15), and retainer (14). Discard packings.
6. Remove bolts (8) and washers (7).
7. Remove transfer pump (5) and gasket (4) from sump assembly (3).
8. Remove plug (16) and packing (17). Discard packing.
9. Install protective cover over cell opening.



(S/N 33108 THRU 33213)
(S/N 36001 AND SUBSEQUENT)

1. Hose
2. Deleted
3. Sump assembly
4. Gasket
5. Transfer pump
6. Electrical leads
7. Washer
8. Bolt
9. Packing
10. Pump drain valve
11. Bolt
12. Washer
13. Packing
14. Retainer
15. Packing
16. Plug
17. Packing
18. Fitting

ICN-412-A-281000-B-97499-00054-A-001-01

Figure 1. Transfer Pump (Sheet 1 of 1)

Transfer Pumps - Installation

Preliminary requirements

Consumables, materials and expendables

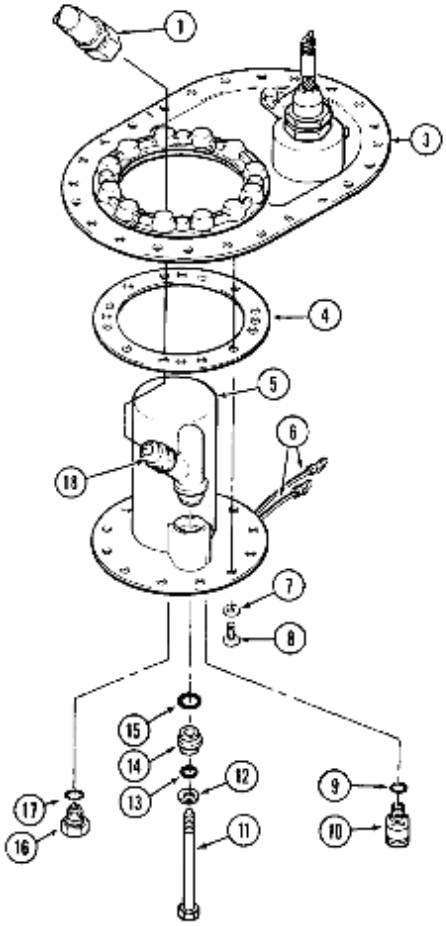
Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	
Lockwire	C-405	AR	

Procedure

NOTE

The following installation procedures for the left and right transfer pumps are the same.

1. Remove protective cover over cell opening.
2. Apply assembly fluid (C-024) to packing (17, Figure 1). Place packing on plug (16). Install plug.
3. Apply assembly fluid (C-024) to packings (13 and 15). Place washer (12), packing (13), retainer (14), and packing (15) on bolt (11).
4. Coat gasket (4) with assembly fluid (C-024). Place gasket on transfer pump (5).
5. Align transfer pump (5) with fitting (18). Install bolt (11) and apply lockwire (C-405).
6. Install bolts (8), washers (7), transfer pump (5), and gasket (4) on sump assembly (3).
7. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).
8. Perform fuel system operational check.
9. Check area for leaks.



(S/N 33108 THRU 33213)
(S/N 36001 AND SUBSEQUENT)

1. Hose
2. Deleted
3. Sump assembly
4. Gasket
5. Transfer pump
6. Electrical leads
7. Washer
8. Bolt
9. Packing
10. Pump drain valve
11. Bolt
12. Washer
13. Packing
14. Retainer
15. Packing
16. Plug
17. Packing
18. Fitting

ICN-412-A-281000-B-97499-00054-A-001-01

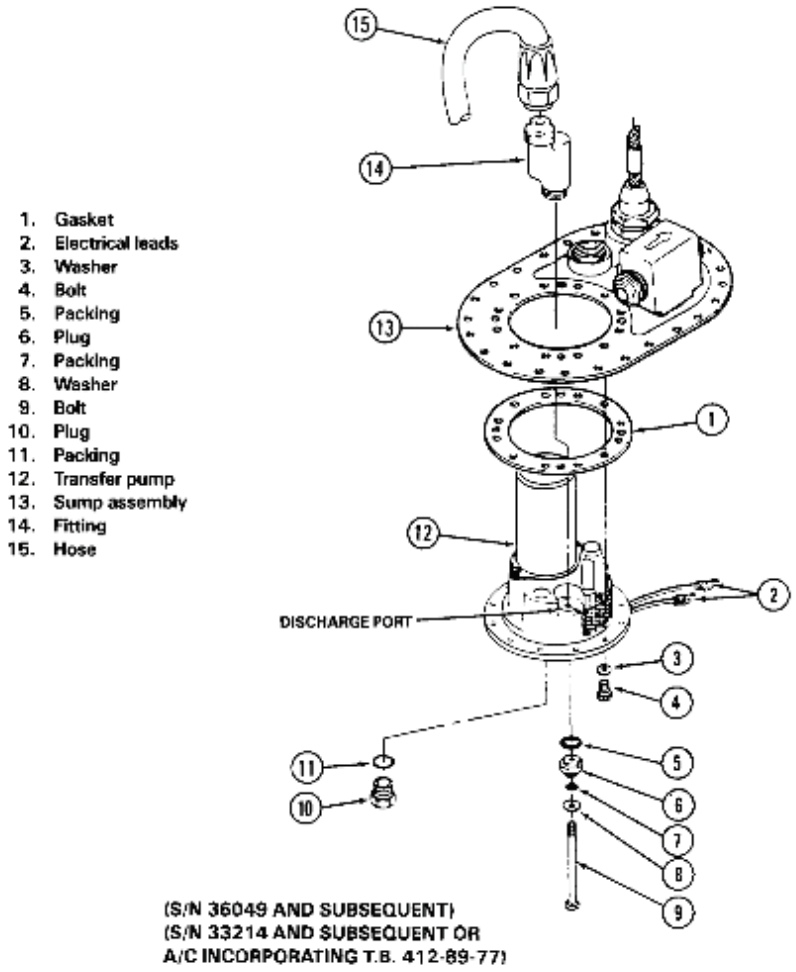
Figure 1. Transfer Pump (Sheet 1 of 1)

Canister Type Transfer Pumps - Description

NOTE

Maintenance procedures for left and right canister type transfer pumps are the same.

The canister type transfer pumps (Figure 1) incorporate an integral cartridge pumping element.



ICN-412-A-281000-B-97499-00055-A-001-01

Figure 1. Canister Type Transfer Pump (Sheet 1 of 1)

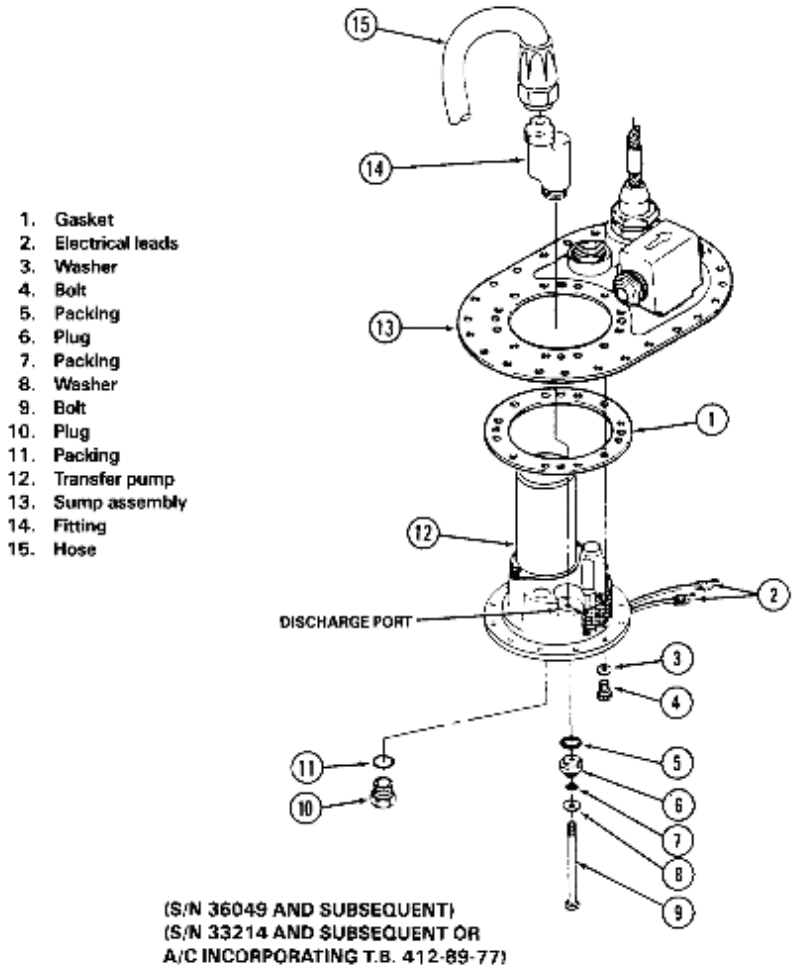
Canister Type Transfer Pumps - Removal

Procedure

NOTE

The following removal procedures for left and right canister type transfer pumps are the same.

1. Disconnect electrical power.
2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).
3. Drain any trapped fuel through sump drain.
4. Disconnect transfer pump electrical leads (2, Figure 1).
5. Remove bolt (9), washer (8), packings (5 and 7), and plug (6).
6. Remove bolts (4) and washers (3).
7. Remove transfer pump (12) and gasket (1) from sump assembly (13).
8. Install protective cover over cell opening.



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Figure 1. Canister Type Transfer Pump (Sheet 1 of 1)

Canister Type Transfer Pumps - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	
Lockwire	C-405	AR	

Procedure

NOTE

The following installation procedures for left and right canister type transfer pumps are the same.

1. Remove protective cover from cell opening.

2. Apply assembly fluid (C-024) to packing (11, Figure 1) and install on plug (10). Install plug.

3. Apply assembly fluid (C-024) to packings (5 and 7) and install on plug (6).

4. Install washer (8) on bolt (9). Insert bolt through plug (6) and into discharge port of pump.

5. Coat gasket (1) with assembly fluid (C-024). Place gasket on transfer pump (12).

6. Position transfer pump (12) on sump assembly (13). Align transfer pump with fitting (14). Tighten bolt (9) and ensure fitting seats in discharge port.

7. Install bolts (4) and washers (3).

8. Connect electrical leads (2).

9. Secure plug (10) and bolt (9) with lockwire (C-405).

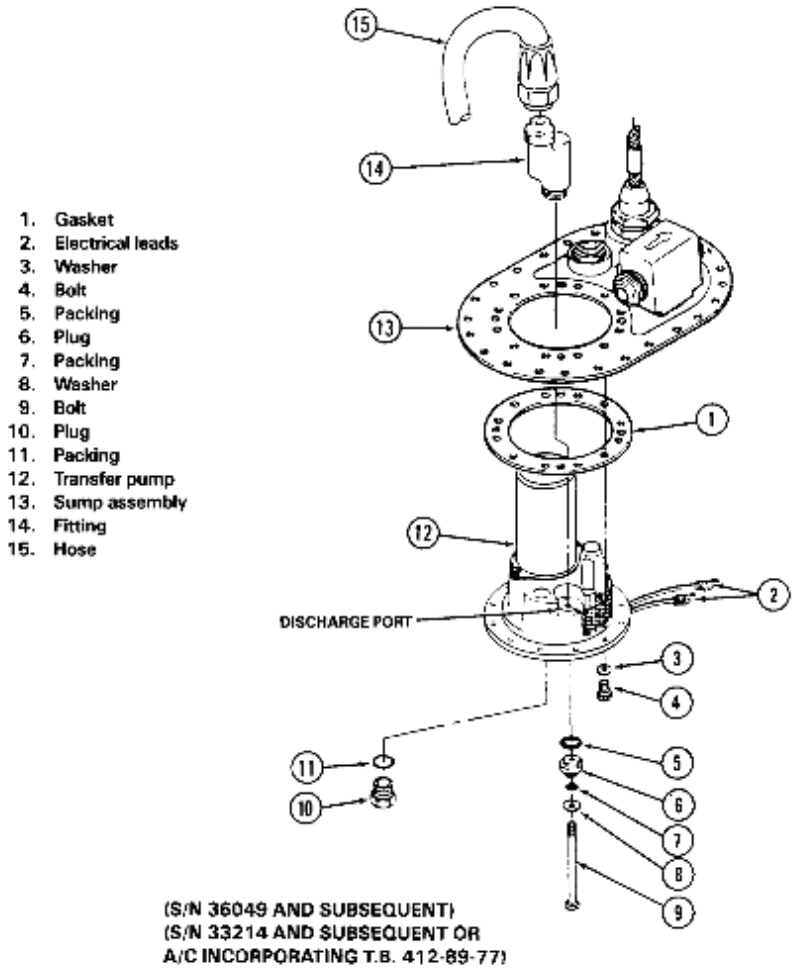
10. Perform system pressure test (412-A-28-10-00-00A-362B-A / 00438).

11. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).

12. Perform operational check on transfer pump (412-A-96-04-00-03A-320B-A / 02108 and 412-A-96-04-00-03A-320C-A / 02109).

13. Check area for leaks.

14. Install sump access door.



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Figure 1. Canister Type Transfer Pump (Sheet 1 of 1)

Canister Type Transfer Pump Cartridge - Description

For removal, cleaning, and installation procedures, refer to 412-A-28-10-00-60A-042A-A / 00518.

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Sump Drain Valves - Description

NOTE

Maintenance procedures for sump drain valve installed on each lower forward cell sump assembly are the same.

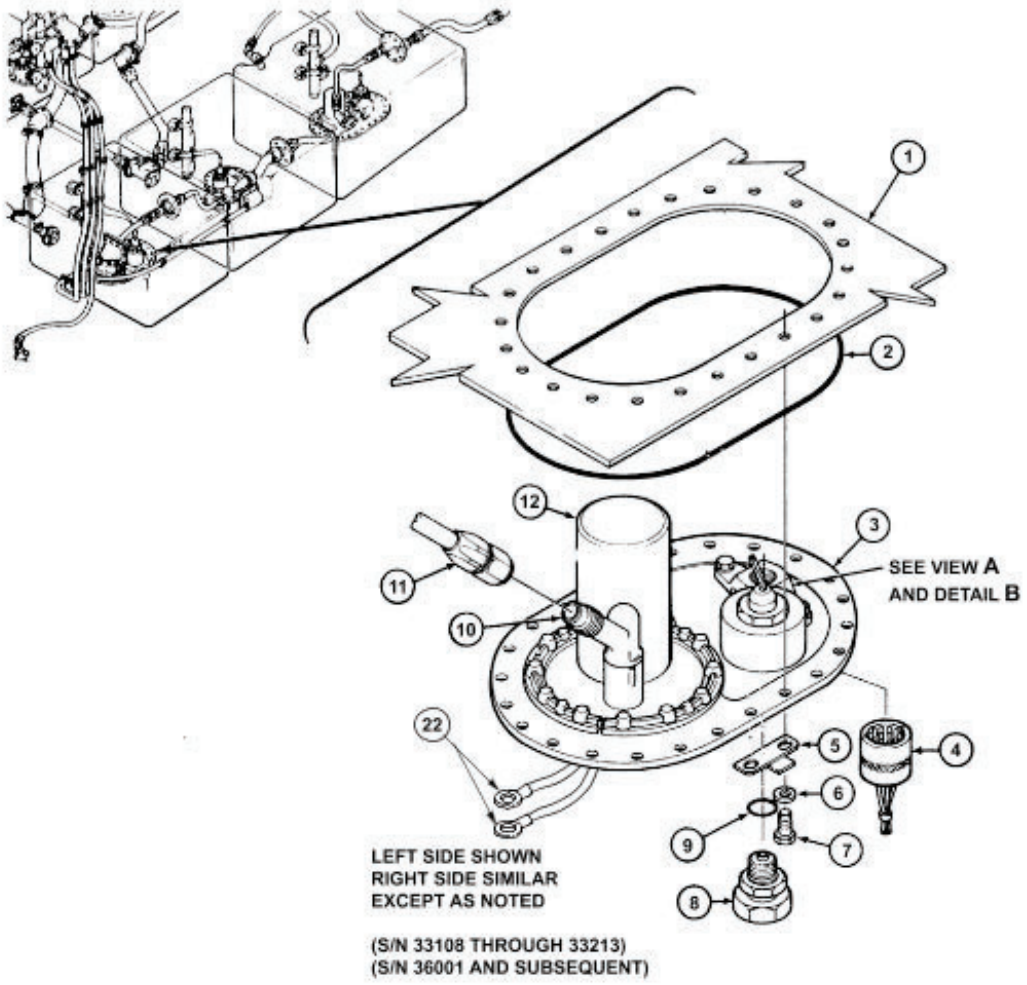
A manually operated sump drain valve is mounted on each lower forward cell sump assembly. These valves provide for draining of any possible accumulation of moisture or other contamination.

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Sump Drain Valves - Removal

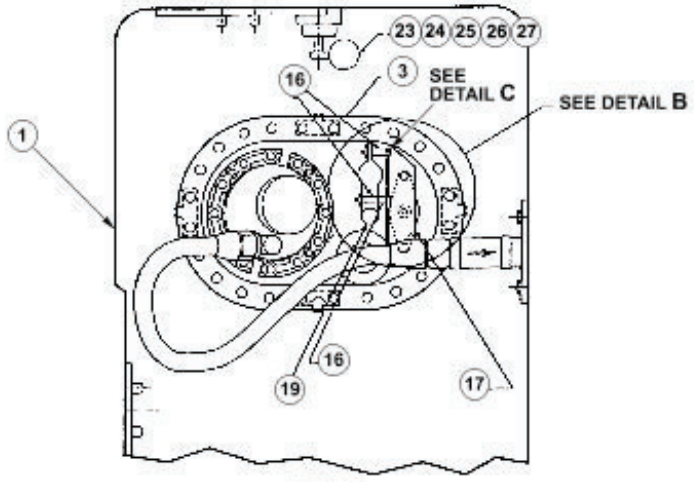
Procedure

1. Disconnect electrical power.
2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).
3. Remove sump access door.
4. Remove lockwire from outer flange of sump drain valve (8, Figure 1).
5. Remove sump drain valve (8) and packing (9) from sump assembly (3). Discard packing.
6. Install protective cover or plug opening.

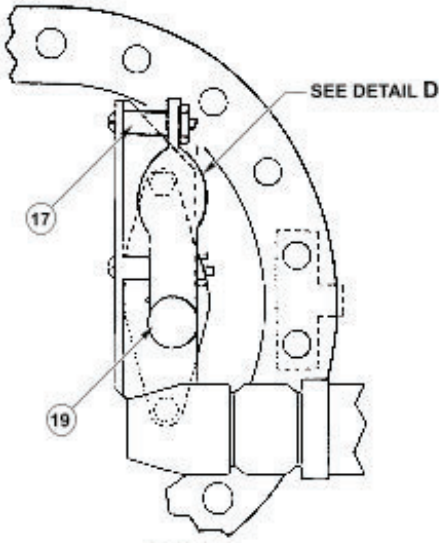


ICN-412-A-281000-B-97499-00051-A-001-01

Figure 1. Lower Forward Cell Sump Assembly (Sheet 1 of 3)



VIEW A
LOOKING DOWN (LHS SHOWN, RHS TYPICAL)
PRE TB 412-95-133



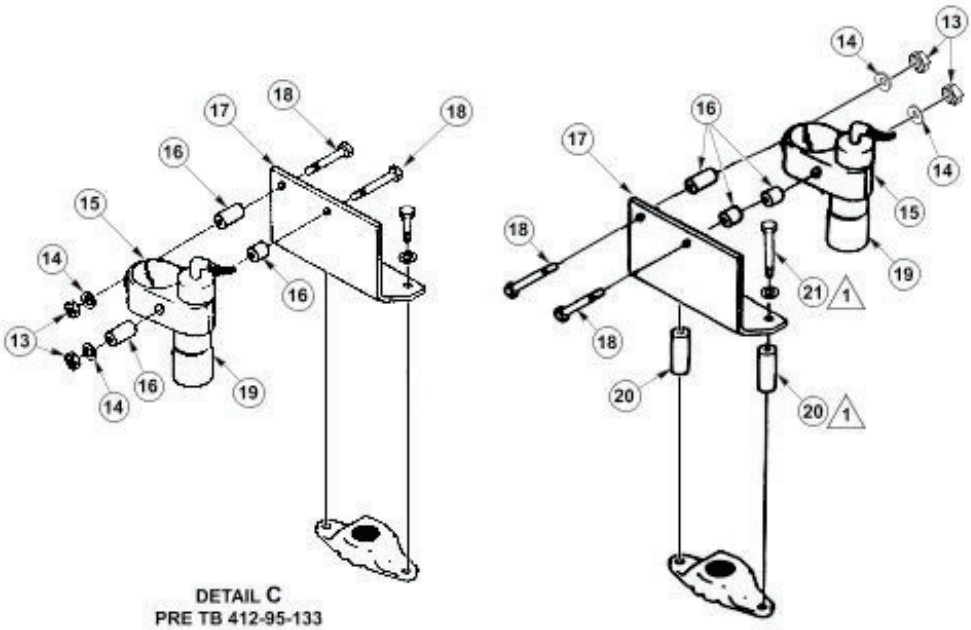
DETAIL B
S/N 36087, 36096 AND 36110 AND SUBSEQUENT
POST TB 412-95-133 (1)

NOTE

(1) On helicopters S/N 33108 through 33213, 36001 through 36086, 36088 through 36095, and 36097 through 36109.

ICN-412-A-281000-B-97499-00052-A-001-01

Figure 1. Lower Forward Cell Sump Assembly (Sheet 2 of 3)



1. Lower forward cell
2. Packing
3. Sump assembly
4. Electrical connector
5. Retainer
6. Washer
7. Bolt
8. Drain valve
9. Packing
10. Fitting
11. Hose
12. Transfer pump
13. Nut
14. Washer
15. Clamp
16. Spacer
17. Bracket
18. Bolt
19. Fuel sensor (thermistor)
20. Spacer (NAS43DD3-96N)
21. Bolt (NAS6203.28)
22. Transfer pump electrical leads
23. Fuel level transmitter
24. Bolt
25. Washer
26. Spacer
27. Mounting bracket

NOTE

1 Right-hand installation only, Post TB 412-95-133.

ICN-412-A-281000-B-97499-00053-A-001-01

Figure 1. Lower Forward Cell Sump Assembly (Sheet 3 of 3)

Sump Drain Valves - Installation

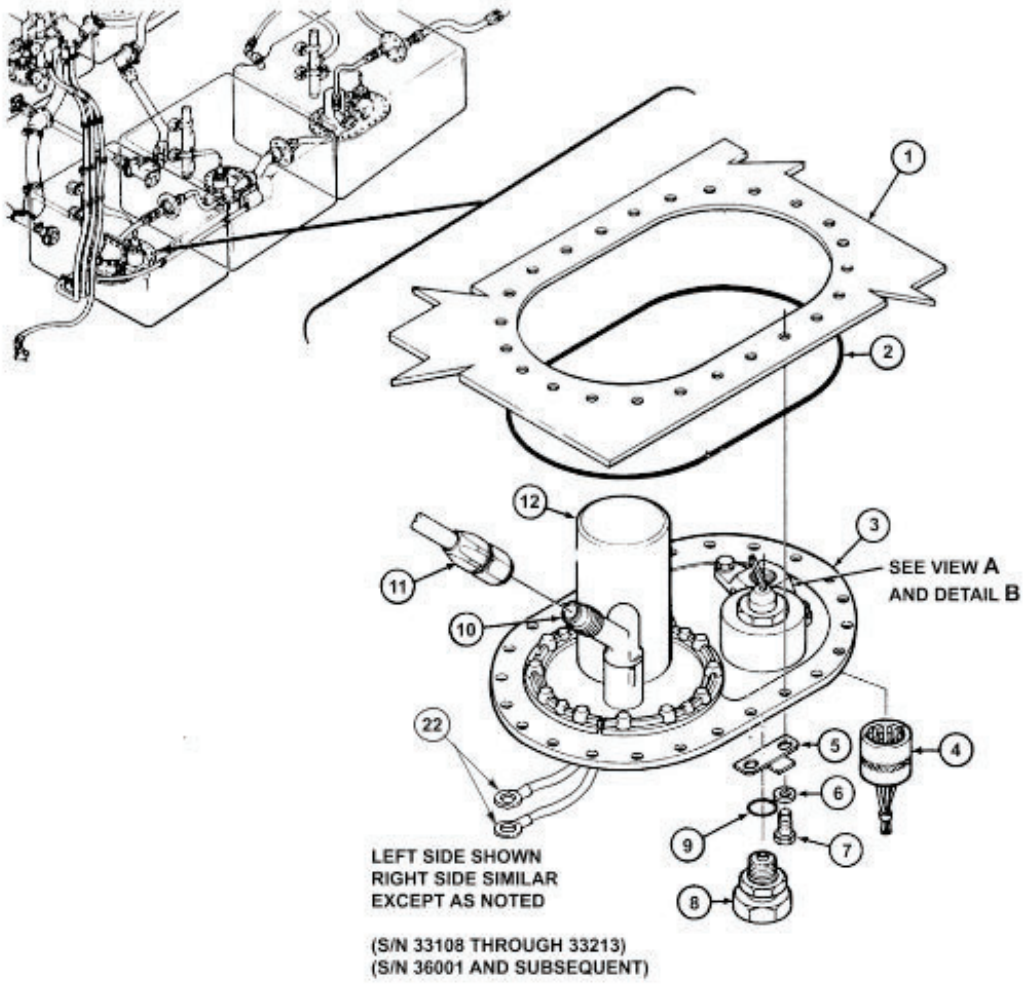
Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	
Lockwire	C-405	AR	

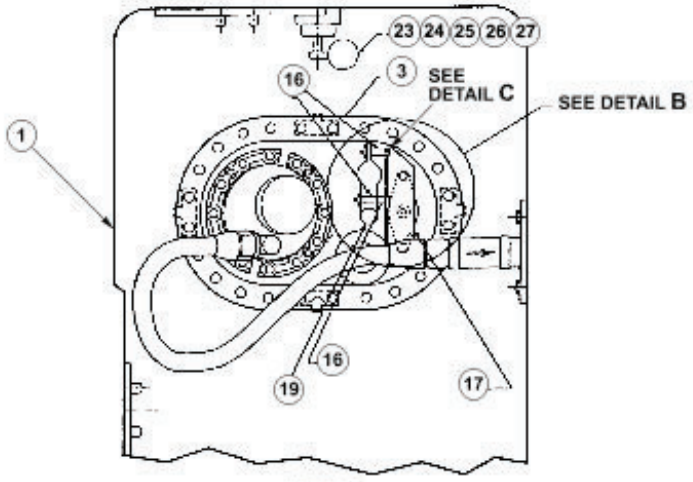
Procedure

1. Remove protective cover from opening.
2. Apply assembly fluid (C-024) to packing (9, Figure 1). Place packing on sump drain valve (8).
3. Install sump drain valve (8) and lockwire (C-405).
4. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).
5. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).
6. Perform fuel system operational check.
7. Check area for leaks.

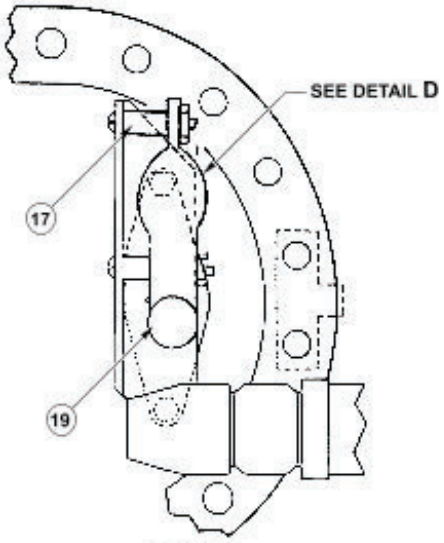


ICN-412-A-281000-B-97499-00051-A-001-01

Figure 1. Lower Forward Cell Sump Assembly (Sheet 1 of 3)



VIEW A
LOOKING DOWN (LHS SHOWN, RHS TYPICAL)
PRE TB 412-95-133



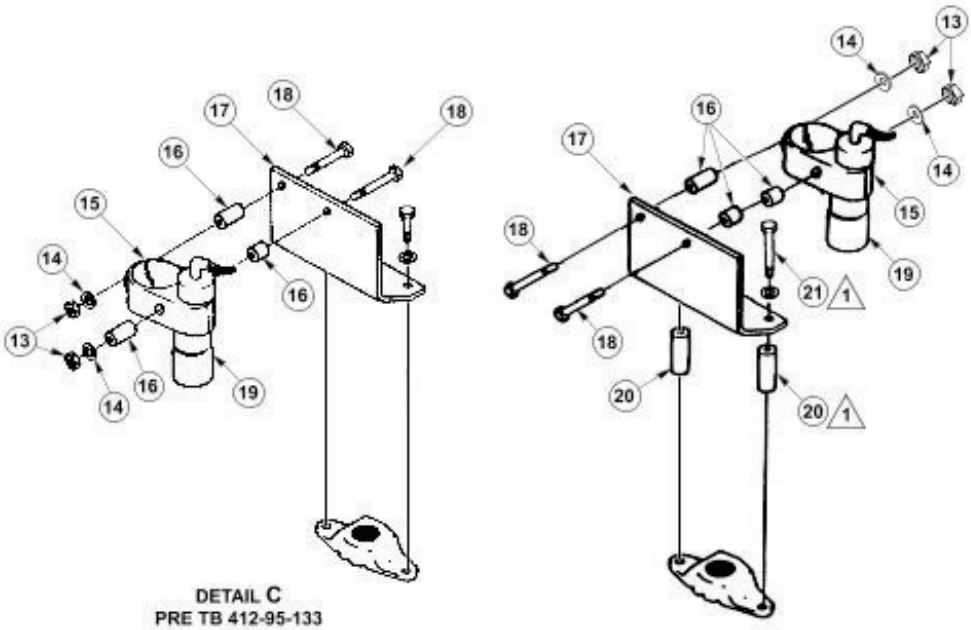
DETAIL B
S/N 36087, 36096 AND 36110 AND SUBSEQUENT
POST TB 412-95-133 (1)

NOTE

(1) On helicopters S/N 33108 through 33213, 36001 through 36086, 36088 through 36095, and 36097 through 36109.

ICN-412-A-281000-B-97499-00052-A-001-01

Figure 1. Lower Forward Cell Sump Assembly (Sheet 2 of 3)



1. Lower forward cell
2. Packing
3. Sump assembly
4. Electrical connector
5. Retainer
6. Washer
7. Bolt
8. Drain valve
9. Packing
10. Fitting
11. Hose
12. Transfer pump
13. Nut
14. Washer
15. Clamp
16. Spacer
17. Bracket
18. Bolt
19. Fuel sensor (thermistor)
20. Spacer (NAS43DD3-96N)
21. Bolt (NAS6203.28)
22. Transfer pump electrical leads
23. Fuel level transmitter
24. Bolt
25. Washer
26. Spacer
27. Mounting bracket

NOTE

1 Right-hand installation only, Post TB 412-95-133.

ICN-412-A-281000-B-97499-00053-A-001-01

Figure 1. Lower Forward Cell Sump Assembly (Sheet 3 of 3)

Lower Forward Cell Check Valves - Description

NOTE

Maintenance procedures for left and right lower forward cell check valves are the same.

Check valves are installed in the output hoses of the transfer pumps. These valves permit forward flow of fuel from lower forward cells to lower middle cells when transfer pumps are operating, but prevent back flow of fuel during shutdown.

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Lower Forward Cell Check Valves - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure

NOTE

The following removal procedures for left and right lower forward cell check valves are the same.

1. Disconnect electrical power.

2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).

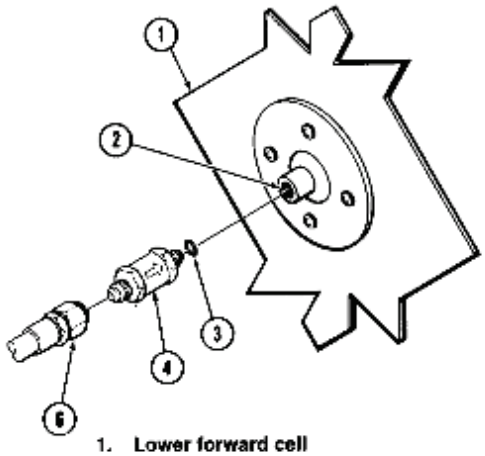
3. Remove access door on underside of helicopter directly under applicable lower forward cell.

4. Remove sump assembly (412-A-28-10-00-41A-520A-A / 00456) to obtain internal access to cell.

5. Disconnect hose (6, Figure 1).

6. Remove check valve (4) and packing (3). Discard packing.

7. Install caps or plugs (C-428) and covers over cell opening.



1. Lower forward cell
2. Lower forward cell / lower middle cell interconnect fitting
3. Packing
4. Check valve
5. Deleted
6. Hose

(S/N 33108 THRU 33213)
(S/N 36001 AND SUBSEQUENT)

ICN-412-A-281000-B-97499-00056-A-001-01

Figure 1. Lower Forward Cell Check Valve (Sheet 1 of 1)

Lower Forward Cell Check Valves - Installation

Preliminary requirements

Consumables, materials and expendables

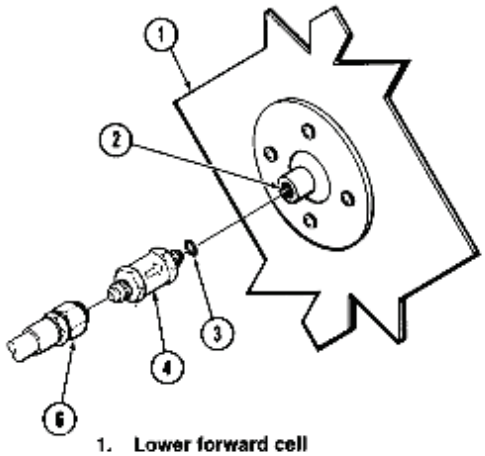
Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

NOTE

The following installation procedures for left and right lower forward cell check valves are the same.

1. Remove protective cover over cell opening.
2. Apply assembly fluid (C-024) to packing (3, Figure 1).
3. Place packing (3) on check valve (4) so that the arrow on the side of the check valve is pointing towards the packing.
4. Install the check valve (4) with arrow pointing toward the lower forward cell/lower middle cell interconnect fitting (2).
5. Connect the hose (6).
6. Install the sump assembly (412-A-28-10-00-41A-720A-A / 00458).



1. Lower forward cell
2. Lower forward cell / lower middle cell interconnect fitting
3. Packing
4. Check valve
5. Deleted
6. Hose

(S/N 33108 THRU 33213)
(S/N 36001 AND SUBSEQUENT)

ICN-412-A-281000-B-97499-00056-A-001-01

Figure 1. Lower Forward Cell Check Valve (Sheet 1 of 1)

Lower Forward Cell/Lower Middle Cell Interconnect Fitting - Description

NOTE

Maintenance procedures for left and right lower forward cell/lower middle cell interconnect fitting are the same.

A lower forward cell/lower middle cell interconnect fitting is located on the aft interconnect port of each lower forward cell. The fitting provides interconnection between lower forward and lower middle cells and serves as a mounting point for the lower forward cell check valve.

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Lower Forward Cell/Lower Middle Cell Interconnect Fitting - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure

NOTE

The following removal procedures for left and right lower forward cell/lower middle cell interconnect fitting are the same.

1. Disconnect electrical power.

2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).

3. Remove cabin floor access panels and access doors on underside of helicopter below the applicable lower forward and lower middle cells.

4. Remove lower forward cell sump assembly (412-A-28-10-00-41A-520A-A / 00456).

5. Remove lower middle cell sump assembly (412-A-28-10-00-51A-520A-A / 00488).

6. In lower middle cell (1), disconnect tube (4, Figure 1) from union (3).

7. Remove union (3) and packing (2). Discard packing.

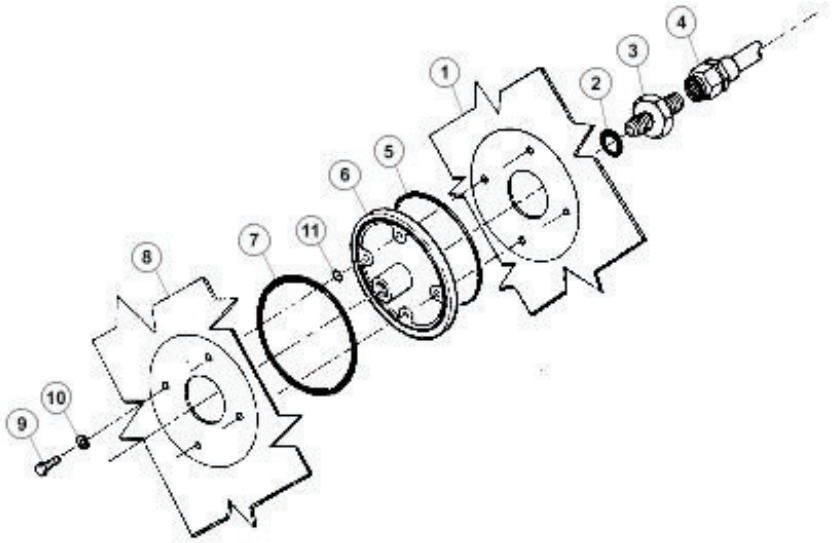
8. In lower forward cell (8), remove forward cell check valve (412-A-28-10-00-46A-520A-A / 00470).

9. Remove bolts (9) and seal (10).

10. Partially collapse lower forward cell (8) and lower middle cell (1) for hand access.

11. Remove packing (5), interconnect fitting (6), and packing (7). Discard packings.

12. Install caps or plugs (C-428) and covers over cell opening.



1. Lower middle cell
2. Packing
3. Union
4. Tube
5. Packing
6. Interconnect fitting
7. Packing
8. Lower forward cell
9. Bolt
10. Seal
11. Packing

(S/N 33108 THROUGH 33213,
S/N 36001 THROUGH 36999, AND
S/N 37002 THROUGH 37999)

ICN-412-A-281000-B-97499-00057-A-001-01

Figure 1. Lower Forward Cell/Lower Middle Cell Interconnect Fitting (Sheet 1 of 1)

Lower Forward Cell/Lower Middle Cell Interconnect Fitting - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

NOTE

The following installation procedures for left and right lower forward cell/lower middle cell interconnect fitting are the same.

1. Remove protective covers from cell openings.

2. Coat packings (5 and 7, Figure 1) with assembly fluid (C-024).

3. Place packings (5 and 7) on each side of interconnect fitting (6).

4. Align fitting with packings (5 and 7) between lower forward cell (8) and lower middle cell (1) walls, and in hole in bulkhead between.

5. From inside the lower forward fuel cell (8), install bolts (9) and seal (10) under bolt heads.

6. Apply assembly fluid (C-024) to packings (11) and install on bolts (9) between forward fuel cell (8) and interconnect fitting (6).

7. Apply assembly fluid (C-024) to packing (2).

8. Install packing (2) on union (3).

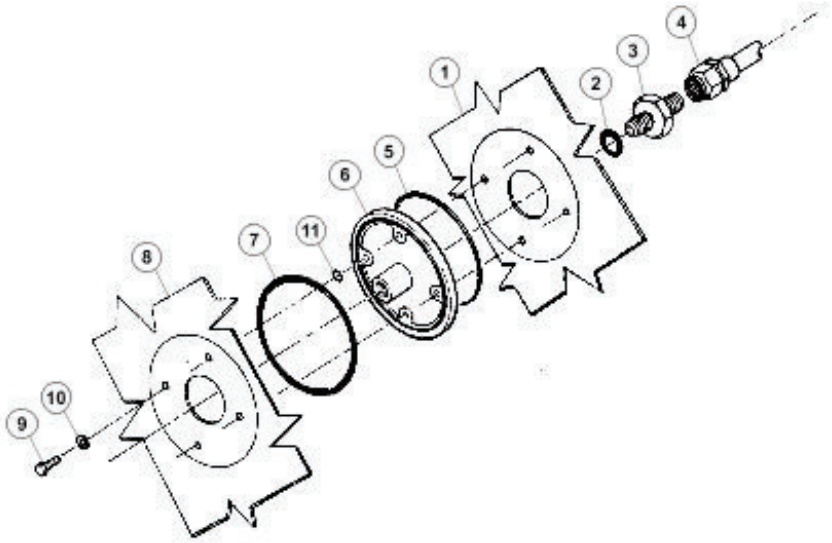
9. Install union (3) into the interconnect fitting (6).

10. Connect tube (4) to union (3).

11. Install lower forward cell check valve (4, Figure 2) (412-A-28-10-00-46A-720A-A / 00471).

12. Install lower middle cell sump assembly (412-A-28-10-00-51A-720A-A / 00490).

13. Install lower forward cell sump assembly (412-A-28-10-00-41A-720A-A / 00458).

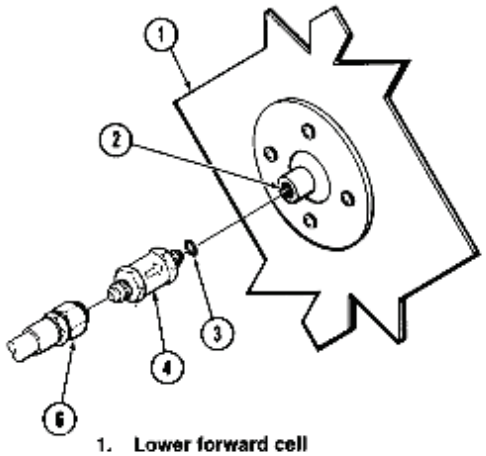


1. Lower middle cell
2. Packing
3. Union
4. Tube
5. Packing
6. Interconnect fitting
7. Packing
8. Lower forward cell
9. Bolt
10. Seal
11. Packing

(S/N 33108 THROUGH 33213,
S/N 36001 THROUGH 36999, AND
S/N 37002 THROUGH 37999)

ICN-412-A-281000-B-97499-00057-A-001-01

Figure 1. Lower Forward Cell/Lower Middle Cell Interconnect Fitting (Sheet 1 of 1)



1. Lower forward cell
2. Lower forward cell / lower middle cell interconnect fitting
3. Packing
4. Check valve
5. Deleted
6. Hose

(S/N 33108 THRU 33213)
(S/N 36001 AND SUBSEQUENT)

ICN-412-A-281000-B-97499-00056-A-001-01

Figure 2. Lower Forward Cell Check Valve (Sheet 1 of 1)

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Lower Middle Cells and Lower Main Cells - Description

NOTE

Lower left middle cell and lower left main cell are connected together by a lower middle cell/lower main cell interconnect fitting prior to installation, and are removed and installed together. The lower right middle cell and lower right main cell are connected together by a lower middle cell/lower main cell interconnect fitting prior to installation, and are removed and installed together. Removal and installation procedures for left and right installations are the same.

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Lower Middle Cells and Lower Main Cells - Removal

Preliminary requirements

Consumables, materials and expendables

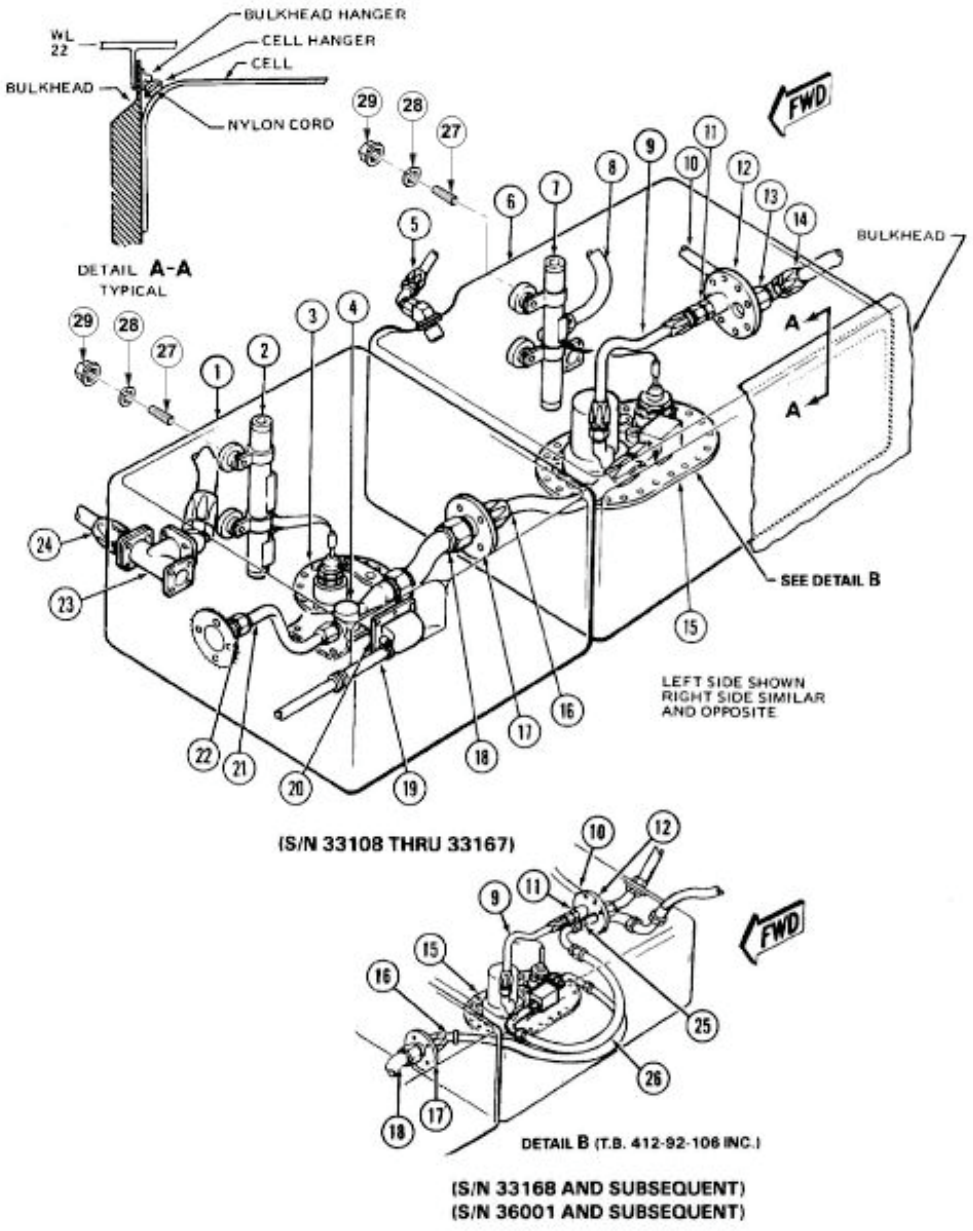
Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure

NOTE

The following removal procedures for left and right installations are the same.

1. Disconnect electrical power.
2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).
3. Remove cabin floor panels above the cells to be removed.
4. Remove access doors on underside of helicopter below all lower cells to be removed.
5. Remove lower forward cell sump assembly for access (412-A-28-10-00-41A-520A-A / 00456).
6. Remove lower middle cell sump assembly (412-A-28-10-00-51A-520A-A / 00488).
7. Remove lower main cell sump assembly (412-A-28-10-00-57A-520A-A / 00509).
8. Remove lower forward cell/lower middle cell interconnect fitting (412-A-28-10-00-47A-520A-A / 00473).
9. Remove lower main cell inlet/outlet fitting (412-A-28-10-00-63A-520A-A / 00529).
10. Remove lower middle cell vent hose (412-A-28-10-00-49A-520A-A / 00482).
11. Remove lower main cell vent hose (412-A-28-10-00-55A-520A-A / 00503).
12. Remove lower middle cell interconnect fitting (23, Figure 1) and interconnect hose (24) from lower middle cell (1) (412-A-28-10-00-50A-520A-A / 00485).
13. Remove lower main interconnect hose (8) from cell (6) (412-A-28-10-00-56A-520A-A / 00506).
14. Install caps or plugs (C-428) on all open end fittings, and install protective covers over cell openings.
15. While holding the setscrews (27) with an allen key, remove the nuts (29) and the washers (28).
16. With the allen key, remove the set screws (27).
17. Remove nylon cord from hangers around top of cells and top of cell cavity.
18. Partially collapse cells. Lift cells evenly from cell cavity.



ICN-412-A-281000-B-97499-00058-A-001-01

Figure 1. Lower Middle Cell/Lower Main Cell (Sheet 1 of 2)

1. Lower middle cell
2. Fuel quantity probe
3. Sump asssembly
4. Ejector pump
5. Vent hose
6. Lower main cell
7. Fuel quantity probe
8. Interconnect hose
9. Hose
10. Hose
11. Reducer
12. Lower main cell inlet/ outlet fitting
13. Union
14. Hose
15. Sump assembly
16. Hose
17. Lower middle cell/lower main cell interconnect fitting
18. Tube
19. Vent hose
20. Retainer
21. Hose
22. Union
23. Lower middle cell interconnect fitting
24. Interconnect hose
25. Union
26. Hose
27. Setscrew
28. Washer
29. Nut

ICN-412-A-281000-B-97499-00059-A-001-01

Figure 1. Lower Middle Cell/Lower Main Cell (Sheet 2 of 2)

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Lower Middle Cells and Lower Main Cells - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Soap	100066	AR	Zodiac Aerospace (www.zodiacaerospace.com)
Cleaning Solution	601199	AR	Zodiac Aerospace (www.zodiacaerospace.com)
Clean cloth	C-516	AR	

Safety conditions



OBEY ALL THE SAFETY PRECAUTIONS WHEN YOU DO MAINTENANCE ON FUEL SYSTEM EQUIPMENT.



OBEY ALL THE FUEL CELL HANDLING PRECAUTIONS DURING CLEANING OR DAMAGE TO THE CELL MAY OCCUR.



DO NOT ALLOW SOLVENTS OR COMPOUNDS TO MAKE CONTACT WITH PARTS MADE OF ELASTOMERIC MATERIAL OR DAMAGE WILL OCCUR.

Procedure

NOTE

Maintenance procedures for the left and right fuel vent manifolds are the same, except where noted.

NOTE

Any evidence of fuel leakage must be identified before cleaning.

NOTE

Refer to the document "FUEL-F6101-SPM20-00" (Zodiac Aerospace), for additional information on the cleaning of the fuel cells.

1. Clean the internal surface of the fuel cell with a clean cloth (C-516) moistened with a cleaning solution (601199). Dry the liner with low-pressure filtered compressed air for 2 to 4 hours, at approximately 68°F (20°C).
2. Clean small areas on the external surface of the fuel cell with a clean cloth (C-516) moistened with a cleaning solution (601199). Dry with a clean cloth (C-516).
3. Clean an extensive area on the external surface of the fuel cell with a solution of soap (100066) and hot water. Rinse with water and dry with a clean cloth (C-516).
4. Dry clean the fittings of the fuel cell with filtered compressed air and a clean cloth (C-516) to remove any blockage or obstruction.
5. Clean fungus growth found on the fuel cell or on the fuel cell cavity per the instructions contained in the BHT-ALL-SPM, Chapter 5, Standard Practices Manual.

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Lower Middle Cells and Lower Main Cells - Inspection

Procedure

NOTE

Maintenance procedures for the left and right fuel vent manifolds are the same, except where noted.

1. Inspect cells for cracks, scuffing, and general condition.

2. Inspect threaded inserts for serviceability. Damage which can be detected visually requires replacement of insert.

3. Inspect hoses for damage and deterioration. Damage which can be detected visually requires replacement of hose.

4. Inspect removed components for damage.

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Lower Middle Cells and Lower Main Cells - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Talcum powder	C-408	AR	
Lacing cord	C-471	AR	

Procedure

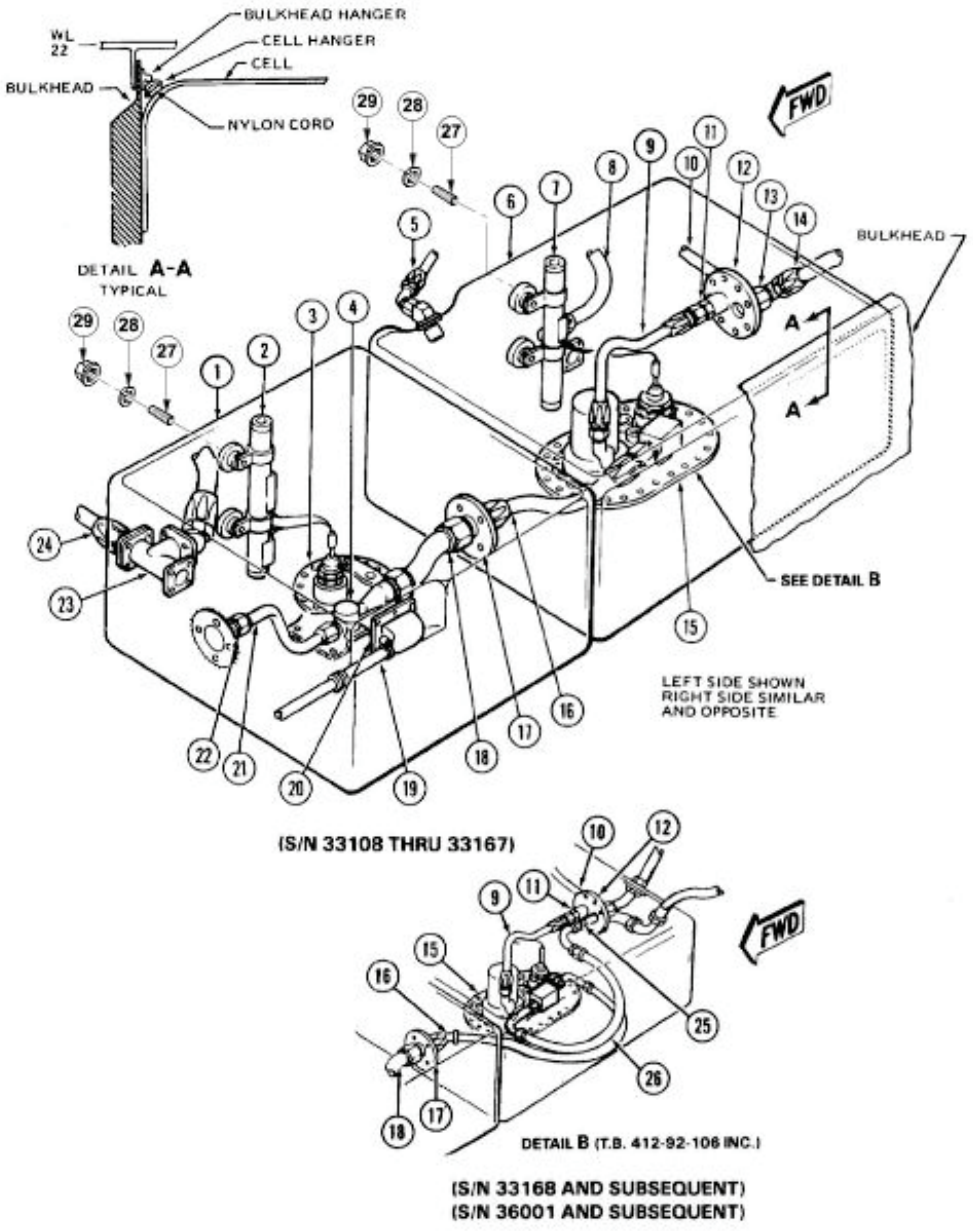


DO NOT HANDLE CELLS WHEN AMBIENT TEMPERATURE IS BELOW 65°F (18°C).

NOTE

The following installation procedures for left and right installations are the same.

1. Remove protective caps, plugs, and covers. Inspect the full cell cavity for clean, smooth condition of all surfaces.
2. Apply talcum powder (C-408) to cell cavity surfaces.
3. *Put the interconnected lower middle cell (1,* Figure 1*) and lower main cell (6) in cell cavity.*
4. Align lower forward cell and lower middle cell interconnect parts with opening in bulkhead.
5. With an allen key, install the setscrews (27) in the lower middle fuel cells (1) and lower main fuel cells (6), through the bulkhead.
6. Make four turns of the setscrews (27) with the allen key.
7. While holding the setscrews (27) in place with the allen key, install the washers (28) and nuts (29). Tighten the nuts.
8. Install lower main cell inlet/outlet fitting (12) (412-A-28-10-00-63A-720A-A / 00530).
9. *Install lower forward cell/lower middle cell interconnect fitting (*412-A-28-10-00-47A-720A-A / 00474*) and lower forward cell check valve (*412-A-28-10-00-46A-720A-A / 00471*).*
10. Make sure that all internal cell components are tightly mounted and all internal hose and tube connections have been tightened.
11. Install lower forward cell sump assembly (3, Figure 2) (412-A-28-10-00-41A-720A-A / 00458).
12. Install lower main cell sump assembly (15, Figure 1) (412-A-28-10-00-46A-720A-A / 00471).
13. Install lower middle cell sump assembly (3) (412-A-28-10-00-51A-720A-A / 00490).
14. Install lower middle cell interconnect fitting (23) or hose (24) (412-A-28-10-00-54A-720A-A / 00501).
15. Install lower main cell interconnect hose (8) (412-A-28-10-00-56A-720A-A / 00507).
16. *Install lower middle cell vent hose (19) (*412-A-28-10-00-49A-720A-A / 00483*).*
17. Install lower main cell vent hose (5) (412-A-28-10-00-55A-720A-A / 00504).
18. Lace and tie a single length of lacing cord (C-471) between hangers around top of cells and cell cavity (Figure 1).
19. Install cabin floor panels. Install access doors on underside of helicopter.
20. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).
21. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).
22. Perform fuel system operational check.
23. Inspect the area for leaks.



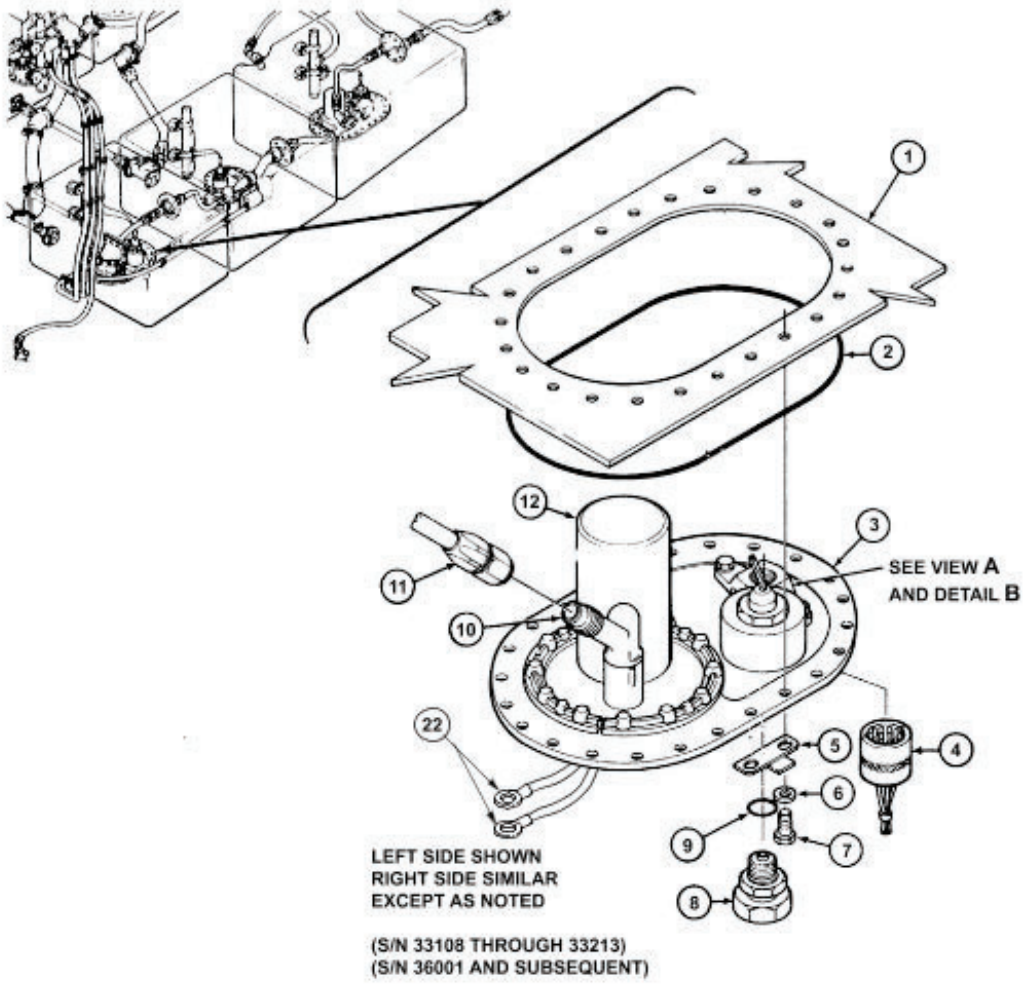
ICN-412-A-281000-B-97499-00058-A-001-01

Figure 1. Lower Middle Cell/Lower Main Cell (Sheet 1 of 2)

1. Lower middle cell
2. Fuel quantity probe
3. Sump asssembly
4. Ejector pump
5. Vent hose
6. Lower main cell
7. Fuel quantity probe
8. Interconnect hose
9. Hose
10. Hose
11. Reducer
12. Lower main cell inlet/ outlet fitting
13. Union
14. Hose
15. Sump assembly
16. Hose
17. Lower middle cell/lower main cell interconnect fitting
18. Tube
19. Vent hose
20. Retainer
21. Hose
22. Union
23. Lower middle cell interconnect fitting
24. Interconnect hose
25. Union
26. Hose
27. Setscrew
28. Washer
29. Nut

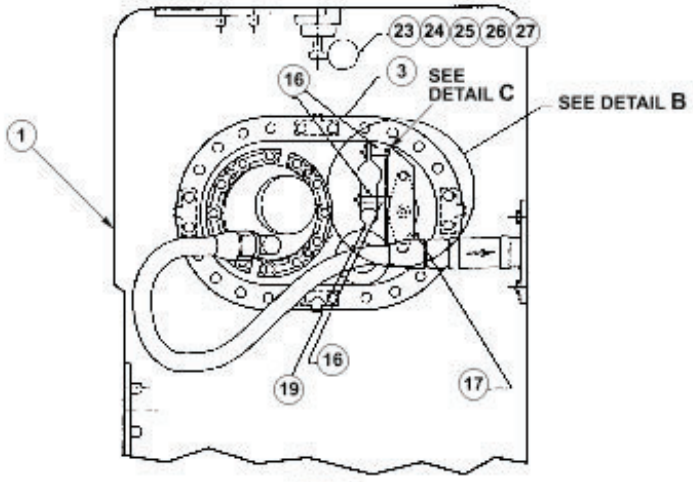
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Figure 1. Lower Middle Cell/Lower Main Cell (Sheet 2 of 2)

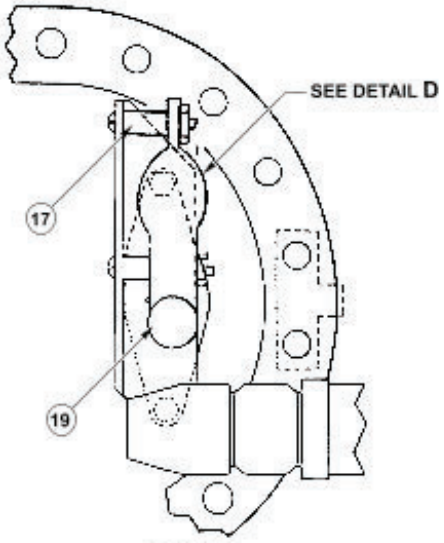


ICN-412-A-281000-B-97499-00051-A-001-01

Figure 2. Lower Forward Cell Sump Assembly (Sheet 1 of 3)



VIEW A
LOOKING DOWN (LHS SHOWN, RHS TYPICAL)
PRE TB 412-95-133



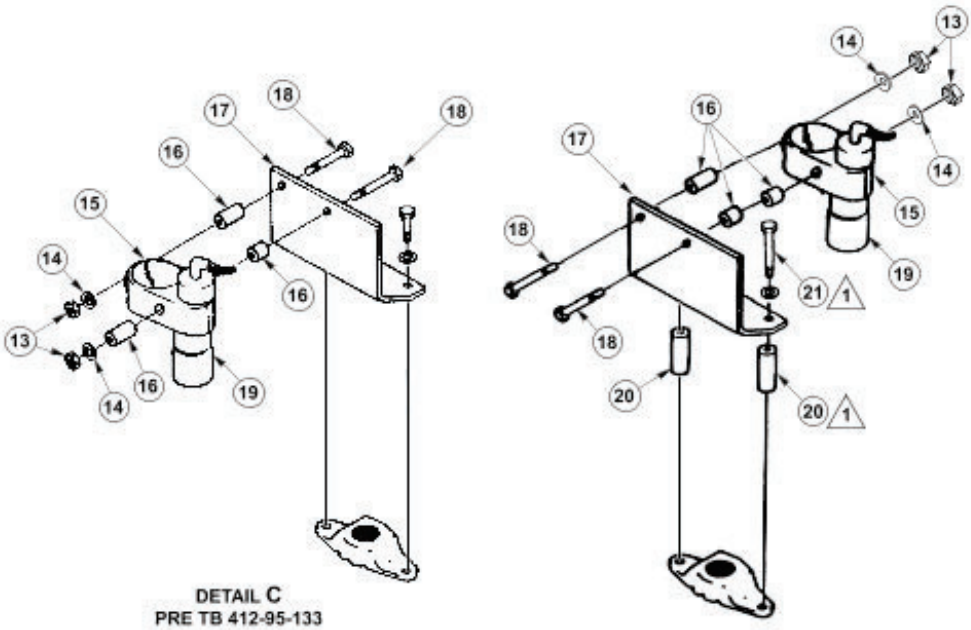
DETAIL B
S/N 36087, 36096 AND 36110 AND SUBSEQUENT
POST TB 412-95-133 (1)

NOTE

(1) On helicopters S/N 33108 through 33213, 36001 through 36086, 36088 through 36095, and 36097 through 36109.

ICN-412-A-281000-B-97499-00052-A-001-01

Figure 2. Lower Forward Cell Sump Assembly (Sheet 2 of 3)



1. Lower forward cell
2. Packing
3. Sump assembly
4. Electrical connector
5. Retainer
6. Washer
7. Bolt
8. Drain valve
9. Packing
10. Fitting
11. Hose
12. Transfer pump
13. Nut
14. Washer
15. Clamp
16. Spacer
17. Bracket
18. Bolt
19. Fuel sensor (thermistor)
20. Spacer (NAS43DD3-96N)
21. Bolt (NAS6203.28)
22. Transfer pump electrical leads
23. Fuel level transmitter
24. Bolt
25. Washer
26. Spacer
27. Mounting bracket

NOTE

1 Right-hand installation only, Post TB 412-95-133.

ICN-412-A-281000-B-97499-00053-A-001-01

Figure 2. Lower Forward Cell Sump Assembly (Sheet 3 of 3)

Forward to Middle Fuel Cell - Leak Check

Procedure

1. Make sure that the fuel indicator shows zero.
2. Open both middle tank sump drain valves.
3. Open both forward sump drain valves.
4. Push “IN” all fuel circuit breakers.
5. Make sure that the fuel valve switches, boost pump switches and transfer pump switches are in “OFF” position.



DO NOT TURN BOOST PUMP OR TRANSFER SWITCHES “ON” UNTIL AFTER MID TANK LEAK CHECK IS COMPLETED.

6. Turn fuel interconnect switch to “OPEN” and make sure that crossfeed switch is on “NORMAL” position.
7. *Fuel the helicopter to 700 pounds (317.5 kg) or 108 gallons (408.8 L). Check that the fuel low caution light goes out at approximately 450 pounds (204 kg) or 70 gallons (265 L).*
8. After 5 to 10 minutes, examine for fuel leaking out of the middle tank sump drain valves. This examines for leaks between the middle tanks and main tanks. Close both forward sump drain valves.
9. Fuel the helicopter to 1200 pounds (544.3 kg) or 184 gallons (696.5 L).
10. After 5 to 10 minutes, examine for fuel leaking out of the middle tank sump drain valves. This examines for leaks between forward and middle tanks. Close both middle tank sump drain valves.

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Lower Middle Cell Vent Hoses - Description

NOTE

Maintenance procedures for left and right lower middle cell vent hoses are the same.

Vent hoses are attached to vent ports on the upper outboard side of each lower middle cell. The vent hoses connect the lower middle cells into the forward venting system. Fume tight boots are installed on hose connections to lower middle cells to prevent escape of fumes into surrounding cell areas.

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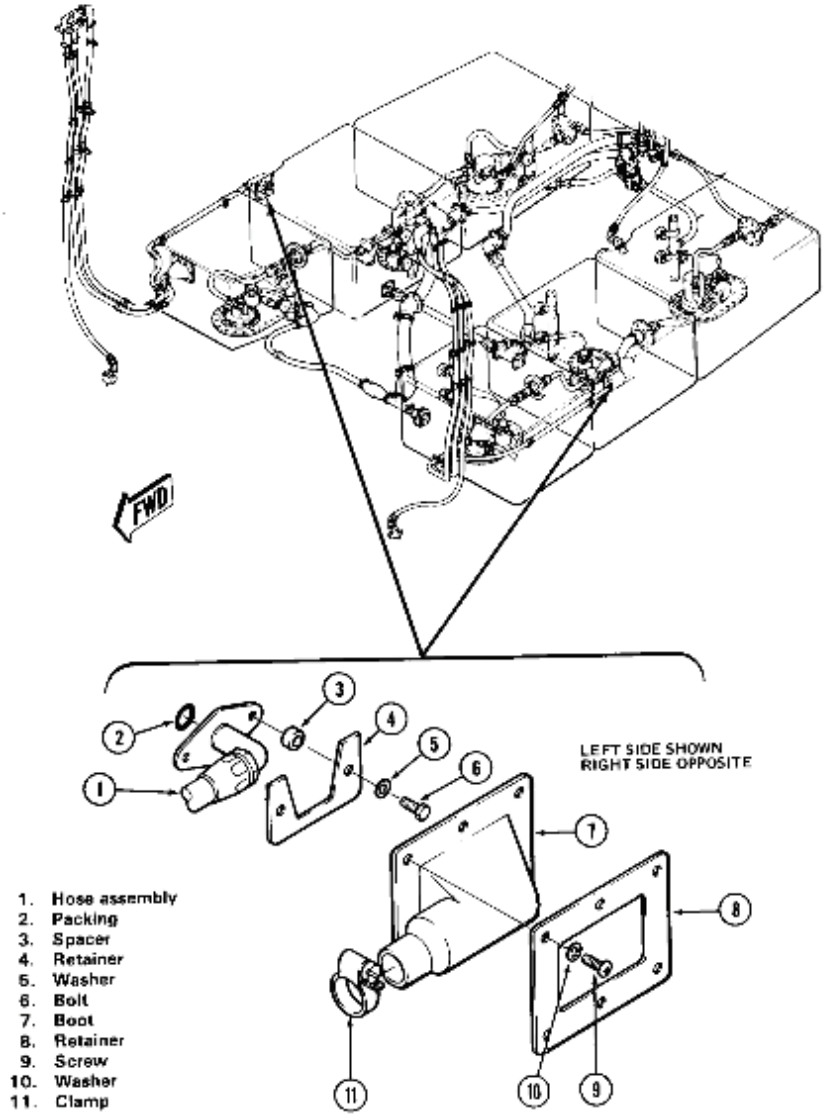
Lower Middle Cell Vent Hoses - Removal

Procedure

NOTE

The following removal procedures for left and right lower middle cell vent hoses are the same.

1. Disconnect electrical power.
2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).
3. Remove floor panel above cell.
4. Loosen clamp (11, Figure 1) and slide forward on hose assembly (1).
5. Remove screws (9) and washers (10).
6. Loosen and move retainer (8) forward on hose assembly (1).
7. Loosen boot (7) and slide over hose assembly (1) end connector and forward on hose assembly to obtain access to hose assembly end connector.
8. Remove bolts (6), washers (5), retainer (4), and spacers (3).
9. Remove hose assembly (1) and packing (2) from lower middle cell vent port. Discard packing.
10. Install protective cover over cell opening.



ICN-412-A-281000-B-97499-00060-A-001-01

Figure 1. Lower Middle Cell Vent Hose (Sheet 1 of 1)

Lower Middle Cell Vent Hoses - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

NOTE

The following installation procedures for left and right lower middle cell vent hoses are the same.

1. Remove protective cover from cell opening.

NOTE

Ensure clamp (11, Figure 1), retainer (8), and boot (7) are on hose assembly (1).

2. Apply assembly fluid (C-024) to packing (2).

3. Place packing (2) and hose assembly (1) end flange in position on lower middle cell vent port.

4. Install screws (6), washers (5), retainer (4), and spacers (3).

5. Position boot (7) in place over hose assembly (1) end flange.

6. Place retainer (8) in place over boot (7).

7. Install screws (10) and washers (9).

8. Position clamp (11) in place over end of boot (7). Tighten clamp.

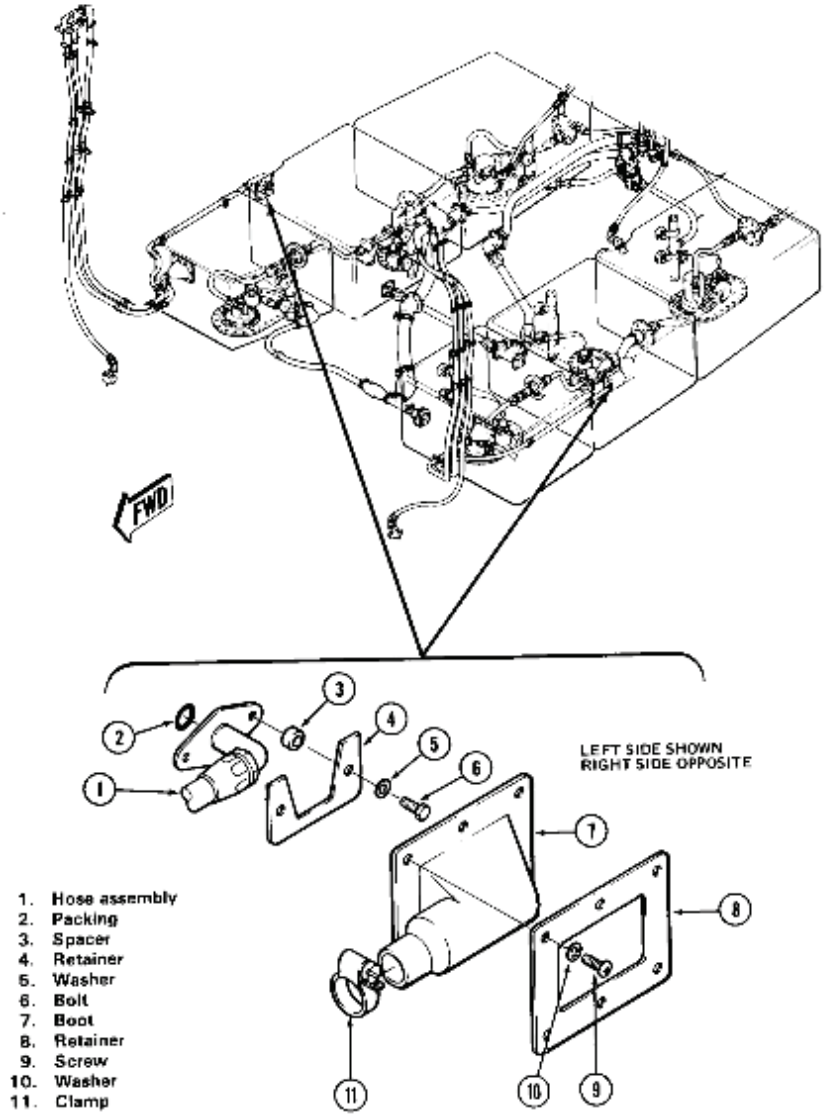
9. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).

10. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).

11. Perform fuel system operational check.

12. Check area for leaks.

13. Install floor panel above cell.



ICN-412-A-281000-B-97499-00060-A-001-01

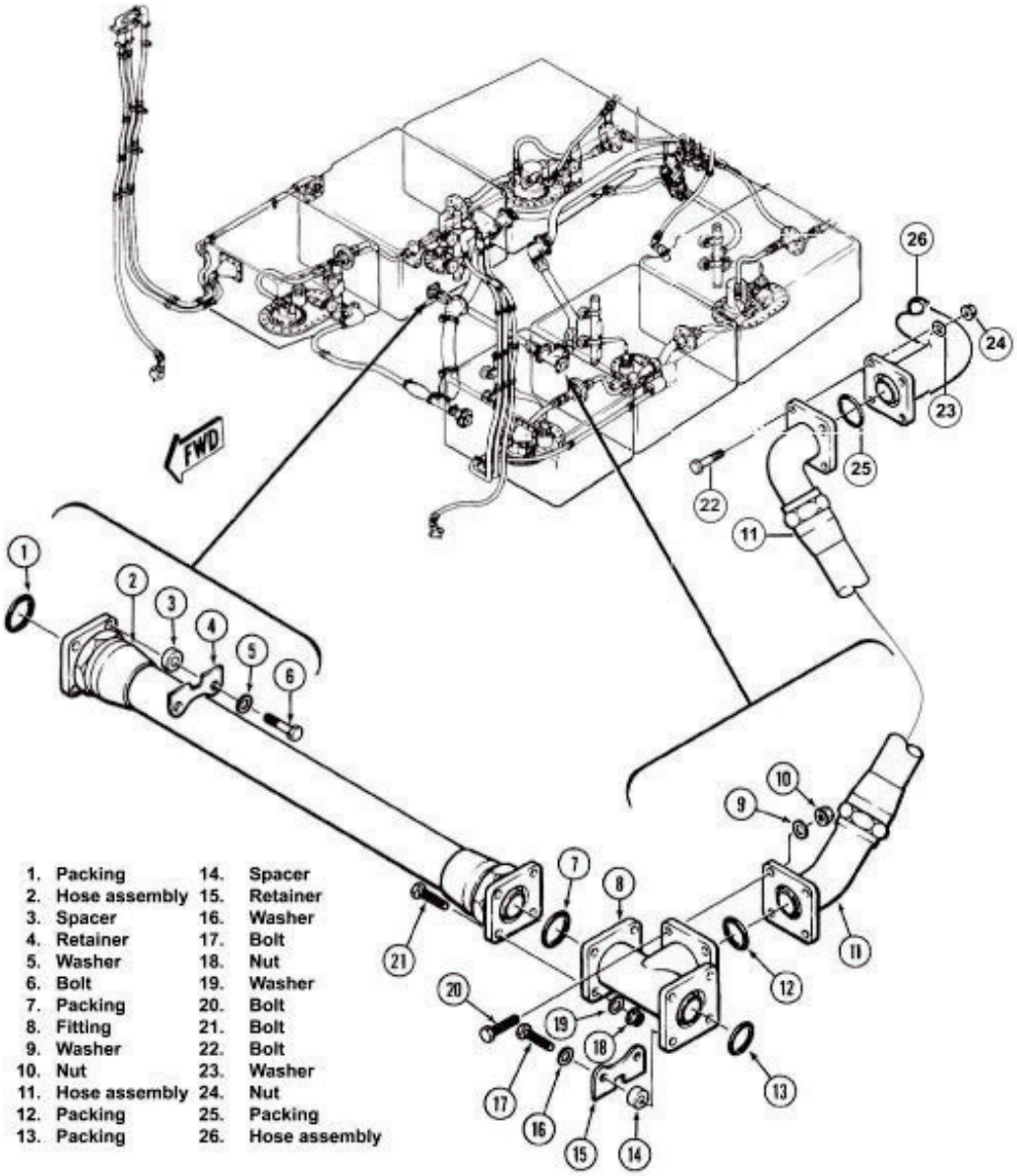
Figure 1. Lower Middle Cell Vent Hose (Sheet 1 of 1)

Lower Middle Cell Interconnect Hose and Fitting - Description

NOTE

Both the interconnect hose and fitting are included in the removal and installation procedures. If both are not to be removed and/or installed, only the applicable portions of the procedures need be performed.

A fitting (8, Figure 1) is connected directly to the inboard interconnect port of the left lower middle cell. An interconnect hose assembly (2) is attached to one port of the fitting and to the inboard interconnect port of the right lower middle cell. An interconnect hose assembly (11) is attached to the remaining port of the fitting (8) and to the upper forward center cell output port.



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Figure 1. Lower Middle Cell Interconnect Hose and Fitting (Sheet 1 of 1)

Lower Middle Cell Interconnect Hose and Fitting - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure

1. Disconnect electrical power.

2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).

3. Remove cabin floor access panels above the lower left and right middle cells.

4. Remove bolts (6, Figure 1), washers (5), retainers (4), and spacers (3) from hose assembly (2).

5. Remove nuts (18), washers (19), and bolts (21) from fitting (8).

6. Remove packings (1 and 7) and hose assembly (2). Discard packings.

7. Remove bolts (20), washers (9), and nuts (10) from fitting (8) and hose assembly (11).

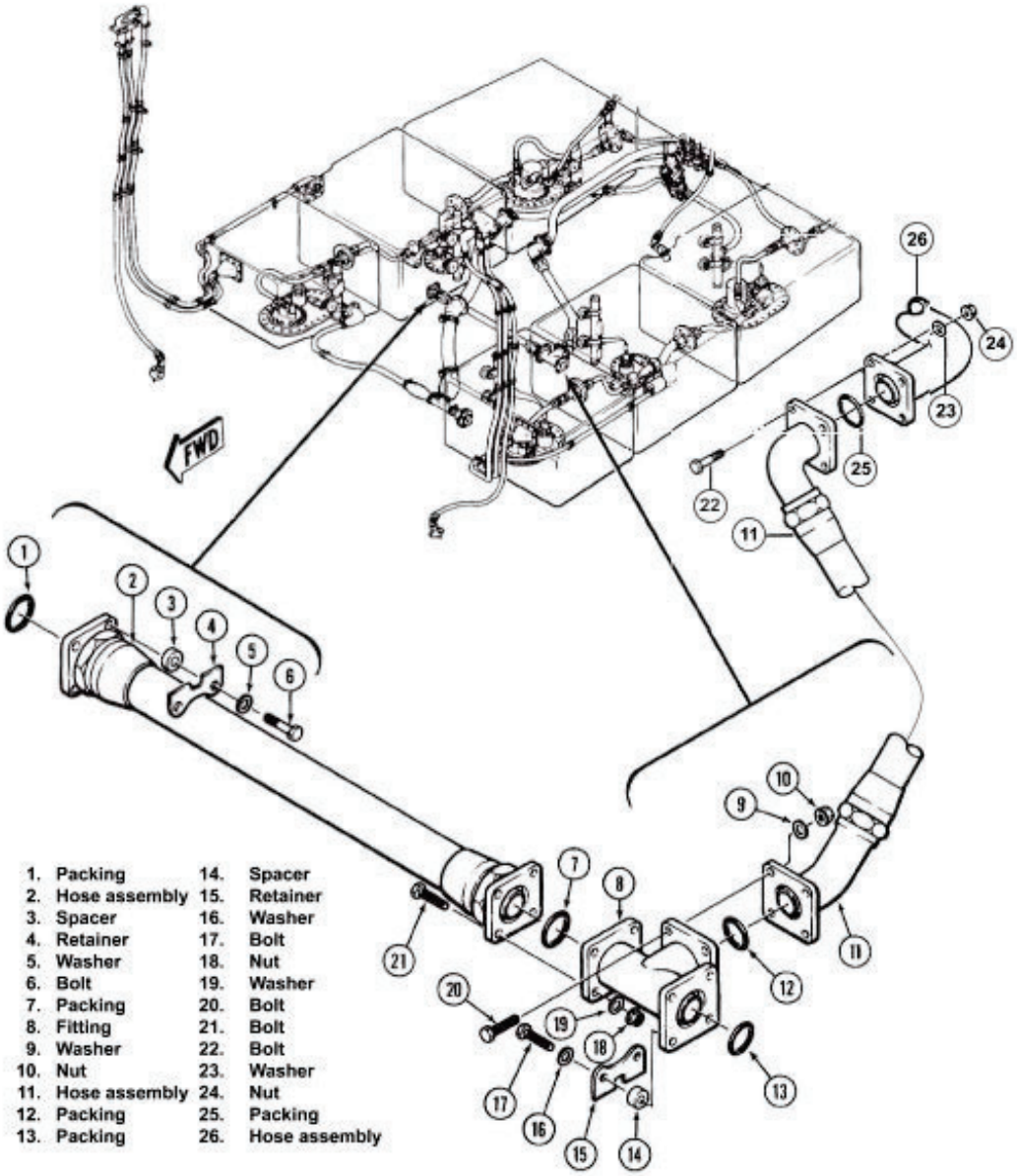
8. Remove bolts (22), washers (23), and nuts (24) from hose assemblies (11 and 26).

9. Remove hose assembly (11) packings (12 and 13). Discard packings.

10. Remove bolts (17), washers (16), retainer (15), and spacers (14) from fitting (8).

11. Remove fitting (8) and packing (12). Discard packing.

12. Install caps or plugs (C-428) on the fuel cell openings, the open end fittings of the hose assemblies (2 and 11), and on the fitting (8).



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Figure 1. Lower Middle Cell Interconnect Hose and Fitting (Sheet 1 of 1)

Lower Middle Cell Interconnect Hose and Fitting - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

1. Apply assembly fluid (C-024) to packings (12 and 13, Figure 1).

2. Remove protective covers from cell opening and hose end. Place packings (7, 12, and 13) on fitting (8).

3. Install bolts (17), washers (16), retainers (15), and spacers (14). Tighten bolts.

4. Install hose assembly (11), bolts (20), washers (9), and nuts (10). Tighten bolts.

5. Place packing (7) on hose assembly (2).

6. Install bolts (21), washers (19), and nuts (18). Tighten bolts.

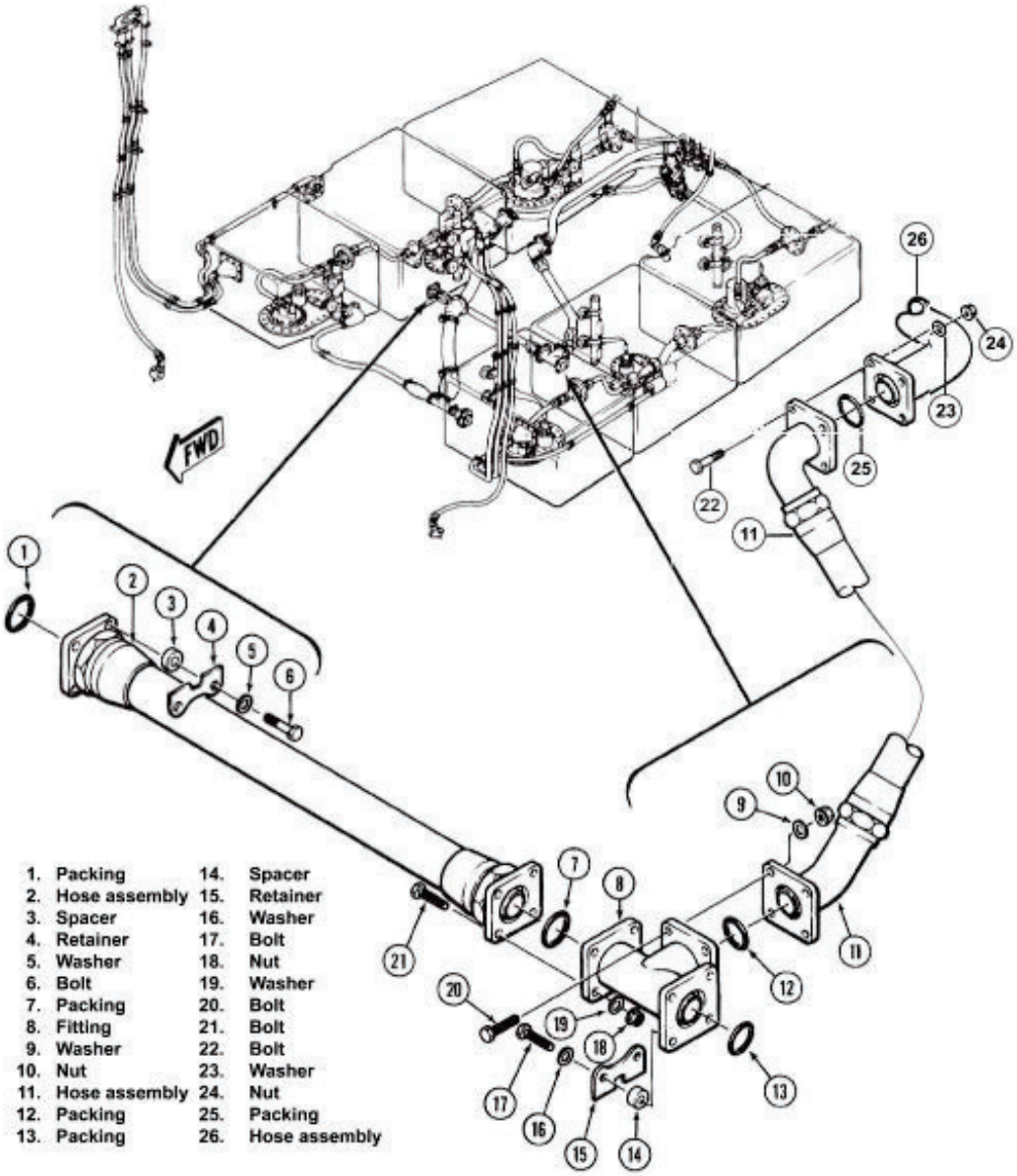
7. Place packing (1) on hose assembly (2).

8. Install bolts (6), washers (5), retainers (4), and spacers (3). Tighten bolts.

9. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).

10. Check area for leaks.

11. Install cabin floor panels.



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Figure 1. Lower Middle Cell Interconnect Hose and Fitting (Sheet 1 of 1)

Lower Middle Cell Sump Assemblies - Description

NOTE

Maintenance procedures are the same for both left and right lower middle cell sump assemblies.

A sump assembly is mounted on the opening on the bottom of each lower middle cell. Removal of the sump assembly provides access for maintenance of the ejector pump, drain valve, and fuel quantity probe. The drain valve can be removed without removing the sump assembly.

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Lower Middle Cell Sump Assemblies - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure

NOTE

The following removal procedures are the same for both left and right lower middle cell sump assemblies.

1. Disconnect electrical power.

2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).

3. Remove sump access door.

4. Disconnect electrical connector (6, Figure 1).

5. Remove bolts (7) with washers (8) and four retainers (9).

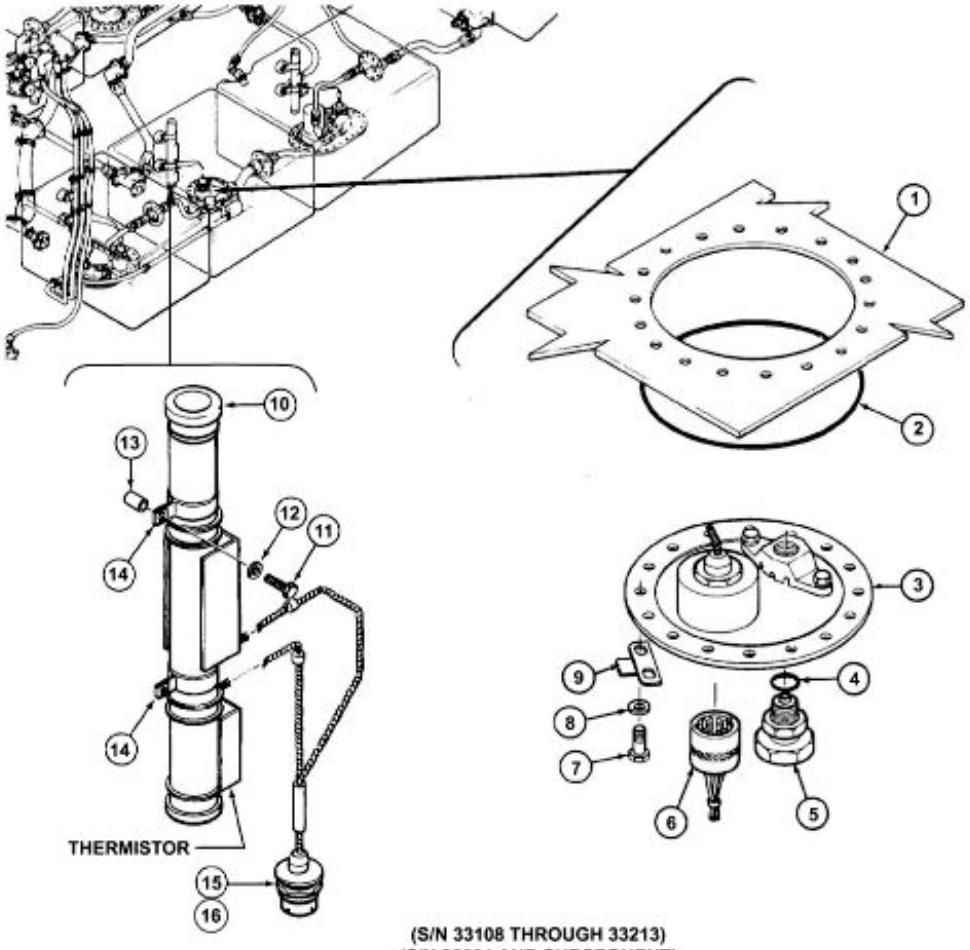
6. Lower sump assembly (3) as needed for access.

7. Remove the bolts (11), washers (12), and spacers (13) from the mounting brackets (14).

8. Remove the fuel quantity transmitter (10) and lay it on the sump assembly (3).

9. Remove sump assembly (3) with fuel quantity transmitter (10) and packing (2). Discard packing.

10. Install caps or plugs (C-428) and covers over cell opening.



(S/N 33108 THROUGH 33213)
(S/N 36001 AND SUBSEQUENT)

1. Lower middle cell
2. Packing
3. Sump assembly
4. Packing
5. Drain valve
6. Electrical connector
7. Bolt
8. Washer
9. Retainer
10. Fuel quantity transmitter
11. Bolt
12. Washer
13. Spacer
14. Mounting bracket
15. Electrical connector
16. Packing

ICN-412-A-281000-B-97499-00062-A-001-01

Figure 1. Lower Middle Cell Sump Assembly (Sheet 1 of 1)

Lower Middle Cell Sump Assemblies - Inspection and Repair

Procedure

NOTE

The following inspection and repair procedures are the same for both left and right lower middle cell sump assemblies.

1. Inspect area where sump assembly mates with cell for damage and cleanliness. Inspect packing groove for rough edges or nicks.

2. Inspect sump assembly for security of mounting.

3. Inspect drain valve for security.

4. Inspect electrical leads and connector for condition.

5. Repair or replace components which fail to meet inspection requirements.

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Lower Middle Cell Sump Assemblies - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

NOTE

The following installation procedures are the same for both left and right lower middle cell sump assemblies.

1. Remove protective covers, caps, and/or plugs from cell opening.

2. Coat packing (2, Figure 1) with assembly fluid (C-024). Place packing in groove around lower middle cell (1) sump opening.

3. Connect internal electrical wiring.

4. Hold lower middle cell sump assembly in position over sump opening in lower middle cell (1).

5. Install the bolts (11) and washers (12) through the mounting brackets (14) of the fuel quantity transmitter (10).

6. Place the spacers (13) on the bolts (11), between the mounting brackets (14) and the lower middle cell (1) inboard wall. Tighten the bolts.

7. Install bolts (7), washers (8), and retainers (9).

8. Connect electrical connector (6).

9. Apply assembly fluid (C-024) to packing (4).

10. Install packing (4) on drain valve (5).

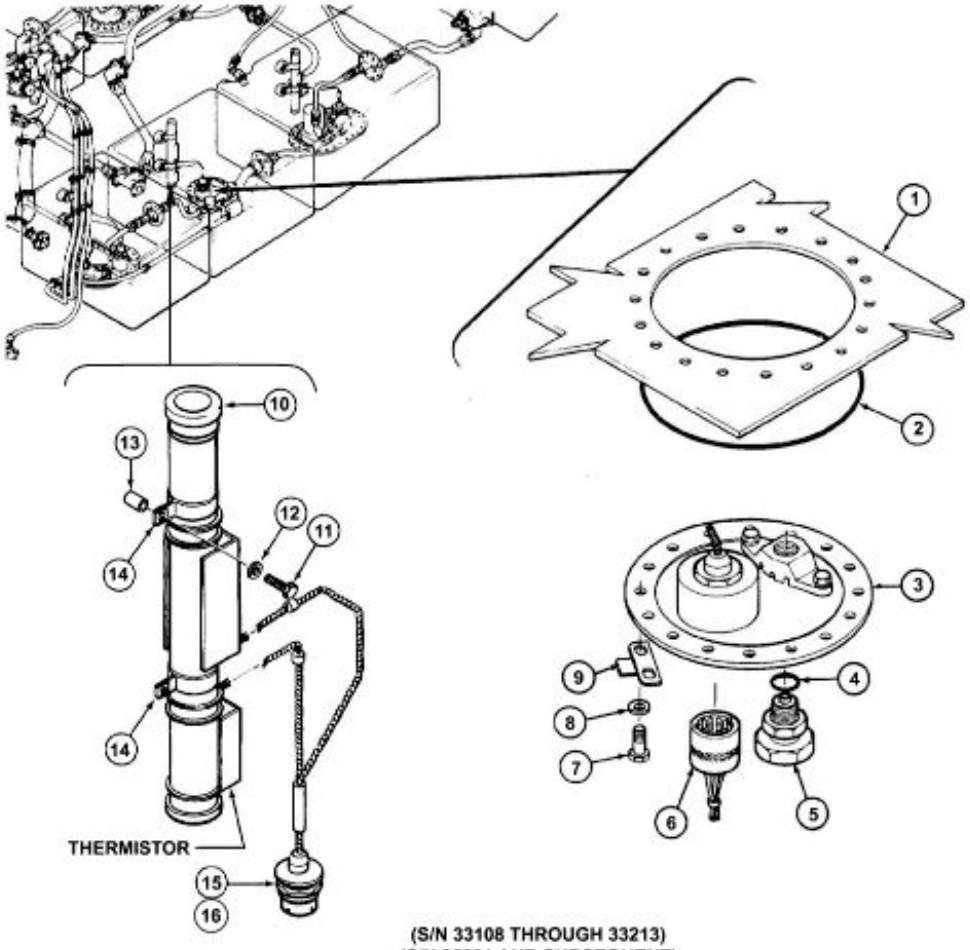
11. Install drain valve (5) on sump assembly (3).

12. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).

13. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).

14. Perform fuel system operational check.

15. Check area for leaks.



(S/N 33108 THROUGH 33213)
(S/N 36001 AND SUBSEQUENT)

1. Lower middle cell
2. Packing
3. Sump assembly
4. Packing
5. Drain valve
6. Electrical connector
7. Bolt
8. Washer
9. Retainer
10. Fuel quantity transmitter
11. Bolt
12. Washer
13. Spacer
14. Mounting bracket
15. Electrical connector
16. Packing

ICN-412-A-281000-B-97499-00062-A-001-01

Figure 1. Lower Middle Cell Sump Assembly (Sheet 1 of 1)

Ejector Pumps - Description

Applicable: SN: 33168-Subsequent:

NOTE

Maintenance procedures for lower middle cell ejector pumps are the same.

An ejector pump is mounted internally in each lower middle cell. The pumps are operated by fuel flow through the hoses attached to the output ports of the transfer pumps. The ejector pumps transfer fuel from the lower middle cells to the lower main cells (aft center cells). The pumps cease operation when fuel flow from the transfer pumps ceases.

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Ejector Pumps - Removal

Preliminary requirements

Consumables, materials and expendables

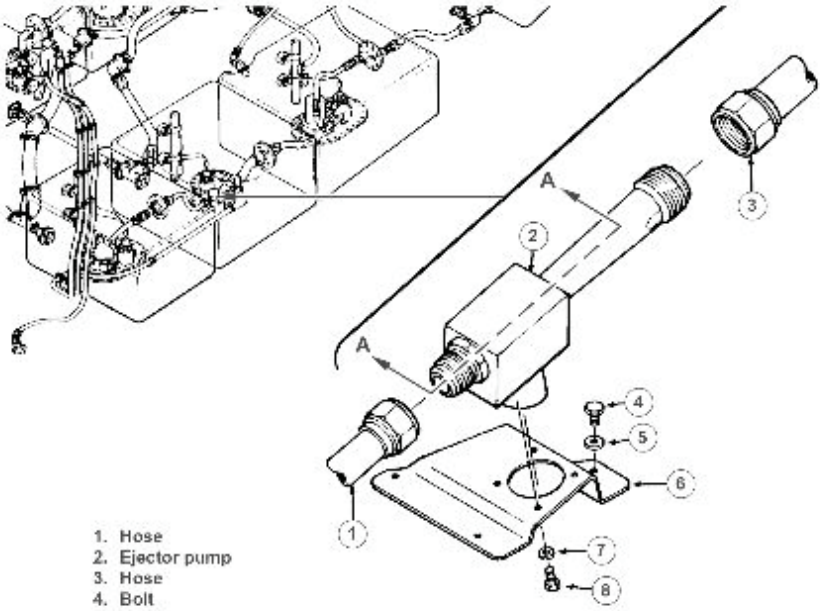
Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

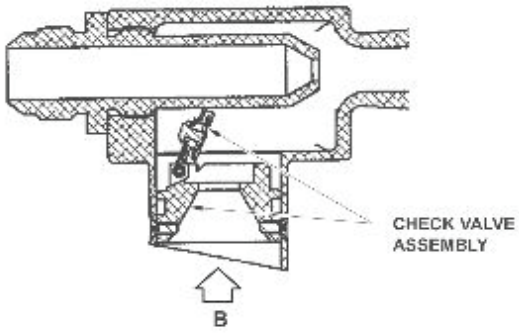
Procedure

NOTE

The following removal procedures for lower middle cell ejector pumps are the same.

1. Remove lower middle cell sump assembly (412-A-28-10-00-51A-520A-A / 00488) to gain access to ejector pump.
2. Disconnect hoses (1 and 3, Figure 1) from ejector pump (2).
3. Remove bolts (4) and washers (5) from mounting bracket (6).
4. Remove ejector pump (2) with mounting bracket (6).
5. Remove bolts (8) and washers (7) securing bracket to pump.
6. Install protective cover over cell opening.
7. Install caps or plugs (C-428) on hoses (1 and 3) end fittings and on fuel ejector pump (2) ports.







ICN-412-A-281000-B-97499-00063-A-001-01

Figure 1. Ejector Pump (Sheet 1 of 1)

Ejector Pumps - Cleaning

Procedure

NOTE

The following cleaning procedures for lower middle cell ejector pumps are the same.

1. Remove any obstructions from openings of ejector pump using compressed air.



DO NOT ENLARGE NOZZLE OPENING OF EJECTOR PUMP WHILE CLEANING.

2. If necessary, clean nozzle opening of ejector pump using lockwire.

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Ejector Pumps - Inspection

Procedure

NOTE

The following inspection procedures for lower middle cell ejector pumps are the same.

1. Inspect ejector pump for mechanical damage, corrosion, looseness of parts, and threads for condition.

2. Inspect ejector pump for contamination.

NOTE

SN: 36360-Subsequent:

The following step is only applicable to ejector pumps incorporating a flapper valve.

3. Inspect ejector pump flapper valve for condition and freedom of movement.

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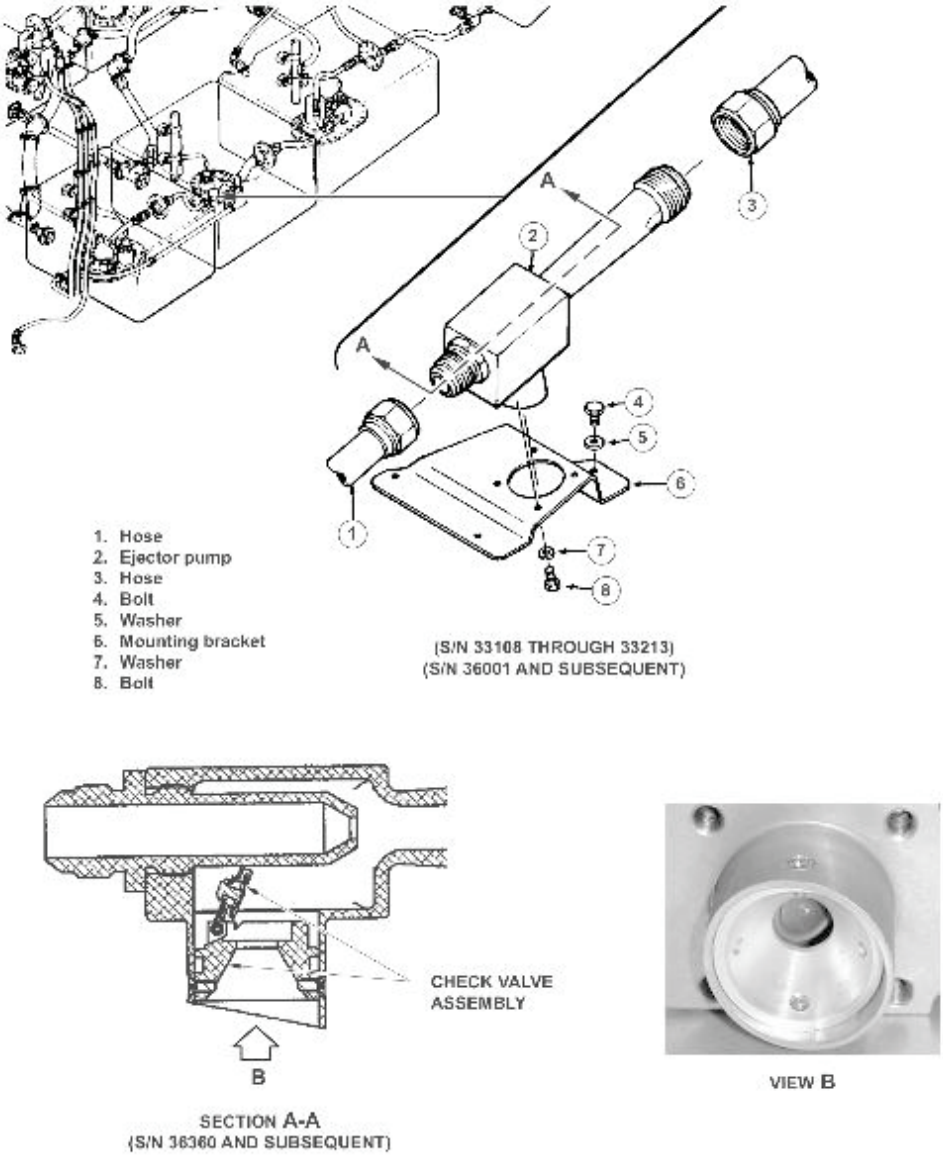
Ejector Pumps - Installation

Procedure

NOTE

The following installation procedures for lower middle cell ejector pumps are the same.

1. Remove protective cover from cell opening.
2. Remove protective caps and/or plugs from hoses (1 and 3) end fittings and from ejector pump (2) ports.
3. Install mounting bracket (6, Figure 1) on ejector pump (2) with bolts (8) and washers (7).
4. Place ejector pump (2) and mounting bracket (6) over mounting bosses in cell. Install bolts (4) and washers (5).
5. Connect hoses (1 and 3). Tighten end fittings.
6. Install sump assembly (412-A-28-10-00-51A-720A-A / 00490).



ICN-412-A-281000-B-97499-00063-A-001-01

Figure 1. Ejector Pump (Sheet 1 of 1)

Sump Drain Valves - Description

NOTE

Maintenance procedures for the lower middle cell drain valves are the same.

A manually operated sump drain valve is mounted on the sump assembly of each lower middle cell. The sump drain valves provide for the draining off of any possible moisture accumulation or other contamination.

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Sump Drain Valves - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

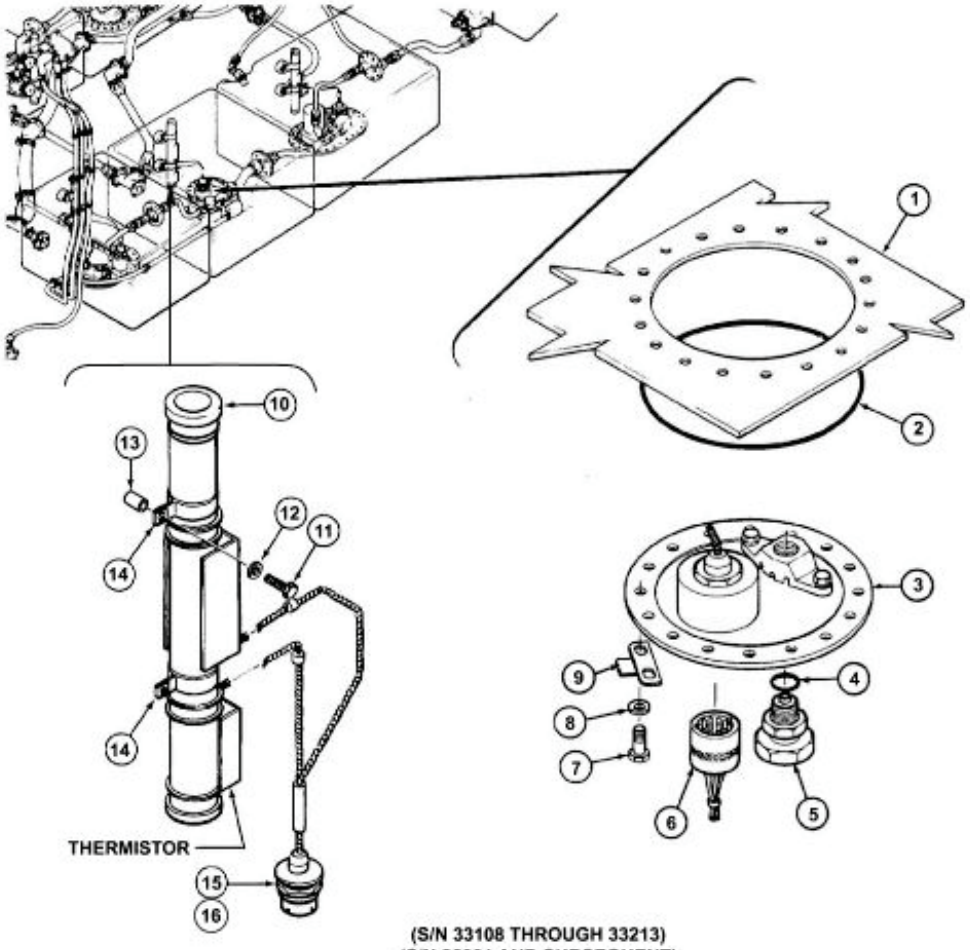
Procedure

1. Disconnect the electrical power.

2. Defuel the system (412-A-12-10-01-00A-221A-A / 00131).

3. Remove the drain valve (5, Figure 1) and packing (4). Discard the packing.

4. Install the caps or plugs (C-428) and covers on the cell opening.



(S/N 33108 THROUGH 33213)
(S/N 36001 AND SUBSEQUENT)

1. Lower middle cell
2. Packing
3. Sump assembly
4. Packing
5. Drain valve
6. Electrical connector
7. Bolt
8. Washer
9. Retainer
10. Fuel quantity transmitter
11. Bolt
12. Washer
13. Spacer
14. Mounting bracket
15. Electrical connector
16. Packing

ICN-412-A-281000-B-97499-00062-A-001-01

Figure 1. Lower Middle Cell Sump Assembly (Sheet 1 of 1)

Sump Drain Valves - Installation

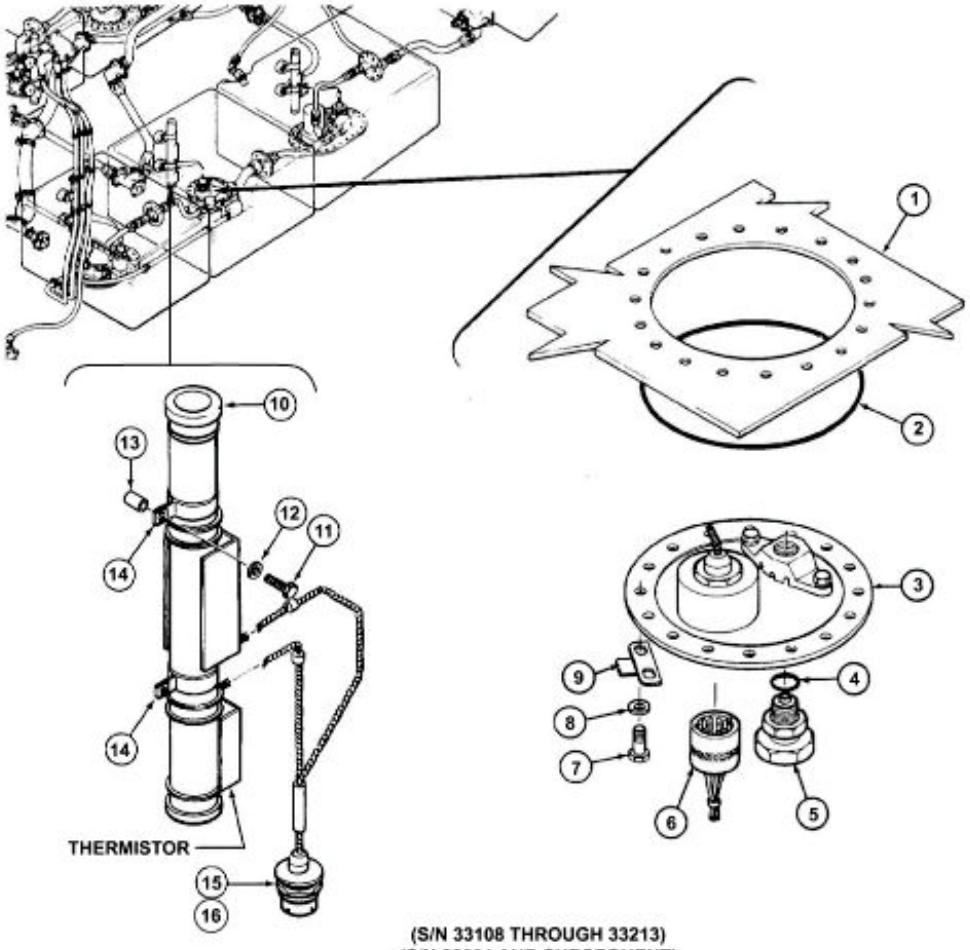
Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

1. Remove protective covers, caps, and/or plugs from cell opening.
2. Apply assembly fluid (C-024) to packing (4, Figure 1).
3. Install packing (4) on drain valve (5).
4. Install drain valve (5) into sump assembly (3).
5. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).
6. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).
7. Perform fuel system operational check.
8. Check area for leaks.



(S/N 33108 THROUGH 33213)
(S/N 36001 AND SUBSEQUENT)

1. Lower middle cell
2. Packing
3. Sump assembly
4. Packing
5. Drain valve
6. Electrical connector
7. Bolt
8. Washer
9. Retainer
10. Fuel quantity transmitter
11. Bolt
12. Washer
13. Spacer
14. Mounting bracket
15. Electrical connector
16. Packing

ICN-412-A-281000-B-97499-00062-A-001-01

Figure 1. Lower Middle Cell Sump Assembly (Sheet 1 of 1)

Lower Middle Cell/Lower Main Cell Interconnect Fittings - Description

NOTE

Maintenance procedures are the same for both the right and left lower middle cell/lower main cell interconnect fittings.

A lower middle cell/lower main cell interconnect fitting is located on the aft interconnect port of each forward middle cell. The fitting provides interconnection between the lower middle and lower main cells.

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Lower Middle Cell/Lower Main Cell Interconnect Fittings - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure

NOTE

The following removal procedures are the same for both the right and left lower middle cell/lower main cell interconnect fittings.

1. Remove floor panel above fitting to be removed.

2. Remove lower middle cell sump assembly (412-A-28-10-00-51A-520A-A / 00488).

3. Remove lower main cell sump (412-A-28-10-00-57A-520A-A / 00509).

4. From the lower main fuel cell (1, Figure 1), disconnect hose (2) from interconnect fitting (4).

5. From the lower middle fuel cell (6), disconnect tube (7) from interconnect fitting (4).

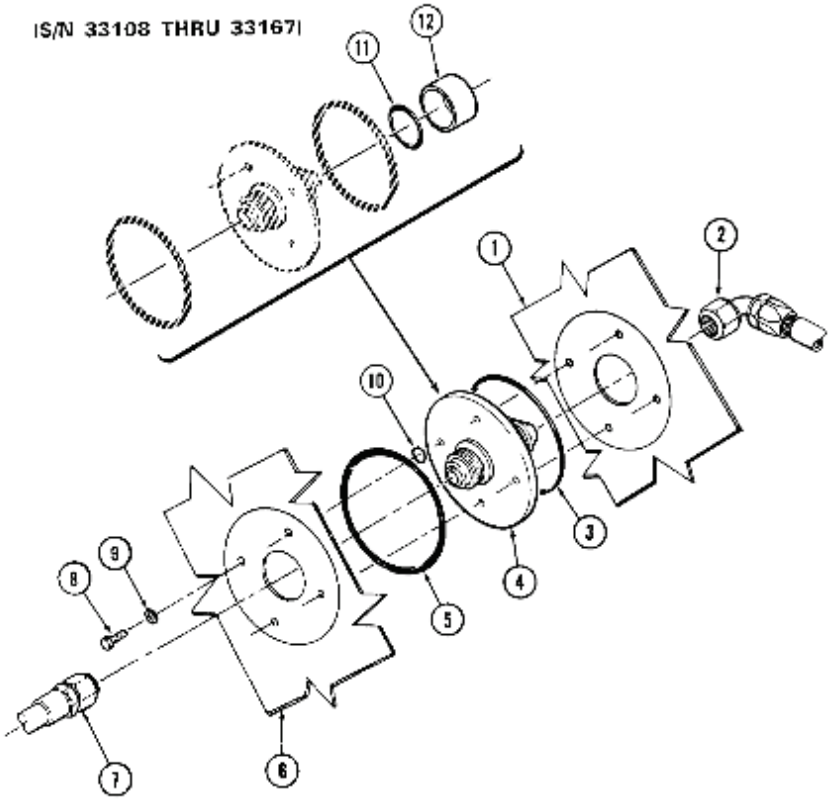
6. From the lower middle fuel cell (6), remove bolts (8) and seals (9). Discard seals.

7. Partially collapse lower middle cell (6). Remove interconnect fitting (4) and packings (3, 5, and 10). Discard packings.

8. Remove sleeve (12) and packing (11), if installed. Discard packing.

9. Install protective cover over cell opening.

10. Install caps or plugs (C-428) on the hose assemblies (2 and 7) end fittings.



(S/N 33168 THRU 33213)
(S/N 36001 AN SUBSEQUENT)

1. Lower main cell
2. Hose
3. Packing
4. Interconnect fitting
5. Packing
6. Lower middle cell
7. Tube
8. Bolt
9. Seal
10. Packing
11. Packing
12. Sleeve

ICN-412-A-281000-B-97499-00064-A-001-01

Figure 1. Lower Middle Cell/Lower Main Cell Interconnect Fitting (Sheet 1 of 1)

Lower Middle Cell/Lower Main Cell Interconnect Fittings - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

NOTE

The following installation procedures are the same for both the right and left lower middle cell/lower main cell interconnect fittings.

1. Remove protective cover from cell opening.

2. Remove the protective caps and/or plugs from the hose assemblies (2 and 7) end fittings.

3. Coat packings (3 and 5, Figure 1) with assembly fluid (C-024).

4. Coat packings (10 and 11), if removed, with assembly fluid (C-024).

5. Install interconnect fitting (4), packings (3 and 5), bolts (8), seals (9), and packings (10) between lower middle cell (6) and lower main cell (1).

6. Connect hose (2), sleeve (12), if removed, and tube (7) to interconnect fitting (4).



DO NOT PERFORM FUEL SYSTEM PRESSURE TEST, SERVICE, OR OPERATIONAL CHECK UNTIL COMPLETION OF Step 6..

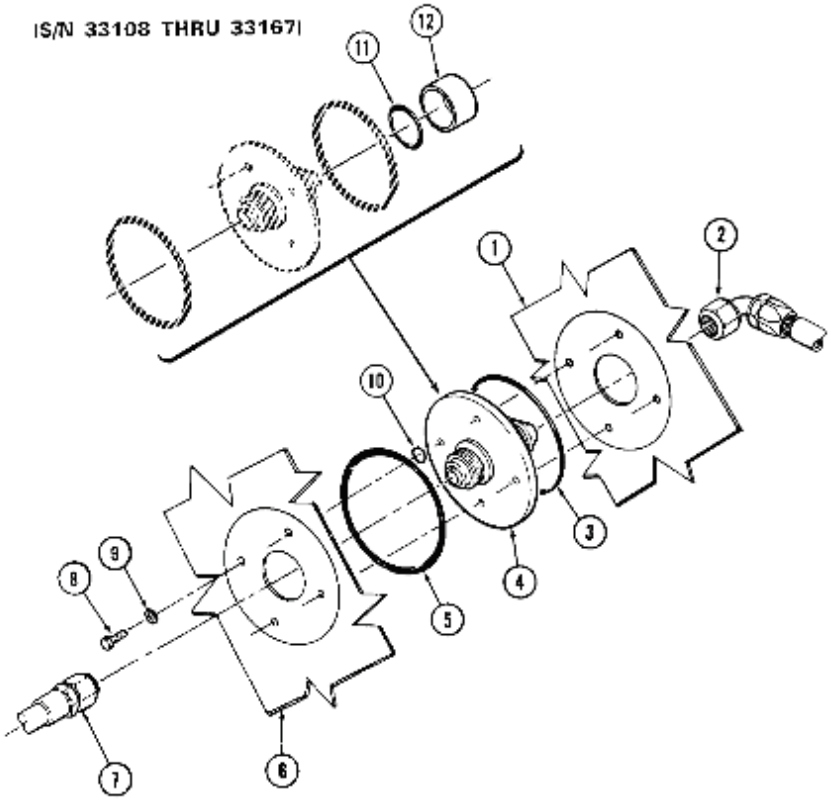
7. Install lower middle cell sump assembly (412-A-28-10-00-51A-720A-A / 00490).

8. Install lower main cell sump assembly (412-A-28-10-00-57A-280A-A / 00510).

9. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).

10. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).

11. Check for area leaks.



(S/N 33168 THRU 33213)
(S/N 36001 AN SUBSEQUENT)

1. Lower main cell
2. Hose
3. Packing
4. Interconnect fitting
5. Packing
6. Lower middle cell
7. Tube
8. Bolt
9. Seal
10. Packing
11. Packing
12. Sleeve

ICN-412-A-281000-B-97499-00064-A-001-01

Figure 1. Lower Middle Cell/Lower Main Cell Interconnect Fitting (Sheet 1 of 1)

Lower Main Cell Vent Valves and Hoses - Description

NOTE

Maintenance procedures are the same for both the right and left lower main cell vent valves.

Vent hoses are attached to the vent ports on the upper inboard side of each lower main cell. The vent hoses connect the lower main cells to the upper cell common venting system. Fume tight boots are installed on the hose connections to the lower main cells to prevent the escape of fumes into the surrounding cell areas.

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Lower Main Cell Vent Valves and Hoses - Removal

Preliminary requirements

Consumables, materials and expendables

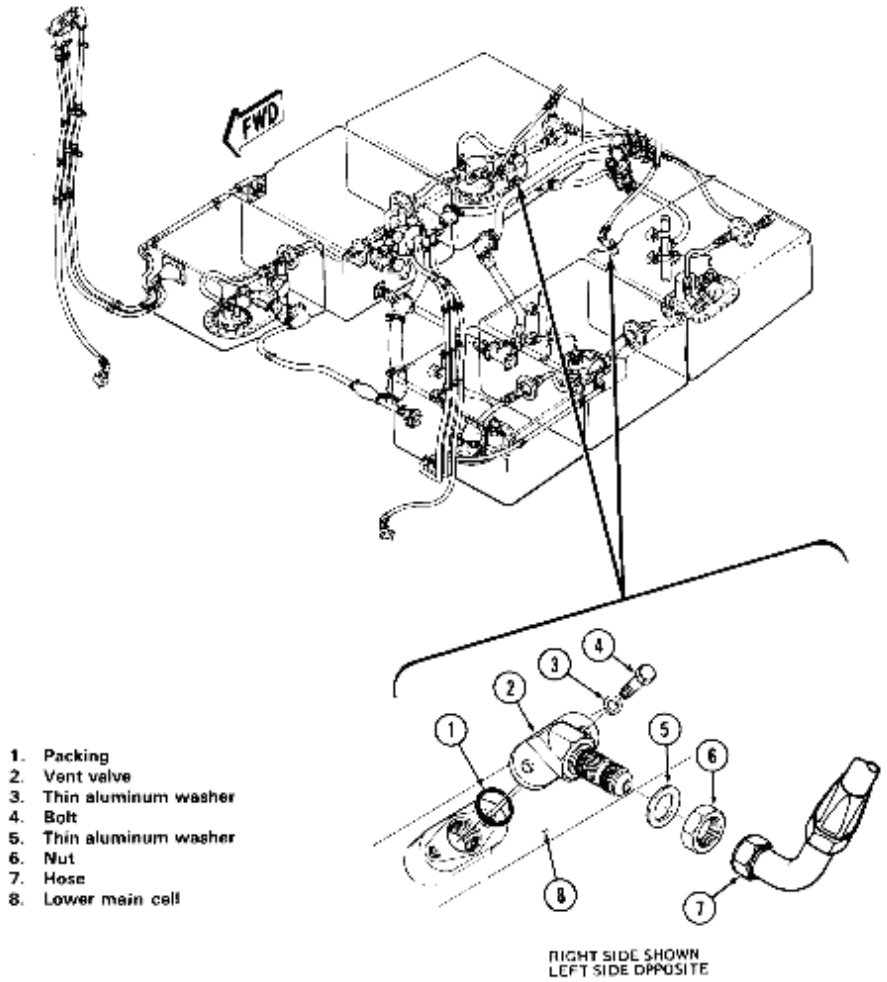
Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure

NOTE

The removal procedures of the right and left lower main cell vent valves are same.

1. Disconnect the electrical power.
2. Defuel the system (412-A-12-10-01-00A-221A-A / 00131).
3. Remove the cabin floor access panels above the lower right and lower left main cells.
4. Disconnect the hose (7, Figure 1).
5. Remove the nut (6) and thin aluminum washers (5).
6. Remove the bolts (4) and thin aluminum washers (3).
7. Remove the vent valve (2) and packing (1) from the lower main cell (8). Discard the packing.
8. Cover the cell opening to prevent cell contamination. Cap the hose (7).
9. Install the caps or plugs (C-428) on open end of the hose (7).



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Figure 1. Lower Main Cell Vent Valve and Hose (Sheet 1 of 1)

Lower Main Cell Vent Valves and Hoses - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

NOTE

The following installation procedures are the same for both the right and left lower main cell vent valves.

1. Remove protective cover from lower main cell (8, Figure 1) and cap from hose (7).

2. Apply assembly fluid (C-024) to packing (1). Place packing and vent valve (2) on lower main cell (8).

3. Install bolts (4) and thin aluminum washers (3).

4. Install thin aluminum washers (5) and nut (6).

5. Connect hose (7).

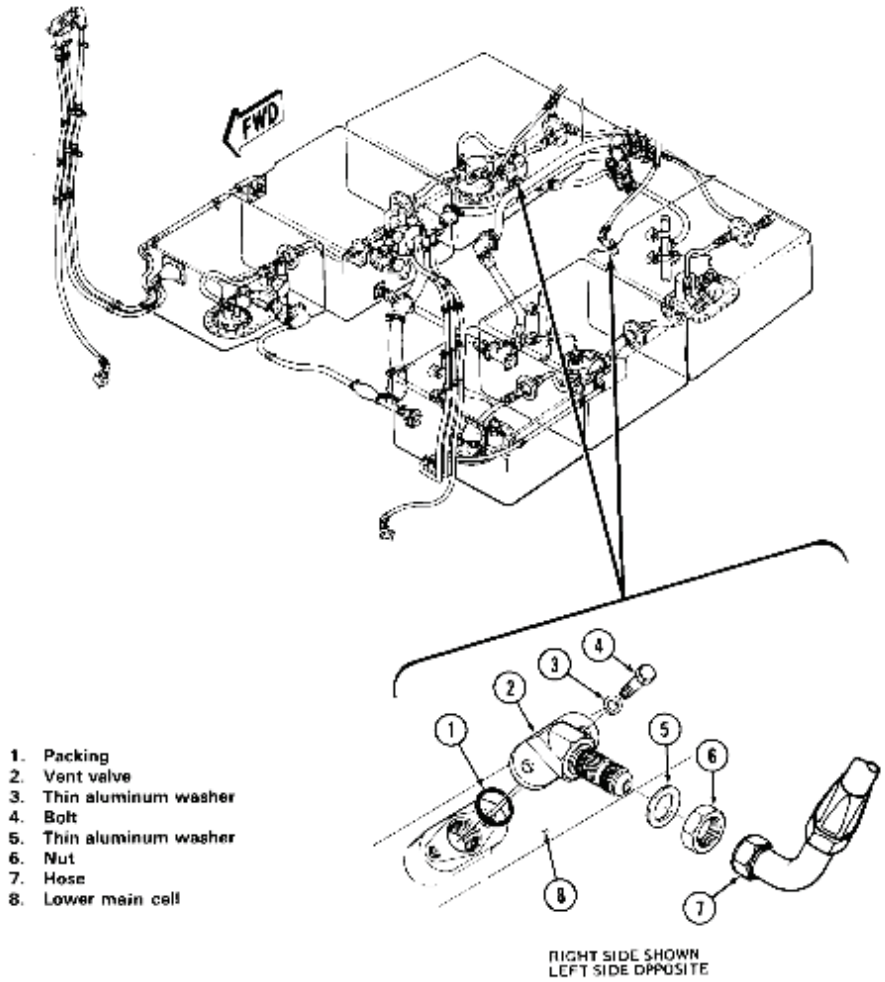
6. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).

7. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).

8. Perform fuel system operational test.

9. Check area for leaks.

10. Install cabin floor access panel.



ICN-412-A-281000-B-97499-00065-A-001-01

Figure 1. Lower Main Cell Vent Valve and Hose (Sheet 1 of 1)

Lower Main Cell Interconnect Hoses - Description

NOTE

Maintenance procedures are the same for both right and left lower main cell interconnect hoses.

An interconnect hose is attached directly to the inboard interconnect port of each lower main cell. The other end of each interconnect hose is attached to a port on the interconnect valve.

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Lower Main Cell Interconnect Hoses - Removal

Preliminary requirements

Consumables, materials and expendables

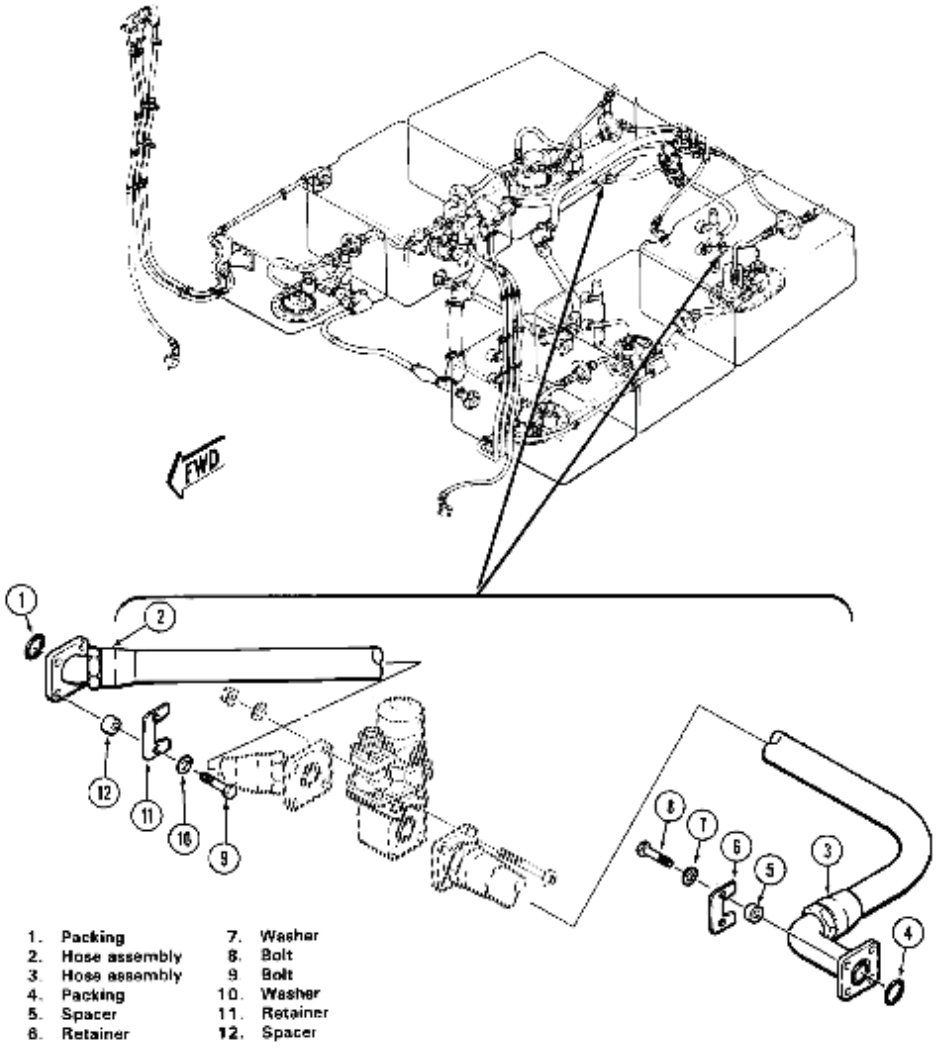
Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure

NOTE

The following removal procedures are the same for both right and left lower main cell interconnect hoses.

1. Disconnect electrical power.
2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).
3. Remove floor panels above hose to be removed.
4. Remove interconnect valve (412-A-28-10-00-64A-520A-A / 00532).
5. Remove bolts (8, Figure 1), washers (7), retainer (6), and spacers (5).
6. Remove hose assembly (3) and packing (4). Discard packing.
7. Install protective cover over cell opening.
8. Install caps or plugs (C-428) on hose assemblies open end fittings.



1. Packing
2. Hose assembly
3. Hose assembly
4. Packing
5. Spacer
6. Retainer
7. Washer
8. Bolt
9. Bolt
10. Washer
11. Retainer
12. Spacer

(S/N 33108 THRU 33213)
(S/N 36001 AND SUBSEQUENT)

ICN-412-A-281000-B-97499-00066-A-001-01

Figure 1. Lower Main Cell Vent Valve and Hose (Sheet 1 of 1)

Lower Main Cell Interconnect Hoses - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

NOTE

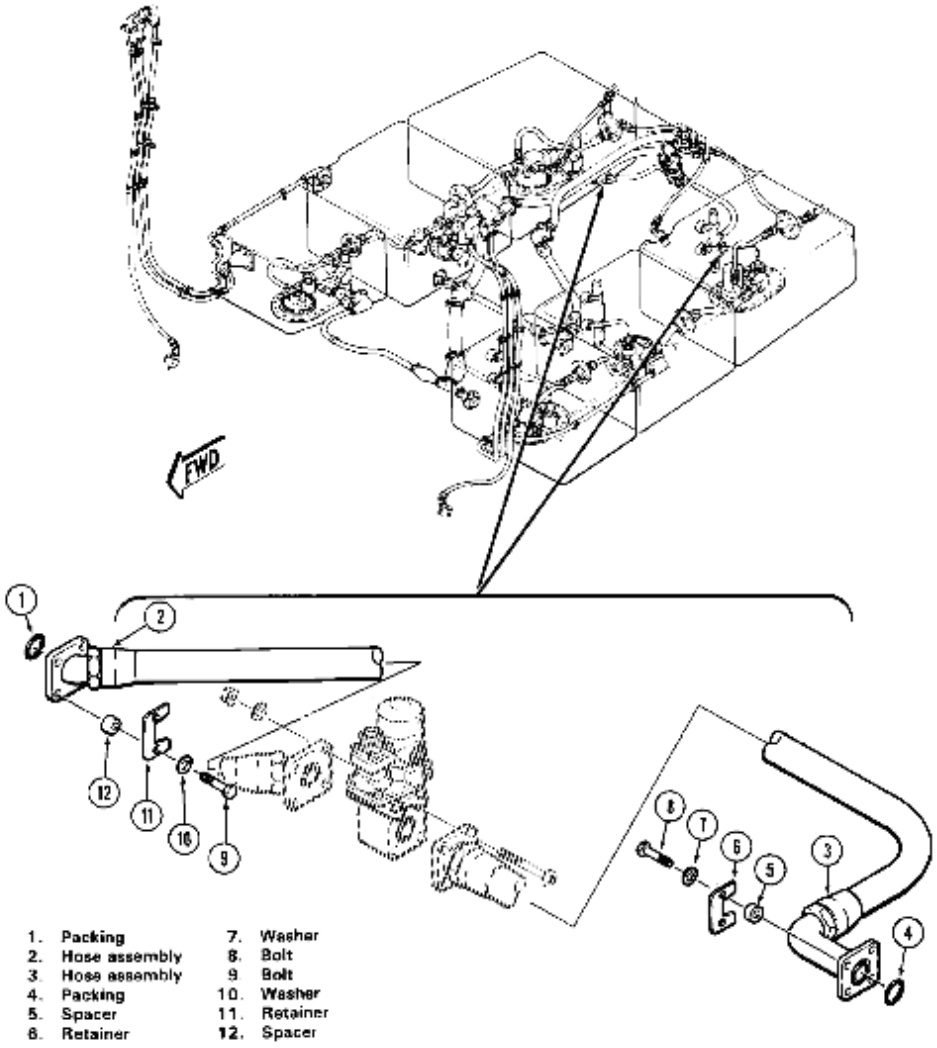
The following installation procedures are the same for both right and left lower main cell interconnect hoses.

1. Remove protective cover and cap from cell opening and hose assembly end.
2. Install interconnect valve (412-A-28-10-00-64A-720A-A / 00533).

NOTE

Do not perform fuel system pressure test, service, or operational check at this time.

3. Apply assembly fluid (C-024) to packing (4, Figure 1). Install hose assembly (3), packing (4), bolts (8), washers (7), retainer (6), and spacers (5).
4. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).
5. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).
6. Perform fuel system operational check.
7. Check area for leaks.



1. Packing
2. Hose assembly
3. Hose assembly
4. Packing
5. Spacer
6. Retainer
7. Washer
8. Bolt
9. Bolt
10. Washer
11. Retainer
12. Spacer

(S/N 33108 THRU 33213)
(S/N 36001 AND SUBSEQUENT)

ICN-412-A-281000-B-97499-00066-A-001-01

Figure 1. Lower Main Cell Vent Valve and Hose (Sheet 1 of 1)

Lower Main Cell Sump Assemblies - Description

NOTE

Maintenance procedures are the same for left and right lower main cell sump assemblies.

A sump assembly is mounted above the opening on the bottom of each lower main cell. Removal of sump assembly provides access for maintenance of boost pump, drain valve, defueling valve, flow switch check valve, fuel quantity probe, and input/output fitting. The boost pump and drain valve can both be removed without removing the sump assembly.

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Lower Main Cell Sump Assemblies - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure

NOTE

The following removal procedures are the same for left and right lower main cell sump assemblies.

1. Disconnect electrical power.

2. Defuel sump system (412-A-12-10-01-00A-221A-A / 00131).

3. Remove sump access door.

4. Disconnect electrical connector (5, Figure 1).

5. Remove bolts (8), washers (7), and retainers (6).

6. Lower sump assembly (3) for access.

7. Remove bolts (23), washers (24), and spacers (26) from the mounting brackets (25) of the fuel quantity probe (22).

8. Remove the fuel quantity probe (22) from the lower main cell (1) inboard the wall and lay it on the sump assembly (3).

Applicable: SN: 33108-33167:

9. For helicopters, disconnect hoses (11 and 16).

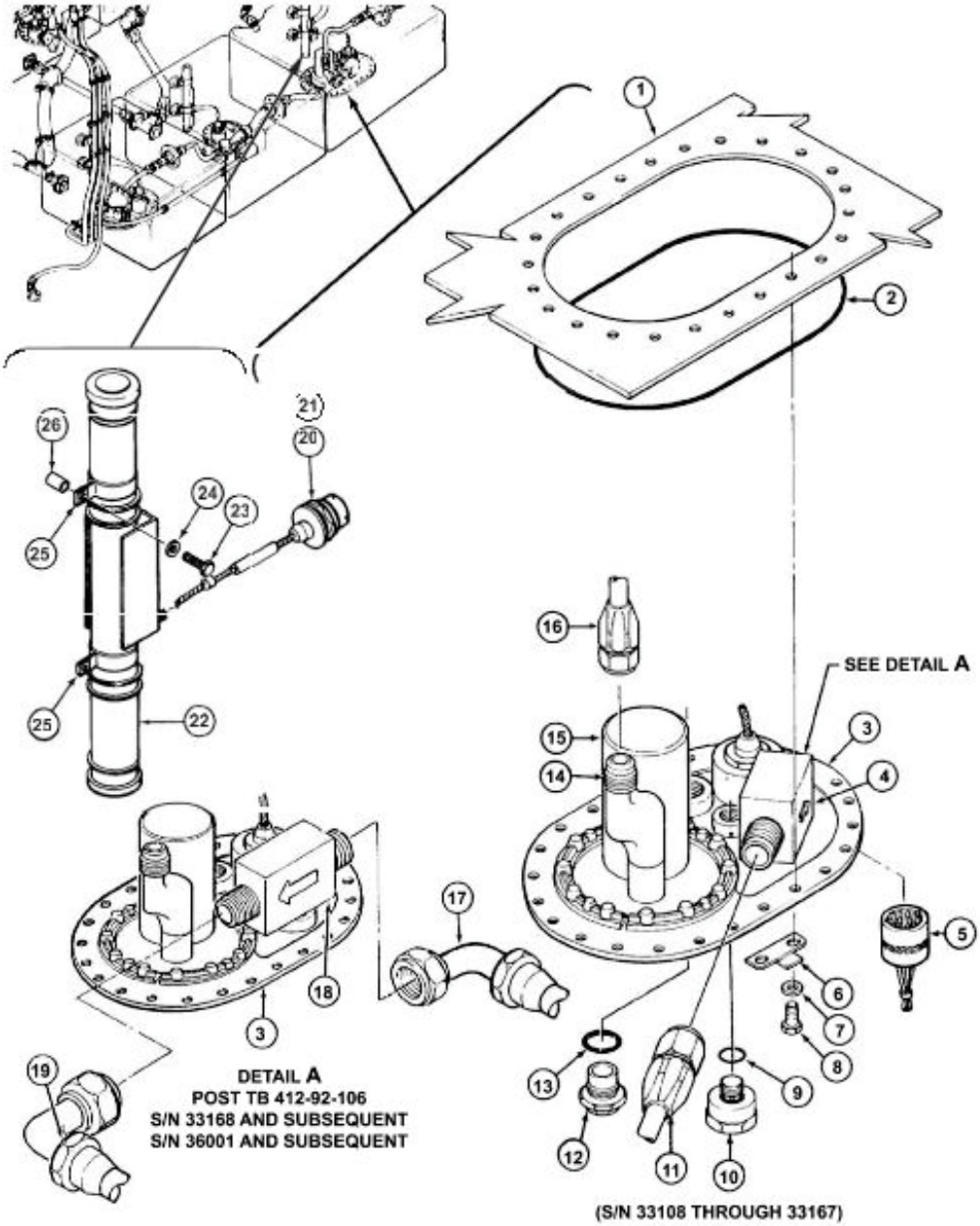
Applicable: SN: 33168-Subsequent or 36001-Subsequent or TB 412-92-106: Post:

10. For helicopters, disconnect hoses (16, 17, and 19).

11. Remove sump assembly (3), with the fuel quantity probe (22) and packing (2). Discard packing.

12. Install protective covers on lower main cell (1) openings.

13. Install protective caps or plugs (C-428) on the applicable hose assemblies (11, 16, 17, and 19) open end fittings.



ICN-412-A-281000-B-97499-00067-A-001-01

Figure 1. Lower Main Cell Sump Assembly (Sheet 1 of 2)

1. Lower main cell
2. Packing
3. Sump assembly
4. Flow switch check valve
5. Electrical connector
6. Retainer
7. Washer
8. Bolt
9. Packing
10. Drain valve
11. Hose
12. Defueling valve
13. Packing
14. Fitting
15. Boost pump
16. Hose
17. Hose
18. Flow switch
19. Hose
20. Electrical connector
21. Packing
22. Fuel quantity probe
23. Bolt
24. Washer
25. Mounting bracket
26. Spacer

ICN-412-A-281000-B-97499-00068-A-001-01

Figure 1. Lower Main Cell Sump Assembly (Sheet 2 of 2)

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Lower Main Cell Sump Assemblies - Inspection

Procedure

NOTE

The following inspection procedures are the same for left and right lower main cell sump assemblies.

1. Inspect area where sump assembly mates with cell for damage and cleanliness. Inspect packing groove.

2. Inspect sump assembly for security of mounting.

3. If installed, inspect drain valve for security.

4. Inspect electrical leads and connector for condition.

5. Repair or replace components failing to meet inspection requirements.

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Lower Main Cell Sump Assemblies - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

NOTE

The following installation procedures are the same for left and right lower main cell sump assemblies.

1. Remove the protective covers, caps, and/or plugs from the lower main cell (1, Figure 1) opening and the applicable hose assemblies (11, 16, 17, and 19) end fittings.

2. Coat the packing (2) with assembly fluid (C-024). Place packing in groove around sump opening in lower main fuel cell.

3. Install the bolts (23) with the washers (24) through the mounting brackets (25) of the fuel quantity probe (22).

4. Install the spacers (26) on the bolts (23) between the inboard wall of the lower main cell (1) and the fuel quantity probe (22) mounting bracket (25). Tighten the bolts.

5. Hold the sump assembly (3) below the lower main cell (1) sump opening.

Applicable: SN: 33108-33167 TB 412-92-106: Pre:

6. Perform the following:

6.1. Connect the hose assembly (11) to the flow switch check valve (4).

Applicable: SN: 36168-Subsequent or 36001-Subsequent TB 412-92-106: Post:

7. Perform the following:

7.1. Connect the hose assemblies (17 and 19) to the flow switch check valve (4).

8. Connect the hose assembly (16) to the fitting (14).

9. Put the sump assembly (3) in position on the opening of the lower main cell (1).

10. Install the bolts (8), washers (7), and retainers (6).

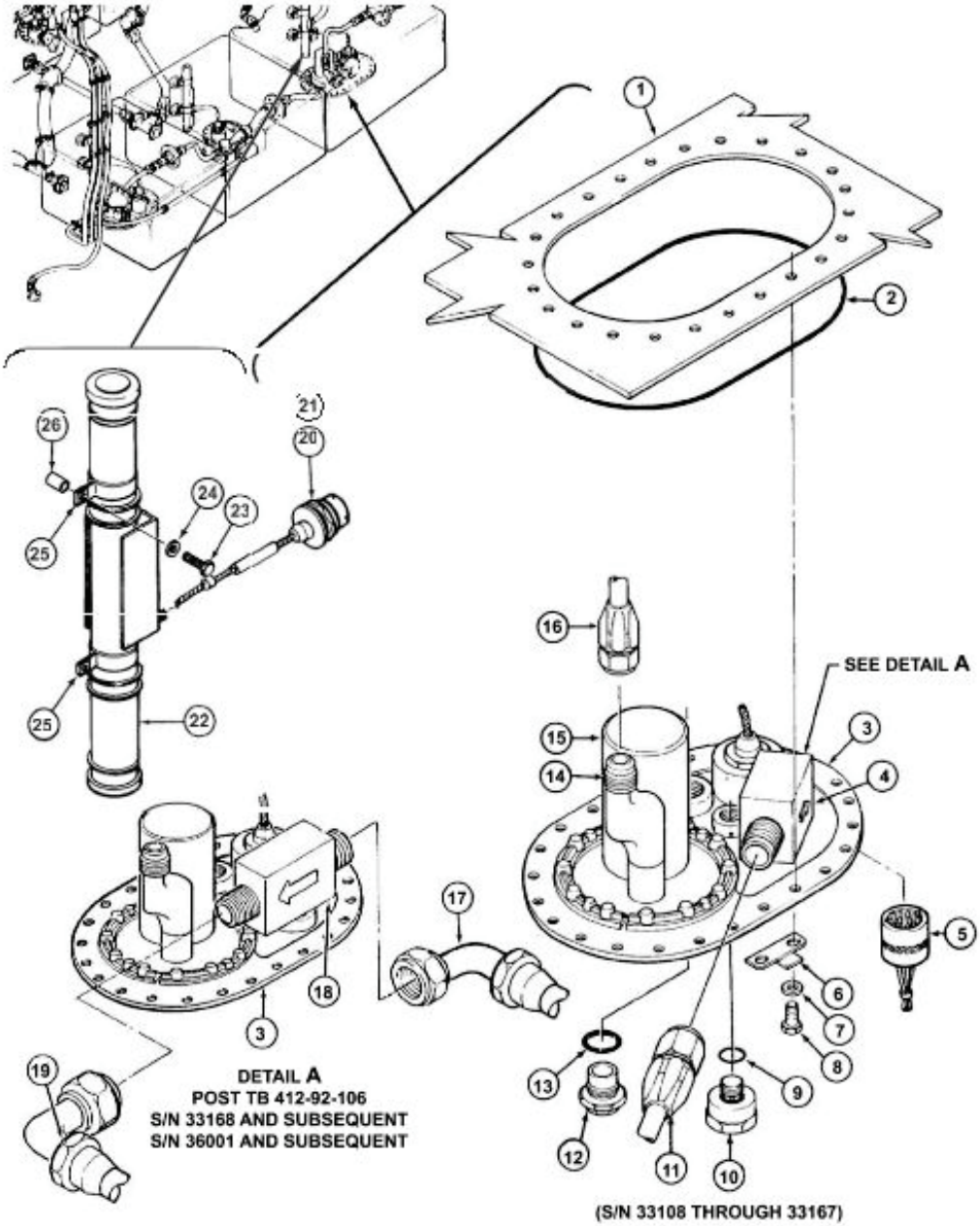
11. Connect the electrical connector (5) on the bottom of the sump assembly (3).

12. Perform the fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).

13. Service the fuel system (412-A-12-00-00-00A-040A-A / 00126).

14. Perform the fuel system operational check (412-A-96-00-00-00A-009A-A / 01870).

15. Check area for leaks.



ICN-412-A-281000-B-97499-00067-A-001-01

Figure 1. Lower Main Cell Sump Assembly (Sheet 1 of 2)

1. Lower main cell
2. Packing
3. Sump assembly
4. Flow switch check valve
5. Electrical connector
6. Retainer
7. Washer
8. Bolt
9. Packing
10. Drain valve
11. Hose
12. Defueling valve
13. Packing
14. Fitting
15. Boost pump
16. Hose
17. Hose
18. Flow switch
19. Hose
20. Electrical connector
21. Packing
22. Fuel quantity probe
23. Bolt
24. Washer
25. Mounting bracket
26. Spacer

ICN-412-A-281000-B-97499-00068-A-001-01

Figure 1. Lower Main Cell Sump Assembly (Sheet 2 of 2)

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Boost Pumps - Description

A boost pump is mounted on the sump assembly of each lower main cell. The boost pumps provide a positive pressure in lines supplying fuel to each power section.

NOTE

Maintenance procedures left and right lower main cell boost pumps are the same.

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Boost Pumps - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure

NOTE

The following removal procedures left and right lower main cell boost pumps are the same.

1. Disconnect electrical power.

2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).

3. Drain any trapped fuel through pump drain valve (10, Figure 1).

4. Remove bolt (11), washer (12), packings (13 and 15) and retainer (14). Discard packings.

5. Remove bolts (8), washers (7), and grounding wire.

6. Disconnect electrical leads (6) from boost pump (5).

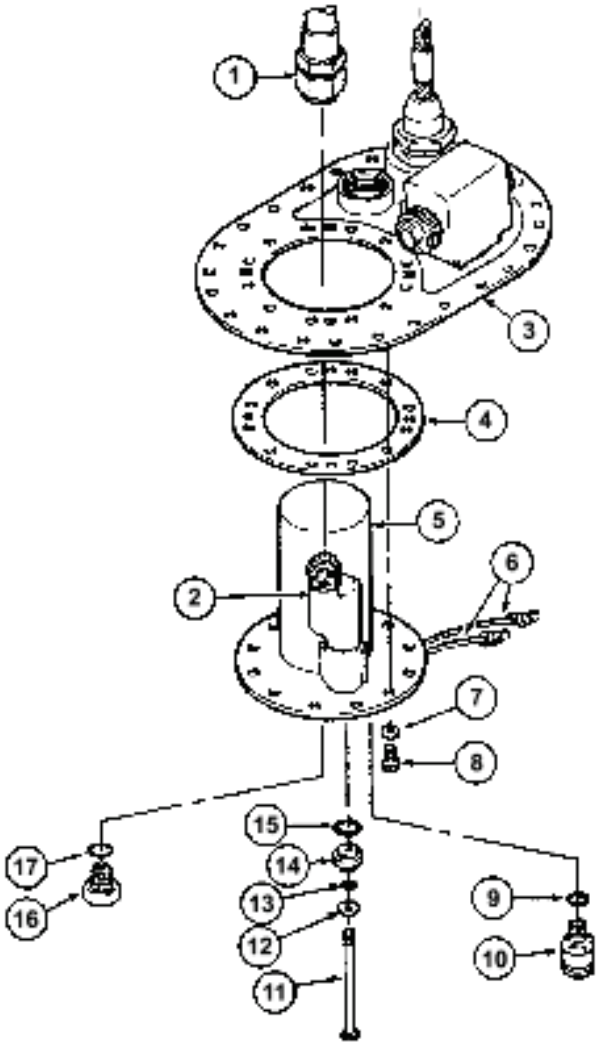
7. Remove boost pump (5) from sump assembly (3).

8. Remove plug (16) and packing (17). Discard packing.

9. Remove pump drain valve (10) and packing (9). Discard packing.

10. Install protective cover over cell opening.

11. Install caps or plugs (C-428) on open end of fitting (2).



(S/N 33108 THROUGH 33213)
(S/N 36001 THROUGH 36048)
(PRE TB 412-89-77)

1. Hose
2. Fitting
3. Sump assembly
4. Gasket
5. Boost pump
6. Electrical leads
7. Washer
8. Bolt
9. Packing
10. Pump drain valve
11. Bolt
12. Washer
13. Packing
14. Retainer
15. Packing
16. Plug
17. Packing

ICN-412-A-281000-B-97499-00069-A-001-01

Figure 1. Boost Pump (Sheet 1 of 1)

Boost Pumps - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	
Lockwire	C-405	AR	

Procedure

NOTE

The following installation procedures left and right lower main cell boost pumps are the same.

1. Remove the protective cover, caps, and/or plugs from the fuel cell opening and the fitting (2, Figure 1).

2. Apply assembly fluid (C-024) to packings (9, 13, 15, and 17).

3. Place packing (9) on pump drain valve (10). Install pump drain valve.

4. Install plug (16) and packing (17).

5. Install washer (12) on bolt (11). Install packings (13 and 15) on retainer (14).

6. Install retainer (14) on bolt (11).

7. Connect electrical leads (6).

8. Coat gasket (4) with assembly fluid (C-024). Install gasket on boost pump (5).

9. Insert boost pump (5) into sump assembly (3).

10. Align boost pump (5) with fitting (2) and install bolt (11) and apply lockwire (C-405).

11. Install ground wire on nearest bolt (8).

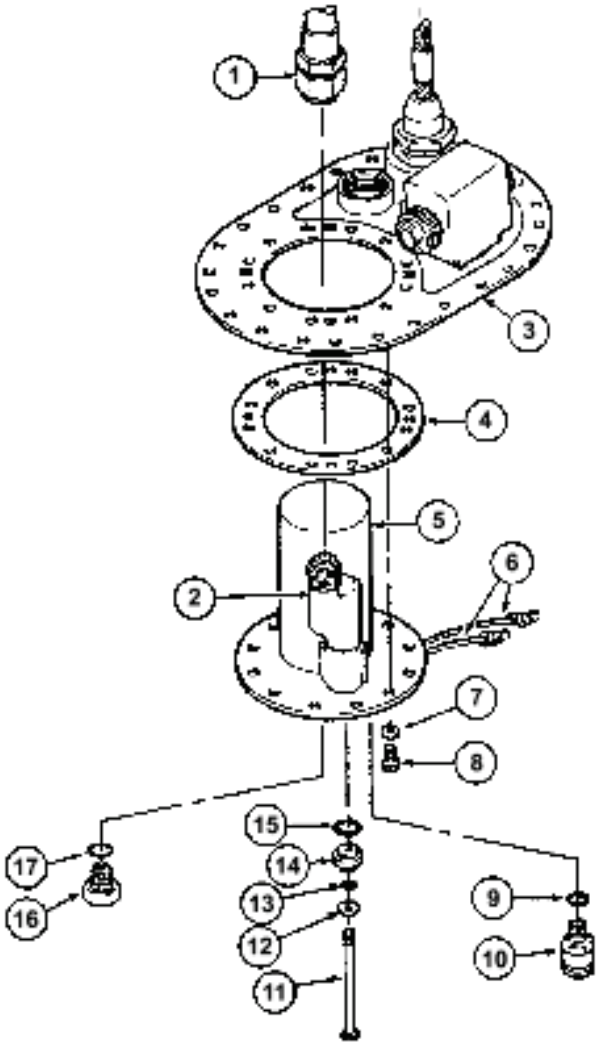
12. Install bolts (8) and washers (7).

13. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).

14. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).

15. Perform fuel system operational check.

16. Check area for leaks.



(S/N 33108 THROUGH 33213)
(S/N 36001 THROUGH 36048)
(PRE TB 412-89-77)

1. Hose
2. Fitting
3. Sump assembly
4. Gasket
5. Boost pump
6. Electrical leads
7. Washer
8. Bolt
9. Packing
10. Pump drain valve
11. Bolt
12. Washer
13. Packing
14. Retainer
15. Packing
16. Plug
17. Packing

ICN-412-A-281000-B-97499-00069-A-001-01

Figure 1. Boost Pump (Sheet 1 of 1)

Canister Type Boost Pump - Description

Applicable: (SN: 33001-33213 or 36001-36048 TB 412-89-77: Post) or SN: 36049-Subsequent:

The canister type boost pumps are electrically operated by 28 Vdc circuits controlled by separate switches. Each pump incorporates an integral cartridge pumping element. These pumps are applicable to helicopters. Have canister type pumps installed as part of basic helicopter configuration.

NOTE

Maintenance procedures for left and right canister type boost pumps are the same.

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Canister Type Boost Pump - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure

NOTE

The following removal procedures for left and right canister type boost pumps are the same.

1. Disconnect electrical power.

2. Defuel system (412-A-12-00-00-00A-040A-A / 00126).

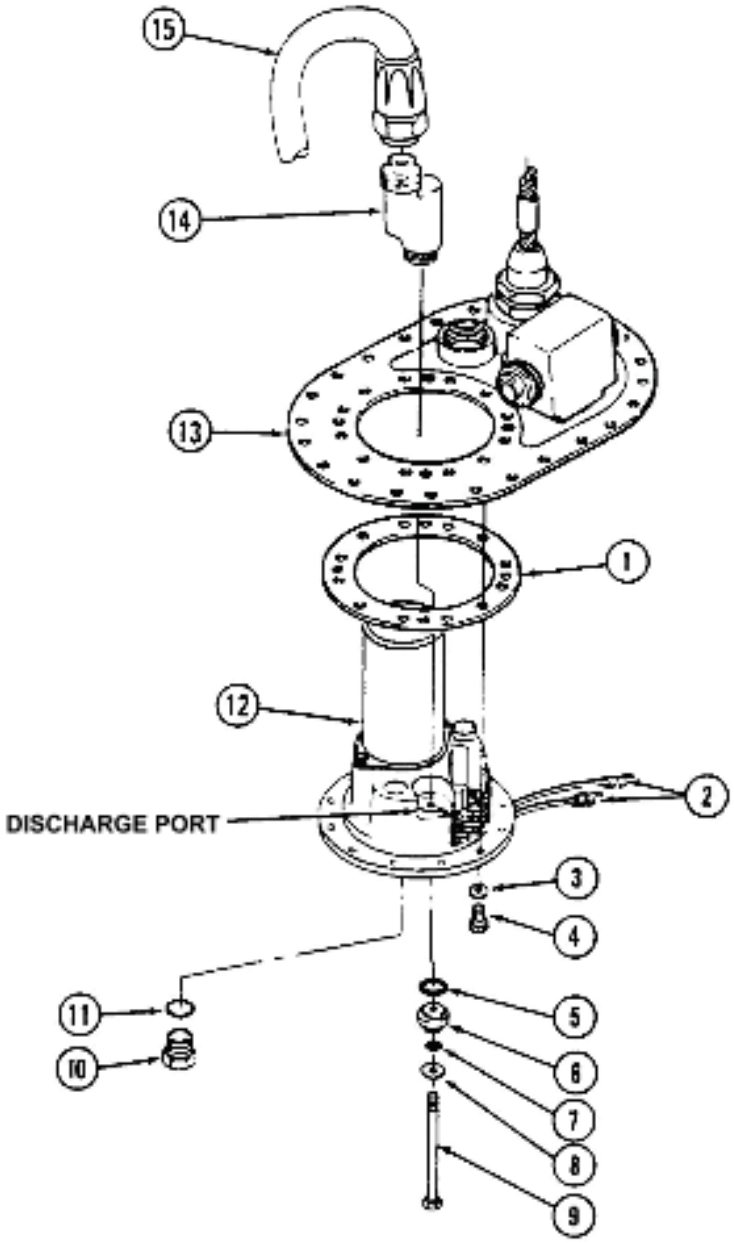
3. Drain trapped fuel through drain plug (10, Figure 1).

4. Disconnect electrical leads (2) from boost pump (12).

5. Remove bolt (9), washer (8), packings (5 and 7), and plug (6). Discard packings.

6. Remove bolts (4) and washers (3). Remove boost pump (12) and gasket (1) from sump assembly (13). Discard gasket.

7. Install protective covers over cell opening and caps or plugs (C-428) on the open end of the fitting (14).



(S/N 36049 AND SUBSEQUENT)
(POST TB 412-89-77)

1. Gasket
2. Electrical leads
3. Washer
4. Bolt
5. Packing
6. Plug
7. Packing
8. Washer
9. Bolt
10. Plug
11. Packing
12. Boost pump
13. Sump assembly
14. Fitting
15. Hose

ICN-412-A-281000-B-97499-00070-A-001-01

Figure 1. Canister Type Boost Pump (Sheet 1 of 1)

Canister Type Boost Pump - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	
Lockwire	C-405	AR	

Procedure

NOTE

The following installation procedures for left and right canister type boost pumps are the same.

1. Remove the protective covers from the cell opening and protective caps and/or plugs from the fitting (14).

2. Apply assembly fluid (C-024) to packing (11, Figure 1) and install on plug (10).

3. Install plug (10) in base of pump at cartridge drain port.

4. Lubricate packings (5 and 7) with assembly fluid (C-024) and install on plug (6).

5. Install washer (8) on bolt (9). Insert bolt through plug (6) and into discharge port of pump.

6. Coat gasket (1) with assembly fluid (C-024). Install gasket (1) on boost pump (12).

7. Position boost pump (12) on sump assembly (13). Align pump with fitting (14). Tighten bolt (9) and ensure fitting seats in discharge port.

8. Install bolts (4) and washers (3).

9. Connect electrical leads (2).

10. Secure plug (10) and bolt (9) with lockwire (C-405).

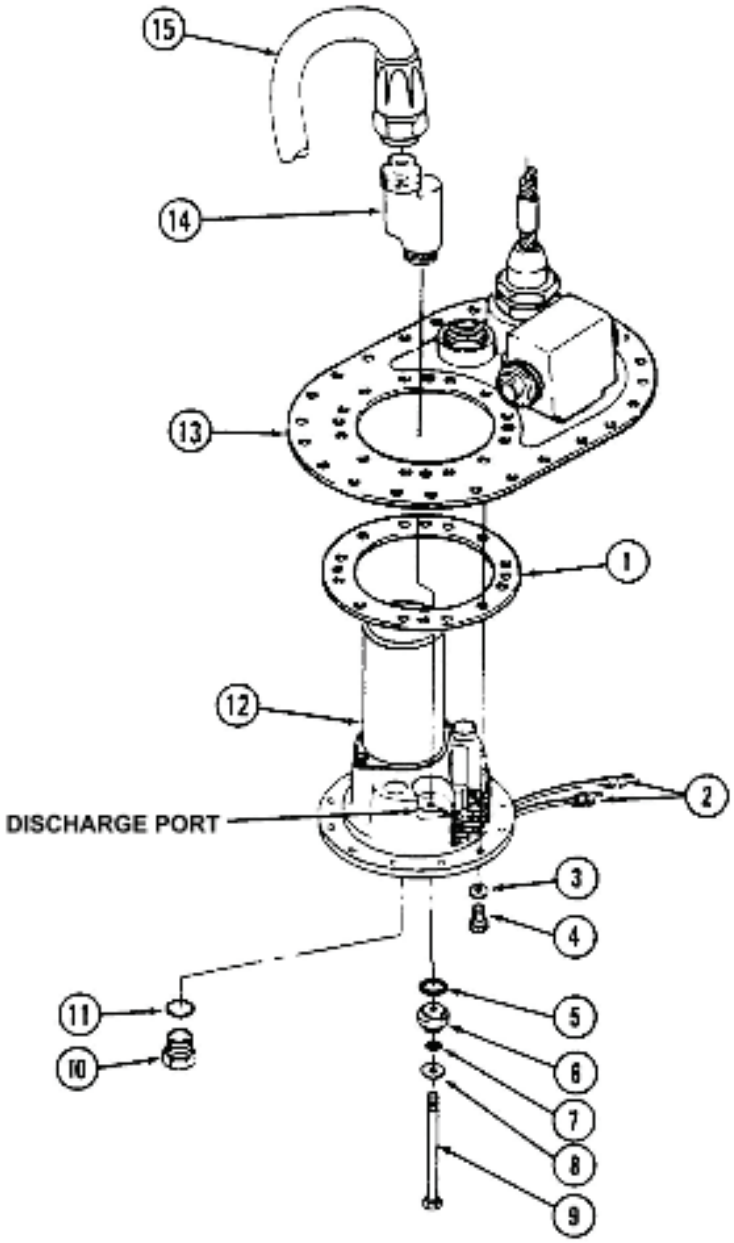
11. Perform system pressure test (412-A-28-10-00-00A-362B-A / 00438).

12. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).

13. Perform operational check on boost pump (12) (412-A-96-04-00-02A-320B-A / 02105 and 412-A-96-04-00-02A-320C-A / 02106

14. Check area for leaks.

15. Install sump access cover.



(S/N 36049 AND SUBSEQUENT)
(POST TB 412-89-77)

1. Gasket
2. Electrical leads
3. Washer
4. Bolt
5. Packing
6. Plug
7. Packing
8. Washer
9. Bolt
10. Plug
11. Packing
12. Boost pump
13. Sump assembly
14. Fitting
15. Hose

ICN-412-A-281000-B-97499-00070-A-001-01

Figure 1. Canister Type Boost Pump (Sheet 1 of 1)

Canister Type Boost Pump Cartridge - Description

The canister type boost pump incorporates an integral cartridge pumping element. Replacement of element reduces damage to fuel cell components and eliminates need to break cell-to-pump seal during routine maintenance.

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Canister Type Boost Pump Cartridge - Removal

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Cartridge Removal Tool	E2-10-1	1	

Procedure

1. Disconnect electrical power.

2. Remove the lockwire.

3. Remove allen screw (4, Figure 1) from pump assembly valve locking bar (5). Rotate bar clear of cartridge.

4. Disconnect black (+) positive lead from cartridge terminal and move wiring clear of cartridge area.

5. Remove drain plug (10, Figure 2) from cartridge center drain port and allow trapped fuel to drain.

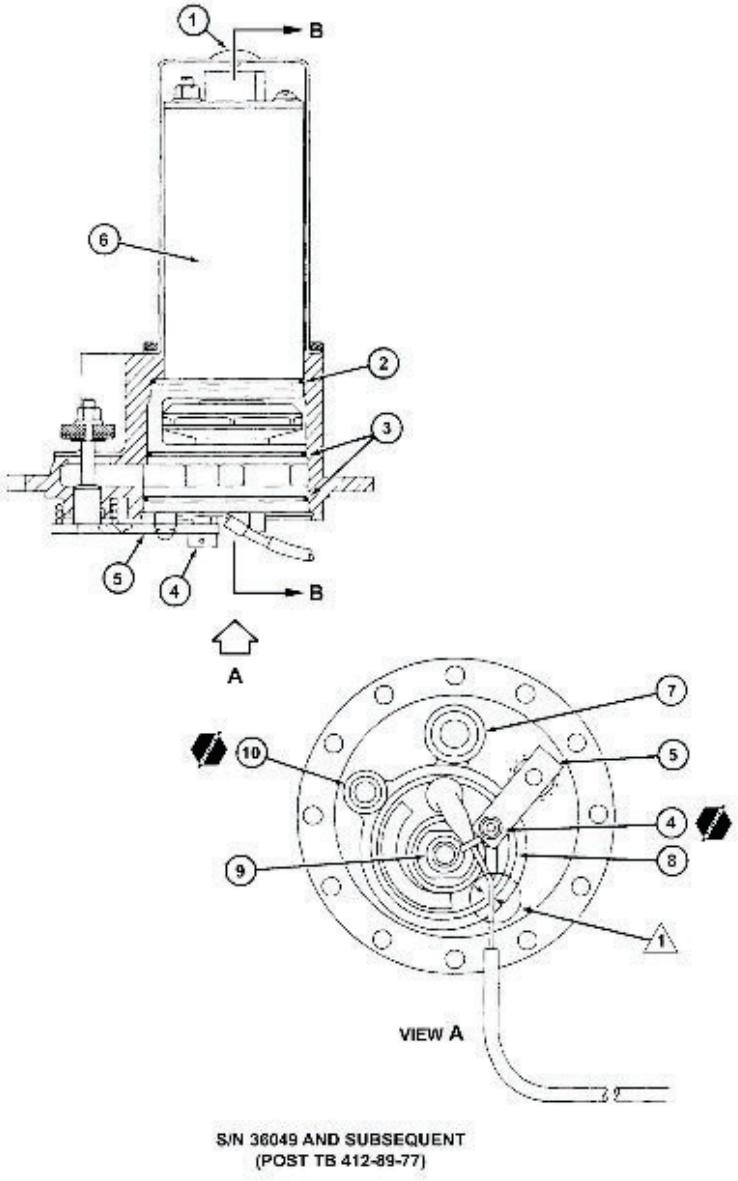
6. Remove retaining ring (8, Figure 1) that secures the cartridge in pump housing.

7. Install cartridge removal tool (E2-10-1) in drain port and secure with check nut. Using weight, impact cartridge from pump housing. Remove tool from cartridge.

NOTE

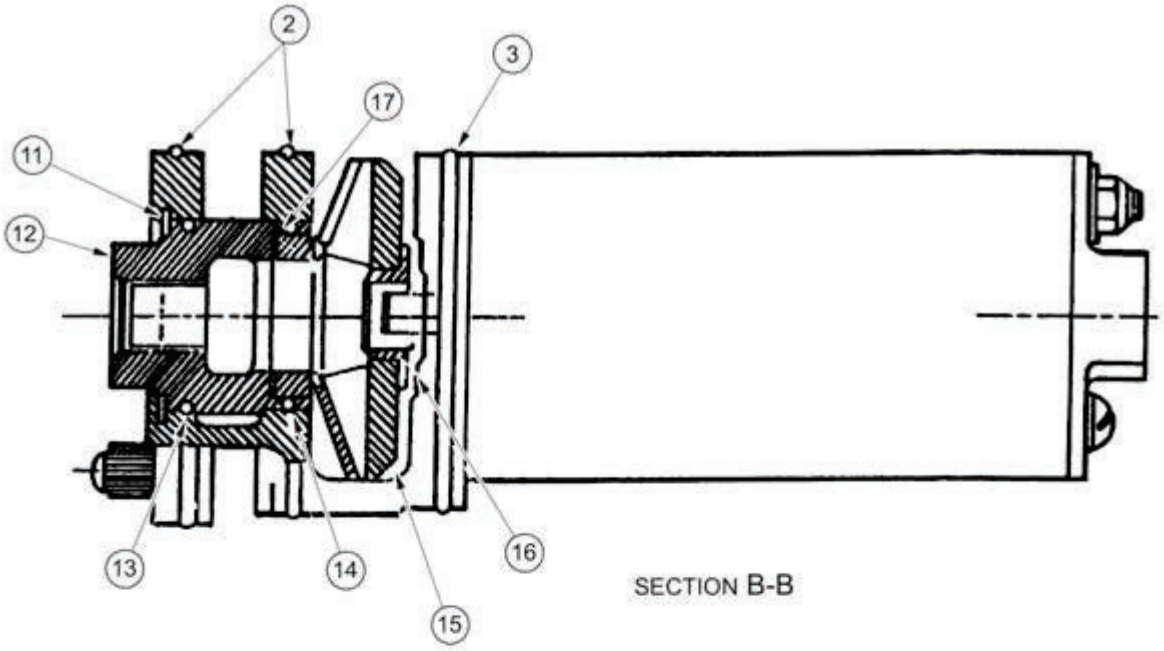
For cartridge repair, return to: Airborne, Aviation Products Group, 711 Taylor Street, Elyria, Ohio 44035.

8. Install protective cover on cartridge opening.



ICN-412-A-281000-B-97499-00071-A-001-01

Figure 1. Boost Pump Cartridge Type (Sheet 1 of 2)



1. Check valve
2. Packing
3. Packing
4. Allen screw
5. Locking bar
6. Cartridge
7. Discharge port
8. Retaining ring
9. Cartridge drain port
10. Seal drain port
11. Spirolox retaining ring
12. Pump cover
13. Pump cover packing
14. Inlet seal packing
15. Impeller
16. Impeller key
17. Inlet seal

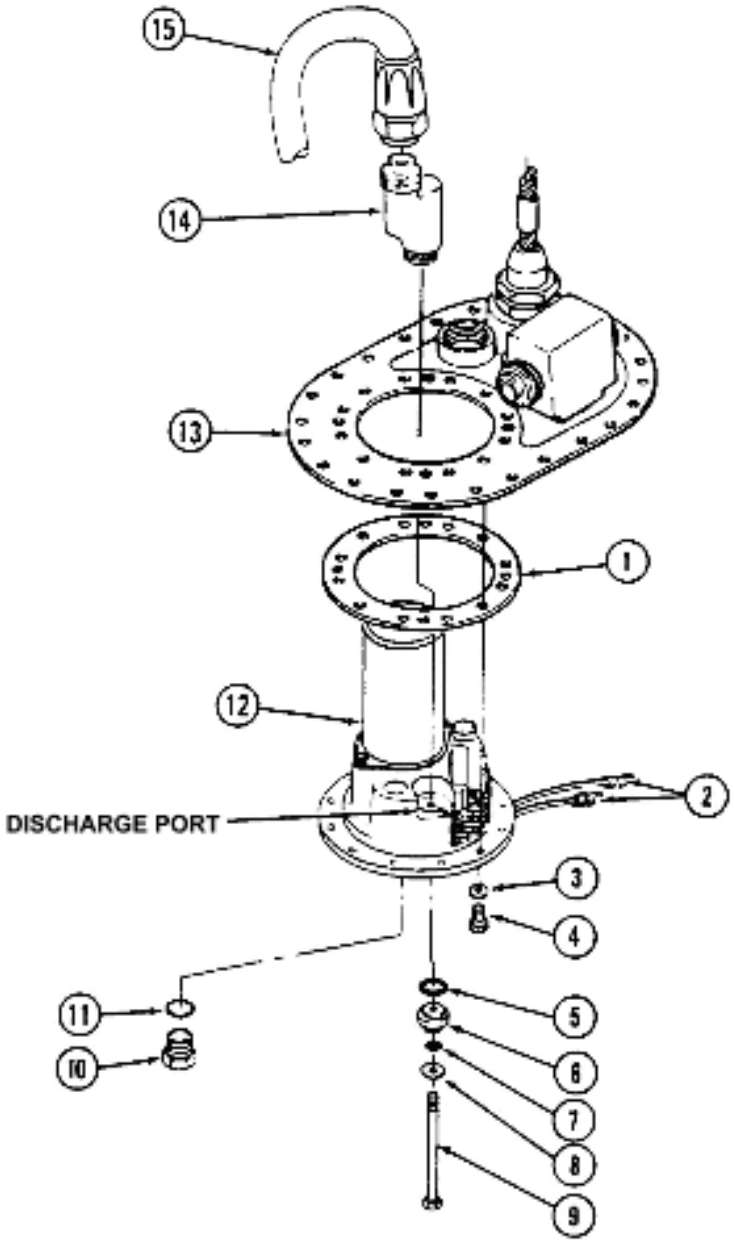
LOCKWIRE (C-405)

NOTE

1. Arrows must align within ±3° for cartridge installation.

ICN-412-A-281000-B-97499-00072-A-001-01

Figure 1. Boost Pump Cartridge Type (Sheet 2 of 2)



(S/N 36049 AND SUBSEQUENT)
(POST TB 412-89-77)

1. Gasket
2. Electrical leads
3. Washer
4. Bolt
5. Packing
6. Plug
7. Packing
8. Washer
9. Bolt
10. Plug
11. Packing
12. Boost pump
13. Sump assembly
14. Fitting
15. Hose

ICN-412-A-281000-B-97499-00070-A-001-01

Figure 2. Canister Type Boost Pump (Sheet 1 of 1)

Canister Type Boost Pump Cartridge - Cleaning

Procedure

1. Remove all foreign matter from cartridge bore, retaining ring groove in pump housing, and from retaining ring.

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Canister Type Boost Pump Cartridge - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	
Lockwire	C-405	AR	

Procedure

1. Remove protective cover from cartridge opening.

2. Lubricate packings (2 and 3, Figure 1) with assembly fluid (C-024) and install on cartridge (6).

3. Using hand pressure, install cartridge (6) in pump housing with arrows aligned on flanges (within ±3°).



DO NOT IMPACT CARTRIDGE OR DAMAGE TO PARTS MAY RESULT.

NOTE

Cartridge removal tool may be used as a workaid during installation.

4. Secure cartridge (6) in pump housing with retaining ring (8).

5. Install drain plug (10, Figure 2) in cartridge drain port with packing (11).

6. Rotate valve locking bar (5, Figure 1) over cartridge and in line with minus (-) terminal. Insert white (-) negative lead terminal under bar, compress spring by pressing up on lever arm at shaft end, and secure both with allen screw (4).

7. Connect black (+) positive lead terminal to (+) terminal on cartridge. Do not overtorque.

8. Secure drain plugs, fitting through bolt, and allen screw with lockwire (C-405).

9. Perform boost pump operational check (412-A-96-04-00-02A-320B-A / 02105 and 412-A-96-04-00-02A-320C-A / 02106).

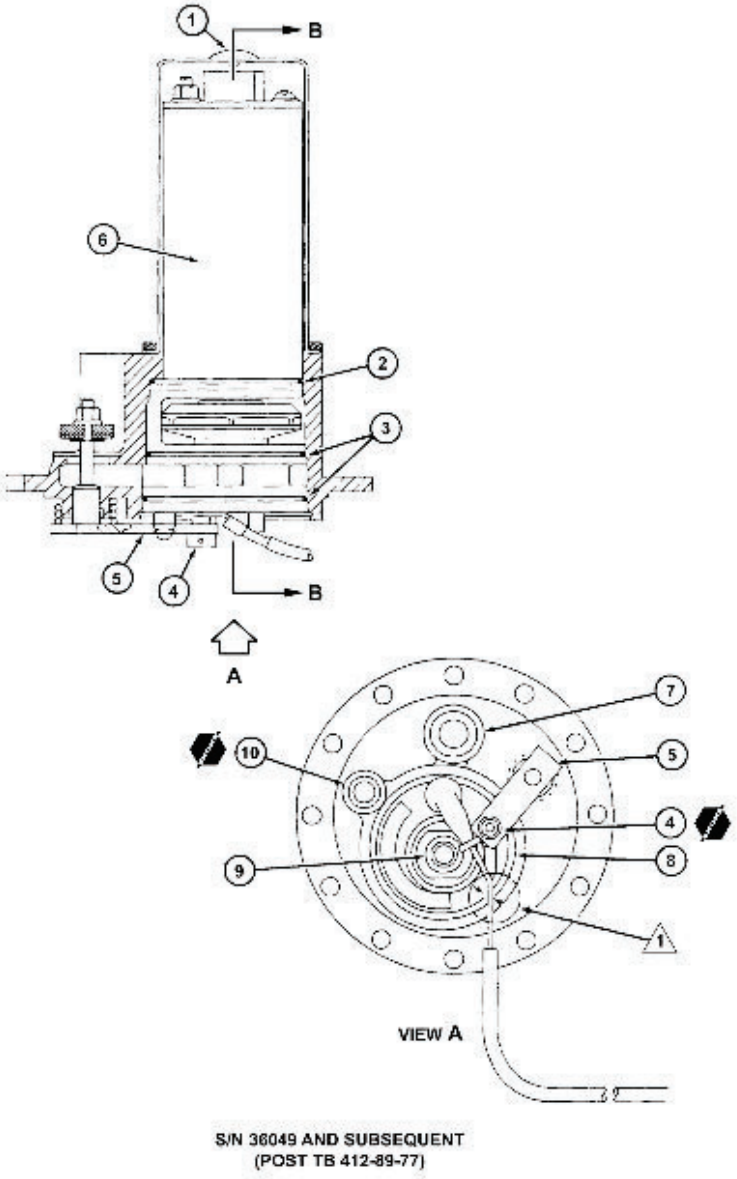


CARTRIDGE PUMPS ARE COOLED AND LUBRICATED BY FUEL WHICH EXITS THROUGH TOP UMBRELLA CHECK VALVE (1) ON PUMP/CARTRIDGE HOUSING. DO NOT RUN DRY.

10. Make sure pumps are covered with fuel. Refuel helicopter, as required.

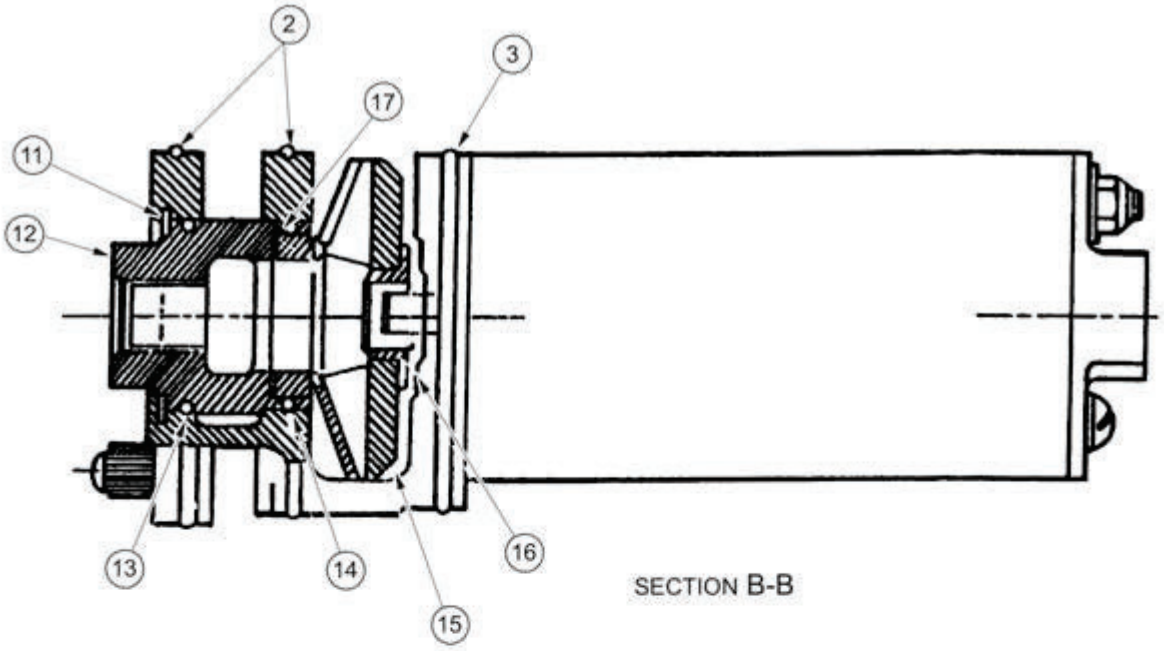
11. Check cartridge/pump for leakage.

12. Install sump access cover.



ICN-412-A-281000-B-97499-00071-A-001-01

Figure 1. Boost Pump Cartridge Type (Sheet 1 of 2)



1. Check valve
2. Packing
3. Packing
4. Allen screw
5. Locking bar
6. Cartridge
7. Discharge port
8. Retaining ring
9. Cartridge drain port
10. Seal drain port
11. Spirolox retaining ring
12. Pump cover
13. Pump cover packing
14. Inlet seal packing
15. Impeller
16. Impeller key
17. Inlet seal

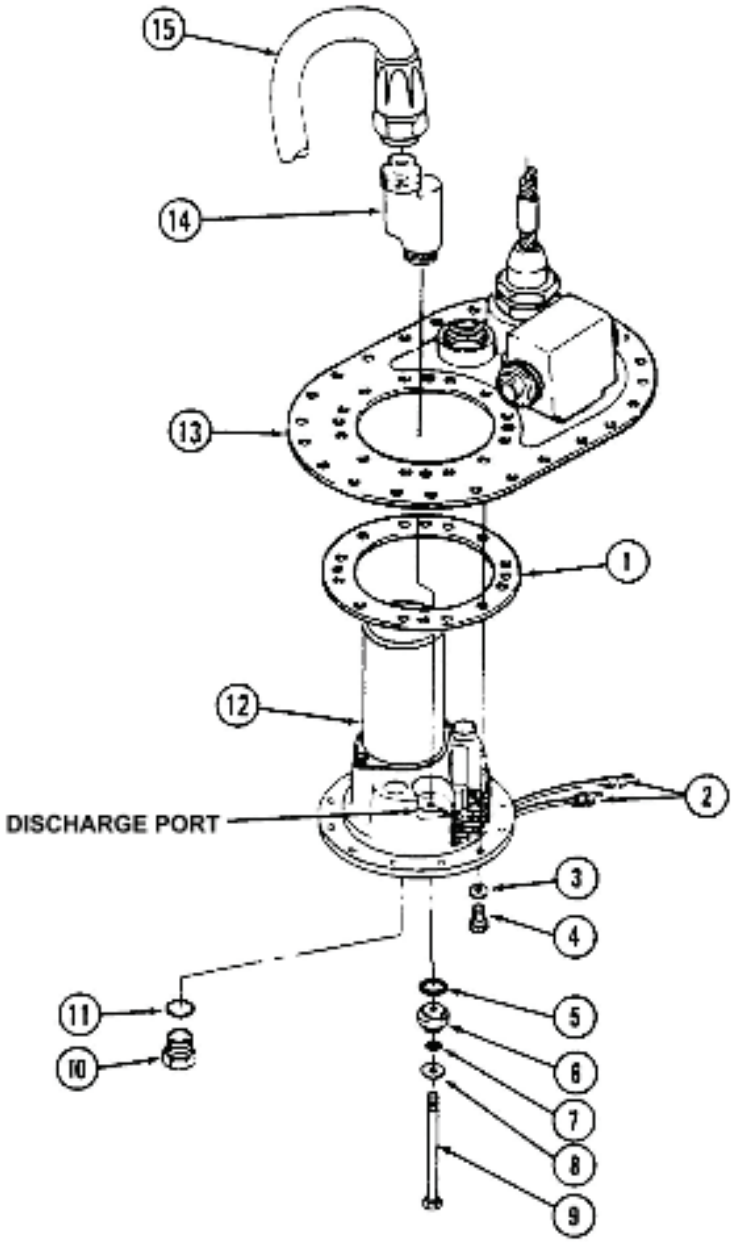
LOCKWIRE (C-405)

NOTE

1. Arrows must align within ±3° for cartridge installation.

ICN-412-A-281000-B-97499-00072-A-001-01

Figure 1. Boost Pump Cartridge Type (Sheet 2 of 2)



(S/N 36049 AND SUBSEQUENT)
(POST TB 412-89-77)

1. Gasket
2. Electrical leads
3. Washer
4. Bolt
5. Packing
6. Plug
7. Packing
8. Washer
9. Bolt
10. Plug
11. Packing
12. Boost pump
13. Sump assembly
14. Fitting
15. Hose

ICN-412-A-281000-B-97499-00070-A-001-01

Figure 2. Canister Type Boost Pump (Sheet 1 of 1)

Flow Switch Check Valves - Description

Applicable: SN: 33168-Subsequent:

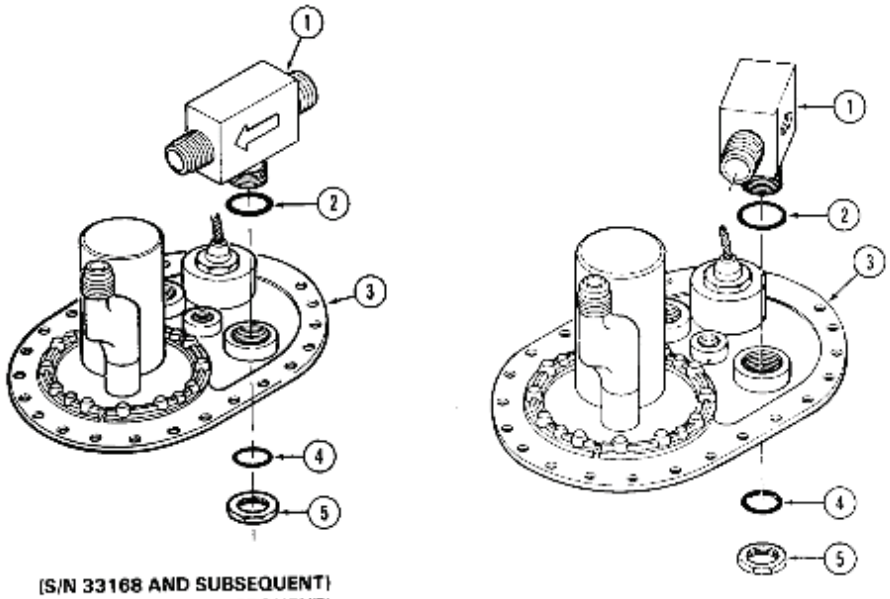
A flow switch check valve is mounted on the sump assembly of each lower main cell and is attached to the end of hose from output port of ejector pump. The flapper type check valve prevents backflow of fuel from lower main cell to the lower middle cell, but allows the forward flow of fuel from the lower middle cell to the lower main cell (aft center cell). The check valve contains a flow switch with contacts that are closed when no fuel flow exists, and open when fuel flow exists. When switch contacts are closed, the FUEL TRANS caution lights will be illuminated, provided fuel is present in the forward or mid tanks.

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Flow Switch Check Valves - Removal

Procedure

1. Remove lower main cell sump assembly (412-A-28-10-00-57A-520A-A / 00509).
2. Remove nut (5, Figure 1) and washer (4).
3. Remove flow switch check valve (1) and packing (2) from sump assembly (3). Discard packing.



1. Flow switch check valve
2. Packing
3. Sump assembly
4. Washer
5. Nut

ICN-412-A-281000-B-97499-00073-A-001-01

Figure 1. Flow Switch Check Valve (Sheet 1 of 1)

Flow Switch Check Valves - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

1. Apply assembly fluid (C-024) to packing (2, Figure 1).

2. Install flow switch check valve (1) and packing (2) through sump assembly (3).

3. Install washer (4) and nut (5). Tighten the nut.

Applicable: SN: 33108-33167:

4. For helicopters, make sure that the directional arrow on the flow switch check valve (1) points towards the aft of the helicopter.

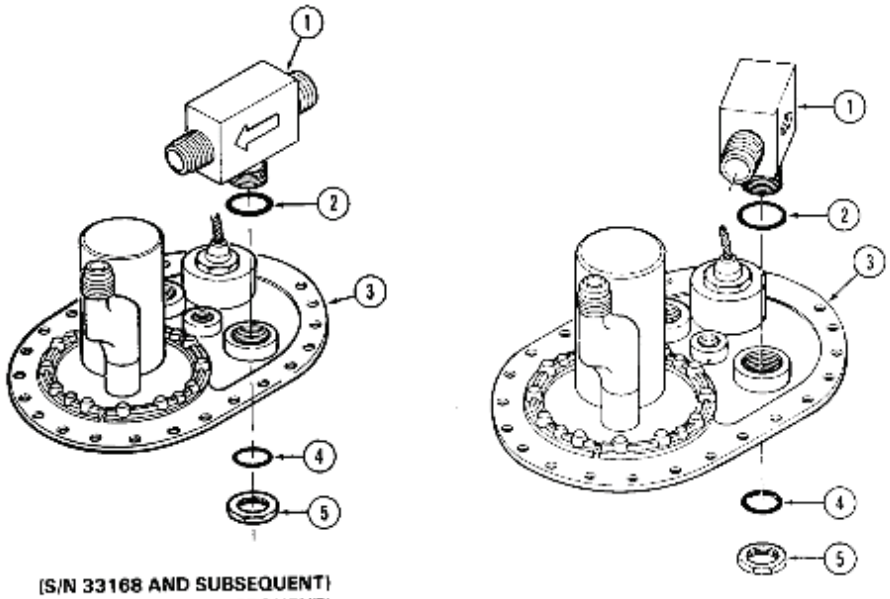
Applicable: (SN: 33108-33167 TB 412-92-106: Post) or SN: 33168-Subsequent or 36001-Subsequent:

5. For helicopters, make sure that the directional arrow on the flow switch check valve (1) points towards the front of the helicopter.

NOTE

Ensure directional arrow is positioned properly and flow switch check valve operates correctly before returning sump assembly to service.

6. Install lower main cell sump assembly (412-A-28-10-00-57A-720A-A / 00511).



(S/N 33168 AND SUBSEQUENT)
(S/N 36001 AND SUBSEQUENT)
(T.B. 412-92-106 INC.)

(S/N 33108 THRU 33167)

1. Flow switch check valve
2. Packing
3. Sump assembly
4. Washer
5. Nut

ICN-412-A-281000-B-97499-00073-A-001-01

Figure 1. Flow Switch Check Valve (Sheet 1 of 1)

Sump Drain Valves - Description

An electrically (or mechanically) operated sump drain valve is mounted on the sump assembly of each lower main cell. The sump drain valve may be electrically operated by pushing a push-button switch, or may be mechanically operated. The sump drain valves provide for draining off possible moisture or other contamination.

NOTE

Maintenance procedures are the same for both the right and left lower main sump drain valves.

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Sump Drain Valves - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure

NOTE

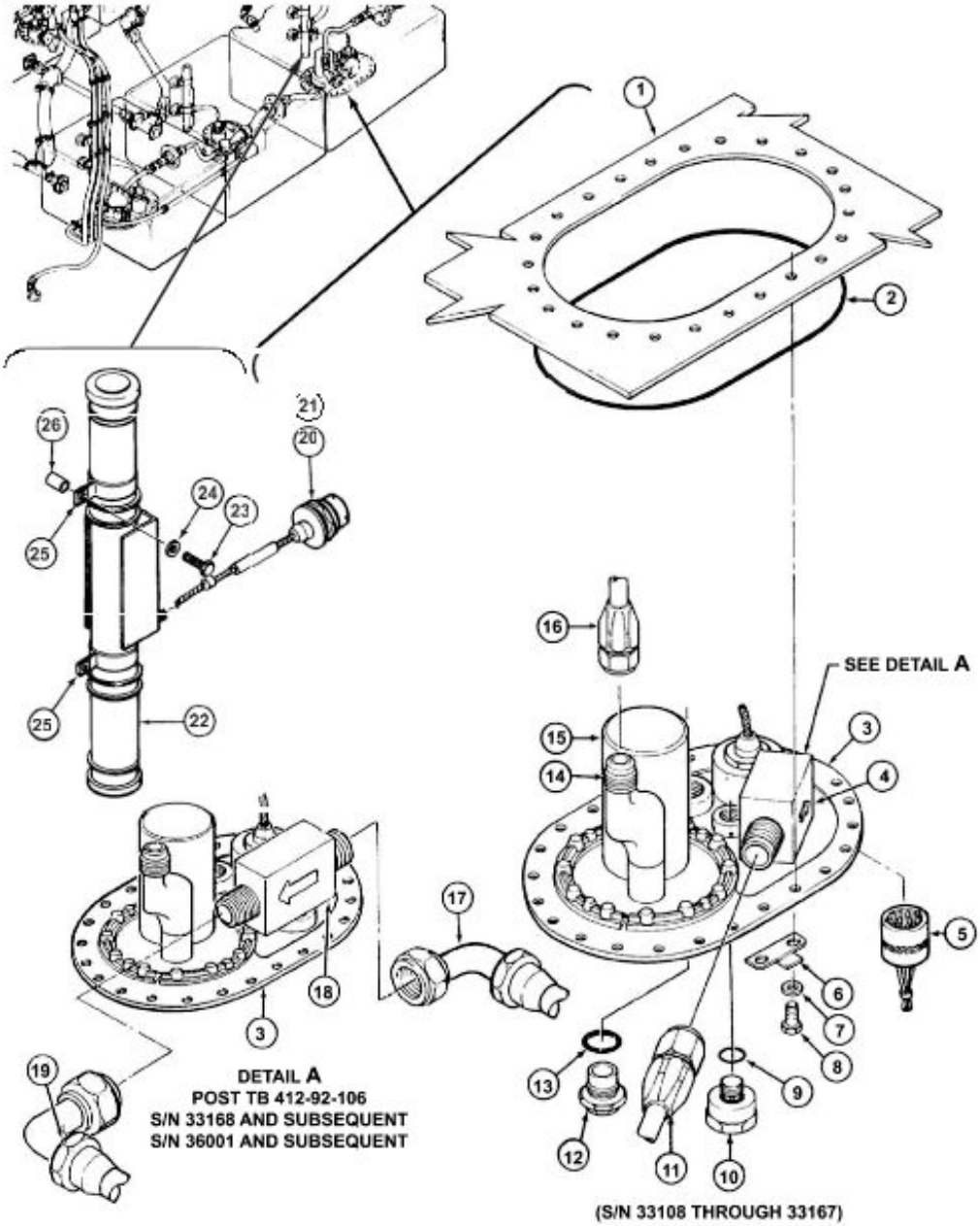
The following removal procedures are the same for both the right and left lower main sump drain valves.

1. Disconnect electrical power.

2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).

3. Remove drain valve (10, Figure 1) and packing (9). Discard packing.

4. Install caps or plugs (C-428) over cell opening.



ICN-412-A-281000-B-97499-00067-A-001-01

Figure 1. Lower Main Cell Sump Assembly (Sheet 1 of 2)

1. Lower main cell
2. Packing
3. Sump assembly
4. Flow switch check valve
5. Electrical connector
6. Retainer
7. Washer
8. Bolt
9. Packing
10. Drain valve
11. Hose
12. Defueling valve
13. Packing
14. Fitting
15. Boost pump
16. Hose
17. Hose
18. Flow switch
19. Hose
20. Electrical connector
21. Packing
22. Fuel quantity probe
23. Bolt
24. Washer
25. Mounting bracket
26. Spacer

ICN-412-A-281000-B-97499-00068-A-001-01

Figure 1. Lower Main Cell Sump Assembly (Sheet 2 of 2)

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Sump Drain Valves - Installation

Preliminary requirements

Consumables, materials and expendables

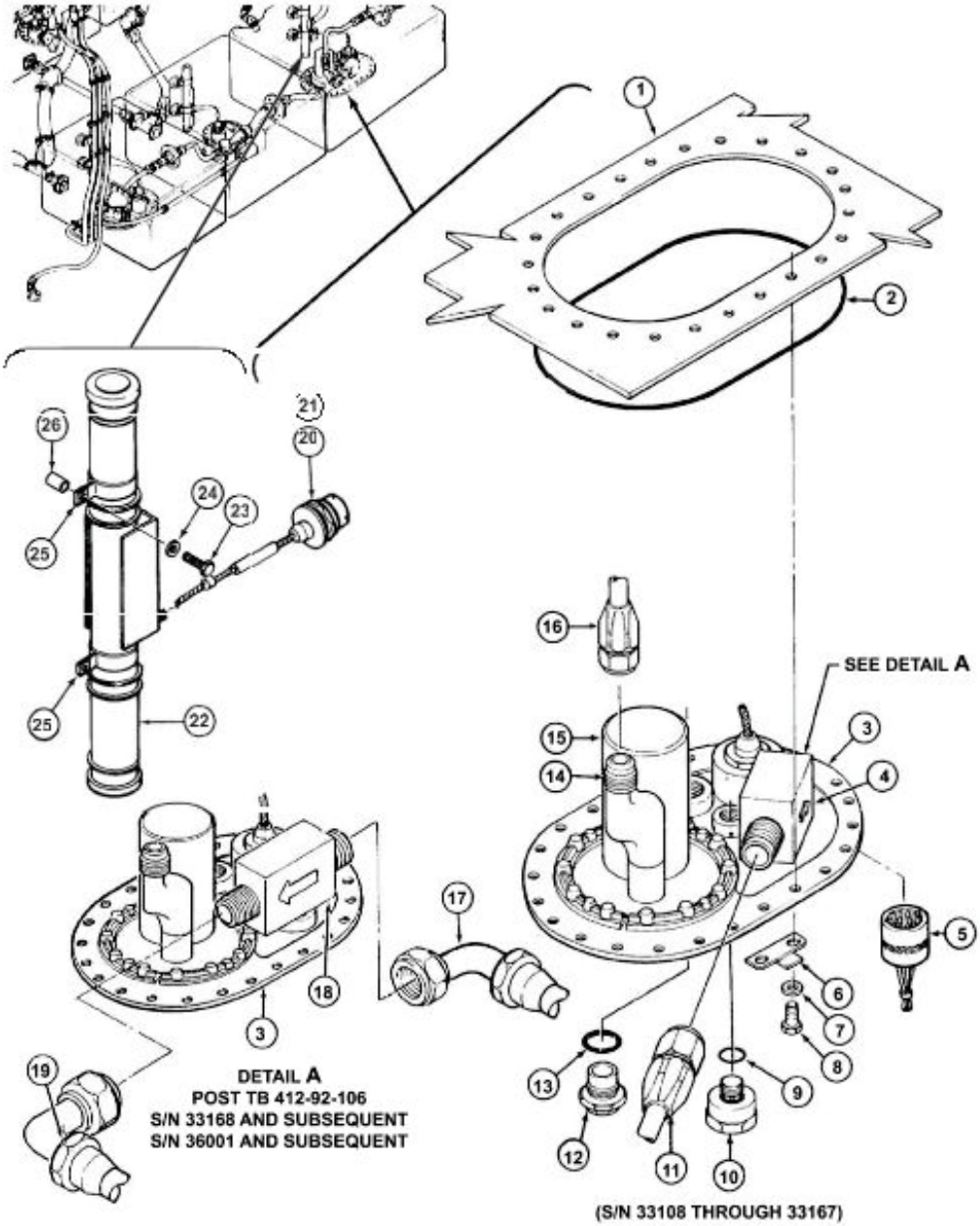
Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

NOTE

The following installation procedures are the same for both the right and left lower main sump drain valves.

1. Remove protective cover, caps and/or plugs from cell opening.
2. Apply assembly fluid (C-024) to packing (9, Figure 1) before installation.
3. Install packing and drain valve (10) in sump assembly (3).
4. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).
5. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).
6. Perform fuel system operational check.
7. Check area for leaks.



ICN-412-A-281000-B-97499-00067-A-001-01

Figure 1. Lower Main Cell Sump Assembly (Sheet 1 of 2)

1. Lower main cell
2. Packing
3. Sump assembly
4. Flow switch check valve
5. Electrical connector
6. Retainer
7. Washer
8. Bolt
9. Packing
10. Drain valve
11. Hose
12. Defueling valve
13. Packing
14. Fitting
15. Boost pump
16. Hose
17. Hose
18. Flow switch
19. Hose
20. Electrical connector
21. Packing
22. Fuel quantity probe
23. Bolt
24. Washer
25. Mounting bracket
26. Spacer

ICN-412-A-281000-B-97499-00068-A-001-01

Figure 1. Lower Main Cell Sump Assembly (Sheet 2 of 2)

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Lower Main Cell Inlet/Outlet Fittings - Description

A fuel inlet/outlet fitting is installed at the back of each lower main fuel cell. These fittings provide attachment points for hoses which transfer fuel from the upper left and the right fuel cells to the lower the main fuel cells, respectively. The fittings also provide the attachment points for hoses which connect to fuel boost pumps, allowing fuel transfer from boost pumps to engine fuel feed lines.

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Lower Main Cell Inlet/Outlet Fittings - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure

1. Disconnect electrical power.

2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).

3. Remove cabin floor access panel located aft of respective lower main cell sump assembly.

NOTE

Refer to Step 4., Step 5., or Step 6. for removal procedures for the specific fuel system installed.

Applicable: SN: 33108-33167:

4. For helicopters, remove lower right main cell inlet/outlet fitting (12, Figure 1, Detail A):

4.1. Disconnect hoses (3 and 9).

4.2. Remove bolts (6), thin aluminum washers (5), and retainers (4).

4.3. Separate inlet/outlet fitting (12) from cell.

4.4. Disconnect hose (16).

4.5. Remove unions (2, 8, and 15). Remove and discard packings (1, 7, and 14).

4.6. Remove inlet/outlet fitting (12) and packing (13) from cell. Discard packing.

4.7. Install caps or plugs (C-428) on the hose assemblies (3, 9, and 16) open end fittings.

4.8. Install protective cover over cell opening.

Applicable: SN: 33108-33167:

5. For helicopters, remove lower left main cell inlet/outlet fitting (24, Figure 1, Detail B):

5.1. Disconnect hoses (22 and 30).

5.2. Remove bolts (27), thin aluminum washers (26), and retainers (25).

5.3. Separate inlet/outlet fitting (24) from cell.

5.4. Disconnect hose (17).

5.5. Remove unions (18, 21, and 29). Remove and discard packings (19, 20, and 28).

5.6. Remove inlet/outlet fitting (24) and packing (23) from cell. Discard packing.

5.7. Install caps or plugs (C-428) on the hose assemblies (22, 30, and 17) open end fittings.

5.8. Install protective cover over cell opening.

Applicable: (SN: 33108-33167 TB 412-92-106: Post) or SN: 33168-Subsequent or 36001-Subsequent:

6. For helicopters, remove the lower left or right main cell inlet/outlet fitting (45, Figure 1, Detail C):

6.1. Disconnect hoses (33, 40, and 43).

6.2. Remove bolts (36), thin aluminum washers (35), and retainers (34).

6.3. Separate inlet/outlet fitting (45) from cell.

6.4. Disconnect hoses (48 and 51).

6.5. Remove unions (32, 42, and 50). Remove and discard packings (31, 41, and 49).

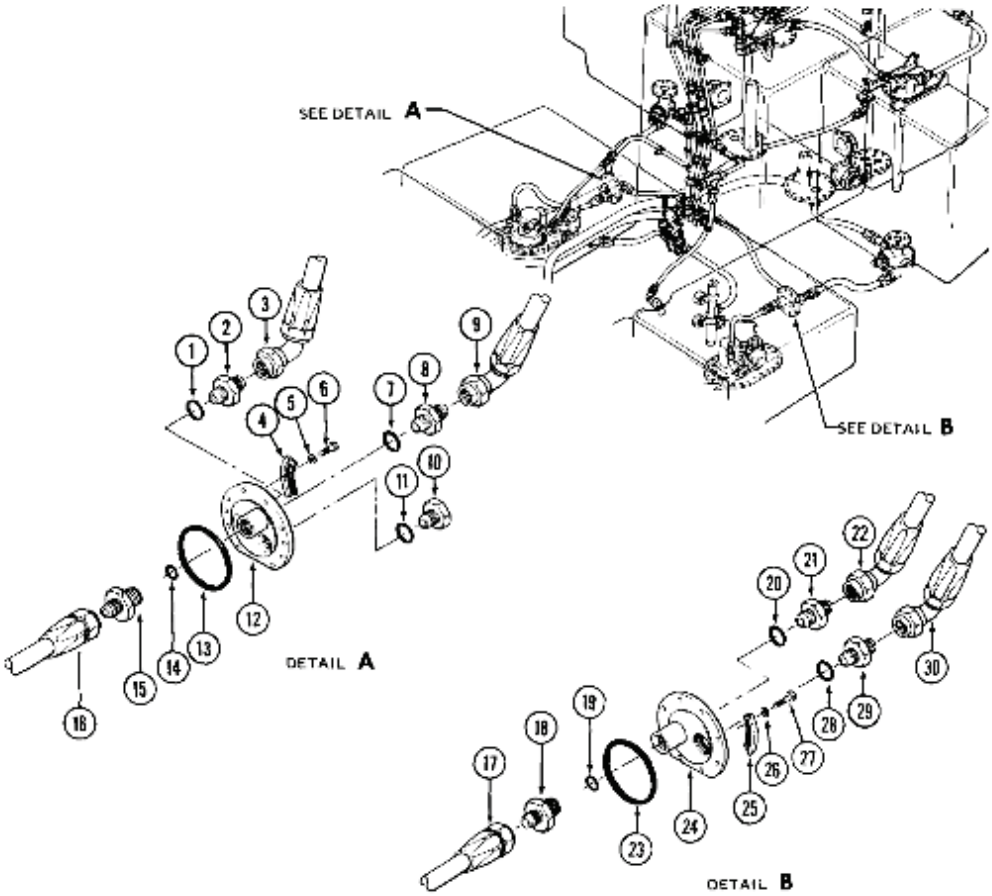
6.6. Remove elbow (39). Remove and discard packing (37).

6.7. Remove reducer (47). Remove and discard packing (46).

6.8. Remove inlet/outlet fitting (45) and packing (44) from cell. Discard packing.

6.9. Install caps or plugs (C-428) on the hose assemblies (33, 40, 43, 48, and 51) open end fittings.

6.10. Install protective cover over cell opening.

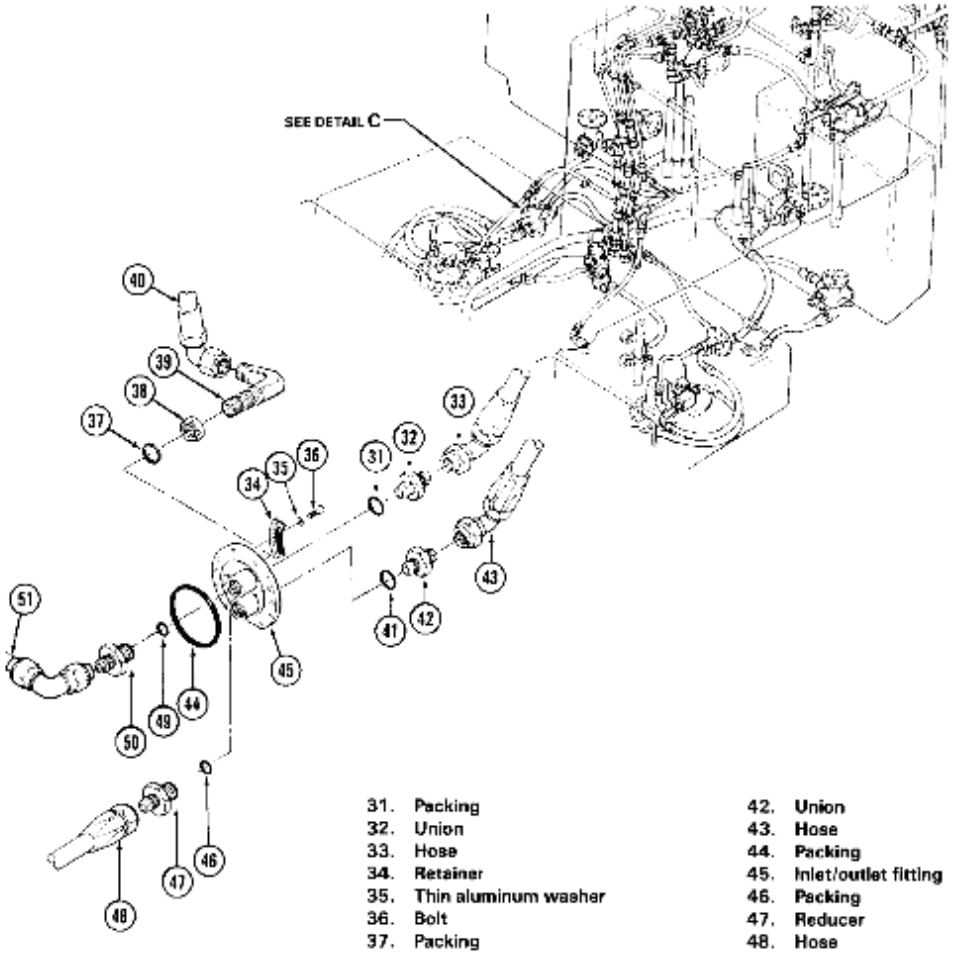


(S/N 33108 THRU 33167)

1. Packing
2. Union
3. Hose
4. Retainer
5. Thin aluminum washer
6. Bolt
7. Packing
8. Union
9. Hose
10. Plug
11. Packing
12. Inlet/outlet fitting
13. Packing
14. Packing
15. Union
16. Hose
17. Hose
18. Union
19. Packing
20. Packing
21. Union
22. Hose
23. Packing
24. Inlet/outlet fitting
25. Retainer
26. Thin aluminum washer
27. Bolt
28. Packing
29. Union
30. Hose

ICN-412-A-281000-B-97499-00074-A-001-01

Figure 1. Lower Main Cell Inlet/Outlet Fitting (Sheet 1 of 2)



DETAIL C

(RIGHT SIDE SHOWN-LEFT OPPOSITE)

(S/N 33168 AND SUBSEQUENT)
(S/N 36001 AND SUBSEQUENT)
(T.B. 412-92-106 INC.)

31. Packing
32. Union
33. Hose
34. Retainer
35. Thin aluminum washer
36. Bolt
37. Packing
38. Nut
39. Elbow
40. Hose
41. Packing
42. Union
43. Hose
44. Packing
45. Inlet/outlet fitting
46. Packing
47. Reducer
48. Hose
49. Packing
50. Union
51. Hose

ICN-412-A-281000-B-97499-00075-A-001-01

Figure 1. Lower Main Cell Inlet/Outlet Fitting (Sheet 2 of 2)

Lower Main Cell Inlet/Outlet Fittings - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

NOTE

Refer to Step 1., Step 2., or Step 3. for installation procedures for the specific fuel system installed.

1. Remove protective cover from the fuel cell opening.

2. Remove caps and/or plugs from the hose assemblies and fittings.

Applicable: SN: 33108-33167:

3. For helicopters, install lower right main cell inlet/outlet fitting (12, Figure 1, Detail A), as follows:

3.1. Apply assembly fluid (C-024) to packings (1, 7, and 14).

3.2. Install packings (1, 7, and 14) on unions (2, 8, and 15).

3.3. Install unions (2, 8, and 15) in inlet/outlet fitting (12).

3.4. Place packing (13) in groove around cell port. Packing may be held in place with assembly fluid (C-024).

3.5. Hold inlet/outlet fitting (12) close to cell port and connect hose (16).

NOTE

Bolts (6) installed in top six holes of inlet/outlet fitting (12) are longer than bolts (6) used in bottom two holes.

3.6. Position inlet/outlet fitting (12) over cell port. Install longer bolts (6), thin aluminum washers (5), and retainers (4) in upper six holes. Install shorter bolts (6) and thin aluminum washers (5) in two lower holes.

3.7. Connect hoses (3 and 9).

3.8. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).

3.9. Check area for leaks.

Applicable: SN: 33108-33167:

4. For helicopters, install lower left main cell inlet/outlet fitting (24, Figure 1, Detail B):

4.1. Apply assembly fluid (C-024) to packings (19, 20, and 28).

4.2. Install packings (19, 20, and 28) on unions (18, 21, and 29).

4.3. Install unions (18, 21, and 29) in inlet/outlet fitting (24).

4.4. Place packing (23) in groove around cell port. Packing may be held in place with assembly fluid (C-024).

4.5. Hold inlet/outlet fitting (24) close to cell port and connect hose (17).

NOTE

Bolts (27) used in the top six holes of the inlet/outlet fitting (24) are longer than bolts (27) used in two bottom holes.

4.6. Position inlet/outlet fitting (24) over cell port. Install longer bolts (27), thin aluminum washers (26), and retainers (25) in the upper six holes. Install shorter bolts (27) and thin aluminum washers (26) in the two lower holes.

4.7. Connect hoses (22 and 30).

4.8. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).

4.9. Check area for leaks.

Applicable: (SN: 33108-33167 TB 412-92-106: Post) or SN: 33168-Subsequent or 36001-Subsequent:

5. For helicopters, install the lower left or right main cell inlet/outlet fitting (24, Figure 1, Detail C):

5.1. Apply assembly fluid (C-024) to the packings (31, 37, 41, 46, and 49).

5.2. Install the packings (31, 41, and 49) on the unions (32, 42, and 50).

5.3. Install the packing (37) on the elbow (39).

5.4. Install the packing (46) on the reducer (47).

5.5. Install the unions (32, 42, and 50), elbow (39), and reducer (47) in the inlet/outlet fitting (45).

5.6. Place the packing (44) in the groove around the cell port. The packing may be held in place with assembly fluid (C-024).

5.7. Hold inlet/outlet fitting (45) close to cell port and connect hoses (48 and 51).

NOTE

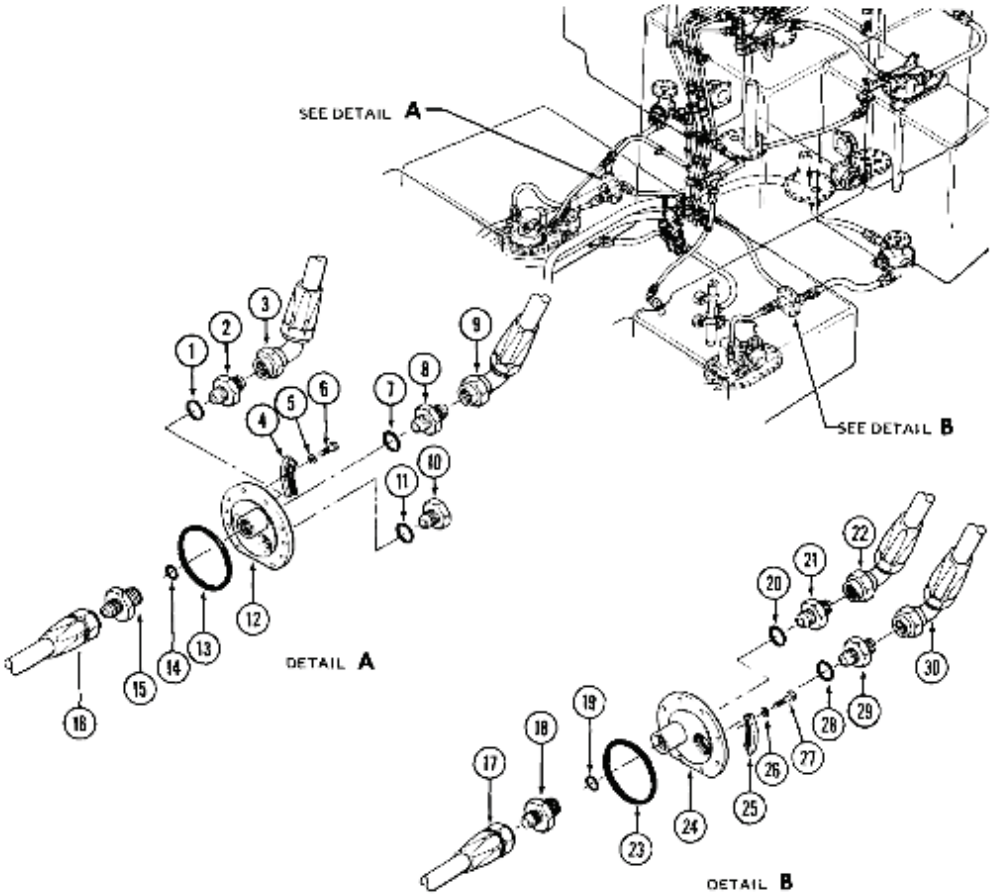
Bolts (36) used in the top six holes of inlet/outlet fitting (45) are longer than bolts (36) used in the two bottom holes.

5.8. Position inlet/outlet fitting (45) over cell port. Install longer bolts (36), thin aluminum washers (35), and retainers (34) in upper six holes and shorter bolts (36) and thin aluminum washers (35) in the two lower bolt holes.

5.9. Connect hoses (33, 40, and 43).

5.10. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).

5.11. Check area for leaks.

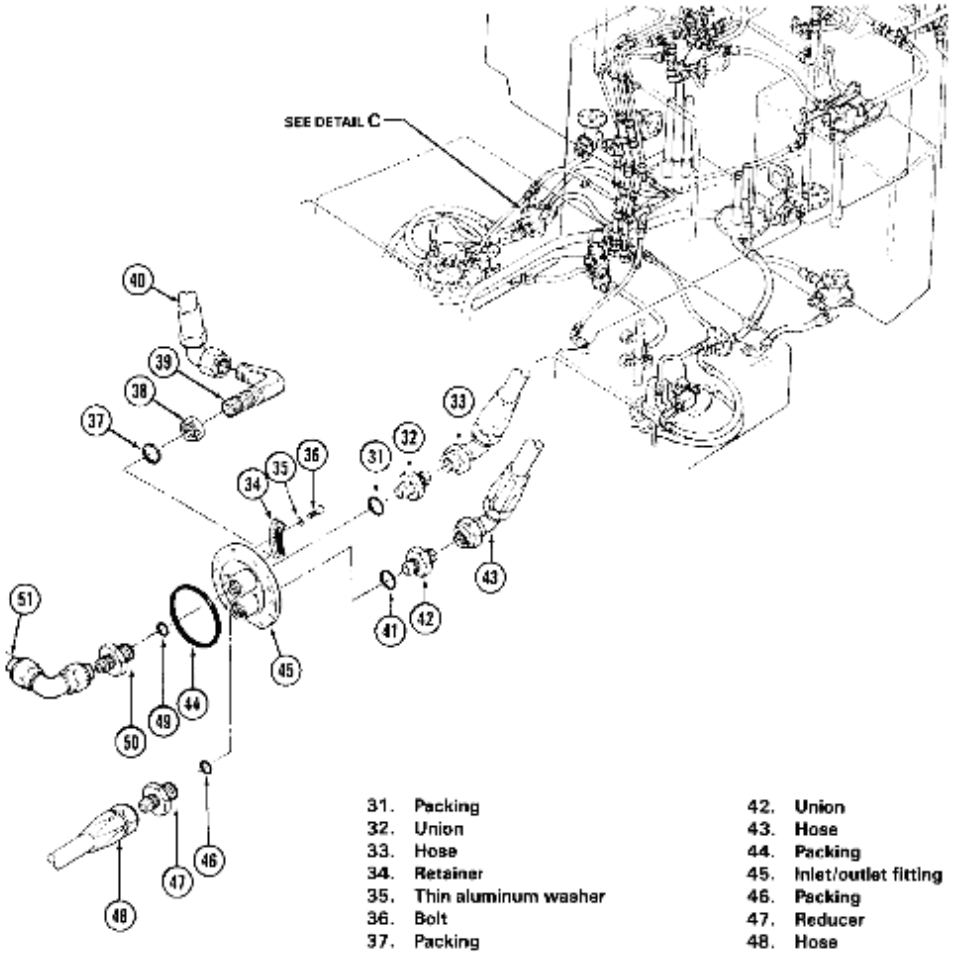


(S/N 33108 THRU 33167)

1. Packing
2. Union
3. Hose
4. Retainer
5. Thin aluminum washer
6. Bolt
7. Packing
8. Union
9. Hose
10. Plug
11. Packing
12. Inlet/outlet fitting
13. Packing
14. Packing
15. Union
16. Hose
17. Hose
18. Union
19. Packing
20. Packing
21. Union
22. Hose
23. Packing
24. Inlet/outlet fitting
25. Retainer
26. Thin aluminum washer
27. Bolt
28. Packing
29. Union
30. Hose

ICN-412-A-281000-B-97499-00074-A-001-01

Figure 1. Lower Main Cell Inlet/Outlet Fitting (Sheet 1 of 2)



DETAIL C

(RIGHT SIDE SHOWN-LEFT OPPOSITE)

(S/N 33168 AND SUBSEQUENT)
(S/N 36001 AND SUBSEQUENT)
(T.B. 412-92-106 INC.)

31. Packing
32. Union
33. Hose
34. Retainer
35. Thin aluminum washer
36. Bolt
37. Packing
38. Nut
39. Elbow
40. Hose
41. Packing
42. Union
43. Hose
44. Packing
45. Inlet/outlet fitting
46. Packing
47. Reducer
48. Hose
49. Packing
50. Union
51. Hose

ICN-412-A-281000-B-97499-00075-A-001-01

Figure 1. Lower Main Cell Inlet/Outlet Fitting (Sheet 2 of 2)

Interconnect Valve - Description

A motor-operated interconnect valve is located in the interconnect line between the right and left lower main cells. The interconnect valve is operated by the FUEL INTCON switch on the engine and fuel control panel. The valve is normally closed to provide isolation of the two power section fuel supplies, but may be opened to permit crossflow when fuel levels are low. Valve opens automatically if low fuel level condition occurs.

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Interconnect Valve - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure

1. Disconnect electrical power.

2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).

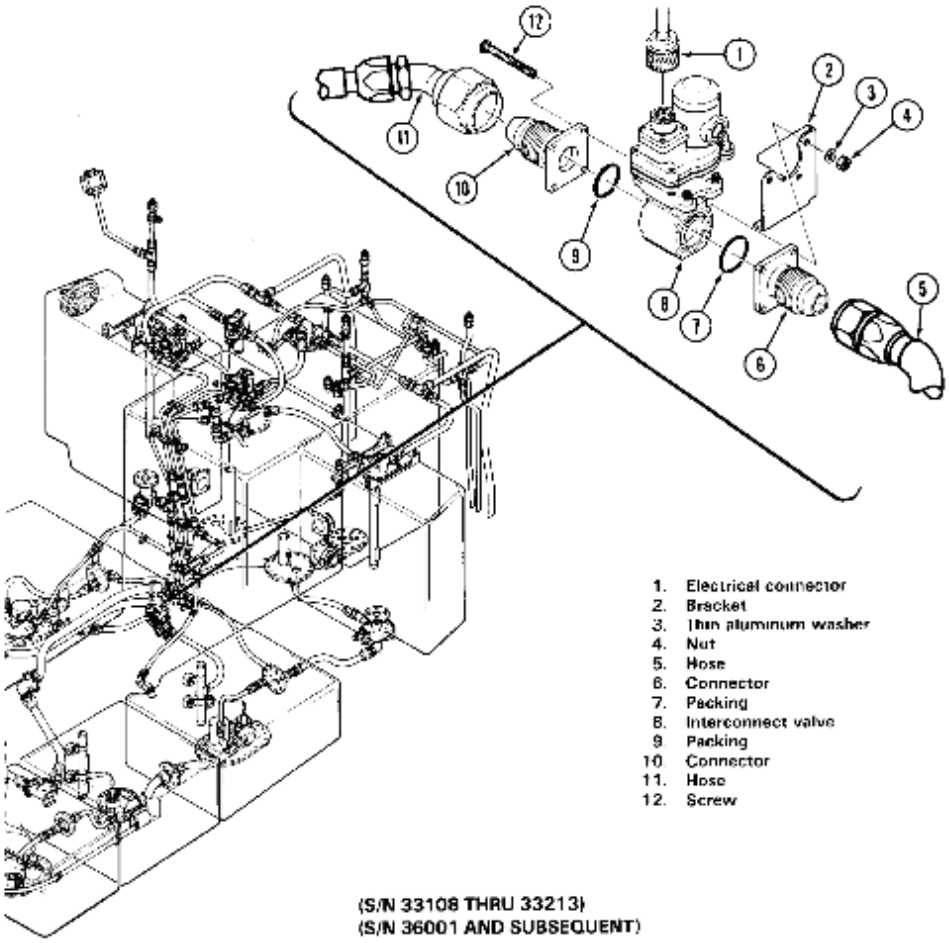
3. Disconnect electrical connector (1, Figure 1).

4. Disconnect hoses (5 and 11).

5. Remove screws (12), thin aluminum washers (3), and nuts (4).

6. Remove interconnect valve (8), connectors (6 and 10), and packings (7 and 9) from bracket (2). Discard packings.

7. Install caps or plugs (C-428) on hose assemblies (5 and 11) open end fittings and on valve openings.



ICN-412-A-281000-B-97499-00076-A-001-01

Figure 1. Interconnect Valve (Sheet 1 of 1)

Interconnect Valve - Installation

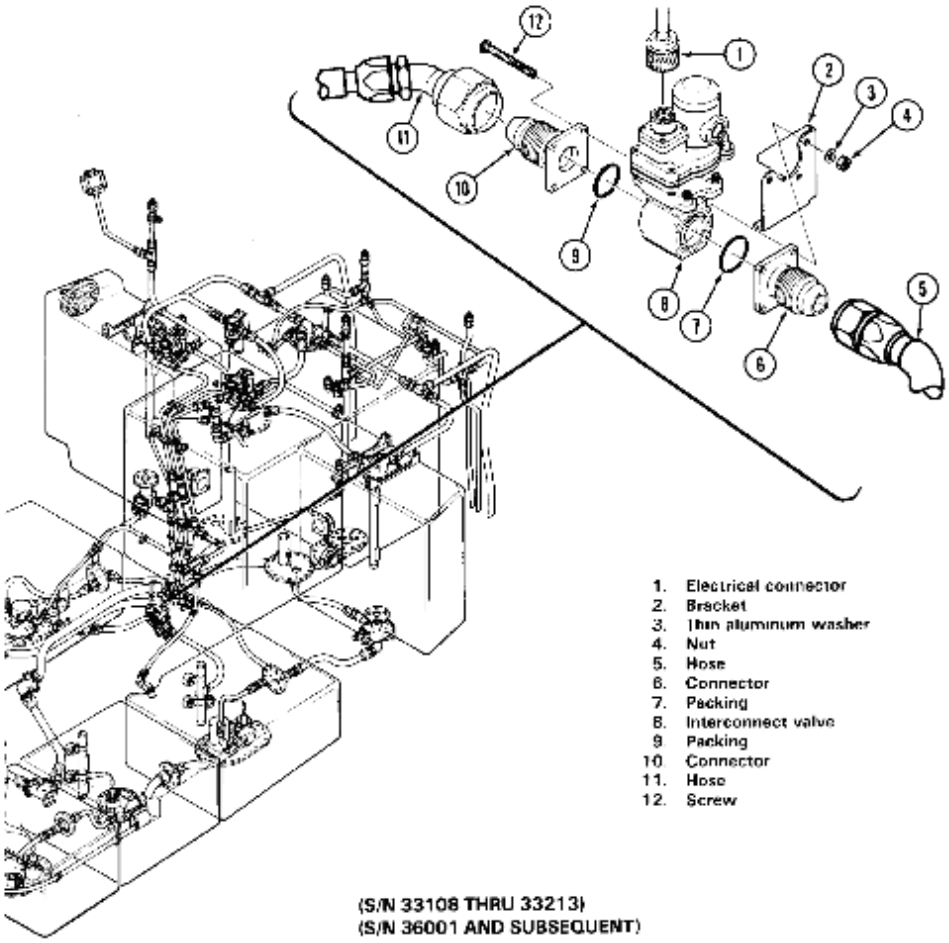
Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

1. Remove the protective caps and/or plugs from the hose assemblies (5 and 11) end fittings, and from the valve openings.
2. Apply assembly fluid (C-024) to packings (7 and 9, Figure 1).
3. Place packings (7 and 9) and connectors (6 and 10) on interconnect valve (8).
4. Position the interconnect valve (8) and connectors (6 and 10) on bracket (2). Install screws (12), thin aluminum washers (3), and nuts (4). Tighten the nuts.
5. Connect electrical connector (1).
6. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).
7. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).
8. Perform fuel system operational check.
9. Check area for leaks.



ICN-412-A-281000-B-97499-00076-A-001-01

Figure 1. Interconnect Valve (Sheet 1 of 1)

Upper Forward Center Cell - Description

The upper forward center cell is located above W.L. 22, between the two main longitudinal beams (at left B.L. 14 and right B.L. 14), and between F.S. 155 and 178. The upper forward center cell has a vent port on top and interconnect ports on the bottom, left, right, and aft sides. Internal access is provided by openings at top and bottom of cell.

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Upper Forward Center Cell - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure



DO NOT HANDLE FUEL CELL WHEN AMBIENT TEMPERATURE IS BELOW 65°F (18°C).

1. Disconnect electrical power.

2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).

3. Remove large access door on underside of helicopter below upper aft center cell.

4. Remove smaller access doors on underside of helicopter below upper forward center cell.

5. Remove access panels in engine compartment floor above upper forward cell, upper aft cell, upper left cell, and upper right cell.

6. Remove upper forward center cell bottom cover (short standpipe) assembly (7, Figure 1) (412-A-28-10-00-66A-520A-A / 00540).

7. Remove upper forward center cell bottom cover (tall standpipe) assembly (8) (412-A-28-10-00-67A-520A-A / 00543).

8. Remove upper aft center cell (412-A-28-10-00-70A-520A-A / 00552).

9. Remove large access panel on bulkhead between upper aft center cell cavity and upper forward center cell cavity.

10. Remove upper left cell and upper right cell top cover assemblies for access (412-A-28-10-00-76A-520A-A / 00574).

11. Working through top vent port of upper left cell, remove lockwire, large nut, and packing from interconnect fitting (6) (412-A-28-10-00-68A-520A-A / 00546). Discard packing.

12. Working through top vent port of upper right cell, remove lockwire, large nut, and packing from interconnect fitting (2) (412-A-28-10-00-68A-520A-A / 00546). Discard packing.

13. Remove upper forward center cell top cover assembly (3) and support assembly (4) (412-A-28-10-00-69A-520A-A / 00549).

14. Install protective covers on cell openings except upper forward center cell openings, and install caps or plugs (C-428) on open lines.

15. Collapse aft upper edge of upper forward center cell downward and inward. Ensure fittings (2, 5, and 6) are clear of bulkhead openings.

16. Position two webbing straps vertically around cell. Tighten straps to compress cell to pass through access openings.

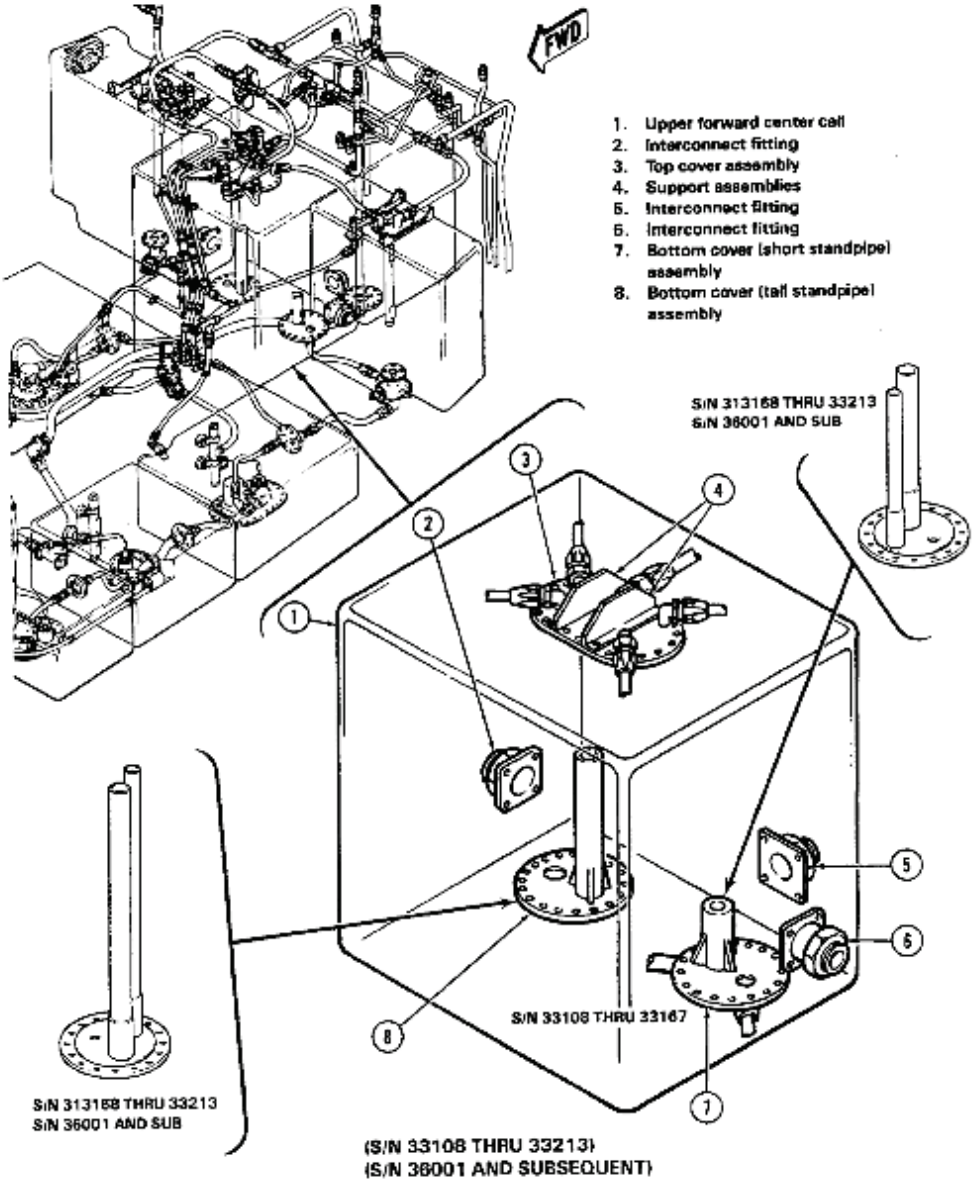
17. Remove cell from upper forward center cell cavity through large access opening in bulkhead between upper forward and upper aft cell cavities, and through large access opening on underside of helicopter below upper aft center cell.

18. Restore cell to original shape. Cover cell openings.



STORE CELL IN A HEATED AREA IF AMBIENT TEMPERATURE IS BELOW 65°F (18°C).

19. Install protective covers on upper forward center cell openings.



ICN-412-A-281000-B-97499-00077-A-001-01

Figure 1. Upper Forward Center Fuel Cell (Sheet 1 of 1)

Upper Forward Center Cell - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Soap	100066	AR	Zodiac Aerospace (www.zodiacaerospace.com)
Cleaning Solution	601199	AR	Zodiac Aerospace (www.zodiacaerospace.com)
Clean cloth	C-516	AR	

Safety conditions



OBEY ALL THE SAFETY PRECAUTIONS WHEN YOU DO MAINTENANCE ON FUEL SYSTEM EQUIPMENT.



OBEY ALL THE FUEL CELL HANDLING PRECAUTIONS DURING CLEANING OR DAMAGE TO THE CELL MAY OCCUR.



DO NOT ALLOW SOLVENTS OR COMPOUNDS TO MAKE CONTACT WITH PARTS MADE OF ELASTOMERIC MATERIAL OR DAMAGE WILL OCCUR.

Procedure

NOTE

Any evidence of fuel leakage must be identified before cleaning.

NOTE

Refer to the document "FUEL-F6101-SPM20-00" (Zodiac Aerospace), for additional information on the cleaning of the fuel cells.

1. Clean the internal surface of the fuel cell with a clean cloth (C-516) moistened with a cleaning solution (601199). Dry the liner with low-pressure filtered compressed air for 2 to 4 hours, at approximately 68°F (20°C).
2. Clean small areas on the external surface of the fuel cell with a clean cloth (C-516) moistened with a cleaning solution (601199). Dry with a clean cloth (C-516).
3. Clean an extensive area on the external surface of the fuel cell with a solution of soap (100066) and hot water. Rinse with water and dry with a clean cloth (C-516).
4. Dry clean the fittings of the fuel cell with filtered compressed air and a clean cloth (C-516) to remove any blockage or obstruction.
5. Clean fungus growth found on the fuel cell or on the fuel cell cavity per the instructions contained in the BHT-ALL-SPM, Chapter 5, Standard Practices Manual.

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Upper Forward Center Cell - Inspection

Procedure



DO NOT HANDLE FUEL CELL WHEN AMBIENT TEMPERATURE IS BELOW 65°F (18°C).

1. Inspect cell for cracks and scuffing. Cracks and severe scuffing are not acceptable.

2. Inspect threaded inserts for damage. Visually detectable damage to threads is not acceptable.

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Upper Forward Center Cell - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	
Lockwire	C-405	AR	
Talcum powder	C-408	AR	

Procedure



DO NOT HANDLE FUEL CELL WHEN AMBIENT TEMPERATURE IS BELOW 65°F (18°C).

1. Inspect cell cavity for cleanliness and smoothness of interior surfaces.
2. Apply talcum powder (C-408) to interior surfaces of cell cavity.
3. Remove protective covers, caps, and/or plugs.
4. Collapse cell and position two webbing straps vertically around cell. Tighten straps to compress cell sufficiently to pass through access openings.
5. Carefully pass cell through large access opening below upper aft cell cavity, through large access opening in bulkhead between upper aft center cell and upper forward center cell cavities, and into upper forward center cell cavity.
6. Remove webbing straps and arrange cell in approximate position in cell cavity.
7. Align interconnect ports and bottom outlet ports with matching cell ports and fuselage openings.

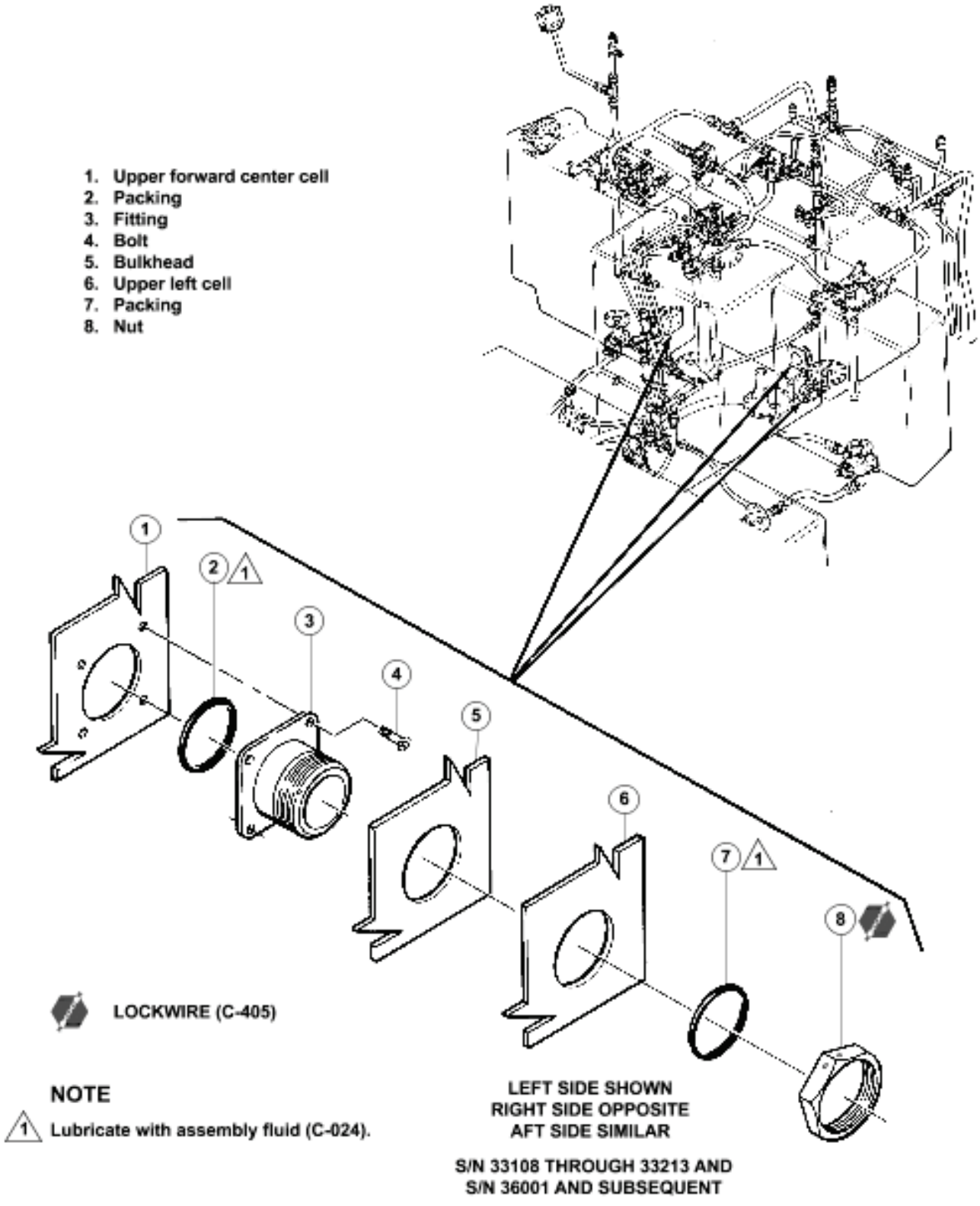
NOTE

Make sure nut (8, Figure 1) contacts packing (7) before contacting upper left cell (6). To prevent deforming of packing between nut and upper left cell, do not apply excessive torque to nut.

8. Lubricate packings (7) with assembly fluid (C-024).
9. Working through top vent port of upper left cell (6), install the packing (7) and the nut (8) on the interconnect fitting (3). Tighten the nut.
10. Secure the nut (8) to the interconnect fitting (3) with lockwire (C-405).
11. Working through the top vent cover of the upper right cell, install the packing (7) and the nut (8) on the interconnect fitting (3). Tighten the nut.
12. Secure the nut (8) to the interconnect fitting (3) with lockwire (C-405).
13. Install upper left cell and upper right cell top cover assemblies (412-A-28-10-00-76A-720A-A / 00575).
14. Install upper forward center cell top cover assembly (412-A-28-10-00-69A-720A-A / 00550).
15. Install upper forward center cell bottom cover (tall standpipe) assembly (412-A-28-10-00-67A-720A-A / 00544).
16. Install upper forward center cell bottom cover (short standpipe) assembly (412-A-28-10-00-66A-720A-A / 00541).
17. Install large access panel on bulkhead between upper forward center cell cavity and upper aft center cell cavity.
18. Install upper aft center cell (412-A-28-10-00-70A-720A-A / 00555).
19. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).
20. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).
21. Perform fuel system operational check.

22. Check area for leaks.

23. Install access panels and doors.



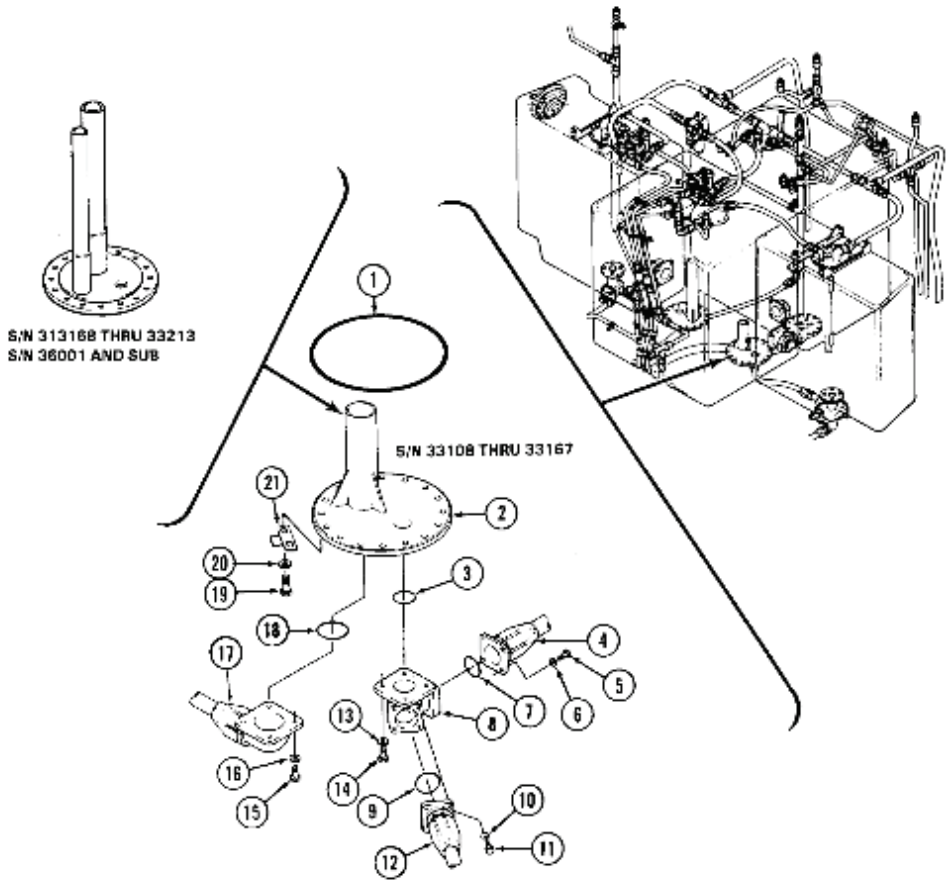
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Figure 1. Upper Forward Center Cell Interconnect Fitting (Sheet 1 of 1)

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Upper Forward Center Cell Bottom Cover (Short Standpipe) Assembly - Description

A bottom cover (short standpipe) assembly (Figure 1) is installed on left bottom interconnect port. Two fittings mounted on bottom cover (short standpipe) assembly provide attachment points for interconnecting hoses from upper aft center cell and upper left cell, and an interconnecting hose to input port of input/output fitting on aft side of left lower main cell.



1. Packing
2. Cover (short stand pipe)
3. Packing
4. Hose assembly
5. Screw
6. Washer
7. Packing
8. Fitting
9. Packing
10. Washer
11. Screw
12. Hose assembly
13. Washer
14. Screw
15. Screw
16. Washer
17. Hose assembly
18. Packing
19. Bolt
20. Washer
21. Retainer

ICN-412-A-281000-B-97499-00079-A-001-01

Figure 1. Upper Forward Center Cell Bottom Cover (Short Standpipe) Assembly (Sheet 1 of 1)

Upper Forward Center Cell Bottom Cover (Short Standpipe) Assembly - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure

1. Disconnect electrical power.

2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).

3. Remove lower fuselage access doors below short standpipe assembly.

4. Remove the screws (5, Figure 1), washers (6), and packing (7) from the fitting (8). Discard the packing.

5. Remove screws (11), washers (10), and packing (9). Discard the packing.

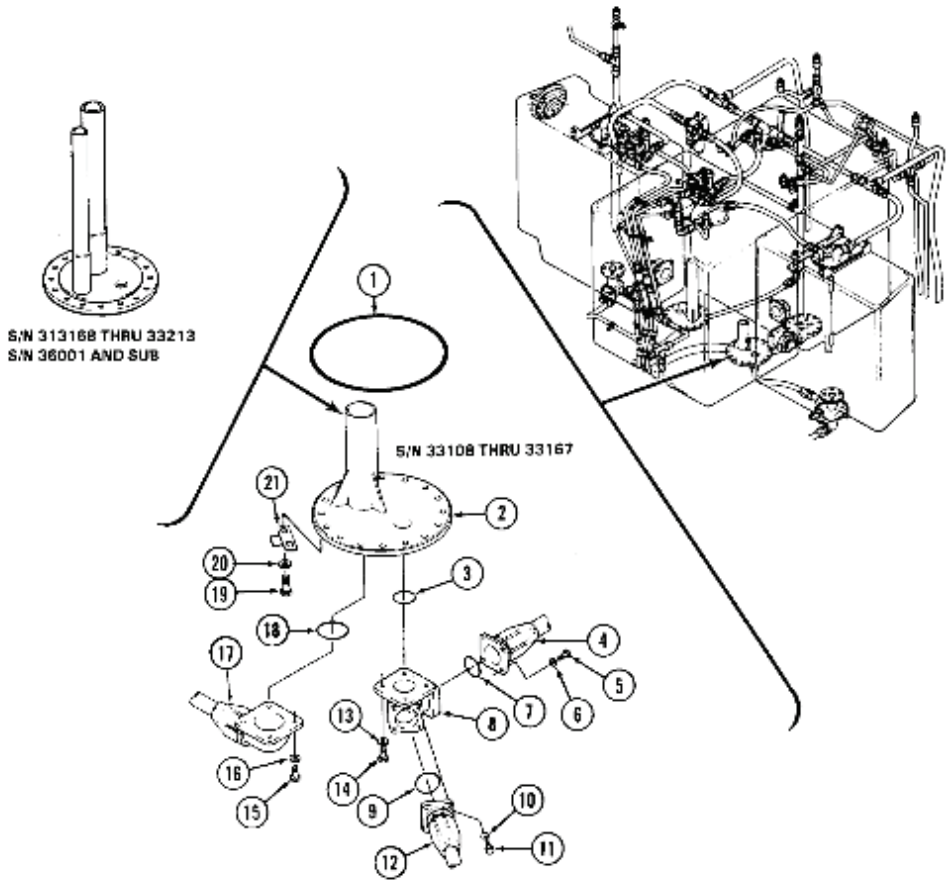
6. Remove screws (15), washers (16), and packing (18). Discard the packing.

7. Remove screws (14), washers (13), fitting (8), and packing (3). Discard the packing.

8. Remove bolts (19), washers (20), retainers (21), cover (2), and packing (1). Discard the packing.

9. Install protective cover over cell opening.

10. Install caps or plugs (C-428) on hose assemblies (4, 12, and 17).



1. Packing
2. Cover (short stand pipe)
3. Packing
4. Hose assembly
5. Screw
6. Washer
7. Packing
8. Fitting
9. Packing
10. Washer
11. Screw
12. Hose assembly
13. Washer
14. Screw
15. Screw
16. Washer
17. Hose assembly
18. Packing
19. Bolt
20. Washer
21. Retainer

ICN-412-A-281000-B-97499-00079-A-001-01

Figure 1. Upper Forward Center Cell Bottom Cover (Short Standpipe) Assembly (Sheet 1 of 1)

Upper Forward Center Cell Bottom Cover (Short Standpipe) Assembly - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

1. Remove protective cover from cell opening.

2. Remove protective caps and/or plugs from the hose assemblies (4, 12, and 17, Figure 1) end fittings.

3. Apply assembly fluid (C-024) to packing (1).

4. Place packing (1) on cover (2).

5. Install bolts (19), washers (20), and retainers (21).

6. Apply assembly fluid (C-024) to packing (3).

7. Install packing (3), screws (14), and washers (13).

8. Remove protective covers from hose assemblies (4 and 12).

9. Apply assembly fluid (C-024) to packings (7 and 9).

10. Install hose assemblies (4 and 12) and packings (7 and 9) with screws (5 and 11) and washers (6 and 10).

11. Remove protective cover from hose assembly (17).

12. Apply assembly fluid (C-024) to packing (18).

13. Install hose assembly (17) and packing (18) with screws (15) and washers (16).

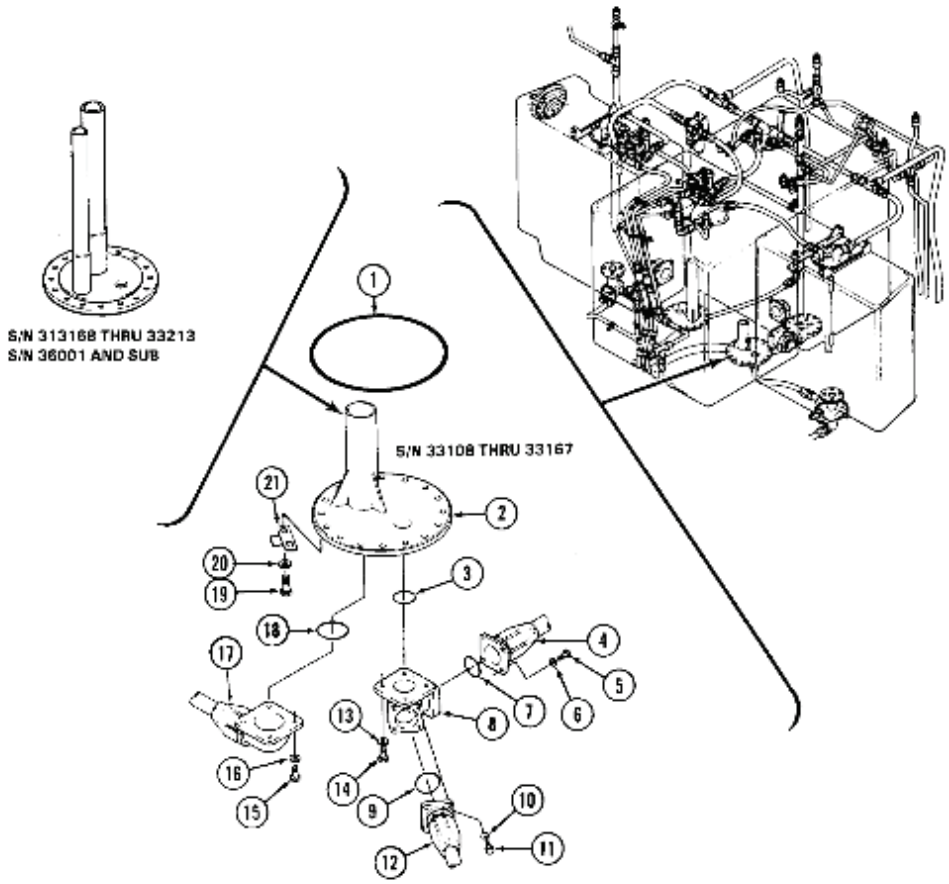
14. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).

15. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).

16. Perform fuel system operational check.

17. Check area for leaks.

18. Install lower fuselage access door.



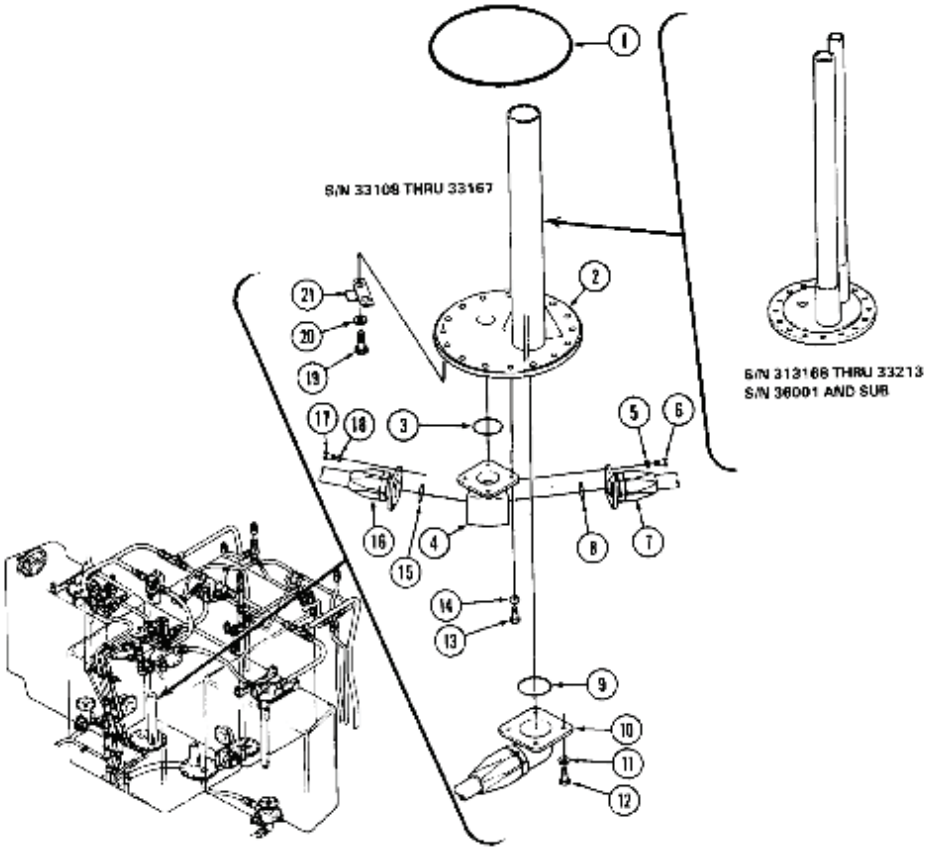
1. Packing
2. Cover (short stand pipe)
3. Packing
4. Hose assembly
5. Screw
6. Washer
7. Packing
8. Fitting
9. Packing
10. Washer
11. Screw
12. Hose assembly
13. Washer
14. Screw
15. Screw
16. Washer
17. Hose assembly
18. Packing
19. Bolt
20. Washer
21. Retainer

ICN-412-A-281000-B-97499-00079-A-001-01

Figure 1. Upper Forward Center Cell Bottom Cover (Short Standpipe) Assembly (Sheet 1 of 1)

Upper Forward Center Cell Bottom Cover (Tall Standpipe) Assembly - Description

A bottom cover (tall standpipe) assembly (Figure 1) is installed on right bottom interconnect port. Two fittings mounted on bottom cover (tall standpipe) assembly provide attachment points for interconnecting hoses from upper aft center cell bottom fitting and upper right cell outlet fitting, and an interconnecting hose to input port of inlet/outlet fitting on aft side of right lower main cell.



1. Packing
2. Cover (long stand pipe)
3. Packing
4. Fitting
5. Washer
6. Screw
7. Hose assembly
8. Packing
9. Packing
10. Hose assembly
11. Washer
12. Screw
13. Screw
14. Washer
15. Packing
16. Hose assembly
17. Screw
18. Washer
19. Bolt
20. Washer
21. Retainer

ICN-412-A-281000-B-97499-00080-A-001-01

Figure 1. Upper Forward Center Cell Bottom Cover (Tall Standpipe) Assembly (Sheet 1 of 1)

Upper Forward Center Cell Bottom Cover (Tall Standpipe) Assembly - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure

1. Disconnect electrical power.

2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).

3. Remove the lower fuselage access door below the tall standpipe assembly.

4. Remove the screws (6 and 17, Figure 1), washers (5 and 18), hose assemblies (7 and 16), and packings (8 and 15). Discard the packings.

5. Remove the screws (13), washers (14), fitting (4), and packing (3). Discard the packing.

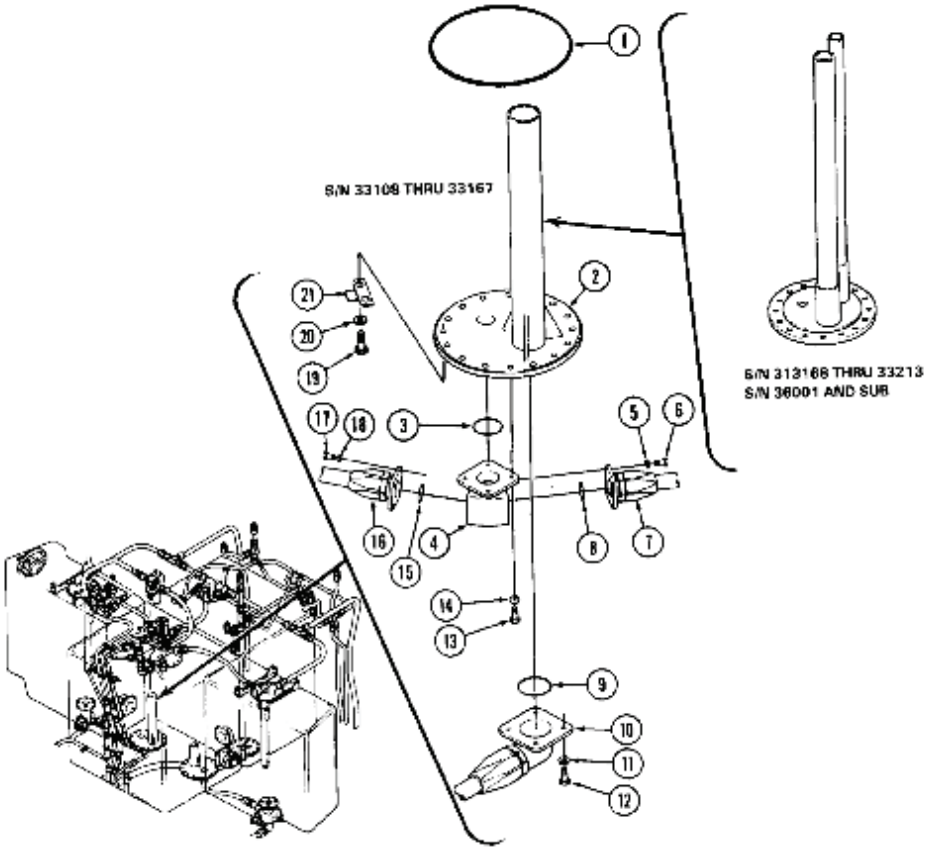
6. Remove the screws (12), washers (11), hose assembly (10), and packing (9). Discard the packing.

7. Remove the bolts (19), washers (20), retainers (21), cover (2), and packing (1). Discard the packing.

8. Install the protective cover over the hose assembly opening.

9. Install the protective cover over the cell opening.

10. Install caps or plugs (C-428) on the hose assemblies (7, 10, and 16) open end fittings.



1. Packing
2. Cover (long stand pipe)
3. Packing
4. Fitting
5. Washer
6. Screw
7. Hose assembly
8. Packing
9. Packing
10. Hose assembly
11. Washer
12. Screw
13. Screw
14. Washer
15. Packing
16. Hose assembly
17. Screw
18. Washer
19. Bolt
20. Washer
21. Retainer

ICN-412-A-281000-B-97499-00080-A-001-01

Figure 1. Upper Forward Center Cell Bottom Cover (Tall Standpipe) Assembly (Sheet 1 of 1)

Upper Forward Center Cell Bottom Cover (Tall Standpipe) Assembly - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

1. Remove protective cover from cell opening.

2. Remove protective caps and/or plugs from hose assemblies (7, 10, and 16, Figure 1) end fittings.

3. Apply assembly fluid (C-024) to packing (1). Place packing on cover (2).

4. Install cover (2) and packing (1) with bolts (19), washers (20), and retainers (21).

5. Apply assembly fluid (C-024) to packing (3). Place packing on fitting (4).

6. Install fitting (4) and packing (3) with screws (13) and washers (14).

7. Apply assembly fluid (C-024) to packings (8 and 15).

8. Install hose assemblies (7 and 16) and packings (8 and 15) with screws (6 and 17) and washers (5 and 18).

9. Apply assembly fluid (C-024) to packing (9).

10. Install hose assembly (10) with packing (9), screws (12), and washers (11).

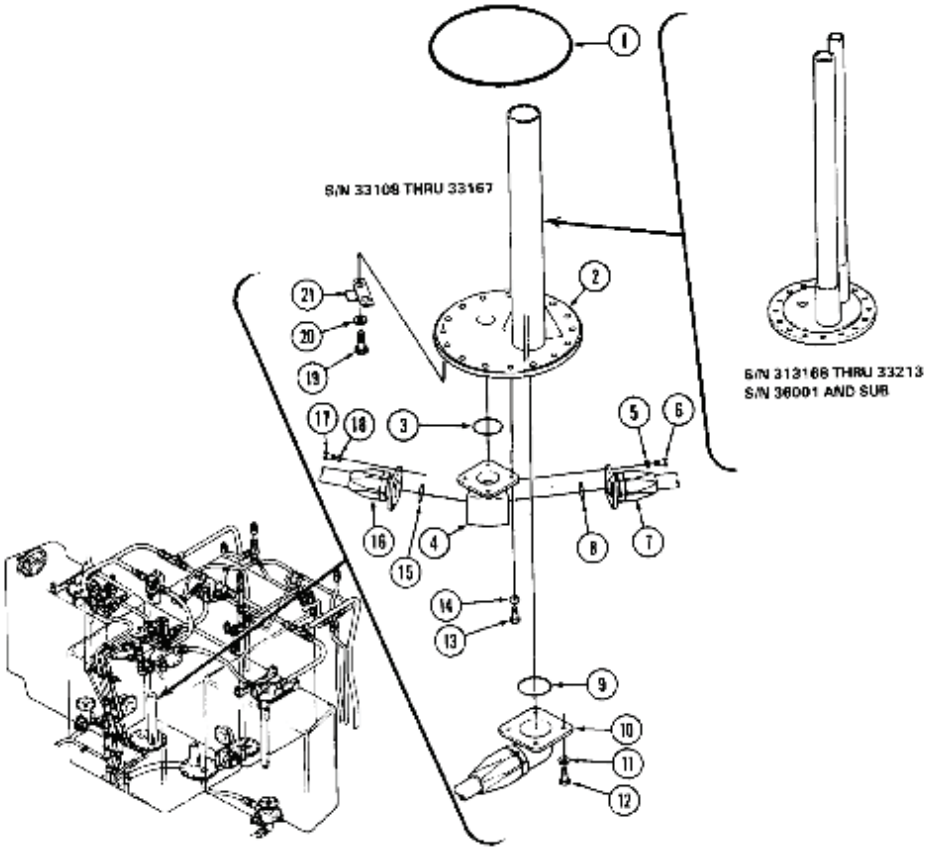
11. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).

12. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).

13. Perform fuel system operational check.

14. Check area for leaks.

15. Install lower fuselage access door.



1. Packing
2. Cover (long stand pipe)
3. Packing
4. Fitting
5. Washer
6. Screw
7. Hose assembly
8. Packing
9. Packing
10. Hose assembly
11. Washer
12. Screw
13. Screw
14. Washer
15. Packing
16. Hose assembly
17. Screw
18. Washer
19. Bolt
20. Washer
21. Retainer

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Figure 1. Upper Forward Center Cell Bottom Cover (Tall Standpipe) Assembly (Sheet 1 of 1)

Upper Forward Center Cell Interconnect Fittings - Description

NOTE

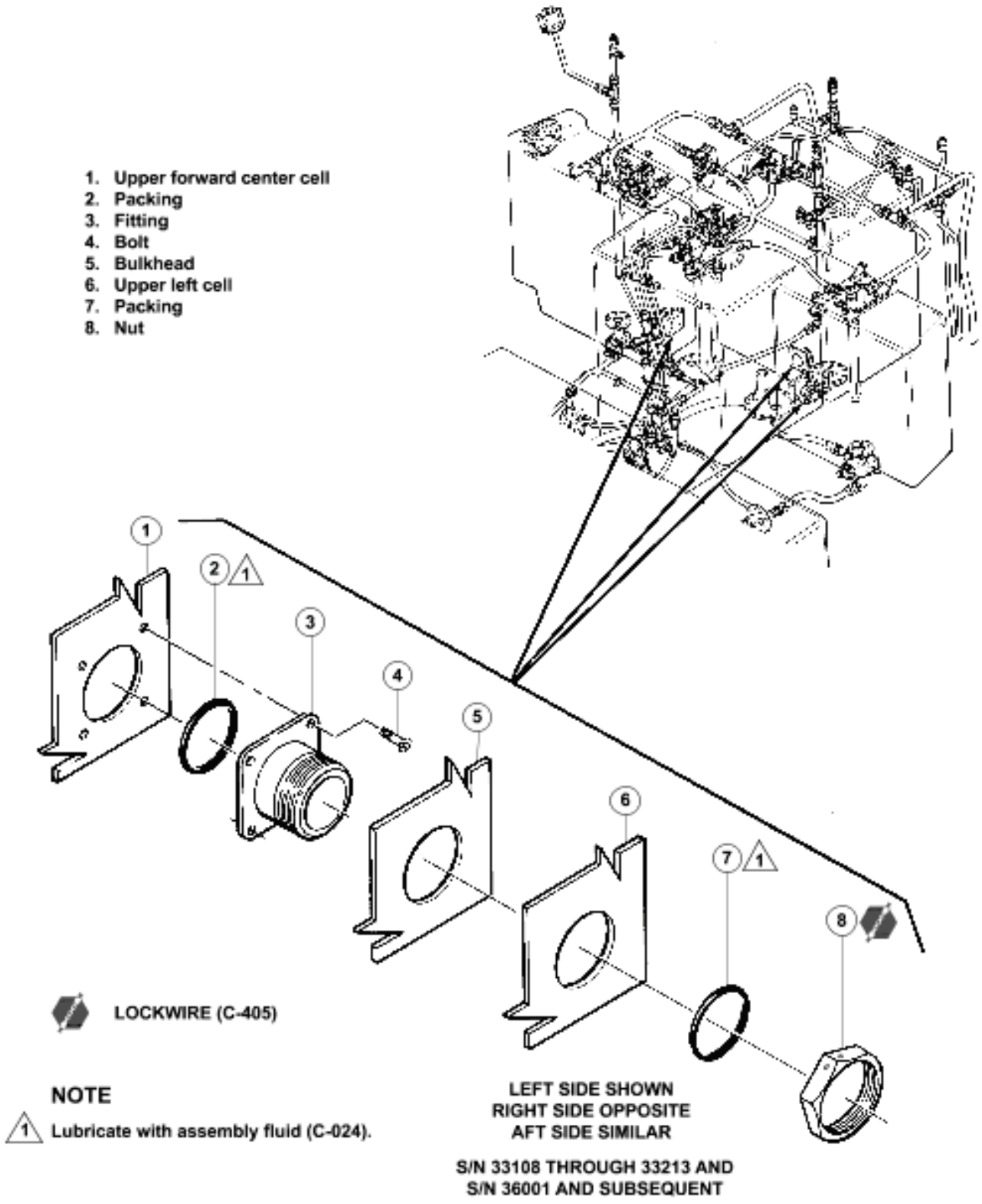
Maintenance procedures are the same for the three upper forward center cell interconnect fittings.

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Upper Forward Center Cell Interconnect Fittings - Removal

Procedure

1. Remove upper right cell top cover assembly (412-A-28-10-00-76A-520A-A / 00574).
2. Remove upper left cell top cover assembly (412-A-28-10-00-76A-520A-A / 00574).
3. Remove upper aft center cell top cover assembly (412-A-28-10-00-73A-520A-A / 00563).
4. Remove lockwire, nut (8, Figure 1), and packing (7). Discard packing.
5. Partially collapse upper forward center cell (1) to gain access to fitting (3).
6. Remove bolts (4), fitting (3), and packing (2). Discard packing.
7. Install protective covers over cell openings.



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Figure 1. Upper Forward Center Cell Interconnect Fitting (Sheet 1 of 1)

Upper Forward Center Cell Interconnect Fittings - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	
Lockwire	C-405	AR	

Procedure

1. Remove protective covers from cell openings.

2. Apply assembly fluid (C-024) to packing (2, Figure 1). Place packing between upper forward center cell (1) and fitting (3).

3. Install bolts (4). Tighten bolts.

4. Install fitting (3) through bulkhead (5) and upper left cell (6).

5. Apply assembly fluid (C-024) to packing (7). Place packing on fitting (3).

NOTE

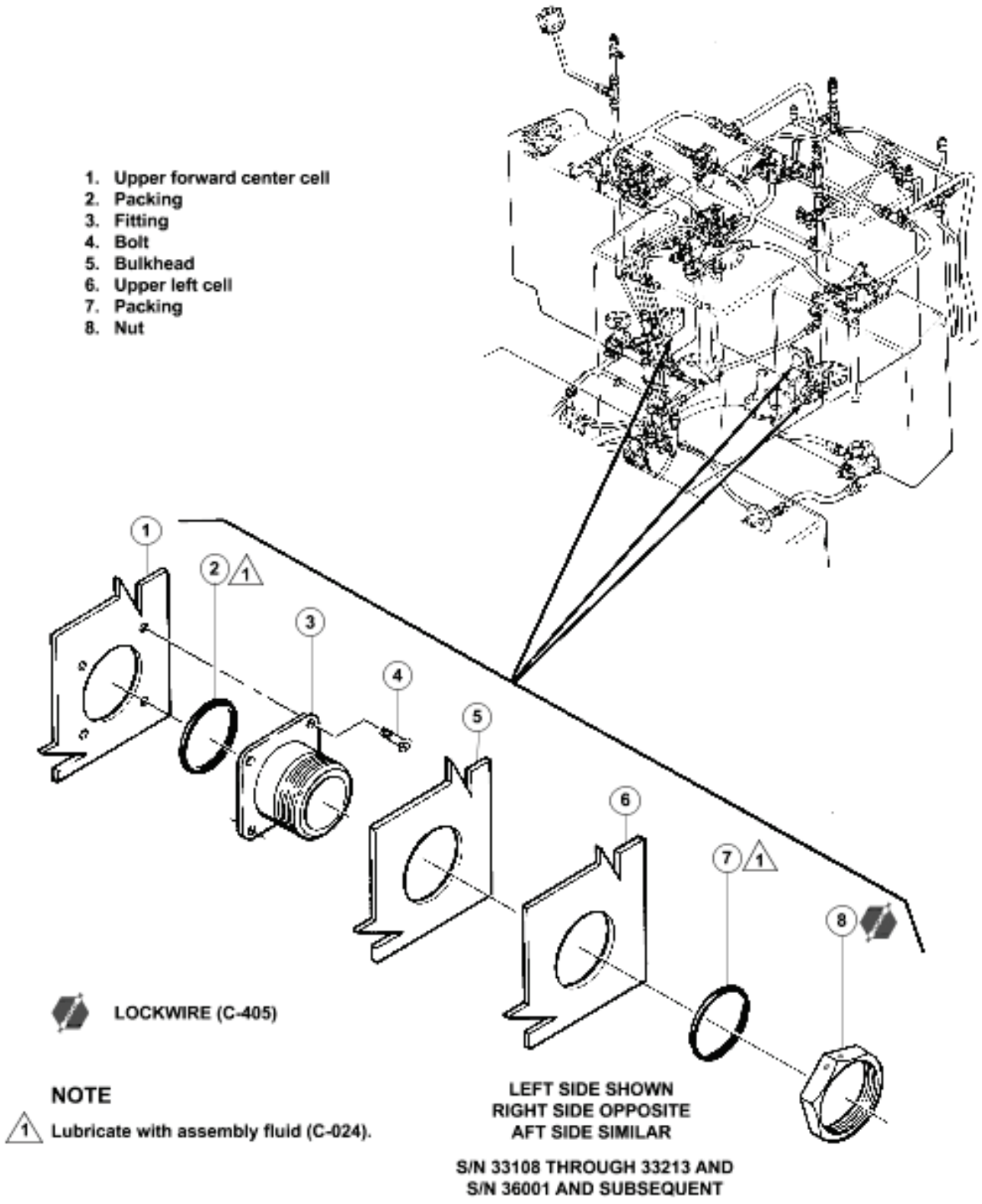
Ensure nut (8) contacts packing (7) before contacting upper left cell (6). To prevent deforming of packing between nut and upper left cell, do not apply excessive torque to nut. Nut is sufficiently tight when there is a 0.020 to 0.030 inch (0.508 to 0.762 mm) gap between nut and fuel cell.

6. Install nut (8) and secure with lockwire (C-405).

7. Install upper right cell top cover assembly (412-A-28-10-00-76A-720A-A / 00575).

8. Install upper left cell top cover assembly (412-A-28-10-00-76A-720A-A / 00575).

9. Install upper aft cell top cover assembly (412-A-28-10-00-73A-720A-A / 00564).



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Figure 1. Upper Forward Center Cell Interconnect Fitting (Sheet 1 of 1)

Upper Forward Center Cell Top Cover Assembly - Description

An upper forward center cell top cover assembly is installed on top port of upper forward center cell. Five fittings are mounted on top cover assembly which provide attachment points for connection of aft vent system hoses.

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Upper Forward Center Cell Top Cover Assembly - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure

1. Disconnect electrical power.

2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).

3. Remove access door above upper forward center cell top cover assembly.

4. Remove screws (3, Figure 1) and washers (2).

5. Remove bolts (4) and washers (5), securing support assembly (1) to cover assembly (20).

6. Disconnect hose assemblies (7, 8, 11, 21, and 24).

7. Remove remaining bolts (4) and washers (5). Lift cover assembly (20) and remove packing (19). Discard packing.

NOTE

SN: 36248-Subsequent:
Washer (13) is not used.

8. To remove elbow (6), remove nut (14), washer (13), and packing (12). Discard packing.

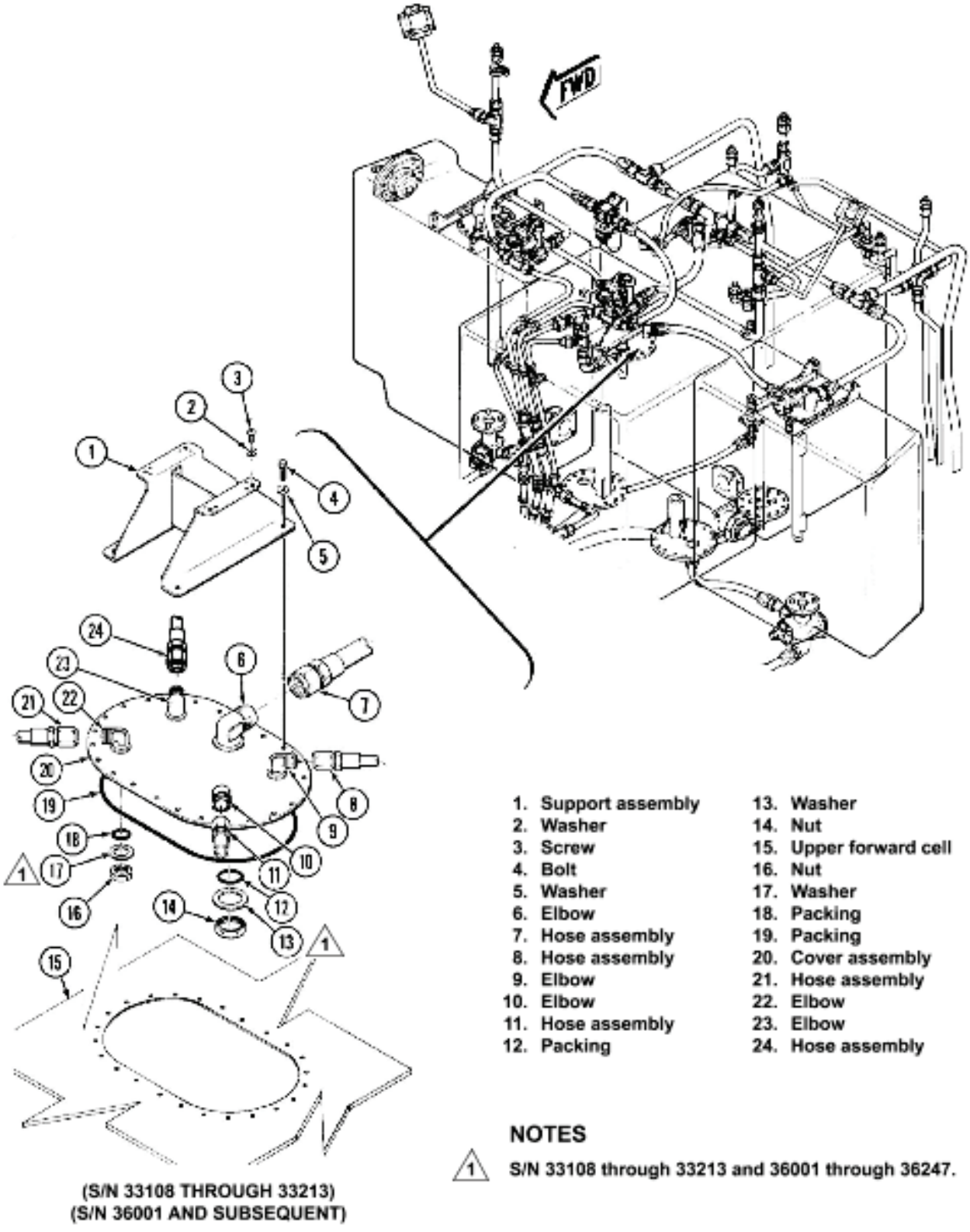
NOTE

SN: 36248-Subsequent:
Washer (17) is not used.

9. To remove elbows (9, 10, 22, and 23), remove nuts (16), washers (17), and packing (18). Discard packing.

10. Install protective cover over cell opening and cap or cover hose assembly openings.

11. Install caps or plugs (C-428) on hose assemblies (7, 8, 11, 21, and 23) open end fittings.



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Figure 1. Upper Forward Center Cell Top Cover Assembly (Sheet 1 of 1)

Upper Forward Center Cell Top Cover Assembly - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

1. Remove protective cover, caps, and/or plugs from cell opening and hose assemblies and fittings.

1.1. Lubricate packing (18, Figure 1) with assembly fluid (C-024).

2. Install elbows (9, 10, 22, and 23) through cover assembly (20).

NOTE

SN: 36248-Subsequent:
Washer (17) is not used.

3. Install packings (18) on elbows (9, 10, 22, and 23) with washers (17) and nuts (16).

3.1. Lubricate packing (12) with assembly fluid (C-024).

4. Install elbow (6) through cover assembly (20).

NOTE

SN: 36248-Subsequent:
Washer (13) is not used.

5. Install packing (12) on elbow (6) with washer (13) and nut (14).

6. Install support assembly (1) on cover assembly (20) with bolts (4) and washers (5).

7. Apply assembly fluid (C-024) to packing (19). Place packing in position on upper forward cell (15).

8. Install remaining bolts (4) and washers (5).

9. Connect hose assemblies (7, 8, 11, 21 and 24). Tighten the hose assemblies end fittings.

10. Install screws (3) and washers (2).

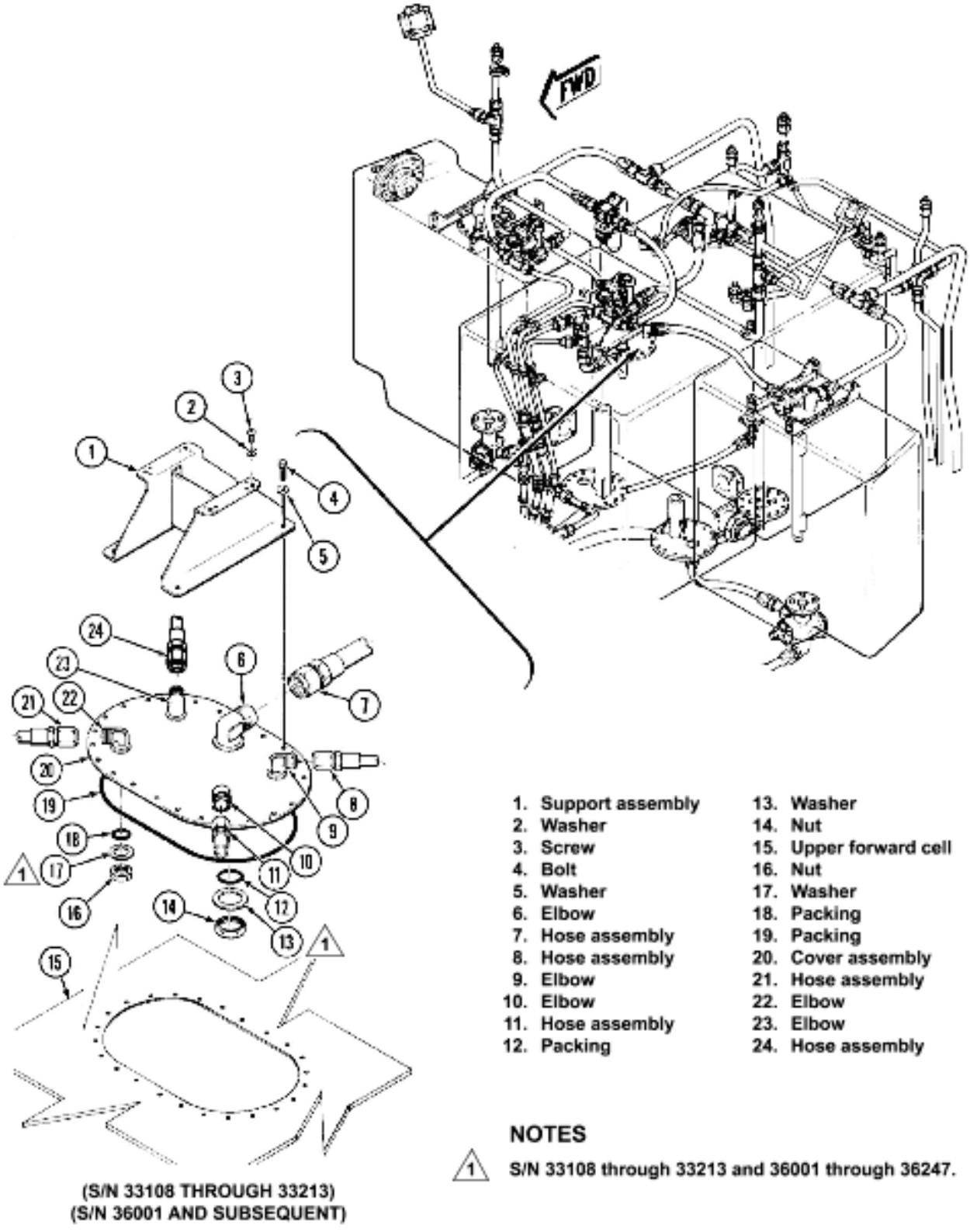
11. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).

12. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).

13. Perform fuel system operational check.

14. Check area for leaks.

15. Install access door over upper forward center cell top cover assembly.



1. Support assembly
2. Washer
3. Screw
4. Bolt
5. Washer
6. Elbow
7. Hose assembly
8. Hose assembly
9. Elbow
10. Elbow
11. Hose assembly
12. Packing
13. Washer
14. Nut
15. Upper forward cell
16. Nut
17. Washer
18. Packing
19. Packing
20. Cover assembly
21. Hose assembly
22. Elbow
23. Elbow
24. Hose assembly

NOTES

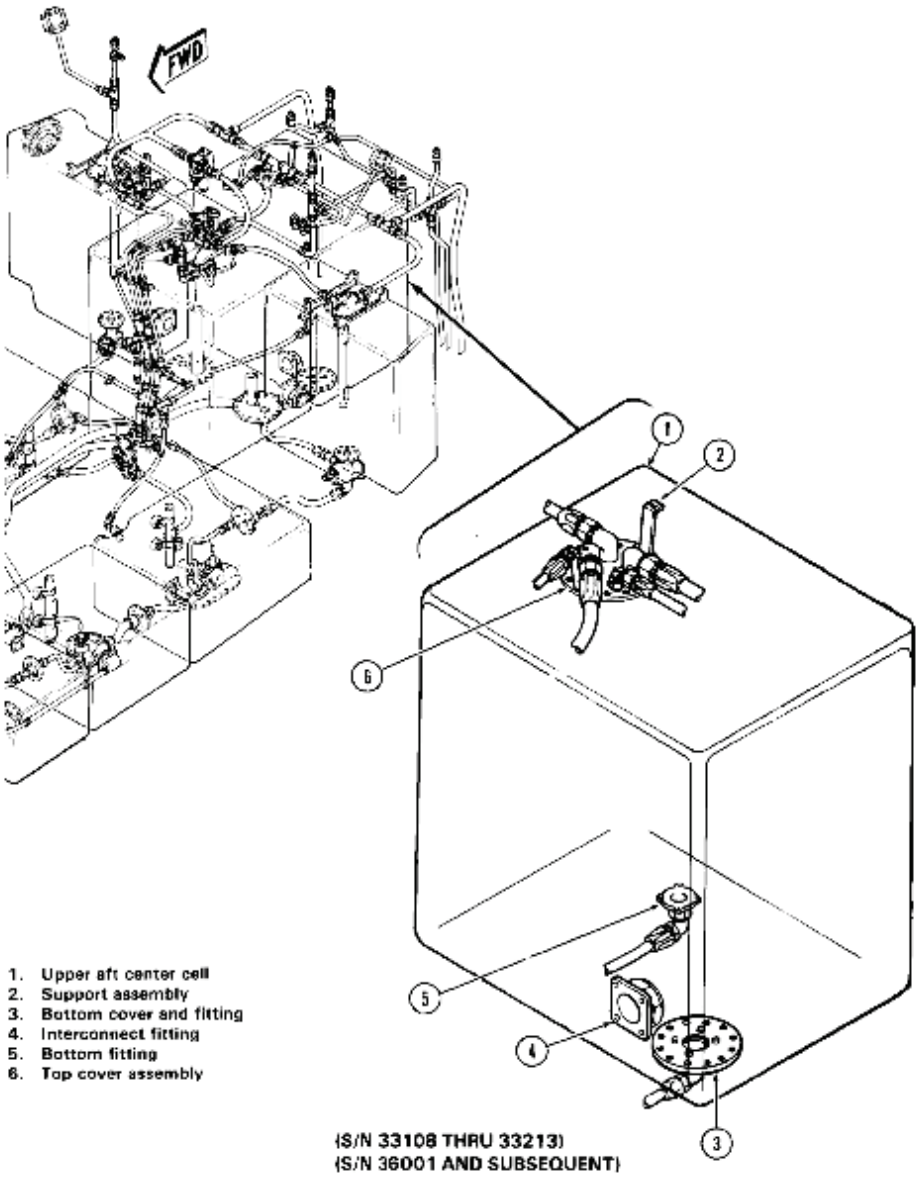
1. S/N 33108 through 33213 and 36001 through 36247.

ICN-412-A-281000-B-97499-00081-A-001-01

Figure 1. Upper Forward Center Cell Top Cover Assembly (Sheet 1 of 1)

Upper Aft Center Cell - Description

The upper aft center cell (Figure 1) is located above WL 22, between the two main longitudinal beams (at left BL 14 and right BL 14), and between FS 178 and 200. The upper aft center cell has a vent port on top and interconnect ports on bottom, left, right, and forward sides.



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Figure 1. Upper Aft Center Cell (Sheet 1 of 1)

Upper Aft Center Cell - Removal

Preliminary requirements

Consumables, materials and expendables

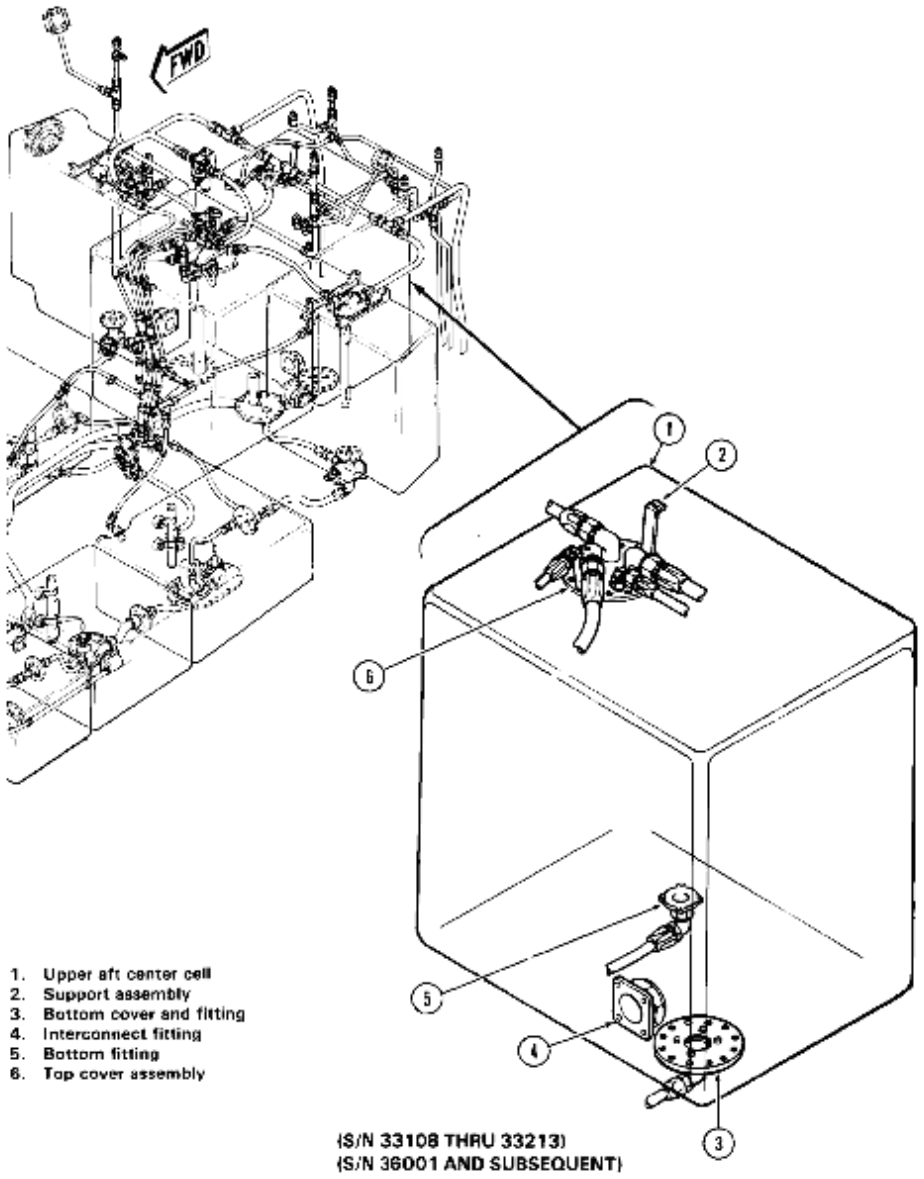
Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure



DO NOT HANDLE FUEL CELL WHEN AMBIENT TEMPERATURE IS BELOW 65°F (18°C).

1. Disconnect electrical power.
2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).
3. Remove large access door on underside of helicopter below upper aft center cell.
4. Remove access panel in engine compartment floor above cell.
5. Remove bottom cover and fitting (412-A-28-10-00-71A-520A-A / 00557).
6. Remove bottom fitting (5, Figure 1) (412-A-28-10-00-72A-520A-A / 00560).
7. Remove support assembly (2) and top cover assembly (6) (412-A-28-10-00-73A-520A-A / 00563).
8. Working through top vent port of cell, disconnect interconnect fitting (4) (412-A-28-10-00-68A-520A-A / 00546).
9. Partially collapse cell. Remove cell through large access opening in bottom of cell cavity and bottom of helicopter.
10. Install protective cover over cell openings and cap or plug open fuel lines.
11. Install caps or plugs (C-428) on hose assemblies open end fittings.



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Figure 1. Upper Aft Center Cell (Sheet 1 of 1)

Upper Aft Center Cell - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Soap	100066	AR	Zodiac Aerospace (www.zodiacaerospace.com)
Cleaning Solution	601199	AR	Zodiac Aerospace (www.zodiacaerospace.com)
Clean cloth	C-516	AR	

Safety conditions



OBEY ALL THE SAFETY PRECAUTIONS WHEN YOU DO MAINTENANCE ON FUEL SYSTEM EQUIPMENT.



OBEY ALL THE FUEL CELL HANDLING PRECAUTIONS DURING CLEANING OR DAMAGE TO THE CELL MAY OCCUR.



DO NOT ALLOW SOLVENTS OR COMPOUNDS TO MAKE CONTACT WITH PARTS MADE OF ELASTOMERIC MATERIAL OR DAMAGE WILL OCCUR.

Procedure

NOTE

Any evidence of fuel leakage must be identified before cleaning.

NOTE

Refer to the document "FUEL-F6101-SPM20-00" (Zodiac Aerospace), for additional information on the cleaning of the fuel cells.

1. Clean the internal surface of the fuel cell with a clean cloth (C-516) moistened with a cleaning solution (601199). Dry the liner with low-pressure filtered compressed air for 2 to 4 hours, at approximately 68°F (20°C)

2. Clean small areas on the external surface of the fuel cell with a clean cloth (C-516) moistened with a cleaning solution (601199). Dry with a clean cloth (C-516).

3. Clean an extensive area on the external surface of the fuel cell with a solution of soap (100066) and hot water. Rinse with water and dry with a clean cloth (C-516).

4. Dry clean the fittings of the fuel cell with filtered compressed air and a clean cloth (C-516) to remove any blockage or obstruction.

5. Clean fungus growth found on the fuel cell or on the fuel cell cavity per the instructions contained in the BHT-ALL-SPM, Chapter 5, Standard Practices Manual.

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Upper Aft Center Cell - Inspection

Procedure



DO NOT HANDLE FUEL CELL WHEN AMBIENT TEMPERATURE IS BELOW 65°F (18°C).

1. Inspect cell for cracks and scuffing. Cracks and severe scuffing are not acceptable.

2. Inspect threaded inserts for damage. Visually detectable damage to threads is not acceptable.

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Upper Aft Center Cell - Installation

Preliminary requirements

Consumables, materials and expendables

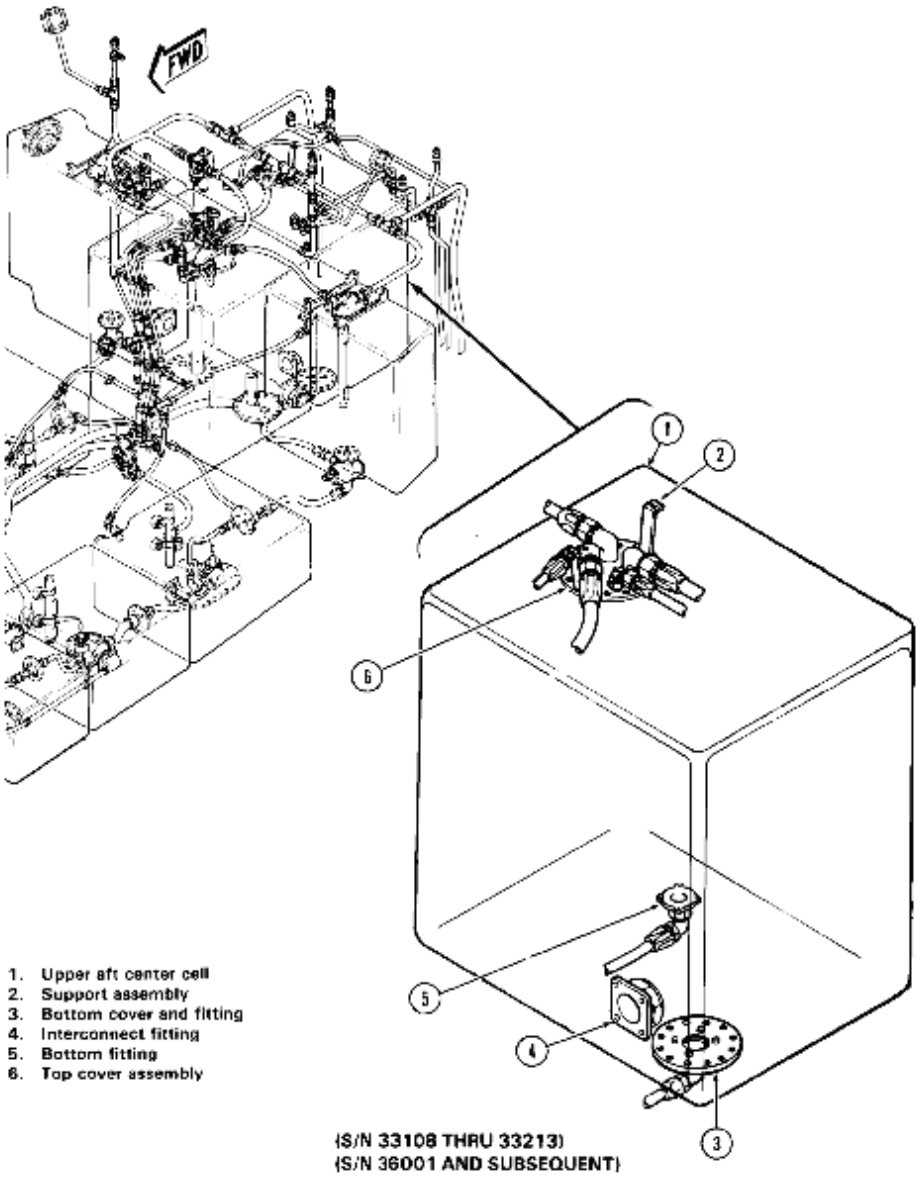
Nomenclature	Identification No.	Qty	Remarks
Talcum powder	C-408	AR	

Procedure



DO NOT HANDLE FUEL CELL WHEN AMBIENT TEMPERATURE IS BELOW 65°F (18°C).

1. Inspect cell cavity for cleanliness and smoothness of interior surfaces.
2. Apply talcum powder (C-408) to interior surfaces of cell cavity.
3. Remove protective covers, caps, and/or plugs.
4. Insert upper aft center cell (1, Figure 1) into cell cavity through large access opening under cell cavity on underside of helicopter. Align cell ports with matching ports and fuselage openings. Align interconnect port to slip over end of upper forward center cell interconnect fitting (4).
5. Working through top vent port, connect interconnect fitting (4) (412-A-28-10-00-68A-720A-A / 00547).
6. Install top cover assembly (6, Figure 1) and support assembly (2) (412-A-28-10-00-69A-720A-A / 00550).
7. Install bottom cover and fitting (3) (412-A-28-10-00-71A-720A-A / 00558).
8. Install bottom fitting (5) (412-A-28-10-00-72A-720A-A / 00561).
9. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).
10. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).
11. Perform fuel system operational check.
12. Check area for leaks.
13. Install access panel and door.



ICN-412-A-281000-B-97499-00082-A-001-01

Figure 1. Upper Aft Center Cell (Sheet 1 of 1)

Upper Aft Center Cell Bottom Cover Assembly and Fitting - Description

A bottom cover and fitting assembly is mounted on the larger bottom port of the upper aft center cell. The fitting provides an attachment point for the outlet hose that connects the upper aft center cell to the left outlet fitting on the bottom of upper forward center cell.

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Upper Aft Center Cell Bottom Cover Assembly and Fitting - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure

1. Disconnect electrical power.

2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).

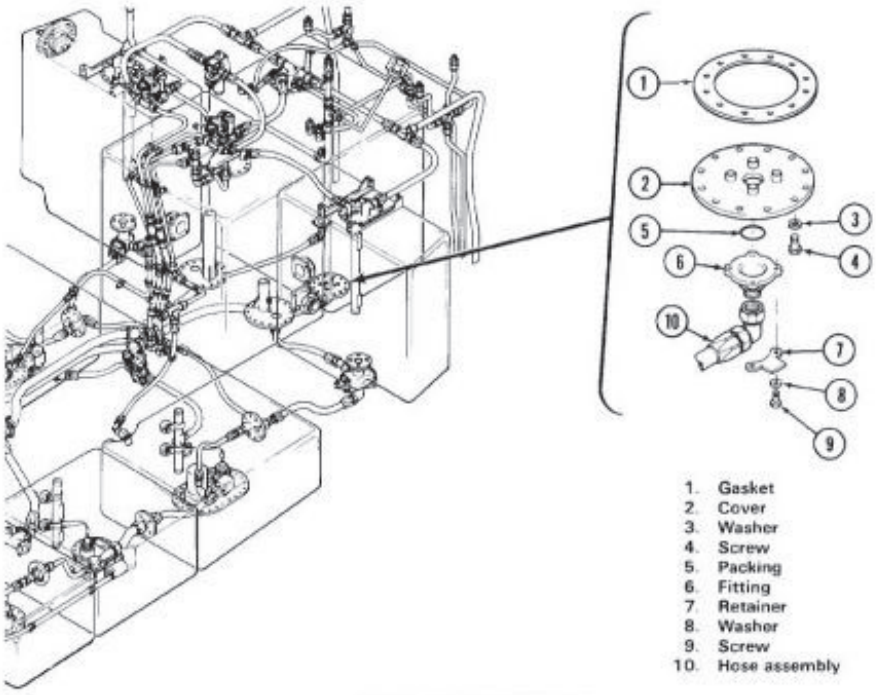
3. Remove access door on underside of helicopter below aft center cell.

4. Disconnect hose assembly (10, Figure 1).

5. Remove screws (9), washers (8), and retainers (7). Remove fitting (6) and packing (5). Discard packing.

6. Remove screws (4) and washers (3). Remove cover (2) and gasket (1). Discard gasket.

7. Install protective cover over cell opening and caps or plugs (C-428) on open fuel lines.



ICN-412-A-281000-B-97499-00083-A-001-01

Figure 1. Upper Aft Center Cell Bottom Cover and Fitting (Sheet 1 of 1)

Upper Aft Center Cell Bottom Cover Assembly and Fitting - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

1. Remove protective covers, caps, and/or plugs.

2. Apply assembly fluid (C-024) to gasket (1, Figure 1).

3. Place gasket (1) on cover (2). Place cover and gasket on flat surface of larger outlet port on bottom of aft center cell.

4. Install screws (4) and washers (3). Tighten screws.

5. Apply assembly fluid (C-024) to packing (5). Place packing on fitting (6).

6. Position fitting (6) on cover (2). Install screws (9), washers (8), and retainers (7). Tighten screws.

7. Connect hose assemblies (10).

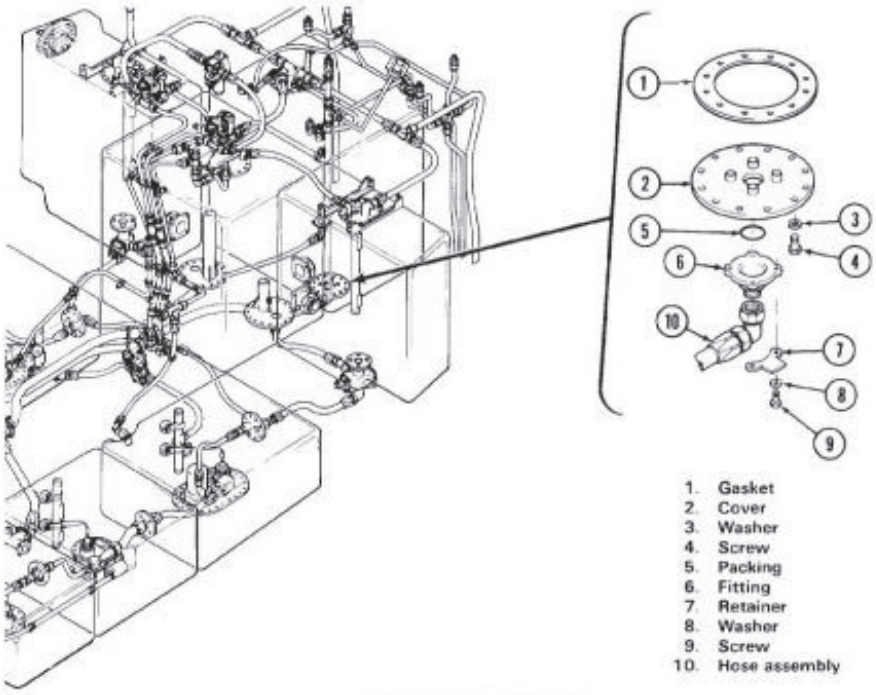
8. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).

9. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).

10. Perform fuel system operational check.

11. Check area for leaks.

12. Install access doors.



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Figure 1. Upper Aft Center Cell Bottom Cover and Fitting (Sheet 1 of 1)

Upper Aft Center Cell Bottom Fitting - Description

A bottom fitting is installed on the smaller bottom port of the upper aft center cell. This fitting provides an attachment point for the hose connecting the upper aft center cell to the right outlet fitting on the bottom of upper forward center cell.

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Upper Aft Center Cell Bottom Fitting - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure

1. Disconnect electrical power.

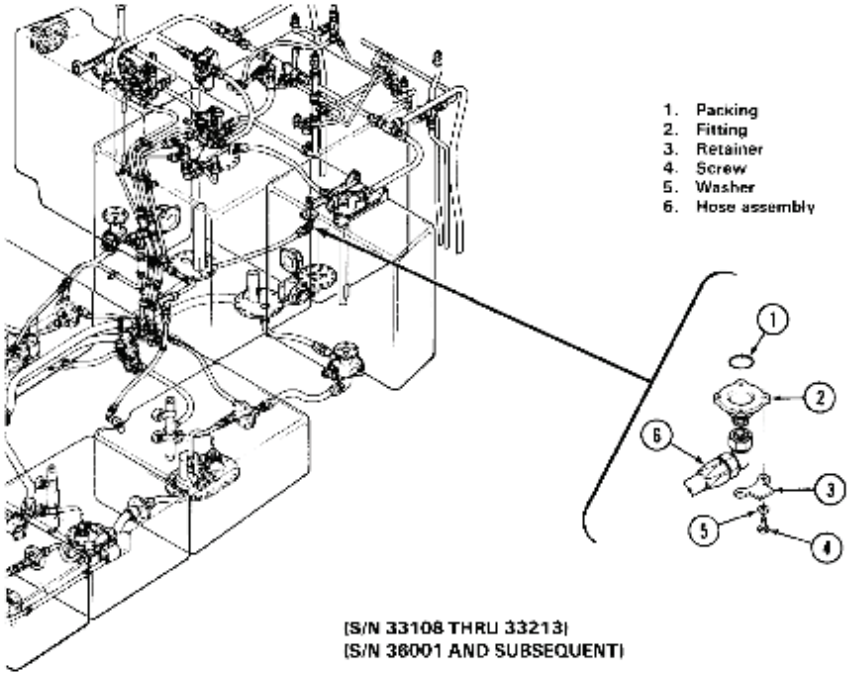
2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).

3. Remove access door on underside of helicopter below aft center cell.

4. Disconnect hose assembly (6, Figure 1).

5. Remove screws (4), washers (5), and retainers (3). Remove fitting (2) and packing (1). Discard packing.

6. Install protective cover over cell opening and caps or plugs (C-428) on open fuel lines.



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Figure 1. Upper Aft Center Cell Bottom Fitting (Sheet 1 of 1)

Upper Aft Center Cell Bottom Fitting - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

1. Remove protective covers, caps, and/or plugs.

2. Apply assembly fluid (C-024) to packing (1, Figure 1). Install packing in groove of fitting (2).

3. Position fitting (2) and packing (1) over outlet port on bottom of cell and install screws (4), washers (5), and retainers (3). Tighten screws.

4. Connect hose assembly (6).

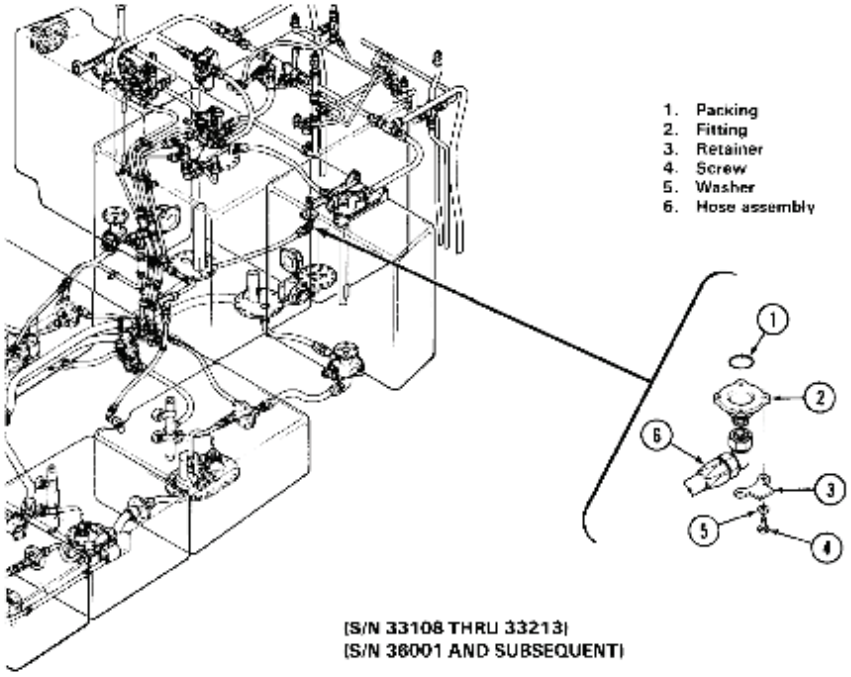
5. Perform fuel system pressure (412-A-28-10-00-00A-362B-A / 00438).

6. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).

7. Perform fuel system operational check.

8. Check area for leaks.

9. Install access door.

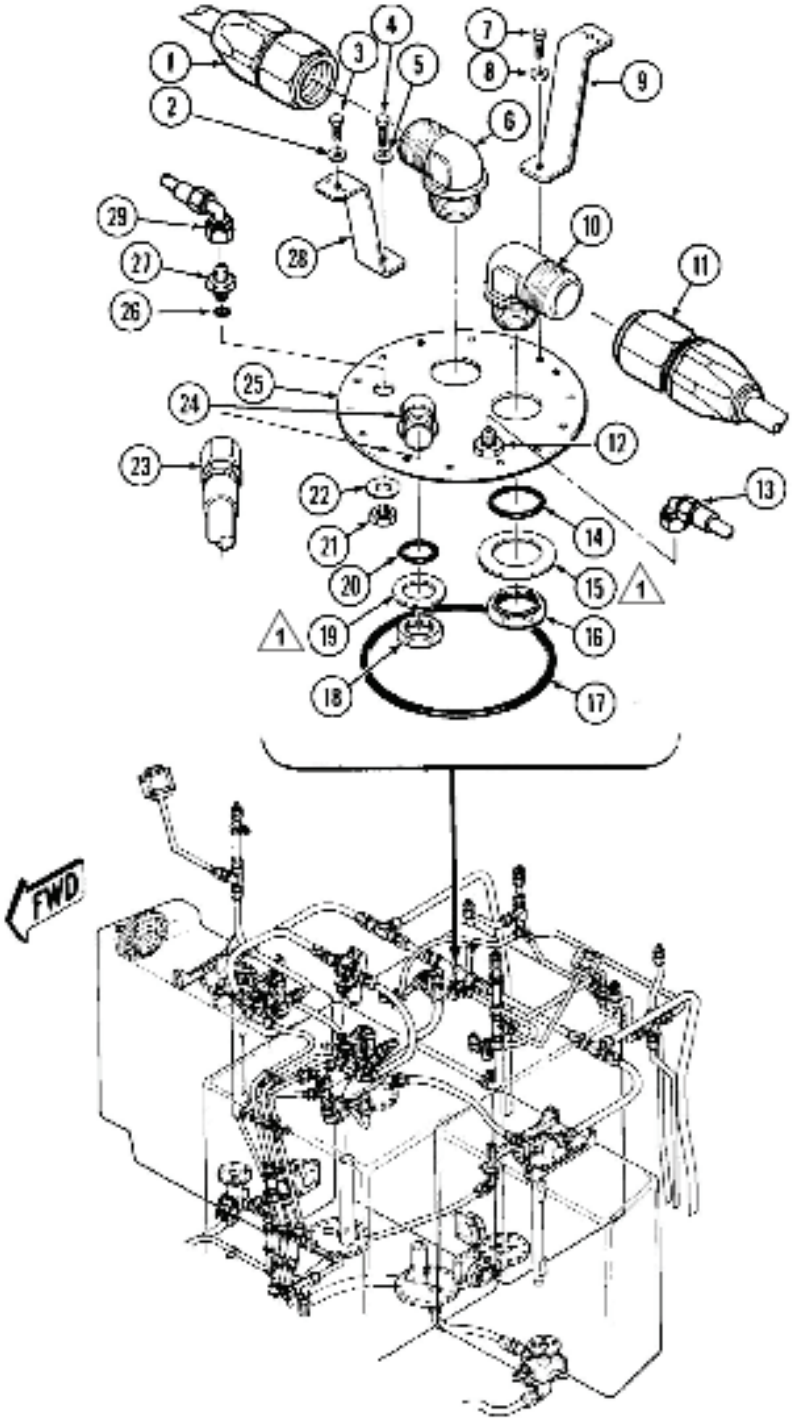


ICN-412-A-281000-B-97499-00084-A-001-01

Figure 1. Upper Aft Center Cell Bottom Fitting (Sheet 1 of 1)

Upper Aft Center Cell Top Cover Assembly - Description

An upper aft center cell top cover assembly (Figure 1) is mounted over the top port of the upper aft center cell. Five fittings are mounted on the top cover assembly which provide attachment points for the connection of aft vent system hoses.



1. Hose assembly
2. Washer
3. Bolt
4. Bolt
5. Washer
6. Elbow
7. Bolt
8. Washer
9. Support
10. Elbow
11. Hose assembly
12. Union
13. Hose assembly
14. Packing
15. Washer
16. Nut
17. Packing
18. Nut
19. Washer
20. Packing
21. Nut
22. Washer
23. Hose assembly
24. Elbow
25. Cover assembly
26. Packing
27. Union
28. Support
29. Hose assembly

NOTE

1 S/N 33108 through 33213 and 36001 through 36247.

ICN-412-A-281000-B-97499-00085-A-001-01

Figure 1. Upper Aft Center Cell Top Cover Assembly (Sheet 1 of 1)

Upper Aft Center Cell Top Cover Assembly - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure

1. Disconnect electrical power.

2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).

3. Remove access panel in engine compartment floor above aft center cell.

4. Remove bolts (3, Figure 1) and washers (2) holding supports (9 and 28) to helicopter structure.

5. Remove bolts (4 and 7) and washers (5 and 8) to remove the supports (9 and 28) from the cover (25).

6. Remove hose assembly (1) from elbow (6).

7. Remove hose assembly (11) from elbow (10).

8. Remove hose assembly (23) from elbow (24).

9. Remove hose assembly (13) from union (12).

10. Remove hose assembly (29) from union (27).

11. Remove the remaining bolts (4 and 7) and washers (5 and 8).

12. Remove cover assembly (25) and packing (17) from cell. Discard packing.

13. Install protective cover over cell opening and caps or plugs (C-428) on all open fuel lines.

NOTE

SN: 36248-Subsequent:
Washer (15) is not used.

14. Remove elbows (6 and 10) from the cover assembly (25) by removing the nuts (16), washers (15), and packings (14). Discard the packings.

NOTE

Packing, washer, and nut for elbows (6 and 10) are shown only once in illustration for clarity.

NOTE

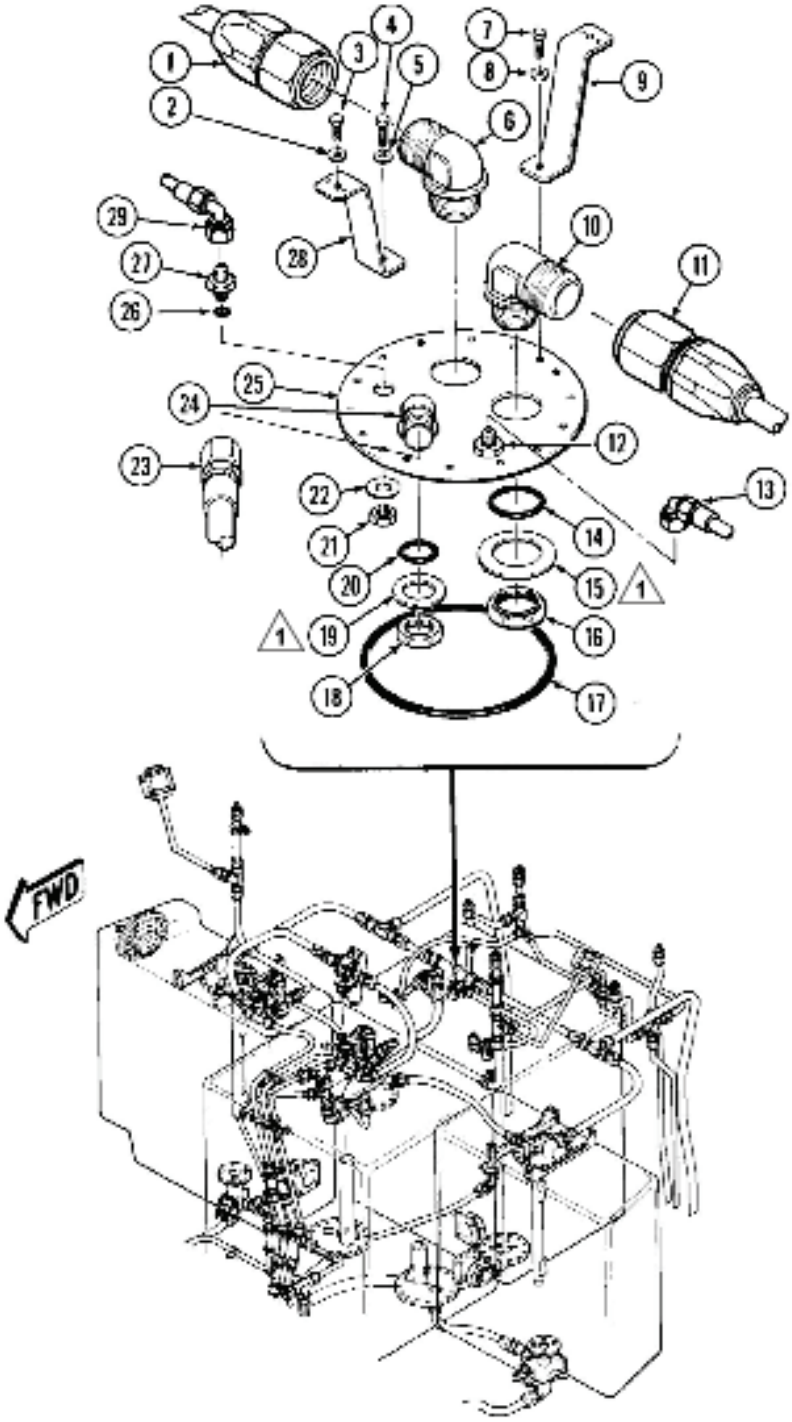
SN: 36248-Subsequent:
Washer (19) is not used.

15. Remove the elbow (24) from the cover assembly (25) by removing nut (18), washer (19), and packing (20). Discard the packing.

NOTE

Packing, washer, and nut for unions (12 and 27) are shown only once in illustration for clarity.

16. Remove the unions (12 and 27) from the cover assembly (25) by removing the nuts (21), washers (22), and packings (26).



1. Hose assembly
2. Washer
3. Bolt
4. Bolt
5. Washer
6. Elbow
7. Bolt
8. Washer
9. Support
10. Elbow
11. Hose assembly
12. Union
13. Hose assembly
14. Packing
15. Washer
16. Nut
17. Packing
18. Nut
19. Washer
20. Packing
21. Nut
22. Washer
23. Hose assembly
24. Elbow
25. Cover assembly
26. Packing
27. Union
28. Support
29. Hose assembly

NOTE

1 S/N 33108 through 33213 and 36001 through 36247.

ICN-412-A-281000-B-97499-00085-A-001-01

Figure 1. Upper Aft Center Cell Top Cover Assembly (Sheet 1 of 1)

Upper Aft Center Cell Top Cover Assembly - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

1. Remove the protective covers from the cell opening.

2. Remove the protective caps and/or plugs from the fuel lines.

3. To install the unions (12 and 27, Figure 1), do the following:

3.1. Apply assembly fluid (C-024) to the packings (26).

3.2. Install the packings (26) on the unions (12 and 27).

3.3. Put the unions (12 and 27) in position on the cover assembly (25) and install the washers (22) and the nuts (21). Tighten the nuts.

4. To install the elbow (24), do the following:

4.1. Put the elbow (24) in position on the cover assembly (25).

4.2. Apply assembly fluid (C-024) to the packing (20), and install the packing on the elbow (24).

NOTE

SN: 36248-Subsequent:

Make sure that the elbow (24) is properly aligned with the hose assembly (23) before tightening the nut (18).

Washer (19) is not used.

4.3. Install the washer (19) and the nut (18) on the elbow (24). Tighten the nut.

5. To install the elbows (6 and 10), do the following:

5.1. Put the elbows (6 and 10) in position on the cover assembly (25).

5.2. Apply assembly fluid (C-024) to the packings (14) and install the packings on the elbows (6 and 10).

NOTE

SN: 36248-Subsequent:

Make sure that the elbow (6) is properly aligned with the hose assembly (1) before tightening the nut (16).

Make sure that the elbow (10) is properly aligned with the hose assembly (11) before tightening the nut (16).

Washer (15) is not used.

5.3. Install the washers (15) and nuts (16) on the elbows (6 and 10). Tighten the nuts.

6. Apply assembly fluid (C-024) to the packing (17).

7. Put the top cover assembly (25) and packing (17) in position on the upper aft center cell.

NOTE

Make sure that the mounting holes for bolts (3) in the supports (9 and 28) are lined up with the mounting holes in the helicopter structure before tightening the bolts (4 and 7).

8. Install the support (9) with the bolt (7) and washer (8). Tighten the bolt.

9. Install the support (28) with the bolt (4) and washer (5). Tighten the bolt.

10. Install the remaining bolts (7) and washers (8) on the cover assembly (25). Tighten the bolts.

11. Connect the hose assembly (11) to the elbow (10). Tighten the end fitting.

12. Connect the hose assembly (13) to the union (12). Tighten the end fitting.

13. Connect the hose assembly (23) to the elbow (24). Tighten the end fitting.

14. Connect the hose assembly (29) to the union (27). Tighten the end fitting.

15. Connect the hose assembly (1) to the elbow (6). Tighten the end fitting.

16. Install the bolts (3) and washers (2) through the helicopter structure and the supports (9 and 28). Tighten the bolts.

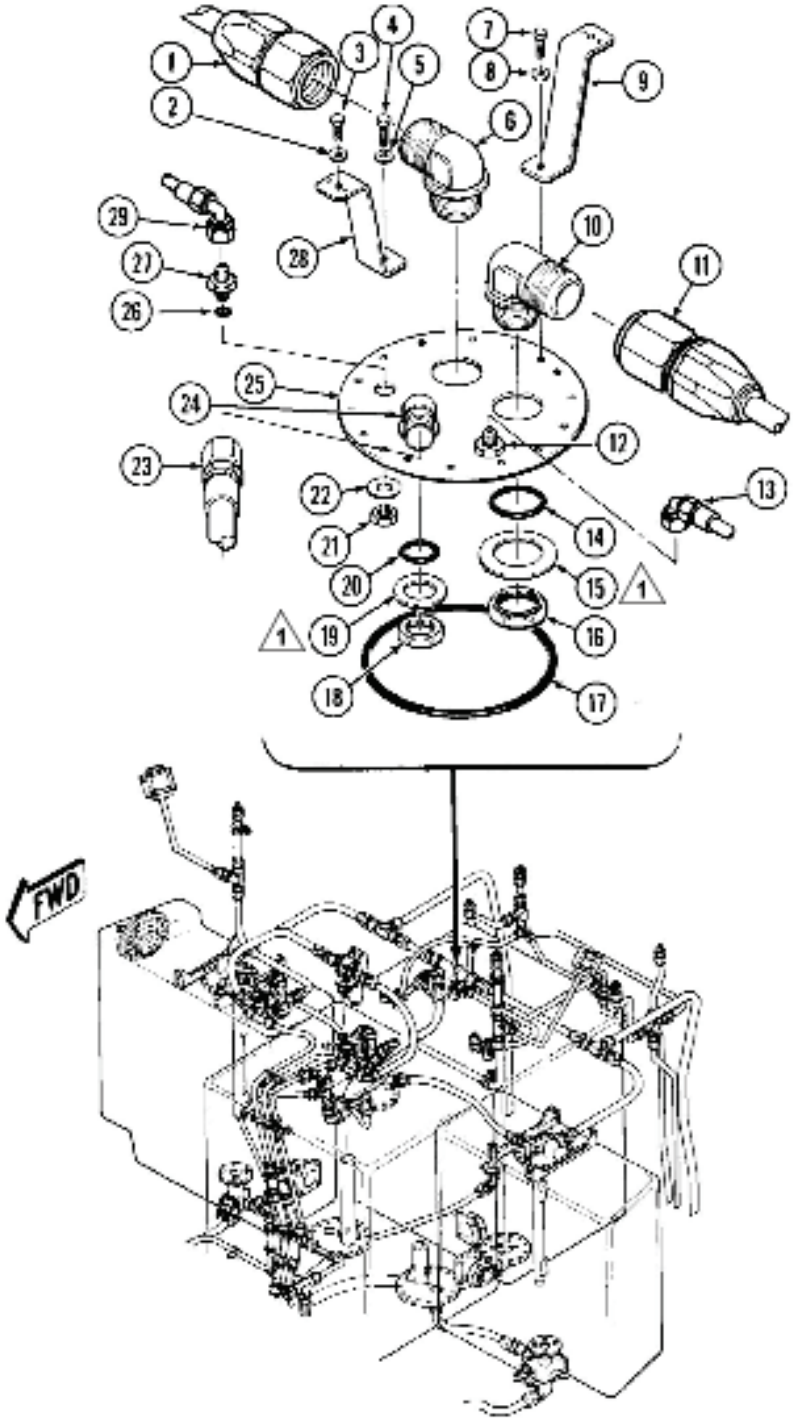
17. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).

18. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).

19. Perform fuel system operational check.

20. Check area for leaks.

21. Install access cover in floor of engine compartment.



1. Hose assembly
2. Washer
3. Bolt
4. Bolt
5. Washer
6. Elbow
7. Bolt
8. Washer
9. Support
10. Elbow
11. Hose assembly
12. Union
13. Hose assembly
14. Packing
15. Washer
16. Nut
17. Packing
18. Nut
19. Washer
20. Packing
21. Nut
22. Washer
23. Hose assembly
24. Elbow
25. Cover assembly
26. Packing
27. Union
28. Support
29. Hose assembly

NOTE

△1 S/N 33108 through 33213 and 36001 through 36247.

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Figure 1. Upper Aft Center Cell Top Cover Assembly (Sheet 1 of 1)

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Upper Left Cell and Upper Right Cell - Description

The upper left and upper right cells are located above WL 22, outboard of the two main longitudinal beams (at left BL 14 and right BL 14), and between FS 166 and 178. The upper left and upper right cells have vent ports on the top and interconnect ports on the bottom and inboard sides. The upper right cell has a port on outboard side for filler cap and adapter assembly.

NOTE

Maintenance procedures for the upper left cell and upper right cell are the same, except for the filler cap and adapter assembly procedures, which are applicable for upper right cell only.

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Upper Left Cell and Upper Right Cell - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure

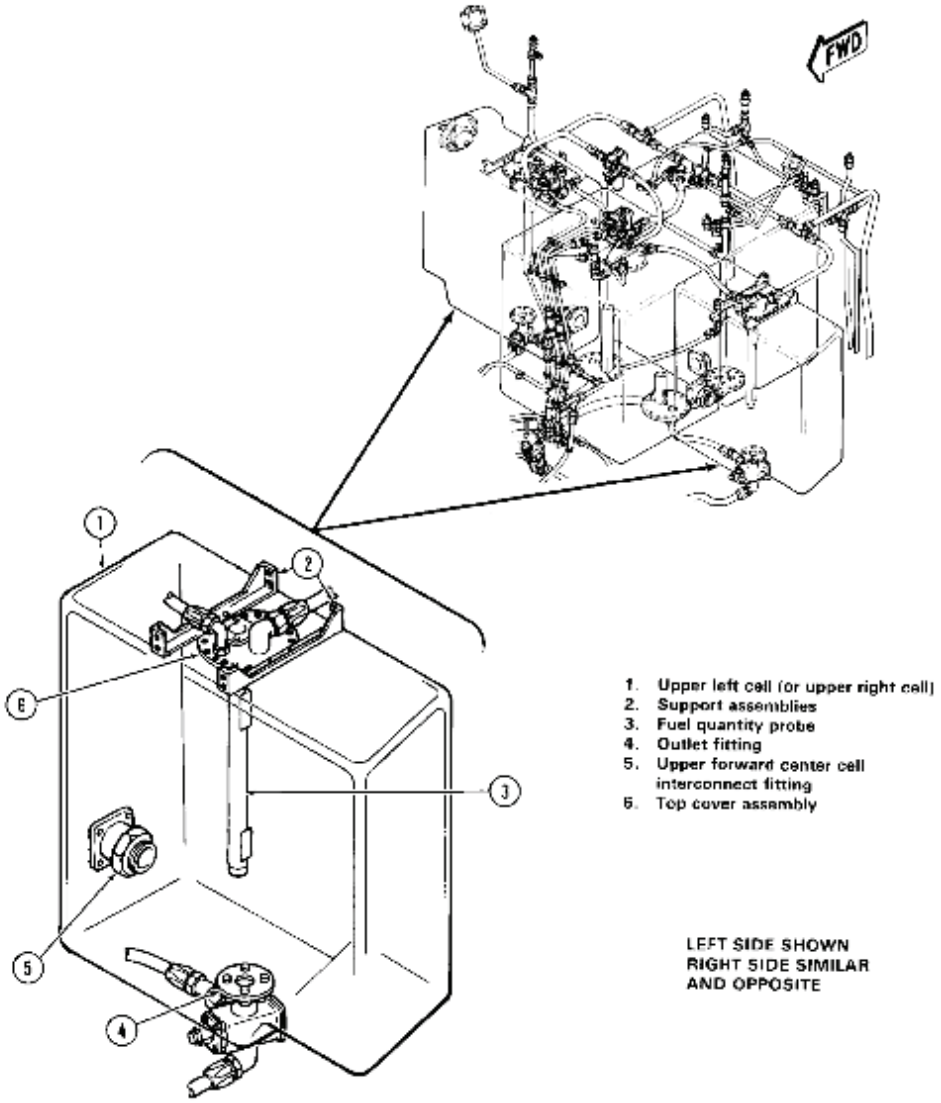


DO NOT HANDLE FUEL CELL WHEN AMBIENT TEMPERATURE IS BELOW 65°F (18°C).

NOTE

The following removal procedures for the upper left cell and upper right cell are the same, except for the filler cap and adapter assembly procedures, which are applicable for upper right cell only.

1. Disconnect electrical power.
2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).
3. Remove access door on underside of helicopter below cell (1, Figure 1).
4. Remove engine compartment floor panel above cell (1).
5. Remove outlet fitting (4) (412-A-28-10-00-75A-520A-A / 00571).
6. Remove support assemblies (2) and top cover assembly (6) (412-A-28-10-00-76A-520A-A / 00574).
7. If removing upper right cell (1), remove filler cap and adapter assembly (412-A-28-10-00-77A-520A-A / 00577 or 412-A-28-10-00-78A-520A-A / 00580).
8. Working through top vent port of cell, disconnect the interconnect fitting (5) (412-A-28-10-00-68A-520A-A / 00546).
9. Partially collapse cell (1). Lift cell from cell cavity.
10. Install protective cover over cell openings and caps or plugs (C-428) open fuel lines.



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Figure 1. Upper Left Cell and Upper Right Cell (Sheet 1 of 1)

Upper Left Cell and Upper Right Cell - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
100066	Commercial	AR	Zodiac Aerospace (www.zodiacaerospace.com)
Cleaning Solution	601199	AR	Zodiac Aerospace (www.zodiacaerospace.com)
Clean cloth	C-516	AR	

Procedure



OBEY ALL THE SAFETY PRECAUTIONS WHEN YOU DO MAINTENANCE ON FUEL SYSTEM EQUIPMENT.



OBEY ALL THE FUEL CELL HANDLING PRECAUTIONS DURING CLEANING OR DAMAGE TO THE CELL MAY OCCUR.



DO NOT ALLOW SOLVENTS OR COMPOUNDS TO MAKE CONTACT WITH PARTS MADE OF ELASTOMERIC MATERIAL OR DAMAGE WILL OCCUR.

NOTE

The following cleaning procedures for the upper left cell and upper right cell are the same, except for the filler cap and adapter assembly procedures, which are applicable for upper right cell only.

NOTE

Any evidence of fuel leakage must be identified before cleaning.

NOTE

Refer to the document "FUEL-F6101-SPM20-00" (Zodiac Aerospace), for additional information on the cleaning of the fuel cells.

1. Clean the internal surface of the fuel cell with a clean cloth (C-516) moistened with a cleaning solution (601199). Dry the liner with low-pressure filtered compressed air for 2 to 4 hours, at approximately 68°F (20°C).
2. Clean small areas on the external surface of the fuel cell with a clean cloth (C-516) moistened with a cleaning solution (601199). Dry with a clean cloth (C-516).
3. Clean an extensive area on the external surface of the fuel cell with a solution of 100066 (Commercial) and hot water. Rinse with water and dry with a clean cloth (C-516).
4. Dry clean the fittings of the fuel cell with filtered compressed air and a clean cloth (C-516) to remove any blockage or obstruction.
5. Clean fungus growth found on the fuel cell or on the fuel cell cavity per the instructions contained in the BHT-ALL-SPM, Chapter 5, Standard Practices Manual.

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Upper Left Cell and Upper Right Cell - Inspection

Procedure



DO NOT HANDLE FUEL CELL WHEN AMBIENT TEMPERATURE IS BELOW 65°F (18°C).

NOTE

The following inspection procedures for the upper left cell and upper right cell are the same, except for the filler cap and adapter assembly procedures, which are applicable for upper right cell only.

1. Inspect cell for cracks and scuffing. Cracks and severe scuffing are not acceptable.
2. Inspect threaded inserts for damage.
3. Damage to threads that is visually detectable is not acceptable.

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Upper Left Cell and Upper Right Cell - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Talcum powder	C-408	AR	

Procedure

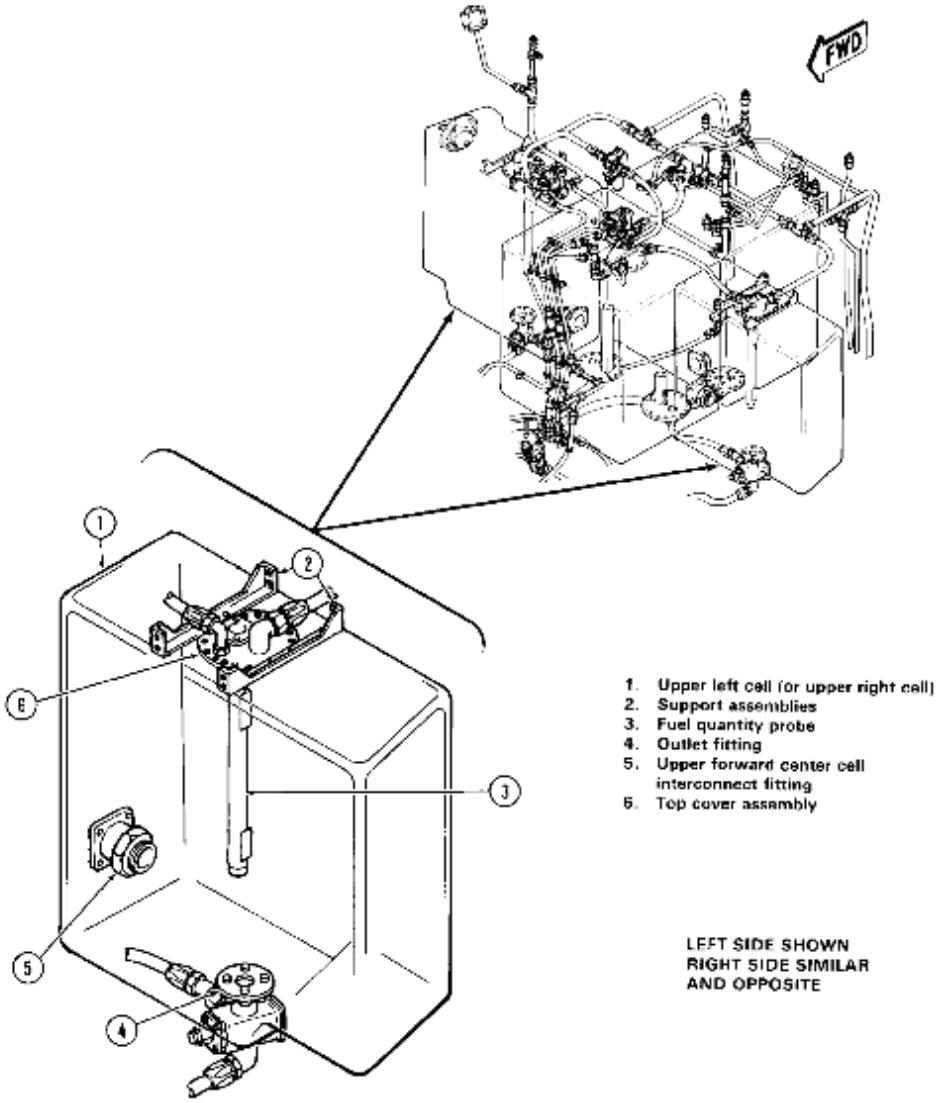


DO NOT HANDLE FUEL CELL WHEN AMBIENT TEMPERATURE IS BELOW 65°F (18°C).

NOTE

The following installation procedures for the upper left cell and upper right cell are the same, except for the filler cap and adapter assembly procedures, which are applicable for upper right cell only.

1. Remove protective covers, caps, and/or plugs.
2. Inspect cell cavity for presence of any foreign objects and/or rough interior surfaces. Remove foreign objects from cell and repair rough surfaces prior to cell installation.
3. Apply talcum powder (C-408) to all interior surfaces of cell cavity.
4. Insert cell (1, Figure 1) into cell cavity through opening above cell in engine compartment floor. Use care to avoid damage to cell. Align cell ports with matching ports and fuselage openings. Align interconnect port to slip over end of upper forward center cell interconnect fitting (5).
5. Working through vent port, connect the upper forward center cell interconnect fitting (5) (412-A-28-10-00-68A-720A-A / 00547).
6. If installing the upper right cell, install the filler cap and adapter assembly (412-A-28-10-00-77A-720A-A / 00578 or 412-A-28-10-00-78A-720A-A / 00581)
7. Install top cover assembly (412-A-28-10-00-73A-720A-A / 00564).
8. Install outlet fitting (4) (412-A-28-10-00-75A-720A-A / 00572).
9. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).
10. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).
11. Perform operational check of fuel quantity indicating system (412-A-96-12-00-11A-273A-A / 02006).
12. Check for leaks.
13. Install access panels and doors.



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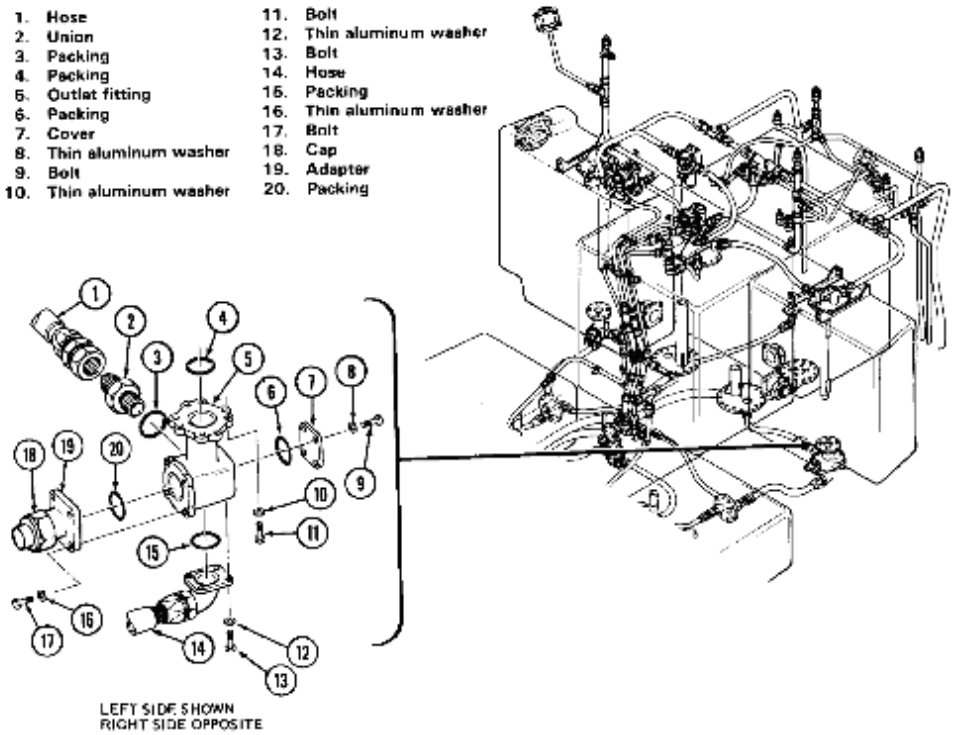
Figure 1. Upper Left Cell and Upper Right Cell (Sheet 1 of 1)

Upper Left Cell and Upper Right Cell Outlet Fittings - Description

NOTE

Maintenance procedures for upper left cell outlet fitting and upper right cell outlet fittings are the same.

An outlet fitting (Figure 1) is installed on bottom port of upper left cell and upper right cell. These fittings provide hose attachment points for transfer of fuel from upper left cell and upper right cell to lower main cell on same side of helicopter.



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Figure 1. Upper Left Cell and Upper Right Cell Outlet Fittings (Sheet 1 of 1)

Upper Left Cell and Upper Right Cell Outlet Fittings - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure

NOTE

The following removal procedures for upper left cell outlet fitting and upper right cell outlet fittings are the same.

1. Disconnect electrical power.

2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).

3. Remove lower access doors.

4. Remove bolts (13, Figure 1), washers (12), hose (14), and packing (15). Discard packing.

5. Disconnect hose (1).

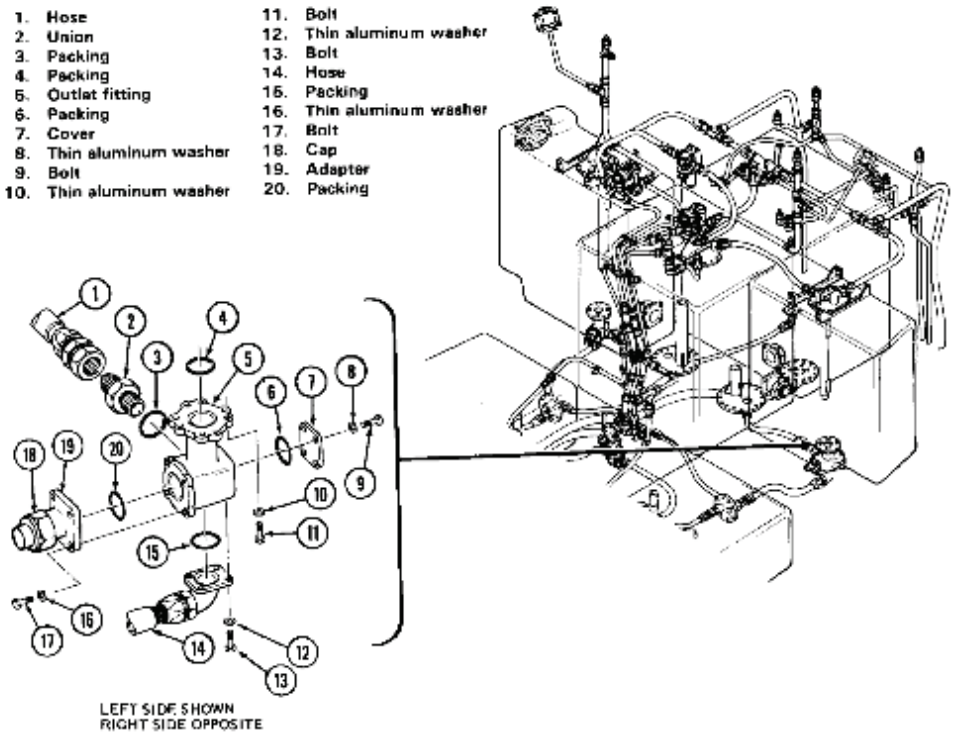
6. Remove union (2) and packing (3). Discard packing.

7. Remove bolts (17), washers (16), adapter (19), and packing (20). Discard packing.

8. Remove bolts (9), thin aluminum washers (8), cover (7), and packing (6). Discard packing.

9. Remove bolts (11), thin aluminum washers (10), outlet fitting (5), and packing (4). Discard packing.

10. Install protective cover over cell opening and caps or plugs (C-428) on open fuel lines.



ICN-412-A-281000-B-97499-00087-A-001-01

Figure 1. Upper Left Cell and Upper Right Cell Outlet Fittings (Sheet 1 of 1)

Upper Left Cell and Upper Right Cell Outlet Fittings - Installation

Preliminary requirements

Consumables, materials and expendables

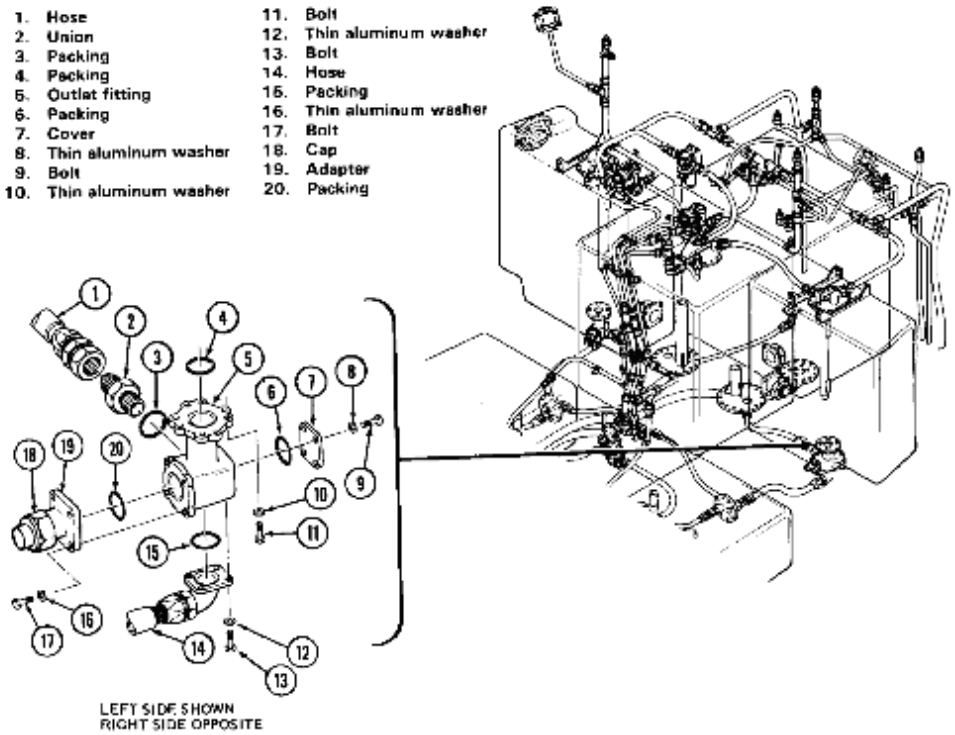
Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

NOTE

The following installation procedures for upper left cell outlet fitting and upper right cell outlet fittings are the same.

1. Remove protective cover, caps, and/or plugs.
2. Apply assembly fluid (C-024) to packings (3, 4, 6, 15, and 20, Figure 1).
3. Position packing (4) and outlet fitting (5) on bottom outlet port of cell.
4. Install bolts (11) and thin aluminum washers (10). Tighten the bolts.
5. Install cover (7), packing (6), thin aluminum washers (8), and bolts (9). Tighten the bolts.
6. Install union (2) and packing (3).
7. Connect hose (1). Tighten the hose end fitting.
8. Install adapter (19) with cap (18), packing (20), thin aluminum washers (16), and bolts (17). Tighten the bolts.
9. Install packing (15), hose (14), thin aluminum washers (12), and bolts (13). Tighten the bolts.
10. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).
11. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).
12. Perform fuel system operational check.
13. Check for leaks.
14. Install access panel in engine compartment floor.



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Figure 1. Upper Left Cell and Upper Right Cell Outlet Fittings (Sheet 1 of 1)

Upper Left Cell and Upper Right Cell Top Cover Assemblies - Description

A top cover assembly is installed on the top port of each upper left and upper right cell. Each top cover assembly includes two fittings which provide attachment points for vent hoses.

NOTE

Maintenance procedures for upper left cell and upper right cell top cover assemblies are the same.

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Upper Left Cell and Upper Right Cell Top Cover Assemblies - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure

NOTE

The following removal procedures for upper left cell and upper right cell top cover assemblies are the same.

1. Disconnect electrical power.

2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).

3. Remove access panel in engine compartment floor above upper left and upper right cells.

4. Disconnect electrical connector from fuel quantity probe (29, Figure 1). Cap electrical connector.

5. Remove fuel quantity probe (29) (412-A-28-10-00-88A-520A-A / 00608).

6. Remove hose assembly (11) from elbow (12).

7. Remove hose assembly (28) from union (27).

8. Remove screws (3 and 8) and washers (2 and 7).

9. Remove bolts (4 and 10) and washers (5 and 9) and remove support assemblies (1 and 6).

10. Remove bolts (16) and washers (17). Remove top cover assembly (18) and packing (19). Discard packing.

NOTE

SN: 36248-Subsequent:

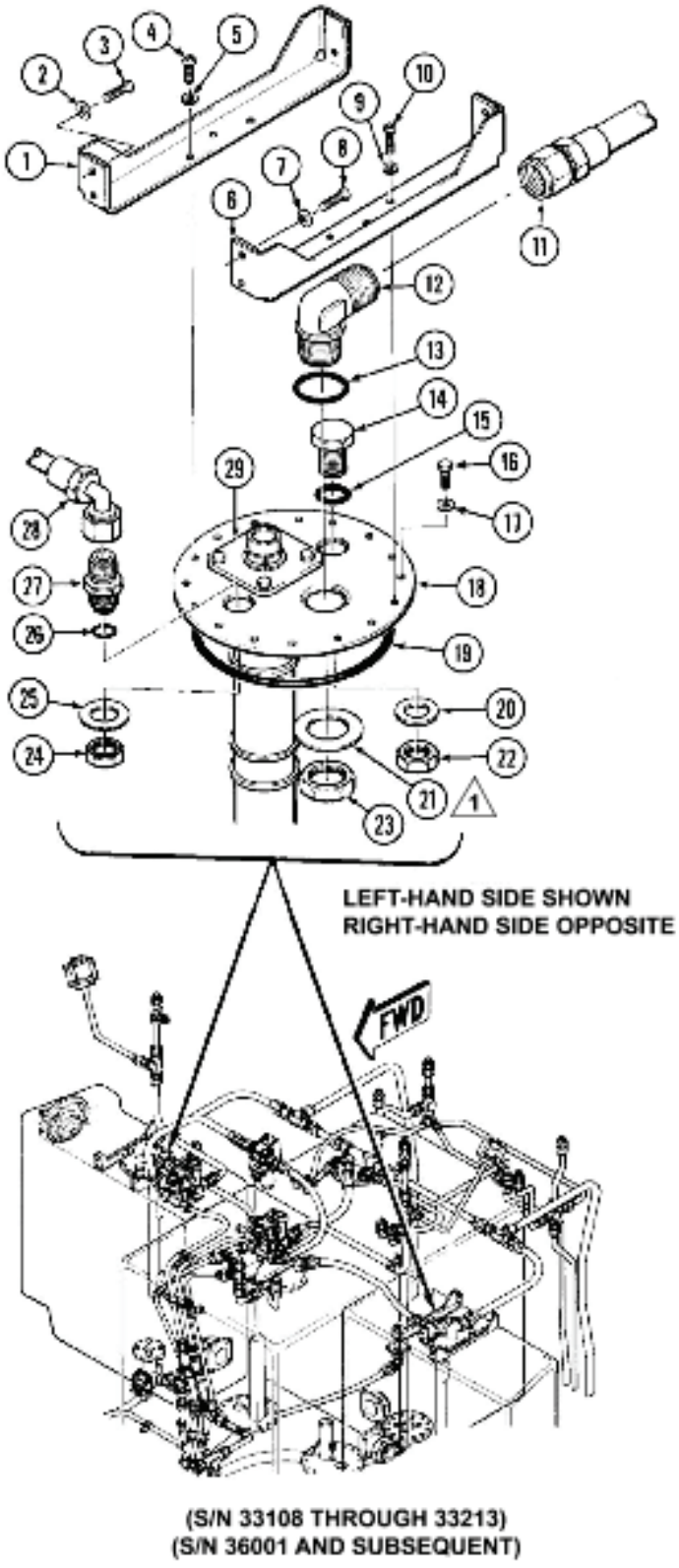
Washer (21) is not used.

11. Remove nut (23) and washer (21). Remove elbow (12) and packing (13) from top cover assembly (18). Discard packing.

12. Remove nut (24) and washer (25). Remove union (27) and packing (26). Discard packing.

13. Remove nut (22) and washer (20). Remove plug (14) and packing (15). Discard packing.

14. Install protective cover over cell opening and caps or plugs (C-428) on all open fuel lines.



1. Support assembly
2. Washer
3. Screw
4. Bolt
5. Washer
6. Support assembly
7. Washer
8. Screw
9. Washer
10. Bolt
11. Hose assembly
12. Elbow
13. Packing
14. Plug
15. Packing
16. Bolt
17. Washer
18. Cover assembly
19. Packing
20. Washer
21. Washer
22. Nut
23. Nut
24. Nut
25. Washer
26. Packing
27. Union
28. Hose assembly
29. Fuel quantity probe installation

NOTE

△1 S/N 33108 through 33213 and 36001 through 36247.

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Figure 1. Upper Left Cell and Upper Right Cell Top Cover Assembly (Sheet 1 of 1)

Upper Left Cell and Upper Right Cell Top Cover Assemblies - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

NOTE

The following installation procedures for upper left cell and upper right cell top cover assemblies are the same.

1. Remove protective covers and caps or plugs.

2. Apply assembly fluid (C-024) to packings (13, 15, 19, and 26, Figure 1).

3. Place packing (15) on plug (14) and install in mounting hole of top cover assembly (18).

4. Install washer (20) and nut (22) on plug (14). Tighten nut.

5. Place packing (26) on union (27) and install in top cover assembly (18).

6. Install washer (25) and nut (24) on union (27). Tighten nut.

7. Place packing (13) on elbow (12) and install in top cover assembly (18).

NOTE

SN: 36248-Subsequent:
Washer (21) is not used.

8. Install washer (21) and nut (23) on elbow (12). Tighten nut.

9. Install packing (19) in groove around cell vent opening.

10. Place top cover assembly (18) on cell vent opening. Install bolts (16) and washers (17). Tighten bolts.

11. Place support assembly (6) on top cover assembly (18). Install bolts (10) and washers (9). Tighten bolts.

12. Place support assembly (1) on top cover assembly (18). Install bolts (4) and washers (5). Tighten bolts.

13. Install screws (8) and washers (7). Tighten screws.

14. Install screws (3) and washers (2). Tighten screws.

15. Connect hose assembly (11) to elbow (12). Tighten the end fitting of hose assembly.

16. Connect hose assembly (28) to union (27). Tighten the end fitting of hose assembly.

17. Install fuel quantity probe (29) (412-A-28-10-00-88A-720A-A / 00609) and connect fuel quantity indicating system connector.

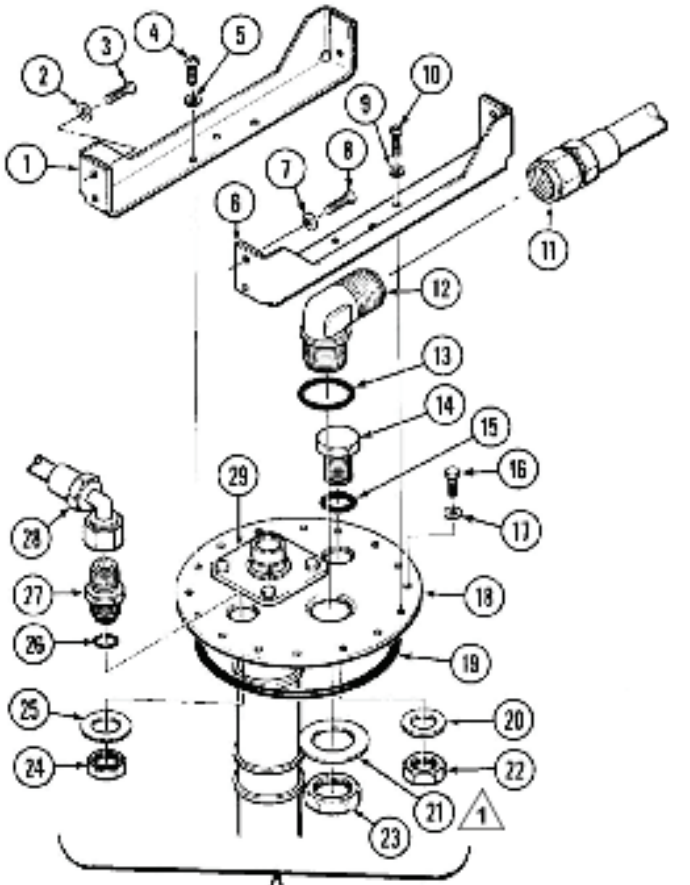
18. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).

19. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).

20. Perform fuel quantity system operational check.

21. Check area for leaks.

22. Install access panel in engine compartment floor above cell area.



1. Support assembly
2. Washer
3. Screw
4. Bolt
5. Washer
6. Support assembly
7. Washer
8. Screw
9. Washer
10. Bolt
11. Hose assembly
12. Elbow
13. Packing
14. Plug
15. Packing
16. Bolt
17. Washer
18. Cover assembly
19. Packing
20. Washer
21. Washer
22. Nut
23. Nut
24. Nut
25. Washer
26. Packing
27. Union
28. Hose assembly
29. Fuel quantity probe installation

NOTE

△1 S/N 33108 through 33213 and 36001 through 36247.

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Figure 1. Upper Left Cell and Upper Right Cell Top Cover Assembly (Sheet 1 of 1)

Filler Cap and Adapter Assembly - Description

NOTE

SN: 33001-33167 TB 412-89-71: Pre:

The maintenance procedures in 412-A-28-10-00-77A-520A-A / 00577 and 412-A-28-10-00-77A-720A-A / 00578 are applicable to helicopters.

The fuel system filler cap and adapter assembly is mounted on the upper outboard side of upper right cell. A spring-loaded plate on the inside of the assembly serves as a fuel seal in the event the cap opens accidentally.

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Filler Cap and Adapter Assembly - Removal

Procedure

NOTE

SN: 33001-33167 TB 412-89-71: Pre:

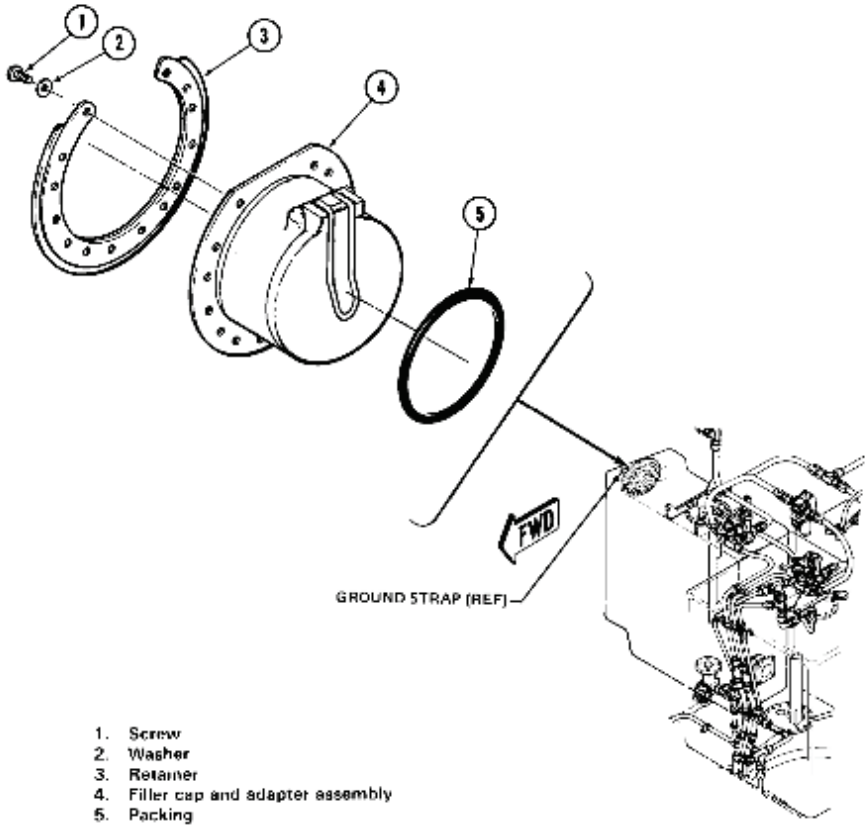
The following removal procedures are applicable to helicopters.

1. Disconnect electrical power.

2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).

3. Remove screws (1, Figure 1), washers (2), retainer (3), filler cap and adapter assembly (4), and packing (5). Discard packing.

4. Install protective cover over cell opening.



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Figure 1. Filler Cap and Adapter Assembly (Helicopters S/N 33001 through 33167 not Modified by TB 412-89-71) (Sheet 1 of 1)

Filler Cap and Adapter Assembly - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

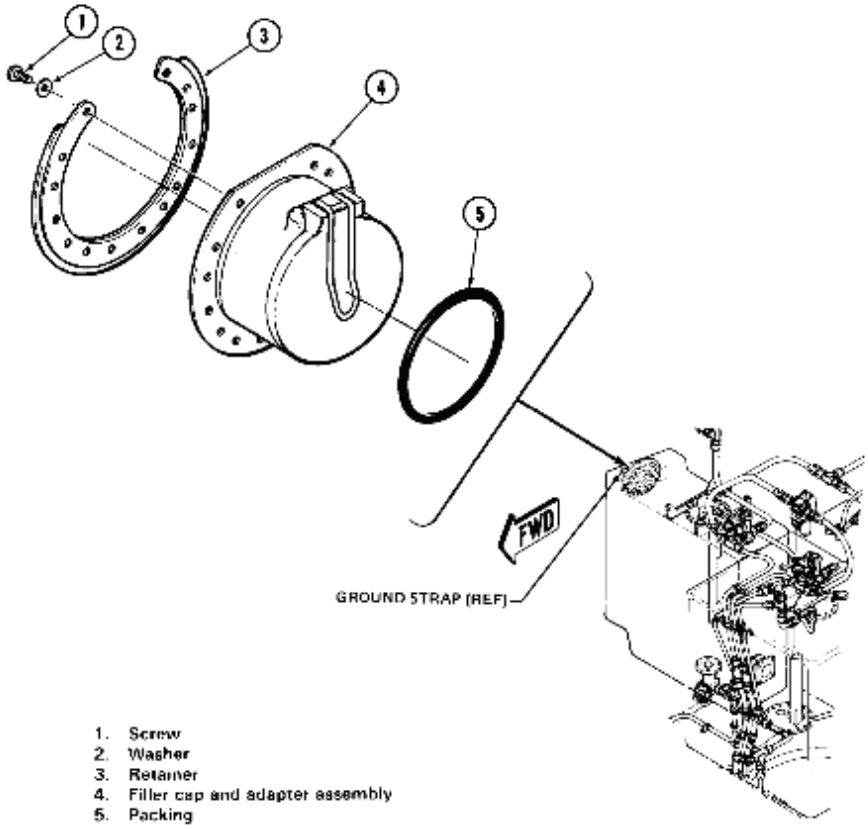
Procedure

NOTE

SN: 33001-33167 TB 412-89-71: Pre:

The following installation procedures are applicable to helicopters.

1. Remove protective cover from cell opening.
2. Coat packing (5, Figure 1) with assembly fluid (C-024). Place packing (5) in groove around cell opening.
3. Install filler cap and adapter assembly (4) and retainer (3) on cell opening.
4. Install screws (1), washers (2), and ground strap (located at second screw hole).
5. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).
6. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).
7. Check area for leaks.



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Figure 1. Filler Cap and Adapter Assembly (Helicopters S/N 33001 through 33167 not Modified by TB 412-89-71) (Sheet 1 of 1)

Improved Filler Cap and Adapter Assembly - Description

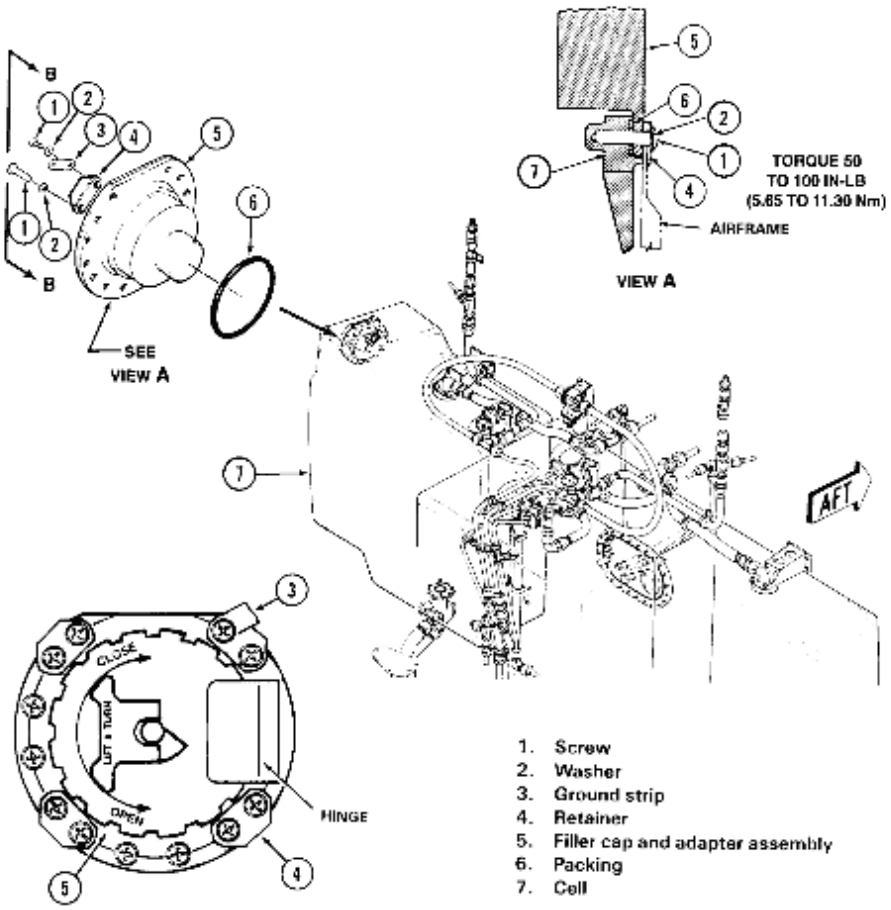
The improved filler cap and adapter assembly is located on the upper outboard side of the upper right cell. This filler cap incorporates a spring-loaded hinge which will minimize fuel and cap loss in the event cap is not locked.

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Improved Filler Cap and Adapter Assembly - Removal

Procedure

1. Disconnect electrical power.
2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).
3. Remove screws (1, Figure 1), washers (2), four retainers (4), filler cap and adapter assembly (5), and packing (6). Discard packing.
4. Install protective cover over cell opening.



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Figure 1. Improved Filler Cap and Adapter Assembly (Helicopters S/N 33001 through 33167 Modified by TB 412-89-71, S/N 33168 and Subsequent, and S/N 36001 and Subsequent) (Sheet 1 of 1)

Improved Filler Cap and Adapter Assembly - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

1. Remove protective cover.

2. Coat packing (6, Figure 1) with assembly fluid (C-024). Place packing (6) in groove around cell opening.

3. Taking care not to disturb packing (6), install cap and adapter assembly (5), with four retainers (4), washers (2), and screws (1). Locate retainers (4) at the 2, 4, 8, and 10 o'clock position as shown in detail B.

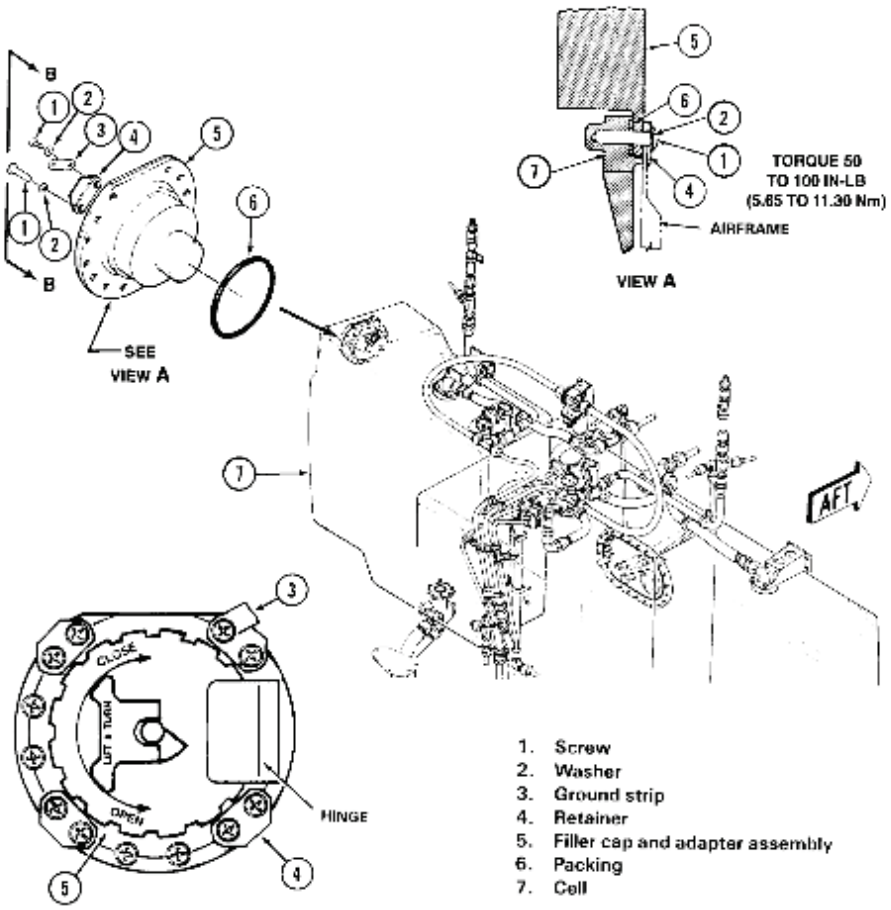
4. Connect ground strip (3) at the one o'clock position.

5. Torque all screws (1) 50 to 100 inch-pounds (5.65 to 11.30 Nm).

6. Pressure test fuel system (412-A-28-10-00-00A-362B-A / 00438).

7. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).

8. Check area for leaks.



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Figure 1. Improved Filler Cap and Adapter Assembly (Helicopters S/N 33001 through 33167 Modified by TB 412-89-71, S/N 33168 and Subsequent, and S/N 36001 and Subsequent) (Sheet 1 of 1)

Engine Fuel Feed Line Pressure Switches - Description

A fuel feed line pressure switch is installed in each power section fuel feed line. These switches are located in front of bulkhead at FS 154. Contacts are closed when no fuel pressure exists and open when normal fuel pressure is present. The FUEL BOOST caution lights illuminate when switch contacts are closed.

NOTE

Maintenance procedures for left and right power section fuel feed line pressure switches (fuel pressure switches) are the same.

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Engine Fuel Feed Line Pressure Switches - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure

NOTE

The following removal procedures for left and right power section fuel feed line pressure switches (fuel pressure switches) are the same.

1. Disconnect electrical power.

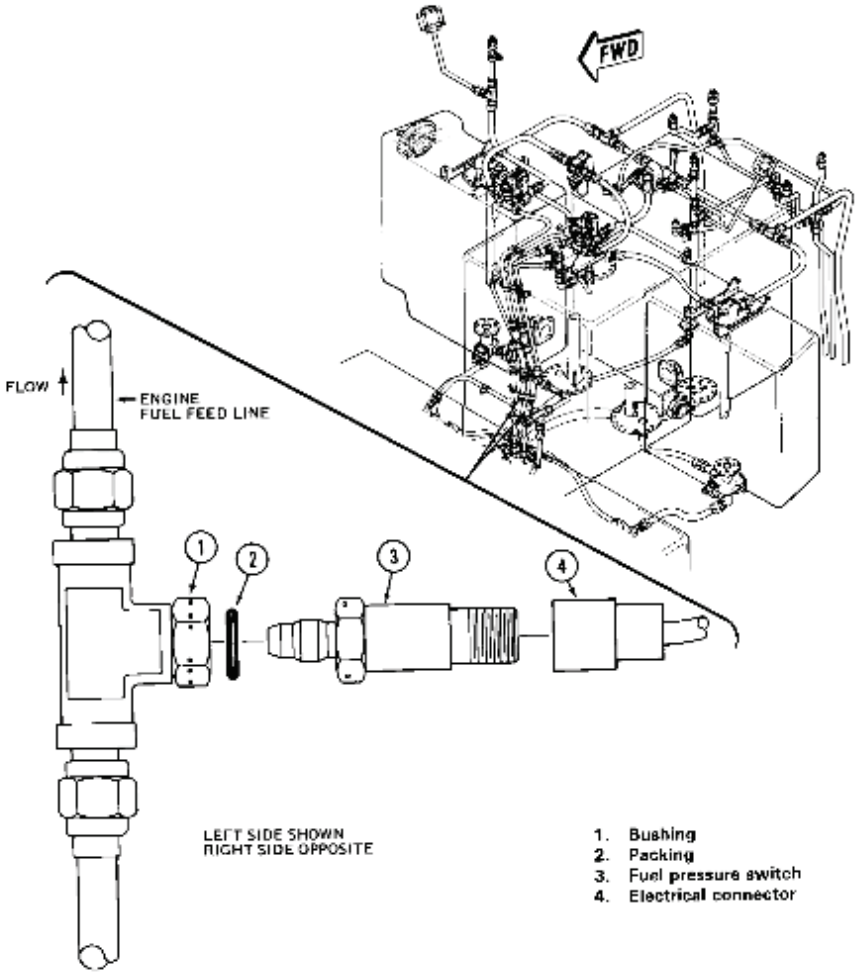
2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).

3. Remove access door over cargo hook opening on bottom of helicopter, if installed.

4. Disconnect electrical connector (4, Figure 1).

5. Remove fuel pressure switch (3) and packing (2) from bushing (1). Discard packing.

6. Install caps or plugs (C-428) on open fuel lines and on the electrical connector (4).



(S/N 33108 THRU 33213)
(S/N 36001 AND SUBSEQUENT)

ICN-412-A-281000-B-97499-00091-A-001-01

Figure 1. Fuel Feed Line Pressure Switch (Sheet 1 of 1)

Engine Fuel Feed Line Pressure Switches - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	
Lockwire	C-405	AR	

Procedure

NOTE

The following installation procedures for left and right power section fuel feed line pressure switches (fuel pressure switches) are the same.

1. Remove caps and/or plugs from open fuel lines and from electrical connector (4, Figure 1).

2. Apply assembly fluid (C-024) to packing (2).

3. Place packing (2) on threaded end of fuel pressure switch (3).

4. Install fuel pressure switch (3).

5. Connect electrical connector (4).

6. Secure fuel pressure switch (3) and bushing (1) to tee fitting using lockwire (C-405).

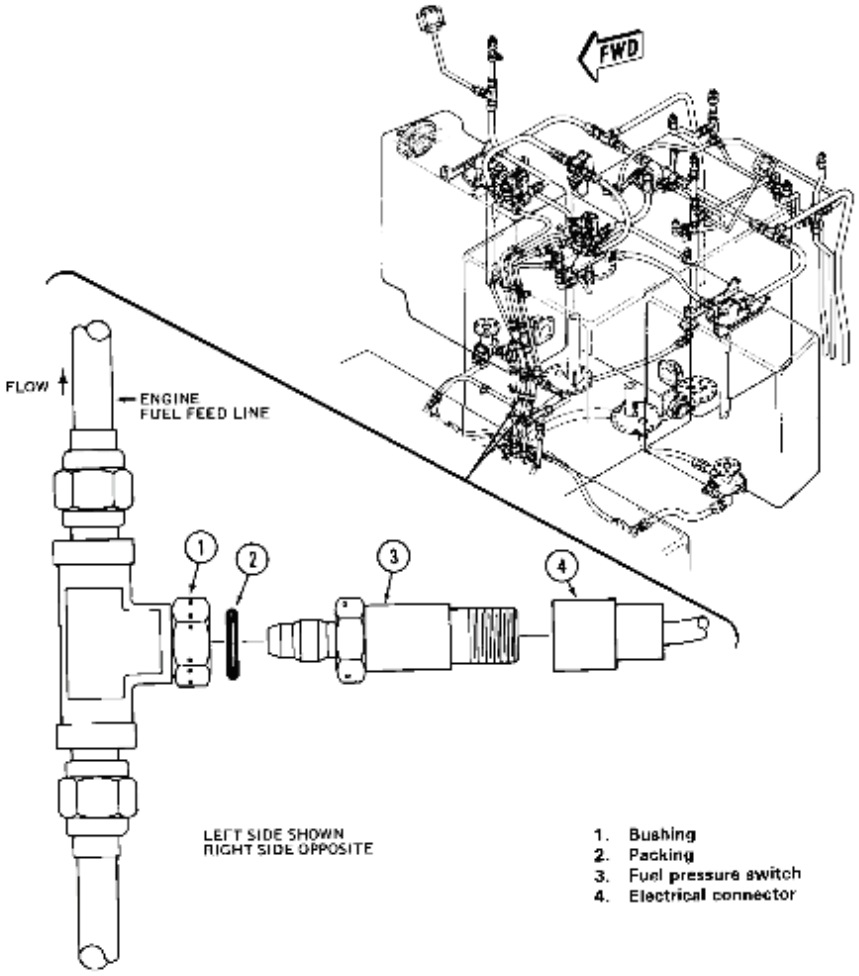
7. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).

8. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).

9. Perform fuel pressure system operational test.

10. Check area for leaks.

11. Install access door over cargo hook opening, if applicable.



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(S/N 36001 AND SUBSEQUENT)

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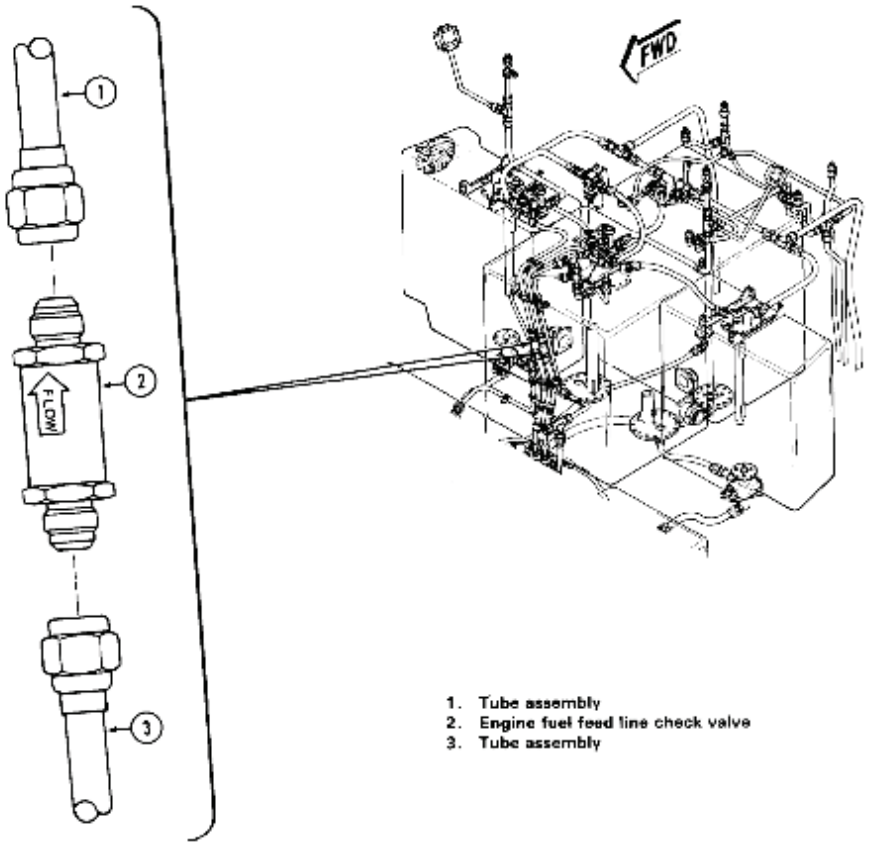
Figure 1. Fuel Feed Line Pressure Switch (Sheet 1 of 1)

Fuel Feed Line Check Valves - Description

A check valve (Figure 1), with internal orifices for thermal relief, is installed in each of power section fuel feed lines. These valves are located in front of bulkhead at FS 154 (forward of upper forward center cell).

NOTE

Maintenance procedures for left and right feed line check valves are the same.



1. Tube assembly
2. Engine fuel feed line check valve
3. Tube assembly

|S/N 33108 THRU 33213|
|S/N 36001 AND SUBSEQUENT|

ICN-412-A-281000-B-97499-00092-A-001-01

Figure 1. Fuel Feed Line Check Valve (Sheet 1 of 1)

Fuel Feed Line Check Valves - Removal

Preliminary requirements

Consumables, materials and expendables

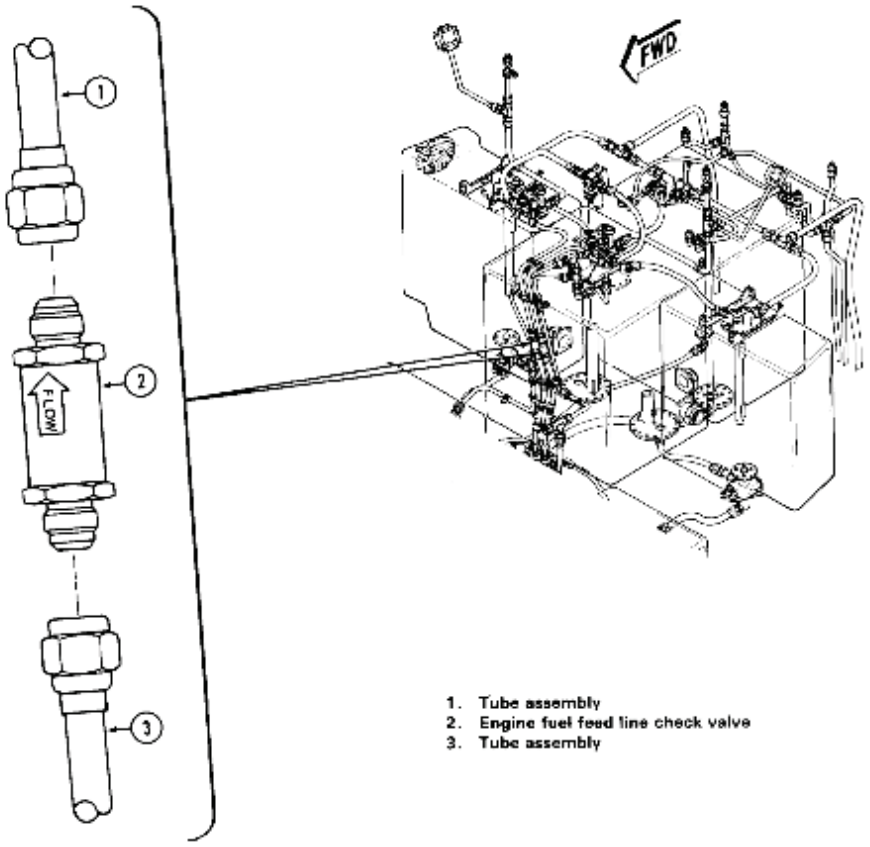
Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure

NOTE

The following removal procedures for left and right feed line check valves are the same.

1. Disconnect electrical power.
2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).
3. Remove access door over cargo hook opening on bottom of helicopter, if installed.
4. Loosen first mounting clamp on fuel feed line below and above engine fuel feed line check valve (2, Figure 1).
5. Disconnect tube assembly (1) from engine fuel feed line check valve (2).
6. Disconnect tube assembly (3) from engine fuel feed line check valve (2).
7. Remove engine fuel feed line check valve (2).
8. Install caps or plugs (C-428) on open fuel lines.



1. Tube assembly
2. Engine fuel feed line check valve
3. Tube assembly

|S/N 33108 THRU 33213|
|S/N 36001 AND SUBSEQUENT|

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Figure 1. Fuel Feed Line Check Valve (Sheet 1 of 1)

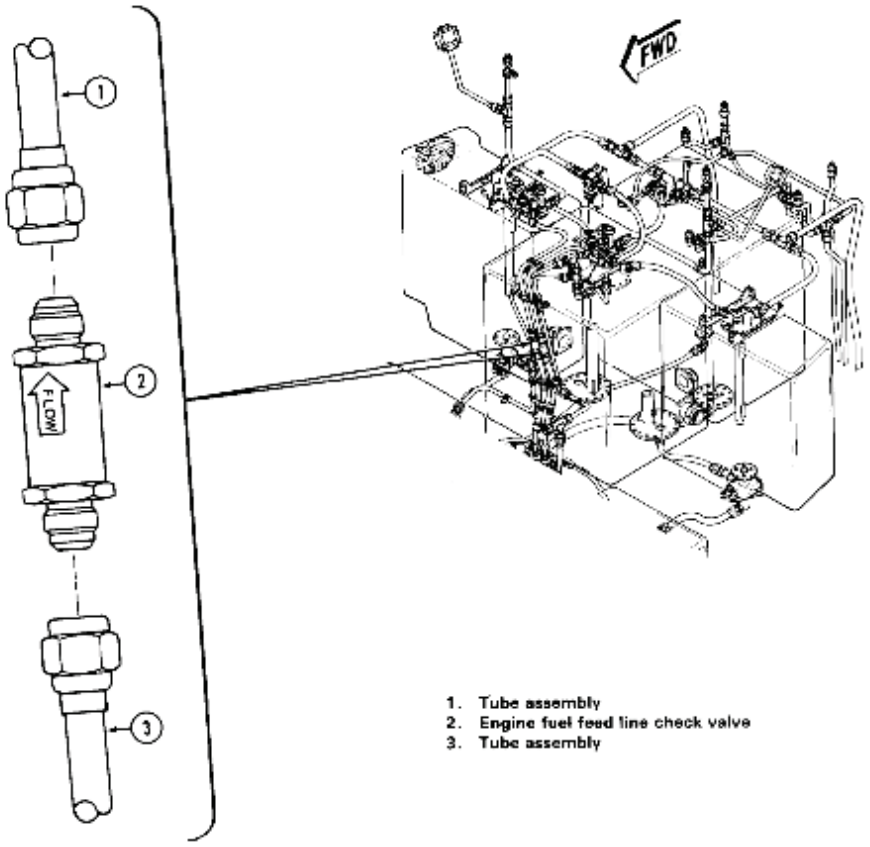
Fuel Feed Line Check Valves - Installation

Procedure

NOTE

The following installation procedures for left and right feed line check valves are the same.

1. Remove protective caps and/or plugs.
2. Hold engine fuel feed line check valve (2, Figure 1) in place near tube assembly (1) and tube assembly (3), with large flow arrow on side of engine fuel feed line check valve pointing up.
3. Connect the tube assembly (1) on top of threaded portion of engine fuel feed line check valve (2). Tighten nut.
4. Connect the tube assembly (3) on bottom threaded portion of engine fuel feed line check valve (2). Tighten nut.
5. Tighten first mounting clamps above and below engine fuel feed line check valve (2).
6. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).
7. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).
8. Check area for leaks.
9. Install access door over cargo hook opening, if applicable.



1. Tube assembly
2. Engine fuel feed line check valve
3. Tube assembly

|S/N 33108 THRU 33213|
|S/N 36001 AND SUBSEQUENT|

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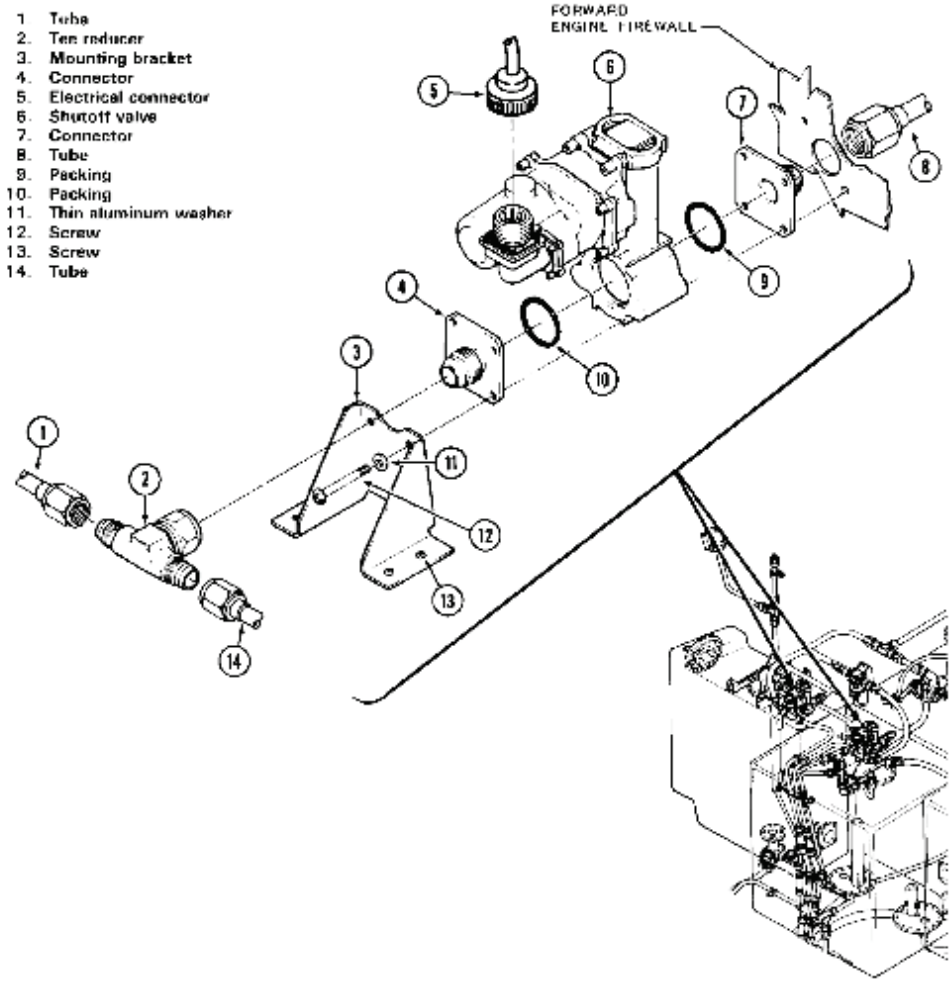
Figure 1. Fuel Feed Line Check Valve (Sheet 1 of 1)

Shutoff Valves - Description

A motor-operated shutoff valve (Figure 1) is installed in each fuel feed line above the fuel feed line check valves. These valves are located on the front of the lower forward engine firewall. The shutoff valves are controlled by the ENGINE 1 and ENGINE 2 FUEL switches on the engine and fuel control panel. The valves have manual override handles which also serve as valve position indicators. Thermal relief valves are incorporated for internal bypass of trapped fuel when system is inoperative.

NOTE

Maintenance procedures for left and right shutoff valves are the same.



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Figure 1. Shutoff Valve (Sheet 1 of 1)

Shutoff Valves - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure

NOTE

The following removal procedures for left and right shutoff valves are the same.

1. Disconnect electrical power.

2. Open engine cowling.

3. Remove access door on side of pylon support in cabin.

4. Disconnect electrical connector (5, caps or plugs (C-428)).

5. Disconnect tubes (1 and 14) from the tee reducer (2).

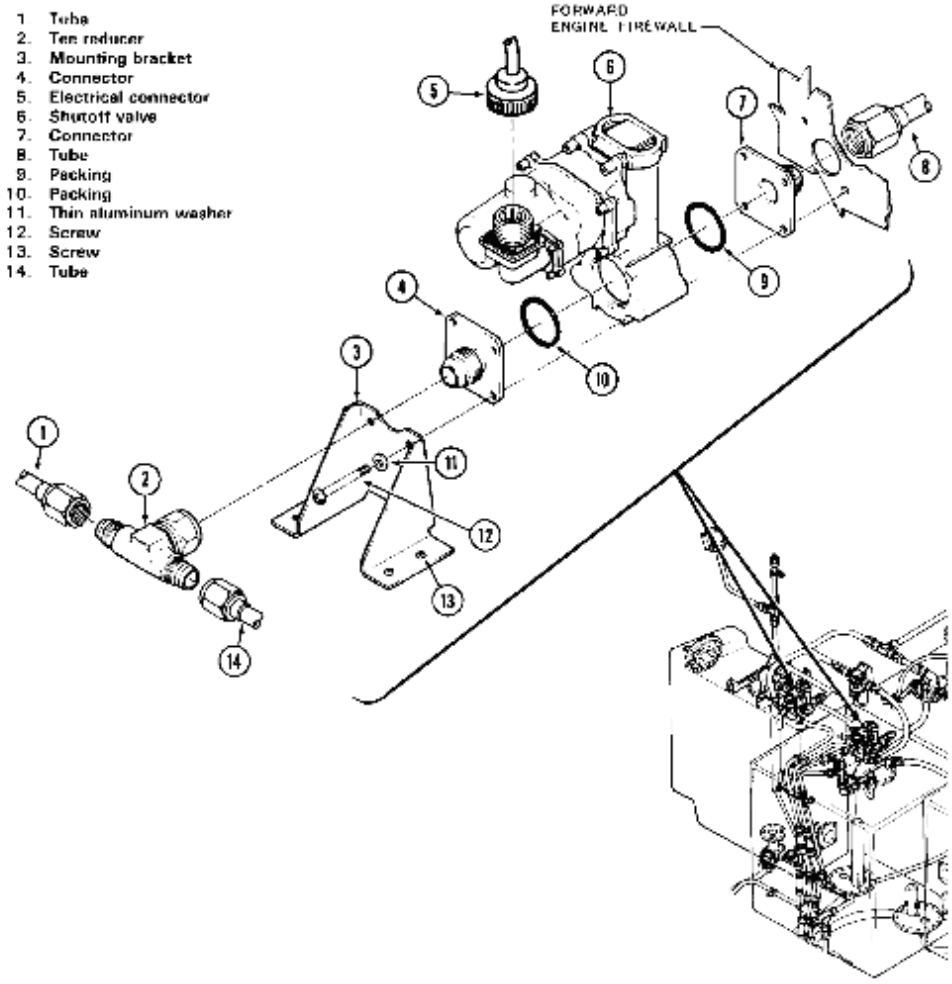
6. If the connector (4) is to be replaced, remove the tee reducer (2).

7. Disconnect tube (8) from connector (7).

8. Remove screws (12) and thin aluminum washers (11).

9. Remove shutoff valve (6), connectors (4 and 7), and packings (9 and 10). Discard packings.

10. Install caps or plugs (C-428) on open ends of tubes (1, 8, and 14).



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Figure 1. Shutoff Valve (Sheet 1 of 1)

Shutoff Valves - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

NOTE

The following installation procedures for left and right shutoff valves are the same.

1. Apply assembly fluid (C-024) to packings (9 and 10, Figure 1).

2. Position packings (9 and 10) and connectors (4 and 7) on body of shutoff valve (6).

NOTE

Screws (13) securing mounting bracket (3) to deck can be loosened as necessary to provide clearance for installation of shutoff valve (6).

3. Position shutoff valve (6) between mounting bracket (3) and firewall.

4. Insert two upper screws (12) and thin aluminum washers (11). Start upper screws into firewall nutplates.

5. Install two lower screws (12) and thin aluminum washers (11). Tighten all four screws (12).

6. Tighten screws (13), if loosened.

7. Install tee reducer (2), if removed.

8. Connect tubes (1 and 14) to tee reducer (2).

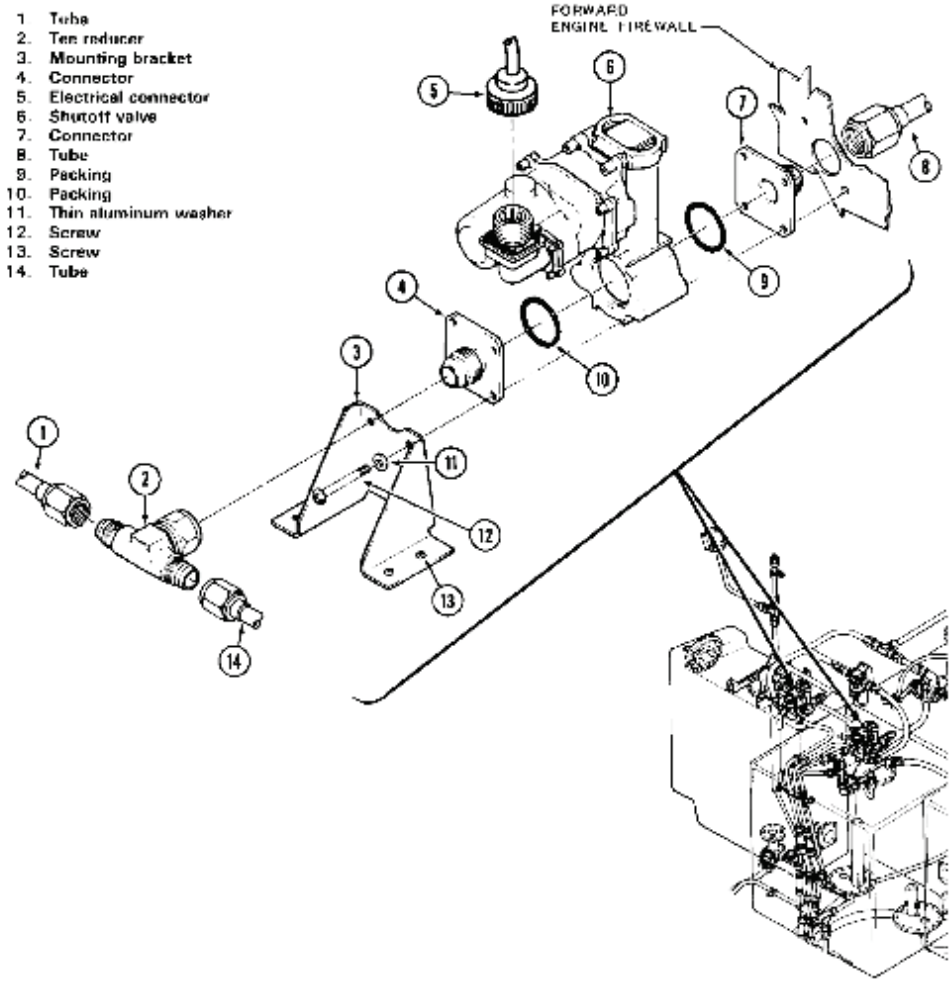
9. Connect tube (8) to connector (7).

10. Connect electrical connector (5).

11. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).

12. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).

13. Check area for leaks.

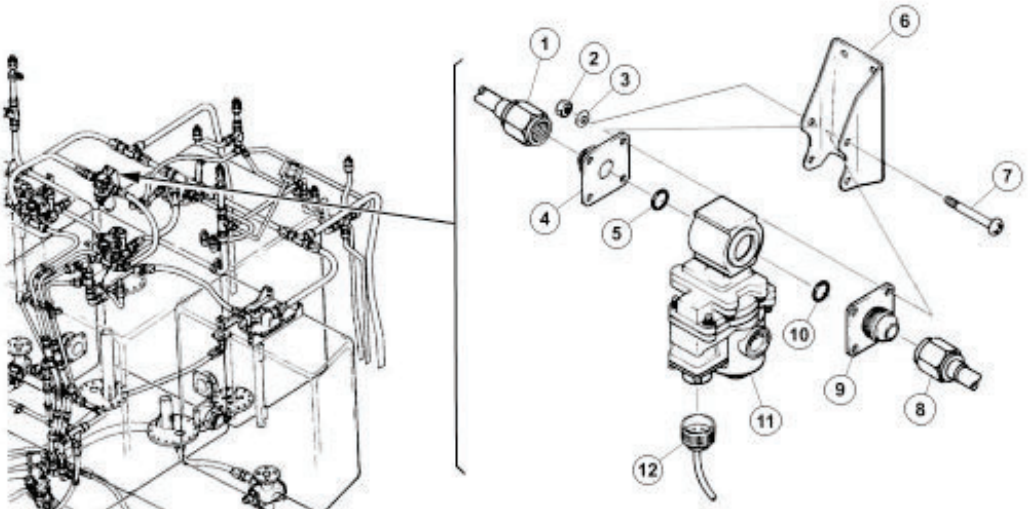


ICN-412-A-281000-B-97499-00093-A-001-01

Figure 1. Shutoff Valve (Sheet 1 of 1)

Crossfeed Valve - Description

A crossfeed valve (Figure 1) is located above and between the shutoff valves on the front of the lower forward engine firewall. This valve is normally closed so fuel is supplied to each power section from the isolated lower main cell on the same side of the helicopter. If boost pump in either lower main cell ceases to function, the crossfeed valve opens automatically and fuel flows from the remaining operating boost pump to both power sections.



1. Tube
2. Nut
3. Thin aluminum washer
4. Connector
5. Packing
6. Bracket
7. Screw
8. Tube
9. Connector
10. Packing
11. Crossfeed valve
12. Electrical connector

(S/N 33108 THROUGH 33213,
S/N 36001 THROUGH 36999, AND
S/N 37002 THROUGH 37999)

ICN-412-A-281000-B-97499-00094-A-001-01

Figure 1. Crossfeed Valve (Sheet 1 of 1)

Crossfeed Valve - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure

1. Disconnect electrical power.

2. Remove transmission cowling.

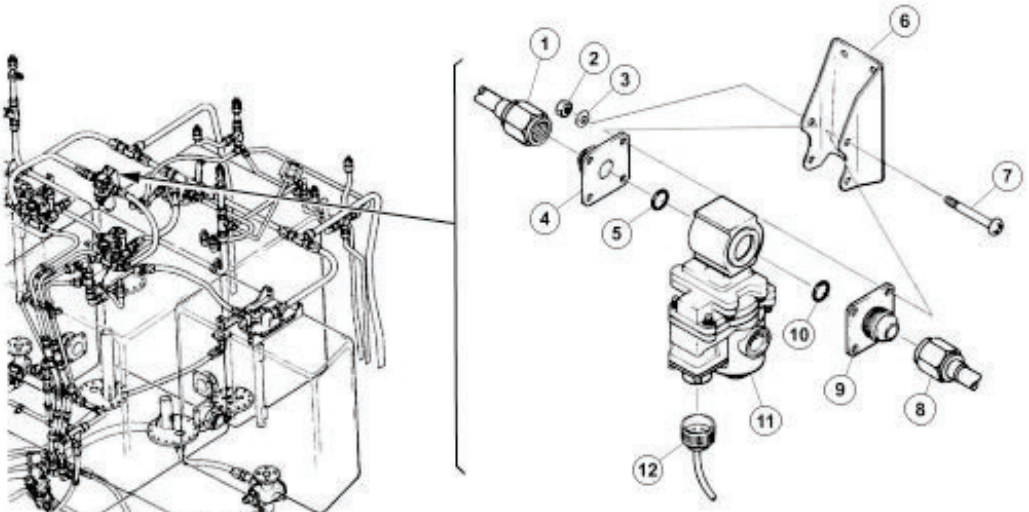
3. Disconnect electrical connector (12, Figure 1).

4. Disconnect tubes (1 and 8) from connectors (4 and 9).

5. Remove screws (7), nuts (2), and thin aluminum washers (3).

6. Remove crossfeed valve (11), connectors (4 and 9), and packings (5 and 10). Discard packings.

7. Install caps or plugs (C-428) to the tubes (1 and 8) open end fittings.



1. Tube
2. Nut
3. Thin aluminum washer
4. Connector
5. Packing
6. Bracket
7. Screw
8. Tube
9. Connector
10. Packing
11. Crossfeed valve
12. Electrical connector

(S/N 33108 THROUGH 33213,
S/N 36001 THROUGH 36999, AND
S/N 37002 THROUGH 37999)

ICN-412-A-281000-B-97499-00094-A-001-01

Figure 1. Crossfeed Valve (Sheet 1 of 1)

Crossfeed Valve - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Assembly fluid	C-024	AR	

Procedure

1. Apply assembly fluid (C-024) to packing (5 and 10, Figure 1).

2. Put the packings (5 and 10) and the connectors (4 and 9) in position on the crossfeed valve (11).

3. Install two forward screws (7), thin aluminum washers (3), and nuts (2). Do not tighten the screws at this time.

4. Put the crossfeed valve (11) and the connectors (4 and 9) in position on the bracket (6).

5. Install two aft screws (7), thin aluminum washers (3), and nuts (2). Tighten the four screws.

6. Connect tubes (1 and 8) to connectors (4 and 9).

7. Connect electrical connector (12).

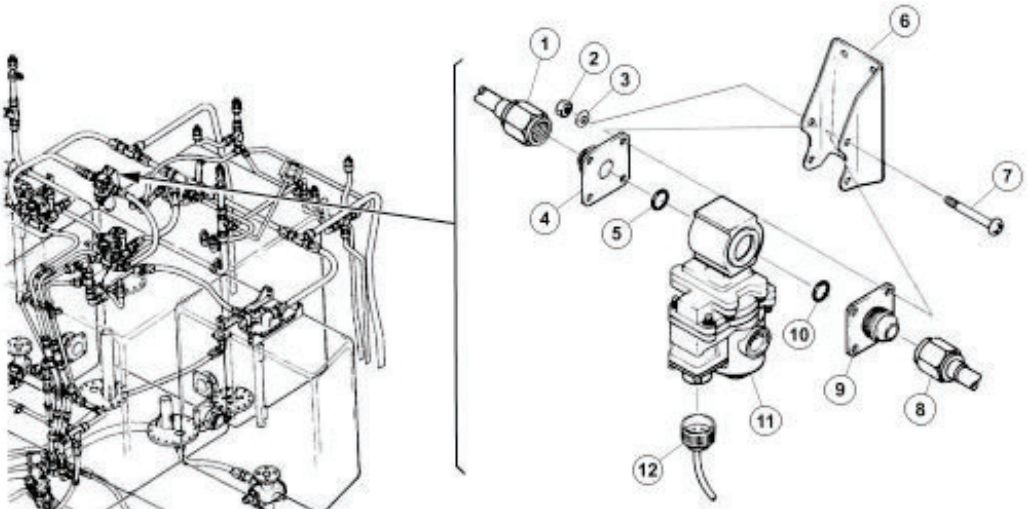
8. Perform operational check of crossfeed valve (11) (412-A-96-04-00-07A-320A-A / 02119).

9. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).

10. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).

11. Check area for leaks.

12. Install transmission cowling.



1. Tube
2. Nut
3. Thin aluminum washer
4. Connector
5. Packing
6. Bracket
7. Screw
8. Tube
9. Connector
10. Packing
11. Crossfeed valve
12. Electrical connector

(S/N 33108 THROUGH 33213,
S/N 36001 THROUGH 36999, AND
S/N 37002 THROUGH 37999)

ICN-412-A-281000-B-97499-00094-A-001-01

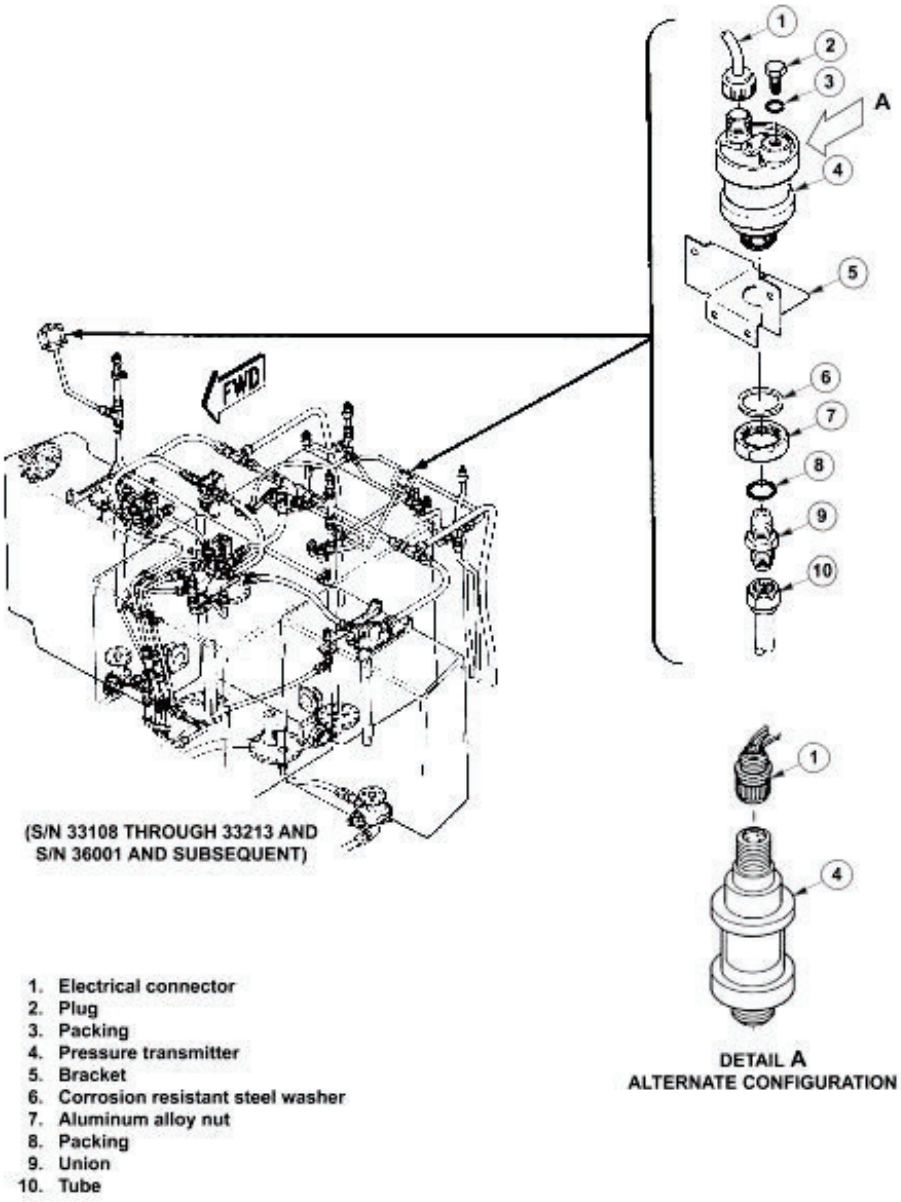
Figure 1. Crossfeed Valve (Sheet 1 of 1)

Fuel Pressure Transmitters - Description

An electrical fuel pressure transmitter (Figure 1) is connected in each power section fuel feed line. The fuel pressure transmitters are sensor units for the fuel pressure indicating system, which provide indicator readings of fuel pressure. The transmitters are mounted on the left and right sides of upper forward engine firewall.

NOTE

Maintenance procedures for left and right fuel pressure transmitters are the same.



ICN-412-A-281000-B-97499-00095-A-001-01

Figure 1. Left and Right Fuel Pressure Transmitter (Sheet 1 of 1)

Fuel Pressure Transmitters - Removal

Preliminary requirements

Consumables, materials and expendables

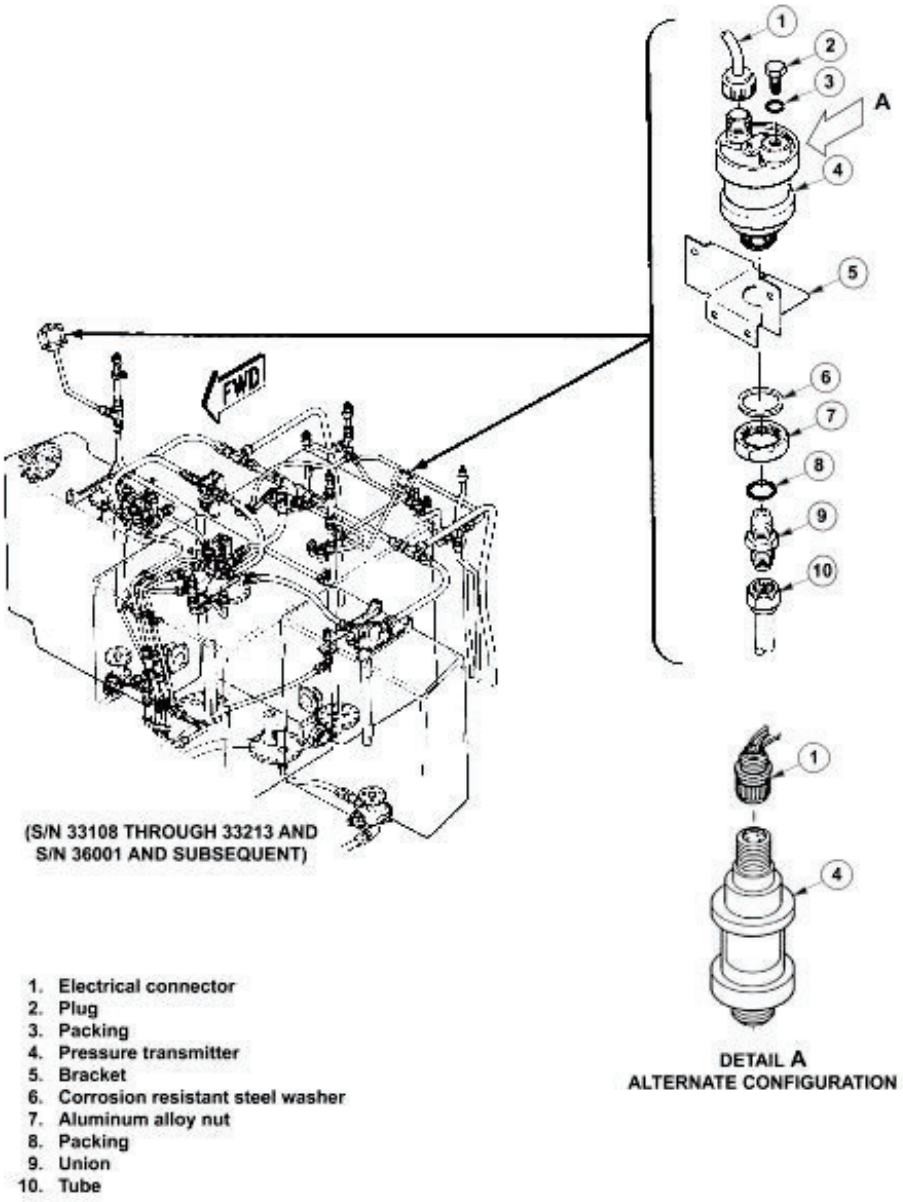
Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure

NOTE

The following removal procedures for left and right fuel pressure transmitters are the same.

1. Disconnect electrical power.
2. Open engine cowling.
3. Disconnect electrical connector (1, Figure 1) from pressure transmitter (4) on forward firewall.
4. Disconnect tube (10) from the union (9).
5. Remove union (9) and packing (8). Discard packing.
6. Remove aluminum alloy nut (7) and corrosion resistant steel washer (6).
7. Remove pressure transmitter (4). If applicable, remove plug (2) and packing (3). Discard packing.
8. Install caps or plugs (C-428) on tube (10) and electrical connector (1).



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Figure 1. Left and Right Fuel Pressure Transmitter (Sheet 1 of 1)

Fuel Pressure Transmitters - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Petrolatum	C-008	AR	
Assembly fluid	C-024	AR	

Procedure

NOTE

The following installation procedures for left and right fuel pressure transmitters are the same.

1. Remove protective caps and/or plugs.

2. Apply assembly fluid (C-024) to packing (3, Figure 1) before installation. If applicable, install plug (2) and packing (3).

3. Position pressure transmitter (4) on bracket (5). Install corrosion resistant steel washer (6) and aluminum alloy nut (7). Tighten the nut.

4. Apply petrolatum (C-008) to packing (8) before installation. Install packing and union (9).

5. Connect tube (10) to union (9). Tighten the tube (10) end fitting.

6. Connect electrical connector (1).

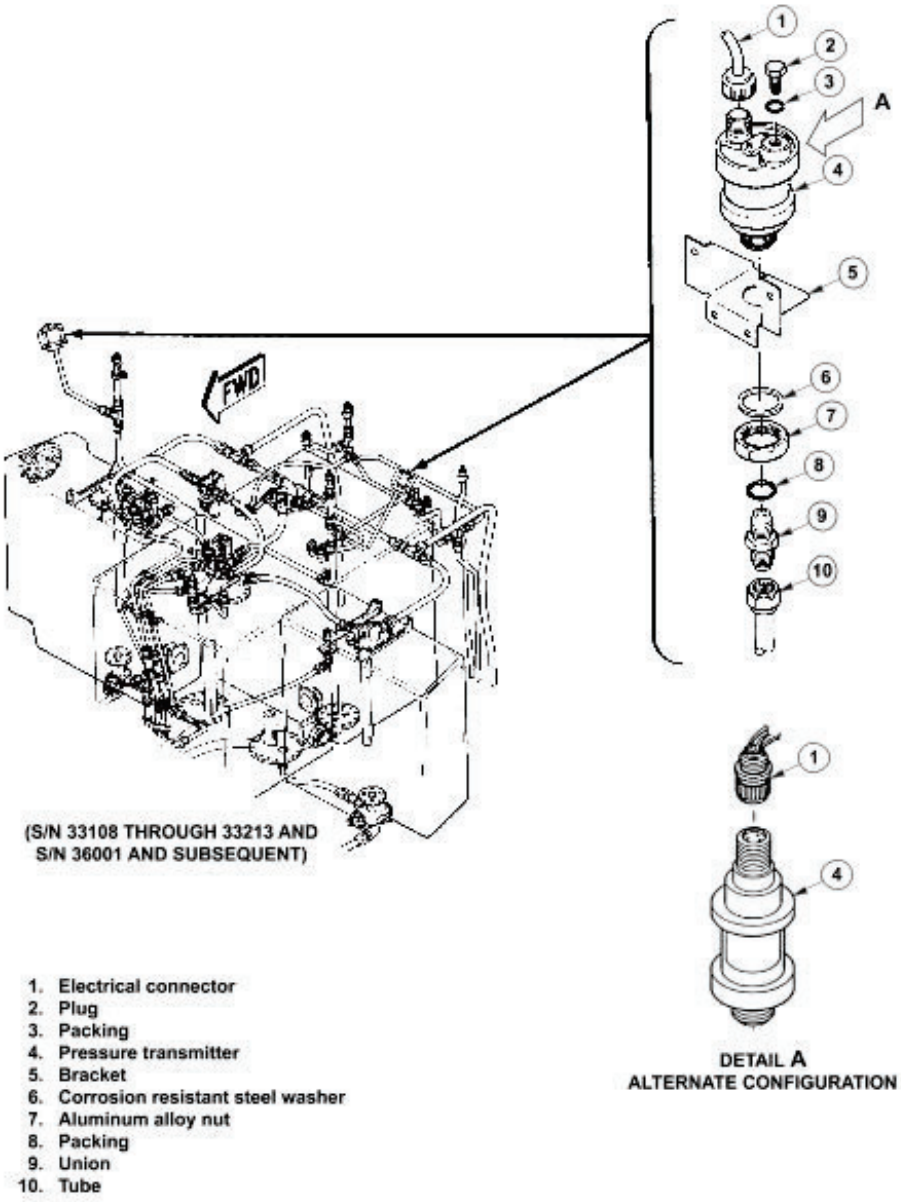
7. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).

8. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).

9. Perform operational check.

10. Check area for leaks.

11. Close engine cowling and install sump access door.



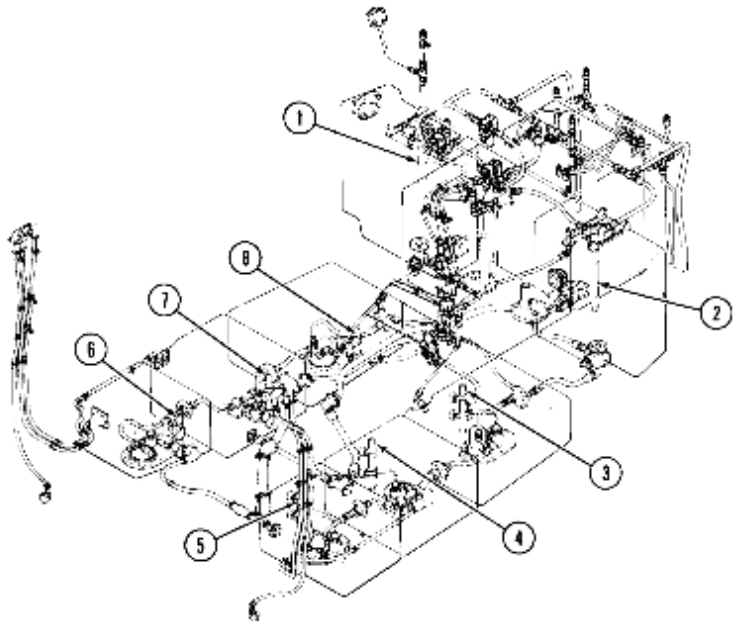
ICN-412-A-281000-B-97499-00095-A-001-01

Figure 1. Left and Right Fuel Pressure Transmitter (Sheet 1 of 1)

Fuel Quantity Probes - Description

Eight cell mounted fuel quantity probes (tank units) (1 through 8, Figure 1) are used to measure quantity (weight) of fuel in system. The probes are variable capacitors in which capacitance varies with changes in dielectric constant of fuel and air dielectric material present between tubes (capacitor plates) of probes. The level and density of fuel in cells determine dielectric constant. Probe capacitance is transmitted to a self-balancing bridge network in the fuel level indicator.

1. Upper right cell fuel quantity probe
2. Upper left cell fuel quantity probe
3. Lower left main cell fuel quantity probe
4. Lower left middle cell fuel quantity probe
5. Lower left forward cell fuel quantity probe
6. Lower right forward cell fuel quantity probe
7. Lower right middle cell fuel quantity probe
8. Lower right main cell fuel quantity probe



(S/N 33108 THRU 33213)
(S/N 36001 AND SUBSEQUENT)

ICN-412-A-281000-B-97499-00096-A-001-01

Figure 1. Fuel Quantity Probes (Sheet 1 of 1)

Lower Forward Cell Fuel Quantity Probes - Description

A fuel quantity probe is mounted on inboard wall of each lower forward cell. The base of each probe supports a fuel level sensor (thermistor) which causes the transfer pump to shut off automatically when cell is emptied. With cell empty, the thermistor also provides a ground path to the signal conditioner to extinguish the associated “no transfer” (NO 1 FUEL TRANS or NO 2 FUEL TRANS) caution light segment. When the thermistor is covered with fuel, the ground path is open, and the caution light remains on as long as the flow switch check valve senses there is no transfer flow.

NOTE

Maintenance procedures for lower left forward and lower right forward cell fuel quantity probes are the same.

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Lower Forward Cell Fuel Quantity Probes - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure

NOTE

The following removal procedures for lower left forward and lower right forward cell fuel quantity probes are the same.

1. Remove sump assembly (2, Figure 1) (412-A-28-10-00-41A-520A-A / 00456).

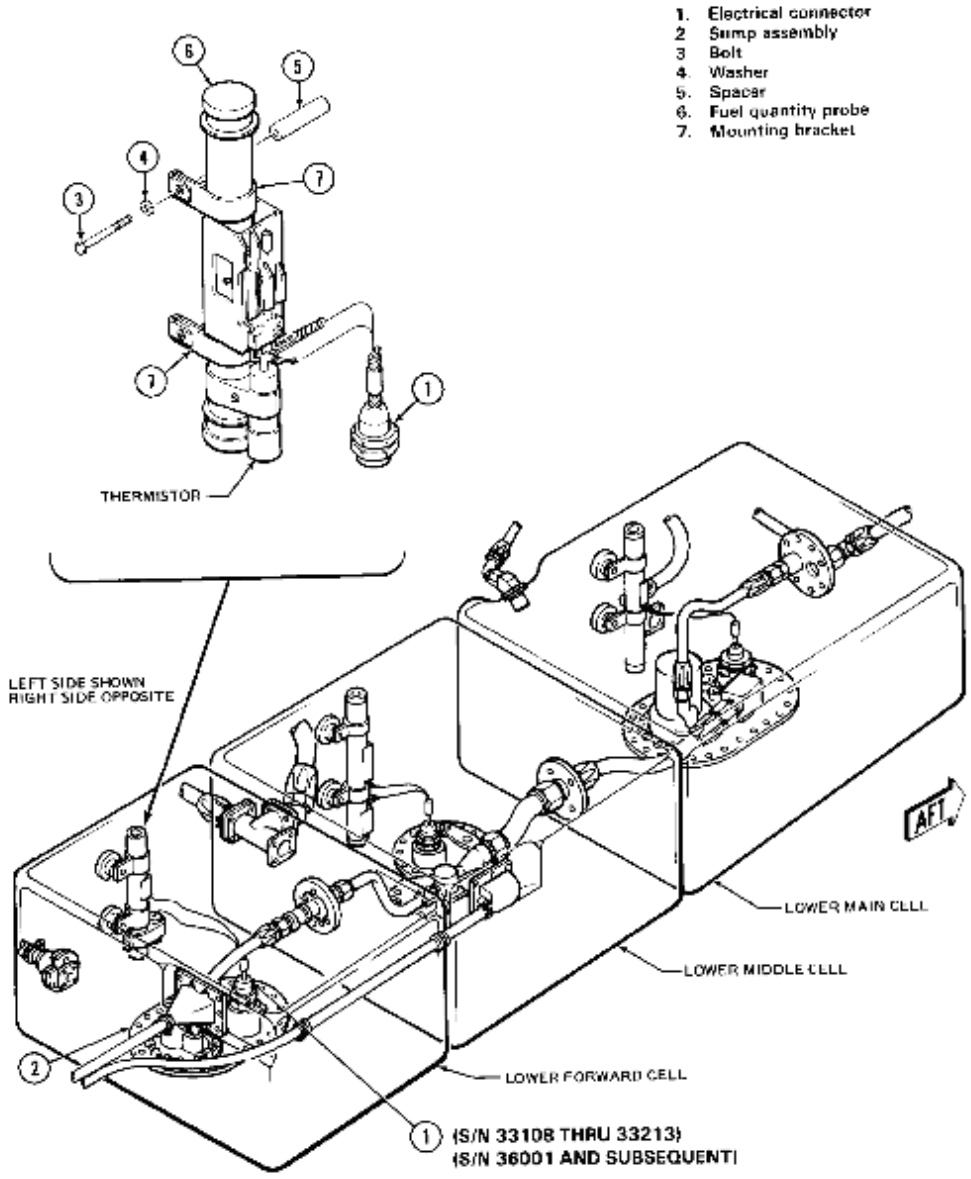
2. Disconnect electrical connector (1).

3. Remove bolts (3), washers (4), and spacers (5).

4. Remove fuel quantity probe (6).

5. Install protective cover over cell sump opening.

6. Install caps or plugs (C-428) on the electrical connector (1).



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Figure 1. Lower Forward Cell Fuel Quantity Probe (Sheet 1 of 1)

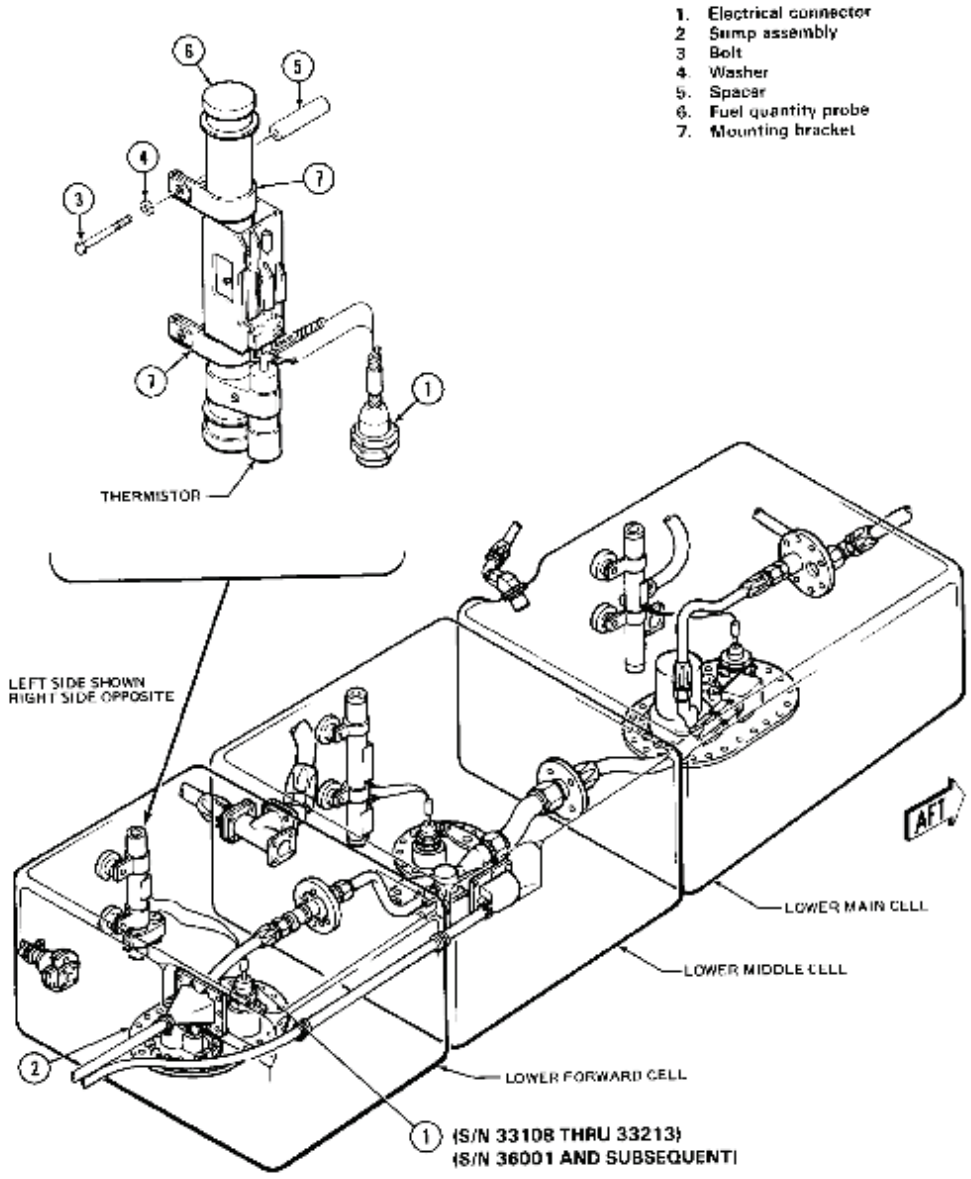
Lower Forward Cell Fuel Quantity Probes - Installation

Procedure

NOTE

The following installation procedures for lower left forward and lower right forward cell fuel quantity probes are the same.

1. Remove the protective caps and/or plugs from the electrical connector (1, Figure 1).
2. Remove protective cover from cell sump opening.
3. Insert bolts (3) with washers (4) through mounting brackets (7).
4. Place spacers (5) on bolts (3) and position fuel quantity probe (6) onto attaching points on inboard wall of cell. Tighten bolts.
5. Connect electrical connector (1).
6. Install sump assembly (2) (412-A-28-10-00-41A-720A-A / 00458).
7. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).
8. Perform operational check of fuel quantity indicating system (412-A-96-12-00-11A-273A-A / 02006).
9. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).
10. Check area for leaks.

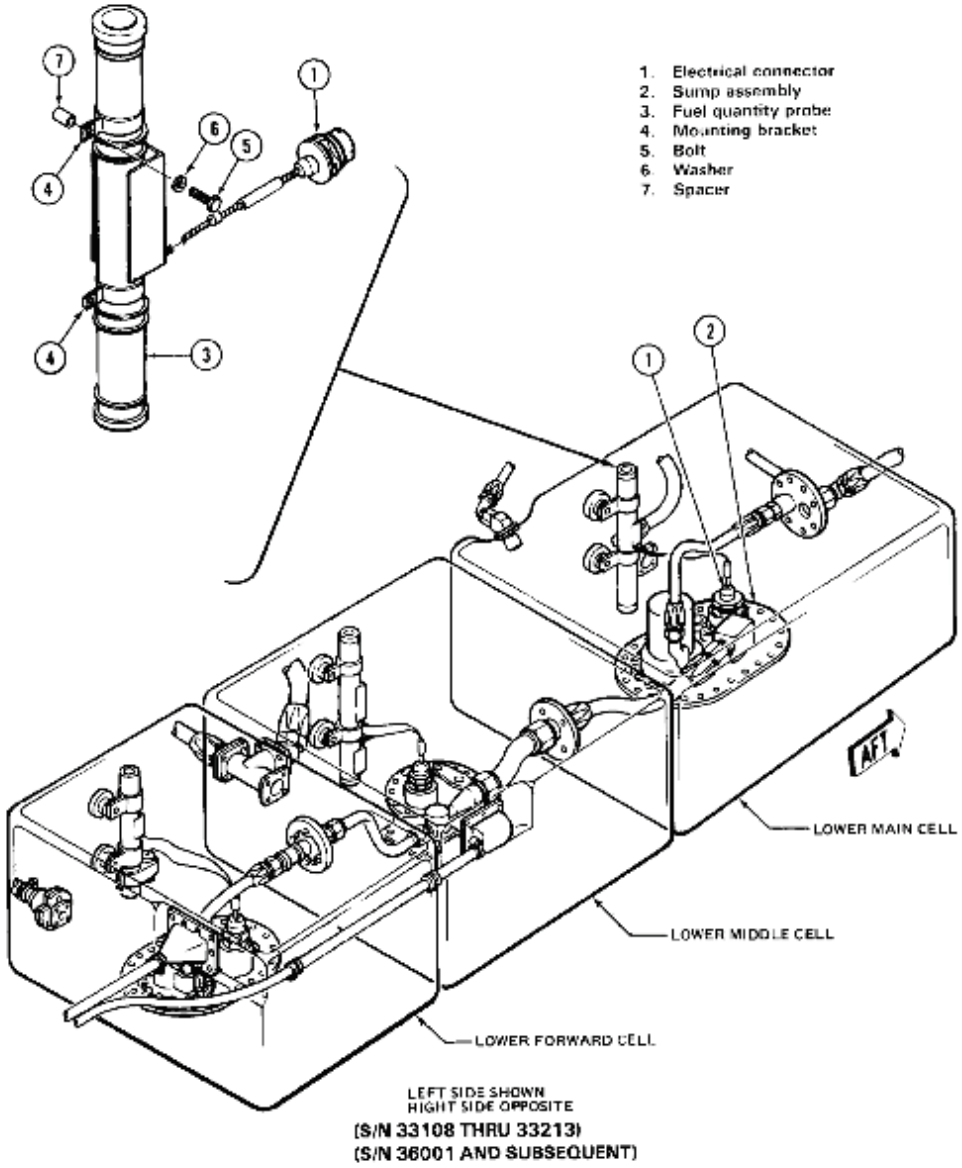


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Figure 1. Lower Forward Cell Fuel Quantity Probe (Sheet 1 of 1)

Lower Middle Cell Fuel Quantity Probes - Description

Each lower middle cell contains a fuel quantity probe (tank unit) (Figure 1) equipped with a fuel level sensor (thermistor) which causes associated no transfer (NO 1 FUEL TRANS or NO 2 FUEL TRANS) caution light segment to be extinguished when fuel flow through flow switch check valve ceases because of cell being empty.



ICN-412-A-281000-B-97499-00099-A-001-01

Figure 1. Lower Main Cell Fuel Quantity Probe (Sheet 1 of 1)

Lower Middle Cell Fuel Quantity Probes - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure

1. Remove sump assembly (7, Figure 1) (412-A-28-10-00-51A-520A-A / 00488).

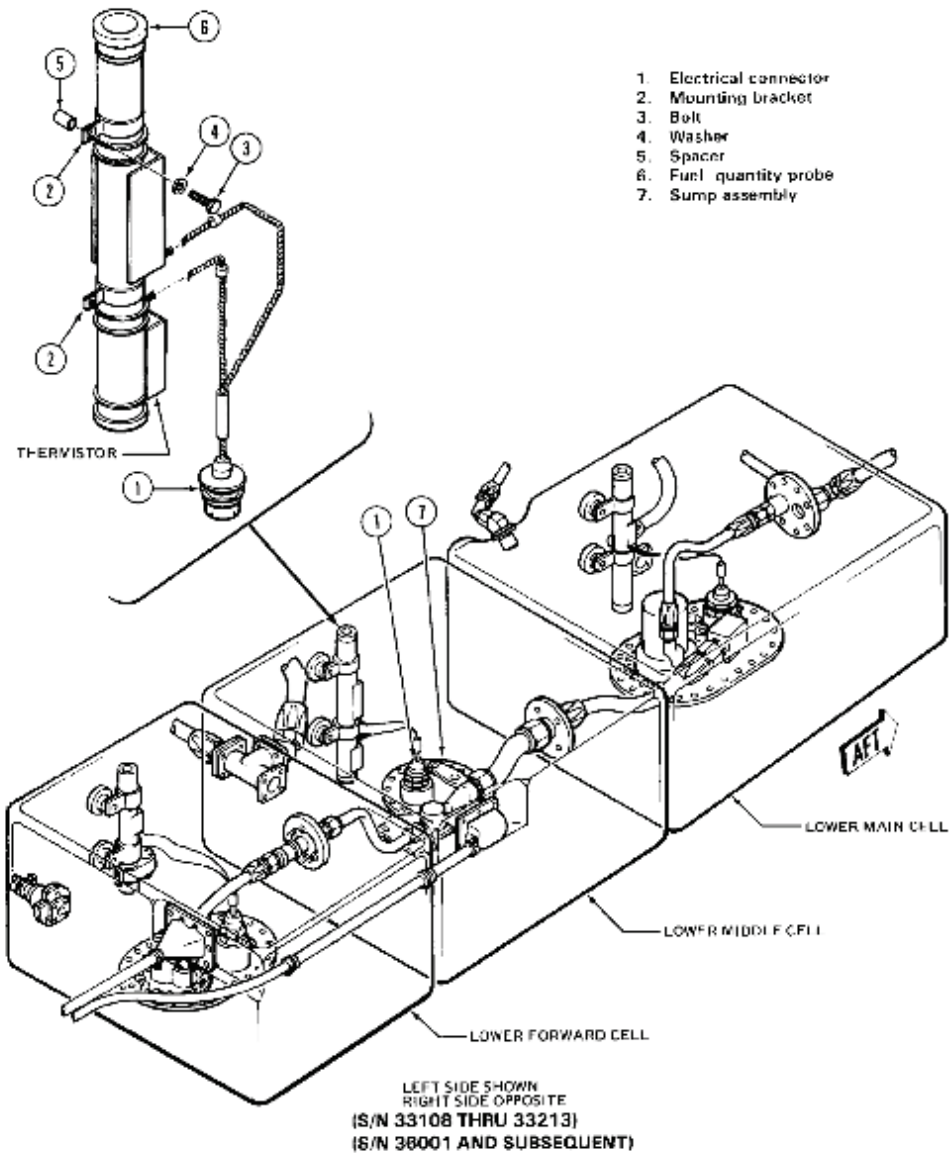
2. Disconnect electrical connector (1).

3. Remove bolts (3), washers (4), and spacers (5).

4. Remove fuel quantity probe (6).

5. Install protective cover over cell sump opening.

6. Install caps or plugs (C-428) on the electrical connector (1).



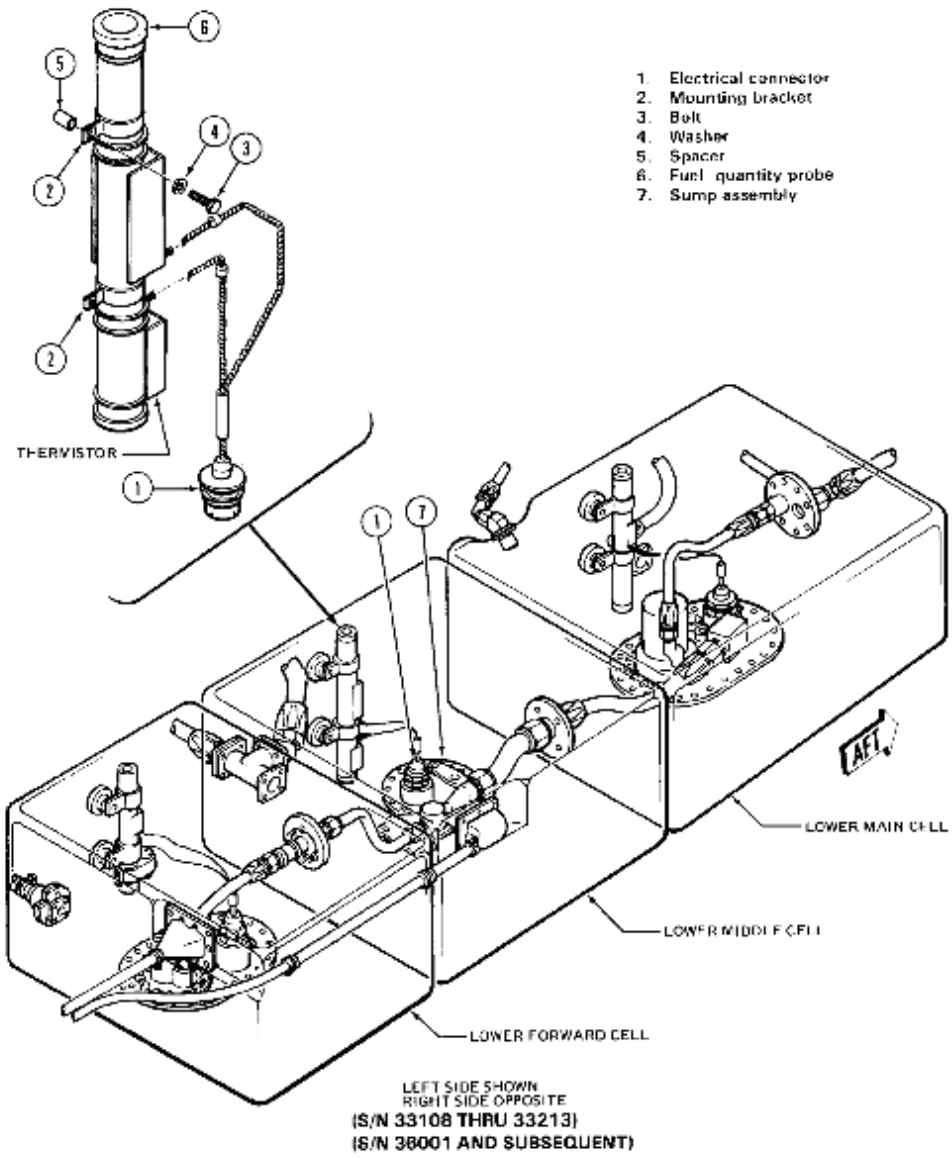
ICN-412-A-281000-B-97499-00098-A-001-01

Figure 1. Lower Middle Cell Fuel Quantity Probe (Sheet 1 of 1)

Lower Middle Cell Fuel Quantity Probes - Installation

Procedure

1. Remove the protective caps and/or plugs from the electrical connector (1, Figure 1).
2. Remove protective cover from cell opening.
3. Insert bolts (3) with washers (4) through mounting brackets (2).
4. Place the spacers (5) on bolts (3) and position fuel quantity probe (6) onto attaching points on inboard wall of cell. Tighten the bolts.
5. Connect electrical connector (1).
6. Install sump assembly (7) (412-A-28-10-00-51A-720A-A / 00490).
7. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).
8. Perform operational check of fuel quantity indicating system (412-A-96-12-00-11A-273A-A / 02006).
9. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).
10. Check area for leaks.

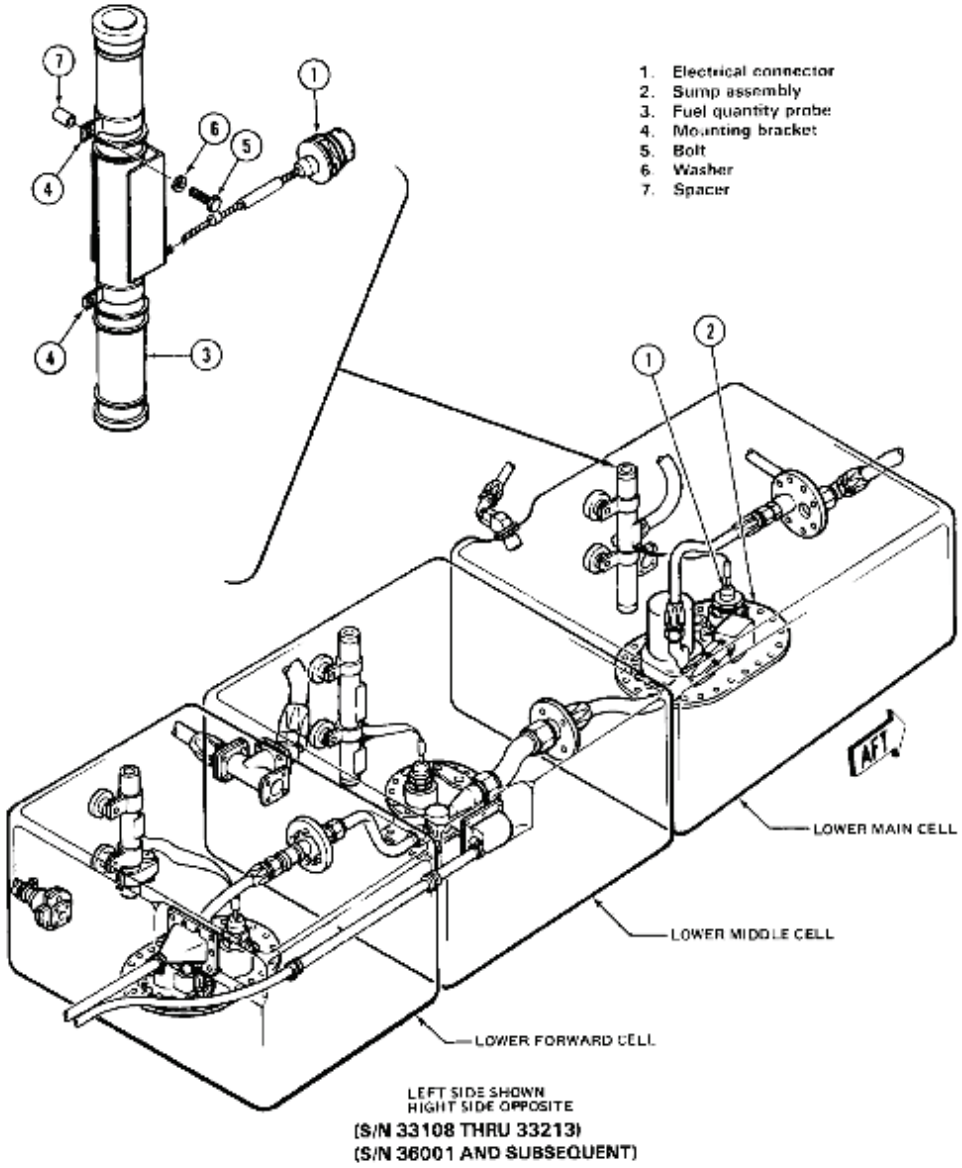


ICN-412-A-281000-B-97499-00098-A-001-01

Figure 1. Lower Middle Cell Fuel Quantity Probe (Sheet 1 of 1)

Lower Main Cell Fuel Quantity Probes - Description

One fuel quantity probe is mounted on the inboard wall of each lower main cell (Figure 1). These probes serve only the quantity measuring function.



ICN-412-A-281000-B-97499-00099-A-001-01

Figure 1. Lower Main Cell Fuel Quantity Probe (Sheet 1 of 1)

Lower Main Cell Fuel Quantity Probes - Removal

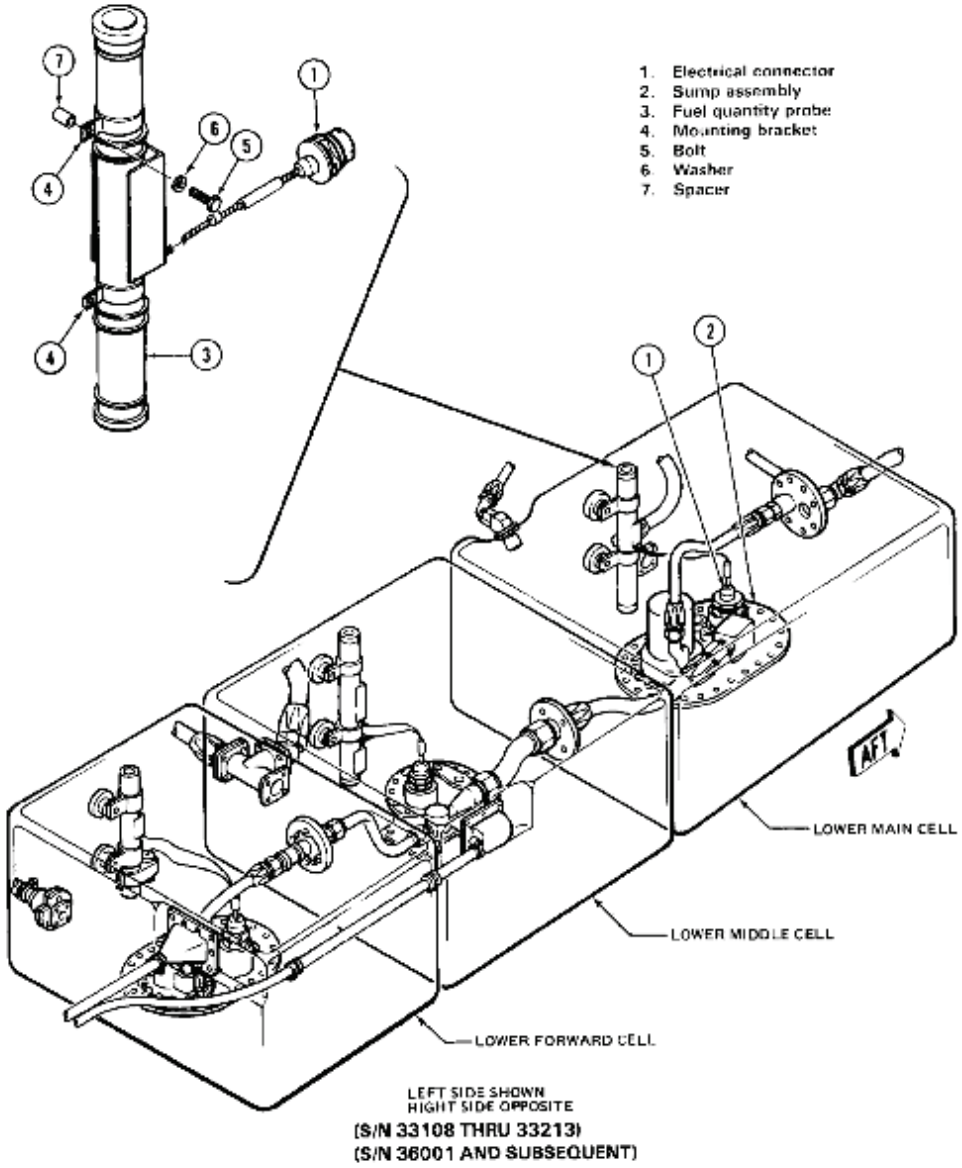
Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure

1. Remove sump assembly (2, Figure 1) (412-A-28-10-00-57A-520A-A / 00509).
2. Disconnect electrical connector (1).
3. Remove bolts (5), washers (6), and spacers (7).
4. Remove fuel quantity probe (3).
5. Install protective cover over cell sump opening.
6. Install caps or plugs (C-428) on the electrical connector (1).



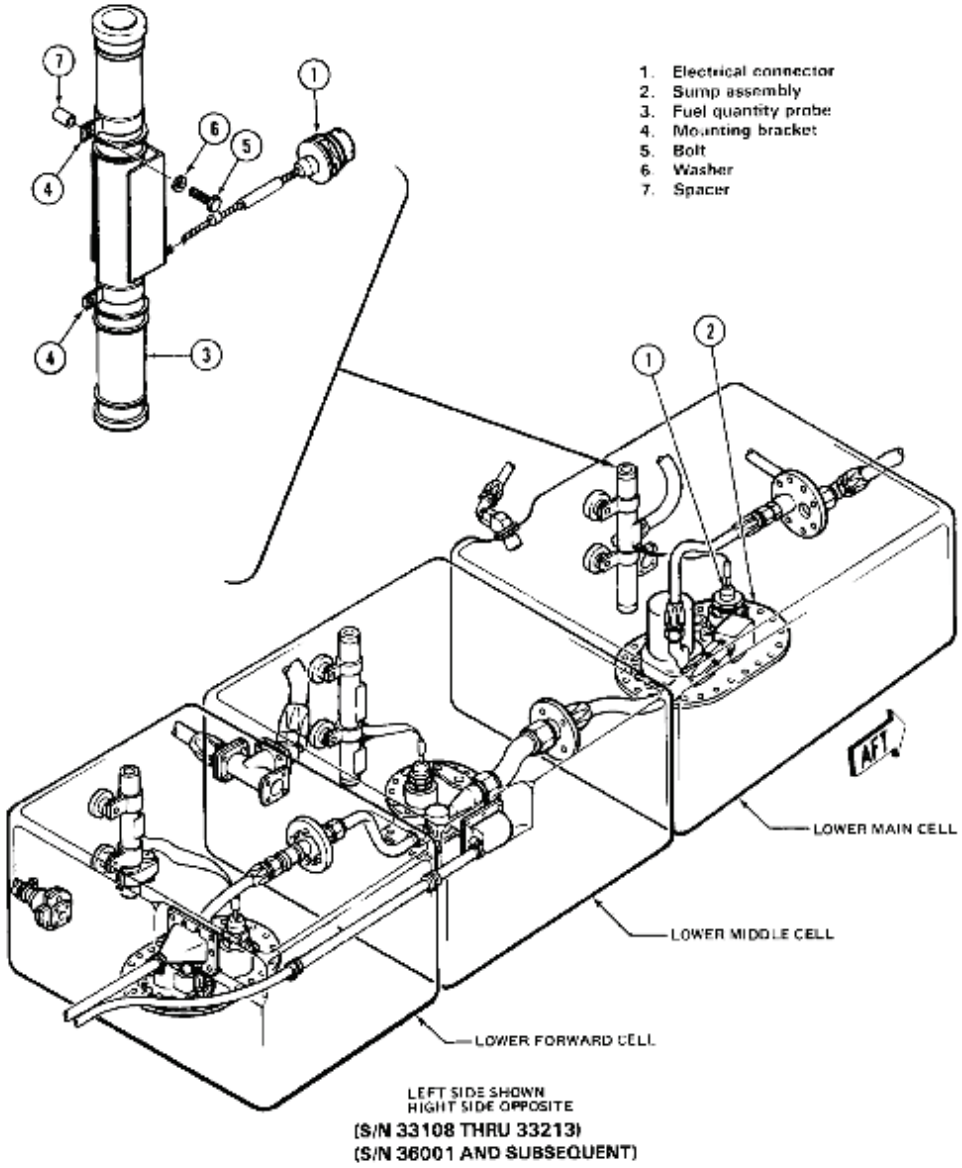
ICN-412-A-281000-B-97499-00099-A-001-01

Figure 1. Lower Main Cell Fuel Quantity Probe (Sheet 1 of 1)

Lower Main Cell Fuel Quantity Probes - Installation

Procedure

1. Remove the protective caps and/or plugs from the electrical connector (1, Figure 1).
2. Remove protective cover from cell sump opening.
3. Insert bolts (5) with washers (6) through mounting brackets (4).
4. Place spacers (7) on bolts (5) and position fuel quantity probe (3) onto attaching points on inboard wall of cell. Tighten bolts (5).
5. Connect electrical connector (1).
6. Install sump assembly (2) (412-A-28-10-00-57A-720A-A / 00511).
7. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).
8. Perform operational check of fuel quantity indicating system (412-A-96-12-00-11A-273A-A / 02006).
9. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).
10. Check area for leaks.



ICN-412-A-281000-B-97499-00099-A-001-01

Figure 1. Lower Main Cell Fuel Quantity Probe (Sheet 1 of 1)

Upper Left and Upper Right Cell Fuel Quantity Probes - Description

The fuel quantity probe in the upper left cell is connected electrically to signal conditioner No. 1 while probe in upper right cell is connected to signal conditioner No. 2. Each probe contains a fuel level sensor (thermistor) at its base that signals the corresponding NO. 1 FUEL LOW or NO. 2 FUEL LOW segment in the caution panel that the cell has become empty.

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Upper Left and Upper Right Cell Fuel Quantity Probes - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Caps or plugs	C-428	AR	

Procedure

1. Disconnect electrical power.

2. Defuel system (412-A-12-10-01-00A-221A-A / 00131).

3. Remove engine deck panel.

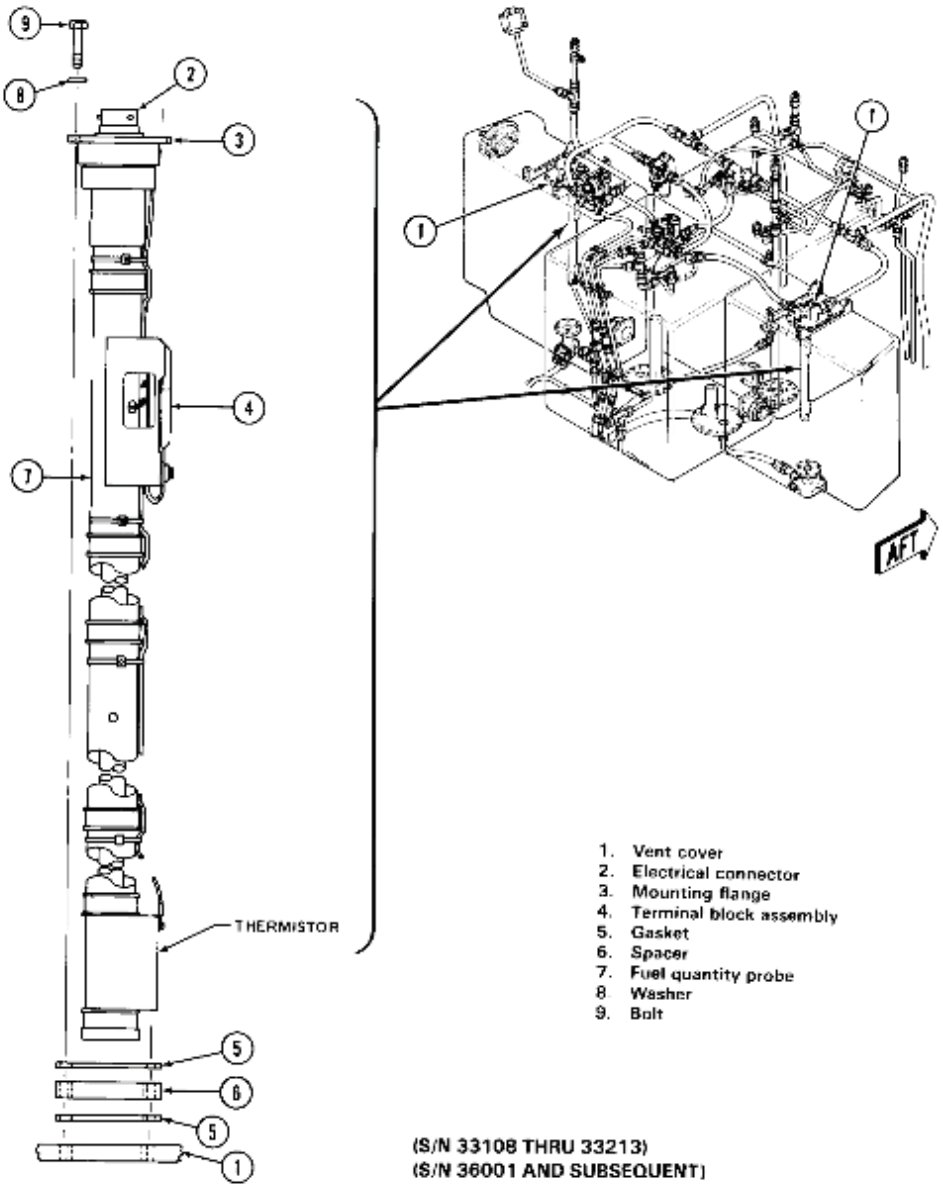
4. Disconnect electrical connector (2, Figure 1).

5. Remove bolts (9) and washer (8).

6. Remove fuel quantity probe (7), gaskets (5), and spacer (6).

7. Install protective cover over cell opening.

8. Install caps or plugs (C-428) on the electrical connector (2).



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Figure 1. Upper Left and Upper Right Cell Fuel Quantity Probe (Sheet 1 of 1)

Upper Left and Upper Right Cell Fuel Quantity Probes - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

1. Remove the protective caps and/or plugs from the electrical connector (2, Figure 1).

2. Remove protective cover from cell opening.

3. Place gasket (5), spacer (6), and a second gasket (5) on fuel quantity probe (7). Position these items near mounting flange (3) of fuel quantity probe (7).

4. Insert fuel quantity probe (7) into cell. Rotate fuel quantity probe (7) so terminal block assembly (4) faces aft.

5. Align holes in mounting flange (3) with those in gasket (5), spacer (6), and vent cover (1).

6. Install bolts (9) with washers (8). Tighten bolts (9).

7. Safety the bolts (9) with lockwire (C-405).

8. Connect electrical connector (2).

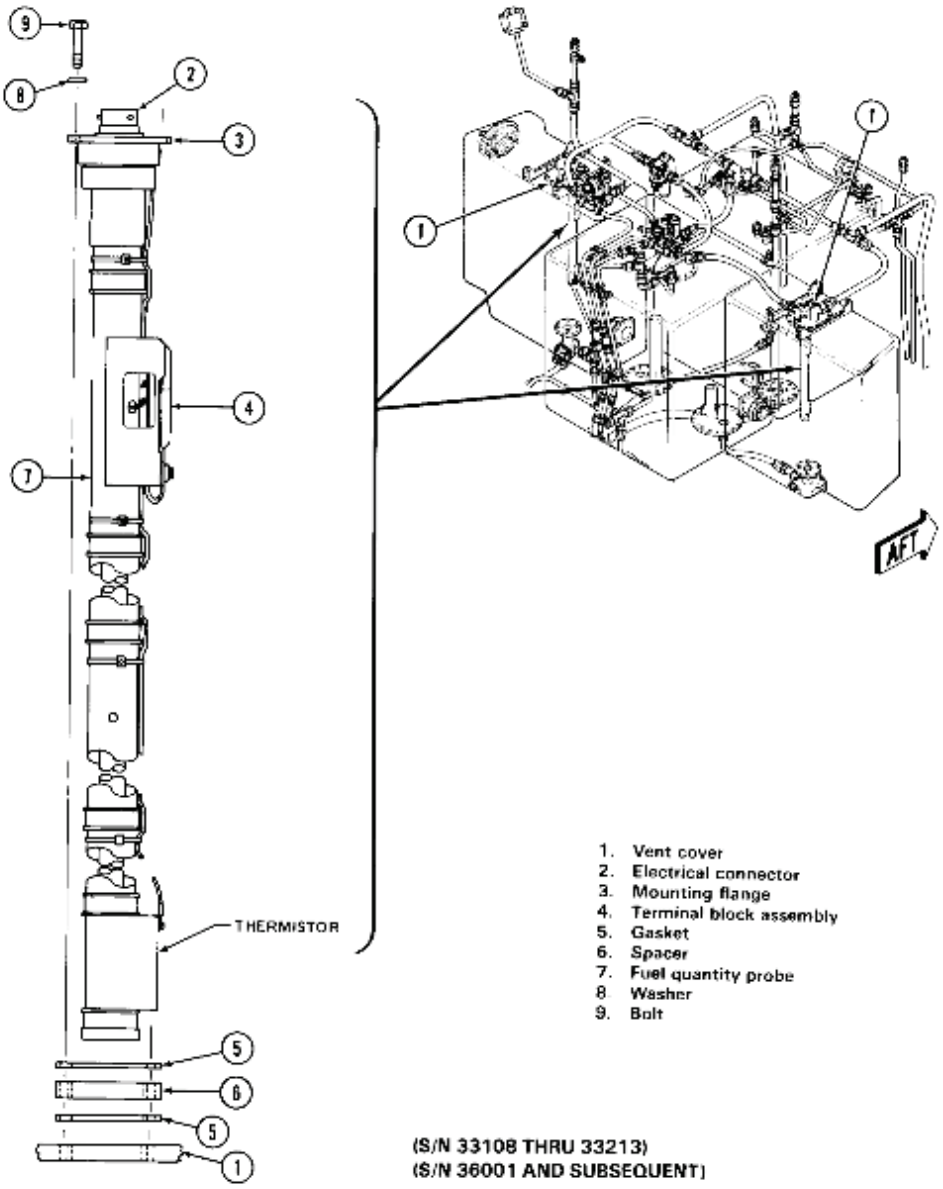
9. Perform fuel system pressure test (412-A-28-10-00-00A-362B-A / 00438).

10. Perform operational check of fuel quantity indicating system (412-A-96-12-00-11A-273A-A / 02006).

11. Service fuel system (412-A-12-00-00-00A-040A-A / 00126).

12. Check area for leaks.

13. Install sump access door and engine deck panel.



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Figure 1. Upper Left and Upper Right Cell Fuel Quantity Probe (Sheet 1 of 1)

CHAPTER 29
HYDRAULIC SYSTEM

MAINTENANCE MANUAL

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List of Effective Data Modules

The listed documents are included in issue 007, dated 2025-11-18, of this publication.

C = Changed data module, N = New data module

Document title	Data module code / Seq number	Issue date	Applicable to
Hydraulic System - Description	DMC-412-A-29-00-00-00A-028A-A / 00611	2023-05-31	ALL
Hydraulic System - General Maintenance Practices Description	DMC-412-A-29-00-00-00A-042A-A / 00612	2023-05-31	ALL
Hydraulic System - Troubleshooting	DMC-412-A-29-00-00-00A-422A-A / 00613	2023-05-31	ALL
Hydraulic System - Leakage Limits	DMC-412-A-29-00-00-00A-042B-A / 00614	2025-05-13	ALL
Hydraulic System - Flushing	DMC-412-A-29-00-00-00A-251A-A / 00615	2023-05-31	ALL
Hydraulic System - Operational Checks (Using Test Stand)	DMC-412-A-29-00-00-00A-320A-A / 00616	2023-05-31	ALL
Hydraulic System - Operational Checks (Using Helicopter Power)	DMC-412-A-29-00-00-00A-320B-A / 00617	2023-05-31	ALL
Hydraulic System Components - Description	DMC-412-A-29-10-00-00A-040A-A / 00618	2023-05-31	ALL
Hydraulic Pump (Hydraulic System No. 1) - Description	DMC-412-A-29-10-00-01A-042A-A / 00619	2023-05-31	ALL
Hydraulic Pump (Hydraulic System No. 1) - Removal	DMC-412-A-29-10-00-01A-520A-A / 00620	2023-05-31	ALL
Hydraulic Pump (Hydraulic System No. 1) - Inspection	DMC-412-A-29-10-00-01A-280A-A / 00621	2023-05-31	ALL
Hydraulic Pump (Hydraulic System No. 1) - Repair	DMC-412-A-29-10-00-01A-685A-A / 00622	2023-05-31	ALL
Hydraulic Pump (Hydraulic System No. 1) - Installation	DMC-412-A-29-10-00-01A-720A-A / 00623	2023-05-31	ALL
Hydraulic Pump (Hydraulic System No. 2) - Description	DMC-412-A-29-10-00-02A-042A-A / 00624	2023-05-31	ALL
Hydraulic Pump (Hydraulic System No. 2) - Removal	DMC-412-A-29-10-00-02A-520A-A / 00625	2023-05-31	ALL
Hydraulic Pump (Hydraulic System No. 2) - Inspection	DMC-412-A-29-10-00-02A-280A-A / 00626	2023-05-31	ALL
Hydraulic Pump (Hydraulic System No. 2) - Repair	DMC-412-A-29-10-00-02A-685A-A / 00627	2023-05-31	ALL
Hydraulic Pump (Hydraulic System No. 2) - Installation	DMC-412-A-29-10-00-02A-720A-A / 00628	2023-05-31	ALL
Hydraulic Reservoir - Description	DMC-412-A-29-10-00-03A-042A-A / 00629	2023-05-31	ALL
Hydraulic Reservoir - Removal	DMC-412-A-29-10-00-03A-520A-A / 00630	2023-05-31	ALL
Hydraulic Reservoir - Inspection	DMC-412-A-29-10-00-03A-280A-A / 00631	2023-05-31	ALL
Hydraulic Reservoir - Repair	DMC-412-A-29-10-00-03A-685A-A / 00632	2023-05-31	ALL
Hydraulic Reservoir - Installation	DMC-412-A-29-10-00-03A-720A-A / 00633	2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Integrated Valve and Filter Assembly - Description	DMC-412-A-29-10-00-04A-042A-A / 00634	2023-05-31	ALL
Integrated Valve and Filter Assembly - Removal	DMC-412-A-29-10-00-04A-520A-A / 00635	2023-05-31	ALL
Integrated Valve and Filter Assembly - Inspection	DMC-412-A-29-10-00-04A-280A-A / 00636	2023-05-31	ALL
Integrated Valve and Filter Assembly - Installation	DMC-412-A-29-10-00-04A-720A-A / 00637	2023-05-31	ALL
Hydraulic Filter Assemblies - Description	DMC-412-A-29-10-00-05A-042A-A / 00638	2023-05-31	ALL
Hydraulic Filter Assembly (212-076-006-007) - Removal	DMC-412-A-29-10-00-05A-520A-A / 00639	2025-05-13	ALL
Hydraulic Filter Assembly (212-076-006-105) - Removal	DMC-412-A-29-10-00-05A-520B-A / 00640	2025-05-13	ALL
Hydraulic Filter Assembly (212-076-006-007) - Inspection	DMC-412-A-29-10-00-05A-280A-A / 00641	2025-05-13	ALL
Hydraulic Filter Assembly (212-076-006-105) - Inspection	DMC-412-A-29-10-00-05A-280B-A / 00642	2025-05-13	ALL
Hydraulic Filter Assemblies (212-076-006-007 and -105) - Cleaning	DMC-412-A-29-10-00-05A-251A-A / 00643	2025-05-13	ALL
Hydraulic Filter Assembly (212-076-006-007) - Installation	DMC-412-A-29-10-00-05A-720A-A / 00644	2025-05-13	ALL
Hydraulic Filter Assembly (212-076-006-105) - Installation	DMC-412-A-29-10-00-05A-720B-A / 00645	2025-05-13	ALL
Elbow Check Valve - Description	DMC-412-A-29-10-00-06A-042A-A / 00646	2023-05-31	ALL
Elbow Check Valve - Removal	DMC-412-A-29-10-00-06A-520A-A / 00647	2023-05-31	ALL
Elbow Check Valve - Inspection	DMC-412-A-29-10-00-06A-280A-A / 00648	2023-05-31	ALL
Elbow Check Valve - Installation	DMC-412-A-29-10-00-06A-720A-A / 00649	2023-05-31	ALL
Inline Check Valve - Description	DMC-412-A-29-10-00-07A-042A-A / 00650	2023-05-31	ALL
Inline Check Valve - Removal	DMC-412-A-29-10-00-07A-520A-A / 00651	2023-05-31	ALL
Inline Check Valve - Inspection	DMC-412-A-29-10-00-07A-280A-A / 00652	2023-05-31	ALL
Inline Check Valve - Installation	DMC-412-A-29-10-00-07A-720A-A / 00653	2023-05-31	ALL
Pressure Switch - Description	DMC-412-A-29-10-00-08A-042A-A / 00654	2023-05-31	ALL
Pressure Switch - Removal	DMC-412-A-29-10-00-08A-520A-A / 00655	2023-05-31	ALL
Pressure Switch - Installation	DMC-412-A-29-10-00-08A-720A-A / 00656	2023-05-31	ALL
Hydraulic Accumulator - Description	DMC-412-A-29-10-00-09A-042A-A / 00657	2023-05-31	ALL
Hydraulic Accumulator - Removal	DMC-412-A-29-10-00-09A-520A-A / 00658	2023-05-31	SN: 33001-33213 or 36001-36627

Document title	Data module code / Seq number	Issue date	Applicable to
Hydraulic Accumulator - Inspection	DMC-412-A-29-10-00-09A-280A-A / 00659	2023-05-31	SN: 33001-33213 or 36001-36627
Hydraulic Accumulator - Installation	DMC-412-A-29-10-00-09A-720A-A / 00660	2023-05-31	SN: 33001-33213 or 36001-36627
Three-Way, Two-Position, Pressure-Operated Valve - Description	DMC-412-A-29-10-00-10A-042A-A / 00661	2023-05-31	ALL
Three-Way, Two-Position, Pressure-Operated Valve - Removal	DMC-412-A-29-10-00-10A-520A-A / 00662	2023-05-31	ALL
Three-Way, Two-Position, Pressure-Operated Valve - Inspection	DMC-412-A-29-10-00-10A-280A-A / 00663	2023-05-31	ALL
Three-Way, Two-Position, Pressure-Operated Valve - Installation	DMC-412-A-29-10-00-10A-720A-A / 00664	2023-05-31	ALL
Hydraulic Check Valve - Description	DMC-412-A-29-10-00-11A-042A-A / 00665	2023-05-31	ALL
Hydraulic Check Valve - Removal	DMC-412-A-29-10-00-11A-520A-A / 00666	2023-05-31	ALL
Hydraulic Check Valve - Inspection	DMC-412-A-29-10-00-11A-280A-A / 00667	2023-05-31	ALL
Hydraulic Check Valve - Repair	DMC-412-A-29-10-00-11A-685A-A / 00668	2023-05-31	ALL
Hydraulic Check Valve - Installation	DMC-412-A-29-10-00-11A-720A-A / 00669	2023-05-31	ALL
Pressure Transmitter - Description	DMC-412-A-29-10-00-12A-042A-A / 00670	2023-05-31	ALL
Pressure Transmitter - Removal	DMC-412-A-29-10-00-12A-520A-A / 00671	2023-05-31	ALL
Pressure Transmitter - Inspection	DMC-412-A-29-10-00-12A-280A-A / 00672	2023-05-31	ALL
Pressure Transmitter - Installation	DMC-412-A-29-10-00-12A-720A-A / 00673	2023-05-31	ALL
Pulsation Damper (Hydraulic System No. 2) - Description	DMC-412-A-29-10-00-13A-042A-A / 00674	2023-05-31	ALL
Pulsation Damper (Hydraulic System No. 2) - Removal	DMC-412-A-29-10-00-13A-520A-A / 00675	2023-05-31	ALL
Pulsation Damper (Hydraulic System No. 2) - Installation	DMC-412-A-29-10-00-13A-720A-A / 00676	2023-05-31	ALL

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Hydraulic System - Description

The hydraulic system is comprised of two independent systems identified as hydraulic system No. 1 and hydraulic system No. 2. Each system has a reservoir, variable-delivery pump, integrated valve and filter assembly, accumulator, and check valves. Each pump is powered by a separate drive quill driven by the transmission. The valve and filter assembly contains filters, solenoid valves, and relief valves.

The three dual hydraulic servo actuators in cyclic and collective flight control systems are operated by both hydraulic systems. Two isolated sets of passages and piston chambers inside the servo actuators maintain the integrity of each system. In the event of a failure in one system, the remaining system will continue to operate the servo actuators.

The directional control (anti-torque) system servo actuator is operated by hydraulic system No. 1 only; therefore, hydraulic system No. 1 has a larger rated volume pump than hydraulic system No. 2.

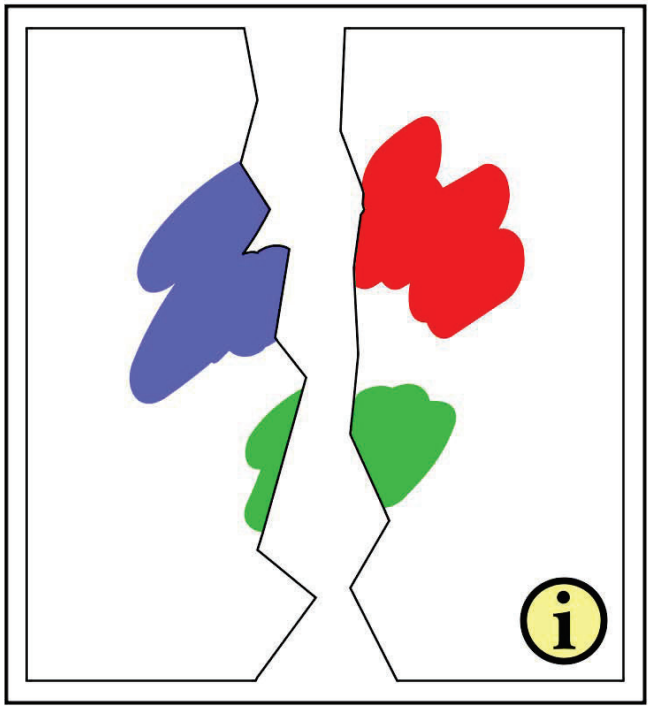
Hydraulic system No. 2 has a pulsation damper between the pump and integrated valve and filter assembly. Hydraulic system No. 1 does not have a pulsation damper.

Both hydraulic systems have a temperature bulb located at the reservoir inlet and a pressure transmitter at the module outlet. These sensors are connected to a hydraulic pressure/temperature gauge on the instrument panel. In addition, a temperature switch at the reservoir inlet causes the appropriate NO.1 HYDRAULIC or NO. 2 HYDRAULIC caution segment on the caution/warning panel to illuminate if hydraulic fluid temperature reaches 190°F (88°C). A pressure switch at the module outlet energizes the same segment if pressure drops below 650 PSI (4481.5 kPa).

Each valve and filter assembly has two red indicator buttons for the two filters, and is also electrically connected to a remote warning indicator located in the cabin nose (visible through the right nose window) forward and below pedestal of the pilot station. If any of the four filters becomes clogged and causes a differential pressure of 70 ±10 PSI (483 ±69 kPa) across the element, this differential pressure will cause the adjacent red button to extend and the remote indicator to change from green to red or, cause the “cats-eye” indicator to turn from black to white, as applicable. The remote indicator will remain red, and the red button will remain extended until the clogged filter element is cleaned or changed and both indicators are reset. The remote indicator is intended to provide an impending bypass indication. If the remote indicator keeps going back to red after being reset and the indicators on the filter and valve assemblies are not extended, the remote indicator is very likely defective. It will be acceptable to return helicopter to service with noted condition, provided the indicators on the filter and valve assemblies are not extended and until such time the remote indicator may be repaired.

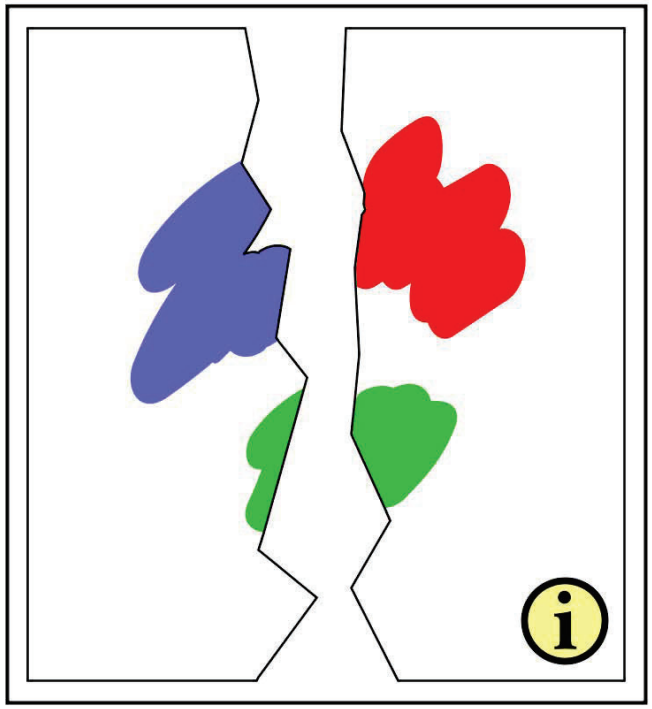
In normal operation of each system, hydraulic fluid is supplied from non-pressurized reservoirs by gravity feed and suction to a transmission driven pump. The pumps are variable delivery types with internal pressure compensation, preset to provide 1000 ±25 PSI (6895 ±172 kPa) output pressure at operating RPM, according to system demands. Pump output is delivered to the module and passes through the pressure filter. A relief valve in the module guards the system against excessive pressure, being set to open at 1100 PSI (7584 kPa). The system solenoid valve is normally de-energized and open to the SYS PRESS outlet of the module but can be electrically energized to the OFF position by means of the HYDR SYS NO. 1 and HYDR SYS NO. 2 switches. Figure 1 provides a hydraulic system schematic.

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Figure 1. Hydraulic System Schematic (Sheet 1 of 2)



ICN-412-A-290000-B-97499-00002-A-001-01

Figure 1. Hydraulic System Schematic (Sheet 2 of 2)

Hydraulic System - General Maintenance Practices Description

The maintenance procedures paragraphs provide troubleshooting, leakage limits, and operational checks using a test stand and using helicopter power.

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Hydraulic System - Troubleshooting

Procedure

1. The following list of probable causes, isolation procedures, and remedies is intended to aid in hydraulic system troubleshooting. The guide should be used with other sources of information such as: hydraulic schematic illustration, electrical diagrams, operational check, and other detailed procedures in this chapter.

NOTE

Perform operational check of hydraulic system following any corrective action or component replacement.

Table 1. Hydraulic System Troubleshooting

PROBABLE CAUSE	ISOLATION PROCEDURE	CORRECTIVE ACTION
Indication of Trouble: One System Lacks Power and Pressure is Low		
Leaks in system.	Visually check for leaks due to loose connections, line rupture, component rupture, or failed seals.	Repair connections; replace faulty lines, hoses, seals, or components. Service system and perform operational check.
Hydraulic pump malfunction.	Check that system can be powered by hydraulic test unit.	Replace hydraulic pump.
System check valve stuck closed in pump pressure line.	If operation on hydraulic test unit is normal but operation with a known serviceable pump remains faulty, check valve may be defective.	Replace check valve.
Reservoir vent obstructed.	If system pressure is intermittently low when operating on pump though normal with test unit, check reservoir for obstructed vent causing pump cavitation.	Clear reservoir vent screen or replace reservoir. Operationally check for possible hydraulic pump damage.
System solenoid valve remaining in Off position.	Electrically check that solenoid is not energized. If not a circuit fault, valve may be mechanically stuck.	Repair electrical circuit, or replace valve and filter assembly.
Pressure-operated shutoff valve in reservoir return line locked in energized (test coupling) position.	If system is normal on test unit but controls are not powered by pump operation, valve may be defective.	Replace valve.
Reservoir return port screen clogged.	Visually check reservoir return screen.	Clean screen or replace reservoir.
Pressure-operated shutoff valve or internal check valve in integrated valve and filter assembly malfunctioning.	Disconnect reservoir return line. With system on, move controls and check for steady flow. If not, valve is faulty.	Replace integrated valve and filter assembly.
Indication of Trouble: Pressure Fluctuation With Controls Motionless		
Airlock in line between pressure transmitter and restrictor.	Operating on ground test unit, briefly loosen line connection on elbow at P port of pressure transmitter. Tighten connector after bleeding, and recheck pressure indication.	If pressure is steady after bleeding line, no further action is required. If fluctuation continues, check transmitter and gauge, replacing faulty units.
Reservoir return screen ruptured, or vent screen partially obstructed.	Check for foaming in reservoir after rapid control movements, as indication of return screen damage. Inspect reservoir screens.	Clear obstructed vent, or replace reservoir.
Indication of Trouble: Filter Indicators Tripped		
Filters clogged.	If indicator in cabin nose is red, check both system filter assemblies for tripped indicators. Manually reset indicators and operate system. Move controls rapidly. If indicators trip again, filters are clogged.	Clean filter elements as required. Reset module indicators and remote indicator.
Indicators defective.	Indicators trip with known serviceable filter elements.	Replace defective integrated valve and filter assembly.

Table 1. Hydraulic System Troubleshooting (continued)

PROBABLE CAUSE	ISOLATION PROCEDURE	CORRECTIVE ACTION
Remote indicator defective.	Indicator in cabin nose is red and neither system filter assembly indicators are extended.	Replace or repair defective remote indicator.
Indication of Trouble: Either Hydraulic System Overheats		
	NOTE Hydraulic system No. 2 operates at slightly higher temperature than hydraulic system No. 1 due to difference in hydraulic pump RPM.	
Hydraulic pump defective, allows increasing pressure.	Use a test gauge to check system pressure, 1050 PSI (7239.5 kPa) maximum.	Replace hydraulic pump.
System relief valve held fully open.	Check for loss of control response, and for flow through relief valve.	Replace integrated valve and filter assembly.
Indication of Trouble: Hydraulics and Master Caution Lights On		
Hydraulic system No. 1 or No. 2 overheating.	Check temperature gauge for temperature higher than 190°F (88°C).	Refer to troubleshooting data for Either Hydraulic System Overheats.
Hydraulic system No. 1 or No. 2 has low pressure.	Check pressure gauge for low pressure below 650 PSI (4481.5 kPa).	Refer to troubleshooting data for One System Lacks Power and Pressure is Low.
Temperature switch malfunctioning.	Temperature gauge indicates normal.	Replace switch of each hydraulic system independently and recheck system.
Pressure switch malfunctioning.	Pressure gauge indicates normal.	Replace switch of each hydraulic system independently and recheck system.
Indication of Trouble: Controls Do Not Operate Smoothly		
Servo or bypass valve sticking.	With hydraulic pressure off, disconnect control tube from servo input lever of dual actuator. Manually check force required to displace lever 0.070 inch (1.78 mm), not to exceed 32 ounces (142 N).	1. Check lever pivot for binding. Clean or replace bolt. 2. Flush system thoroughly. Recheck operation. 3. Replace seals on servo or bypass valves, or replace actuator.
Control tube bearing rotational force excessive.	With hydraulic pressure off, disconnect control tubes from cyclic and collective control cylinder servo input levers. Manually check force required to rotate inner race (ball) of the bearing, not to exceed 4 inch-ounces (0.028 Nm) (412-A-67-01-00-00A-042A-A / 01322).	Replace control tube.
Binding in flight control linkage.	Check linkage components for freedom of motion. Check for loose clamps and dislocated boots on control linkage under aft center floor panel of cabin.	Replace defective parts.
Indication of Trouble: Inadequate Control Response		
Internal leakage past piston in actuator.	Check actuator is only partially powered.	Replace servo actuator.
Check valve stuck closed in pressure port of actuator or in return port of integrated valve and filter assembly.	Substitute valve with one known to be serviceable. Operationally check system.	Replace check valve.
Indication of Trouble: Reservoir Overflows When Using Ground Test Unit		
Test unit pressure too low.	Check test unit setting.	Increase test unit pressure above 550 PSI (3790 kPa).
Pressure-operated shutoff valve (212-076-008) malfunction.	Check for increased overflow when flight controls are moved.	Replace valve.

Table 1. Hydraulic System Troubleshooting (continued)

PROBABLE CAUSE	ISOLATION PROCEDURE	CORRECTIVE ACTION
System check valve (212-076-009) in pump pressure line open.	Check for overflow with no control motion. Rotor may rotate slowly when on ground cart.	Replace valve.
Pump check valve (205-076-365) malfunction	Check for increased overflow when pressure on the ground cart is increased.	Replace check valve.
Indication of Trouble: High Frequency Vibrations Due to Tail Rotor Controls		
Stiffness or binding in action of tail rotor control hydraulic cylinder.	1. Cylinder input lever binding on spacer at bellcrank. (Move lever laterally to check.)	1. Polish spacer O.D. until free in lever bearing.
	2. Cylinder trunnions binding in mounting plates. (Move cylinder laterally 0.003 to 0.050 inch (0.076 to 1.270 mm).	2. Polish trunnion plate bushings I.D. for free fit.
	3. Cylinder input lever binding at pivot.	3. Repair cylinder assembly per overhaul instructions.
Cylinder input lever bearing rotational force excessive.	Disconnect anti-torque control cylinder input lever from bellcrank. Check cylinder lever bearing rotational force not to exceed 8 inch-ounces (0.055 Nm) (412-A-67-12-00-02A-280A-A / 01406).	Repair lever bearing or replace cylinder (412-A-67-12-02-02A-520A-A / 01573).
Binding in tail rotor controls.	1. Check pitch control tube for binding in bearing in left side of tail rotor gearbox.	1. Remove pitch change tube. Clean I.D. of bearing and O.D. of tube. Reinstall and recheck.
	2. Check lever on left side of tail rotor gearbox for stiffness of lower end bearing in alignment motion. If stiffness occurs, twist attached link to turn nut end of bolt upward. Pull nut downward with a spring scale. If force to move bearing exceeds 10 pounds (4.536 kg), lever is unserviceable.	2. Replace unserviceable lever.
Indication of Trouble: Motoring Servo Actuator in Tail Rotor Controls		
Servo or bypass valves sticking.	Disconnect anti-torque control cylinder input lever from bellcrank. Manually check force to operate lever at bearing, not to exceed 24 inch-ounces (0.165 Nm) (412-A-67-12-00-02A-280A-A / 01406).	Replace cylinder (412-A-67-12-02-02A-520A-A / 01573).
Indication of Trouble: Collective Stick Light or Heavy on Downstroke		
Collective minimum friction incorrect.	Check friction preload on collective stick.	Adjust minimum friction (412-A-67-01-00-01A-276A-A / 01323).
Balance spring on collective cylinder defective.	Check condition of balance spring on input lever of collective hydraulic cylinder.	Replace defective spring.
Indication of Trouble: Inability to Make Normal Right and Left Turns in Flight		
Tail rotor controls rigging incorrect.	Check tail rotor rigging.	Rig tail rotor controls.
Indication of Trouble: Binding in Tail Rotor Movement		
Pitch change tube binding in bearing at left side of tail rotor gearbox.	Manually check tube for free movement through full range of travel.	Clean surfaces of tube and bearing.
Binding bearing in lower end of lever on left side of tail rotor gearbox.	Manually check bearing.	Replace lever.

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Hydraulic System - Leakage Limits

The following provides guidelines for allowable external leakage of in-service hydraulic system components and suggested methods of measuring such leakage.

Scope - Limits described are only for in-service helicopter hydraulic system components. Intent is to minimize replacement of hydraulic components which are still serviceable.

These limits may differ from those contained in various specifications for components, which are intended to control quality, assembly, and proper functioning. Components in service sometimes develop leakage rates in excess of specification limits, without necessarily becoming detrimental to the system or failing to allow for reliable operation.

These limits are not to be used as a basis for the acceptance or rejection of components of any bench functional test or of any systems on new helicopters.

These limits are not applicable to self-contained closed-compartment hydraulic units such as viscous dampers, liquid springs, or oleo struts.

Causes of Leakage - Some seepage is normally present, since static or dynamic seals are not functionally perfect, due to such causes as follows:

A film of hydraulic fluid being retained by metal surfaces, such as piston rods, and thus carried past seals. This film is necessary for seal lubrication.

Pressure and temperature variations affecting seals.

Seals tending to take a permanent set after a period of time.

Classification of Leakage - External leakages of hydraulic fluid can be broadly classified as excessive or allowable.

Excessive Leakage: Fluid leakage such that hydraulic reservoir level may be dangerously lowered or depleted during normal operation, or fire hazard created, or airworthiness of helicopter otherwise compromised.

Allowable Leakage: Fluid leakage such that quantity lost is insignificant, no detrimental effect on helicopter operation, and correction does not warrant maintenance time involved.

General: Leakage usually shows as a seepage, stain, or wet area. It is possible for allowable leakage or seepage to collect in a cavity or depression in adjacent structure over a period of time and falsely indicate excessive leakage. Accumulation on a flat area or a white-painted surface often appears to be excessive, though actually being allowable. It is also possible to have enough components with allowable leakages that the combined leakage is excessive.

Leakage Checks - Measurement of leakage rates, for classification according to Table 1, are performed as follows:

When hydraulic systems have remained in a static, unpressurized condition for an appreciable period of time, leakage checks should not be performed immediately after starting. Activate systems and cycle components several times. Wipe off any accumulated hydraulic fluid before making leakage checks.

Where location of a component does not permit direct observation, measure leakage on a flat surface, either part of structure below the component or a panel temporarily positioned for that purpose. Wipe surface clean and place a drop of fluid on area, allow to stabilize, then outline area with soft lead pencil before wiping off fluid. Pressurize system and cycle component to observe leakage rate, comparing wetted surface to marked one-drop area.

Where fluid leaking from a component can be directly observed, pressurize and cycle the component until a drop falls free. Continue operating, observing time until next drop to determine leakage rate.

For tests requiring long periods of time and whereby fluid can drop, wipe surface clean and dry without using a solvent. Use a clean blotter or white cloth after system has operated or has been idle for the required period of time.

Table 1. Permitted Leakage for the In-Service Hydraulic Components

COMPONENT	FUNCTION	LEAK TYPE	PERMITTED LEAKAGE RATE
FLIGHT CONTROL ACTUATORS	Rod seal	D	1 drop/20 full stroke cycles
		S-D	1 drop/15 minutes
	End cap	S	2 drops/day
	Valve input	D	1 drop/5 cycles
		S-D	1 drop/5 minutes
	Pressure Switch	S-D	1 drop/5 minutes
	Valve body (weep holes)	S-D	1 drop/5 minutes
PUMPS	Output shaft	D	8 drops/minute
		S-D	1 drop/minute
	Housing (mating surfaces)	D	2 drops/day
VALVES	Body (weep hole)	S	2 drops/day
	Manual stem	D	1 drop/5 cycles
		S-D	1 drop/15 minutes
	Dump valve	S	2 drops/day
FITTINGS	Flared or flareless	S	None
	Compression seals	S	1 drop/30 minutes (less if readily accessible)
CHECK VALVE	Pressure-operated	D	2 drops/25 cycles
		S	None at static pressures of 5 to 1000 PSI (34.5 to 6890.0 kPa)

NOTES:

1. Leak Types:
- *D = Dynamic*
- *S = Static*
- *S-D = Static leakages through dynamic seal*

2. Approximately 20 drops = 1 cubic centimeter

3. Components are in static condition, such as a parked helicopter, are permitted maximum leakage of two drops per seal or packing per day.

Hydraulic System - Flushing

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

Procedure



CLEANING OF HYDRAULIC COMPONENTS SHALL BE ACCOMPLISHED WITH drycleaning solvent (C-304) ONLY. DO NOT USE ALCOHOL.

1. Flush hydraulic system No. 1 and/or hydraulic system No. 2 when evidence of contamination is present in system.
2. Remove, clean, and install reservoir filler strainer and return port.
3. Remove hydraulic system pump, attach a tag stating reason for removal, and forward hydraulic pump to overhaul facility.
4. Remove hydraulic pump supply hose and hydraulic pump pressure hose and flush with drycleaning solvent (C-304). Install hoses.
5. Remove and clean filter bowls. Replace filter elements and install filter bowls.
6. Remove flight control cylinders. Tag cylinders with a statement giving reason for removal and forward cylinders to overhaul facility. Connect pressure and return hoses with reducers (MS21916D6-4).
7. If hydraulic system No. 1 is contaminated, remove directional control cylinder. Attach a tag stating reason for removal, and forward cylinder to overhaul facility. Connect pressure and return hose with reducers (MS21916D6-4).
8. Connect hydraulic test stand to appropriate ground test connections under left rear door of helicopter.



THROUGHOUT THE OPERATIONAL CHECKS, DO NOT OPERATE BELOW 550 PSI. HOWEVER, IF IT BECOMES NECESSARY TO REDUCE SYSTEM PRESSURE TO ZERO, DE-ENERGIZE TEST UNIT FLOW TO HELICOPTER, AND THEN REDUCE SYSTEM PRESSURE. FAILURE TO FOLLOW THE ABOVE INSTRUCTIONS WILL RESULT IN OVERFILLING OF SYSTEM RESERVOIR. TESTS REQUIRING DEACTIVATION OF A SYSTEM SHALL BE ACCOMPLISHED BY DISCONNECTING TEST UNIT FROM THAT SYSTEM.

9. Set test unit pressure relief valve for a cracking pressure of 2100 PSI (14 478.99 kPa) and set test unit to pressure sufficient to maintain a 6.0 gallons (22.7 liters) per minute minimum flow through system.

NOTE

Test unit reservoir shall be used in this flushing procedure.

10. Maintain pressure setting for at least 5 minutes while observing all portions of system for external leakage. Take appropriate action to correct any leakage.
11. Stop fluid flow and decrease pressure. Disconnect hydraulic test stand.
12. Install replacement directional control cylinder and cyclic and collective control cylinders.
13. Install replacement hydraulic pump.
14. Install new hydraulic filter elements.
15. Service reservoir (412-A-12-10-02-00A-040A-A / 00138).
16. Ground run helicopter and operationally check all components (412-A-96-03-03-05A-320A-A / 02100). Check for leaks and indications of tripped clogged filter indicators.

17. Remove, clean, and install reservoir return port strainer.

Hydraulic System - Operational Checks (Using Test Stand)

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

1. Provide a portable hydraulic test unit conforming to the following requirements:

1.1. Thoroughly cleaned and serviced with same type hydraulic fluid used in system.

1.2. Equipped with a 10-micron absolute maximum particle size metal filter through which all fluid passes before leaving unit and a 10-micron filter on return line to test unit before fluid returns to test unit.

1.3. Capable of producing pressure to 1400 PSI (9652.66 kPa) and a minimum flow rate of 6.0 gallons (22.7 L) per minute.

1.4. Equipped with a calibrated pressure gauge with minimum 1500 PSI (10342.14 kPa) capacity.

2. Visually inspect entire hydraulic system of helicopter to ensure all lines and components are secure and appear capable of operation. Obtain access by opening transmission cowling, doors on front and sides of pylon structure in cabin, and door to fuselage compartment in which tail rotor control actuator is located.

3. Prepare portable test unit for operation: Relief valve set for 1300 PSI (8963.18 kPa) pressure; pump set for at least 6 gallons per minute (22.7 L/min) flow.

4. Remove test coupling access doors below left cargo door opening, at approximately station 150.0 for hydraulic system No. 1 and station 134.0 for hydraulic system No. 2.



THROUGHOUT OPERATIONAL CHECKS, DO NOT OPERATE BELOW 550 PSI. IF NECESSARY TO REDUCE SYSTEM PRESSURE TO ZERO, DE-ENERGIZE TEST UNIT FLOW TO HELICOPTER BEFORE REDUCING SYSTEM PRESSURE. FAILURE TO DO THIS MAY RESULT IN OVERFILLING SYSTEM RESERVOIR. TESTS REQUIRING DEACTIVATION OF A SYSTEM SHALL BE ACCOMPLISHED BY DISCONNECTING TEST UNIT FROM THAT SYSTEM.

5. Connect test unit to both systems. If test unit does not have dual hoses, perform test on each system in turn. Apply 1000 PSI (6894.76 kPa) pressure for at least 15 minutes, meanwhile performing the following:

5.1. Observe all parts of system for evidence of leakage, taking corrective action as necessary.

5.2. Slowly cycle all controls through full motions. Verify that all moving parts have clearance to preclude fouling or binding. Give particular attention to flexible connections to ensure hoses are not pinched or twisted and vibration does not tend to loosen fittings.

5.3. Work any air from system by actuating each of the flight controls through at least 10 full strokes, cyclic and collective, on both systems, and tail rotor pedals in hydraulic system No. 1.

6. Check system pressure gauges as follows:

6.1. Close HYD NO. 1 SYS and HYD NO. 2 SYS circuit breakers. Place BATTERY BUS 1 and BATTERY BUS 2 switches ON. Energize inverters with INV 1 and INV 2 switches.

6.2. Slowly decrease test unit pressure to 550 PSI (3447.38 kPa) observing both hydraulic system pressure gauges operate smoothly throughout range and NO. 1 HYDRAULIC and NO. 2 HYDRAULIC caution lights illuminate when pressure drops below 650 PSI (4481.59 kPa).

6.3. Slowly increase test unit pressure to 1000 PSI (6894.76 kPa), observing both hydraulic system pressure gauges operate smoothly throughout the range and the caution lights extinguish when pressure rises above 750 PSI (5171.06 kPa).

7. Check single system operation, if not already accomplished in performing Step 5., as follows:

7.1. With test unit connected to hydraulic system No. 2 only, apply 1000 PSI (6894.76 kPa) pressure from test unit. Verify HYDR SYS 2 pressure gauge indicates 1000 PSI (6894.76 kPa) while HYDR SYS 1 gauge indicates 0. Move flight controls. Verify cyclic and collective controls are smooth, positive, and fully powered, and directional (tail rotor) controls are not powered.

7.2. With test unit connected to hydraulic system No. 1 only, apply 1000 PSI (6894.76 kPa) test unit pressure. Verify HYDR SYS 1 pressure gauge indicates 1000 PSI (6894.76 kPa) while HYDR SYS 2 gauge indicates 0. Move flight controls. Verify cyclic, collective, and directional controls are smooth, positive, and fully powered.

7.3. Shut off test unit flow and reduce pressure to 0.

NOTE

If operational checkout is satisfactory, proceed to Step 9.. When further checks for proper operation of pressure-operated shutoff valves and system relief valves are required, perform Step 8..

8. With hydraulic test unit connected to one system only, monitor pressure gauge of test unit while performing following checks:

8.1. Apply 550 PSI (3447.38 kPa) test unit pressure. Station a person to check action of flight controls and slowly increase test unit pressure. Controls should become fully powered (directional controls will not be powered if hydraulic system No. 2 is being checked) before pressure exceeds 900 PSI (6205.28 kPa), indicating pressure-operated shutoff valve in system integrated valve and filter assembly has opened.



DO NOT INCREASE PRESSURE ABOVE 1400 PSI (9652.66 KPA) WHILE PERFORMING THE FOLLOWING STEP.

8.2. Stop all movement of flight controls. Continue increasing pressure. Listen to, or feel, integrated valve and filter assembly, for evidence of system relief valve opening between 1100 and 1300 PSI (7584.23 and 8963.78 kPa), and being fully open by 1400 PSI (9652.66 kPa).

8.3. Slowly reduce pressure. Observe for evidence of system relief valve closing at not less than 1065 PSI (7342.92 kPa).

8.4. Continue reducing pressure while checking action of flight controls. Controls should lose hydraulic power between 600 and 750 PSI (4136.85 and 5171.07 kPa) indicating pressure-operated shutoff valve has closed. The appropriate NO. 1 HYDRAULIC or NO. 2 HYDRAULIC caution light will illuminate.

8.5. Shut off test unit flow and reduce pressure to zero. Remove filter bowls and drain all fluid. Clean filter elements and filter bowls. Install elements and tighten filter bowls 100 to 140 inch-pounds (11.30 to 15.82 Nm) and secure to housing with lockwire (C-405).

8.6. Apply 1000 PSI (6894.76 kPa) test unit pressure and bleed air from system by cycling all controls through full travel a minimum of ten cycles.



IF HYDRAULIC TEST STAND REMAINS CONNECTED AFTER IT IS SHUT OFF, GRAVITY WILL CAUSE HYDRAULIC RESERVOIRS TO DRAIN INTO TEST STAND.

8.7. Shut off test unit flow and reduce pressure to zero. Note NO. 1 HYDRAULIC/NO. 2 HYDRAULIC light illuminates when pressure drops below 650 PSI (4481.59 kPa). If required, connect hydraulic test stand to other system and repeat checks. Disconnect hydraulic test stand.

9. When checks are completed, place BATTERY BUS 1 and BATTERY BUS 2 and INV 1 and INV 2 switches OFF. Open circuit breakers.

NOTE

For complete electrical functional check of hydraulic systems, refer to 412-A-96-05-00-02A-320E-A / 02138.

10. Install access doors.

11. Verify hydraulic fluid level and replenish fluid as required (412-A-12-10-02-00A-292A-A / 00140).

Hydraulic System - Operational Checks (Using Helicopter Power)

Procedure

1. Start either power section (BHT-412-FM). After 10 main rotor revolutions, shut down power section.

2. Replenish hydraulic fluid as required.

3. Start power sections and operate at idle speed.



WHILE GROUND RUNNING HELICOPTER, EXERCISE CAUTION TO AVOID MOVING FLIGHT CONTROLS TO EXTREME POSITIONS.

4. Cycle collective, cyclic, and anti-torque controls as follows:

4.1. Slowly cycle collective cylinder a minimum of 25 times by moving collective control up and down through approximately 2.0 inches (50.8 mm) of travel.

4.2. With rotor RPM set at 100%, cycle cyclic cylinder by slowly displacing cyclic stick in any direction to a maximum of 2.5 inches (63.5 mm) from stick center and then back to center. Repeat a minimum of 25 times.

4.3. Slowly cycle antitorque cylinder a minimum of 15 times by moving pedals left and right through approximately 2.0 inches (50.8 mm) of travel.

5. The collective, cyclic, and antitorque controls should feel smooth and positive after cycling. If control feedback of erratic cylinder operation is present, air may still remain in system. Repeat cylinder cycling as required.

6. Shut down engine.

7. Check for leaks and replenish fluid as required (412-A-12-10-02-00A-292A-A / 00140).

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Hydraulic System Components - Description

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Drycleaning solvent	C-304	AR	

Procedure



CAP OR PLUG ALL LINES, HOSES, AND PORTS AS COMPONENTS ARE REMOVED.



CLEANING OF HYDRAULIC COMPONENTS SHALL BE ACCOMPLISHED WITH drycleaning solvent (C-304) ONLY. DO NOT USE ALCOHOL.

1. The maintenance procedures paragraphs provide maintenance information on the various hydraulic system components.

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Hydraulic Pump (Hydraulic System No. 1) - Description

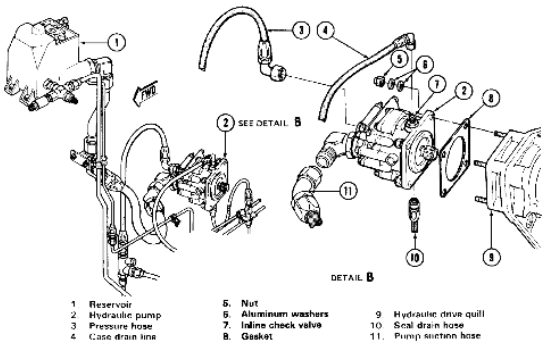
Hydraulic system No. 1 hydraulic pump is installed on the transmission sump case. The hydraulic pump is an axial piston type rated at 6.3 gallons per minute (23.84 L/min) at 1000 ±25 PSI (6894.76 ±172 kPa) at operating speed.

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Hydraulic Pump (Hydraulic System No. 1) - Removal

Procedure

1. Remove forward pylon fairing and access panel on right side of pylon support structure inside cabin.
2. Loosen connections of hoses (3, 4, 10, and 11, Figure 1) on hydraulic pump (2) to break torque but not to point of leaking.
3. Remove nuts (5), aluminum washers (6), and hydraulic pump (2) from transmission studs. Place pump, with hoses (3, 4, 10, and 11) attached, in container large enough to hold fluid contents of reservoir (1) and hoses.
4. Disconnect hoses (3, 4, 10, and 11) from hydraulic pump (2), draining fluid from reservoir (1) and hoses.
5. Cap or plug hoses (3, 4, 10, and 11), as applicable.
6. If replacing hydraulic pump (2), remove inline check valve (7) and fittings from pump for reuse on new hydraulic pump. Discard gasket (8).
7. Plug ports in hydraulic pump (2) to prevent contamination.



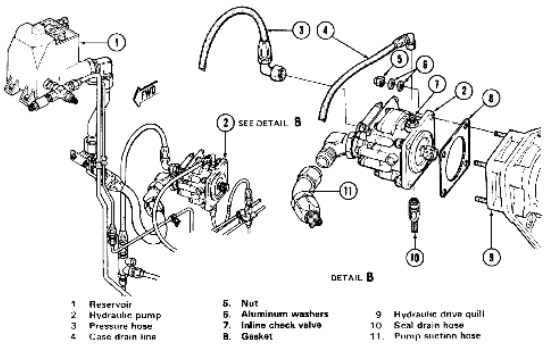
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Figure 1. Hydraulic Pump (Hydraulic System No. 1) (Sheet 1 of 1)

Hydraulic Pump (Hydraulic System No. 1) - Inspection

Procedure

1. Inspect hydraulic pump (2, Figure 1) for damage and leakage.
2. Inspect fittings for damaged threads.
3. Inspect inline check valve (7) for damage and proper operation.
4. Inspect splines and mounting studs for damage.



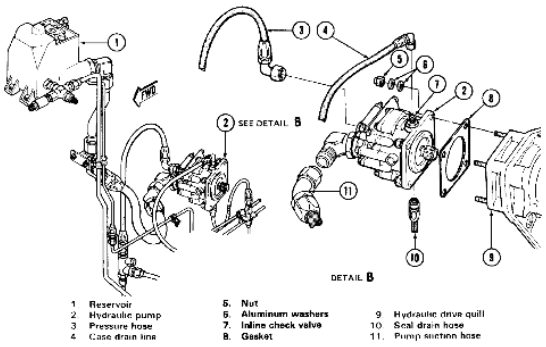
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Figure 1. Hydraulic Pump (Hydraulic System No. 1) (Sheet 1 of 1)

Hydraulic Pump (Hydraulic System No. 1) - Repair

Procedure

1. Replace damaged studs and fittings as necessary.
2. Replace inline check valve (7, Figure 1) if check valve shows evidence of malfunction.
3. Replace damaged hydraulic pump (2).



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Figure 1. Hydraulic Pump (Hydraulic System No. 1) (Sheet 1 of 1)

Hydraulic Pump (Hydraulic System No. 1) - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Hydraulic fluid	C-002	AR	Refer to Chapter 12 for information on usage and mixing of hydraulic fluid (C-002).
Hydraulic fluid	C-072	AR	Refer to Chapter 12 for information on usage and mixing of hydraulic fluid (C-072).
Grease	C-007	AR	

Procedure



DO NOT LUBRICATE SPLINES FOR HYDRAULIC PUMPS P/N 731634 (PV3-044-8A), P/N 66003-01 (AP2V-77-01), OR 212-076-010-101 WITH NON-METALLIC COUPLING INSTALLED.

1. Apply a thin coat of grease (C-007) to splines of hydraulic pump (2, Figure 1).



ENSURE HYDRAULIC PUMP CASE IS FULL PRIOR TO INSTALLATION.

2. Fill hydraulic pump (2) with hydraulic fluid (C-002) or hydraulic fluid (C-072) through case drain port.

3. Install fittings and inline check valve (7) into hydraulic pump (2) if not previously accomplished.

4. Position new gaskets (8) and hydraulic pump (2) on studs in hydraulic drive quill (9). Install aluminum washers (6) and nuts (5). Tighten nuts evenly.

5. Connect hoses (3, 4, 10, and 11).

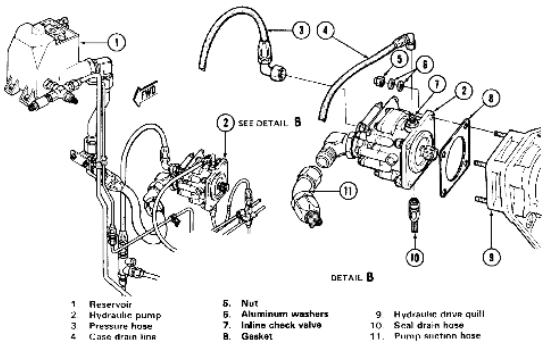
NOTE

If hydraulic pump 212-076-010-001 is being installed as an alternate to hydraulic pump PV3-044-8, install reducer MS21916D8-6 and packing MS28778-8 in pressure port of hydraulic pump.

6. Service reservoir (1) (412-A-12-10-02-00A-040A-A / 00138). Install forward pylon fairing.

7. Perform an operational check (412-A-29-00-00-00A-320A-A / 00616 and 412-A-29-00-00-00A-320B-A / 00617).

8. Install access panel on pylon support inside cabin.



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Figure 1. Hydraulic Pump (Hydraulic System No. 1) (Sheet 1 of 1)

Hydraulic Pump (Hydraulic System No. 2) - Description

Hydraulic system No. 2 hydraulic pump is installed on front of transmission main case. The pump is an axial piston type rated at 5.6 gal/min (21.2 L/min) at 1000 ±25 PSI (6894.76 ±172.4 kPa) at operating speed.

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Hydraulic Pump (Hydraulic System No. 2) - Removal

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Hydraulic Suction Gun	Commercial	1	

Procedure

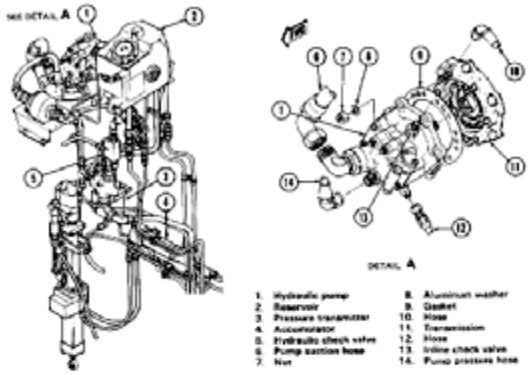
1. Remove forward pylon fairing.

2. Remove hydraulic fluid from hydraulic system No. 2 reservoir (2, Figure 1) using a hydraulic suction gun.

3. Disconnect hoses (6, 10, 12, and 14).

4. Remove nuts (7), aluminum washers (8), and hydraulic pump (1).

5. Plug open hoses (6, 10, 12, and 14) to prevent contamination.



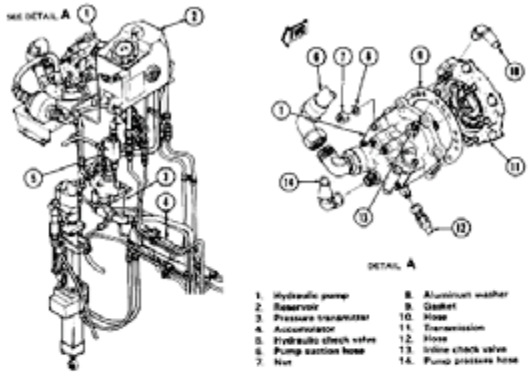
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Figure 1. Hydraulic Pump (Hydraulic System No. 2) (Sheet 1 of 1)

Hydraulic Pump (Hydraulic System No. 2) - Inspection

Procedure

1. Inspect hydraulic pump (1, Figure 1) for damage and signs of leakage.
2. Inspect fittings for damaged threads.
3. Inspect inline check valve (13) for damage and proper operation.
4. Inspect splines and studs for damage.



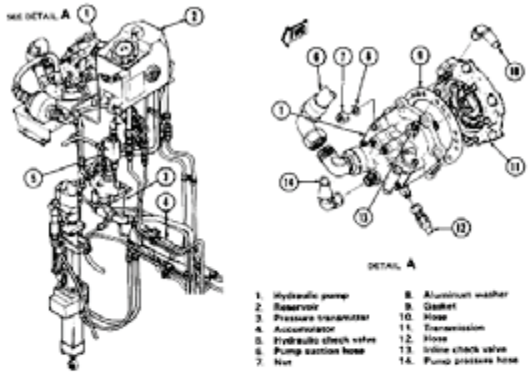
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Figure 1. Hydraulic Pump (Hydraulic System No. 2) (Sheet 1 of 1)

Hydraulic Pump (Hydraulic System No. 2) - Repair

Procedure

1. Replace damaged studs, fittings, and inline check valve (13, Figure 1).
2. Replace damaged hydraulic pump (1).



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Figure 1. Hydraulic Pump (Hydraulic System No. 2) (Sheet 1 of 1)

Hydraulic Pump (Hydraulic System No. 2) - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Hydraulic fluid	C-002	AR	Refer to Chapter 12 for information on usage and mixing of hydraulic fluid (C-002).
Hydraulic fluid	C-072	AR	Refer to Chapter 12 for information on usage and mixing of hydraulic fluid (C-072).
Grease	C-007	AR	

Procedure



DO NOT LUBRICATE PUMP SPLINE FOR 212-076-011-101 WITH NON-METALLIC DRIVE SPLINES.

1. Apply a thin coat of grease (C-007) to splines of hydraulic pump (1, Figure 1).



ENSURE HYDRAULIC PUMP CASE IS FULL PRIOR TO INSTALLATION.

2. Fill hydraulic pump (1) with hydraulic fluid (C-002) or hydraulic fluid (C-072) through case drain port.

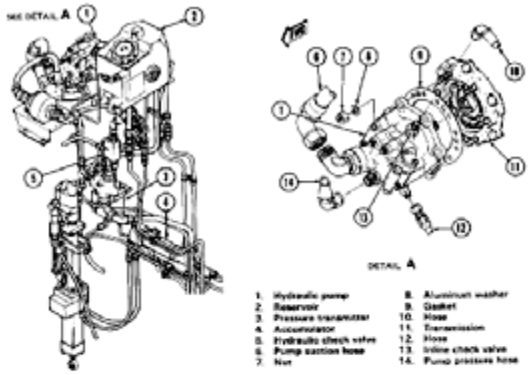
3. Remove caps and connect hoses (6, 10, 12, and 14).

4. Place new gasket (9) and hydraulic pump (1) over studs on transmission (11) case and secure with aluminum washers (8) and nuts (7).

5. Service reservoir (2) (412-A-12-10-02-00A-040A-A / 00138).

6. Perform operational check (412-A-29-00-00-00A-320A-A / 00616 and 412-A-29-00-00-00A-320B-A / 00617).

7. Install forward pylon fairing.



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Figure 1. Hydraulic Pump (Hydraulic System No. 2) (Sheet 1 of 1)

Hydraulic Reservoir - Description

Hydraulic system No. 1 (Figure 1) and hydraulic system No. 2 (Figure 2) have separate reservoirs installed forward of the transmission. The reservoirs have strainer screens under the filler caps to prevent entry of foreign particles when fluid is added. A second strainer screen is installed in the return fitting on each reservoir to strain fluid returning to the reservoir. Each reservoir has a sight glass for visual indication of fluid level. Decals on reservoirs provide reservoir capacity, filling instructions, and type of hydraulic fluid required. Each reservoir has a vent screen.

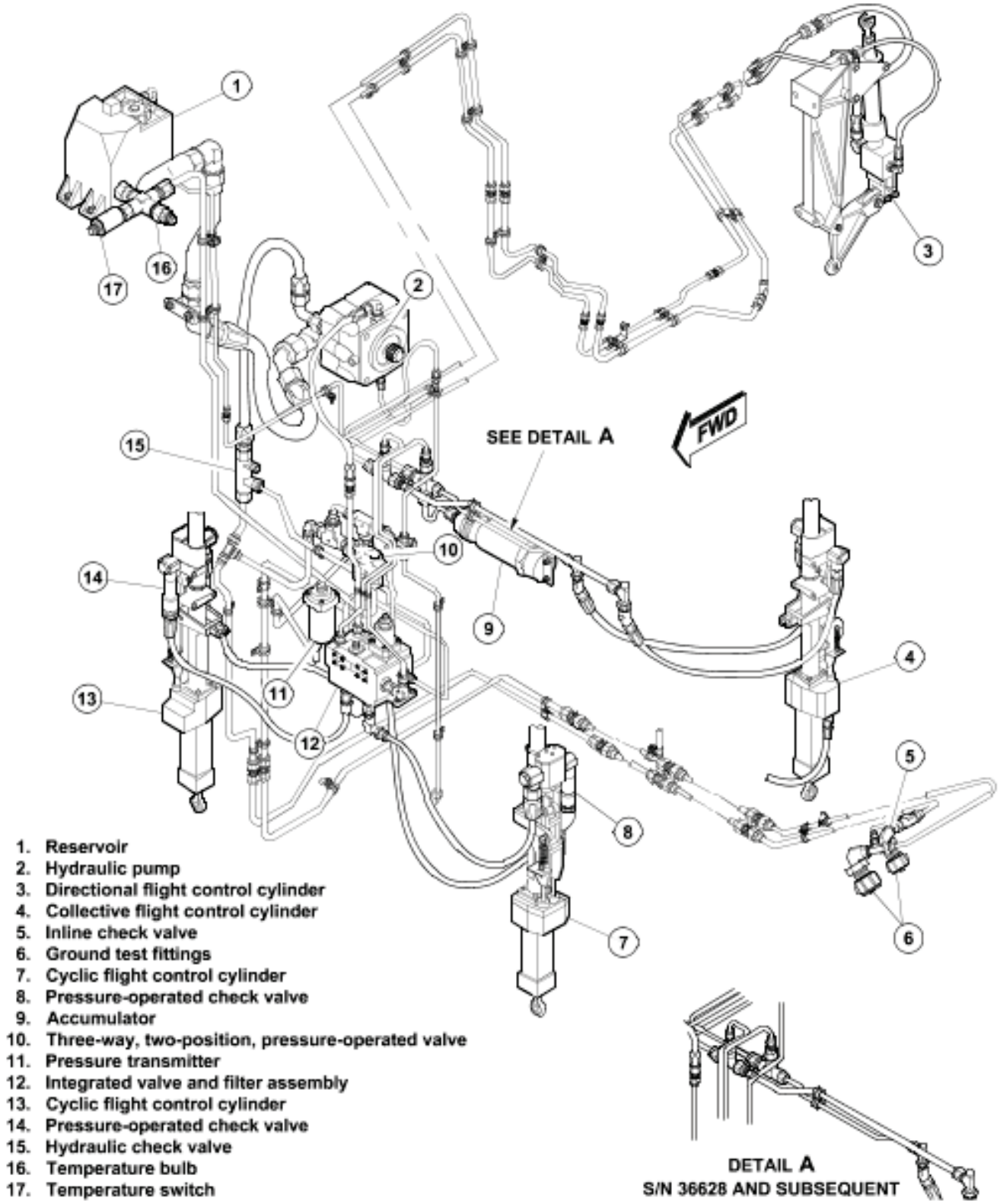
Hydraulic system No. 1 and hydraulic system No. 2 each have temperature bulbs installed in the return fitting of the reservoir. Temperature sensed by temperature bulbs is transmitted to instruments on pilot instrument panel.

Both hydraulic systems No 1 and No. 2 each have a temperature switch installed at the return fitting of reservoir. The switches cause the NO. 1 HYDRAULIC and/or NO. 2 HYDRAULIC caution segment (respectfully) and MASTER CAUTION segment to illuminate in the event hydraulic fluid temperature exceeds 190°F (88°C).

In the event NO. 1 HYDRAULIC and/or NO. 2 HYDRAULIC caution segment illuminates, the pilot can verify whether fault is high temperature or low pressure in hydraulic system No. 1 or hydraulic system No. 2 from separate instrument readings.

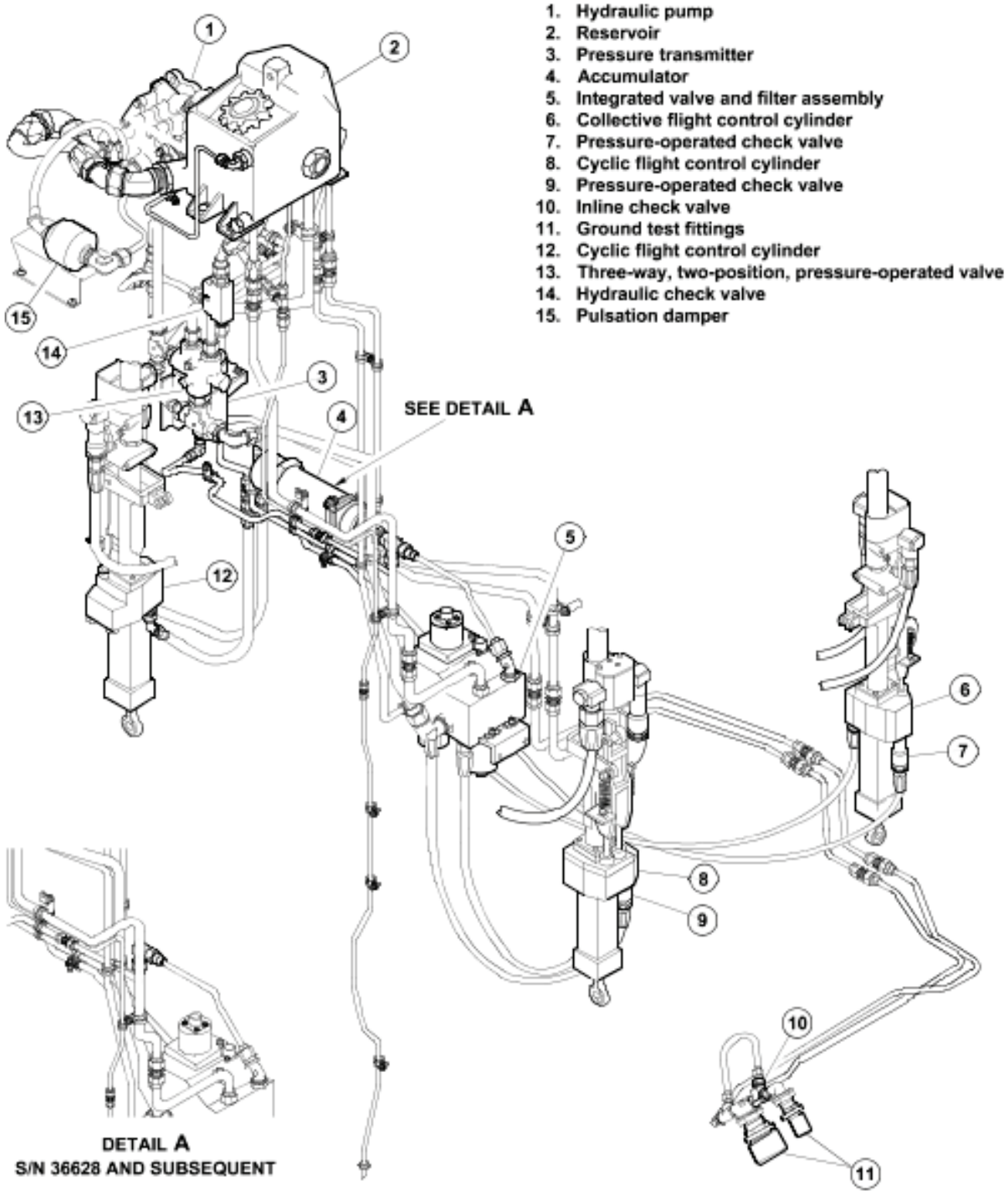
NOTE

Maintenance procedures are the same for hydraulic systems No. 1 and No. 2.



ICN-412-A-291000-B-97499-00007-A-001-01

Figure 1. Hydraulic System No. 1 (Sheet 1 of 1)



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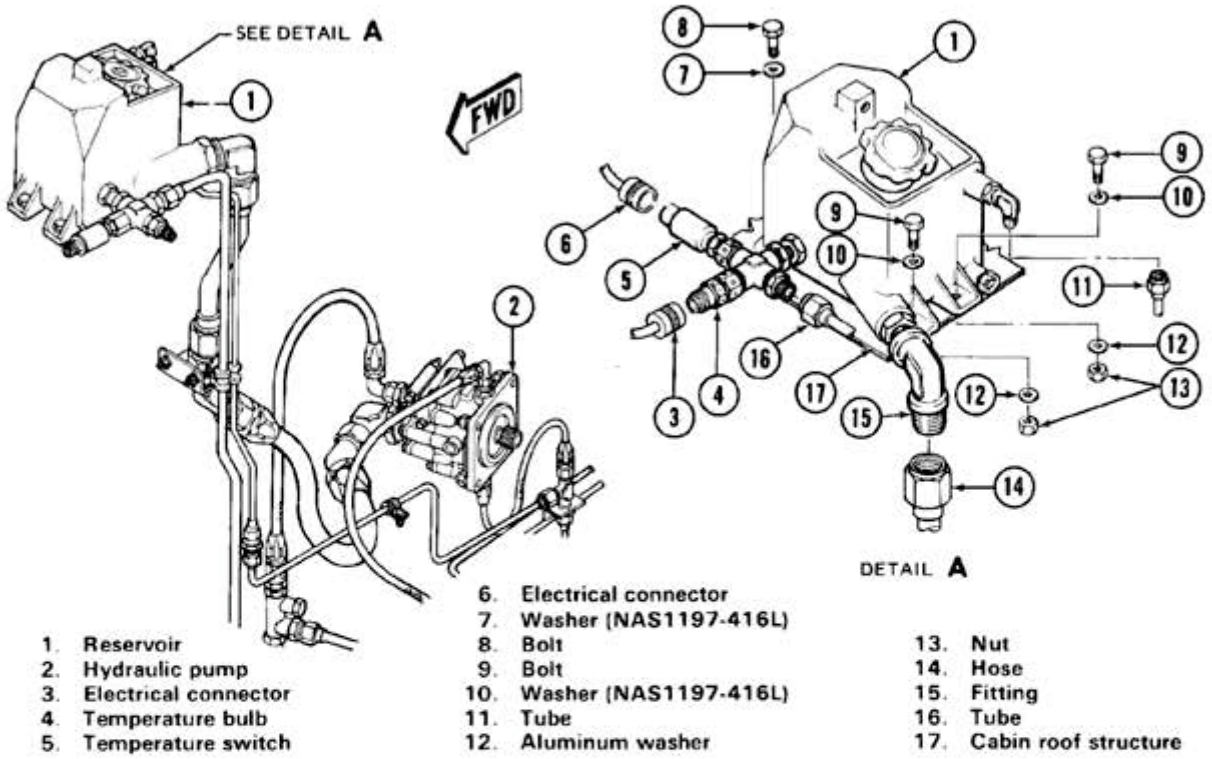
Figure 2. Hydraulic System No. 2 (Sheet 1 of 1)

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Hydraulic Reservoir - Removal

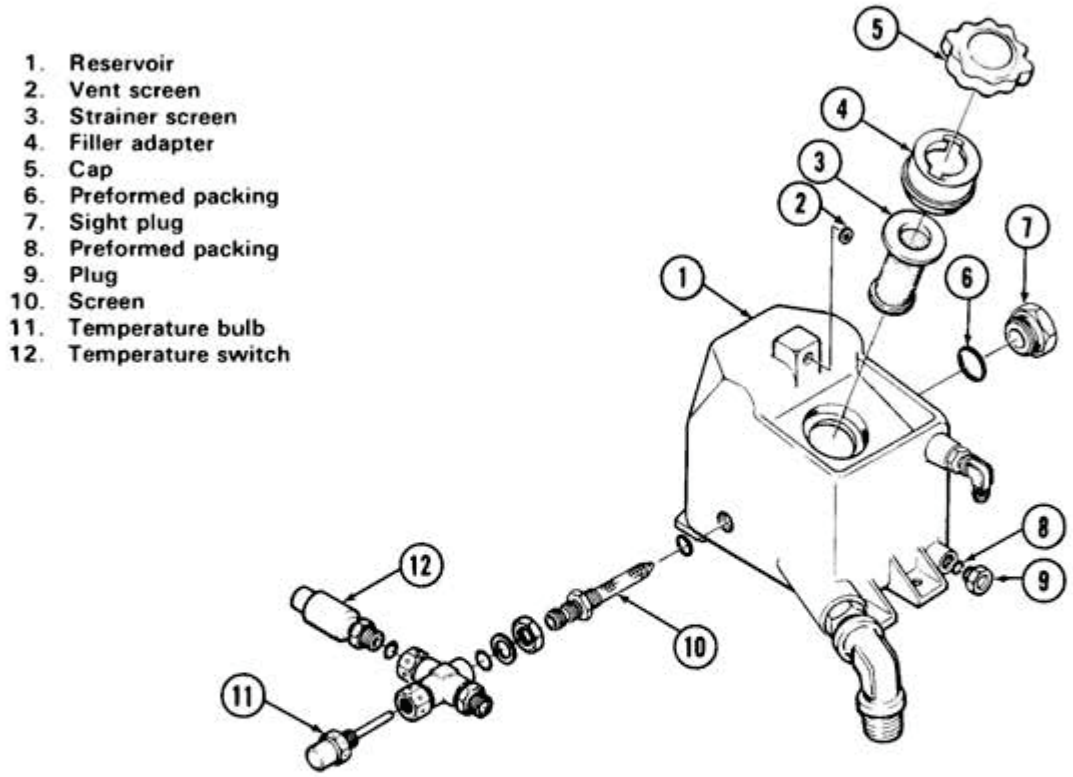
Procedure

1. Remove forward pylon fairing on cabin roof.
2. Disconnect electrical connectors (3 and 6, Figure 1).
3. Remove majority of fluid from reservoir (1) by removing temperature bulb (4). Catch fluid in suitable container.
4. Disconnect tubes (11 and 16) and hose (14), and catch remaining fluid.
5. If replacing reservoir (1), or removing for cleaning because of contamination, remove fittings and filter screen (10, Figure 2).
6. Remove bolts (8 and 9, Figure 1). Remove reservoir (1).
7. Plug open tubes (11 and 16) and hose (14) to prevent contamination.



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Figure 1. Hydraulic Reservoir No. 1 (Sheet 1 of 1)



ICN-412-A-291000-B-97499-00012-A-001-01

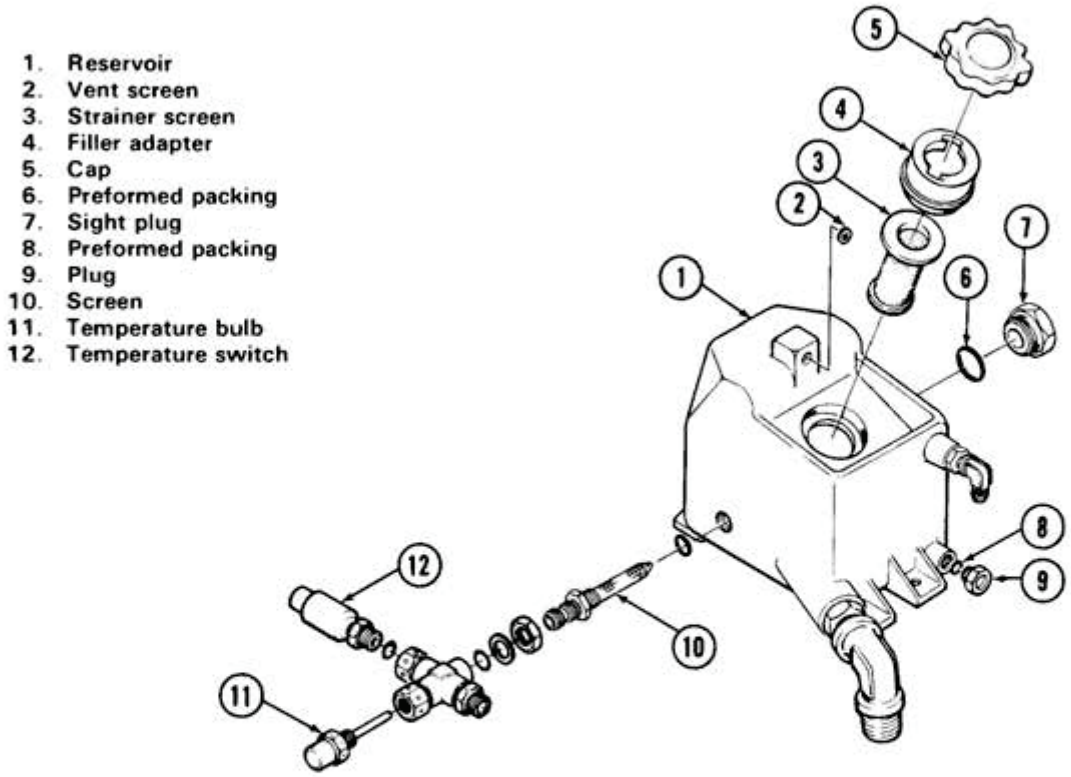
Figure 2. Hydraulic Reservoir Assembly (Sheet 1 of 1)

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Hydraulic Reservoir - Inspection

Procedure

1. Inspect reservoir (1, Figure 1) for mechanical and corrosion damage and evidence of leakage.
2. Inspect cap (5), filler adapter (4), and strainer screen (3) for mechanical and corrosion damage.
3. Inspect fluid level sight plug (7) for cracks, crazing, staining, or other defects that would affect visibility.
4. Inspect vent screen (2) for distortion, corrosion, cracks, and tears. Damage to vent screen (2) resulting in enlarged holes is not acceptable.
5. Remove screen (10), and inspect for distortion, corrosion, cracks, and tears. Damage to screen (10) resulting in enlarged holes is not acceptable.
6. Inspect bosses for damaged seal contact surfaces and thread damage.
7. Inspect temperature bulb (11) and temperature switch (12) for damage.



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Figure 1. Hydraulic Reservoir Assembly (Sheet 1 of 1)

Hydraulic Reservoir - Repair

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Hydraulic fluid	C-002	AR	Refer to Chapter 12 for information on usage and mixing of hydraulic fluid (C-002) and hydraulic fluid (C-072).
Hydraulic fluid	C-072	AR	Refer to Chapter 12 for information on usage and mixing of hydraulic fluid (C-072) and hydraulic fluid (C-002).
Zinc chromate primer	C-201	AR	
Lockwire	C-405	AR	

Procedure

1. Replace unserviceable filler cap strainer screen (3, Figure 1) as follows:

1.1. Unscrew filler adapter (4) from filler opening, and lift out strainer screen (3).

1.2. Clean threads of filler adapter (4) and opening.

1.3. Place strainer screen (3) into reservoir (1).

1.4. Coat filler adapter (4) threads with zinc chromate primer (C-201) and screw filler adapter into reservoir (1).

2. Replace unserviceable vent screen (2) as follows:



EXERCISE CARE TO PREVENT FOREIGN DEBRIS FROM ENTERING RESERVOIR (1) WHILE VENT SCREEN (2) IS REMOVED.

2.1. Carefully drill out area where vent screen (2) is staked in place.

2.2. Position new vent screen (2) in reservoir (1) and stake in three places to secure vent screen in reservoir.

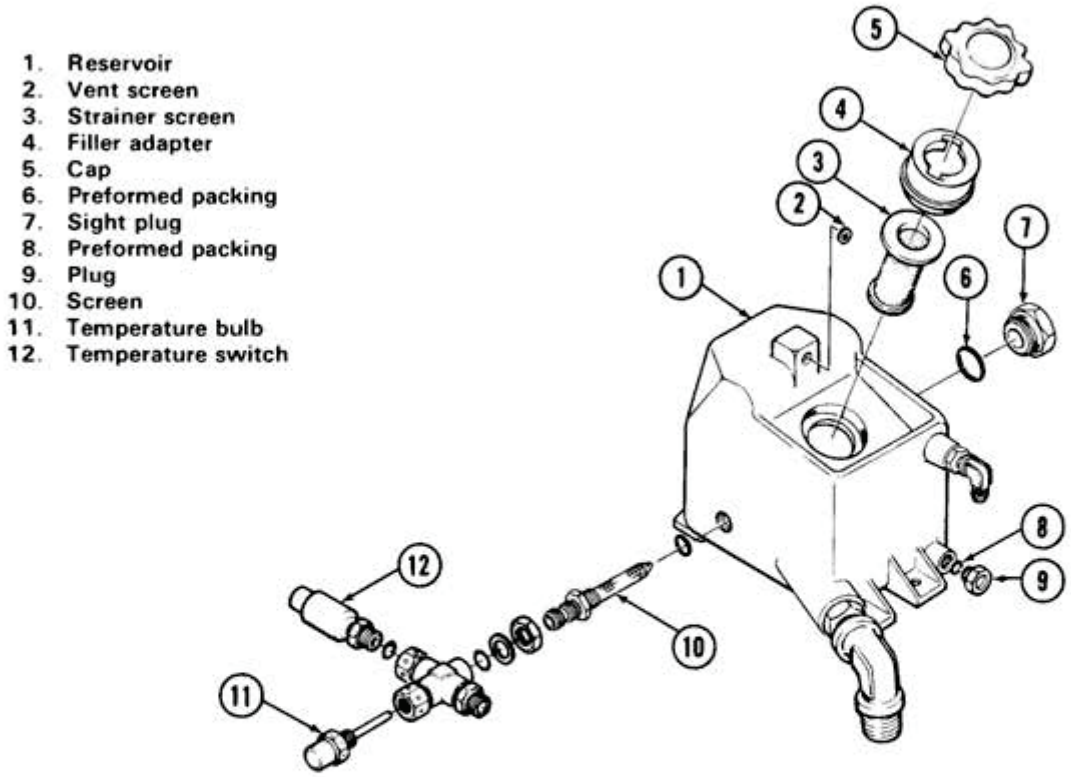
3. Replace unserviceable sight plug (7) as follows:

3.1. Remove old sight plug (7).

3.2. Lubricate new preformed packing (6) with hydraulic fluid (C-002) or hydraulic fluid (C-072) and position packing on serviceable sight plug (7).

3.3. Install sight plug (7) and lockwire (C-405) to fitting in adjacent boss.

4. Replace damaged screen (10), damaged or defective temperature bulb (11), and temperature switch (12).



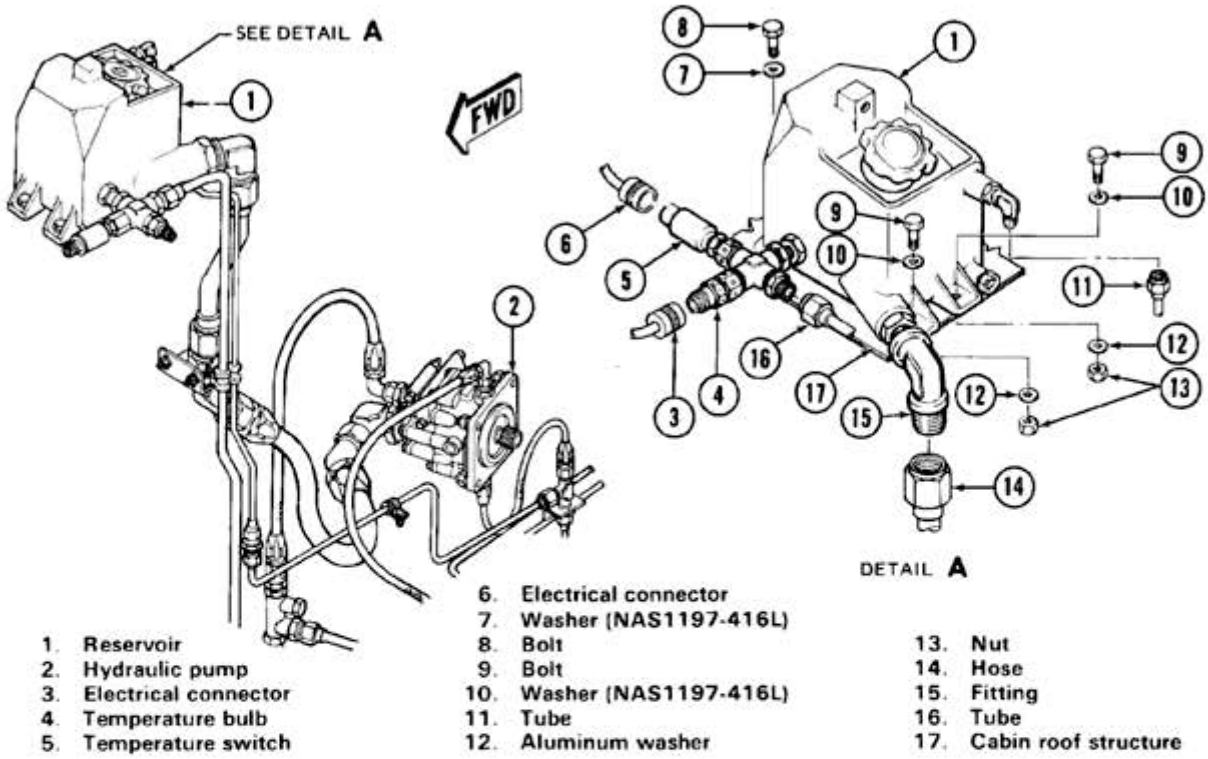
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Figure 1. Hydraulic Reservoir Assembly (Sheet 1 of 1)

Hydraulic Reservoir - Installation

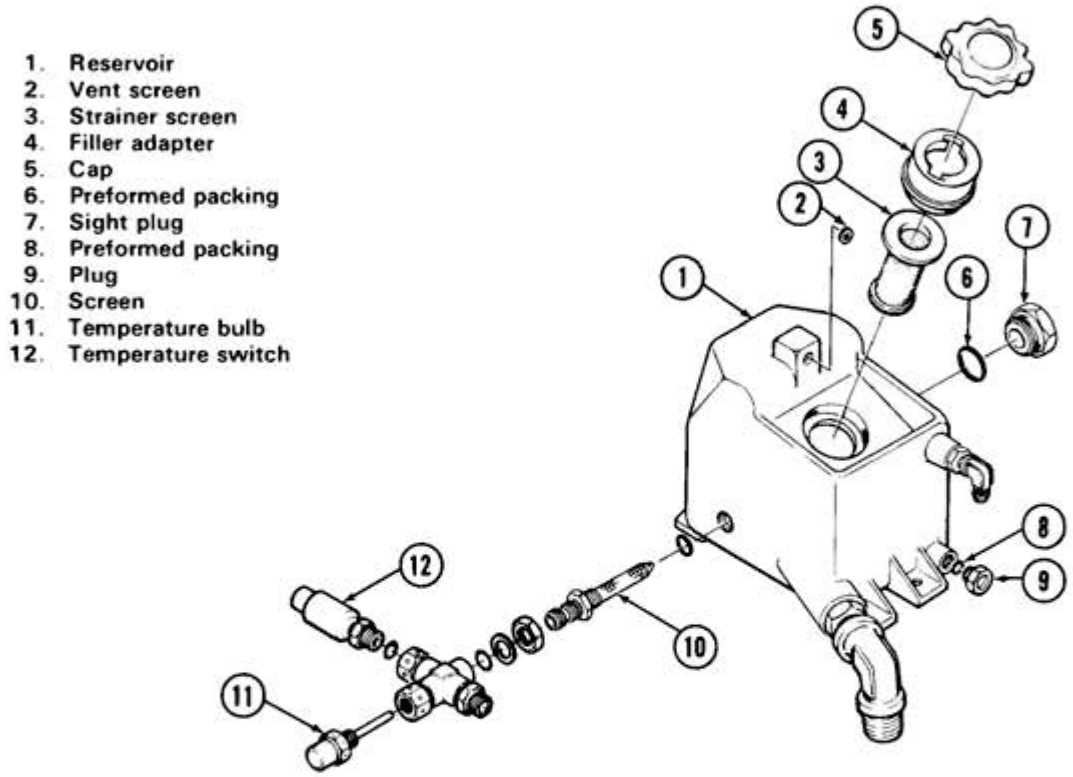
Procedure

1. Position reservoir (1, Figure 1) on cabin roof.
2. Install bolts (8) and washers (7). Install bolts (9), washers (10 and 12), and nuts (13).
3. Install filter screen (10, Figure 2) and fittings.
4. Remove caps and connect tubes (11 and 16, Figure 1) and hose (14).
5. Connect electrical connectors (3 and 6) to temperature bulb (4) and temperature switch (5).
6. Service reservoir (1) (412-A-12-10-02-00A-040A-A / 00138).
7. Perform ground run and verify system functions normally and no leaks are present.
8. Install forward pylon fairing.



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Figure 1. Hydraulic Reservoir No. 1 (Sheet 1 of 1)



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Figure 2. Hydraulic Reservoir Assembly (Sheet 1 of 1)

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Integrated Valve and Filter Assembly - Description

NOTE

Except as noted, the maintenance procedures in 412-A-29-10-00-04A-520A-A / 00635, 412-A-29-10-00-04A-280A-A / 00636 and 412-A-29-10-00-04A-720A-A / 00637 are the same for hydraulic systems No. 1 and No. 2.

Hydraulic system No. 1 (Figure 1) and hydraulic system No. 2 (Figure 2) have separate integrated valve and filter assemblies. Hydraulic system No. 1 assembly is located in the center of the compartment below the transmission. Hydraulic system No. 2 assembly is located on the left side of the compartment. Each assembly comprises a valve body with the following components installed: pressure filter, return filter, pressure relief valve, thermal relief valve, return shutoff valve, clogged filter indicators, clogged filter switch, and a solenoid-operated valve which allows fluid to flow in either direction in event of system failure. The pressure and return filters are completely separate. Each filter is provided with an indicator button which extends at 70 PSI (483 kPa) differential pressure across the filter. This indicator gives a visual indication of filter clogging. The indicator button trips an electrical switch on the integrated valve and filter assembly which causes an indicator located in the helicopter nose compartment to change from green to red. The indicator is visible through the right nose window.

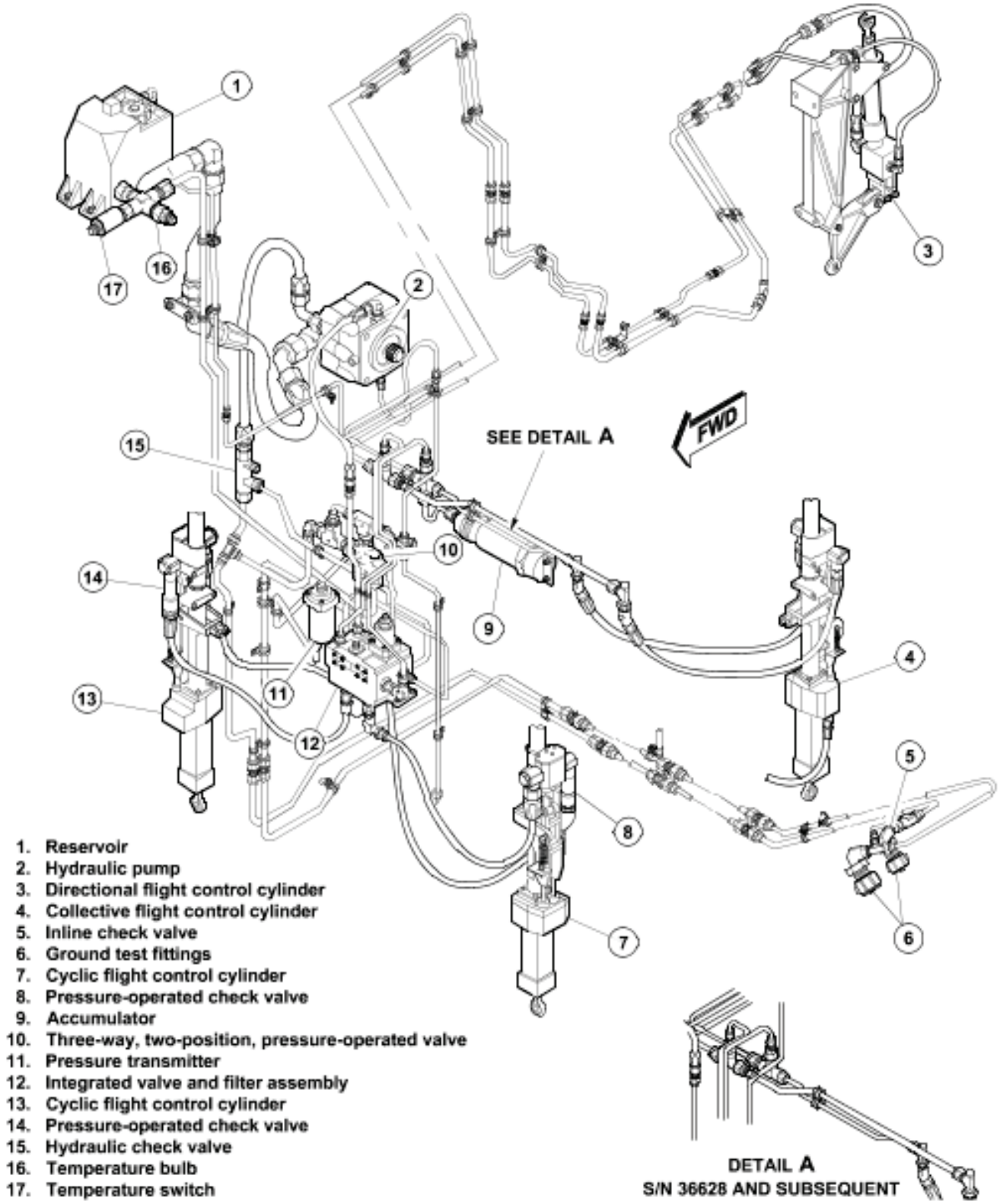
The system pressure relief valve opens at 1100 PSI (7584.23 kPa) to allow fluid to return to reservoir.

The return shutoff valve is pressure-operated. When system pressure decreases, the shutoff valve closes to maintain pressure in cyclic, collective, and directional servo actuators.

The thermal relief valve prevents damage to hydraulic system components caused by excessive pressure buildup due to trapped hydraulic fluid when helicopter is not operating.

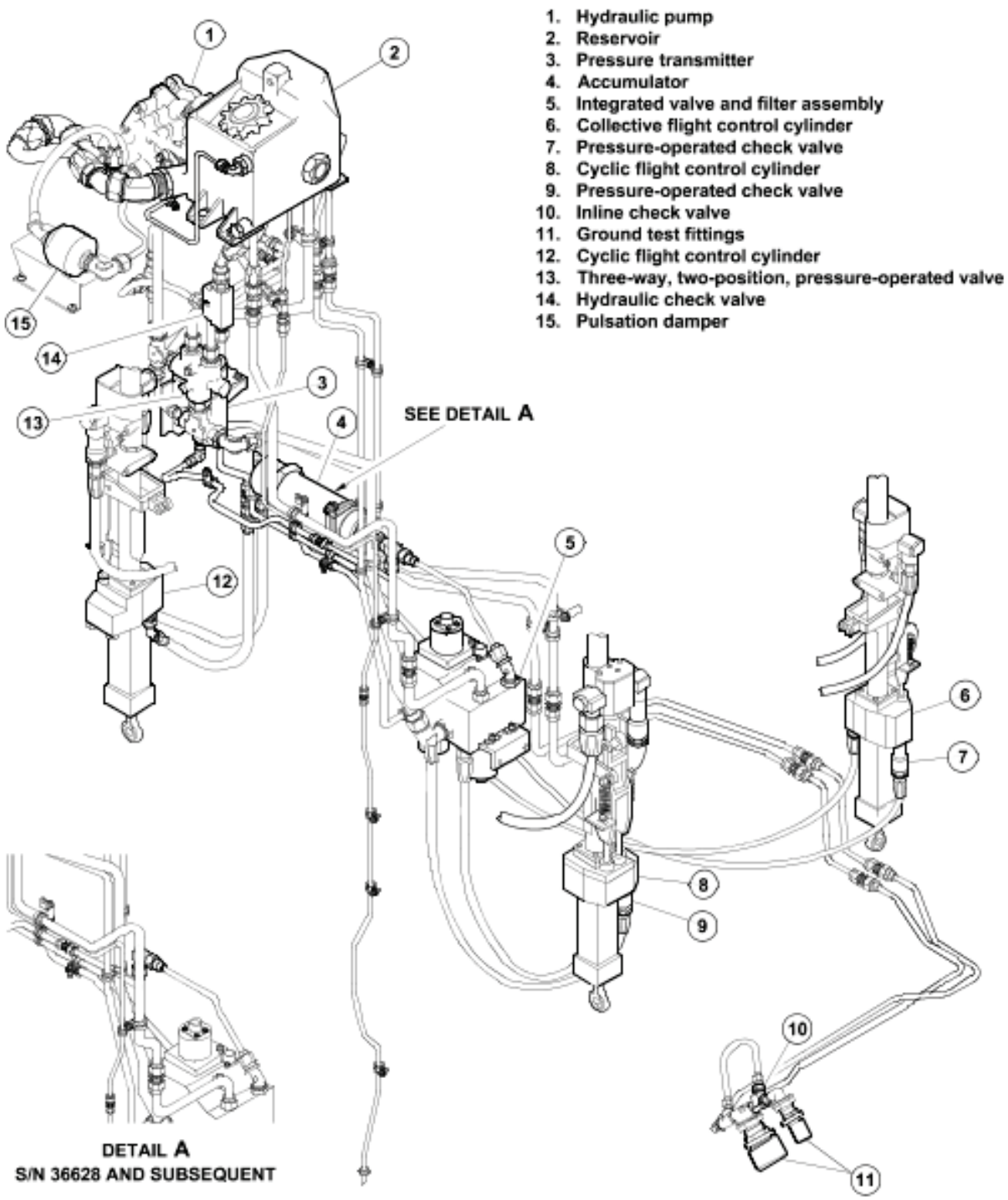
The solenoid-operated system bypass valve is normally de-energized. In the event of the failure of hydraulic system No. 1, the pilot can position HYDR SYS NO. 1 switch to OFF. This will cause bypass valve to directly connect pressure port to return port to prevent hydraulic lock within hydraulic system No. 1.

Operation of the corresponding valve in hydraulic system No. 2 is the same except the pilot positions HYDR SYS NO. 2 switch to OFF.



ICN-412-A-291000-B-97499-00007-A-001-01

Figure 1. Hydraulic System No. 1 (Sheet 1 of 1)



ICN-412-A-291000-B-97499-00008-A-001-01

Figure 2. Hydraulic System No. 2 (Sheet 1 of 1)

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Integrated Valve and Filter Assembly - Removal

Procedure

NOTE

The following removal procedures are the same for hydraulic systems No. 1 and No. 2.

1. Remove access panel on forward side of pylon support inside cabin, for hydraulic system No. 1.

NOTE

Disconnecting tube (4, Figure 1) will drain fluid from reservoir. Provide a suitable container to catch fluid.

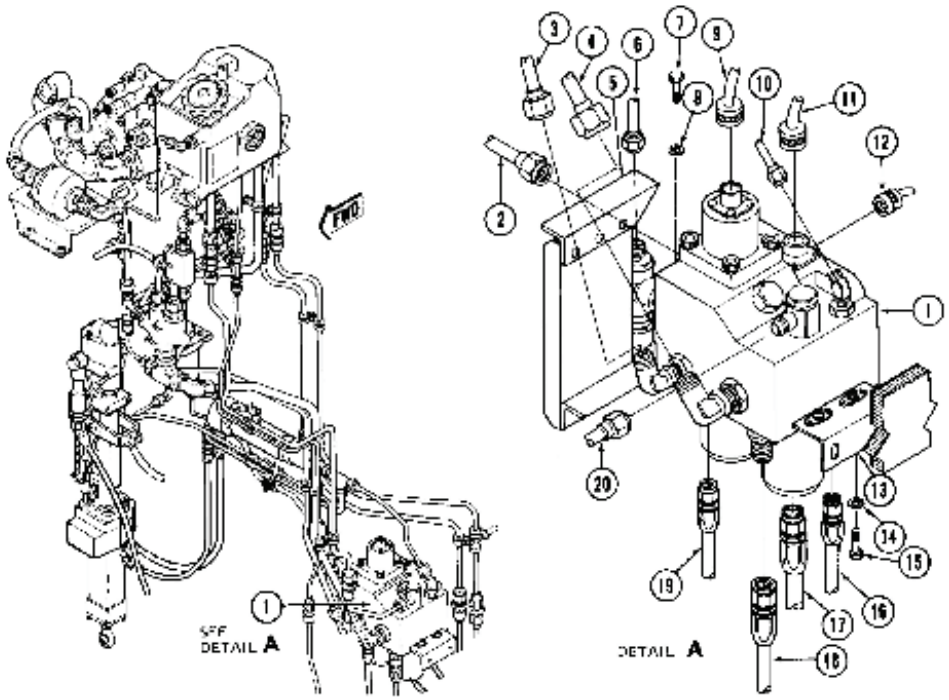
2. Disconnect tube (4).

3. Disconnect tubes and hoses (2, 3, 6, 10, 16, 17, 18, 19, and 20). Remove clamps and loosen opposite ends of tubes as needed.

4. Disconnect electrical connectors (9, 11, and 12).

5. Remove bolts (7 and 15). Remove integrated valve and filter assembly (1).

6. Plug open lines to prevent contamination.



1. Integrated valve and filter assembly
2. Tube
3. Tube
4. Tube
5. Support
6. Tube
7. Bolt
8. Thin aluminum washer
9. Electrical connector
10. Tube
11. Electrical connector
12. Electrical connector
13. Bracket
14. Thin aluminum washer
15. Bolt
16. Hose
17. Hose
18. Hose
19. Hose
20. Tube

ICN-412-A-291000-B-97499-00013-A-001-01

Figure 1. Integrated Valve and Filter (Sheet 1 of 1)

Integrated Valve and Filter Assembly - Inspection

Procedure

NOTE

The following inspection procedures are the same for hydraulic systems No. 1 and No. 2.

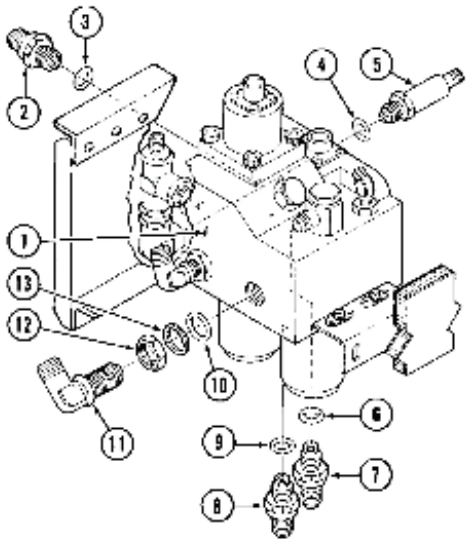
1. Inspect integrated valve and filter assembly (1, Figure 1) for damage and signs of leakage. Inspect clogged filter indicators for tripped condition.

2. Inspect inline check valves (2, 7, and 8) and elbow check valve (11) for damage.

3. Inspect pressure switch (5) for damage.

4. Inspect fittings for damaged threads.

1. Integrated valve and filter assembly
2. Inline check valve
3. Preformed packing
4. Preformed packing
5. Pressure switch
6. Preformed packing
7. Inline check valve
8. Inline check valve
9. Preformed packing
10. Preformed packing
11. Elbow check valve
12. Nut
13. Retainer



ICN-412-A-291000-B-97499-00015-A-001-01

Figure 1. Integrated Valve and Filter Assembly (Sheet 1 of 1)

Integrated Valve and Filter Assembly - Installation

Procedure

NOTE

The following installation procedures are the same for hydraulic systems No. 1 and No. 2.

1. Position integrated valve and filter assembly (1, Figure 1). Install four bolts (7 and 15) and thin aluminum washers (8 and 14).

2. Connect electrical connectors (9, 11, and 12).

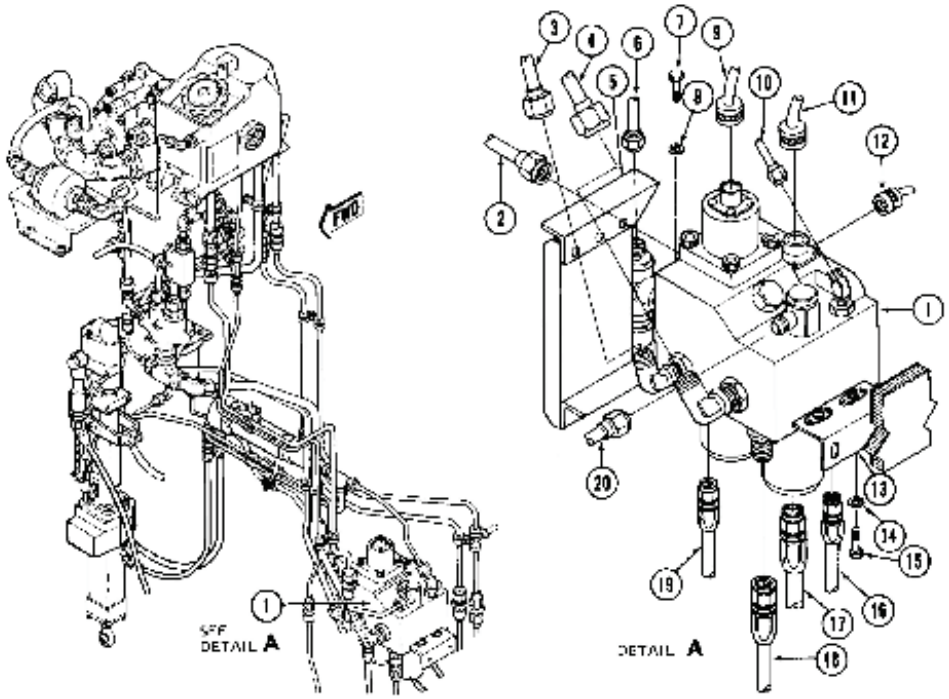
3. Connect tubes/hoses (2, 3, 6, 10, 16, 17, 18, 19, and 20). Install clamps. Tighten fittings at opposite ends of all tubes.

4. Connect tube (4).

5. Service reservoir (412-A-12-10-02-00A-040A-A / 00138).

6. Perform ground run and verify system functions normally and no leaks are present.

7. Install access panels.



1. Integrated valve and filter assembly
2. Tube
3. Tube
4. Tube
5. Support
6. Tube
7. Bolt
8. Thin aluminum washer
9. Electrical connector
10. Tube
11. Electrical connector
12. Electrical connector
13. Bracket
14. Thin aluminum washer
15. Bolt
16. Hose
17. Hose
18. Hose
19. Hose
20. Tube

ICN-412-A-291000-B-97499-00013-A-001-01

Figure 1. Integrated Valve and Filter (Sheet 1 of 1)

Hydraulic Filter Assemblies - Description

The pressure and return filter assemblies are located on the bottom of the integrated valve and filter assembly valve body. Each filter assembly contains: a filter bowl, a filter element, two packings, and a retainer. The hydraulic filter element removes the particles from the hydraulic fluid. These particles come from usual, and sometimes unusual, wear of hydraulic components. Examine, clean, or replace filter element when the impending bypass indicator button extends (pops), or at recommended inspection intervals (412-A-05-00-00-00A-009A-A / 00014).

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Hydraulic Filter Assembly (212-076-006-007) - Removal

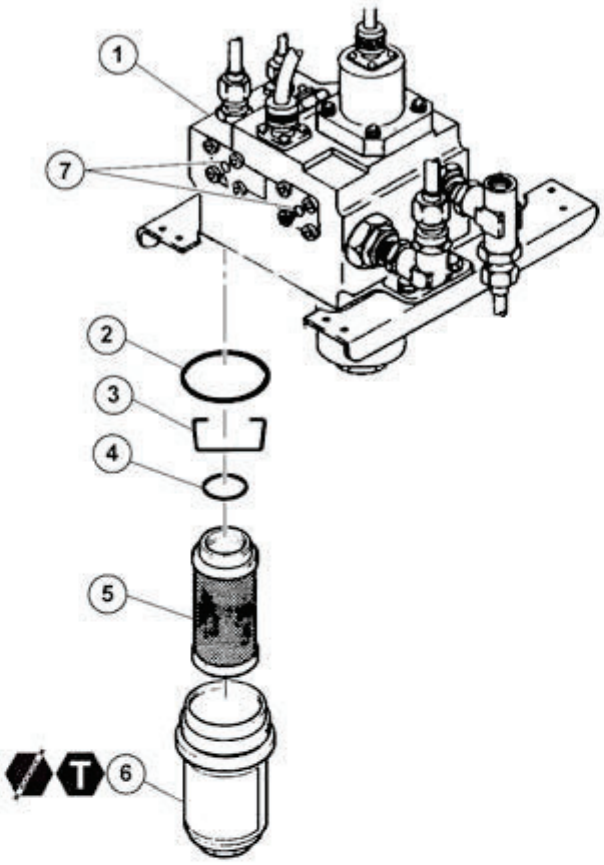
Procedure

1. Remove the forward pylon fairing (412-A-71-10-00-01A-520A-A / 01605).
2. Remove lockwire from filter bowl (6, Figure 1) and valve body (1).
3. Remove filter bowl (6).

NOTE

If filter element is to be used again, put it in a new, clean plastic bag to prevent contamination.

4. Remove retainer (3) and filter element (5).
5. Remove and discard packings (2 and 4).



INTEGRATED VALVE AND FILTER ASSEMBLY
(212-076-006-007)

1. Valve body
2. Packing
3. Retainer
4. Packing
5. Filter element (Cleanable element 205-076-034-007)
6. Filter bowl
7. Indicator button
8. Deleted

LOCKWIRE (C-405)

T 100 TO 140 IN-LBS
(11 TO 16 Nm)

ICN-412-A-291000-B-97499-00014-A-002-01

Figure 1. Hydraulic Filter Assembly (212-076-006-007) (Sheet 1 of 1)

Hydraulic Filter Assembly (212-076-006-105) - Removal

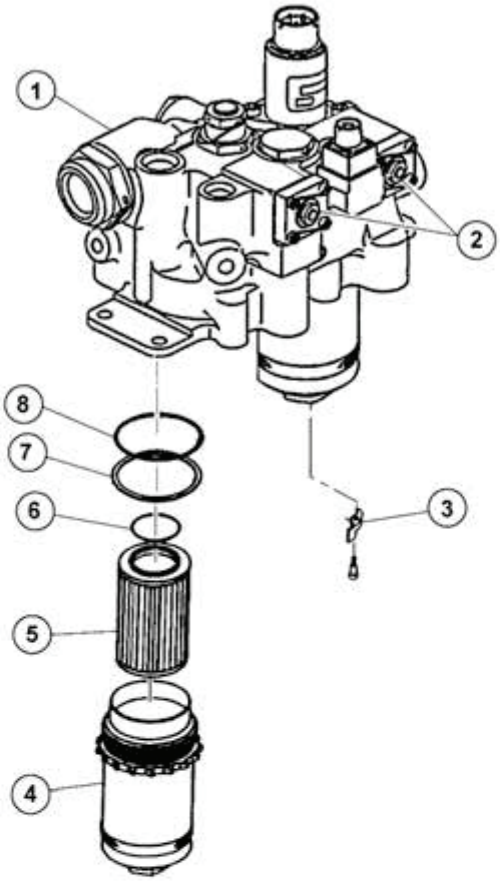
Procedure

1. Remove the forward pylon fairing (412-A-71-10-00-01A-520A-A / 01605).
2. *Press on ratchet lever (3,* Figure 1*) and remove filter bowl (4).*

NOTE

If filter element is to be used again, put it in a new, clean plastic bag to prevent contamination.

3. *Remove retainer (7) and filter element (5).*
4. *Remove and discard packings (6 and 8).*



INTEGRATED VALVE AND FILTER ASSEMBLY
(212-076-006-105)

1. Valve body
2. Indicator button
3. Ratchet lever
4. Filter bowl
5. Filter element (Non cleanable element 205-076-034-003)
   Filter element (Cleanable element 205-076-034-007)
6. Packing
7. Retainer
8. Packing

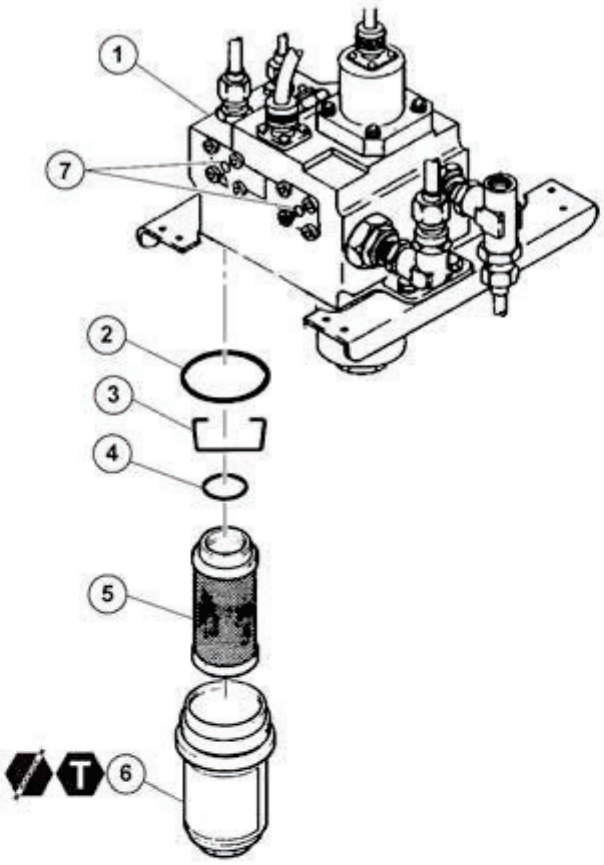
ICN-412-A-291000-B-97499-03024-A-001-01

Figure 1. Hydraulic Filter Assembly (212-076-006-105) (Sheet 1 of 1)

Hydraulic Filter Assembly (212-076-006-007) - Inspection

Procedure

1. Examine the hydraulic fluid in filter bowl (6, Figure 1) for condition and for particles.
2. If the hydraulic fluid is dirty, discolored, or contains bubbles, flush the system (412-A-29-00-00-00A-251A-A / 00615).
3. Examine valve body (1), retainer (3), and filter bowl (6) for scratches, nicks, dents, thread damage, and damage to the lockwire holes.
4. If the lockwire holes in the valve body (1) are damaged, replace the valve body.
5. Examine impending bypass indicator button (7) to make sure it is not in the extended position.
6. Examine the filter element (5) for general condition. Pay particular attention to damage to the filtering surfaces, and bonding of the end caps to the filtering core.
7. Replace filter element (5) if any abnormality is found.
8. Inspect the filter element (5) core for metal, brass, and aluminum contamination.
9. Replace cleanable filter element (5) 205-076-034-007 if contamination is found and flush the hydraulic system.



INTEGRATED VALVE AND FILTER ASSEMBLY
(212-076-006-007)

1. Valve body
2. Packing
3. Retainer
4. Packing
5. Filter element (Cleanable element 205-076-034-007)
6. Filter bowl
7. Indicator button
8. Deleted

LOCKWIRE (C-405)

T 100 TO 140 IN-LBS
(11 TO 16 Nm)

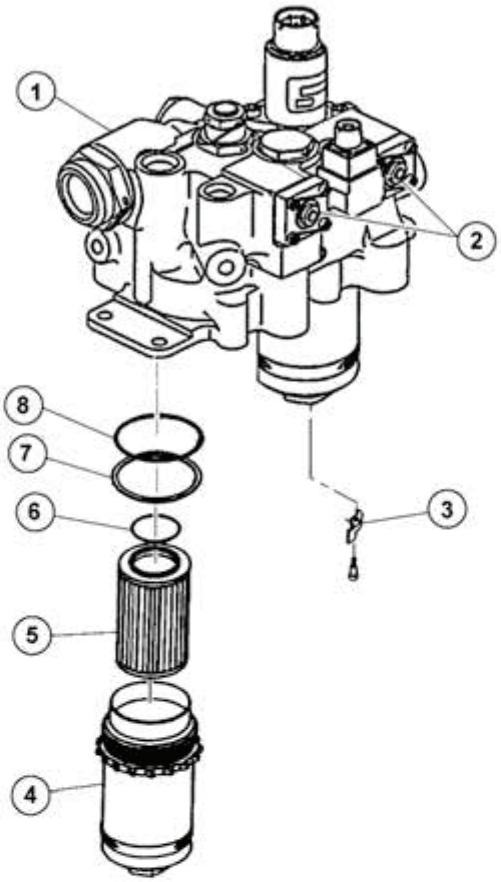
ICN-412-A-291000-B-97499-00014-A-002-01

Figure 1. Hydraulic Filter Assembly (212-076-006-007) (Sheet 1 of 1)

Hydraulic Filter Assembly (212-076-006-105) - Inspection

Procedure

1. *Examine the hydraulic fluid in filter bowl (4, Figure 1) for condition and for particles.*
2. If the hydraulic fluid is dirty, discolored, or contains bubbles, flush the system (412-A-29-00-00-00A-251A-A / 00615).
3. *Examine valve body (1), retainer (7), and filter bowl (4) for scratches, nicks, dents, and thread damage.*
4. *Examine ratchet lever (3) for wear and security of installation.*
5. *Examine ratchet teeth on filter bowl (4) for wear.*
6. *Examine impending bypass indicator button (2) to make sure it is not in the extended position.*
7. *Examine the filter element (5) for general condition. Pay particular attention to damage to the filtering surfaces, and bonding of the end caps to the filtering core.*
8. *Replace filter element (5) if any abnormality is found.*
9. *Inspect the filter element (5) for metal, brass, and aluminum contamination.*
10. *Replace non-cleanable filter element (5) 205-076-034-003 or cleanable filter element 205-076-034-007 if contamination is found and flush the hydraulic system.*



INTEGRATED VALVE AND FILTER ASSEMBLY
(212-076-006-105)

1. Valve body
2. Indicator button
3. Ratchet lever
4. Filter bowl
5. Filter element (Non cleanable element 205-076-034-003)
   Filter element (Cleanable element 205-076-034-007)
6. Packing
7. Retainer
8. Packing

ICN-412-A-291000-B-97499-03024-A-001-01

Figure 1. Hydraulic Filter Assembly (212-076-006-105) (Sheet 1 of 1)

Hydraulic Filter Assemblies (212-076-006-007 and -105) - Cleaning

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Hydraulic fluid	C-002	AR	Refer to Chapter 12 for information on usage and mixing of hydraulic fluid (C-002).
Hydraulic fluid	C-072	AR	Refer to Chapter 12 for information on usage and mixing of hydraulic fluid (C-072).
Turbine fuel	C-003	AR	
Drycleaning solvent	C-304	AR	
Detergent	C-355	AR	

Procedure



CLEAN HYDRAULIC COMPONENTS WITH drycleaning solvent (C-304) ONLY. DO NOT USE ALCOHOL.

NOTE

If non-cleanable 205-076-034-003 filter element (5, Figure 2) is installed, discard contaminated filter element (5).

Examine filter element 205-076-034-007 (5, Figure 1 or Figure 2) as applicable for 'X's inscribed on the filter element (5). If there are three 'X's inscribed on filter element (5), discard filter element (5).

1. *Use a suitable clean container large enough to hold cleanable filter element 205-076-034-007 (5).*

2. Pour enough of one of the following fluids to cover filter element (5): hydraulic fluid (C-002) or hydraulic fluid (C-072), drycleaning solvent (C-304), 0.5% of detergent (C-355) dissolved in clean warm water, or turbine fuel (C-003).

3. Cap or plug both ends of the filter element (5) to prevent contamination migration into the filter element.

NOTE

Discard cleanable filter element 205-076-034-007 after three cleaning events.

4. Insert cleanable filter element (5) only into container and vigorously agitate the filter in the cleaning fluid.

5. Attach an air hose with a nozzle and rubber grommet to the filter element (5) to prevent air from escaping.



USE NO MORE THAN 10 TO 15 PSI (68.95 TO 103.42 KPA) AIR PRESSURE SO AS NOT TO DAMAGE THE FILTER ELEMENT.

6. Using no more than 10 to 15 PSI (68.95 to 103.42 kPa) air pressure, slowly backflush the cleaning fluid through the filter element (5) a maximum of three times until there is no indication of contamination.

NOTE

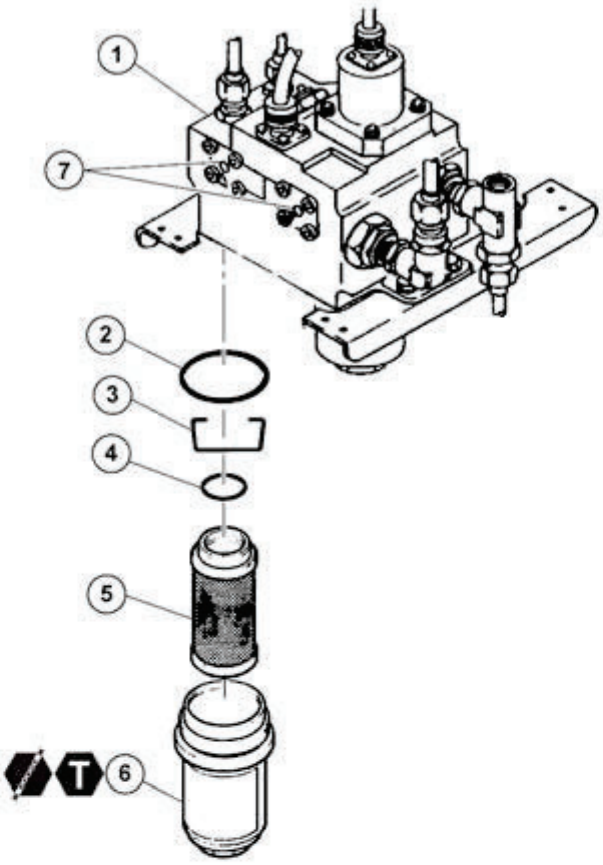
Discard cleanable filter element 205-076-034-007 after three cleaning events.

7. Make an 'X' on bottom of filter element (5) for each cleaning event.

8. Rinse the filter element (5) with clean hydraulic fluid (C-002) or hydraulic fluid (C-072) to remove traces of the cleaning fluid.

9. Dry filter element (5) using filtered, compressed air. If element is not going to be installed immediately, put it in a new plastic bag to prevent contamination.

10. Rinse filter bowl (5) with clean hydraulic fluid (C-002) or hydraulic fluid (C-072).



INTEGRATED VALVE AND FILTER ASSEMBLY
(212-076-006-007)

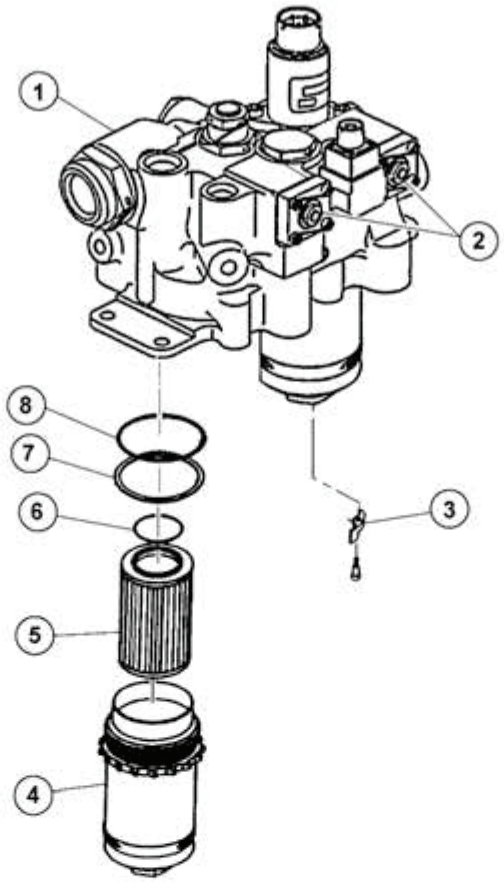
1. Valve body
2. Packing
3. Retainer
4. Packing
5. Filter element (Cleanable element 205-076-034-007)
6. Filter bowl
7. Indicator button
8. Deleted

LOCKWIRE (C-405)

T 100 TO 140 IN-LBS
(11 TO 16 Nm)

ICN-412-A-291000-B-97499-00014-A-002-01

Figure 1. Hydraulic Filter Assemblies (212-075-006-007) (Sheet 1 of 1)



INTEGRATED VALVE AND FILTER ASSEMBLY
(212-076-006-105)

1. Valve body
2. Indicator button
3. Ratchet lever
4. Filter bowl
5. Filter element (Non cleanable element 205-076-034-003)
   Filter element (Cleanable element 205-076-034-007)
6. Packing
7. Retainer
8. Packing

ICN-412-A-291000-B-97499-03024-A-001-01

Figure 2. Hydraulic Filter Assemblies (212-075-006-105) (Sheet 1 of 1)

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Hydraulic Filter Assembly (212-076-006-007) - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Hydraulic fluid	C-002	AR	Refer to Chapter 12 for information on usage and mixing of hydraulic fluid (C-002).
Hydraulic fluid	C-072	AR	Refer to Chapter 12 for information on usage and mixing of hydraulic fluid (C-072).
Lockwire	C-405	AR	

Procedure



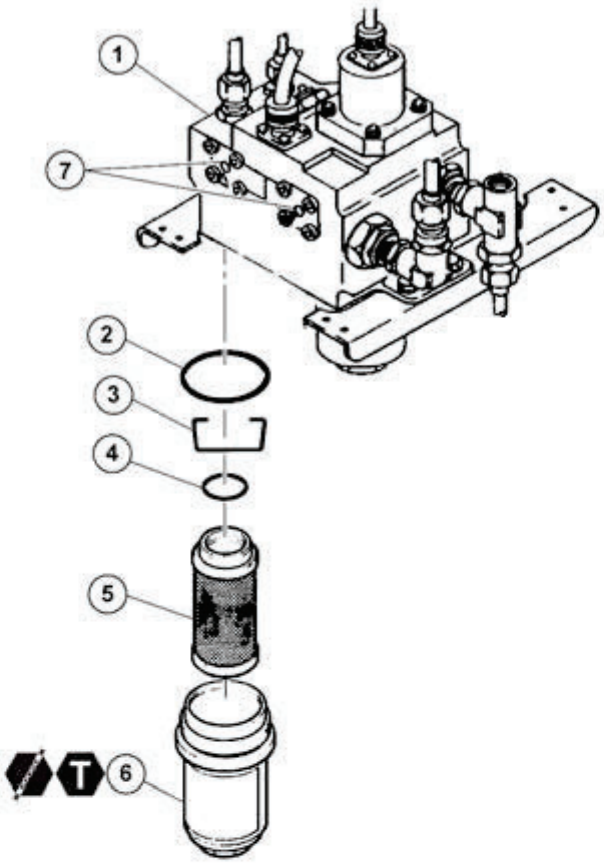
DO NOT USE NON-CLEANABLE 205-076-034-003 FILTER ELEMENT.

1. Lubricate packing (2, Figure 1) with hydraulic fluid (C-002) or hydraulic fluid (C-072).
2. Install packing (2) and in valve body (1).
3. Lubricate packing (4) with clean hydraulic fluid (C-002) or hydraulic fluid (C-072).
4. Install packing (4) in filter element (5).
5. Install filter element (5) and retainer (3) into filter bowl (6).
6. Fill filter bowl (6) half full with clean hydraulic fluid (C-002) or hydraulic fluid (C-072).
7. Lubricate threads of filter bowl (6) with clean hydraulic fluid (C-002) or hydraulic fluid (C-072).
8. Install filter bowl (6) in valve body (1).
9. Torque the filter bowl (6) .



DO NOT EXCESSIVELY TIGHTEN THE LOCKWIRE ON FILTER BODY AND FILTER BOWL. DAMAGE TO THE LOCKWIRE HOLES CAN OCCUR.

10. Carefully secure filter bowl (6) to valve body (1) with lockwire (C-405).
11. Install the forward pylon fairing (412-A-71-10-00-01A-720A-A / 01607).



INTEGRATED VALVE AND FILTER ASSEMBLY
(212-076-006-007)

1. Valve body
2. Packing
3. Retainer
4. Packing
5. Filter element (Cleanable element 205-076-034-007)
6. Filter bowl
7. Indicator button
8. Deleted

LOCKWIRE (C-405)

T 100 TO 140 IN-LBS
(11 TO 16 Nm)

ICN-412-A-291000-B-97499-00014-A-002-01

Figure 1. Hydraulic Filter Assembly (212-076-006-007) (Sheet 1 of 1)

Hydraulic Filter Assembly (212-076-006-105) - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Hydraulic fluid	C-002	AR	Refer to Chapter 12 for information on usage and mixing of hydraulic fluid (C-002).
Hydraulic fluid	C-072	AR	Refer to Chapter 12 for information on usage and mixing of hydraulic fluid (C-072).

Procedure



THE NON-CLEANABLE 205-076-034-003 AND CLEANABLE 205-076-034-007 FILTER ELEMENTS ARE INTERCHANGEABLE. INTERMIXING OF DIFFERENT ELEMENT TYPES IS NOT PERMITTED.

1. *Lubricate packing (8, Figure 1) with hydraulic fluid (C-002) or hydraulic fluid (C-072).*

2. *Install packing (8) and retainer (7) in valve body (1).*

3. *Lubricate packing (6) with clean hydraulic fluid (C-002) or hydraulic fluid (C-072).*

4. *Install packing (6) in filter element (5).*

5. *Install filter element (5) into filter bowl (4).*

6. *Fill filter bowl (4) half full with clean hydraulic fluid (C-002) or hydraulic fluid (C-072).*

7. *Lubricate threads of filter bowl (4) with clean hydraulic fluid (C-002) or hydraulic fluid (C-072).*

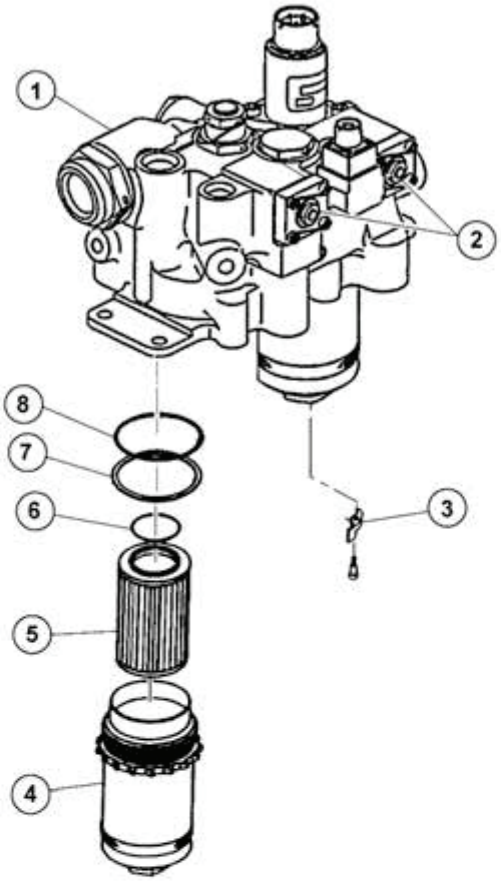
8. *Install filter bowl (4) in valve body (1) until the flange on the bowl touches the filter head. Turn the bowl counterclockwise until the ratchet lever (3) engages the serrations on the bowl.*

9. *Make sure the ratchet lever (3) is well engaged in the ratchet teeth of filter bowl (4).*

DELETED

10. Install the forward pylon fairing (412-A-71-10-00-01A-720A-A / 01607).

11. During initial run-up, check for leaks.



INTEGRATED VALVE AND FILTER ASSEMBLY
(212-076-006-105)

1. Valve body
2. Indicator button
3. Ratchet lever
4. Filter bowl
5. Filter element (Non cleanable element 205-076-034-003)
   Filter element (Cleanable element 205-076-034-007)
6. Packing
7. Retainer
8. Packing

ICN-412-A-291000-B-97499-03024-A-001-01

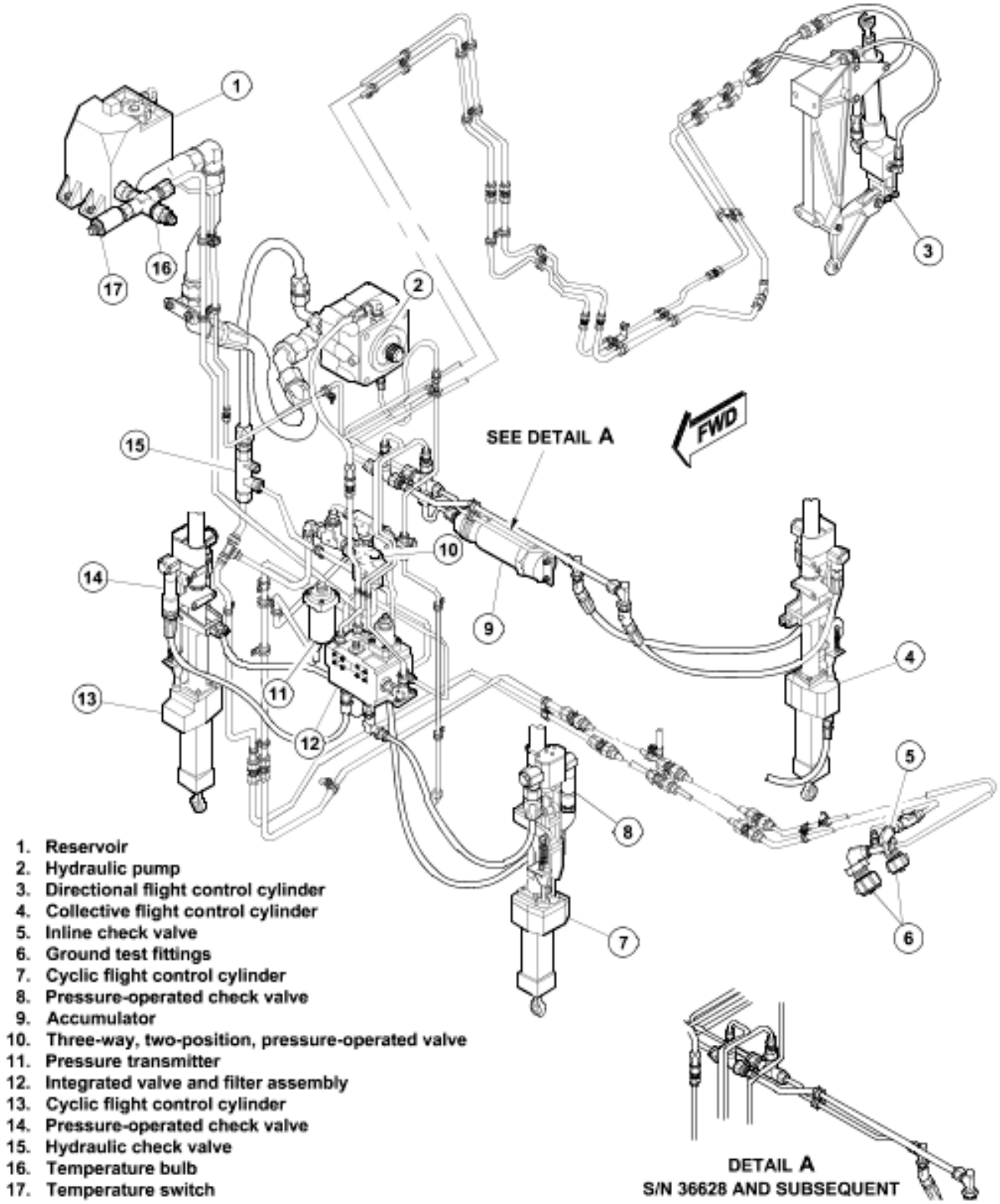
Figure 1. Hydraulic Filter Assembly (212-076-006-105) (Sheet 1 of 1)

Elbow Check Valve - Description

NOTE

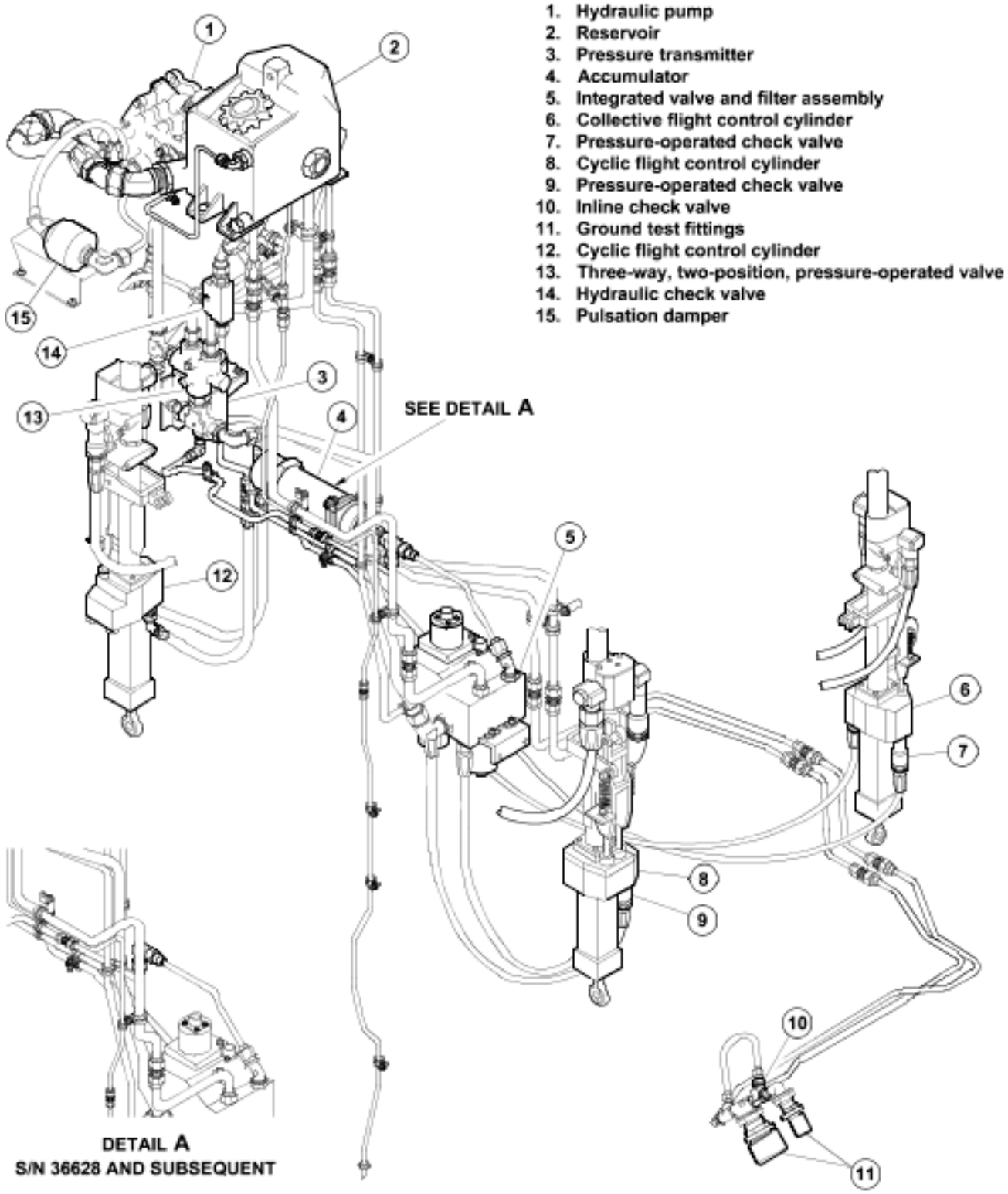
The maintenance procedures in 412-A-29-10-00-06A-520A-A / 00647, 412-A-29-10-00-06A-280A-A / 00648 and 412-A-29-10-00-06A-720A-A / 00649 are the same for hydraulic systems No. 1 and No. 2.

Hydraulic system No. 1 (Figure 1) and hydraulic system No. 2 (Figure 2) each have elbow check valves installed at the outlet port of the integrated valve and filter assembly. The check valves prevent loss of hydraulic fluid when filter is removed for inspection.



ICN-412-A-291000-B-97499-00007-A-001-01

Figure 1. Hydraulic System No. 1 (Sheet 1 of 1)



ICN-412-A-291000-B-97499-00008-A-001-01

Figure 2. Hydraulic System No. 2 (Sheet 1 of 1)

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Elbow Check Valve - Removal

Procedure

NOTE

The following removal procedures are the same for hydraulic systems No. 1 and No. 2.

1. Remove access panel on forward side of pylon support structure.

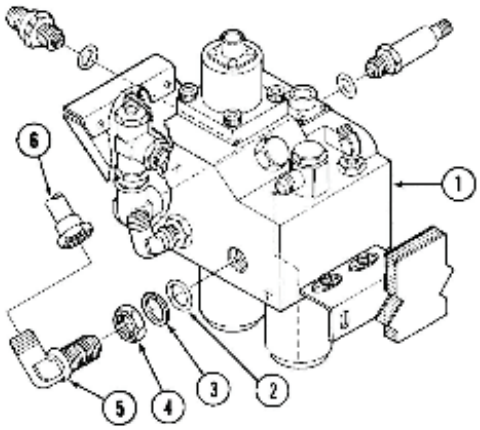
NOTE

Disconnecting tube (6, Figure 1) will drain fluid from reservoir. Provide a suitable container to catch fluid.

2. Disconnect tube (6).

3. Loosen nut (4). Remove elbow check valve (5). Remove and discard preformed packings (2).

4. Plug open tube (6) to prevent contamination.



1. Integrated valve and filter assembly
2. Preformed packing
3. Retainer
4. Nut
5. Elbow check valve
6. Tube

ICN-412-A-291000-B-97499-00016-A-001-01

Figure 1. Elbow Check Valve (Sheet 1 of 1)

Elbow Check Valve - Inspection

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Hydraulic Hand Pump	Commercial	1	

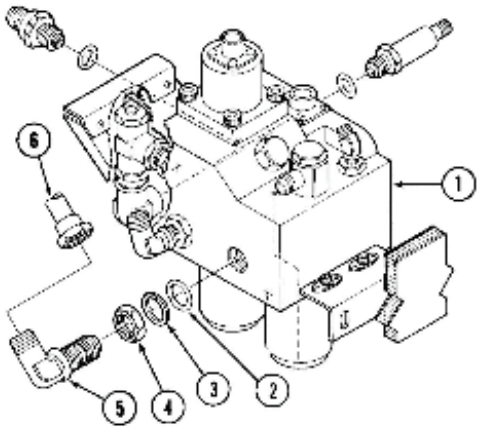
Procedure

NOTE

The following inspection procedures are the same for hydraulic systems No. 1 and No. 2.

1. Inspect elbow check valve (5, Figure 1) for damage.

2. Attach elbow check valve (5) to a hydraulic hand pump. Verify fluid passes in direction of arrow, and is blocked in opposite direction.



1. Integrated valve and filter assembly
2. Preformed packing
3. Retainer
4. Nut
5. Elbow check valve
6. Tube

ICN-412-A-291000-B-97499-00016-A-001-01

Figure 1. Elbow Check Valve (Sheet 1 of 1)

Elbow Check Valve - Installation

Procedure

NOTE

The following installation procedures are the same for hydraulic systems No. 1 and No. 2.

1. Install nut (4, Figure 1) all the way into elbow check valve (5).

2. Install retainer (3) into elbow check valve (5).

3. Lubricate new preformed packing (2) with same type of hydraulic fluid used in system (Chapter 12). Position packing on unthreaded area of elbow check valve (5).

4. Install elbow check valve (5) into integrated valve and filter assembly (1) until preformed packing (2) contacts integrated valve and filter assembly.

5. Connect tube (6).

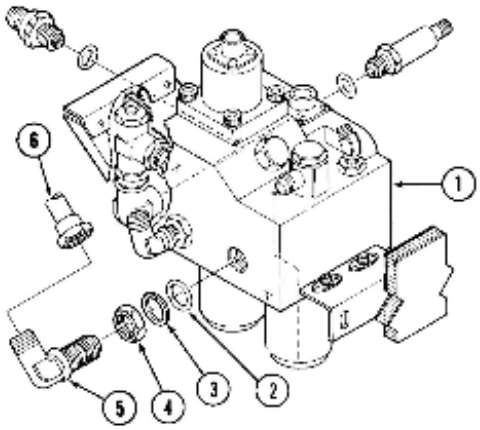
6. Tighten nut (4).

7. Install access panel.

8. Service reservoir (412-A-12-10-02-00A-040A-A / 00138).

9. Perform ground run and verify system functions normally and no leaks are present.

10. Install access panel.



1. Integrated valve and filter assembly
2. Preformed packing
3. Retainer
4. Nut
5. Elbow check valve
6. Tube

ICN-412-A-291000-B-97499-00016-A-001-01

Figure 1. Elbow Check Valve (Sheet 1 of 1)

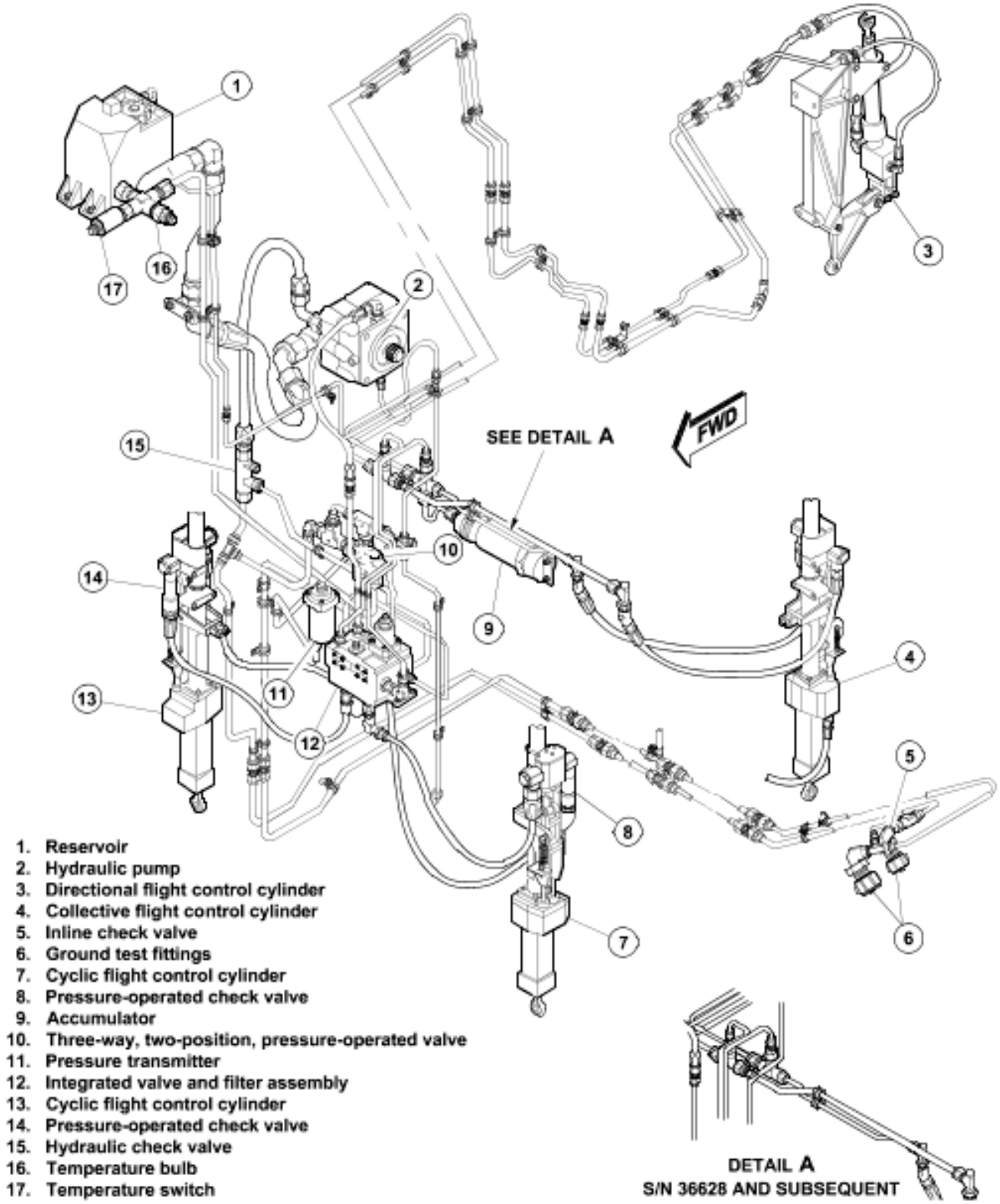
Inline Check Valve - Description

NOTE

The maintenance procedures in 412-A-29-10-00-07A-520A-A / 00651, 412-A-29-10-00-07A-280A-A / 00652 and 412-A-29-10-00-07A-720A-A / 00653 are the same for hydraulic systems No. 1 and No. 2.

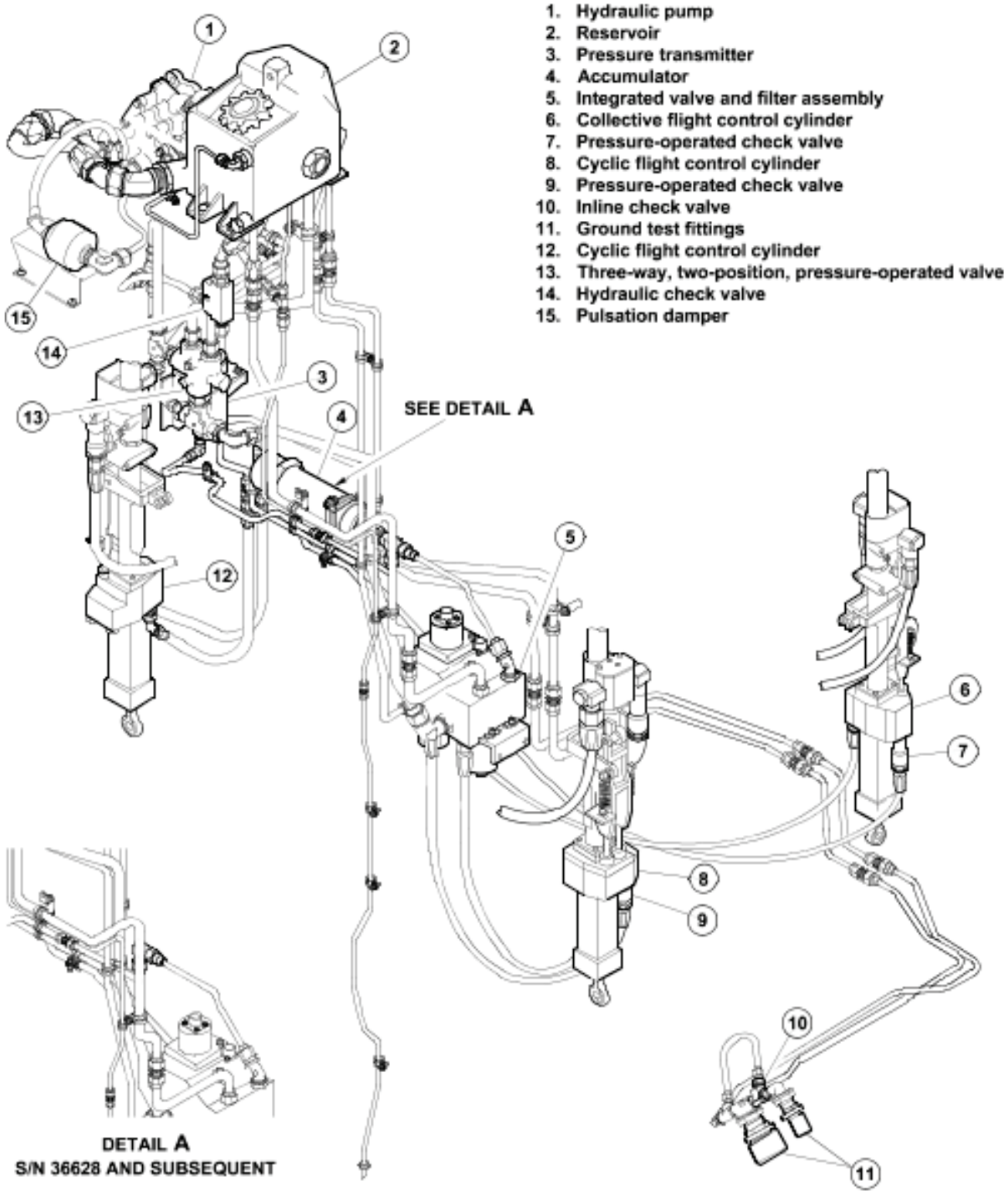
Hydraulic system No. 1 (Figure 1) contains seven inline check valves as follows: three in return ports on integrated valve and filter assembly, one between ground test connections, one between ground test pressure connection and pump pressure line, one at directional control actuator inlet, and one in hydraulic pump case drain outlet.

Hydraulic system No. 2 (Figure 2) contains six inline check valves as follows: three in return ports on integrated valve and filter assembly, one between ground test connections, one between ground test pressure connection and pump pressure line, and one in hydraulic pump case drain outlet.



ICN-412-A-291000-B-97499-00007-A-001-01

Figure 1. Hydraulic System No. 1 (Sheet 1 of 1)



ICN-412-A-291000-B-97499-00008-A-001-01

Figure 2. Hydraulic System No. 2 (Sheet 1 of 1)

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Inline Check Valve - Removal

Procedure

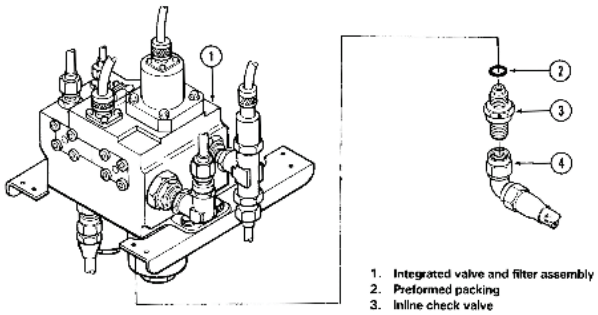
NOTE

The following removal procedures are the same for hydraulic systems No. 1 and No. 2.

1. Remove access panels as necessary to reach inline check valve (3, Figure 1).

2. Disconnect hose (4) from inline check valve (3).

3. Remove inline check valve (3). Remove and discard preformed packing (2).



ICN-412-A-291000-B-97499-00017-A-001-01

Figure 1. Inline Check Valve (Sheet 1 of 1)

Inline Check Valve - Inspection

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Hydraulic Hand Pump	Commercial	1	

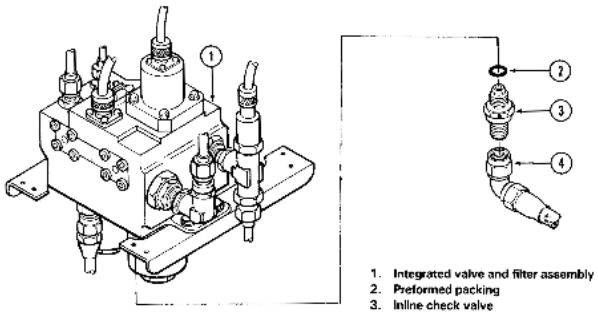
Procedure

NOTE

The following inspection procedures are the same for hydraulic systems No. 1 and No. 2.

1. Inspect inline check valve (3, Figure 1) for damage.

2. Use a hydraulic hand pump to verify inline check valve (3) will pass fluid in direction of arrow and blocks fluid in opposite direction.



ICN-412-A-291000-B-97499-00017-A-001-01

Figure 1. Inline Check Valve (Sheet 1 of 1)

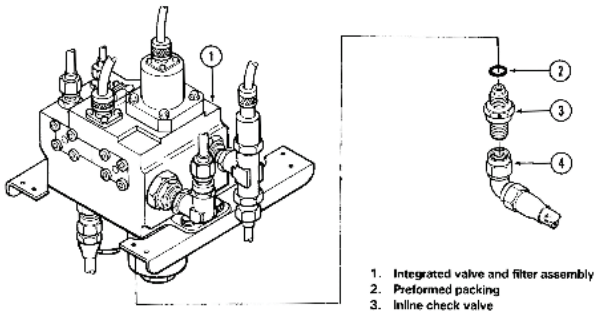
Inline Check Valve - Installation

Procedure

NOTE

The following installation procedures are the same for hydraulic systems No. 1 and No. 2.

1. Lubricate new preformed packing (2, Figure 1) with hydraulic fluid (412-A-12-10-02-00A-040A-A / 00138) and install on inline check valve (3).
2. Install inline check valve (3) into integrated valve and filter assembly (1), carefully observing arrow and direction of fluid flow.
3. Connect the hose (4) to the inline check valve (3).
4. Service the reservoir (412-A-12-10-02-00A-040A-A / 00138).
5. Perform a ground run and verify that the system functions normally and no leaks are present.
6. Install the access panels.



ICN-412-A-291000-B-97499-00017-A-001-01

Figure 1. Inline Check Valve (Sheet 1 of 1)

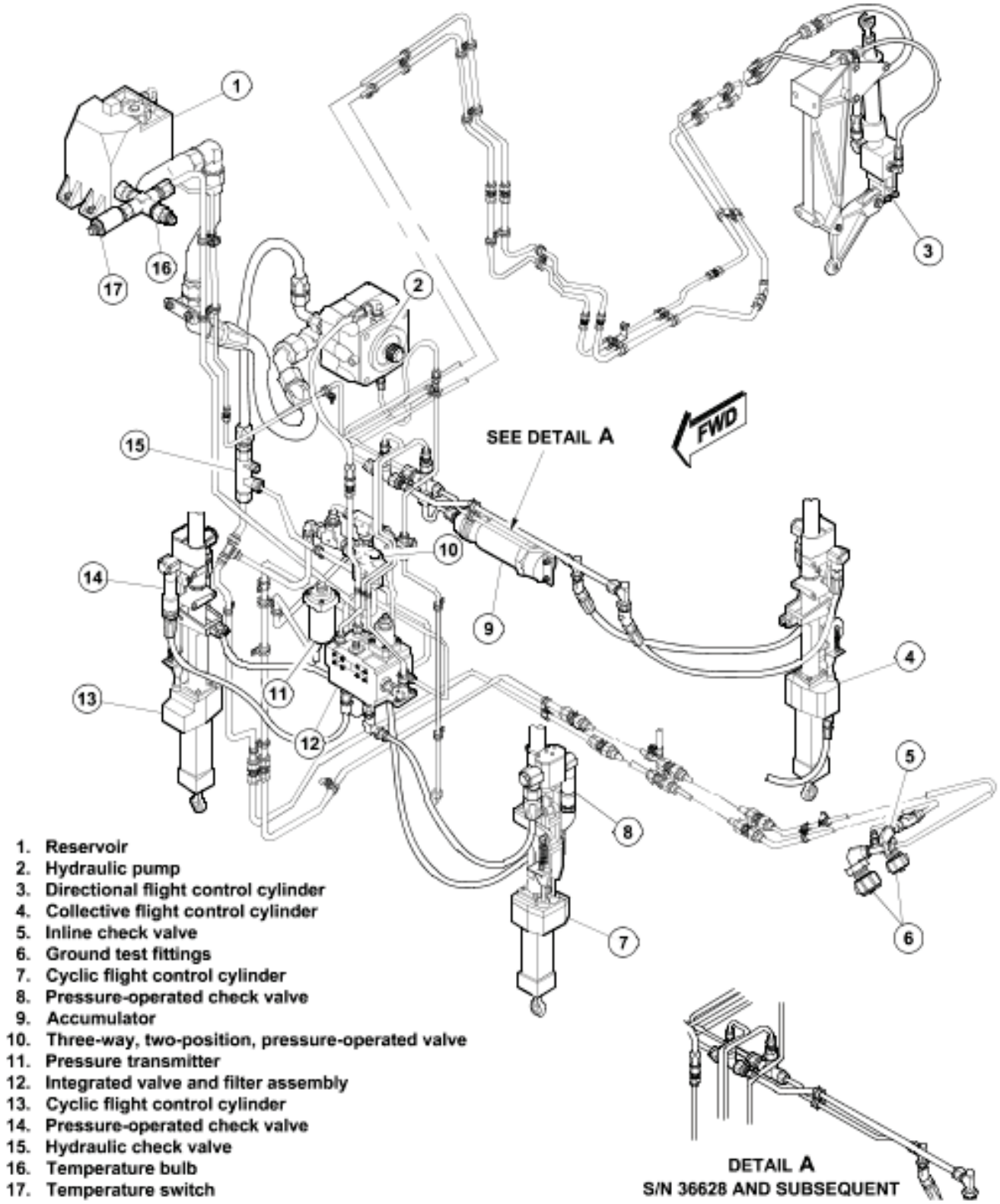
Pressure Switch - Description

NOTE

The maintenance procedures in 412-A-29-10-00-08A-520A-A / 00655 and 412-A-29-10-00-08A-720A-A / 00656 are the same for hydraulic systems No. 1 and No. 2.

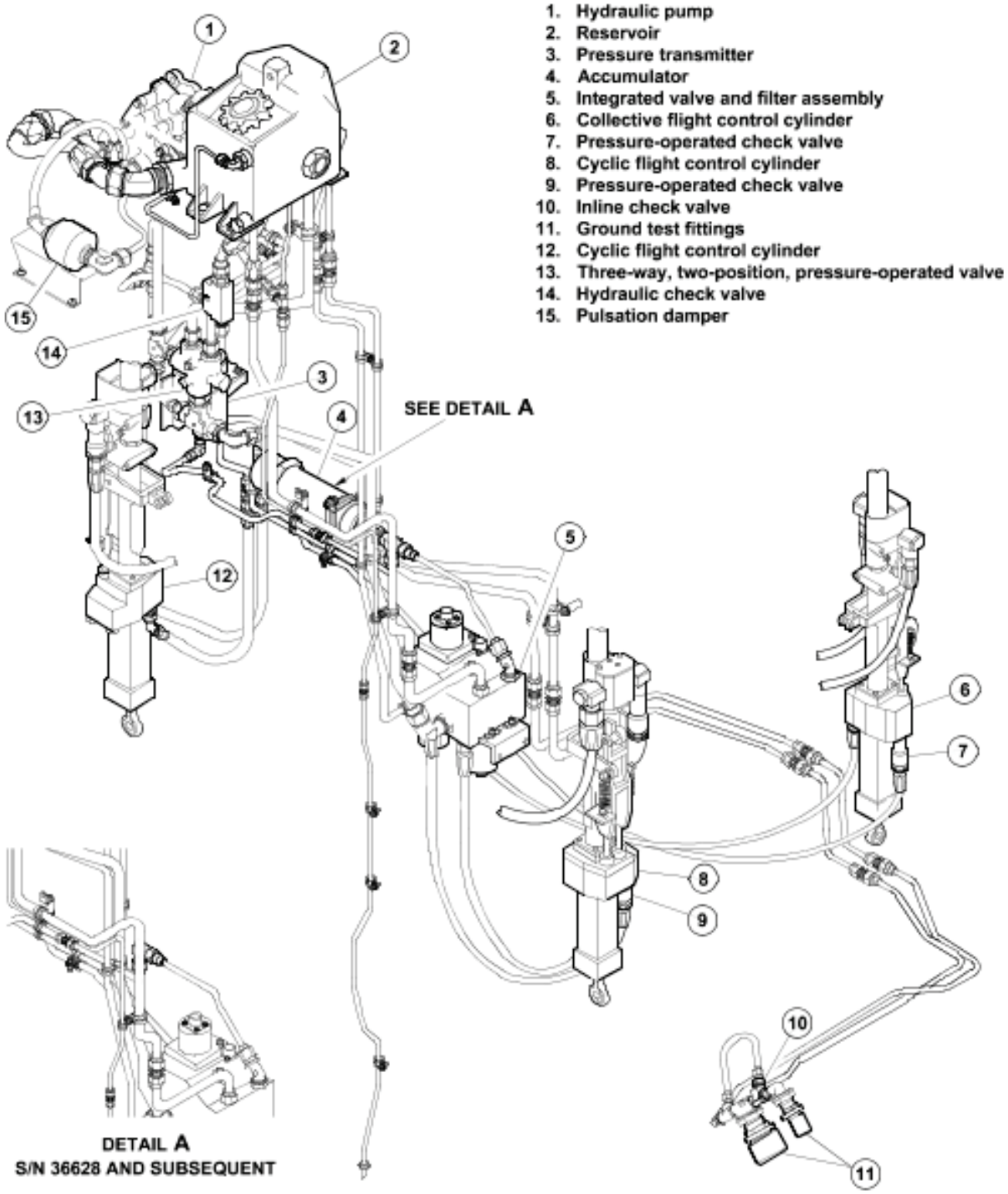
Hydraulic system No. 1 (Figure 1) and hydraulic system No. 2 (Figure 2) each have a pressure operated switch installed in the integrated valve and filter assembly. The switches cause the appropriate caution segment (NO. 1 HYDRAULIC and/or NO. 2 HYDRAULIC) to illuminate in the event pressure in the affected system decreases below 650 PSI (4481.59 kPa).

In the event the caution segment(s) illuminates, the pilot can verify whether fault is low pressure or high temperature in the hydraulic system No. 1 or hydraulic system No. 2 from separate instrument readings.



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Figure 1. Hydraulic System No. 1 (Sheet 1 of 1)



ICN-412-A-291000-B-97499-00008-A-001-01

Figure 2. Hydraulic System No. 2 (Sheet 1 of 1)

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Pressure Switch - Removal

Procedure

NOTE

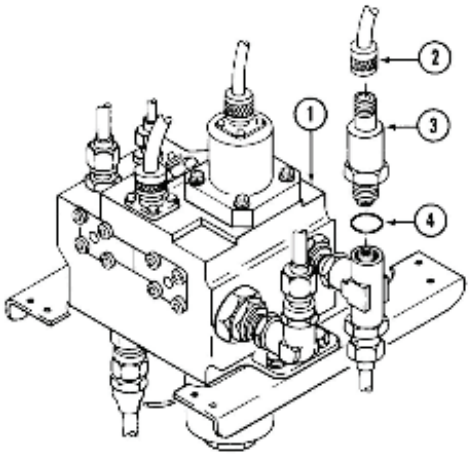
The following removal procedures are the same for hydraulic systems No. 1 and No. 2.

1. Remove the access panel on the forward side of the pylon support structure inside the cabin.

2. Disconnect the electrical connector (2, Figure 1).

3. Remove the pressure switch (3). Remove and discard the preformed packing (4).

4. Plug the open fitting to prevent contamination.



1. Integrated valve and filter assembly
2. Electrical connector
3. Pressure switch
4. Preformed packing

ICN-412-A-291000-B-97499-00018-A-001-01

Figure 1. Pressure Switch (Sheet 1 of 1)

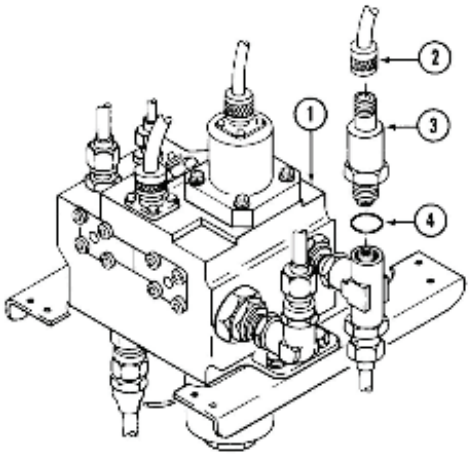
Pressure Switch - Installation

Procedure

NOTE

The following installation procedures are the same for hydraulic systems No. 1 and No. 2.

1. Lubricate a new preformed packing (4, Figure 1) with hydraulic fluid (412-A-12-10-02-00A-040A-A / 00138) and install on the pressure switch (3).
2. Install the pressure switch (3) into the fitting.
3. Connect the electrical connector (2).
4. Service the reservoir (412-A-12-10-02-00A-040A-A / 00138).
5. Perform a ground run and verify that the system functions normally and no leaks are present.
6. Install the access panel.



1. Integrated valve and filter assembly
2. Electrical connector
3. Pressure switch
4. Preformed packing

ICN-412-A-291000-B-97499-00018-A-001-01

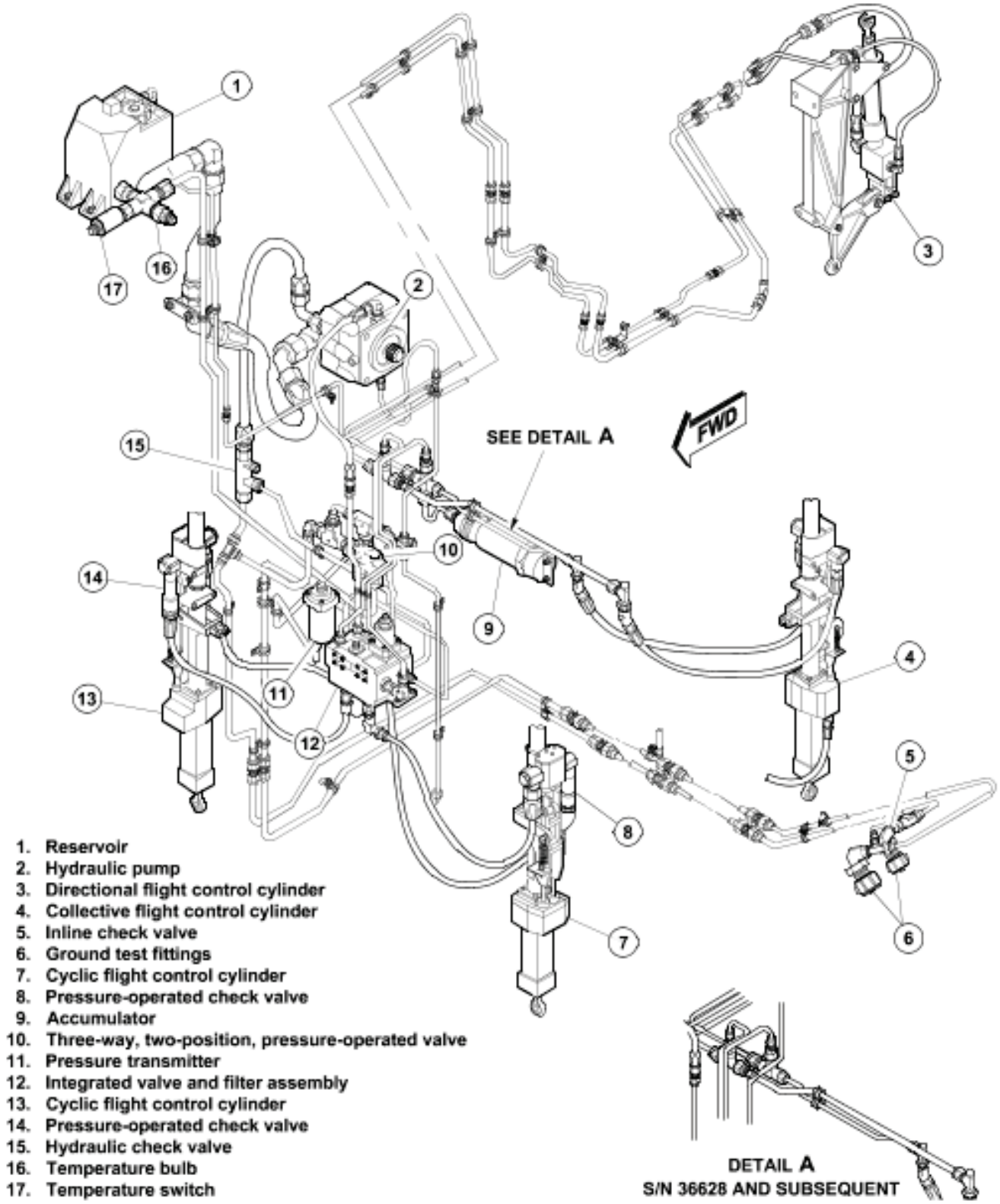
Figure 1. Pressure Switch (Sheet 1 of 1)

Hydraulic Accumulator - Description

NOTE

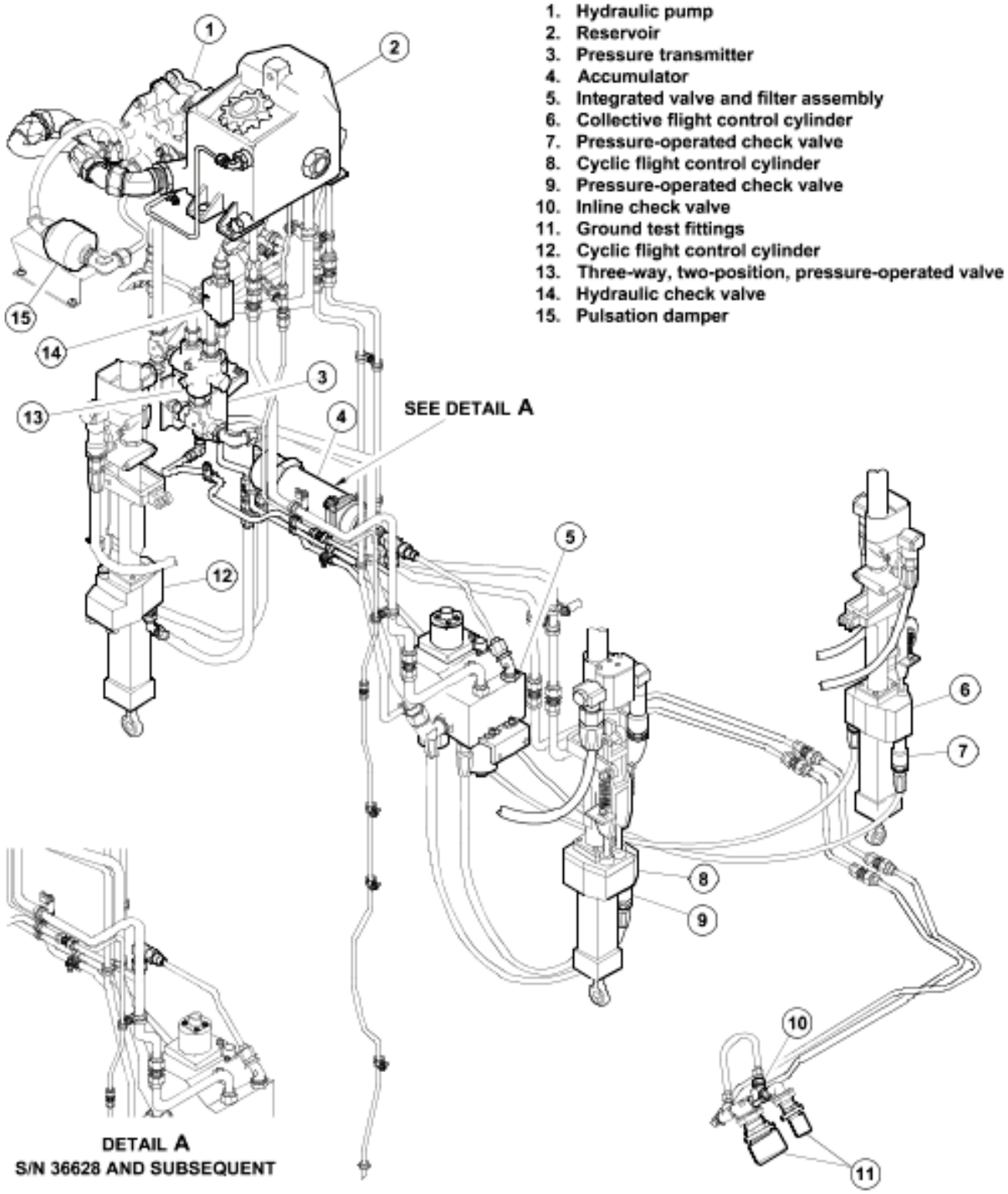
The maintenance procedures in 412-A-29-10-00-09A-520A-A / 00658, 412-A-29-10-00-09A-280A-A / 00659 and 412-A-29-10-00-09A-720A-A / 00660 are the same for hydraulic systems No. 1 and No. 2.

Hydraulic system No. 1 (Figure 1) and hydraulic system No. 2 (Figure 2) have separate accumulators installed on the aft side of the lift beam below the transmission. The source of energy for the accumulators is a spring loaded piston, which is compressed when the hydraulic pump is operating. In the event of pump failure, the spring loaded piston maintains pressure in the system for a limited time.



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Figure 1. Hydraulic System No. 1 (Sheet 1 of 1)



ICN-412-A-291000-B-97499-00008-A-001-01

Figure 2. Hydraulic System No. 2 (Sheet 1 of 1)

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Hydraulic Accumulator - Removal

Procedure

NOTE

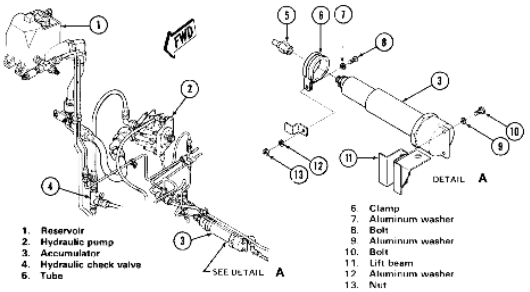
The following removal procedures are the same for hydraulic systems No. 1 and No. 2.

1. Disconnect the tube (5, Figure 1).

2. Plug the tube (5) to prevent contamination.

3. Remove the bolts (8 and 10). Remove the accumulator (3).

4. Remove the clamp (6) from the accumulator (3).



ICN-412-A-291000-B-97499-00019-A-001-01

Figure 1. Hydraulic Accumulator (Sheet 1 of 1)

Hydraulic Accumulator - Inspection

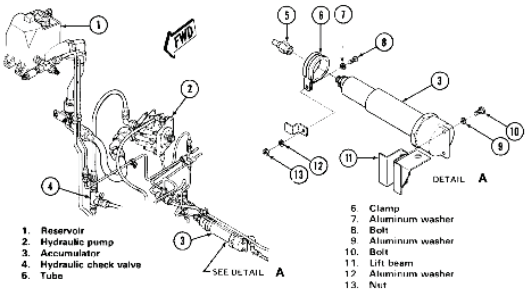
Procedure

NOTE

The following inspection procedures are the same for hydraulic systems No. 1 and No. 2.

1. Inspect the accumulator (3, Figure 1) for damage and signs of leakage.

2. Inspect for fluid leaking from the weep hole in the spring end of the accumulator (3). If fluid is leaking from the weep hole, reseal the unit (BHT-412-CR&O).



ICN-412-A-291000-B-97499-00019-A-001-01

Figure 1. Hydraulic Accumulator (Sheet 1 of 1)

Hydraulic Accumulator - Installation

Procedure

NOTE

The following installation procedures are the same for hydraulic systems No. 1 and No. 2.

1. Install the clamp (6, Figure 1) on the accumulator (3).

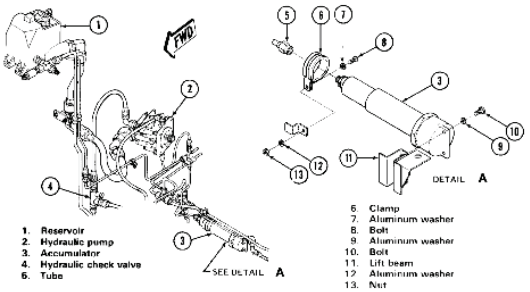
2. Secure the accumulator (3) to the lift beam (11) with the bolts (10) and the aluminum washers (9).

3. Install bolt (8), aluminum washers (7 and 12), and nut (13).

4. Connect tube (5).

5. Service reservoir (412-A-12-10-02-00A-040A-A / 00138).

6. Perform ground run and verify system functions normally and no leaks are present.



ICN-412-A-291000-B-97499-00019-A-001-01

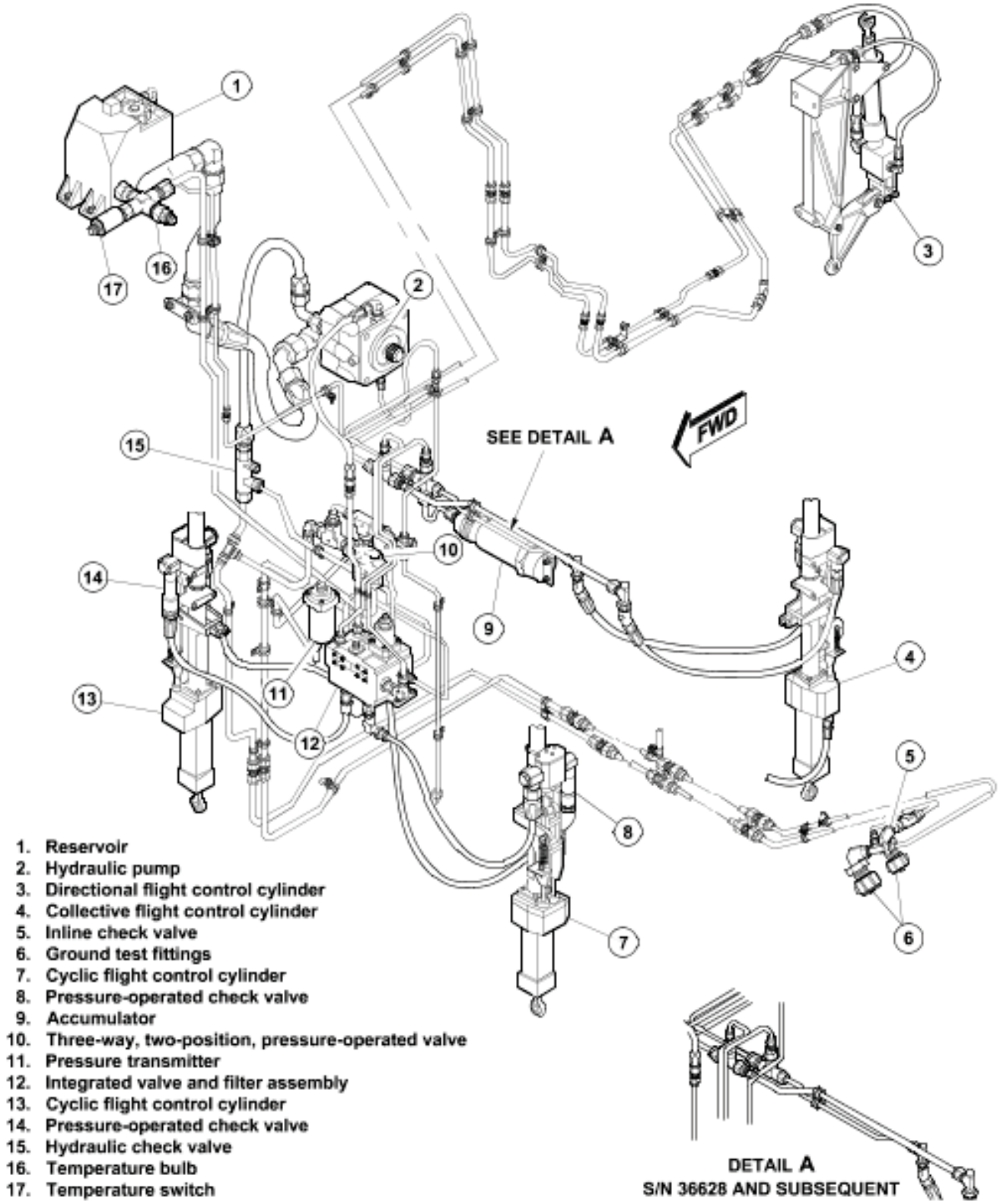
Figure 1. Hydraulic Accumulator (Sheet 1 of 1)

Three-Way, Two-Position, Pressure-Operated Valve - Description

NOTE

The maintenance procedures in 412-A-29-10-00-10A-520A-A / 00662, 412-A-29-10-00-10A-280A-A / 00663 and 412-A-29-10-00-10A-720A-A / 00664 are the same for hydraulic systems No. 1 and No. 2.

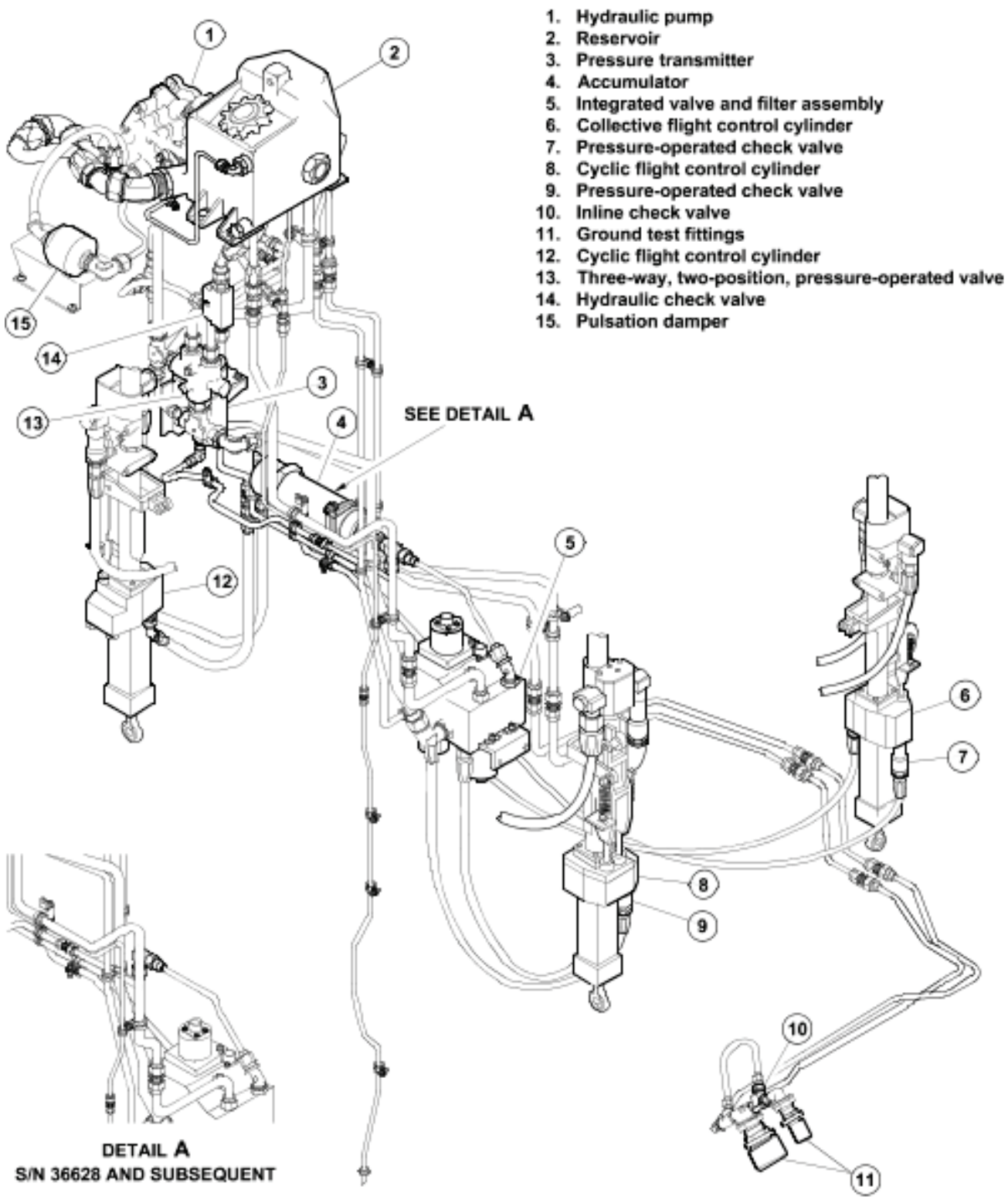
Hydraulic system No. 1 (Figure 1) and hydraulic system No. 2 (Figure 2) have separate three-way, two-position, pressure-operated valves. The valve port systems return fluid to the ground test run fitting when 90 to 360 PSI (621 to 2482 kPa) pressure is applied to ground test pressure connection. This prevents over filling reservoir. The valves are located in the compartment below the transmission.



1. Reservoir
2. Hydraulic pump
3. Directional flight control cylinder
4. Collective flight control cylinder
5. Inline check valve
6. Ground test fittings
7. Cyclic flight control cylinder
8. Pressure-operated check valve
9. Accumulator
10. Three-way, two-position, pressure-operated valve
11. Pressure transmitter
12. Integrated valve and filter assembly
13. Cyclic flight control cylinder
14. Pressure-operated check valve
15. Hydraulic check valve
16. Temperature bulb
17. Temperature switch

ICN-412-A-291000-B-97499-00007-A-001-01

Figure 1. Hydraulic System No. 1 (Sheet 1 of 1)



ICN-412-A-291000-B-97499-00008-A-001-01

Figure 2. Hydraulic System No. 2 (Sheet 1 of 1)

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Three-Way, Two-Position, Pressure-Operated Valve - Removal

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Hydraulic Suction Gun	Commercial	1	

Procedure

NOTE

The following removal procedures are the same for hydraulic systems No. 1 and No. 2.

1. Remove forward pylon fairing.

2. Remove hydraulic fluid from reservoir (2, Figure 1) with a hydraulic suction gun.

3. Remove access panel on forward side of pylon support.

NOTE

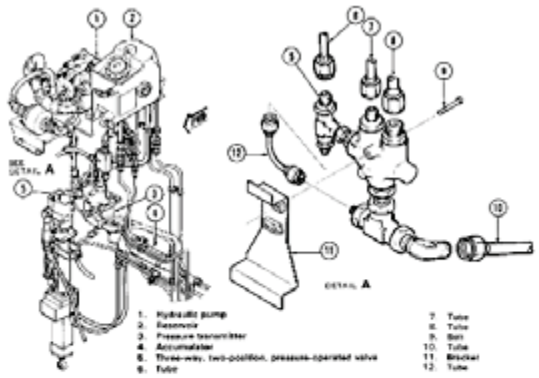
Removing tube (10) will drain reservoir. Provide suitable container to catch fluid.

4. Disconnect tube (10).

5. Disconnect tubes (6, 7, and 8).

6. Remove bolts (9) and three-way, two-position, pressure-operated valve (5).

7. Cap or plug open tubes to prevent contamination.



ICN-412-A-291000-B-97499-00020-A-001-01

Figure 1. Three-Way, Two-Position, Pressure-Operated Valve (Sheet 1 of 1)

Three-Way, Two-Position, Pressure-Operated Valve - Inspection Procedure

NOTE

The following inspection procedure is same for hydraulic systems No. 1 and No. 2.

1. Inspect valve body for damaged threads.

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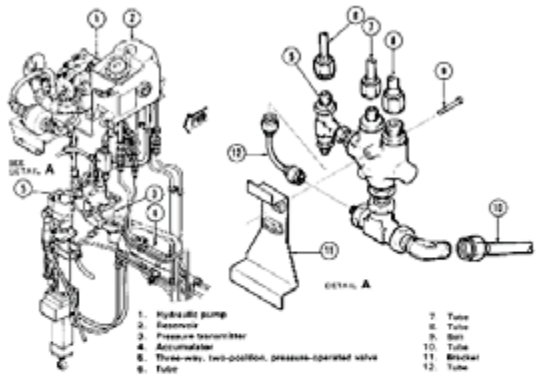
Three-Way, Two-Position, Pressure-Operated Valve - Installation

Procedure

NOTE

The following installation procedures are the same for hydraulic systems No. 1 and No. 2.

1. Install fittings and reducers on three-way, two-position, pressure-operated valve (5, Figure 1). Install tube (12) if removed.
2. Place three-way, two-position, pressure-operated valve (5) against bracket (11) and loosely install bolts (9).
3. Connect tubes (6, 7, 8, and 10).
4. Tighten bolts (9).
5. Service reservoir (412-A-12-10-02-00A-040A-A / 00138).
6. Install pylon fairing.
7. Perform ground run and verify system functions normally and no leaks are present.



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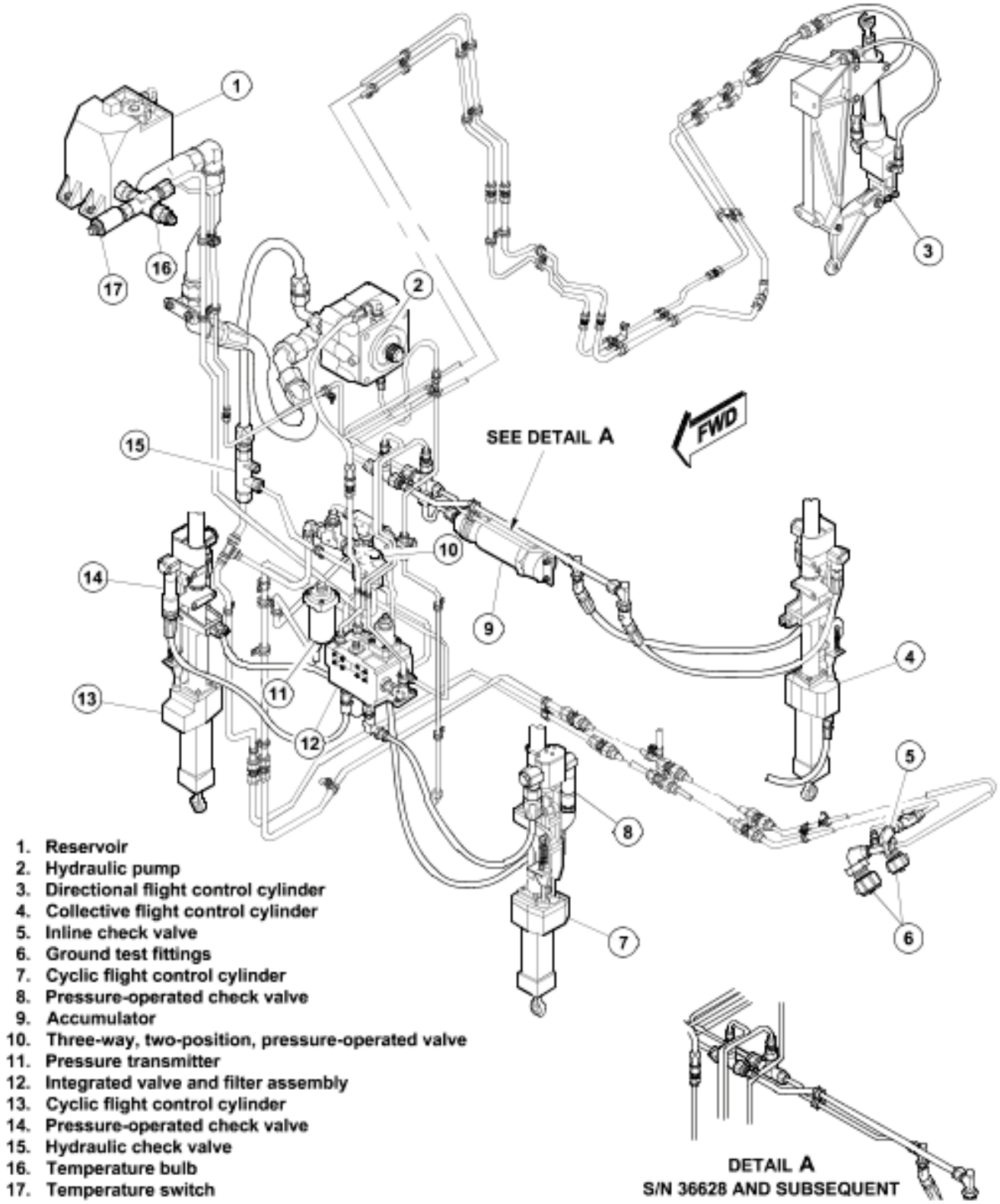
Figure 1. Three-Way, Two-Position, Pressure-Operated Valve (Sheet 1 of 1)

Hydraulic Check Valve - Description

NOTE

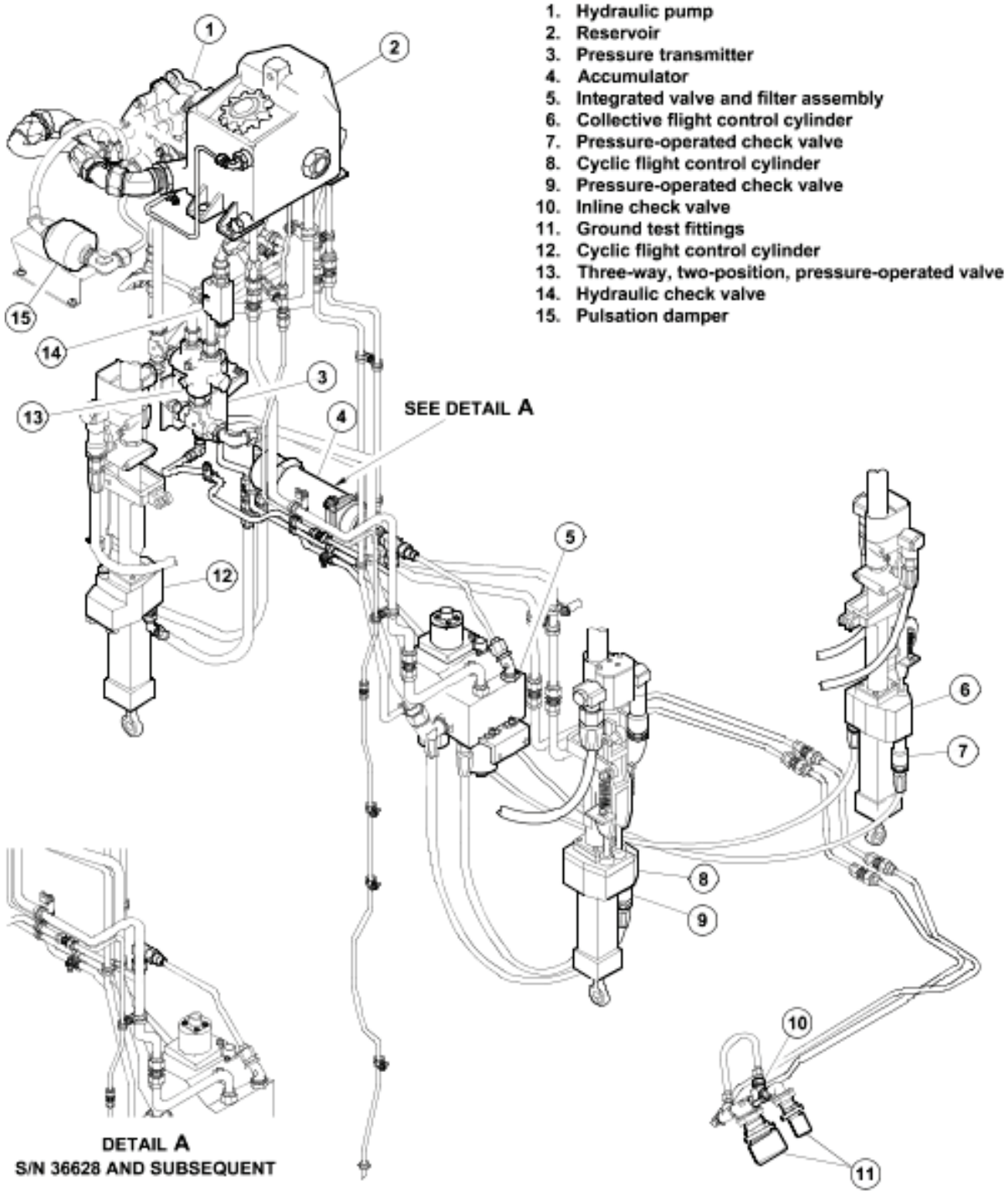
The maintenance procedures in 412-A-29-10-00-11A-520A-A / 00666, 412-A-29-10-00-11A-280A-A / 00667, 412-A-29-10-00-11A-685A-A / 00668 and 412-A-29-10-00-11A-720A-A / 00669 are the same for hydraulic systems No. 1 and No. 2.

Hydraulic system No. 1 (Figure 1) and hydraulic system No. 2 (Figure 2) have separate hydraulic check valves installed between the pump and the integrated valve and filter assembly in the compartment below the transmission. The valves prevent the pump from being pressurized and the reservoir from being overfilled when the ground test connection is used.



ICN-412-A-291000-B-97499-00007-A-001-01

Figure 1. Hydraulic System No. 1 (Sheet 1 of 1)



ICN-412-A-291000-B-97499-00008-A-001-01

Figure 2. Hydraulic System No. 2 (Sheet 1 of 1)

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Hydraulic Check Valve - Removal

Procedure

NOTE

The following removal procedures are the same for hydraulic systems No. 1 and No. 2.

1. Remove access panel on forward side of pylon.

NOTE

When disconnecting pump pressure hose (4, Figure 1), provide suitable container to catch fluid.

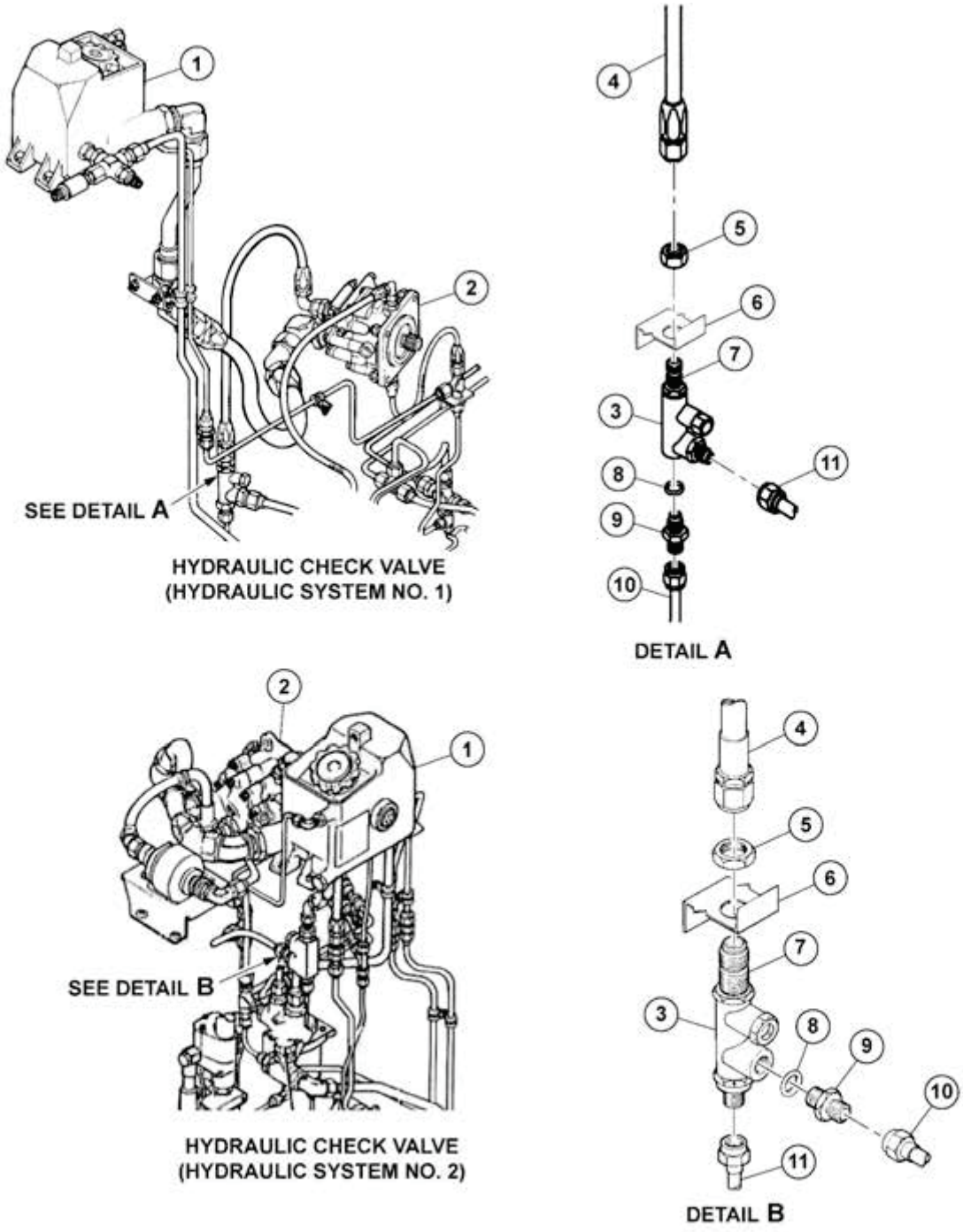
2. Disconnect pump pressure hose (4).

3. Disconnect tubes (10 and 11).

4. Remove nut (5) and hydraulic check valve (3).

5. Remove and discard preformed packing (8).

6. Cap or plug open tubes and hose to prevent contamination.



1. Reservoir
2. Hydraulic pump
3. Hydraulic check valve
4. Pump pressure hose
5. Nut
6. Structure
7. Union
8. Preformed packing
9. Inline check valve
10. Tube
11. Tube

ICN-412-A-291000-B-97499-00021-A-001-01

Figure 1. Hydraulic Check Valve (Sheet 1 of 1)

Hydraulic Check Valve - Inspection

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Hydraulic Hand Pump	Commercial	1	

Procedure

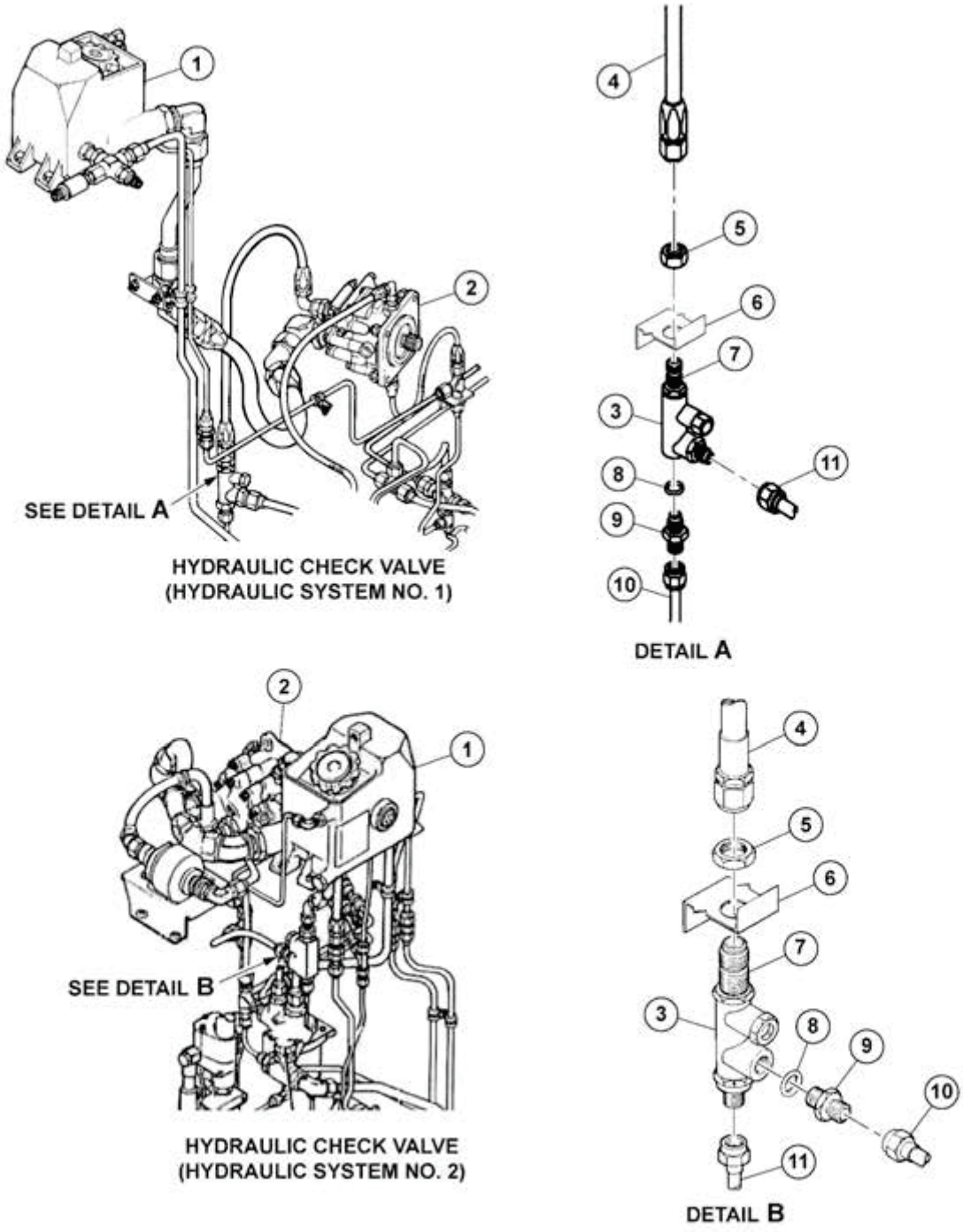
NOTE

The following inspection procedures are the same for hydraulic systems No. 1 and No. 2.

1. Inspect inline check valve (9, Figure 1) for proper operation and damage using hydraulic hand pump.

2. Inspect fittings for damage.

3. Inspect hydraulic check valve (3) for damage, signs of leakage, and proper operation.



1. Reservoir
2. Hydraulic pump
3. Hydraulic check valve
4. Pump pressure hose
5. Nut
6. Structure
7. Union
8. Preformed packing
9. Inline check valve
10. Tube
11. Tube

ICN-412-A-291000-B-97499-00021-A-001-01

Figure 1. Hydraulic Check Valve (Sheet 1 of 1)

Hydraulic Check Valve - Repair

Procedure

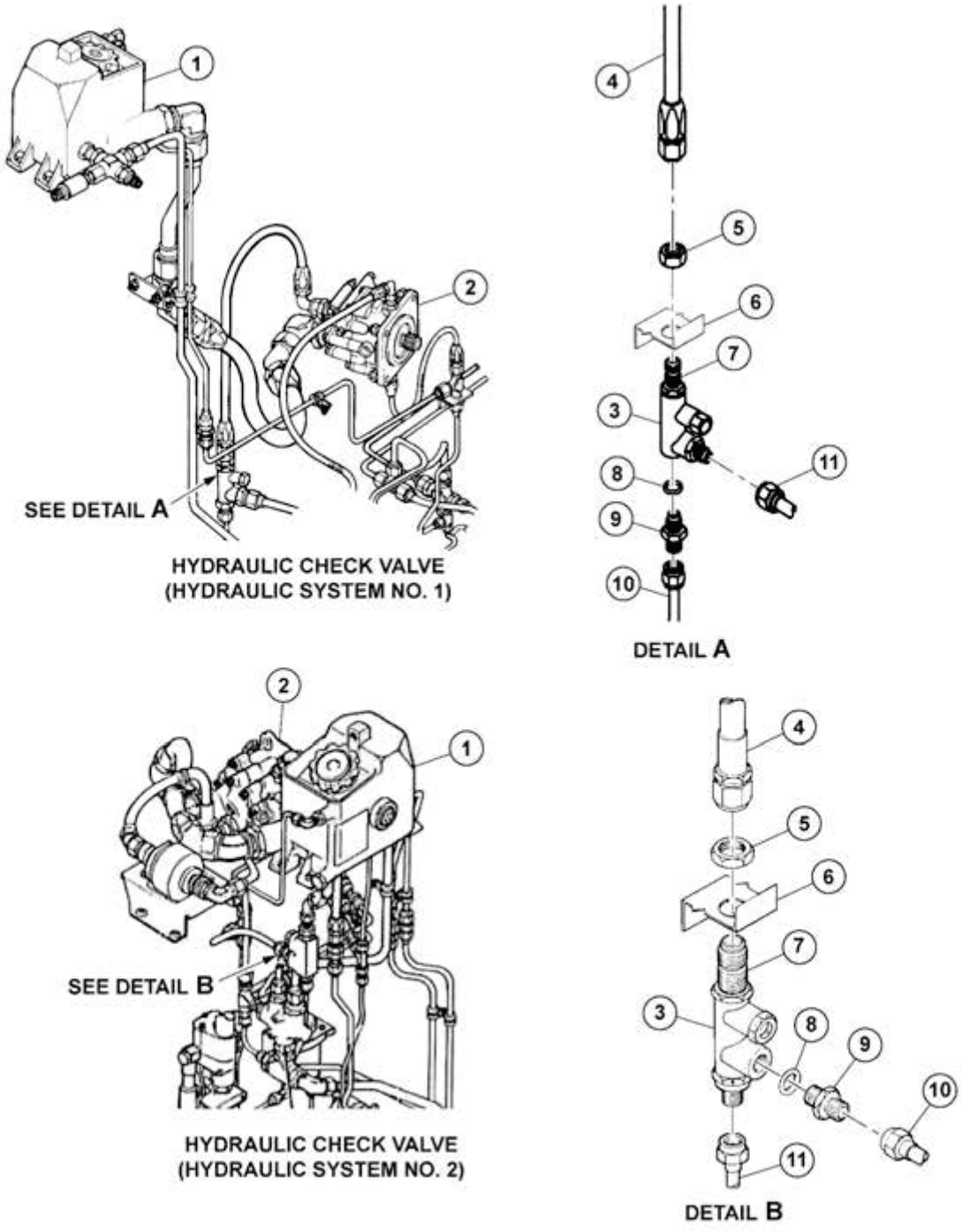
NOTE

The following repair procedures are the same for hydraulic systems No. 1 and No. 2.

1. Replace inline check valve (9, Figure 1) if defective.

2. Replace damaged fittings.

3. Replace hydraulic check valve (3) if defective.



1. Reservoir
2. Hydraulic pump
3. Hydraulic check valve
4. Pump pressure hose
5. Nut
6. Structure
7. Union
8. Preformed packing
9. Inline check valve
10. Tube
11. Tube

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Figure 1. Hydraulic Check Valve (Sheet 1 of 1)

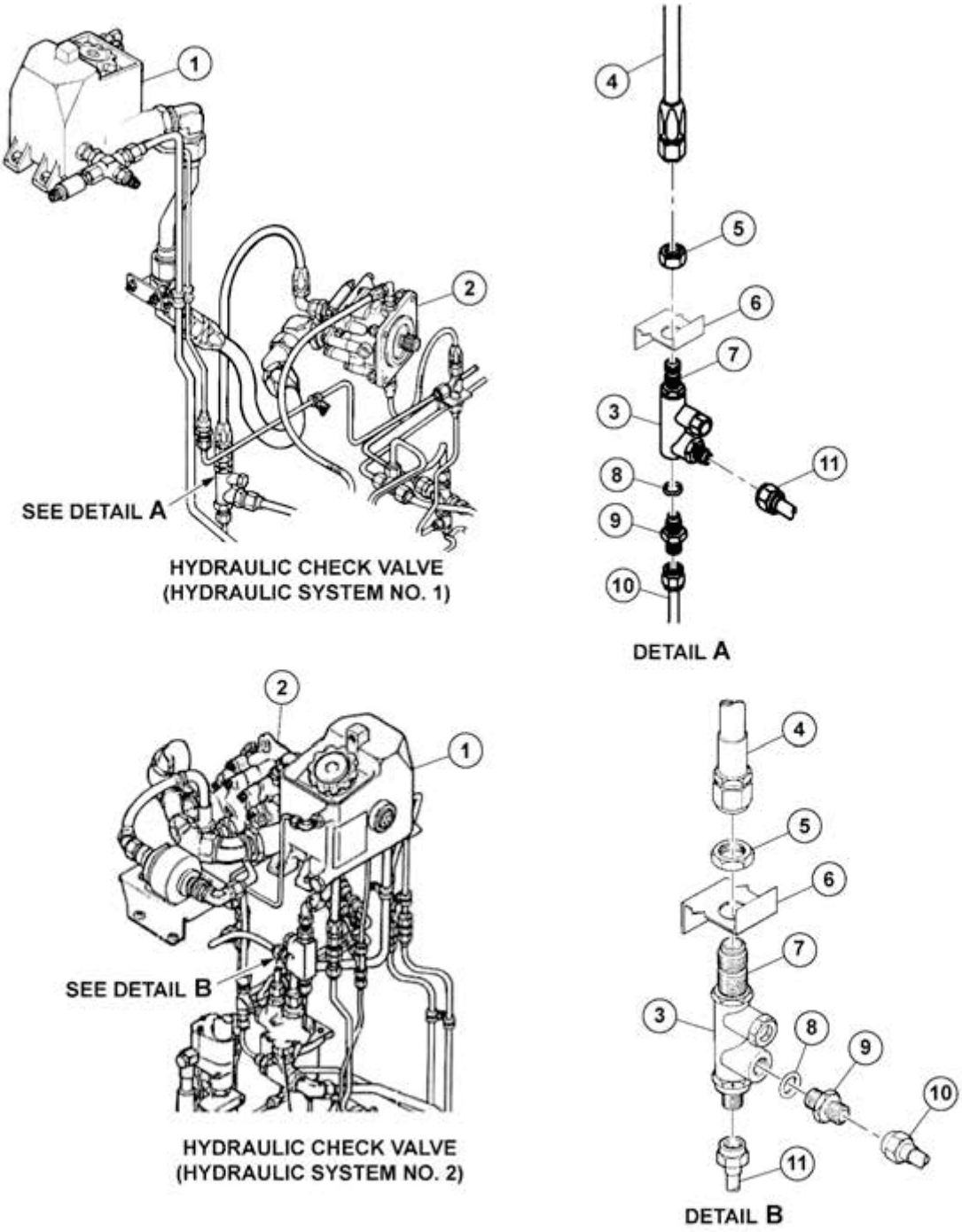
Hydraulic Check Valve - Installation

Procedure

NOTE

The following installation procedures are the same for hydraulic systems No. 1 and No. 2.

1. Install new fittings and preformed packing (8, Figure 1) and inline check valve (9) if hydraulic check valve (3) is being replaced.
2. Place union (7) through structure (6) and loosely install nut (5).
3. Connect pump pressure hose (4) and tubes (10 and 11).
4. Tighten nut (5).
5. Service reservoir (412-A-12-10-02-00A-040A-A / 00138).
6. Perform ground run and verify system functions normally and no leaks are present.
7. Install access panel.



1. Reservoir
2. Hydraulic pump
3. Hydraulic check valve
4. Pump pressure hose
5. Nut
6. Structure
7. Union
8. Preformed packing
9. Inline check valve
10. Tube
11. Tube

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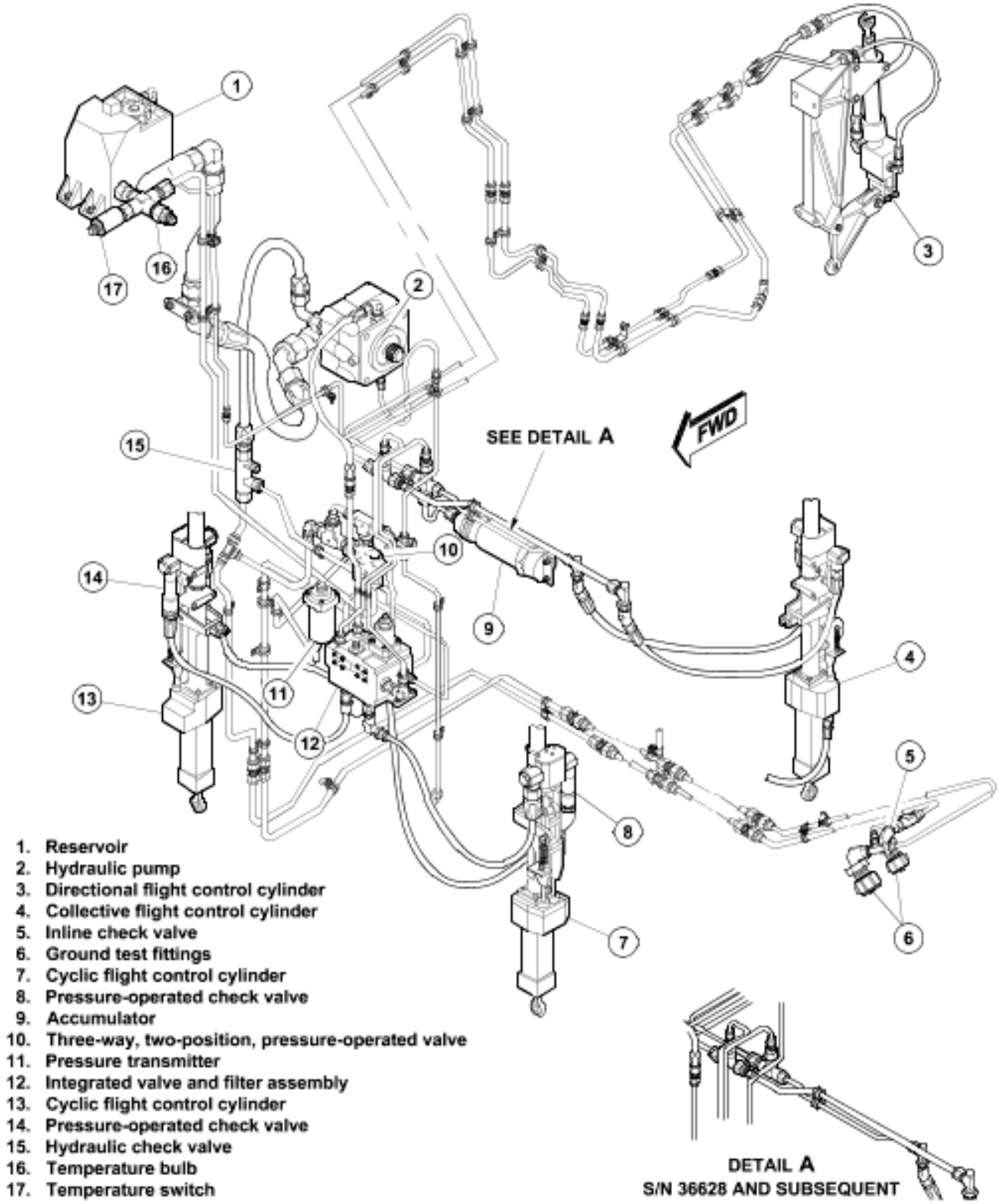
Figure 1. Hydraulic Check Valve (Sheet 1 of 1)

Pressure Transmitter - Description

NOTE

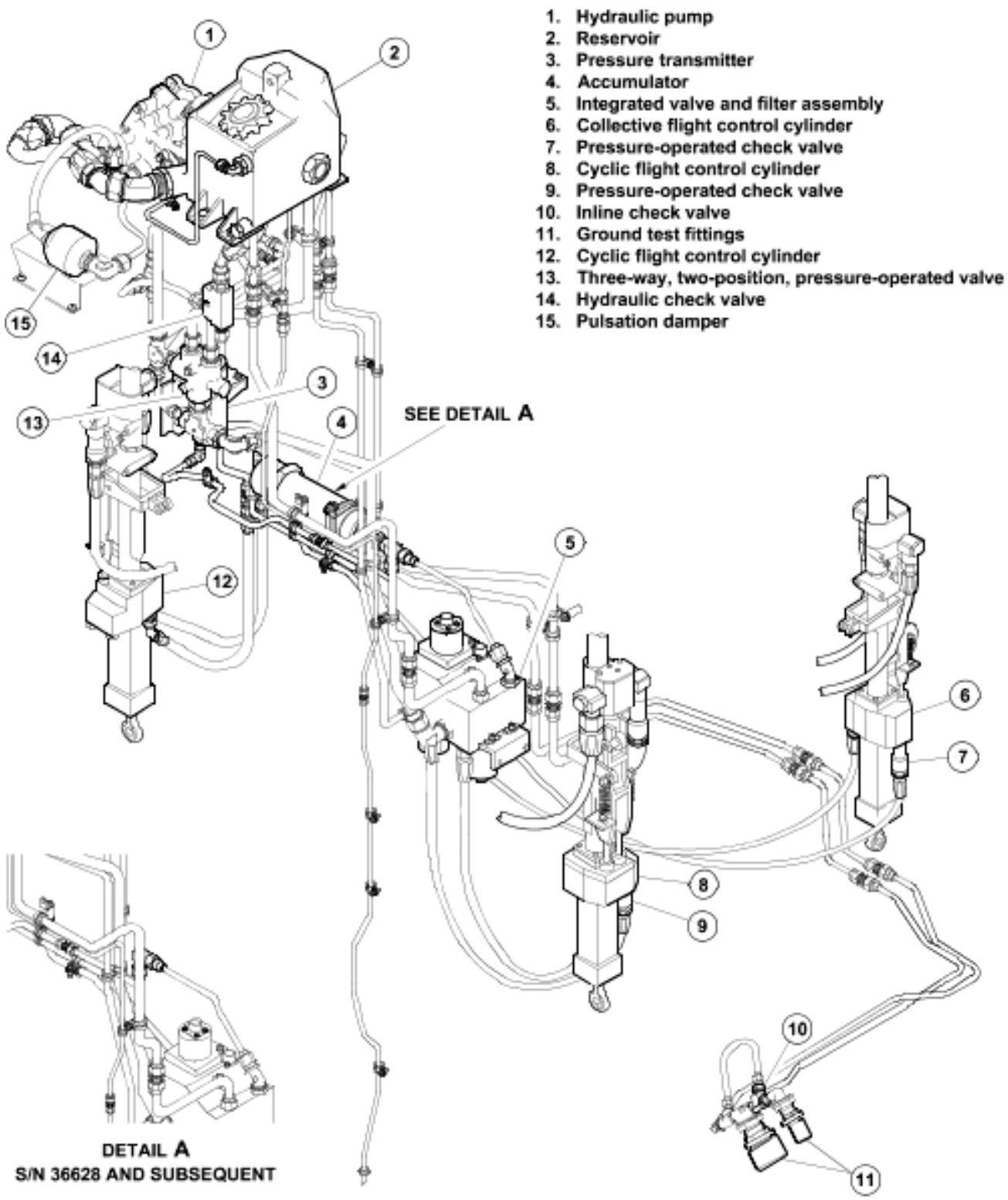
Except as noted, the maintenance procedures in 412-A-29-10-00-12A-520A-A / 00671, 412-A-29-10-00-12A-280A-A / 00672 and 412-A-29-10-00-12A-720A-A / 00673 are the same for hydraulic systems No. 1 and No. 2.

Hydraulic system No. 1 (Figure 1) and hydraulic system No. 2 (Figure 2) have pressure transmitters installed in the compartment below the transmission. Pressure sensed by the transmitters is transmitted to instruments on the pilot instrument panel.



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Figure 1. Hydraulic System No. 1 (Sheet 1 of 1)



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Figure 2. Hydraulic System No. 2 (Sheet 1 of 1)

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Pressure Transmitter - Removal

Procedure

NOTE

The following removal procedures are the same for hydraulic systems No. 1 and No. 2.

1. Remove access panel on forward side of pylon structure for hydraulic system No. 1.

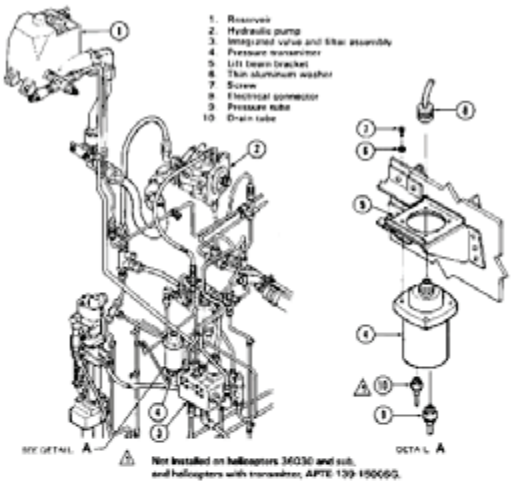
2. Disconnect electrical connector (8, Figure 1).

3. Disconnect tube(s) (9 and 10, if installed).

4. Plug open tubes to prevent contamination.

5. Remove screws (7), thin aluminum washers (6), and pressure transmitter (4).

6. Remove restrictor from pressure transmitter (4) (hydraulic system No. 1).



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Figure 1. Pressure Transmitter (Sheet 1 of 1)

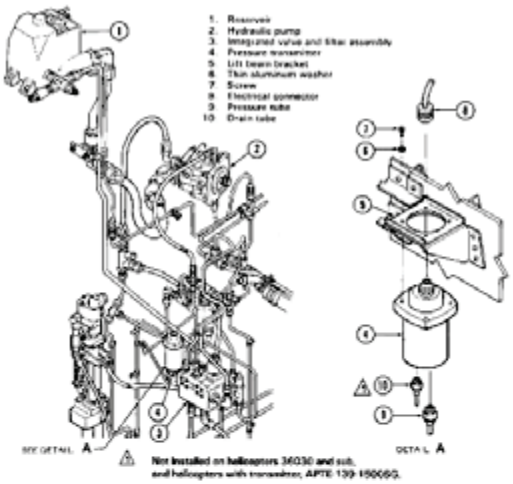
Pressure Transmitter - Inspection

Procedure

NOTE

The following inspection procedure is same for hydraulic systems No. 1 and No. 2.

1. Inspect pressure transmitter (4, Figure 1) for damage or signs of leakage.



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Figure 1. Pressure Transmitter (Sheet 1 of 1)

Pressure Transmitter - Installation

Procedure

NOTE

The following installation procedures are the same for hydraulic systems No. 1 and No. 2.

NOTE

Hydraulic system No. 1 restrictor is installed on pressure transmitter.

1. Place pressure transmitter (4, Figure 1) into lift beam bracket (5) and secure with screws (7) and thin aluminum washers (6).

2. Install restrictor on pressure transmitter (4) (hydraulic system No. 1).

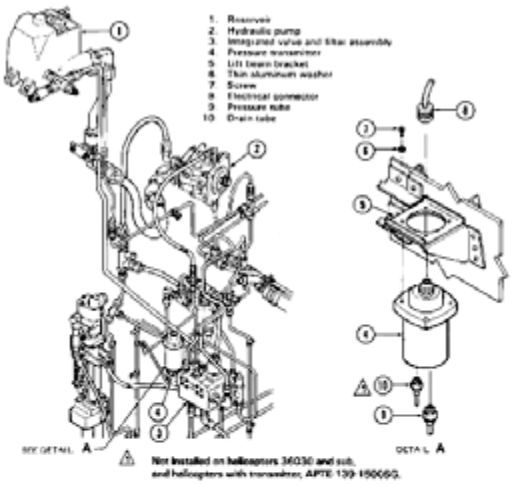
3. Connect tube(s) (9 and 10, if installed).

4. Connect electrical connector (8).

5. Service reservoir (1) (412-A-12-10-02-00A-040A-A / 00138).

6. Perform ground run and verify system functions normally and no leaks are present.

7. Install access panels.

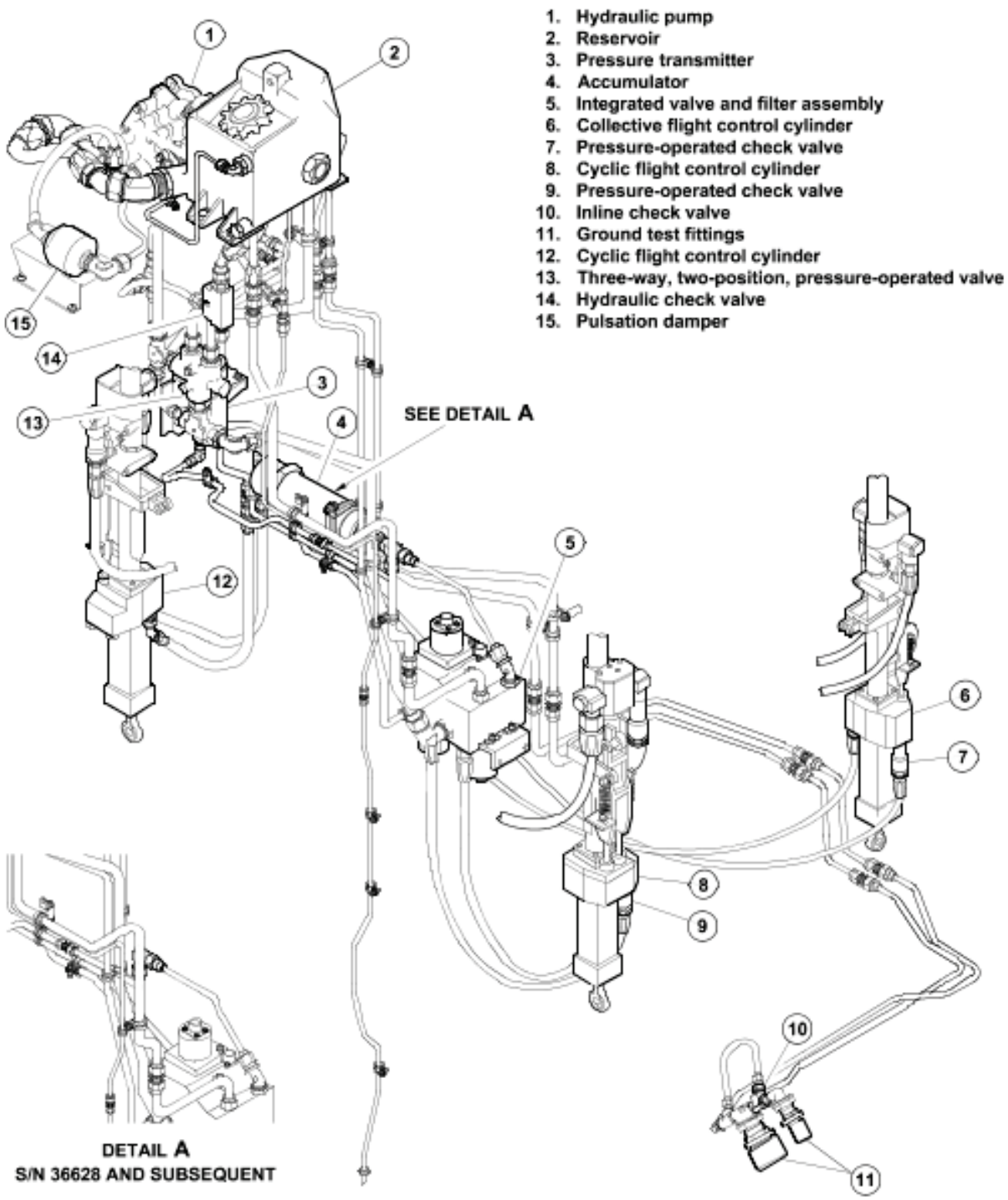


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Figure 1. Pressure Transmitter (Sheet 1 of 1)

Pulsation Damper (Hydraulic System No. 2) - Description

Hydraulic system No. 2 (Figure 1) has a pulsation damper installed in the pump pressure line to dampen hydraulic pressure pulsations, thereby reducing system induced noise.



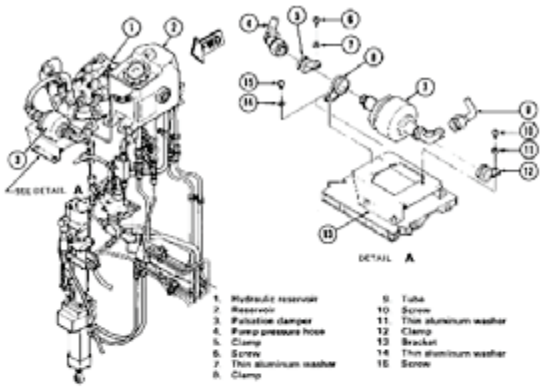
ICN-412-A-291000-B-97499-00008-A-001-01

Figure 1. Hydraulic System No. 2 (Sheet 1 of 1)

Pulsation Damper (Hydraulic System No. 2) - Removal

Procedure

1. Remove forward pylon fairing.
2. Disconnect pump pressure hose (4, Figure 1) and tube (9).
3. Plug open tube and hose to prevent contamination.
4. Remove screws (6, 10, and 15), thin aluminum washers (7, 11, and 14), and remove pulsation damper (3).
5. Remove clamps (5, 8, and 12) from pulsation damper (3).



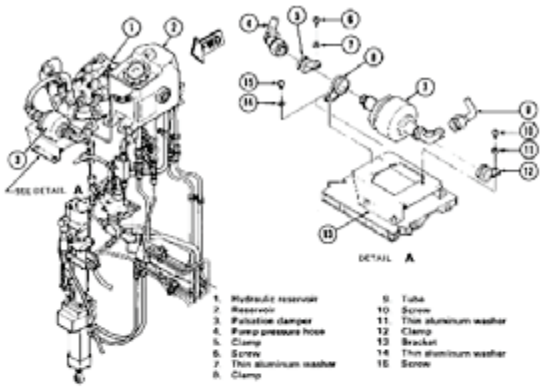
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Figure 1. Pulsation Damper (Hydraulic System No. 2) (Sheet 1 of 1)

Pulsation Damper (Hydraulic System No. 2) - Installation

Procedure

1. Place clamps (5, 8, and 12, Figure 1) on pulsation damper (3).
2. Position pulsation damper (3) on bracket (13). Install screws (6, 10, and 15) and thin aluminum washers (7, 11, and 14).
3. Connect pump pressure hose (4) and tube (9).
4. Service hydraulic reservoir (1) (412-A-12-10-02-00A-040A-A / 00138).
5. Perform ground run and verify system functions normally and no leaks are present.
6. Install forward pylon fairing.



ICN-412-A-291000-B-97499-00023-A-001-01

Figure 1. Pulsation Damper (Hydraulic System No. 2) (Sheet 1 of 1)

CHAPTER 30
ICE AND RAIN PROTECTION

MAINTENANCE MANUAL

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List of Effective Data Modules

The listed documents are included in issue 007, dated 2025-11-18, of this publication.

C = Changed data module, N = New data module

Document title	Data module code / Seq number	Issue date	Applicable to
Ice and Rain Protection - Description	DMC-412-A-30-00-00-00A-028A-A / 00678	2023-05-31	ALL
Windshield Wiper Assembly - Description	DMC-412-A-30-40-00-00A-042A-A / 00679	2023-05-31	ALL
Windshield Wiper Assembly - Troubleshooting	DMC-412-A-30-40-00-00A-422A-A / 00680	2023-05-31	ALL
Windshield Wiper Assembly - Inspection	DMC-412-A-30-40-00-00A-280A-A / 00681	2023-05-31	ALL
Windshield Wiper Assembly - Adjustment	DMC-412-A-30-40-00-00A-271A-A / 00682	2023-05-31	ALL
Windshield Wiper Blade - Removal	DMC-412-A-30-40-00-00A-520A-A / 00683	2023-05-31	ALL
Windshield Wiper Blade - Installation	DMC-412-A-30-40-00-00A-720A-A / 00684	2023-05-31	ALL
Windshield Wiper - Removal	DMC-412-A-30-40-00-01A-520A-A / 00685	2023-05-31	ALL
Windshield Wiper - Installation	DMC-412-A-30-40-00-01A-720A-A / 00686	2023-05-31	ALL
Windshield Wiper Arm - Removal	DMC-412-A-30-40-00-02A-520A-A / 00687	2023-05-31	ALL
Windshield Wiper Arm - Installation	DMC-412-A-30-40-00-02A-720A-A / 00688	2023-05-31	ALL
Windshield Wiper Head Guard - Removal	DMC-412-A-30-40-00-03A-520A-A / 00689	2023-05-31	ALL
Windshield Wiper Head Guard - Installation	DMC-412-A-30-40-00-03A-720A-A / 00690	2023-05-31	ALL
Windshield Wiper Converter and Motor - Removal	DMC-412-A-30-40-00-04A-520A-A / 00691	2023-05-31	ALL
Windshield Wiper Converter and Motor - Installation	DMC-412-A-30-40-00-04A-720A-A / 00692	2023-05-31	ALL
Anti-icing, De-icing and Snow - Removal	DMC-412-A-30-40-00-00A-520B-A / 00693	2023-05-31	ALL

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Ice and Rain Protection - Description

The ice and rain protection consists of pilot and copilot windshield wipers.

Additional ice protection is provided by:

Heating and ventilation system (412-A-21-20-00-07A-042A-A / 00211).

Winterization heater (412-A-21-40-00-00A-028A-A / 00222).

Pitot static system heaters (412-A-95-11-07-00A-042A-A / 01815).

Anti-icing, de-icing and snow removal (412-A-30-40-00-00A-520B-A / 00693).

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Windshield Wiper Assembly - Description

Each windshield wiper assembly consists of a converter and motor, arm, and wiper. The converter and motor drives the arm and the arm drives the wiper. The WIPER SEL switch (overhead console) permits selection of PILOT, COPILOT or BOTH wipers. Separate switches for pilot and copilot windshield wipers are located in the overhead console. Each switch has the following positions: PARK, OFF, LOW, MEDIUM, AND HIGH. PARK is a spring loaded position (returns to OFF) for stowing wiper to a location where vision is not obstructed.

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Windshield Wiper Assembly - Troubleshooting

Procedure

1. For troubleshooting procedures, refer to Table 1.

Table 1. Windshield Wipers - Troubleshooting

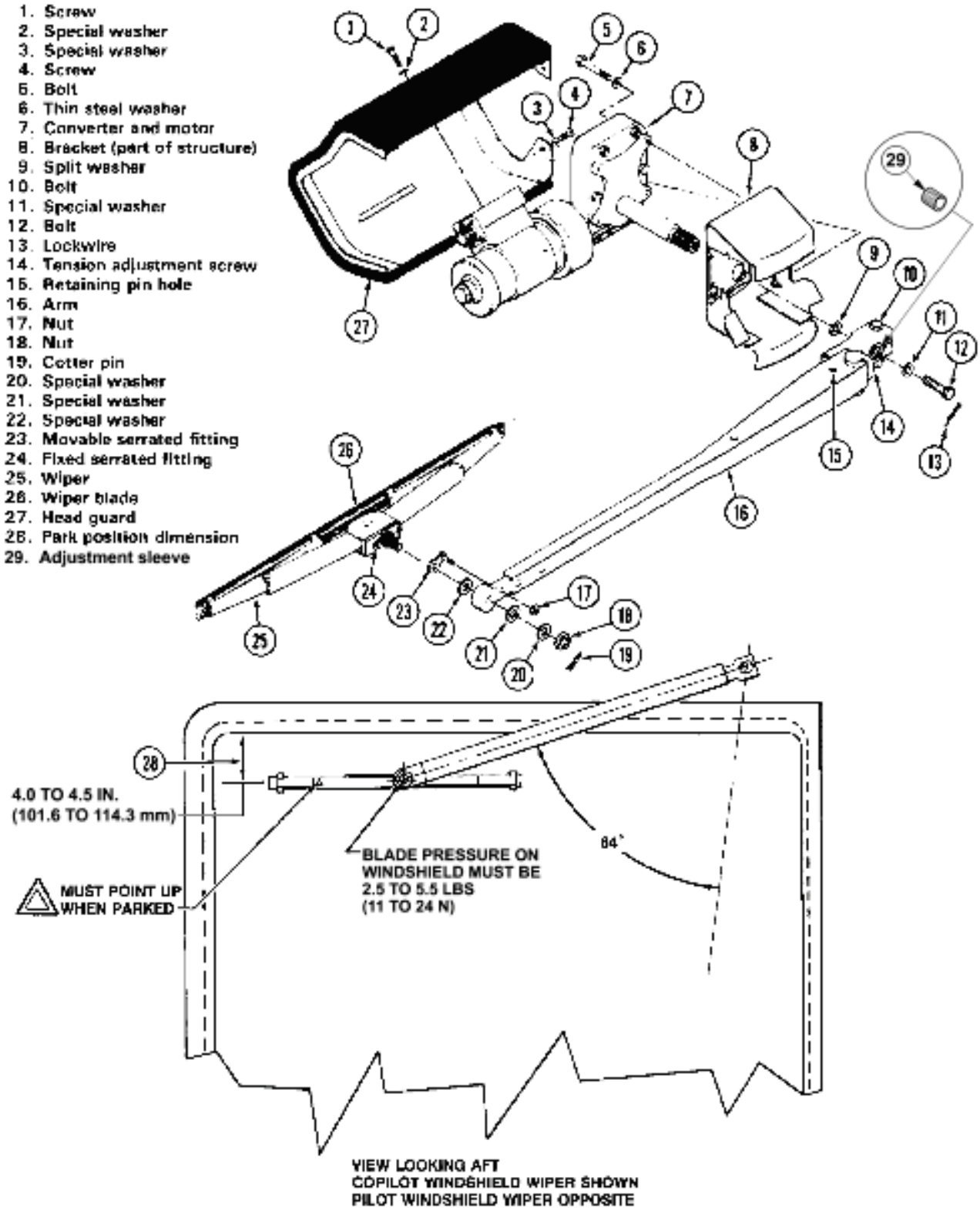
PROBABLE CAUSE	ISOLATION PROCEDURE	REMEDY
BLADE DOES NOT CLEAN WINDSHIELD		
Damaged or deteriorated blade.	Inspect blade.	Replace blade (412-A-30-40-00-00A-520A-A / 00683 and 412-A-30-40-00-00A-720A-A / 00684).
Improper blade pressure.	Check blade pressure.	Adjust blade pressure (412-A-30-40-00-00A-271A-A / 00682).
CONVERTER AND MOTOR INOPERATIVE		
Opened circuit breaker.	Reset circuit breaker. If circuit breaker will not reset, check breaker and associated wiring.	Replace circuit breaker or make required circuit repairs (412-A-96-07-00-01A-320A-A / 02149).
Damaged wiper switch.	Inspect for obvious damage. Make continuity check of switch.	Replace switch (412-A-96-07-00-01A-320A-A / 02149).
Broken electrical wire or loose connections.	Check connections. Make continuity check of circuit.	Replace wire if necessary. Tighten connections or replace connectors or terminals as required (412-A-96-07-00-01A-320A-A / 02149).
Converter and motor inoperative.	Substitute serviceable unit in place of suspected inoperative unit and make operational check.	Replace converter and motor (412-A-30-40-00-04A-520A-A / 00691 and 412-A-30-40-00-04A-720A-A / 00692).

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Windshield Wiper Assembly - Inspection

Procedure

1. Inspect wiper blade (26, Figure 1) for deteriorated rubber.
2. Inspect wiper blade (26) for correct pressure on windshield (412-A-30-40-00-00A-271A-A / 00682).
3. Inspect opposite wiper blade in similar manner.



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Figure 1. Windshield Wiper (Sheet 1 of 1)

Windshield Wiper Assembly - Adjustment

Preliminary requirements

Support equipment

Nomenclature	Identification No.	Qty	Remarks
Spring Scale	Commercial	1	0 - 10 Pounds (0 - 5 kg)
Wrench	XW20509	1	
Wrench	02314-0023-001	1	

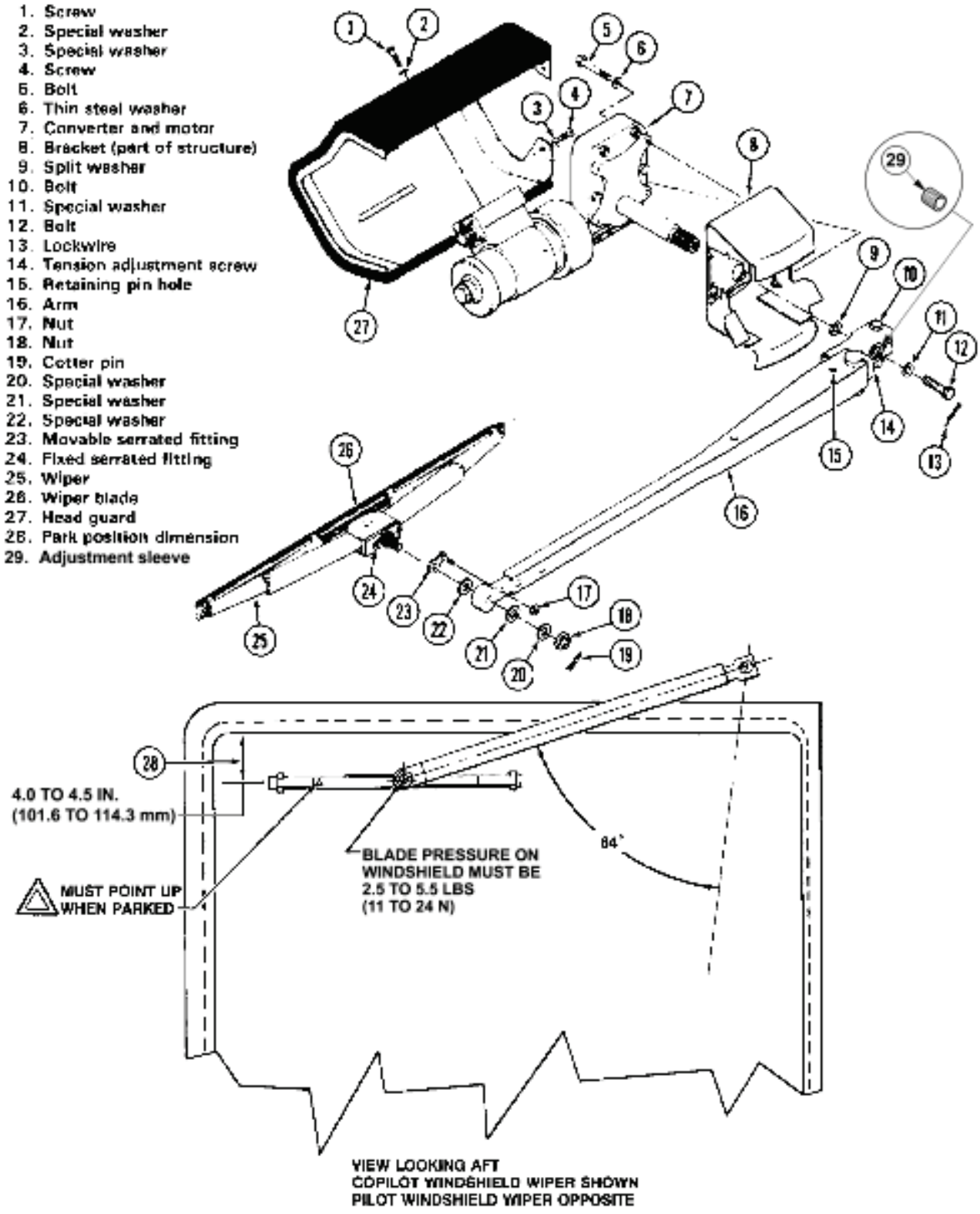
Procedure

1. Adjustment to wiper is limited to adjusting wiper blade pressure on windshield.

1.1. Attach a spring scale to arm (16, Figure 1) at wiper (25) attach point.

1.2. Verify wiper blade (26) pressure is 2.5 to 5.5 pounds (1.2 to 2.4 kg).

1.3. If necessary, increase blade pressure by turning tension adjustment screw (14) counterclockwise with wrench (XW20509), clockwise to decrease blade pressure.



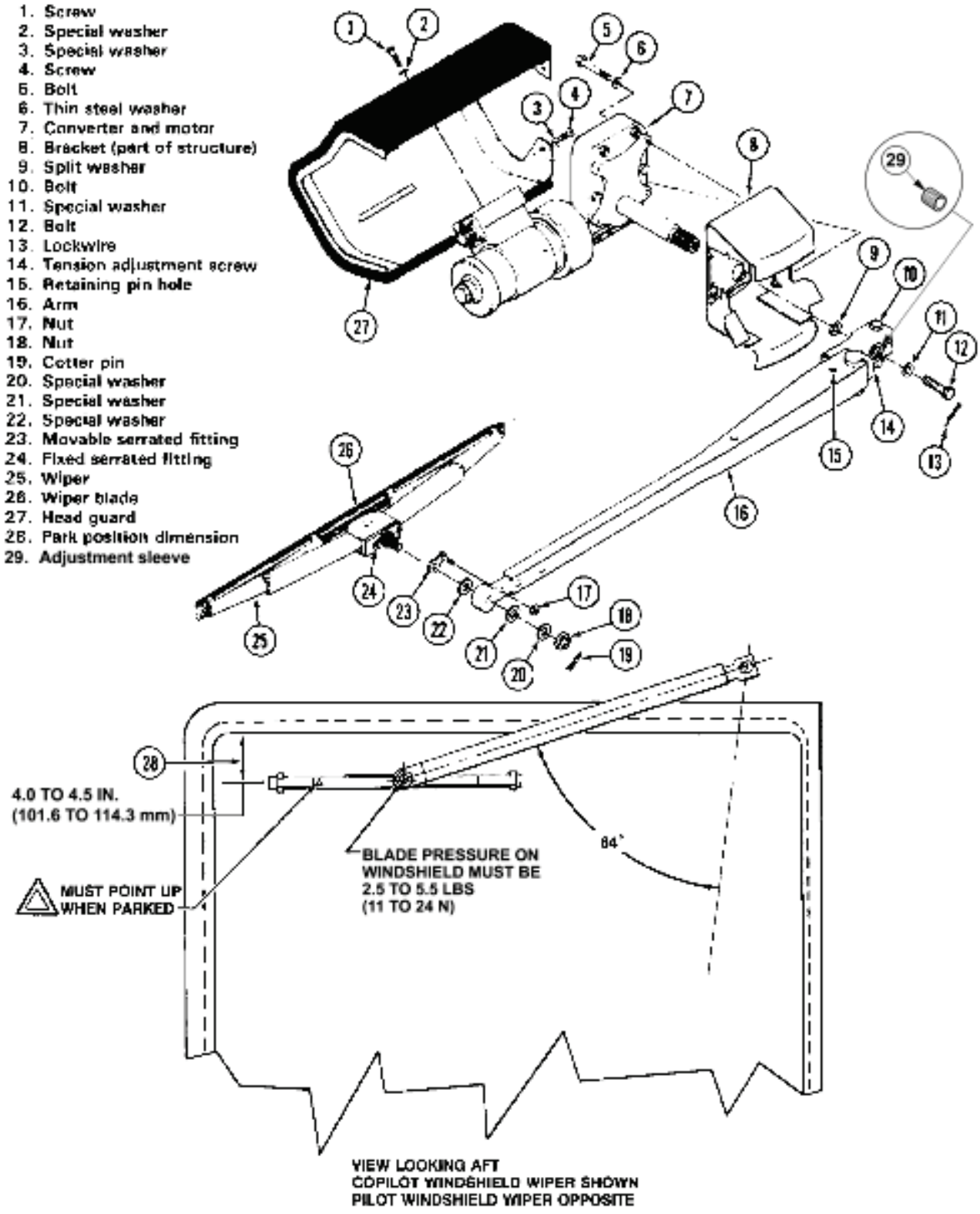
ICN-412-A-304000-B-97499-00001-A-001-01

Figure 1. Windshield Wiper (Sheet 1 of 1)

Windshield Wiper Blade - Removal

Procedure

1. Raise arm (16, Figure 1). Insert 0.125 inch (3.175 mm) diameter pin in retaining pin hole (15).
2. Bend tabs on wiper (25). Slide wiper blade (26) out of wiper (25).
3. Remove opposite blade in a similar manner.



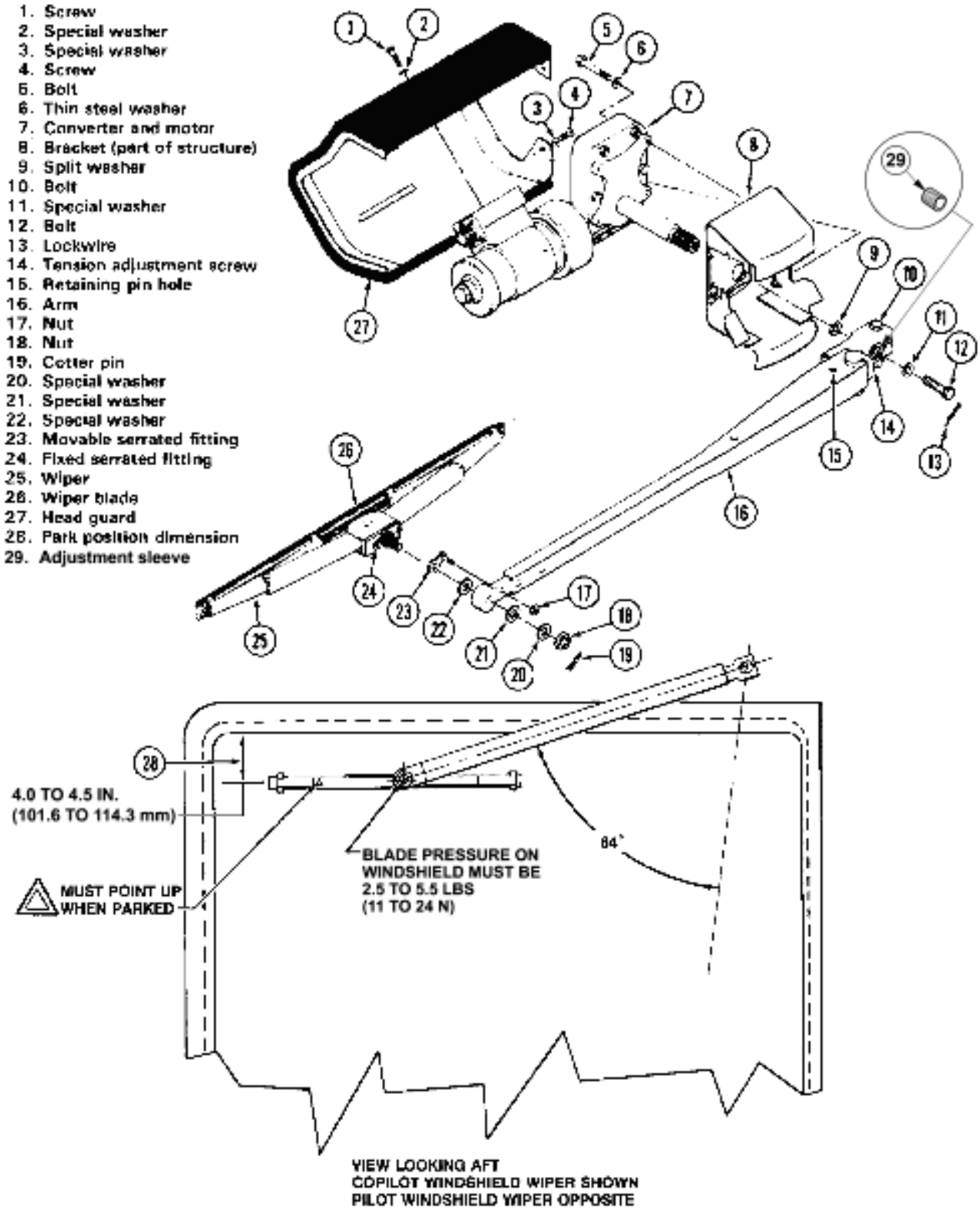
ICN-412-A-304000-B-97499-00001-A-001-01

Figure 1. Windshield Wiper (Sheet 1 of 1)

Windshield Wiper Blade - Installation

Procedure

1. Raise arm (16, Figure 1). Insert 0.125 inch (3.175 mm) diameter pin in retaining pin hole (15).
2. Slide wiper blade (26) on wiper (25).
3. Bend tabs on wiper (25) to secure wiper blade (26).
4. Remove pin from retaining pin hole (15).
5. Adjust wiper blade (26) pressure on windshield (412-A-30-40-00-00A-271A-A / 00682).
6. Install opposite blade in similar manner.



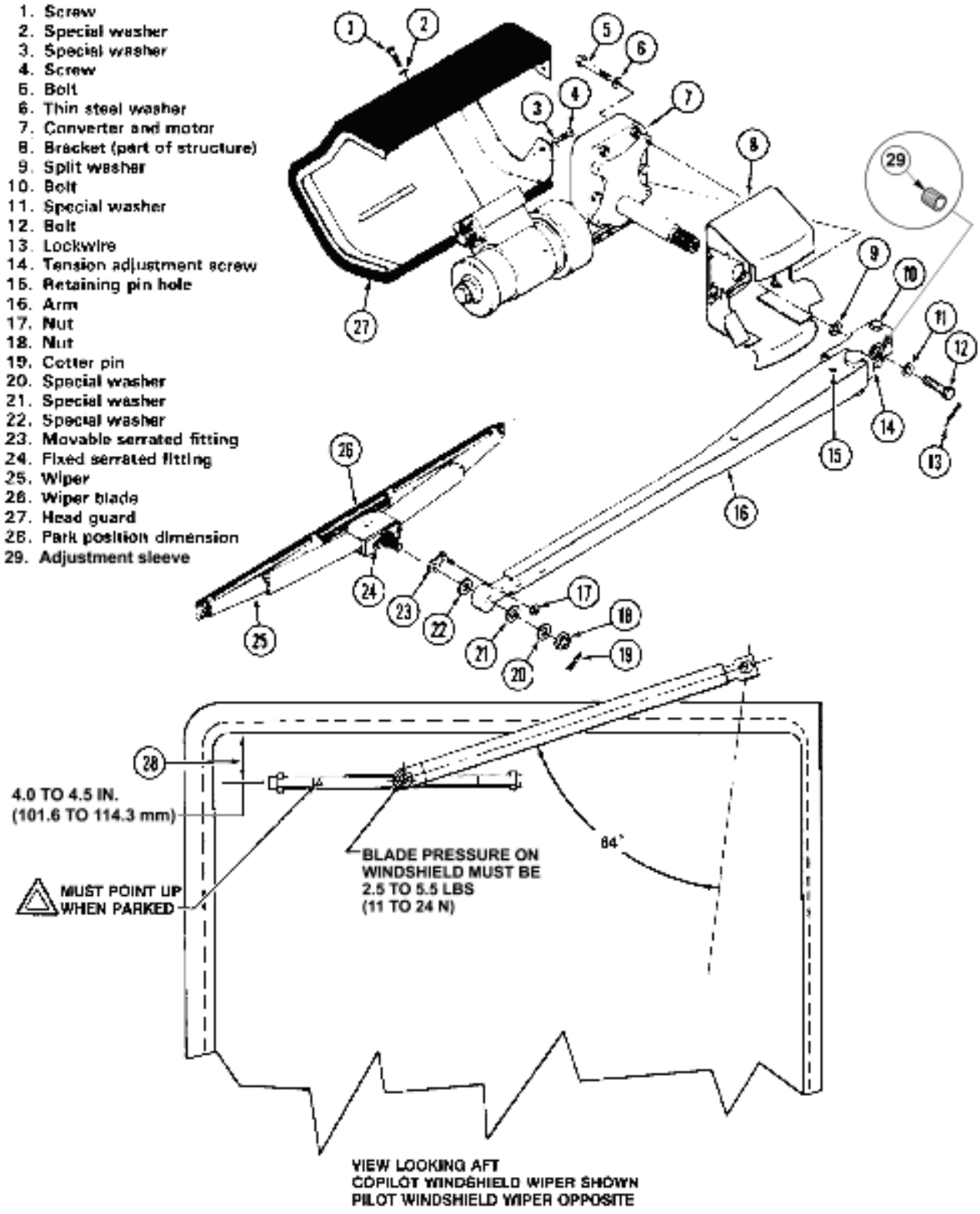
ICN-412-A-304000-B-97499-00001-A-001-01

Figure 1. Windshield Wiper (Sheet 1 of 1)

Windshield Wiper - Removal

Procedure

1. Raise arm (16, Figure 1). Insert 0.125 inch (3.175 mm) diameter pin in retaining pin hole (15).
2. Remove cotter pin (19) and nut (18). Retain attaching hardware. Remove wiper (25).
3. Remove nut (17), movable serrated fitting (23), and special washer (22).
4. Remove opposite wiper in a similar manner.



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Figure 1. Windshield Wiper (Sheet 1 of 1)

Windshield Wiper - Installation

Procedure

1. Raise arm (16, Figure 1). Insert 0.125 inch (3.175 mm) diameter pin in retaining pin hole (15).
2. If not previously accomplished, install fitting (23), special washer (22), and nut (17) on arm (16).

NOTE

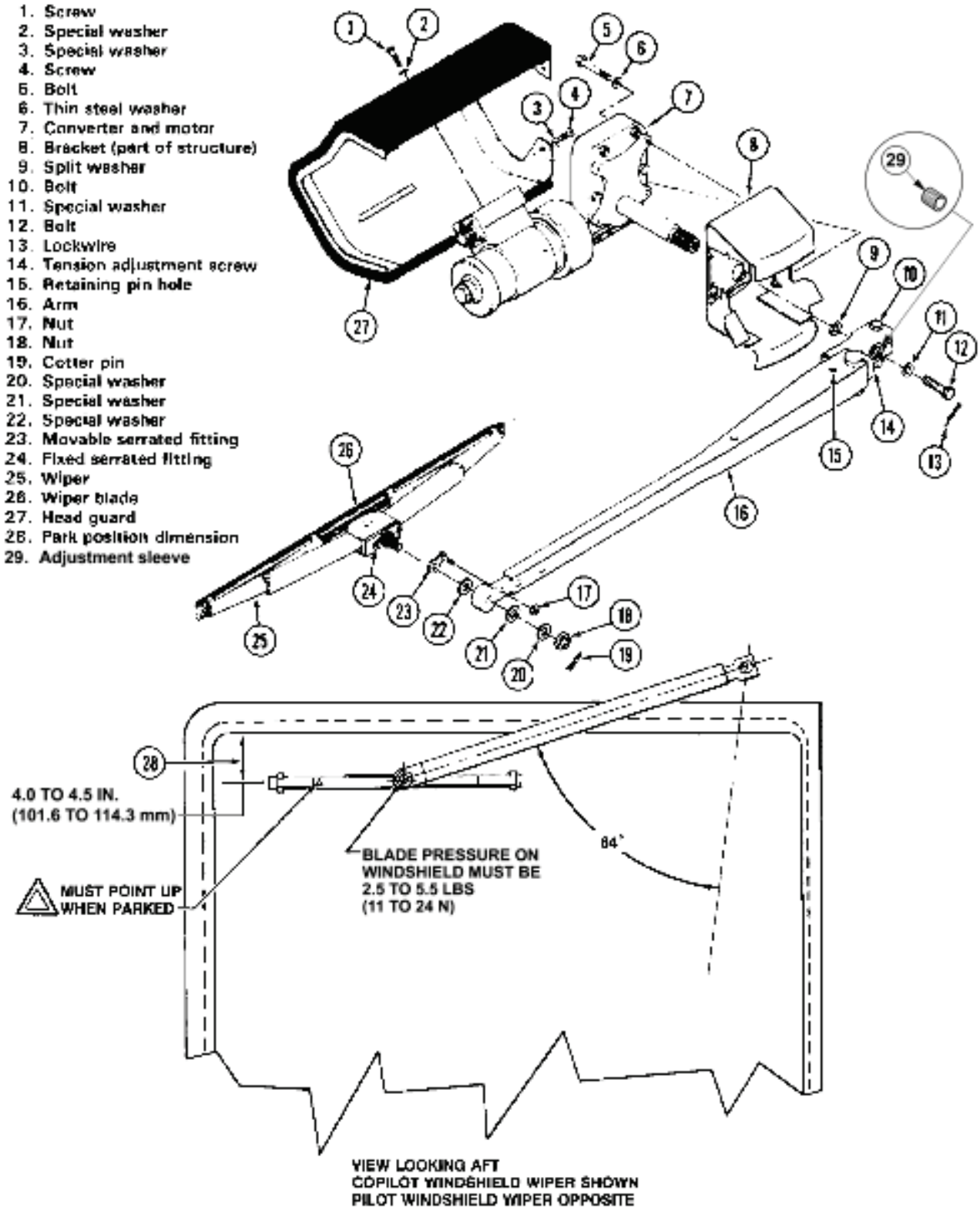
Wiper (25) shall be positioned per metal stamp on wiper, "Δ MUST POINT UP WHEN PARKED."

3. Place threaded shaft of wiper (25) through movable serrated fitting (23), special washer (22), and arm (16). Install special washers (21 and 20), and nut (18). Do not tighten nut at this time.
4. Remove pin from retaining pin hole (15).

NOTE

If necessary to obtain alignment of castellations in nut (18) for installation of cotter pin (19), use thin or standard aluminum washers (21 or 22) as required.

5. Align wiper (25) parallel to top of windshield to the park position dimension (28). Tighten nut (18). Install cotter pin (19).
6. Adjust wiper blade (26) pressure on windshield (412-A-30-40-00-00A-271A-A / 00682).
7. Install opposite wiper in a similar manner.



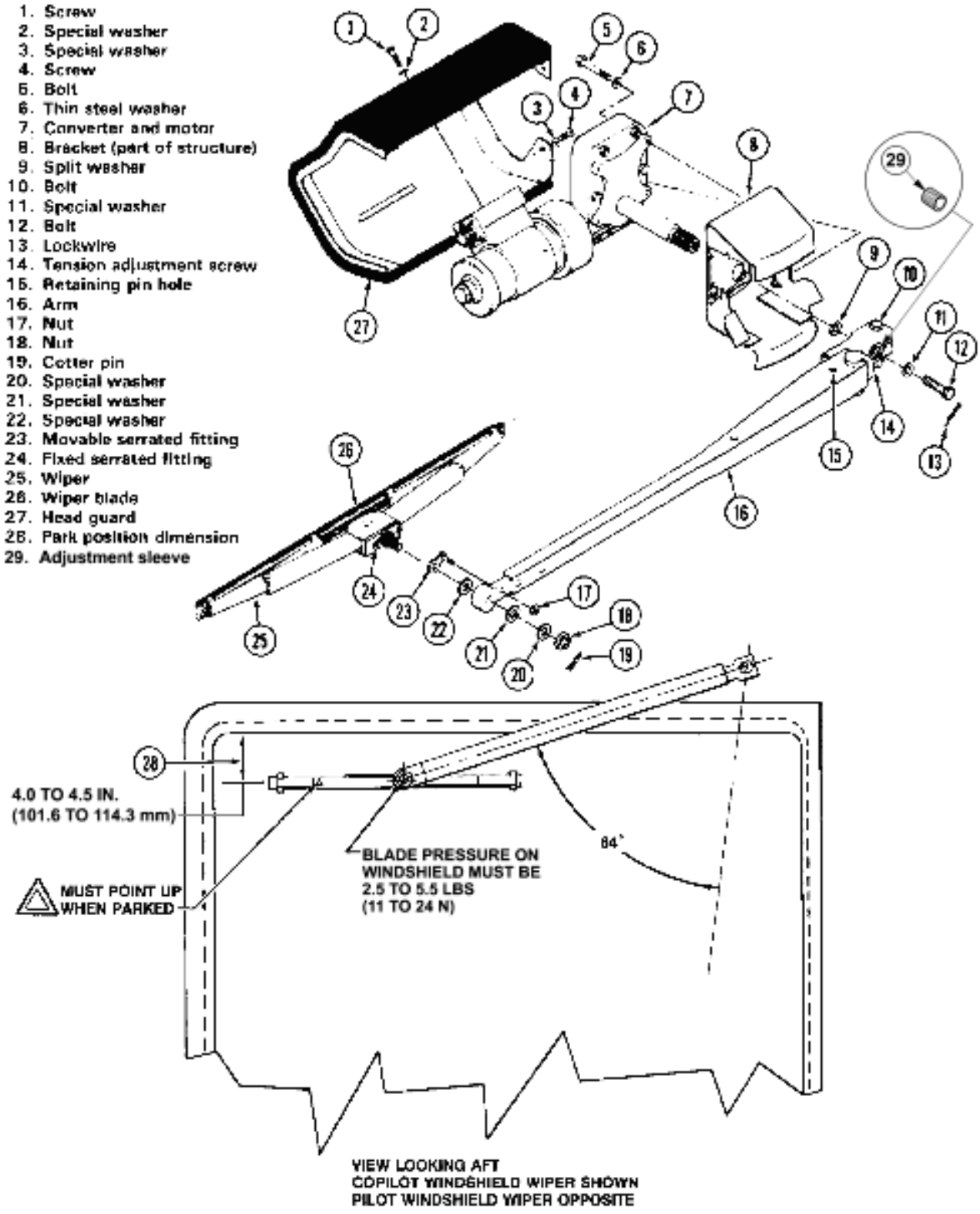
ICN-412-A-304000-B-97499-00001-A-001-01

Figure 1. Windshield Wiper (Sheet 1 of 1)

Windshield Wiper Arm - Removal

Procedure

1. Raise arm (16, Figure 1). Insert 0.125 inch (3.175 mm) diameter pin in retaining pin hole (15).
2. Remove cotter pin (19) and nut (18). Retain attaching hardware. Remove wiper (25).
3. Loosen bolt (10). Remove bolt (12). Remove arm (16). Retain washers (9 and 11).
4. Remove opposite arm in a similar manner.



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Figure 1. Windshield Wiper (Sheet 1 of 1)

Windshield Wiper Arm - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

1. Straighten arm (16, Figure 1). Insert 0.125 inch (3.175 mm) diameter pin in retaining pin hole (15).

2. Place split washer (9) and arm (16) on converter and motor (7) serrated shaft.

NOTE

When adjusting park position dimension (28) take up converter assembly backlash by applying light upward (towards top of windshield) pressure on windshield wiper arm (16).

3. Adjust arm (16) as follows:

3.1. Operate converter (7) to Park position, as shown in Figure 1.

NOTE

Removable adjustment sleeve (29) in arm (16) hub has serrations every 7.2° inside the sleeve and on every 7.06° on the outside.

3.2. Adjust arm (16) wiper attach hole from top of windshield by removing and repositioning adjustment sleeve (29) to achieve 4.0 to 4.5 inches (101.6 to 114.3 mm) dimension, as shown in Figure 1.

4. Install special washers (11), bolt (12), and lockwire (C-405) on converter and motor (7) serrated shaft. Tighten bolt (10).

5. Install wiper (25) (412-A-30-40-00-01A-720A-A / 00686).

6. Adjust wiper blade (26) pressure on windshield (412-A-30-40-00-00A-271A-A / 00682).

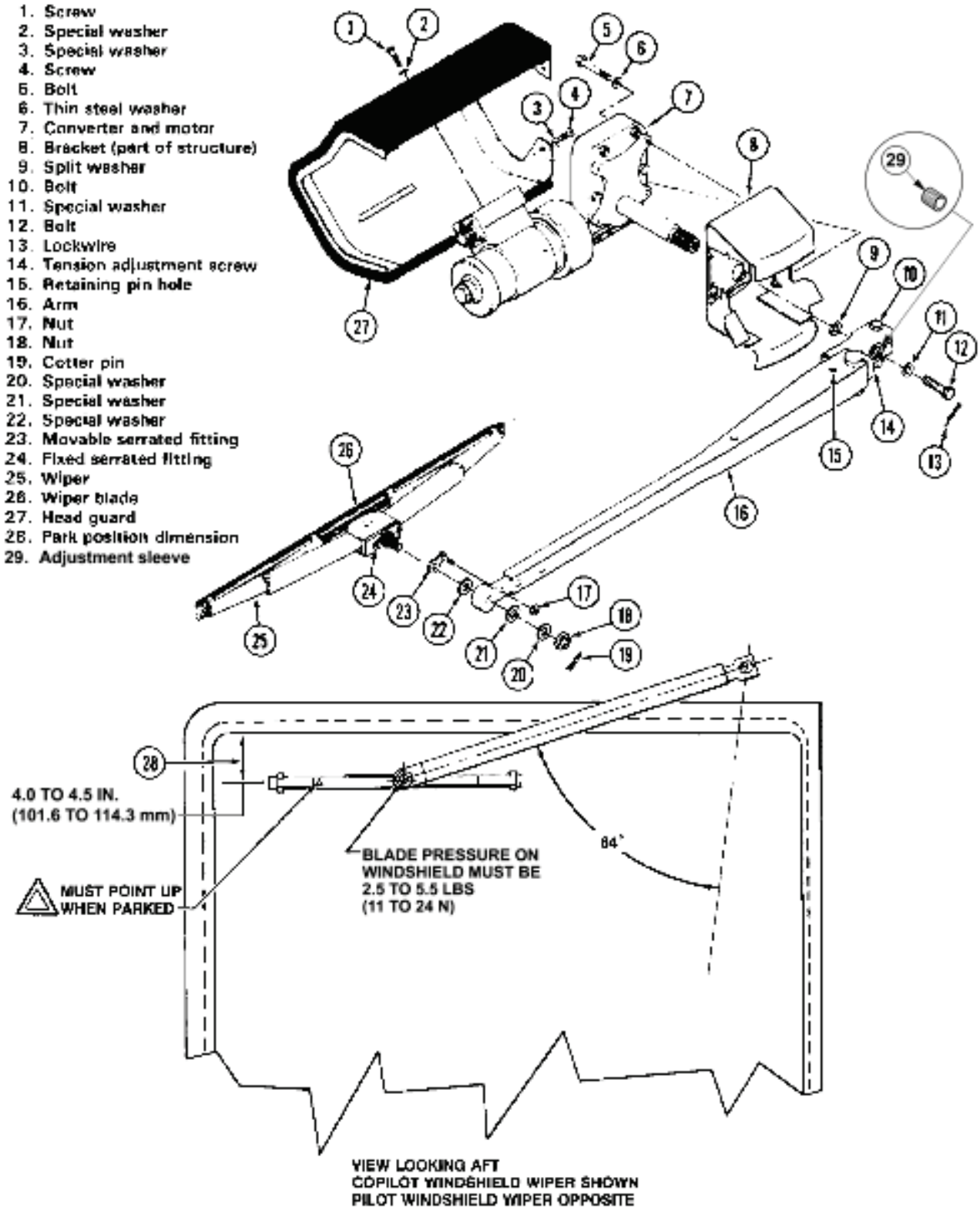


DO NOT OPERATE WIPER ON DRY WINDSHIELD. IT CAN CAUSE DAMAGE TO THE WINDSHIELD.

7. Apply power to system.

8. Check 64° sweep of arm (16) by momentarily rotating WIPER control switch to LOW.

9. Install opposite arm in a similar manner.



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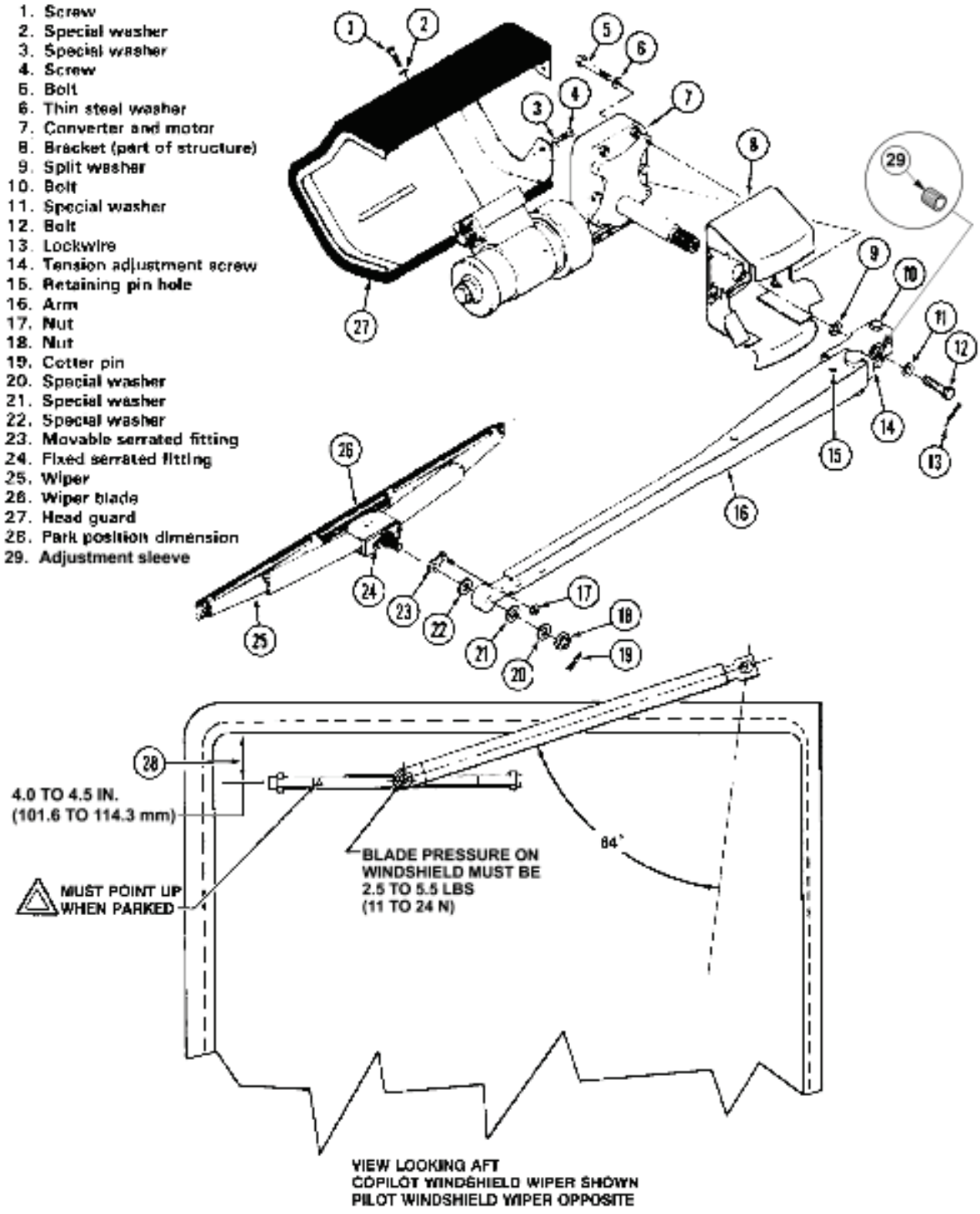
Figure 1. Windshield Wiper (Sheet 1 of 1)

Windshield Wiper Head Guard - Removal

Procedure

1. Remove screws (1 and 4, Figure 1). Retain special washers (2 and 3).

2. Remove head guard (27).



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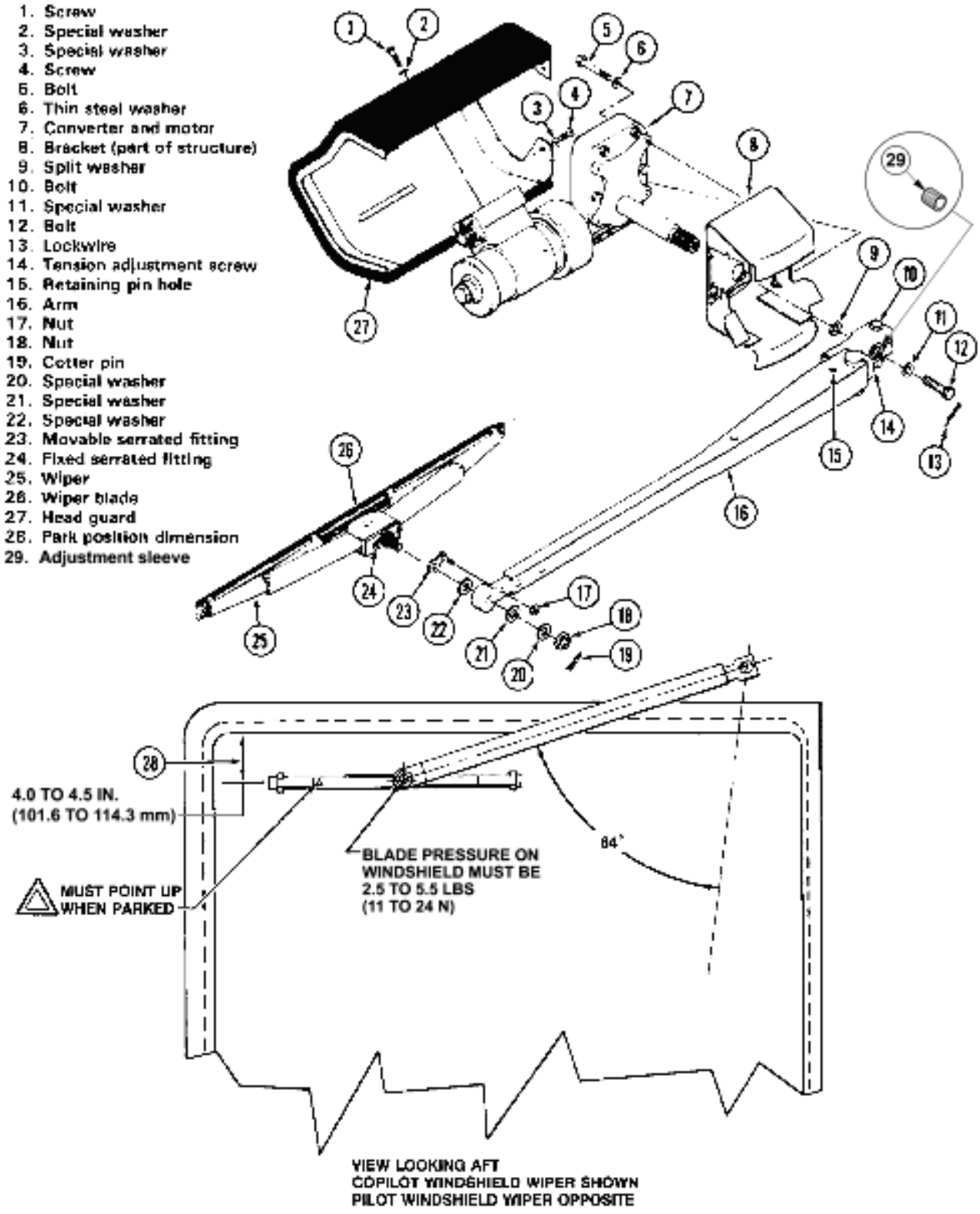
Figure 1. Windshield Wiper (Sheet 1 of 1)

Windshield Wiper Head Guard - Installation

Procedure

1. Position headguard (27, Figure 1) on converter and motor (7).

2. Install two special washers (2 and 3) and screws (1 and 4).



ICN-412-A-304000-B-97499-00001-A-001-01

Figure 1. Windshield Wiper (Sheet 1 of 1)

Windshield Wiper Converter and Motor - Removal

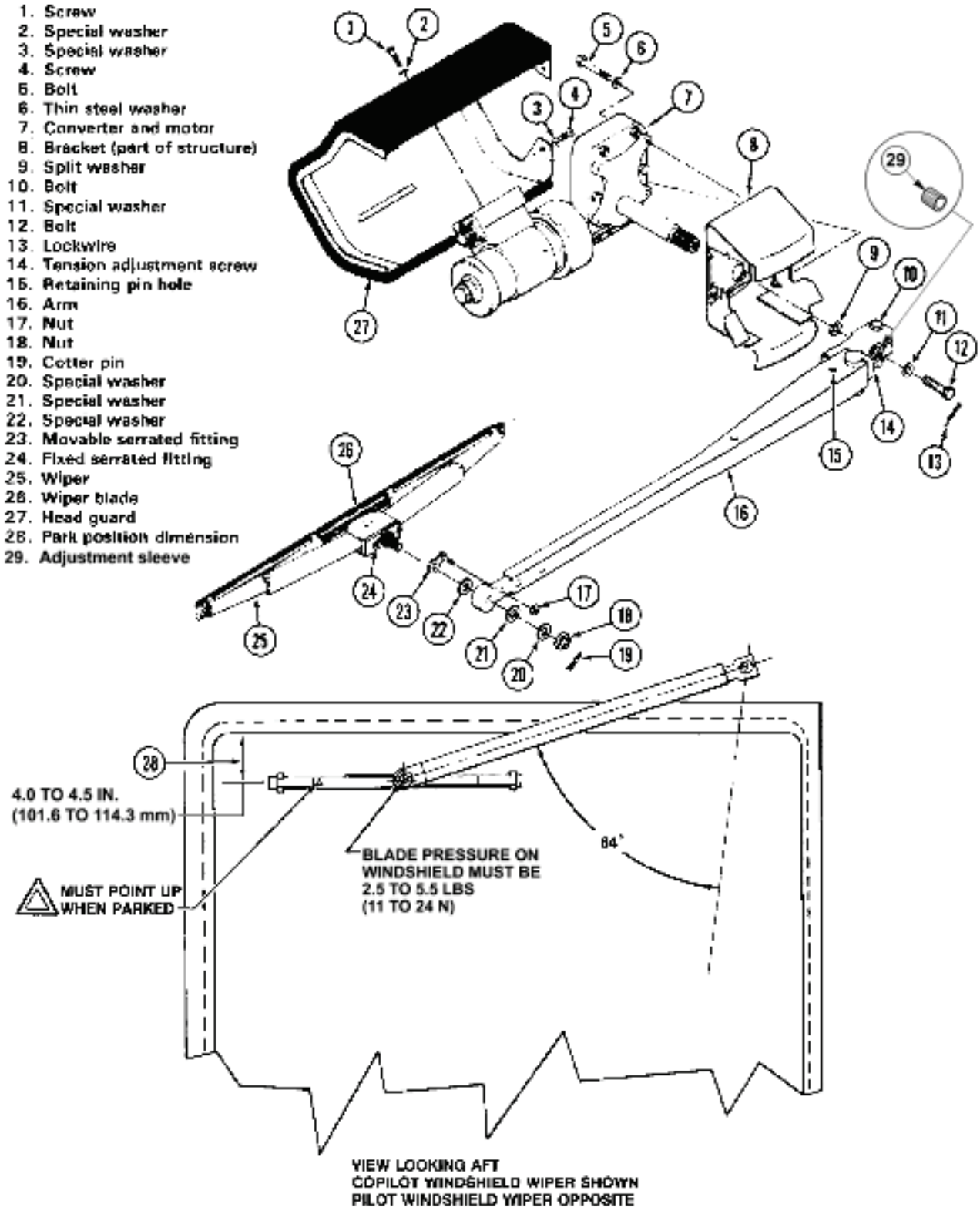
Procedure

1. Raise arm (16, Figure 1). Insert 0.125 inch (3.175 mm) diameter pin in retaining pin hole (15).
2. Loosen bolt (10). Remove bolt (12). Remove arm (16) and wiper (25) as a single unit. Retain washers (9 and 11).
3. Remove screws (1 and 4). Retain special washers (2 and 3). Remove head guard (27).

NOTE

BATTERY BUS 1 and BATTERY BUS 2 switches shall be OFF.

4. Place BATTERY BUS 1 and BATTERY BUS 2 switches OFF.
5. Disconnect electrical connector from converter and motor (7). Cover connector and receptacle with caps or tape.
6. Remove bolts (5) and thin steel washers (6). Remove converter and motor (7).
7. Remove opposite converter and motor in similar manner.



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Figure 1. Windshield Wiper (Sheet 1 of 1)

Windshield Wiper Converter and Motor - Installation

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
Lockwire	C-405	AR	

Procedure

1. Position converter and motor (7, Figure 1) serrated shaft in windshield structure hole. Install thin steel washers (6) and bolts (5).

NOTE

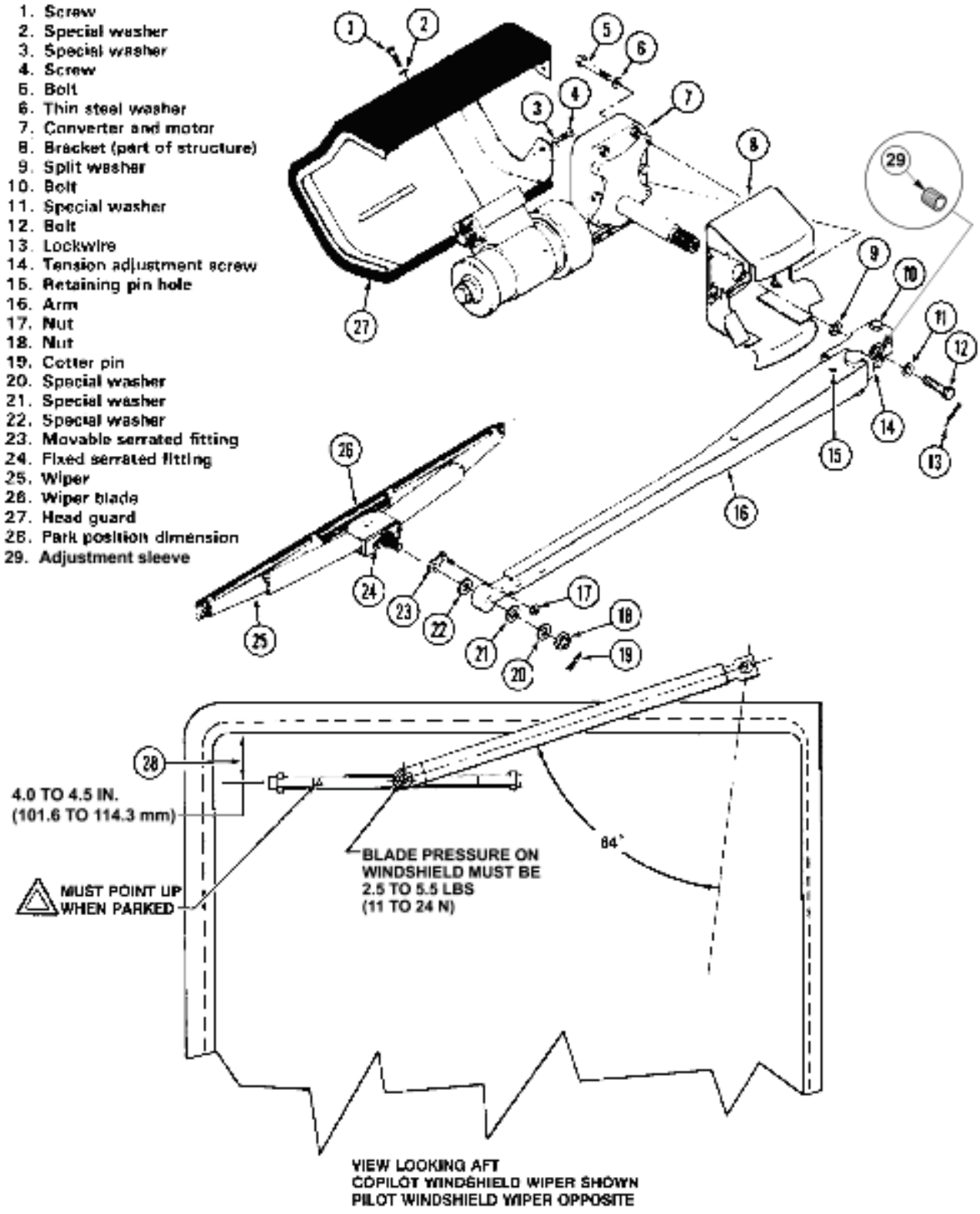
BATTERY BUS 1 and BATTERY BUS 2 switches shall be OFF.

2. Place BATTERY BUS 1 and BATTERY BUS 2 switches OFF.
3. Install electrical connector on converter and motor (7).
4. Position head guard (27). Install special washers (2), screws (1), special washers (3), and screws (4).
5. Position split washer (9) and arm (16) with wiper (25) attached, on converter and motor (7) serrated shaft.
6. Hold wiper (25) in park position dimension (28). Tighten bolt (10). Install special washer (11), bolt (12), and secure with lockwire (C-405).
7. Remove pin from retaining pin hole (15).
8. Adjust wiper blade (26) pressure on windshield (412-A-30-40-00-00A-271A-A / 00682).



DO NOT OPERATE WIPERS ON DRY WINDSHIELD. IT CAN CAUSE DAMAGE TO THE WINDSHIELD.

9. Apply power to system.
10. Perform windshield wiper operation check (412-A-96-07-00-01A-320A-A / 02149).
11. Install opposite converter and motor in similar manner.



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Figure 1. Windshield Wiper (Sheet 1 of 1)

Anti-icing, De-icing and Snow - Removal

Preliminary requirements

Consumables, materials and expendables

Nomenclature	Identification No.	Qty	Remarks
De-icing/anti-icing fluid	C-394	AR	

Procedure



de-icing/anti-icing fluid (C-394) CAN BE HARMFUL OR FATAL IF SWALLOWED. AVOID PROLONGED OR REPEATED BREATHING OF VAPOR. DO NOT TAKE INTERNALLY, IF SWALLOWED, INDUCE VOMITING. CALL A PHYSICIAN.



IT IS VERY DIFFICULT TO PREDICT ACCURATELY HOW LONG de-icing/anti-icing fluid (C-394) ONCE APPLIED, WILL PREVENT ICE BUILDUP ON SURFACES. THE QUANTITY OF FLUID APPLIED, AMBIENT TEMPERATURES, THE RATE OF CONTINUED PRECIPITATION, WIND SPEED, AND OTHER FACTORS AFFECT THE LENGTH OF TIME THE FLUID WILL PREVENT THE REACCUMULATION OF FROST AND ICE. ICE-FREE SURFACES AND JOINTS CAN ONLY BE ASSURED THROUGH AN EFFECTIVE ANTI-ICING/DE-ICING OPERATION AND CAREFUL VISUAL INSPECTION OF THE HELICOPTER.

1. De-icing.

1.1. Remove heavy snow deposits by brushing prior to spraying.

1.2. Dilute de-icing/anti-icing fluid (C-394) concentrate with 50% water to produce a 50% solution. A 40% may be used for light icing.



DO NOT APPLY HOT (180°F) SOLUTION FOR DE-ICING DIRECTLY TO COLD WINDSHIELDS AND WINDOWS. USE FLUID AT OR NEAR 70°F.

1.3. Heat solution up to 180°F for most efficient and effective ice removal from airframe. Allow solution to cool before applying to windshield or windows.

1.4. Apply the diluted solution to helicopter by spraying, using suitable spraying equipment. Use a course stream to loosen and remove ice.

2. Anti-icing: Spray or wipe concentrated de-icing/anti-icing fluid (C-394) onto helicopter surfaces to prevent ice and frost from sticking.

NOTE

The concentrated fluid will freeze at approximately -25°F. Some ice or slush may form at or below -25°F.

3. Removal: Anti-icing/de-icing solutions and residue may be removed from helicopter by washing with fresh water or with a water/detergent solution followed by a rinse using fresh water.

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CHAPTER 32
LANDING GEAR

MAINTENANCE MANUAL

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List of Effective Data Modules

The listed documents are included in issue 007, dated 2025-11-18, of this publication.

C = Changed data module, N = New data module

Document title	Data module code / Seq number	Issue date	Applicable to
Landing Gear - General Description	DMC-412-A-32-00-00-00A-028A-A / 00695	2023-05-31	ALL
Landing Gear Assembly - Description	DMC-412-A-32-01-00-00A-042A-A / 00696	2023-05-31	ALL
Landing Gear Assembly - Removal	DMC-412-A-32-01-00-00A-520A-A / 00697	2023-05-31	ALL
Landing Gear Assembly - Inspection	DMC-412-A-32-01-00-00A-280A-A / 00698	2023-05-31	ALL
Landing Gear Assembly - Repair	DMC-412-A-32-01-00-00A-685A-A / 00699	2023-05-31	ALL
Landing Gear Assembly - Installation	DMC-412-A-32-01-00-00A-720A-A / 00700	2023-05-31	ALL
Skid Tubes - Description	DMC-412-A-32-01-01-00A-042A-A / 00701	2023-05-31	ALL
Skid Tubes - Removal	DMC-412-A-32-01-01-00A-520A-A / 00702	2023-05-31	ALL
Skid Tubes - Inspection	DMC-412-A-32-01-01-00A-280A-A / 00703	2023-05-31	ALL
Skid Tubes - Patch Repair	DMC-412-A-32-01-01-00A-685A-A / 00704	2023-05-31	ALL
Skid Tubes - Corrosion Repair	DMC-412-A-32-01-01-00A-685B-A / 00705	2023-05-31	SN: 38001-38999 or 39101-39999
Skid Tubes - Walkway Repair	DMC-412-A-32-01-01-00A-685C-A / 00706	2023-05-31	SN: 38001-38999 or 39101-39999
Skid Tubes - Installation	DMC-412-A-32-01-01-00A-720A-A / 00707	2023-05-31	ALL
Skid Shoes - Description	DMC-412-A-32-01-02-00A-042A-A / 00708	2023-05-31	ALL
Skid Shoes - Removal	DMC-412-A-32-01-02-00A-520A-A / 00709	2023-05-31	ALL
Skid Shoes - Inspection	DMC-412-A-32-01-02-00A-280A-A / 00710	2023-05-31	ALL
Skid Shoes - Repair	DMC-412-A-32-01-02-00A-685A-A / 00711	2023-05-31	ALL
Skid Shoes - Installation	DMC-412-A-32-01-02-00A-720A-A / 00712	2023-05-31	ALL
Fixed Step - Description	DMC-412-A-32-01-03-00A-042A-A / 00713	2023-05-31	ALL
Fixed Step - Removal	DMC-412-A-32-01-03-00A-520A-A / 00714	2023-05-31	ALL
Fixed Step - Installation	DMC-412-A-32-01-03-00A-720A-A / 00715	2023-05-31	ALL
Eyebolt (Ground Handling Wheels Attach Points) - Description	DMC-412-A-32-01-04-00A-042A-A / 00716	2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Eyebolt (Ground Handling Wheels Attach Points) - Removal	DMC-412-A-32-01-04-00A-520A-A / 00717	2023-05-31	ALL
Eyebolt (Ground Handling Wheels Attach Points) - Inspection and Repair	DMC-412-A-32-01-04-00A-280A-A / 00718	2023-05-31	ALL
Eyebolt (Ground Handling Wheels Attach Points) - Installation	DMC-412-A-32-01-04-00A-720A-A / 00719	2023-05-31	ALL
Landing Gear Crosstube Assemblies - Removal	DMC-412-A-32-01-05-00A-520A-A / 00720	2023-05-31	ALL
Landing Gear Crosstube Assemblies - Inspection	DMC-412-A-32-01-05-00A-280A-A / 00721	2023-05-31	ALL
Landing Gear Crosstube Assemblies - Repair	DMC-412-A-32-01-05-00A-685A-A / 00722	2023-05-31	ALL
Landing Gear Crosstube Assemblies - Installation	DMC-412-A-32-01-05-00A-720A-A / 00723	2023-05-31	ALL
Aft Crosstube Support Beam - Description	DMC-412-A-32-01-06-00A-042A-A / 00724	2023-05-31	ALL
Aft Crosstube Support Beam - Removal	DMC-412-A-32-01-06-00A-520A-A / 00725	2023-05-31	ALL
Aft Crosstube Support Beam - Inspection and Repair	DMC-412-A-32-01-06-00A-280A-A / 00726	2023-05-31	ALL
Aft Crosstube Support Beam - Installation	DMC-412-A-32-01-06-00A-720A-A / 00727	2023-05-31	ALL
Crosstube Supports - Description	DMC-412-A-32-01-07-00A-042A-A / 00728	2023-05-31	ALL
BHT Crosstube Forward Supports - Description	DMC-412-A-32-01-07-01A-042A-A / 00729	2023-05-31	ALL
Two-piece Support with U-Clamps (Crosstubes 205-050-400-055 and 205-050-403-031) - Removal	DMC-412-A-32-01-07-02A-520A-A / 00730	2023-05-31	ALL
One-piece Riveted Support (Crosstubes 412-050-040-101 and 412-050-046-101) - Removal	DMC-412-A-32-01-07-03A-520A-A / 00731	2023-05-31	ALL
BHT Crosstube Forward Supports - Inspection and Repair	DMC-412-A-32-01-07-01A-280A-A / 00732	2023-05-31	ALL
Two-piece Support with U-Clamps (Crosstubes 205-050-400-055 and 205-050-403-031) - Installation	DMC-412-A-32-01-07-02A-720A-A / 00733	2023-05-31	ALL
One-piece Riveted Support (Crosstubes 412-050-040-101 and 412-050-046-101) - Installation	DMC-412-A-32-01-07-03A-720A-A / 00734	2023-05-31	ALL
BHT Crosstube Aft Supports - Maintenance	DMC-412-A-32-01-08-00A-913A-A / 00735	2023-05-31	ALL
BHT Aft Crosstube Supports - Removal	DMC-412-A-32-01-08-01A-520A-A / 00736	2023-05-31	ALL
Abrasion Strip (Larson L101) - Removal	DMC-412-A-32-01-08-02A-520A-A / 00737	2023-05-31	ALL
BHT Aft Crosstube Supports - Inspection and Repair	DMC-412-A-32-01-08-01A-280A-A / 00738	2023-05-31	ALL
BHT Aft Crosstube Supports - Installation	DMC-412-A-32-01-08-01A-720A-A / 00739	2023-05-31	ALL
AAI Crosstube Aft (BL 0.0) and Forward (BL 14.0) Supports - Removal	DMC-412-A-32-01-09-00A-520A-A / 00740	2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
AAI Crosstube Aft (BL 0.0) and Forward (BL 14.0) Supports - Inspection and Repair	DMC-412-A-32-01-09-00A-280A-A / 00741	2023-05-31	ALL
AAI Crosstube Aft (BL 0.0) and Forward (BL 14.0) Supports - Installation	DMC-412-A-32-01-09-00A-720A-A / 00742	2023-05-31	ALL
AAI Crosstube Aft Clamp Assembly (BL 14.0) - Removal	DMC-412-A-32-01-09-01A-520A-A / 00743	2023-05-31	ALL
AAI Crosstube Aft Clamp Assembly (BL 14.0) - Inspection and Repair	DMC-412-A-32-01-09-01A-280A-A / 00744	2023-05-31	ALL
AAI Crosstube Aft Clamp Assembly (BL 14.0) - Installation	DMC-412-A-32-01-09-01A-720A-A / 00745	2023-05-31	ALL
Tail Skid - Description	DMC-412-A-32-20-00-00A-042A-A / 00746	2023-05-31	ALL
Tail Skid - Removal	DMC-412-A-32-20-00-00A-520A-A / 00747	2023-05-31	ALL
Tail Skid - Inspection and Repair	DMC-412-A-32-20-00-00A-280A-A / 00748	2023-05-31	ALL
Tail Skid - Installation	DMC-412-A-32-20-00-00A-720A-A / 00749	2023-05-31	ALL
Passenger Step (Electrically Power Operated) - Description	DMC-412-A-32-02-00-00A-042A-A / 00750	2023-05-31	ALL
Passenger Step - Removal	DMC-412-A-32-02-00-00A-520A-A / 00751	2023-05-31	ALL
Passenger Step - Inspection	DMC-412-A-32-02-00-00A-280A-A / 00752	2023-05-31	ALL
Passenger Step - Repair	DMC-412-A-32-02-00-00A-685A-A / 00753	2023-05-31	ALL
Passenger Step - Installation	DMC-412-A-32-02-00-00A-720A-A / 00754	2023-05-31	ALL
Ground Handling Wheels - Description	DMC-412-A-32-03-00-00A-042A-A / 00755	2023-05-31	ALL
Ground Handling Wheels - Inspection	DMC-412-A-32-03-00-00A-280A-A / 00756	2023-05-31	ALL
Ground Handling Wheels - Lubrication	DMC-412-A-32-03-00-00A-240A-A / 00757	2023-05-31	ALL
Wheel and Tire Assembly - Removal	DMC-412-A-32-03-00-01A-520A-A / 00758	2023-05-31	ALL
Wheel and Tire Assembly - Installation	DMC-412-A-32-03-00-01A-720A-A / 00759	2023-05-31	ALL
Tubeless Tire - Removal	DMC-412-A-32-03-00-02A-520A-A / 00760	2023-05-31	ALL
Tubeless Tire - Installation	DMC-412-A-32-03-00-02A-720A-A / 00761	2023-05-31	ALL
Tube Tire - Removal	DMC-412-A-32-03-00-03A-520A-A / 00762	2023-05-31	ALL
Tube Tire - Installation	DMC-412-A-32-03-00-03A-720A-A / 00763	2023-05-31	ALL
Pump Assembly - Description	DMC-412-A-32-03-00-04A-042A-A / 00764	2023-05-31	ALL
Pump Assembly - Removal	DMC-412-A-32-03-00-04A-520A-A / 00765	2023-05-31	ALL

Document title	Data module code / Seq number	Issue date	Applicable to
Pump Assembly - Installation	DMC-412-A-32-03-00-04A-720A-A / 00766	2023-05-31	ALL
Decals - Removal and Installation	DMC-412-A-32-03-00-05A-920A-A / 00767	2023-05-31	ALL
Ram Assembly and Trunnion - Removal	DMC-412-A-32-03-00-06A-520A-A / 00768	2023-05-31	ALL
Ram Assembly and Trunnion - Installation	DMC-412-A-32-03-00-06A-720A-A / 00769	2023-05-31	ALL
Axles - Removal	DMC-412-A-32-03-00-07A-520A-A / 00770	2023-05-31	ALL
Axles - Installation	DMC-412-A-32-03-00-07A-720A-A / 00771	2023-05-31	ALL
Bushings - Removal	DMC-412-A-32-03-00-08A-520A-A / 00772	2023-05-31	ALL
Bushings - Installation	DMC-412-A-32-03-00-08A-720A-A / 00773	2023-05-31	ALL

Landing Gear - General Description

A high crosstube landing gear kit may be installed (BHT-412-SI-40).

An emergency flotation auxiliary equipment kit may be installed (BHT-412-SI-2, BHT-412-SI-39, and BHT-412-SI-60).

An auxiliary, electrically operated passenger step may be installed on standard skid gear.

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Landing Gear Assembly - Description

The landing gear is fabricated from aluminum alloy tubes. The gear consists of two skids attached on the ends of two arched crosstubes. The forward crosstube is fitted with two supports or one upper riveted support and lower abrasion strip, which contact mating supports in helicopter structure. The forward crosstube is secured to the helicopter structure with two cap fittings. The aft crosstube has a support beam that contacts mating supports in helicopter structure and is secured to helicopter with two cap fittings similar to forward crosstube. The function of the support beam on aft crosstube is to reduce the possibility of ground resonance.

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Landing Gear Assembly - Removal

Procedure



A HOIST WITH SUFFICIENT CAPACITY TO LIFT ENTIRE HELICOPTER SHALL BE USED TO RAISE HELICOPTER CLEAR OF LANDING GEAR WHEN REMOVING OR INSTALLING LANDING GEAR AS AN ASSEMBLY.

NOTE

The landing gear can be removed as a complete assembly or individual components may be removed separately.

1. Disconnect power step electrical connector (31, Figure 1) at bottom of fuselage.

NOTE

After incorporation of TB 412-98-151, aft cap assemblies with a non-removable bumper will have shims attached at the bolt-up faying surface.

Superficial damage to cap fittings (21, 22), 32, and 45) is acceptable without repair. Loose, missing, and/or deteriorated rubber pads on cap fittings are cause for replacement of cap fittings.

2. Support the weight of the helicopter until it is light on the skids and remove bolts (34, 36, 38, 39, 41, and 43). Remove forward cap fitting (32).

3. Remove forward cap fitting (45) in the same manner outlined in previous step.

4. Remove bolts (10, 12, 14, 15, 17, and 19). Remove aft cap fitting (21).

5. Remove aft cap fitting (22) in the same manner outlined in previous steps.

6. If applicable, remove the weight-on-gear switch (3420S1) (60) as follows:

6.1. Remove the screws (58) and washers (59) from bracket assemblies (56). Lower the support assembly (57) and weight-on-gear switch (3420S1) (60).

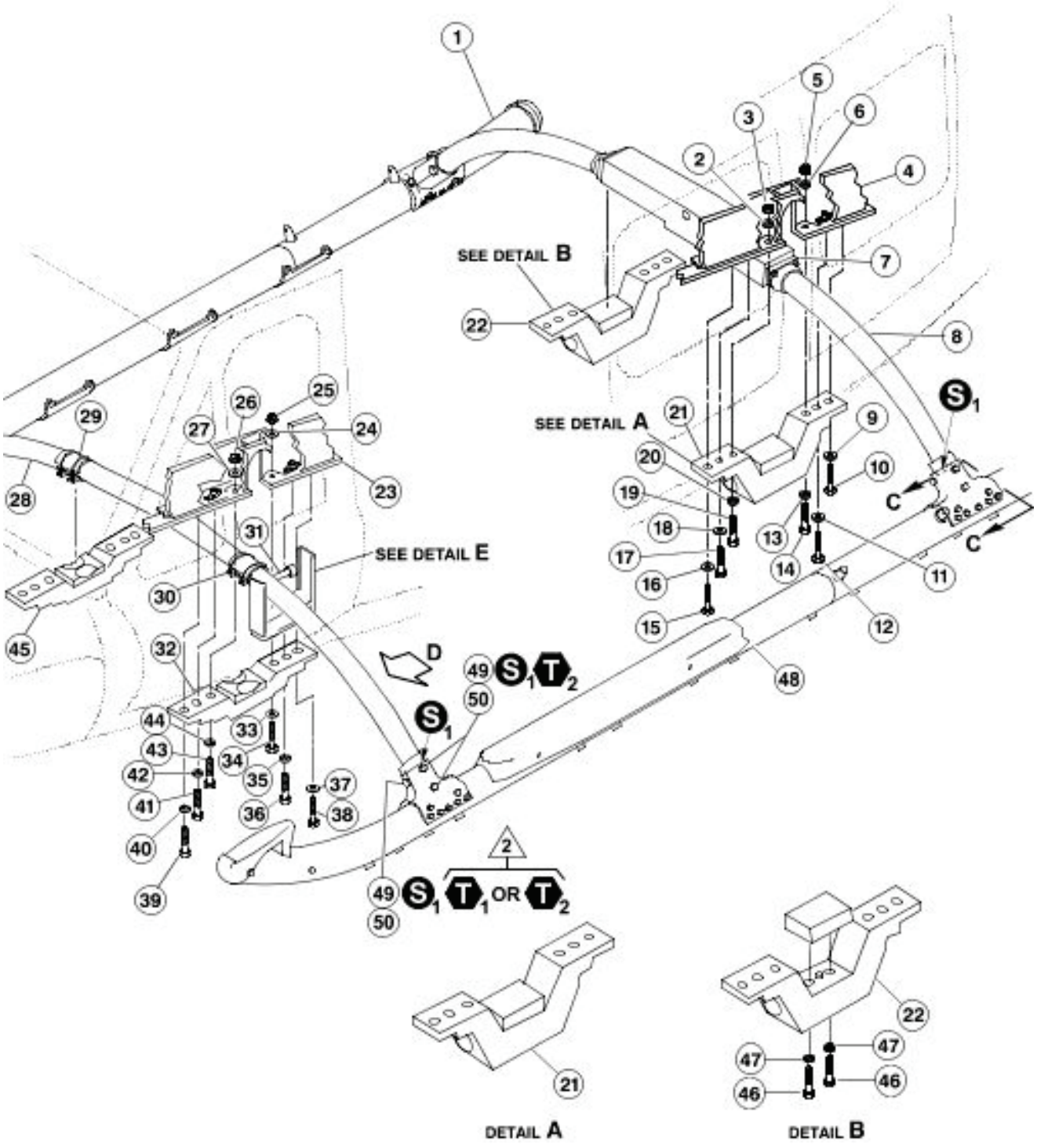
6.2. Using a cable tie or lockwire, temporarily secure the support assembly (57) and weight-on-gear switch (3420S1) (60) to one of the bracket assemblies (56). Make sure it is out of the way of the forward crosstube (28).

7. Raise the helicopter slowly while monitoring the forward and aft crosstubes coming out of the front and aft airframe tunnels, to prevent damage.

8. Raise the helicopter until it is clear of skid landing gear and remove landing gear.

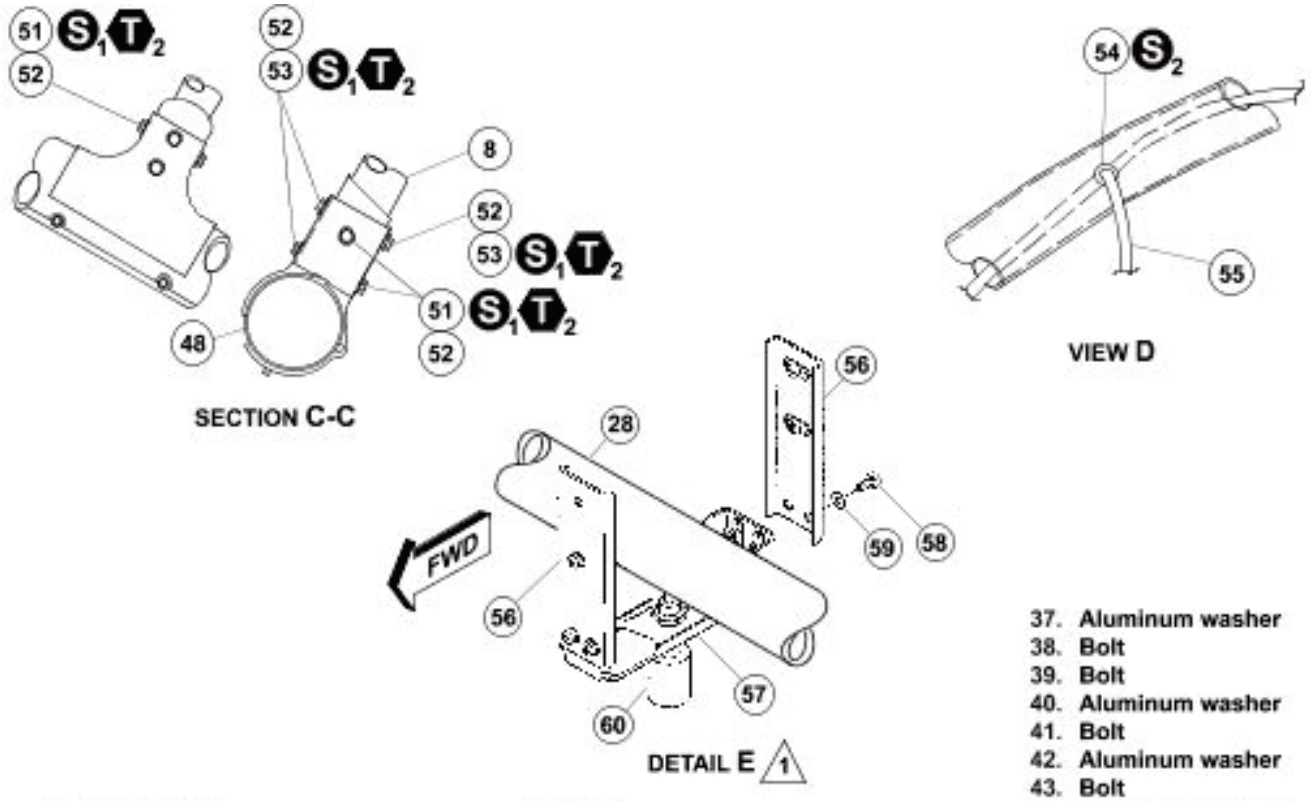
9. If necessary, remove the skid tubes (2 and 9, Figure 2) and the associated hardware from the crosstubes (412-A-32-01-01-00A-520A-A / 00702).

10. If necessary, remove the aft crosstube support beam (6) and attaching hardware from aft crosstubes (412-A-32-01-06-00A-520A-A / 00725).



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Figure 1. Landing Gear (Standard) (Sheet 1 of 2)



1. Landing gear
2. Aluminum washer
3. Nut
4. Landing gear support structure
5. Nut
6. Aluminum washer
7. Aft crosstube support beam
8. Aft crosstube
9. Aluminum washer
10. Bolt
11. Aluminum washer
12. Bolt
13. Aluminum washer
14. Bolt
15. Bolt
16. Aluminum washer
17. Bolt
18. Aluminum washer
19. Bolt
20. Aluminum washer
21. Aft cap fitting
22. Aft cap fitting
23. Forward crosstube support structure
24. Aluminum washer
25. Nut
26. Nut
27. Aluminum washer
28. Forward crosstube
29. Bearing and retaining support
30. Bearing and retaining support
31. Power step electrical connector
32. Forward cap fitting
33. Aluminum washer
34. Bolt
35. Aluminum washer
36. Bolt
37. Aluminum washer
38. Bolt
39. Bolt
40. Aluminum washer
41. Bolt
42. Aluminum washer
43. Bolt
44. Aluminum washer
45. Forward cap fitting
46. Bolt
47. Washer
48. Skid tube
49. Bolt
50. Washer
51. Bolt
52. Washer
53. Bolt
54. Grommet
55. Cable assembly
56. Bracket assembly
57. Support assembly
58. Screw
59. Washer
60. Weight-on-gear switch (3420S1)

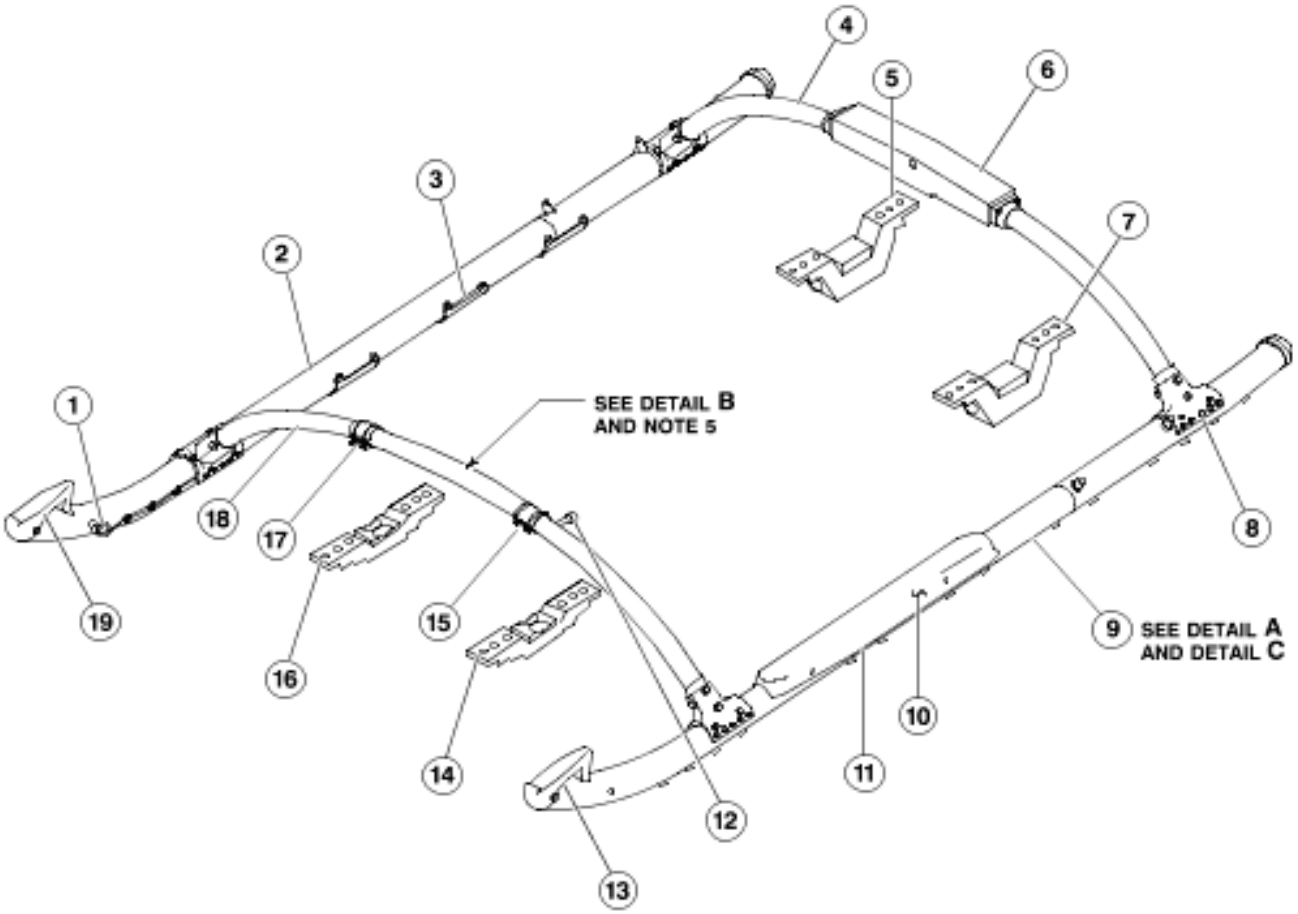
S_1 SEALANT (C-392) S_2 SEALANT (C-308) T_1 50 TO 70 IN-LBS (5.65 TO 7.91 Nm) T_2 160 TO 190 IN-LBS (18.08 TO 21.47 Nm)

NOTES

1. Weight-on-gear switch 3420S1 installed on helicopters S/N 36462 and subsequent. It is also installed on helicopters S/N 36248 through 36461 with the weather radar kit (412-705-034) or weight-on-gear retrofit kit (BHT-412-SI-84) installed.
2. For 1/4-28 bolts, torque per T_1 and for 3/8-24 bolts, torque per T_2.

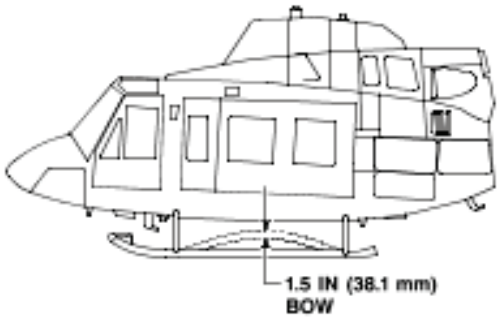
ICN-412-A-320100-B-97499-00002-A-001-01

Figure 1. Landing Gear (Standard) (Sheet 2 of 2)



STANDARD SKID LANDING GEAR ASSEMBLY
412-050-001-101, -103, -105, -107, -109 and -111

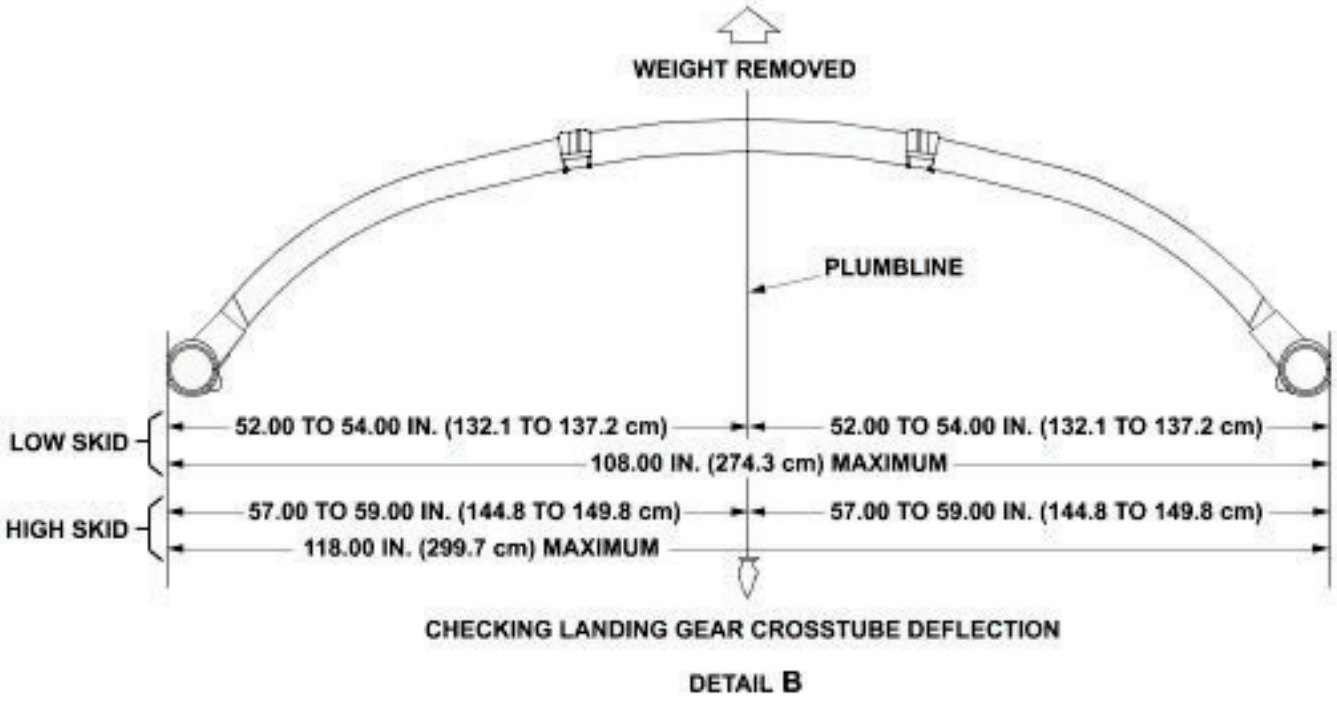
1. Tow ring
2. Skid tube
3. Skid shoe
4. Crosstube
5. Landing gear cap fitting
6. Aft crosstube support beam
7. Landing gear cap fitting
8. Saddle
9. Skid tube
10. Passenger step
11. Skid shoe
12. Passenger step electrical connector
13. Step
14. Landing gear cap fitting
15. Landing gear bearing and retaining support
16. Landing gear cap fitting
17. Landing gear bearing and retaining support
18. Crosstube
19. Step



DETAIL A

ICN-412-A-320100-B-97499-00003-A-001-01

Figure 2. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 1 of 9)

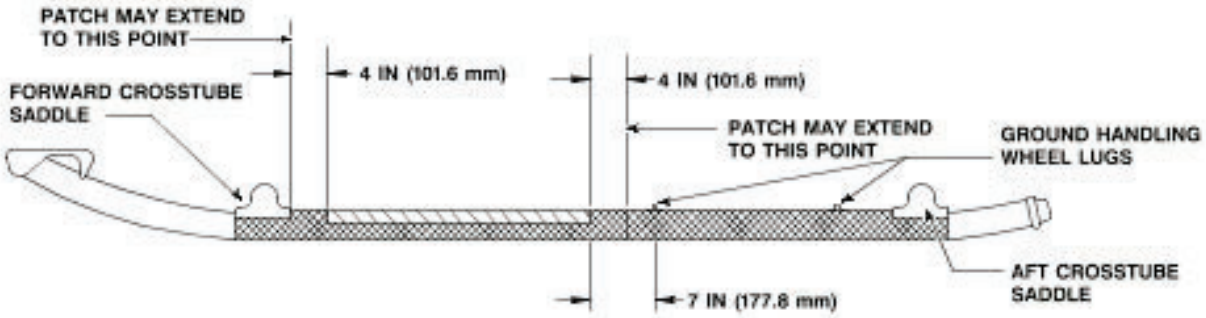


NOTES

1. Deformation (distortion) of either crosstube in excess of limits is cause for replacement of crosstube(s).
2. Crosstube may be deflected on one side only. If limit is exceeded, crosstube is considered scrapped and must be replaced.

ICN-412-A-320100-B-97499-00004-A-001-01

Figure 2. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 2 of 9)



DETAIL C
412-050-023

DAMAGE LOCATION SYMBOLS

TYPE OF DAMAGE	MAXIMUM DEPTHS AND REPAIR AREAS ALLOWED		
CRACKS ALLOWED	None	None	None
DENTS AND HOLES	None	4 inches (101.6 mm) Refer to paragraph 32-9.	Not Critical

ICN-412-A-320100-B-97499-00005-A-001-01

Figure 2. Landing Gear Assembly and Attaching Parts - Damage Limits (Sheet 3 of 9)